

IFSR International Series on Systems Science and Engineering

Robert Rosen

Anticipatory Systems

Philosophical, Mathematical,
and Methodological Foundations

Second Edition

 Springer

Anticipatory Systems

Volume 1

International Federation for Systems Research International Series on Systems Science and Engineering

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Volumes 1–6 were published by Pergamon Press

Robert Rosen

Anticipatory Systems

Philosophical, Mathematical,
and Methodological Foundations

Second Edition

With Contributions by Judith Rosen, John J. Kineman,
and Mihai Nadin

 Springer

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(Deceased)

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Addendum 1: Autobiographical Reminiscences

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ISSN 1574-0463

ISBN 978-1-4614-1268-7

e-ISBN 978-1-4614-1269-4

DOI 10.1007/978-1-4614-1269-4

Springer New York Dordrecht Heidelberg London

Library of Congress Control Number: 2011942919

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Printed on acid-free paper

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Foreword

This manuscript was written during the months between January and June in 1979. The writing was done under difficult circumstances. At that time, I had some reason to fear that I would no longer be able to personally pursue a program of active scientific research. Yet at the time, I felt (and still do) that I had arrived upon the threshold of some entirely new perspectives in the theory of natural systems, and of biological systems in particular. The issues involved were of sufficient interest and importance for me to wish them to be pursued by others, if they so desired. On the other hand, my own outlook and development are different from other people's, and have with justification been called idiosyncratic. Thus I resolved to try, while I knew I was still able, to set down a coherent narrative development, beginning with the most elementary matters, which I hoped would permit others to understand why I was doing the things I was doing.

This volume is the result. I have organized it around the concept of anticipation, which is fundamental in its own right, and which connects naturally to a cluster of other concepts lying at the heart of natural science and of mathematics. Strictly speaking, an anticipatory system is one in which present change of state depends upon future circumstances, rather than merely on the present or past. As such, anticipation has routinely been excluded from any kind of systematic study, on the grounds that it violates the causal foundation on which all of theoretical science must rest, and on the grounds that it introduces a telic element which is scientifically unacceptable. Nevertheless, biology is replete with situations in which organisms can generate and maintain internal predictive models of themselves and their environments, and utilize the predictions of these models about the future for purpose of control in the present. Many of the unique properties of organisms can really be understood only if these internal models are taken into account. Thus, the concept of a system with an internal predictive model seemed to offer a way to study anticipatory systems in a scientifically rigorous way.

This book is about what else one is forced to believe if one grants that certain kinds of systems can behave in an anticipatory fashion. I begin with an autobiographical account of how I came to believe that anticipatory systems were important, and why. I then proceed to an extensive discussion of the central concept

of anticipation, namely, the modeling relation between a system and a model, or a system and an analog. In the process of exploring this relation, and by exhibiting manifold and diverse examples of this relation, I try to establish the basic threads between the concepts of science and mathematics. I hope that this discussion will be of interest in its own right, and will also serve to illustrate that the approaches I take are not really as idiosyncratic as they might at first appear. Only when this background is laid do I turn to the concept of anticipation itself, and explore some of the elementary properties of systems which can anticipate. Here I well realize, the surface is barely scratched. But this was as far as I had been able to progress at the time, and the point of the book is to guide the reader to the surface.

For a variety of external reasons, the manuscript was not published immediately. However, none of the fundamental material has changed in the intervening five years, and I have not altered the original text. I have, however, added an appendix. It has turned out that the initial anxieties which generated the book were unfounded, and since that time, I have been able to push some of the fundamental implications of the relation between system and model much further. In the appendix, some of these more recent developments are sketched, and some of their rather startling implications are pointed out. I hope to enlarge this appendix into a separate monograph in the near future, giving full details to justify what is merely asserted therein.

I would like to take this occasion to thank the many colleagues and friends who have lent precious moral support as a beacon through dark times. Among them I may mention J.F. Danielli, George Klir, I.W. Richardson, Otto Röessler, Ei Teramoto, and Howard Pattee. I hope that this volume will justify at least some small part of their exertions on my behalf over the years.

Robert Rosen

[Note: Judith Rosen would like to add her thanks to Dr. Aloisius Louie for his early help in her efforts to republish this book and to Mr. Pete Giansante, for his assistance and encouragement towards that goal, as well. An enormous debt of gratitude is owed to Dr. George Klir, who once again was instrumental in helping to ensure that this work is made accessible and will now remain so for posterity.]

Preface to the First Edition

The present volume is intended as a contribution to the theory of those systems which contain internal predictive models of themselves and/or of their environment, and which utilize the predictions of their models to control their present behavior.

Systems of this type have a variety of properties which are unique to them, just as “closed-loop” systems have properties which make them different from “open-loop” systems. It is most important to understand these properties, for many reasons. We shall argue that much, if not most, biological behavior is model-based in this sense. This is true at every level, from the molecular to the cellular to the physiological to the behavioral. Moreover, model-based behavior is the essence of social and political activity. An understanding of the characteristics of model-based behavior is thus central to any technology we wish to develop to control such systems, or to modify their model-based behavior in new ways.

The essential novelty in our approach is that we consider such systems as single entities, and relate their overall properties to the character of the models they contain. There have, of course, been many approaches to planning, forecasting, and decision-making, but these tend to concentrate on tactical aspects of model synthesis and model deployment in specific circumstances; they do not deal with the behavioral correlates arising throughout a system simply from the fact that present behavior is generated in terms of a predicted future situation. For this reason, we shall not at all be concerned with tactical aspects of this type; we do not consider, for instance, the various procedures of extrapolation and correlation which dominate much of the literature concerned with decision-making in an uncertain or incompletely defined environment. We are concerned rather with global properties of model-based behavior, irrespective of how the model is generated, or indeed of whether it is a “good” model or not.

From the very outset, we shall find that the study of such global aspects of model-based behavior raises new questions of a basic epistemological character. Indeed, we shall see that the utilization of predictive models for purposes of present control confront us with problems relating to causality. It has long been axiomatic that system behavior in the present must never depend upon future states or future inputs; systems which violate this basic axiom are collectively called *anticipatory*, and are

routinely excluded from science. On the other hand, the presence of a predictive model serves precisely to pull the future into the present; a system with a “good” model thus behaves in many ways like a true anticipatory system. We must thus reconsider what is meant by an anticipatory system; the suggestion arising from the present work is that model-based behavior requires an entirely new paradigm, which we call an “anticipatory paradigm”, to accommodate it. This paradigm extends (but does not replace) the “reactive paradigm” which has hitherto dominated the study of natural systems, and allows us a glimpse of new and important aspects of system behavior.

The main theoretical questions with which we deal in the present work are the following: (a) What is a model? (b) What is a *predictive* model? (c) How does a system which contains a predictive model differ in its behavior from one which does not? In the process of exploring these questions, starting from first principles, we are led to a re-examination of many basic concepts: time, measurement, language, complexity. Since the modeling relation plays a central role in the discussion, we provide numerous illustrations of it, starting from models arising entirely within symbolic systems (mathematics) through physics, chemistry and biology. Only when the modeling relation is thoroughly clarified can we begin to formulate the basic problems of model-based behavior, and develop some of the properties of systems of the kind with which we are concerned.

It is a pleasure to acknowledge the assistance of many friends and colleagues who have aided me in developing the circle of ideas to be expounded below. A primary debt is owed to my teacher, Nicolas Rashevsky, who above all set an example of fearlessness in entering territory which others thought forbidden. An equally important debt is owed to Robert Hutchins, and to the Center which he created; it was there that I was first forced to confront the nature of anticipatory behavior. A third debt is to my colleagues at the Center for Theoretical Biology: James F. Danielli, Howard Pattee, Narendra Goel, and Martynas Ycas, for their intellectual stimulation and support over the years. Gratitude must also be expressed to Dalhousie University, where thought and the leisure to think are still valued, and especially to my colleague I.W. Richardson.

Robert Rosen

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Preface to the Second Edition: The Nature of Life

What you are holding in your hands is a book about biology. About LIFE. It is a book that describes and elucidates, in scientific terms, the causal forces underlying the unique nature of all living organisms. It presents a set of rigorous and logical interlocking ideas which make clear what is generating the familiar behavior patterns we can observe in ourselves and all around us, everyday, and which characterize “life” as a systemic quality or feature of a system.

Life is Anticipatory

Living organisms have the equivalent of one “foot” in the past, the other in the future, and the whole system hovers, moment by moment, in the present – always on the move, through time. The truth is that *the future* represents as powerful a causal force on current behavior as the past does, for all living things. And information, which is often presumed to be a figment of the human mind or at least unique to the province of human thought and interaction, is actually an integral feature of life, itself – even at the most fundamental level: that of system organization.

These are all findings which are described and elaborated here and, in my opinion, this particular book represents the most important of my father’s scientific discoveries. The development of this aspect of his larger theoretical work on relational complexity (lately being called “Relational Science”) is able to explain what have, heretofore, remained the most baffling and inexplicable of organismic living behaviors. It also demonstrates how and why the clues have been missed by science for so long. The current, purely reactive, paradigm for science is able to do many things but it cannot be expected to help us adequately with problems and questions pertaining to living systems if living systems are not merely reactive. This becomes a critical issue because science represents the set of tools humanity uses for exploring and understanding ourselves, our universe, and our place in the web of life here on Earth. We need to be able to trust our tools to help us solve problems in the biosphere without generating side effects worse than the problems, themselves.

Every living organism comes into being with a system-based value for *health*. Emerging from that value for health is a guiding principle that is equally individual: *optimality*. The functional capabilities of life – metabolism and repair – are entirely guided by these two values. What is clear from the outset is that the values pre-exist the business of living. Perhaps *time* is not quite as linear as we have always presumed it to be. My father's view, in fact, was that, "Time is complex."

According to Robert Rosen, the means by which a living system is internally guided and controlled involves encoded information acting as an interactive set of models – of self, of environment, and of relations between the two... through time. These models have the capacity to *predict* next behavior (of self and/or of native environment) based on current behavior. The fact that these are model predictions, as opposed to prescience or prophesy, can be proven by studying the peculiar error behaviors that arise when the encoded information being used in the models no longer accurately represents the systems it was encoded from. For example, if the environment changes, quickly and radically, from the way it has been throughout the recent evolutionary past, the models will no longer be able to reliably predict next behavior of environment from current behavior.

This is what happens when an organism is moved from its native environment and transplanted to some new environment, as in the case of "annual" plants and flowers sold in my neighborhood in Western New York State. Or tropical plants and trees grown indoors in pots, here. Many of these plants are not really annuals, but perennials which can live for many years, sometimes decades and (in the case of trees) centuries. However, they come from environments that never have had a winter like ours, and therefore have no information about it, either. Native plants begin to enter dormancy in mid to late August, here, triggered by various environmental cues such as changing day/night length, but the "annuals" bloom merrily away in pots and in the garden right up until the first freeze kills them outright.

Another proof that these are models can be demonstrated by observing that they can be "fooled." The horticultural industry uses this situation to very good effect: Producing Easter lilies blooming in time for the Easter market, Poinsettias blooming in time for Christmas, and so on. All that is required to trigger initiation of the bloom cycle is to mimic the behavior of their native environment just prior to their natural bloom time. If we have figured out what the triggers or cues are, we can merely mimic those and achieve our ends. It can be quite specific: day or night length, soil or air temperature, temperature differential between day and night, a period of dryness of a specific length... each organism has its own triggers based on the encoded information in its models. All "instinctive" behavior of living organisms is based on the activity of such internal predictive models, generated from encoded information within their own systemic organization. To observe and learn about the annual migration of Monarch butterflies in North America gives us enough evidence to put us in awe of just how detailed the encoded information can be and how powerful is the guiding action of these internal models on the behavior patterns of all living things.

There are stark dangers embodied in this situation, however, which will become clear as you read this book. The dangers stem from the fact that many of the encoded models (and/or the information from which they are constructed) are either not able to be changed within a single organism's life time or else they change too slowly to be able to avoid disaster in a rapidly changing world. **There is no way to know, from within a model, that the system it was encoded from has changed radically. The model will keep on making predictions using wrong information – and the organism will still be guided, partly or entirely, by those predictions.** If the predictions are inappropriate, the behavior will similarly be inappropriate – perhaps to the point of mismatches that prove lethal to the organism. Because organism species within an ecosystem are so interlinked in their requirements and dependencies, the death of significant numbers of one species can initiate further rapid changes in the behavior of the local environment, which can ultimately cause rapidly escalating cascades of extinctions.

This is the Achilles heel, the innate vulnerability, of all anticipatory systems. **With human-induced changes to the composition of Earth's atmosphere happening at an unprecedented pace over the past two hundred years, and the further unknown changes which are likely to be caused by them, we would do well to pay very careful attention to the warning that is inherent in these facts.** Any model-based guidance system will only be as good as the encoded information it uses. If a model is constructed using inaccurate information, its predictions will be unreliable – and this is as true in science as it is in the guided behavior patterns of organisms.

Indeed, because human beings are living organisms, we are also anticipatory systems. The lessons we can learn and apply from this will impact everything. Something as ordinary and commonplace as *how we construct our food pyramid* should be based on what our bodies have encoded as “food”. My hope (my prediction) is that the ramifications of these ideas can expand the paradigm of science itself. The effects of doing so would benefit everything from medical science to psychology, social science, political science, economics – in fact: anything that involves human physiology, human thought and learning, and human interactions with each other or with the biosphere.

The nature of the human mind, in particular, has so far eluded most of the attempts we have made in science to understand it. We seek to comprehend both the nature of our own consciously aware mind and its origins – how it came into being. The final ground-breaking aspect of this book is that it realizes, within the fundamental theory being developed here, that the similarity between life and mind is simply that both are anticipatory systems. The peculiar, anticipatory nature of the mind, at the behavioral and physiological level, was described philosophically a couple thousand years ago (for example, in Buddhist teachings like the Satipathanna Sutta) but it has never been explainable via science, until now. Anticipatory Systems Theory elucidates how it can be that both body and mind run on the same principles of model-based guidance and control.

In that light, it becomes clear that the only reason the human mind can anticipate is because life was already *that way*. Thus, the human mind is merely an evolutionary concentration of the same information-sifting, encoding, and model-

building capacities of living system organization. It is a model-building tool that is capable of error-checking and re-encoding faulty models – but in real time versus evolutionary time. The accuracy and speed of an organism’s ability to model and predict would naturally be something that selection pressures could act on. The boon to survival that human intelligence and imagination represents – allowing us to work around all manner of physical limitations in our environments and ourselves – certainly correlates with how overpopulated humanity has become.

However, the recognition in recent years of the need for human societies to be sustainable in order for us to maintain our own health and well-being, over time, brings with it an awareness that there is an acute and urgent need for scientific answers that we can rely on pertaining to biological questions, problems, and issues. When time is short, trial and error is hardly the most efficient or productive mode for sorting out our choices. Because our own welfare is inextricably bound up with the welfare of the biosphere, humanity will need to consider a much larger set of values for optimality than we have ever needed to use in the past. This book will be essential for helping humanity expand the scientific paradigm in such a way that it can finally be trusted to answer biological questions *accurately* and give us scientific models and model predictions that can *reliably help us in choosing the most optimal pathways towards a healthy and sustainable future*.

It is for all these reasons and more that I have worked to get this book republished in the form of an expanded Second Edition. An entire new area of science has already begun to spring up around this work, but without access to the theoretical underpinnings to guide its growth, I fear that it will be prone to develop improperly. I also wanted to include some of the new science, recruiting scientists I know personally who are developing it; allowing them to describe what they are doing and show the applicability of it. The true test of any theory is to put it into practice and see if it holds – see if it generates results and check that the results are beneficial. I think the evidence is conclusive. I leave it to the reader to decide whether I have my own models properly encoded, or not.

If it is true that knowledge is power, then this book is powerful, indeed. Use it wisely, and well.

Judith Rosen

[Note to Readers: Do not be intimidated by the mathematical notation in this book! In discussions with my father on this subject, he said that the mathematics represent *additional illustration of ideas already described in prose*. It was his form of “bullet-proofing” as well as whatever value could be made available to readers from absorbing the same ideas in a different way. I specifically asked him whether one needed to understand the math to fully comprehend the work and he said, “No.” Therefore, if advanced math is something you have not been trained for, concentrate on the prose and ignore the mathematical illustrations. You have Robert Rosen’s own assurances that you will not be missing any essential information by doing so.]

Prolegomena: What Speaks in Favor of an Inquiry into Anticipatory Processes?

Mihai Nadin

The book you hold in your hands is part of a larger intellectual endeavor. It can be read and understood as a stand-alone monograph. Yet to fully realize its predicaments and appropriately understand it, the reader would be well advised to become familiar with the entire trilogy: *Fundamentals of Measurement and Representation in Natural Systems* (FM, 1978); *Anticipatory Systems. Philosophical, Methodological and Mathematical Foundations* (AS, 1985); and *Life Itself. A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life* (LI, 1991). In their unity, they define Robert Rosen, their author, as an original and provocative scholar. Each of the three books is a self-contained monograph; but only understood in their unity, indeed, as a trilogy, do they provide the reader the opportunity to perceive their interconnectedness. To measure life is different from all other forms of measurement; to understand life is to acknowledge complexity. And once complexity is accounted for, anticipatory processes, corresponding to a comprehension of causality that transcends determinism, can be accounted for as a characteristic of the dynamics of the living.

Preliminaries

Mathematics and biology are subjects that rarely blend. This does not have to be so. Quite a few mathematicians and biologists have brought up their own frustration with each other's methods. Indeed, some biologists – at least those with a “traditional” background – would confirm that their passion for science found its reward in a domain of inquiry that was less mathematical than others. Computation, which after all is automated mathematics, is affecting a change, but not yet in the sense that biologists suddenly become passionate mathematicians. Robert Rosen's entire work – and in particular his contributions to the understanding of what defines the living – utilizes mathematics and biology. This is a difficult position: mathematicians are not necessarily conversant in matters outside their knowledge domain (which they consider universal); biologists prefer descriptions different

from those specific to mathematics. Given the difference between mathematical and biological knowledge, and given the fact that the respective practitioners are not conversant in each other's "language," Rosen, educated and fluent in both mathematics and biology, had a tough time in seeing his work accepted, and an even tougher time in having it properly understood.

These preliminary remarks are undoubtedly connected to the choice I made to write *Prolegomena* – not an Introduction – to the second edition of Rosen's AS. It was Immanuel Kant who, after his disappointment with a superficial and less than competent review of his *Kritik der reinen Vernunft* (1781; *Critique of Pure Reason*, as it is called in English), wrote *Prolegomena to Any Future Metaphysics That Will Be Able to Present Itself as a Science* (1783). Among the many questions he raised, "How can the intuition of the object occur before the experience of the object?" is of particular significance to the subject of anticipation. *Prolegomena* are to be understood as preliminaries for defining a subject. As a matter of fact, Kant, who himself indulged in mathematics, paid quite a bit of attention to *prolepsis*, which predates the word *antecipere* and the subject of *anticipation*. It represents the pre-conception, i.e., the preliminary understanding that eventually leads to our knowledge of the world (cf. Epicurus and the so-called *Stoa* – phases in Stoicism – whose ideas were conveyed indirectly through Roman texts). Although the concept of *prolepsis* is associated with anticipation, I will not venture to claim that anticipation as a knowledge domain originates 300 years BCE, or even from Kant. His *Prolegomena* are rather an inspiration for framing Rosen's AS. We know how the first edition was reviewed; we also know how it fared and, moreover, why a second edition is necessary. (Incidentally, Kant produced a second edition of his *Critique of Pure Reason* in 1787, i.e., six years after the first. The *Prolegomena* were very useful in clarifying some of the ideas and misconceptions of his book on reason.)

In the spirit in which Kant was mentioned, it is significant that Rosen's "trilogy" Measurement – Anticipation – Life (cf. also Louie 2008, p. 290) be juxtaposed to that of the famous idealist philosopher: *Pure Reason* – *Practical Reason* – *Judgment* (Kant 1781, 1788, 1790). The architecture of the arguments, leading to a coherent view, can only be properly recognized if we look at the whole. Admiring the entrance to a cathedral (or any other architectural creation) is never so convincing as the exploration of the entire edifice and the experience of the space and time it encapsulates.

In these preliminaries, I would also like to associate Kant's question regarding the role of intuition, as an anticipation, to some recent scientific reports. "Adaptive prediction of environmental changes by microorganisms" (Mitchell et al. 2009); or "Early-warning signals for critical transitions" (Scheffer et al. 2009) or "Stimulus Predictability Reduces Responses in Primary Visual Cortex" (Alink et al. 2010), to name recent works, are answers, even if only indirect, to his question. They qualify, however, as examples, not as foundational theories, of how far research in anticipation reaches out in addressing various aspects of life and living. The fact that even the most rudimentary forms of life (microorganisms) display anticipatory characteristics clearly aligns with Rosen's description in AS of the monocoell and its phototropic behavior. Furthermore, perception processes, in particular those related

to seeing, allow for an optimum in brain functioning (“energy saving” feature). They correspond to our understanding of anticipation as an underlying process from which the evolutionary process benefits.

But before Kant, the future – which is essentially the focus of anticipation (regardless of how we define it) – was of interest to the most intriguing mathematics. Were mathematicians such as Blaise Pascal, Pierre de Fermat, or Jakob Bernoulli addicted to gambling (of one sort or another), or actually intrigued by the possibility of delivering predictive or even anticipatory descriptions? Winning the lottery could be rewarding; but so can the prediction of a successful course of action. Bernoulli’s *Ars conjectandi* (which [Collani 2006](#)) brought to my attention as a precursor of anticipatory modeling), and for that matter game theory, in particular the work of John Forbes Nash, Jr. years later (which afforded him a Nobel Prize) are yet other examples impossible to ignore as we define a context for understanding what anticipation is. The cultural context is very rich and obviously extends to art and literature. The examples mentioned are only indicative of the variety of perspectives from which we can examine the subject of anticipation. With the exception of Bernoulli’s book, whose object is conjecture in its broadest sense, in our days we encounter a rich variety of examples of particular predictive efforts. Humankind is pursuing increasingly risky endeavors: space exploration, genetic engineering, deep ocean oil exploration, financial instruments (e.g., derivatives). Therefore, the need to develop predictive procedures for evaluating the outcome increases.

Particular perspectives and specific areas of investigation (cognitive science, artificial intelligence, ALife, synthetic life, economics, aesthetics, among others) – which will be mentioned in the Prolegomena – are rather illustrative of the ubiquity of anticipatory processes. But they are not a substitute for a scientific foundation, which can only be multidisciplinary. Rosen’s *AS* conjures not only philosophy, mathematics, and methodology (identified as such in his book’s subtitle), but also developments in all those sciences that made possible the current focus on the living. Therefore it qualifies as a theoretic contribution upon which the community of researchers interested in anticipation can further build. For instance, no one can ignore that advances in understanding a variety of forms of creativity (scientific, technological, artistic, etc.) have allowed for a higher acceptance of the notions associated with anticipatory systems. However, such progress has not yet resulted in a better understanding of creativity as an expression of anticipation.

In March 2008, Springer Publishers contacted me regarding its intention to publish a second edition of *AS*. Since that time, the “state of the art” in anticipation research has continued to change on account of a rapidly increasing body of experimental evidence (cf. [Nadin 2010a](#), [Nadin 2011a](#), [2012a](#)), as well as research on the conceptual level. It became obvious to me that to write an Introduction in the classical form would not do justice to the book, to the author, to the subject. And it would not contribute to the type of research that led Rosen to write his book as part of the trilogy mentioned above. *Anticipatory Systems* is being republished in a new context. It can, and should be, read having *FM* and *LI* in mind. Albeit, the second edition is not only the occasion to correct errors (characteristic of publications in the pre-word processing age), but also to understand the impact of the notions it

advanced. The assignment implied a commitment I was not sure that I was willing to make, even after having argued in favor of a second edition of *AS*. My own research (2009a, 2010b, 2010c) takes the best of my time and effort. Many colleagues (whose opinion I asked for) guided my own decision-making and generously provided extremely useful references. In the meanwhile, and on the initiative of George Klir, General Editor of the Series, I prepared a special issue on Anticipation for the *International Journal of General Systems* (Nadin 2010a) – the Journal that previously published Rosen’s articles. This was followed by another thematic issue on anticipation (Nadin 2011b, Nadin 2012b). And when, finally, the Prolegomena became an inescapable project, I was able to build upon the effort made so far in understanding Rosen’s work, the context in which it was published, its impact. Distinguished colleagues who respect Rosen’s work helped. Given my own long-term dedication to the subject, it is clear that these Prolegomena (in the Kantian tradition) are meant to provide a context in which the work of many, together with my own, will be related to that of Robert Rosen.

Allow me a presumption here: The fact that there is so much to consider in defining the context would have pleased him. I wish that Robert Rosen himself could have written the preface to this second edition of his book; but I will not, by any means, second-guess him.

The Path to the Publication of *AS*

The perspective of time and the evidence of increasing interest from the scientific community in understanding anticipatory processes speak in favor of describing the perspective from which anticipation was initially defined. Moreover, the ever-increasing variety of understandings associated with the concept invites clarification. In a nutshell, this is all I want to provide in these Prolegomena. Twenty-seven years ago, Robert Rosen’s book on anticipation first reached its readers. It was actually written in the first six months of 1979, but did not have an easy path on the way to publication. My own book, *Mind – Anticipation and Chaos* (1991) introducing the concept of anticipation was published six years later (it was also written well before publication).

Rosen’s scientific legacy is the result of scientific commitment that excluded compromise as a path to acceptance or recognition. It also excluded self-delusion: In his will, Rosen left his work (and library) to Judith, the daughter who stood by his side until his untimely death, and who does her very best to share what she learned from her father with those who are interested. The testament came with the following proviso: “You don’t have to do anything with it, Jude. In my opinion, either it continues based on its own merits or it dies a natural death. If it can’t stand on its own merits, then it doesn’t deserve to,” (Rosen 2008). For the reader inclined to see in the publication of the Second Edition of *AS*, a contradiction of Robert Rosen’s position, only this need be said: Many young (and not so young) researchers asked for it. Interest in someone’s writing is an acknowledgment of merit. It is worthwhile mentioning that the book continues to be frequently cited

(sometimes through secondary sources). Given the limited access to the original book, misinterpretations, prompted by less than adequate citations (errors in the first edition transmitted from one quoting author to another) sometimes led to confusion. One of those who helped me in preparing this text (Olaf Wolkenhauer, Systems Biology, Rostock University) recently wrote to me: “We are getting tired of guessing what Rosen meant.” At least as far as *AS* is concerned, no guessing should replace consulting the work itself.

How *AS* came to be published deserves a short note because science and integrity, associated with originality, cannot be conceived as independent of each other. Based on some accounts, Rosen’s manuscript was reviewed by an old nemesis of Rashevsky, and publication was quashed. Six years later, it eventually became the first volume in the *International Series on Systems Science and Engineering*, published by Pergamon Press, under the heading of the International Federation for Systems Research. The Federation itself (founded in 1980) was no less controversial. In the spirit of the activity of those who set the foundation for a system’s approach (Wiener 1948, Ashby 1956, Bertalanffy 1968), it challenged excessive specialization, advancing a holistic view of the world. George Klir, its first president, practiced a systems approach focused on knowledge structures. He met Robert Rosen in 1970, after visiting von Bertalanffy at SUNY–Buffalo (where von Bertalanffy, professor in the School of Social Sciences introduced him to Rosen). They remained in touch, and between 1971 and 1972 they explored the feasibility of a new journal. In 1974, the first issue of the *International Journal of General Systems* was published. In acknowledging Rosen (“He was in some sense a co-founder of the Journal”), Klir makes note of the fact that Rosen was an active Member of the Board and published constantly from the first issue on. In 1975, an opening at the Department of Systems Science at SUNY–Binghamton prompted Klir to recommend to his Dean, Walter Lowen, that the University recruits Robert Rosen. (Rosen described a visit by Klir and Lowen to Buffalo in 1974.) When asked, Rosen, a scientist of high integrity, would not consider moving without his closest collaborators (Howard Pattee and Narendra Goel) at the Center for Theoretical Biology at SUNY–Buffalo. In the end, as the Center was abolished in 1975, Rosen accepted the generous offer from Dalhousie University (Killam Professor, “like a five-year sabbatical”) in Nova Scotia, Canada, while Pattee and Goel moved to Binghamton; they all remained in touch. In 1978, Klir published Rosen’s *FM* in the General Systems Research Series (also a first volume in the series) of Elsevier. Trusting Klir, Rosen finally submitted *AS* to him. Klir realized that it was a difficult text, by no means typical of publications on biology, systems theory, mathematics, or philosophy – disciplines that were integrated in this text – of that time, or of any time, for that matter. His interest in Robert Rosen’s work was the result of numerous interactions in which scholarship and character proved flawless. One of the reasons I accepted to write these Prolegomena is because George Klir deserves our respect for his dedication to scientific integrity. He realized the need for this second edition and helped in making it happen.

As far as my inquiry regarding the beginnings of Rosen’s interest in the subject of anticipation shows, the early 1970s were pretty much the time of his attempts to specifically address the subject. But let’s be more precise, not for the sake of

archival passion, rather in order to place the subject in the larger framework of his research. Since 1957, when Rosen joined Rashevsky's Committee on Mathematical Biology at the University of Chicago, he was prepared to make relational biology the reference for his own theoretic work. Rosen's discovery of the "(M,R)-Systems" was the starting point. In order to define the living in its concrete embodiment as organism, he advanced a class of relational models called "(M,R)-Systems". M stands for metabolism; R for repair; and the system defines relational cell models that describe organisms. In this view, the object of inquiry of biology is the class of material realizations of a particular relational structure expressed in the "(M,R)-Systems". From here on, a large body of publications testifies to the intellectual effort of defining what life itself is. A material system is an organism if and only if it is closed to efficient causation (Rosen 1991). It is worth noting that the definition is focused on causality (in the Aristotelian tradition of material, formal, efficient, and final causes); also, that life is embodied in organisms. The matter of organisms is important; but what defines life is *what* organisms actually do – moreover, *why* they do it. In his *LI*, Rosen dealt with the "necessary condition, not sufficient one, for a material system to be an organism." Complexity, which is characteristic of life, and which in the final analysis explains anticipation, "is the habitat of life ... not life itself." Let us take note that complexity is defined ontologically – as pertaining to existence – not epistemologically – as pertaining to our knowledge of what exists. Rosen ascertained that something else is needed to characterize what is alive from what is complex (Rosen 2000). The simplest "(M,R)-system" is one of replication, repair and metabolism entailing one another. It is, of course, a formal representation of the living (organism, cell).

Between *AS* and his next book, Rosen's original contributions result from a process of discovery impressive in its breadth. His subject was neither more nor less than what life is, after all. The manuscript of *AS* submitted to Klir was conceived over a long time, with many of its hypotheses subjected to discussions in various colloquia, seminars, and conferences. The original manuscript (typed) – preserved by his most talented student (Rosen's own characterization of Aloisius Louie) – went through a judicious editing process. In science, new perspectives rarely break through on account of their soundness. And even less on account of their novelty. Like anyone else in society, scientists are captive to their respective views and quite often unwilling to allow opposing views to enter the arena of public debate. A rather nasty review of one of Rosen's books, *Dynamical System Theory in Biology*, Vol. I, 1970 (Cohen 1971), made Rosen give up publication of the second volume (the manuscript still exists), and even decide to never again write or publish textbooks. The review states, "Physics envy is the curse of biology," (p. 675). Rosen was not in the envy business. Respectful of physics, and working closely with physicists, he rather questioned a physics-based foundation of biology and tried to advance an alternative foundation. Let it be spelled out: In this respect, his approach and my own have much in common.

This brings up the second reason for accepting Springer's invitation: Robert Rosen himself. No, I was not one of his students or colleagues. And I have to confess that when I first heard his name – brought to my attention in the spring

of 1992 by a former graduate student of mine, Jeffrey Nickerson (currently Director of the Center for Decision Technologies, Howe School of Technology Management, Stevens Institute of Technology, Castle Point-on-Hudson, NJ) – I could not relate it to anything of interest to me at that time. My book, *Mind – Anticipation and Chaos*, the result of inquiries into cognitive processes and semiotics, gave me the illusion of being the first to project the notion of anticipation into scientific dialog. In the 1970s, my interest in the scientific foundation of aesthetics (information aesthetics, in particular) prompted the realization that artistic endeavors, and by extension creativity, imply anticipatory processes (cf. Nadin 1972). But, as I would discover, that was also the time when Rosen started his own inquiry into the relevance of predictive models in society. I was wrong in my illusion of being the first, but not unhappy that an intellectual of his worth considered anticipation a subject of scientific interest. The relative simultaneity of our independent interest in the subject is probably not accidental. In the second part of these Prolegomena, I shall return to the “broader picture,” i.e., to all those persons who at that particular time (after a world war with millions of victims) were asking questions pertinent to a future that seemed unstable (the cold war, nuclear weapons, economic and political instability). It is by no means incidental that such matters preoccupied researchers in the “free world” (as the West defined itself) as well as in the self-described “communist paradise” (the Eastern Block and the Soviet Union). That was a time of many events in which the future begged for some acknowledgment. My library research (at Brown University, Ohio State University, and even at OCLC in Columbus, Ohio) on the key words *anticipation*, *anticipatory systems*, *anticipatory processes* did not identify anticipation as a subject. Nickerson’s own inquisitive energy helped me discover Rosen’s book, which I wish I had read before writing mine. The result of this late discovery was my letter of March 24, 1995 to Rosen. He answered, “I wasn’t aware of your book either. It looks beautiful. I think books should be works of art.” He went on to say:

I’ve obliquely been continuing my work on anticipation, which as you know is a very large subject; in fact, a whole new way of looking at the world. It played a major role in developing a concept I called *complexity* [italics his].... I have another book (*Life Itself*, Columbia University Press, 1991) in which this aspect is developed.

We exchanged copies of our books; and I was hoping that we would meet. It was not to be. During my sabbatical at Stanford University (Spring, 1999), Dr. Daniel Dubois, of the now firmly established Center for Hyperincursion and Anticipation in Ordered Systems (CHAOS), at the University of Liège, Belgium, invited me to prepare a presentation for the Conference on Computing Anticipatory Systems (CASYS, 1999). I dedicated my text to Robert Rosen’s memory. (Later, during a lovely private conversation, Dubois mentioned that he was not aware of Rosen’s premature death, although he knew that Rosen had been severely ill.) Yes, I still feel a sense of loss. I owe it to his memory to write these Prolegomena, even though (as stated above) I wish that he himself could have prefaced this second edition.

It is probably less important to the science to which we are dedicated than to maintaining a sense of academic interconnectedness that such considerations

be shared. The reason I brought them up is straightforward: Many in academia respected Robert Rosen, but the extent to which academia accepted his work by no means reflected an understanding of its originality. It is worth mentioning that Mickulecky – who identifies himself as a colleague – called him “the Newton of biology,” (Mickulecky 2007), but it is also inconsequential. So is the title “Biology’s Einstein” (Staiger 1990). Hyperbole does not carry value judgments, rather emotional content. As we shall see, only two authors reviewed *AS*: one was a graduate student, stimulated to do so by George Klir; the other was a Hungarian scientist interested in control theory. A prior book (*Theoretical Biology and Complexity*, 1985b), edited by Rosen, in which Rosen’s own text brought up anticipation, was also barely reviewed, and mostly misunderstood. Even now, when anticipation is no longer a concept prohibited in scientific discourse, hundreds of articles are published with nominal reference to *AS*, and only very rarely with the understanding of Rosen’s comprehensive epistemology. This should not be construed as an expression of anti-Rosen sentiment. Both hyperbole and marginal acknowledgment define yet another reason to give researchers the foundational work that gives meaning to their inquiry and helps in advancing research in the field. The very active discussion on the Rosen mailing list (rosen@home.ease.lsoft.com) on aspects of Rosen’s writings – Rosen would at most have glanced at such mailing lists – will probably benefit from the publication of this book.

In the Perspective of Relational Biology

The purpose of these Prolegomena is by no means to explain *AS*. It speaks for itself, in a crisp but not facile language. After so many years since its publication, it continues to stimulate contributions – theoretic and experimental – of significance. But the context has changed. The life sciences pretty much take the lead today; no need to envy physics. (In a recent book, Louie smuggled in a polemic note: “Biology-envy is the curse of computing science;” cf. 2009, p. 261.) The pendulum has swung from the obsession with physics and chemistry – no less respectable and reputable in our time – to infatuation with genes, cells (stem cells more than any other), protein folding, and synthetic life. What has not changed is the need to understand the fundamental characteristics of the living. And this is what distinguishes Rosen from most researchers in this domain. These Prolegomena to the second edition of *AS* actually report on the state of the art in anticipation research. In other words, they present the indisputable impact that *AS* has had, regardless of whether those who speak “prose” (to bring up Molière’s character, Monsieur Jourdain) know that they do it – i.e., are aware of Rosen’s work – or do not – i.e., have never heard of him, or discard his work as speculative. For someone outside academic dialog, this could appear as an attempt to see how ideas percolate. The psychology of the process (the obsession with “who was first,” “who owns what,” or “whose understanding,” etc.) is a subject for others, especially because this never affords better understanding of the subject.

While he was working on *AS*, which was *in status nascendi* for almost 15 years, Rosen pursued a very ambitious research agenda. His doctoral thesis, which advanced the “(M,R)-Systems” opened a new perspective within which anticipation is only one aspect. This needs to be brought up since in his autobiographical notes, Rosen made a specific reference: “... the “(M,R)-Systems” have an inherent anticipatory aspect, built into their organization.” Still, as we read his *Autobiographical Reminiscences* (2006a), it becomes clear that in implicit terms, the entire focus on relational biology, in line with Rashevsky’s view, is conducive to a line of inquiry that ultimately questions the centuries-old reductionist-deterministic foundations of biology. This is the crux of the matter. Expressed otherwise, the seed of inquiry leading to anticipation is housed in the new perspective from which the “(M,R)-Systems” are derived as a dynamic description of the living cell, obviously contrasted to the atomic model inspired by physics, which is reactive in nature. The biologist, the mathematician, and the philosopher fuse into a new type of scientist. He is no longer willing to further build on the Cartesian foundation, but rather taking up the challenge of submitting an alternative fundamental understanding. Rosen himself brought up the work of Schrödinger (1944), Wiener (1948), and Shannon (1948), as well as game theory, and especially Systems Theory, in particular Bertalanffy and Ashby. Let me quote: “To me, though, and in the light of my own imperative, all those things were potential colors for my palette, but not the palette itself” (Rosen 2006b).

On previous occasions, I pointed out that a name is missing here, and not through some malicious intention: Walter M. Elsasser. Rosen was aware (and respectful) of Elsasser’s work and occasionally referenced his work. The reason we must evoke Elsasser is to provide a broader view of the work in which Rosen was engaged. Educated as a physicist, Elsasser made it the major focus of his work (after he arrived in the USA) to challenge the reductionist understanding of the living. His book, *The Physical Foundation of Biology* (1958, also by Pergamon Press) followed by *Atom and Organism* (1966) and *The Chief Abstraction of Biology* (1975) was a daring attempt to look at what makes life what we know it to be. I would like to point out that *Reflections on a Theory of Organisms* (1987) was, by no coincidence, almost simultaneous with Rosen’s *AS*. (His Measurement book and Elsasser’s book on abstractions of biology appeared in 1978 and 1975, respectively, from the same publisher.) I already confessed that I was unaware of Rosen when my book, *Mind – Anticipation and Chaos*, appeared. I was unaware of Elsasser, as well. During my sabbatical at the University of California–Berkeley (Electrical Engineering and Computer Science), his work was brought to my attention by Dr. Harry Rubin, a distinguished molecular biologist, who wrote the Introduction to the 1998 edition. Elsasser opposed a holistic view to the reductionist model. He proceeds from within the physics to which he remained loyal (given his formative years, cf. *Memoirs of a Physicist in the Atomic Age*, 1977). Under the guidance of, or in interaction with, Werner Heisenberg, Wolfgang Pauli, Albert Einstein, John von Neumann, Hans Bethe, Max Born, Arnold Sommerfeld, and, not to be omitted, Erwin Schrödinger – whose questions “What Is Life?” he adopted, Elsasser tried to provide a foundation for biology as strict as that of physics. Their interaction stimulated the reformulation of fundamental biological questions.

At this juncture, I felt the need to establish the context within which Rosen's contributions were made, in particular those leading to defining anticipatory processes. To this context belongs the activity of the Center for the Study of Democratic Institutions – the brainchild of Robert M. Hutchins, who for many years served as President of the University of Chicago. As seductive as it is, the history of the Center would take us off course from our present aim. It suffices to mention two things. First: Rosen explicitly acknowledged the impact of the Center as his book came together: "The original germinal ideas of which this volume is an outgrowth were developed in 1972, when the author was in residence at the Center for the Study of Democratic Institutions" (1985a). Second: as a member of the Center, Rosen contributed discussion papers over many years. These texts are highly significant to our better understanding of the implications of his research on anticipation. While the fundamental questions led to a new perspective, none of the hypotheses advanced remain exercises in formal biology. Rosen was a very engaged individual; in his own way, he was an activist. He lived his time; he wanted to understand change; he obliged explanations, and even methods of improvement. It was unfair of many of the commentators of his work to see in him a rather esoteric researcher, disconnected from reality, only because his arguments were often accompanied by or articulated in the extremely abstract language of mathematics, in particular, category theory (which he adopted for his uses in biology almost as soon as it appeared). He was a beneficiary of the two "fathers" of category theory: Eilenberg at Columbia University, and MacLane at the University of Chicago (1945, 1950). Subjects such as planning, management, political change, and stable and reliable institutions informed presentations he made at the Center during the 1971–72 academic year, when he was a Visiting Fellow. In an article published seven years later (1979), Rosen made this explicit:

I have come to believe that an understanding of anticipatory systems is crucial not only for biology, but also for any sphere in which decision making based on planning is involved. These are systems which contain predictive models of themselves and their environment, and employ these models to control their present activities (p. 11).

Hutchins conceived the Center as an intellectual community united in the Dialog:

Its members talk about what ought to be done. They come to the conference table as citizens, and their talk is about the common good [...] The Center tries to think about the things it believes its fellow citizens ought to be thinking about (as cited by Rosen 1985a, p. 3).

Among those working at the Center were political scientists, journalists, economists, historians, even philosophers, but not natural scientists. Still, Hutchins' vision and position of principle gave Rosen's presence meaning, as he himself noted, quoting from this visionary intellectual: "Science is not the collection of facts or the accumulation of data. A discipline does not become scientific merely because its professors have acquired a great deal of information." And yet another:

The gadgeteers and data collectors, masquerading as scientists, have threatened to become the supreme chieftains of the scholarly world. As the Renaissance could accuse the Middle Ages of being rich in principles and poor in facts, we are now entitled to inquire whether we are not rich in acts and poor in principles (p. 12).

This was not different from Elsasser's arguments, and was so close to Rosen's own thinking that Rosen realized that working on a theory of biological systems allowed him to formulate the characteristics of biology as an "autonomous science," for which he would then suggest means to formalize. He made a major observation:

The physical structures of organisms play only a minor and secondary role [...] The only requirement which physical structure must fulfill is that it allow the characteristic behaviors themselves to be manifested. Indeed, if this were not so, it would be impossible to understand how a class of systems as utterly diverse in physical structure as that which comprises biological organisms could be recognized as a unity at all.

The Social and the Biological

Rashevsky's relational biology, which Rosen helped develop, stands in contrast to the then dominant analytical approach. Rosen's approach was to focus on functional aspects, on understanding behaviors. John Wilkinson, a Senior Fellow at the Center, extended the invitation to Rosen hoping that his own focus on structure would benefit from interaction with a person focused on function. The parallels between biological processes and social structures led to one of those questions that only Rosen would formulate: "What would it mean if common models of organization could be demonstrated between social and biological structures?" (p. 13). It was very enticing for him to see a variety of disciplines finally cooperating, as it was a challenge to characterize the dynamics of life without having to account for underlying causal structures. In societal situations, the aggregate behavior, involving a multitude of processes, appear quite differently to an observer than to those involved. No less enticing were the considerations regarding the use of social experience as a means for deriving biological insights, and reciprocally, the possibility to develop insights into properties of social systems by building upon biological experiences. Rosen confessed:

In short, the Center seemed to provide me with both the opportunity and the means to explore this virgin territory between biology and society, and to determine whether it was barren or fertile. I thus almost in spite of myself found that I was fulfilling an exhortation of Rashevsky, who had told me years earlier that I could not be a true mathematical biologist until I had concerned myself (as he had) with problems of social organization (p. 14).

From this attempt to establish homologies between social and biological organization, Rosen expanded to predictive models, and realized that the stimulus-reaction explanatory concept could not account for situations in which subjects predict consequences of their own actions, moreover, for situations in which a course of action is changed not as a result of stimuli, but in accordance with a subject's predictive model. The switch from descriptions limited to reactive behavior to the much richer descriptions of what he termed anticipatory behavior resulted from a different understanding of the living. That the agency through which predictions are made turns out to be a model corresponds to the fundamental contributions Rosen

made in defining the “(M,R)-Systems”. It was noted by Kercel (2002, 2007), among others, that Rosen’s epistemology defines properties of logical and mathematical structures. *Impredicativity* is such a property – every functional aspect of the model is contained within another functional component. As we shall see shortly, this is the case of the system and its model unfolding in faster than real time, i.e., definitory of the model, not of reality as such. This means that once we acknowledge the complexity of natural systems, we need the appropriate concepts to describe them, under the assumption that a natural system whose entailment structure is congruent with an impredicative model. But as pervasive as anticipatory behavior was, it was not yet operational in the sense of being easy to translate into a coherent theory, and even less in applications to problems of forecasting and policymaking that were the focus of the Center. The “fortuitous chain of circumstances” described in Rosen’s paper explain why his involvement with the Center can be characterized as yet another element of the context that inspired him, as well as others (members of the Center or not), in questioning the entire analytical foundation of reductionism and determinism.

Readers of *AS* already know that major consideration is given to causality, in particular to phenomena that involve purpose or, alternatively, a goal. It is not the intention of these Prolegomena to offer a *compte rendu* of the book, or of Rosen’s view of anticipation. As already stated, the hope is to provide here a frame of reference. Therefore, from among all those whom Rosen named in his autobiographical notes (and in many other writings), we need to highlight Aristotle, whose work on a typology of causes remains a constant reference. *Causa finalis*, eliminated from scientific vocabulary with the same epistemological fury as vitalism was, found a new champion in Rosen. His strict understanding of dynamics implies finality.

One final note about the onset of Rosen’s work on anticipatory behavior: “Planning, Management, Policies and Strategies: Four Fuzzy Concepts” (1972) was the first of a number of working papers that define his research agenda at the Center. The first lines of this paper could have been written in our days:

It is fair to say that the mood of those concerned with the problems of contemporary society is apocalyptic. It is widely felt that our social structure is in the midst of crises, certainly serious, and perhaps ultimate. [...] The alternative to anarchy is management; and management implies in turn the systematic implementation of specific plans, programs, policies and strategies (p. 1).

Rosen brought up the conceptual requirements for a methodology (“a ‘plan for planning’”) that would allow avoidance of “an infinite and futile anarchic regress.” Given the audience at the Center, i.e., none with a background in mathematics, biology, systems theory, or the like, Rosen built his arguments in favor of defining anticipatory behavior in an almost pedantic manner. But in essence, it was within this context that the major ideas of his future book on anticipatory systems were articulated. The intellectual profile of his listenership and the broad goals of the Center, which Rosen explicitly adhered to, had an impact on formulations, examples, and the general tone. Aware of the fact that “how planning could go wrong” was on the minds of his Fellows at the Center, he explicitly addressed

the question, taking note of the fact that a system's integrated perspective is not bulletproof, just as "the defect of any part of a sensory mechanism in an organism leads to a particular array of symptoms," (1972, p. 3 and 1974, p. 250). At the center of his conception is the "Principle of Function Change": the same structure is capable of simultaneously manifesting a variety of functions. Rosen remained fully dedicated to his research in the foundations of biology and, in a broader sense, to the broader task of reconsidering the reactive paradigm. He was aware of the need to focus on what a model is, and to further define the relation between a biological entity and its model; that is, the relation between something represented and its representation. It is within this broader realm that Rosen realized the urgency of understanding how an open system (the natural system) and its model – always less open – understood in their relative unity, eventually make predictions possible.

Natural and Formal Complexity

Anticipatory Systems followed another foundational text: "Organisms as Causal Systems which are Not Mechanisms: An Essay into the Nature of Complexity" (1985b). By no means to be ignored, the other two contributions – "The Dynamics of Energetics of Complex Real Systems" (by I.W. Richardson) and "Categorical System Theory" (by A.H. Louie) – make it clear that Rosen's research reached another level; those who worked with him were encouraged to examine the various implications of higher complexity definitory of the living. The distinction between simple systems (or mechanisms) and organisms came clearly into focus. Rosen denied, in very clear formulations, that biology is nothing more than a particular case for physics (p. 166), and argued in favor of a mathematical language appropriate to the task, which is, in his view, category theory. Concluding remark: "Complex systems, unlike simple ones, admit a category of final causation, or anticipation, in a perfectly rigorous and nonmystical way," (p. 166). It is in this very well organized essay, of a clarity not frequently matched in his very rich list of articles and books, that Rosen defined a fundamentally new perspective. Schrödinger's question "What is Life?" – which became the focus of his work – led to his description of "relational biology," a concept originating, as already mentioned, with Rashevsky (1954), and which led to the realization that only after abstracting "away the physics and the chemistry," (Rosen 1985b, p. 172) can we reach the organizational features common to all living systems. Rashevsky used graphs, whose "nodes were biological functions" and whose directed edges were "relations of temporal or logical procedure," (p. 172). But, as Rosen noticed, his mentor was ahead of his time:

... the time was quite wrong for his new relational ideas to find any acceptance anywhere. In biology, the 'golden age' of molecular biology was just beginning; experimentalists had no time or use for anything of this kind. Those who considered themselves theorists either were preoccupied with the reductionist modeling that Rashevsky had earlier taught them or were bemused by seductive ideas of 'information theory,' games theory, cybernetics, and the like, regarded Rashevsky and his ideas as generally archaic because he did not take direct cognizance of their enthusiasms (p. 173).

A great deal of effort was spent on defining the “(M,R)-Systems”, in particular on replication mechanisms inherent in the organizational features represented. However, the centerpiece, and appropriately so, is the modeling relation between a natural system and a formal one. Any reader of *AS* would be well advised to read Rosen’s essay (even though its main line of argument reverberates in the book). It is here that the intrinsic limitations of the Newtonian paradigm are spelled out in detail. And it is here, as well, that the major subject of causality, including the teleological, is addressed up front (cf. p. 192). Moreover, it is here that the “mathematical image of a complex system” comes into focus, and becomes subject to mathematical category theory (although the author did not specifically apply it here).

From the very rich text, I would like to refer to Rosen’s considerations on information, specifically, on an alternate approach that relates to his preoccupation with measurement. He defined information as “anything that is or can be the answer to a question” (cf. p. 197). This prompts the observation that formal logic (“including mathematics,” as he put it) does not account for the interrogative. Therefore, information cannot be formally characterized. Rosen used the formalism of implications (If A, then B) in order to eventually formulate a variational form (If δA , then δB) that brought up measurement: “If (initial conditions), then (meter reading)?” and an associated formulation on variations: If (I make certain assumptions), then (what follows?). This is, in his words, “analogous to prediction” (p. 199). The conclusion is powerful: “When formal systems (i.e., logic and mathematics) are used to construct images of what is going on in the world, then interrogations and implications become associated with ideas of causality (p. 199). The reader is encouraged to realize that this is exactly why the Newtonian paradigm cannot accept Aristotle’s *causa finalis*: a (formally) logical system that does not have what it takes to represent interrogation, cannot account for information that always involves a telic aspect: the “What for?” of information. The conclusion is simple (and elegantly formulated):

Like early man, who could see the earth every evening just by watching the sky but could not understand what he was seeing, we have been unable to understand what every organism is telling us. It cannot be stressed strongly enough that the transition from simplicity to complexity is not merely a technical matter to be handled within the Newtonian paradigm; complexity is not just complication but a whole new theoretical world, with a whole new physics associated with it (p. 202).

There is an old question that informs these Prolegomena: If a tree falls in the woods and no one is around to hear it, does it make a sound? In semiotics, one of the research fields from within which my own notion of anticipation took shape, nothing is a sign unless interpreted as a sign. The noise caused by the falling tree is a physical phenomenon corresponding to friction. It propagates at a distance that corresponds to the energy involved (the falling of a huge tree can be heard at a farther distance than the falling of a bush). The energy dissipated in the process can be measured exactly. In trying to define natural law, Laplace (1820, as quoted in [Rosen 1985a](#), p. 9) convincingly described the kind of inferences possible in the reductionist world: An intelligence knowing, at a given instant in time, all forces

acting in nature, as well as the momentary position of all things of which the universe consists, would be able to comprehend the motions of the largest bodies of the world, as well as the lightest atoms in one single formula. To him, nothing would be uncertain, both past and future would be present in this eyes. In the years in which Rosen challenged a description of the world that simply does not account for the richness of life, I myself referred not only to Laplace, but to *Ecclesiastes*: “There is no new thing under the sun” (Nadin 1987). Everything is given, and with it, the laws describing it. We need only a good machine to reconstitute the past from the energy that preserved the noise of the falling tree, as it preserves all the thoughts ever expressed by those speaking to each other. The oscillations of air molecules could help us hear what Aristotle said, and even Socrates, whose words, we assume, Plato wrote down (or made up). My text on the mind, coming together within the timeframe when anticipation was becoming a necessary construct for understanding how minds interact, challenges the acceptance of reductionism while actually having as its object a novel, Umberto Eco’s *The Name of the Rose*. The detective story was probably written from end to beginning, or so it seems. It has a clear final cause, and it offered the author, a distinguished historian of the Middle Ages, the occasion to pose questions relevant to how representations are elaborated. Is there something there – a person, a landscape, or a process – that we simply describe, draw, take a picture of? Or do we actually notice that what is alive induces changes in the observing subject that eventually result in a representation? Even in the universe of physics, the static notion of representation was debunked as quantum mechanics postulated that to measure is to disturb. Rosen (1978, and myself, Nadin 1959) in other ways, said: “to measure is to BE disturbed.” That is, the dynamics of the measured affects the dynamics of the measuring device. Every interpretation is the result of interactions.

Together with the rather impressive number of Center presentations (published in the *International Journal of General Systems*; see references for details) that preceded the book, *AS* invites consideration of its echo in the scientific community. Please remember the question, “If a tree falls in the woods and no one hears the noise, does the event register as directly consequential?” It also invites consideration of Rosen’s essay, “Organisms as causal systems which are not mechanisms” (1985b) in the *Theoretical Biology and Complexity* volume. Let’s be up front: *AS* prompted two reviews: one by Minch (1986), at that time a graduate student at Binghamton, and one by Vámos (1987) of the Technical University of Budapest (Hungary). The Essay volume attracted René Thom, the distinguished mathematician (Catastrophe Theory is associated with his name), Lee Segel (Weizmann Institute of Science, Israel), Lev Ginzburg (SUNY–Stony Brook), and P.T. Saunders (King’s College) to review it. In the perspective of time, this is rather little given the significance of the work. But it is also telling in respect to the difficult cognitive challenge that the work posed, and still poses. Minch, now a respected researcher in his own right, might not have fully realized the impact of the radical ideas that Rosen advanced, but everything in the review is evidence of solid judgment and the desire to understand. He stated: “The essential difference between reactive and anticipatory systems is that reactive control depends on correction of an existing deviation, while

anticipatory control depends on preventions of a predicted deviation” (cf. p. 405). For anyone trying today to convey to the scientific establishment why the study of anticipation is relevant, this sentence from Minch’s review says it all. Minch thoroughly referred to the *modeling relation* – “between a natural system and a formal system” – and to their linkage. He was able to realize the importance of a new understanding of time (“In particular, he shows how we can view models and systems as parameterized by different times” p. 406). The review deserves to be quoted in more detail than appropriate in Prolegomena. The book as a whole, Minch states, is both radical and profound.

It is radical because it not only develops and propounds a paradigm, which is very different from the traditional, but also finds inadequacies in the epistemological roots of science, and overcomes these inadequacies. It is profound because of the depth of the discussion and the extent of its implications (p. 408).

Vámos could not find anything new. In the Essay reviews, Thom admired “an extremely interesting piece of epistemological thinking,” as well as the discussion on causality, “The rediscovery of Aristotelian causality theory, after centuries of blind positivist rejection, has to be hailed as one of the major events in modern philosophy of science.” Neither Segel (“I oppose his urgings to go beyond the evolving state description that was so successful in particle physics,” 1987), nor Ginzburg (1986) realized the significance of Rosen’s model. Saunders, “fascinated by the third chapter (i.e., Rosen’s text) is also taken by the novelty of the approach to causality and the non-Newtonian dynamic system.

Various Understandings of Anticipation

At this moment, the reader of the Prolegomena dedicated to what supports the science that Rosen attempted to initiate might call into question the wisdom of printing a second edition of a book less than enthusiastically received since its beginning – but out of print, nevertheless. I hasten to add that my own survey of anticipation-pertinent scientific publications has resulted in a very interesting observation: Very few mainstream researchers quote Rosen directly; secondary sources, in articles inspired by Rosen’s work, are usually quoted. Rosen is present, i.e., his ideas are either continuously reinvented – I can imagine him smiling about this – or, better yet, there is a definite Rosen presence even in research that is ultimately divergent from his understanding of anticipation. Here I refer explicitly to various attempts to get machines to anticipate one way or another – a subject to which I must return since there is so much and often very good work to survey. The destiny of Rosen’s writings, not unlike his actual destiny, might be a good subject for a novel or a movie. After all, John Simon’s apt description (*New York Magazine*, 1990), inspired by Richard Nelson’s play *Some Americans Abroad*, was an eye-opener to many who still maintain an idealized view of academia, or are convinced that science is a *sui generis* immaculate conception enterprise in which the best always wins.

The intensity of hatred, infighting, and back-stabbing increases with the marginality of a profession; hence, for rivalry, animus, and vindictiveness, there is no business like education. Anyone who has put in time in academia has witnessed intrigue to make a Renaissance Italian dukedom or a Reformation German principality look like an idyll by Theocritus.

Rosen faced bureaucratic imbecility a bit more than his peers did, and became the victim of that infighting and back-stabbing. He distanced himself as much as possible from this behavior. But after all is said and done, and while the difficulties he faced affected him, this is not what defines Rosen as a scholar. It would be unfair to his legacy to put more weight on the unfairness he faced than on the original thinking that defines his contribution. No scientist of integrity will lightly challenge the fundamental epistemological assumptions informing the dominant understanding of life within and outside the scientific community. Generation after generation, we were all educated, and continue to be educated, in the Cartesian understanding. A highly successful body of knowledge testifies to the revolutionary power of this explanatory model of the world. Still, before Rosen, and after Rosen, positions were articulated in which Cartesian doubt – *Dubito ergo sum* – is expressed, moreover in which alternative explanations are advanced (cf. [Wigner 1961, 1982](#)). On several occasions, I presented such views ([Nadin 2000, 2003, 2009b](#), for example), informed by Rosen's work and by other attempts to free science from a limiting view of how knowledge is acquired.

Rosen's realization of the limits of the reaction paradigm is part of his broad conception of the living. Our ability to gain knowledge about it is affected by the Cartesian perspective. To transcend this view, scientists ought to "discard knowledge" (as Niels Bohr put it), and to see the world anew. In a letter (1993, cf. *Praefatio*, [Louie 2009](#)), Rosen alludes to how the "official" position of science often leads to opportunistic positions: "The actual situation reminds me of when I used to travel in Eastern Europe in the old days, when everyone was officially a Dialectical Materialist, but unofficially, behind closed doors, nobody was a Dialectical Materialist." Due to my own life story, I happen to know what this means. My own understanding of anticipation was informed by semiotics – which Rosen and some of his colleagues considered worthy of their attention (they referred mainly to the symbol). At the time (1980–1985), I was teaching at the Rhode Island School of Design and advised some very creative students. I also served as adjunct professor at Brown University, working with the Semiotics Group and students in Computer Science. Cognitive science, in particular Libet's work on readiness potential ([Libet et al. 1983](#)), also informed my research. As Eminent Scholar in Art and Design Technology (Ohio State University, 1985–1989), I was confronted by what Rosen would have qualified as the "mechanistic view of creativity," characteristic of the early days of fascination with computers and their applications in modeling, simulation, and animation. This was the place where "flying logos" originated, where computer graphics – machine-generated imagery and computer-supported animation – made national headlines. The effort received the usual funding (National Science Foundation, Department of Defense, and the like), prompted less by scientific significance and more by media attention. Ever

since my formative years in Romania, first at the Polytechnic Institute and then as a graduate and doctoral student at the University of Bucharest, the question that shaped my intellectual profile concerned creativity, in particular: Can machines be creative? Better yet: Is creativity an expression of deterministic processes?

Today I know that this is a typical Elsasser problem, as it is also a Rosen problem. And it lies at the confluence of the physical (i.e., matter), as a substratum of all there is, and the dynamics of the living. As I wrote my first computer graphics programs (late 1960s), I realized that they encapsulated knowledge of visual expression and representation, and allowed for high performance of repetitive tasks. But the essence of any creative act is that it results in something that has never existed, not in the mindless reproduction of what is already available.

At Ohio State, I could have easily fallen prey to the comfort of a prestigious tenured position. Playing the usual funding games – if you correctly assess the direction from which the “funding winds” blow and don’t challenge established science, you get to “kiss the girl” – and making the conference rounds was part of it. But I did not want that. My concern with the multitude aspects of creativity brought me close to the then still incipient interest in the role of the brain, and into the mind-brain discussion. Intellectually, this was the hottest subject, probably because it took place at the confluence of disciplines (and had a clear European flair, as with Eccles).

I left Ohio State – probably as Rosen left Chicago or Buffalo – but not before accepting the invitation of the Graduate School to give a lecture. My take on anticipation, which was the focus of this lecture, was based on Libet’s measurements. I was not interested in free will, as I did not focus on how synapses actually take place. The scientific question I tried to answer was, “How do minds anticipate?” that is, how brain activity is triggered before an action, not in reaction to something else. “How does the brain know in advance that I will move my arm or scratch my head, or avoid a collision?” In addressing the question, I used the mathematical model of dynamic systems, and I advanced some hypotheses: The mind controls the brain; actually interactions of minds make anticipation possible. Anticipation can be described as an attractor within a space of many possible configurations. The lecture devoted some consideration to senescence (in particular, dementia), and even more to creativity. This lecture, which somehow paralleled Rosen’s lectures at the Center in Santa Barbara, eventually became a manuscript in search of a publisher. (I wish I had known George Klir at that time.) The text also defined my priorities as a researcher. Indeed, I did not want to write programs to be used for more flying logos and commercial art garbage. Repetitive tasks were not for me, just as the automation of bad taste, along with the production of mediocrity for satisfying such bad taste, never captured my interest. To give up an endowed chair sounds almost heroic when one thinks about the uncertainties in academia. But it saved my life as a scientist.

At this time, I was back in New York City and had the opportunity to see a fascinating show at the New York Academy of Sciences. Tod Siler, whose artwork was being exhibited, was a Ph.D. candidate at the Massachusetts Institute of Technology, combining art and brain science. His show seemed to reverberate my own thoughts. It seemed to me that if I could find a publisher for my book, his

images would be congenial to my ideas. Fortuitous events brought me to Belser Presse (Stuttgart/Zurich), which produced a series dedicated to scientific texts of broader relevance than the usual monographs. The series “Milestones in Thought and Discovery” published Leibniz’s *Two Letters on the Binary Number System and Chinese Philosophy* (1968); Heisenberg’s *The Laws of Nature and the Structure of Matter* (1967); and the testimony of Furrer (1988), the first German astronaut. These titles were selected by an extremely competent scientific advisory board. The members took a long time deliberating whether my book and Siler’s images should be published, but in the end an exquisite limited edition was printed. It became the preferred gift among an elite, but not necessarily the subject of scientific debate, as was my hope.

And as my new academic endeavors would have it – teaching in Germany for ten years – my focus on anticipation – in the meanwhile informed by Rosen’s work and influenced by my contact with him – led to more theoretic work. It also led to attempts to test hypotheses in various fields of anticipation expression: communication, design, architecture, human-computer interaction, the various arts. In 2002, the antÉ – Institute for Research in Anticipatory Systems was incorporated, and one of its first projects was a hybrid publication: book (*Anticipation – The end is where we start from*, Nadin 2003), Website (a knowledge base for the community of researchers interested in this area), and a DVD (presenting examples of anticipation ranging from chess to a simple protein folding game). This was the first time that some of Rosen’s work was made available in digital format.

With my appointment as Ashbel Smith Professor at the University of Texas at Dallas, the Institute found a new host, and new fields of inquiry. I would not have accepted the endowed chair in Dallas if the terms of my employment had not specifically spelled out a framework of activity corresponding to my total dedication to research in anticipation. Interestingly enough, Rosen himself worked for a year in Dallas (actually in Arlington, part of the greater Dallas–Fort Worth Metroplex) – he did not like it – before his stint in Kyoto. More to the point, his writing on senescence, and my work in addressing the loss of anticipation in the aging somehow fused in the Project *Seneludens* (Nadin 2004). Through brain plasticity (stimulated by involvement in games with a cognitive and physical component), anticipatory characteristics, vital to maintaining balance and a variety of actions, can be maintained.

The Institute also organized three international symposia: Vico’s *Scienza Nuova*, 2005, *Anticipation and Risk Assessment*, 2006 (see Nadin 2009a), and *Time and the Experience of the Virtual*, 2008. A special issue of the new *Journal of Risk and Decision Analysis* is dedicated to Risk and Anticipation. The University gave serious consideration to acquiring Rosen’s library; my own desire to organize an annual Robert Rosen Memorial Lecture will one day be realized.

These details should be read as testimony to a very dedicated interest in a subject that will always have a reference in Rosen’s AS. Maybe the best way to conclude these remarks is to set side-by-side one of Rosen’s definitions and one of mine:

Rosen	Nadin
An anticipatory system is a system whose current state is determined by a (predicted) future state.	An anticipatory system is a system whose current state is determined not only by a past state, but also by possible future states.

As subtle as the difference is, it only goes to show that could Rosen be around, he would continue to work on foundations as well as on applications.

Distinguishing between prediction and anticipation is the subject that could be of further help in defining anticipatory processes. Prediction and anticipation are not interchangeable. Predictions are expressions of probabilities, i.e., description based on statistical data and on generalizations (that we call scientific laws). While not unrelated to probabilities, anticipations involve possibilities. Zadeh’s genius in defining possibility is expressed in the accepted dicta: Nothing is probable unless it is possible. Not everything possible is probable. The model of itself, which unfolds in faster than real time, in Rosen’s definition (1985a) is driven by both probability realizations and possibility projections. It is in respect to this fundamental distinction that I submitted the thesis according to which the complementary nature of the living – physical substratum and specific irreducible dynamics – is expressed in the complementary nature of anticipatory processes (Nadin 2003, 2009c).

A Broader Context – Awareness of Anticipation

The perspective of time and the evidence of increasing interest from the scientific community in understanding anticipatory processes speak in favor of describing the premises for the initial definition of anticipation. The work (1929) of Alfred North Whitehead (1861–1947) advanced the idea that every process involves the past and the anticipation of future possibilities. This thought is part of a larger philosophic tradition sketched out in the attempt to indentify early considerations on the subject. Indeed, let us be aware of the variety of understandings associated with the concept, because otherwise there is a real risk of trivializing anticipation before we know what it is. Burgers (1975) was inspired by Whitehead. Although he came from physics, Burgers brought up choice and the preservation of freedom as coextensive with anticipation. Bennett, an anthropologist, saw anticipation as “the basis for adaptation” (1976a, p. 847). In his book (1976b), the same broad considerations made the object of an entire chapter (VII), in which Whitehead’s notion of anticipation, extended to the entire realm of reality, is limited to living systems. Both Burgers and Bennett are part of the larger context in which anticipation slowly became part of the vocabulary of science and philosophy at the end of the last century.

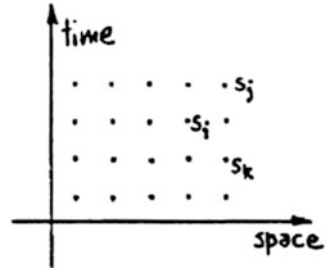
Another area of investigation that leads to explicit considerations of anticipation is psychology. Not unexpectedly, “work and engineering psychology” (Hacker

1978) comes to the realization that goal-oriented activities are expressions of anticipation. The fact that Marx, writing about work as goal oriented (in *Das Kapital*, 1867) becomes a reference in this respect has to do with more than the “style” of scientific writing in Eastern Europe. You needed to provide a good quote from the “classics” (Marx, Engels, Lenin, etc.), and Hacker provided several (“The ideal anticipation of the product as goal” is yet another example; 1978) before you could present your own ideas. Hacker distinguishes between the physical effort of labor and the spiritual effort. The future, embodied in the result of the work defines the various stages of the productive effort (“inverse causality,” 1978, p. 108). Around the time when Rosen advanced his research, Winfried Hacker (currently emeritus of the Technical University in Dresden, Germany) was examining anticipatory expression in work, learning, and playing. Action regulation theory guides his approach. Hacker also analyzed the role of memory and defined perspective memory as relevant to anticipation (1978, p. 292 and following on the sensomotoric). He made very convincing references to [Vygotsky](#) (1934/1964), [Leontiev](#) (1964), and [Galperin](#) (1967), who considered the role of language as an anticipatory expression in various human activities. In a comprehensive study of the many aspects under which anticipation is acknowledged, [Vygotsky and Luria](#) (1970) will indeed have to be mentioned. In his Ph.D. thesis, [Volpert](#) (1969) convincingly brought up the sensory-motoric aspects of anticipation-guided learning. The most relevant contributions, still waiting to be placed in the proper context, are those of Bernstein on the “physiology of activeness” (his terminology). He focused on motor tasks and advanced (beginning 1924) the idea that “Anticipation, or presentiment” (as he calls it) is based on a probabilistic model of the future (cf. [Bernstein](#) 1924, 1935, 1936, 1947, 2004). It is very possible that once research in anticipatory processes expands to include the history of ideas leading to new hypotheses, such and similar contributions will prove significant for the future development of the field. Let me explain this statement through an example.

Feynman, famous for his contributions to quantum electrodynamics (which earned him a Nobel prize, in 1965, shared with Julian Schwinger and Sin-Itiro Tomonaga) is, probably more by intuition than anything else, part of the scientific story of anticipation. The depth of Feynman’s involvement with the subject of computation is significant because he, as opposed to everyone else, brought up anticipation, however indirectly, not from biology, but from computation – that is, exactly where Rosen would have claimed that it cannot be found.

In an article entitled “Simulating Physics with Computers,” Feynman (1982) made relatively clear that he was aware of the **distinction between what is represented (Nature – his spelling with a capital N, and nothing else, since physics always laid claim upon it), and the representation (computation)**. The physical system can be simulated by a machine, but that does not make the machine the same as what it simulates. Not unlike other scientists, Feynman focused on states: the space-time view, “imagining that the points of space and time are all laid out, so to speak, ahead of time.” The computer operation would be to see how changes in the space-time view take place. This is what dynamics is. His drawing is very intuitive (Fig. 1).

Fig. 1 Feynman's (1999) original state diagram



The state s_i at space-time coordinate i is described through a function F (Feynman did not discuss the nature of the function): $s_i = F_i(s_j, s_k, \dots)$. The deterministic view – i.e., the past affects the present – would result, as he noticed, in the cellular automaton: “the value of the function at i only involves the points behind in time, earlier than this time i .” However – and this is the crux of the matter – “just let’s think about a more general kind of computer... whether we could have a wider case of generality, of interconnections.... If F depends on all the points both in the future and the past, what then?” obviously, this is no longer the computation Rosen referred to.

Had Feynman posed his rhetoric question within the context of research in anticipation, the answer would be: If indeed F depends on all the points both in the future and the past, then $\rightarrow$ Anticipation. Defining an anticipatory system as one whose current state depends not only on a previous state and the current state, but also on possible future states, we are in the domain of anticipation. Feynman would answer: “That could be the way physics works” (his words in the article cited).

There is no reason to fantasize over a possible dialog – what he would say, his way of speculating (for which he was famous), and what Rosen would reply. But there is a lot to consider in regard to his own questions. After all, anticipatory computation would at least pose the following questions:

1. “If this computer were laid out, is there in fact an organized algorithm by which a solution could be laid out, that is, computed?”
2. “Suppose you know this function F_i and it is a function of the variables in the future as well. How would you lay out numbers so that they automatically satisfy the above equation?”

These, again, are Feynman’s words, his own questions. To make it crystal clear: the questions Feynman posed fit the framework of anticipation and computing. However, Feynman was not even alluding to a characteristic of a part of Nature – the living – to be affected not only by its past, but also by a possible future realization. Feynman’s focus was on quantum computation, and therefore the argument developed around probability configurations. When he wrote about simulating a probabilistic Nature by using a probabilistic computer, he realized that the output of such a machine “is not a unique function of the input,” that is, he realized the non-deterministic nature of the computation.

As we shall see, where Feynman's model and considerations on anticipation and computing, related to the work of Rosen and Nadin, diverge is not difficult to define. For him, as for all those – from Aristotle to Newton (*Philosophiæ Naturalis Principia Mathematica*) to Einstein – who made physics the fundamental science it is, there is an all-encompassing Nature, and physics is the science of Nature. In other words, physics would suffice in explaining anticipatory processes or in computationally simulating them.

Svoboda (1960, cf. Klir 2002b) published a “model of the instinct of self-preservation” in which the subject is a computer itself. Its own functioning models self-preservation under external disturbances. A probabilistic description based on inferences from past experiences quantifies its predictive capability. Pelikan (1964) further elaborated Svoboda's original idea. Probably, as we advance in our understanding of anticipation, there will be more contributions that, in retrospect, will deserve our attention. For example, in 1950 (cf. Gabel and Walker 2006) Buckminster Fuller outlined a class in anticipatory design (taught at MIT in 1956). In this class, “Eight Strategies for Comprehensive Anticipatory Design Science” were spelled out. Fuller took a broad view of what it means to introduce new artifacts into reality. There is a lot to consider in terms of how they change the given environment and the behavior of individuals and communities. Teleology, i.e., the goal-driven aspect of design, is to be understood in relation to what he called “precession”: the sequence of steps that lead from the assumed goal (subject to continuous reevaluation) to the end result (itself subject to further improvement).

The American economist Willford Isbell King (at one time Chairman of the Committee for Constitutional Government) published *The Causes of Economic Fluctuations: Possibilities of Anticipation and Control* (1938). The circumstances (in particular the Great Depression) explain the subject and hope. The same title could be used in our days. Fluctuations continue to haunt us, and predictive models developed so far are not very helpful when it comes to avoid dire consequences. At about the same time (1937, actually), George Shackle, under the supervision of Friedrich von Hayek, finished his dissertation, which led to his first book (1938). **Expectation, as a particular form of anticipation is connected to his future contributions to defining uncertainty.** Let us take note that in examining the time vector from the beginning of an action (threshold) and the time vector from the end of the action in reverse, Shackle noticed that we never have enough knowledge in order to understand the consequences of our actions. Of interest to us today is Shackle's understanding of possibility, and the contradistinction to probability. A short quote is indicative of the anticipation implications of his writing: “It is the degree of surprise to which we expose ourselves when we examine an imagined happening as to its possibility...” (cf. Klir 2002a for an in-depth analysis). As far as I was able to establish, Shackle did not use the word *anticipation*, but he referred to imagination as guiding choices (1979). His conceptual contribution in understanding imagination as related to the space of possibilities will surely lead to more elaborations of interest to research in anticipation.

Possibility and its relation to probability that was of interest to Shackle (cf. 1961) will have to wait for a more comprehensive approach until Lotfi Zadeh (1978), and

subsequently many distinguished followers, gave it a foundation. Zadeh himself arrived at possibility via fuzzy sets. In June (2009a), Zadeh, continuing his tireless investigation of the realm of knowledge he opened when introducing fuzzy sets, made note of the fact that judgment, perception, and emotions play a prominent role in what we call economic, legal, and political systems. Many years ago, Zadeh (1979/1996) invoked the views of Shackle, among others, as an argument in introducing information granularity. This time, acknowledging complexity – which, as we shall see, is the threshold above which anticipatory behavior becomes possible – Zadeh took a look at a world represented not with the sharp pen of illusory precision, but with the spray can (“spray pen geometry”). Where others look for precision, Zadeh, in the spirit in which Shackle articulated his possibilistic views, wants to capture processes unfolding under uncertainty. We realize, at least intuitively, that anticipations (like imagination) are always of a fuzzy nature, and it seems to me that Zadeh’s new work, intensely discussed on the BISC mailing list, will make the scientific community even more aware of this condition.

It is significant that economics prompts the type of questions that unite the early considerations of King (1938) and Shackle (1938) with Klir’s considerations (2002a) and Zadeh’s (Zadeh et al. 2009b) very recent attempts to extend fuzzy logic. Questions pertinent to economics (and associated fields of inquiry) will undoubtedly further stimulate anticipation research. We want to know what the possibilities for success are, or at least what it takes, under circumstances of uncertainty, to avoid irreversible damage to our well-being, and often to society’s.

Where Do We Go From Here?

The knowledge base reflecting the multitude of perspectives from which individual authors proceeded as they pursued the subject of anticipation continues to grow. This is no longer a preliminary stage, although it would be illusory to assume that a well-defined description has been established. Despite the urgency of providing an anticipatory perspective to the fast dynamics of our time, there are no university classes dedicated to it, and no research initiatives specifically informed by this perspective. Everyone hopes for good predictions; money is spent on funding predictive methods, as though prediction could substitute for anticipation. Still, the subject quite often percolates among the many research themes associated in some ways with cognitive science, computer science, artificial intelligence, and even ALife.

With Rosen’s concept of anticipation as a reference, the following consideration will provide pointers to relevant research. A distinction will be made between studies pursuing Rosen’s theoretic outline, and studies defining the field in ways other than his own; or better yet, what he called pseudo-anticipation. No author could claim credit for a full account. We can more easily find what we look for, but at times to formulate the question is more challenging than to advance a hypothesis as an answer. Example: Ishida and Sawada (2004) report on a very simple experiment

of human hand movement in anticipation of external stimulus. Unfortunately, while actually reporting on anticipation, the authors never name the concept as such. (It is from this experience that I discovered how many Japanese scientists would be happy to have access to a new edition of AS.) In other cases, anticipation, the word, is present, but the results presented have actually nothing to do with it. I prefer not to single out an example because, after all, there is nothing to object to in what is presented, but rather to the use of a concept that has had a precise meaning ever since Rosen's AS and other contributions (mentioned in the previous section of the Prolegomena). The fact that the scientific community at large has not embraced the view reflected in Rosen's particular interpretation, or in definitions congruent with his, means only that more has to be done to disseminate the work, in conjunction with its understanding. Einstein's assessment – No problem can be solved from the same consciousness that created it – is relevant not only for those willing to step out from their epistemological cocoon, but also for those who literally cannot find useful answers within the epistemology they practice.

Classical research in psychology – in particular, on receptive-effector anticipation (Bartlett 1951) – prepared the way for perceptual control theory (PCT) initiated by William T. Powers (1973, 1989, 1992) around the notion of organisms as controllers. All this is obviously different from Hacker's work in what used to be the German Democratic Republic. Kelly's (1955) constructivist position is based on validation in terms of predictive utility. Coherence is gained as individuals improve their capacity to anticipate events. Since the premise is that knowledge is constructed, validated anticipations enhance cognitive confidence and make further constructs possible. In Kelly's terms (also in Mancuso and Adams-Weber 1982), anticipation originates in the mind and is geared towards establishing a correspondence between future experiences and predictions related to them. The fundamental postulate of this theory is that **our representations lead to anticipations**, i.e., alternative courses of action. Since states of mind somehow represent states of the world, anticipation adequacy remains a matter of validation through experience.

Anticipation of moving stimuli (cf. Berry et al. since 1999) is recorded in the form of spike trains of many ganglion cells in the retina. Known retinal processing details, such as the contrast-gain control process, suggest that there are limits to what kind of stimuli can be anticipated. Researchers report that variations of speed, for instance, are important; variations of direction are less significant. That vision remains an area of choice in identifying anticipation is no surprise. An entire conference (University of Dundee, 2003) was dedicated to Eye Movements – considered “a window on mind and brain” – while the European project MindRaces (2009): *from reactive to anticipatory cognitive embodied systems* encouraged studies in this field, given its applied nature (Pezzulo 2007a, 2007b). Balkenius and Johansson (2007) contributed to the project the research of anticipatory models in gaze control, integrating reactive, event-driven and continuous-model-based location of target. Obviously, learning in their view is rather different from Rosen's notion, which predates AI's focus on learning in our days. It is encouraging to notice that the recognition of the role of learning extends to their particular domains of interest.

Arguing from a formalism, such as Rosen used, to existence is definitely different from arguing from existence (seeing, hearing, binding of the visual and aural, etc.) to a formalism. A vast amount of work (concerning tickling, e.g., [Blakemore et al. 1998](#); posture control, e.g., [Gahery 1987](#), [Melzer et al. 2001](#), [Adkin et al. 2002](#)); gait control ([Sahyoun, et al. 2004](#)) exemplifies the latter. In the same category, reference can be made to [König, and Krüger 2006](#) on the subject of predictions about future stimuli (the frog spotting a flying insect, and the process of filling in the informational gap in order to define the position where its tongue will capture it in a swift move). The authors allude to patterns of behavior. Such patterns are also suggested in the research of [Roth \(1978\)](#) as he analyzes the prey catching behavior of *Hydromantes genei*.

The very encouraging aspect in such research is that measurements of trigger-based experiments reveal what happens before the trigger (obviously if the measurement itself is not set off by the trigger). In other words, what happens in anticipation of stimuli (can be guessing, prediction, noise, for example), not as a result of them begins to be examined. Preparation (cf. [Gahery 1987](#)) is part of the anticipatory process. I doubt that a theory of anticipation, or at least some amendments to the available theories, could emerge from these rich sets of data. But such experimental evidence is encouraging first and foremost because it consistently supports the fundamental idea expressed in Rosen's modeling relation: **If a modeling relation between a natural system and a formal description can be established, the formal description (of vision processes, of tickling, of tactility, of sound and image binding, etc.) is a model, and the domain knowledge is a realization of such a description subject to further investigation. Moreover, arguing from computation – which is more and more a gnoseological mode – might impress through even broader sets of data and much more detail, but still not substitute for the lack of a theoretic foundation. As impressive as applications in neural networks ([Homan 1997](#), [Knutson et al. 1998](#), [Kursin 2003](#), [Tsirigotis et al. 2005](#)), artificial intelligence ([Ekdahl et al. 1995](#), [Davidsson 1997](#)), adaptive learning systems ([Butz et al. 2003](#)), among others, are, they can at most make us even more aware of the need to define our terminology and practice scientific discipline.** [Rosen \(1991, p. 238\)](#) pointed out quite clearly that the more constrained a mechanism, the more programmable it is. Albeit, reaction is programmable, even if at times it is not exactly a trivial task to carry out. Modeling and simulation, which are intensive computational tasks, are no more anticipatory than any other mechanisms. They embody the limitation intrinsic in the epistemological horizon in which they were conceived. Neural networks and anticipation, followed by impressive achievements in animation and robot motion planning ([Balkenius et al. 1994](#), [Christensen and Hooker 2000](#), [Fleischer et al. 2003](#)), only allow us to realize again the difference between purposive activities (where there is a finality; a “final cause”) and deterministic activities, of a different causal condition.

This observation brings up the effort known under the name *CASYS conferences* (organized by Daniel M. Dubois, in Liège, Belgium since 1997). Dubois builds upon [McCulloch, and Pitts \(1943\)](#) “formal neuron” and on von Neumann's suggestion that a hybrid digital-analog neuron configuration could explain brain dynamics

(cf. 1951). It is tempting to see the hybrid neuron as a building block of a functional entity with anticipatory properties. But from the premise on, Rosen followed a different path, quite convincingly, that recursions could not capture the nature of anticipatory processes (since the “heart of recursion is the conversion of the present to the future”). Neither could incursion and hyperincursion (an incursion with multiple solutions that Dubois advanced) satisfy the need to allow for a vector pointing from the future to the present. Rosen warned about the non-fractionability of the “(M,R)-Systems”; and this is of consequence to the premise adopted in Dubois’ work. When Dubois (2000) defines “...the main purpose...is to show that anticipation is not only a property of biosystems, but is also a fundamental property of physical systems,” he argues with Rosen’s fundamental ideas from a position that basically ignores the distinction between the ontological (of which anticipation is a characteristic) and the epistemological. Within science, difference of views is perfectly acceptable, provided that the concepts are coherently defined. This provision is ultimately not met. For particular applications, Dubois’ take is quite convincing. His conference usually serves as a good opportunity for featuring contributions inspired by his work. Addressing issues of autonomous systems (i.e., they self-regulate), Collier (2008) builds on Dubois’ conjecture in addressing autonomy and viability. Suffice it to say that such a contribution is in itself relevant for the richness of the dialog that Rosen’s book and its subsequent interpretations triggered.

Chrisley (2002) is among those aware of the contradictory situation in which proponents of “computing anticipatory systems” are. He is explicit: One can go further and inquire as to the extent to which such causal anticipatory systems are computational. This is very important since, as Chrisley notes, the model is essential, not the data. The so-called “transduction of present data into future data (i.e., into predictions) through the agency of a model of the world “does not turn the probabilistic prediction into anticipations. Indeed, the anticipation, expressed in action, is, after all, part of the system from which it originates. In order to address this aspect, Dubois (2000) distinguished between weak anticipation – more or less along Rosen’s idea of a model-based process – and strong anticipation “when the system uses itself for the construction of its future states,” (Dubois and Leydesdorff 2004). Leydesdorff (a distinguished researcher of social systems examined from an anticipatory perspective) argues that “the social system can be considered as anticipatory in the strong sense,” (it constructs its future). Leydesdorff (2008) further enlarges upon this observation in examining intentionality in social interactions.

Along this line, I want to mention some very convincing attempts to relate perception and motoric response (Steckner 2001) to address issues of predictive model generation (Riegler 2001), to associate anticipation with decision-making processes (Nadin 2009a), to deal with interaction as it results in a variety of anticipatory processes (Kinder 2002). Riegler (2004) focuses on “Who (or what) constructs anticipation?” It is, as he defines it, a challenge to the implicit assumption of Rosen’s model-based definition. The “decision maker” whom he is after remains an open question. In the area of applied interest (automobile driving, assessing the impact of emerging technologies, extreme events assessment, the whole gamut

of applications in the *MindRace* program), authors from various backgrounds (Munduteguy and Darses 2007, Myers 2007, Nadin 2005, Nadin 2007; Pezzulo 2007a, 2007b, Butz 2007, among others) advance cognitive and information processing models that contribute to the current interest in pseudo-anticipatory processes. It is significant that Zadeh (2001, 2003) made the connection between computing with perceptions and anticipation. Ernst von Glasersfeld (1995) framed anticipation in the constructivist theory of cognition. Pribam (2000) considered free will and its relation to the brain as an anticipatory system. Klir (2002b) evaluated the role of anticipation in Intelligent Systems.

More recently, within the same interest in fundamental aspects of the subject, issues related to health have been examined from the perspective of anticipation (Berk et al. 2008, research at the Oak Crest Health Research Institute, Loma Linda, CA). The hypothesis that major neuro-motoric disorders (Parkinson's disease, in particular) are the result of skewed anticipation was advanced in an application to an NIH Pioneer grant (Nadin 2007). In respect to this hypothesis, the issue of time and timescale was brought up (Rosen and Kineman 2004), while brain imaging (Haynes 2008) allowed a very telling visualization of decision-making processes.

Beyond Reductionism

Examples – focused on incipient anticipation or on more recent research in anticipatory systems – are only additional arguments to the Prolegomena addressing the justification for a science of anticipatory systems. Very cognizant of the need for a broad foundation, Rosen often referred to Kant. My own decision to follow the example of Kant's *Prolegomena* was also informed by this fact. Evidence from experiments, which has multiplied beyond what was imaginable during Rosen's life (because today experimentation is richer than ever before), places the subject of anticipation in what Kant called the *apodictic*: certain beyond dispute. But the same holds true for reductionism in physics. Rosen was fully aware of this epistemological conundrum. Accordingly, he tried to justify the legitimacy of a science of anticipation as part of a broader science – that of organisms (the “living sciences” or “life sciences” of our days). There is no doubt that *What is Life?* – characterized as a “Fair Scientific Question” – turned out to be for him the “central question of biology (Rosen 1991, p. 25), and the pinnacle of his entire activity. Physics itself would change on account of our better understanding of life.

One does not have to agree with Rosen's concept. However, judgments about it have little meaning if they do not relate his notion of anticipation to his comprehensive view of the living, including the meaning of experimental evidence. Even with the best intentions – there is no reason to expect anything but integrity in the work of researchers – the error of taking a notion out of its broader context can occur. Recently, as I detailed an example I've given many times, I caught myself in such an error. Indeed, in examining the physics of a stone falling, we will notice

that there is no anticipation possible. We can predict the various parameters of the process. A cat falling is an example of “anticipation at work.” But the sentence, “The same stone, let fall from the same position will fall the same way,” does not automatically translate into “The same cat will never fall the same way;” because the cat, after falling once, will no longer be the same. Over time, stones change as well, in the sense that their physical characteristics change; but the time scale of change in the physical is different from the time scale at which the living changes. Moreover, stones do not learn. The cat after the experience is a different cat, a bit older, and richer in experience. This example illustrates how the measurement aspect, the anticipation aspect (influenced by the experience of measurement), and the overarching understanding of the living (as open, learning, on faster time scale, etc.) are tightly connected.

Within this understanding, life emerges beyond the threshold of complexity. Impredicativity and non-fractionability are related because they describe the living in its unity. Therefore, now that the second edition of Rosen’s AS is available, the reader would be well advised to read it in conjunction with *What is Life?* And with Rosen’s book on Measurement. This is an opportunity that the original readers (in 1985) did not have (at least for six years, until *Life Itself* was published). Only in associating all three books can one derive that kind of knowledge alluded to in Kant’s Prolegomena (*On the possibility of metaphysics*).

In reporting on the rich variety or research directions in anticipation today, I was looking less to finding arguments in favor of discipline and more in the direction of acknowledging variations in the meaning of the concept. Rosen’s strict terminological discipline should, of course, not be construed as a declaration of ownership, or, as we shall see in the final part, as an ultimate truth. My own view of anticipation probably differs from his. Those who have tried to approach issues of anticipation have rarely been aware of each other’s work, and almost never of Rosen’s trilogy. My conversations with quite a number of German, Russian, Czech, Japanese, and some French researchers confirmed that Rosen’s work was hardly known to them as part of a larger view of what life is.

The understanding of anticipatory processes as definitory of the living is shared by a minority of those pursuing the subject. But this is science, always subject to subsequent revisions and re-definitions, not religion or a dogmatic pursuit of pure terminology. It would benefit no one to proceed in an exclusionary manner. Knowledge is what we are about; and in the long run, our better understanding of the world and of ourselves, as expressed in new practical experiences, is the final arbiter. In this sense, it can prove useful to our understanding of Rosen’s contribution and the richness of attempts not aligned with his rigorous science, to shortly acknowledge yet another fascinating scientist whose work came close to some of Rosen’s interrogations: Heinz von Foerster. I was unable to find out whether the two of them met. Von Foerster was associated with the University of Chicago for a while; his Biological Computer Lab at the University of Illinois-Urbana Champaign could not have escaped Rosen’s attention. Moreover, his original writings (in establishing Second Order Cybernetics) definitely caught Rosen’s attention. Von Foerster himself was aware of Rosen’s work and found the subject of anticipation

Fig. 2 Rosen's model of anticipation

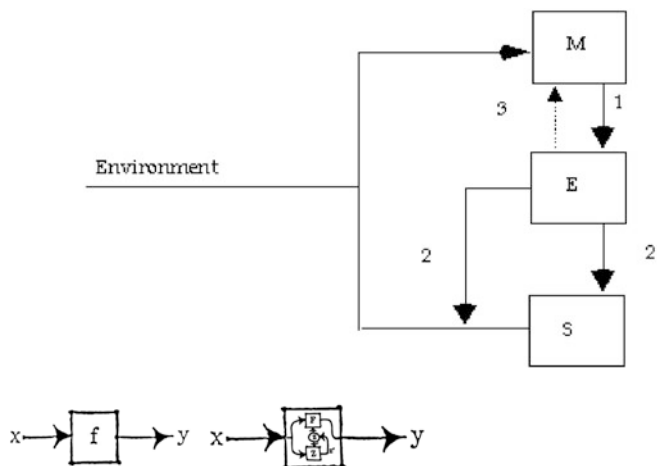


Fig. 3 “Roadsigns Definitions Postulates Aphorisms, etc.” [sic]. Heinz von Foerster (1995)

very close to his own views of the living and on the constructivist Condition of Knowledge. In his famous formulation, “Die Ursache liegt in der Zukunft” (The cause lies in the future), von Foerster gives what is probably the most concise (or expressive) definition of anticipation. But what prompts my decision to bring up von Foerster is the striking analogy between Rosen’s anticipatory model (1985a, p. 13) and von Foerster’s (2002) concept of non-trivial machines (Figs. 2 and 3).

Let us only make note of the fact that non-trivial machines are dependent on their own history (which is the case with Model M in Rosen’s model), cannot be analytically determined, and are unpredictable (cf. 2002, p. 58).

If the suggestion holds – and we should dedicate more time to it – it is quite clear how from Rosen’s original definition of anticipation many more were derived as alternative, non-trivial, machines (in von Foerster’s sense). This brings up important epistemological questions, such as hybrid computation (the human being or a living entity connected somehow to computers), or even quantum computation in the sense Feynman defined it; interactive computation, membrane computations, DNA computation. It is necessary to realize that we are focused on non-algorithmic data processing, sometimes combining analog and digital representations. Howard Pattee, Rosen’s colleague at the Center for Theoretical Biology in Buffalo – and with whom he had many passionate discussions – still cannot accept Rosen’s intransigence in dealing with von Neumann’s universal constructor – a construct that could achieve unlimited complexity. Pattee is willing to concede that formally von Neumann’s model was incomplete. But he argues that ultimately, von Neumann and Rosen agreed (“life is not algorithmic”), moreover, that self-assembly processes characteristic of the living do not require complete genetic instructions. The reason I bring up this point is rather practical, and Pattee (2007) expressed it convincingly: we should avoid getting diverted from Rosen’s arguments only because, at times, they do not conform with the accepted notions (in this case, von Neumann’s replication scheme).

Rosen (1966) was firmly opposed to von Neumann's understanding that there is a "threshold of complexity" that can be crossed by finite iterations (analogous to the notion that infinity can be reached from a finite number, simply by adding one more). Rosen brought up the need to account for the characteristics of the organism as evolvable, adaptive. Nevertheless, in hindsight we can say that both realized, although in different ways, that if complexity is addressed from an informational perspective, we end up realizing that life is ultimately not describable in algorithmic terms. Chu and Ho (2006) noticed that in Rosen's view, "living systems are not realizable in computational universes." – whatever such computational universes might be. They provided a critical (negative) assessment of Rosen's proof, which Louie (2007) convincingly refuted. Louie's argument in some ways confirms that non-algorithmic self-assembly (epigenetic progresses) is of such a condition that it does not require full descriptions either of the functions or of the information involved in living processes.

Computation and Anticipatory Processes

Given the implications of this observation, let us take a closer look at what it means. Along the line of the Church-Turing thesis – i.e., that every physically realizable process is computable – von Neumann (1963, p. 310) went on out a limb and stated, "You insist that there is something a machine cannot do. If you will tell me precisely what it is that a machine cannot do, I can always make a machine which will do just that." If von Neumann was convinced that telling *precisely* what it is a machine cannot do – emphasis on *precisely* – is a given, he was not yet disclosing that *telling precisely* might after all require infinite strings, and thus make the computation to be driven by such a description impossible (*intractable*, in computer science lingo). It is easy to show that if you completely map a process, even the simplest computation can reproduce its function. But can we completely map even simple biological functions? And given the fact that "To live is to change" (cf. Chorda 2010), to map completely is, in the final analysis, to create a *living representation*, a virtual process in which *matter* is replaced by *information*. Actually, von Neumann should have automatically thought of Gödel (and maybe he did) in realizing that a complete description, which would have to be non-contradictory, would be impossible. Descriptions, in words (as he expected, cf. "anything that can be completely and unambiguously put into words..."), or in some other form (e.g., numbers), are, in the final analysis, semiotic entities. They stand as signs for something else (the represented), and in the process of interpretation they are understood as univocally or ambiguously defined (Nadin 1991).

Representations of the world are always incomplete; they are not fragments of the world. It is such incomplete representations that are processed in an algorithm-driven computation or in some non-algorithmic computational process. Until the development of brain imaging, we could not capture the change from sensorial

energy to the cerebral re-presentational level. And even with images of the brain, semiotic processes are still not quantified. What is quantified are information processes, because information was conveniently defined in relation to energy (cf. [Shannon and Weaver 1949](#)). And we are able to measure energy quite precisely. But meaning is more than information; things make sense or not, not on account of bits and bites, but rather on qualitative changes. It is the re-presentation of things, not things themselves, that is subject to processing and understanding. Re-presentations (like a picture of a stone, or its weight, or the chemical formula/formulae describing its composition) are renewed presentations (of the stone) as signs, which means: dematerialized, extracted from the thermodynamic context, from the dynamics in which they are involved. Interpretations are attempts to associate a sign (a semiotic entity) to an object (a physical entity) and to conjure the consequences that the sign might have on our activity. Re-presentations can be of various degrees of ambiguity – from very low (indexical signs, as marks left by the object represented) to very high (symbols, i.e., conventions). Lightning arouses a sense of danger associated with phenomena in the world. The black cat can bring up false associations (superstitions) with dangers in the world. They are of different levels of ambiguity. The living can handle them quite well, even if, at times, in a manner we qualify as irrational (cf. [Dennett 1991](#) on anticipation).

Machines operate also on representations. But if we expect a certain output – such as the visualization of a process, or the processing of a matrix (a mathematical entity that does not correspond to processes in the world) – we have to provide representations that are unambiguous. Machines do not *dis-ambiguate* representations. For this reason, we conceive, design, and deploy artificial languages of no or very low ambiguity. The living operates, most often effectively, with representations regardless of their ambiguity. The machine is “protected” from ambiguity. (We endow machines with threshold identifiers: is the ignition turned on or not? Neither intermediate values nor intentions count! Ambiguity would be a source of error in their functioning.) Von Neumann’s claim that he could conceive a computation for any *precisely* described entity (i.e., complete description) means nothing more than that he proceeds to segregate between the semiotic of the unambiguous and the semiotics of ambiguity. In von Neumann’s thinking, to be precise means to be also unambiguous (in addition to reducing the measurement error to zero).

Moreover, computational reductionism does not acknowledge the fundamental role of time in the dynamics of the living. Time is reduced to interval. There is a clock that keeps track of sequential processes; and a clock is necessary in order to support the rigid synchronicity of parallel computation. It has been repeatedly demonstrated that an anticipatory system has at least two clocks, i.e., correlated processes unfolding at different times scales ([Nadin 2009b](#)). At various levels of the living, several clocks are at work: some very fast (at nanosecond speed); others in the domain of the “gravitational” clock; and yet others are very slow. Therefore, Rosen’s model unfolding at faster than “real time” is probably a distributed anticipatory process with many models operating at various time scales. [Rosen and Kineman \(2004\)](#) examine the characteristics of complexity in (Robert) Rosen’s view, realizing

correctly the central role played by the modeling relation. “The internal predictive models” are, in their view, hypotheses about future behavior. Finally, Feynman’s understanding of the integration of past, present, and future in the computation (meant to simulate Nature) is probably closer to Rosen’s understanding of anticipation than is von Neumann’s implicit anticipatory dynamics.

With all these considerations in mind, the reader of AS in its second edition should now be in a better position to understand that at the level of algorithm-driven machines (digital or not), anticipation is not possible. Such simple machines operate in the interval domain of causes and effects, in a non-ambiguous manner. As [Verbos \(2010\)](#) noticed, “The brain possesses a kind of general ability to learn to predict.”

Rosen – and not only Rosen (even von Neumann, cf. [Pattee 2007](#)) – ascertained that the living is not representable through a computational process. There are good reasons to accept that the living is not reducible to a deterministic machine, no matter how sophisticated such a machine might be. As we have seen with Feynman’s take on quantum computation, the situation changes: this is qualitatively a different machine; it is a *sui generis* process. Actually, quantum mechanics as a construct originates in response to the deterministic view of nature. Therefore, the argument that a quantum type of computational process can represent life is somehow circular: the quantum representation was constructed in order to overcome the limitations inherent in the classical deterministic concept of machines. The same holds true for DNA computing.

Be this as it may – and with the warning that such discussions cannot be relegated to well-intentioned half-baked scientists (cf. [Penrose 1989](#)) – the question to be posed is, after all, not whether we can ever come up with machines that are not machines in the sense propagated since Descartes (and de la Mettrie), but rather: If enough computational resources are available – theoretically an inexhaustible amount – wouldn’t the aggregate computation be sufficient to become a “life-like” process? Brute force computation – the relatively common practice in almost all instances of computation used to deal with complicated processes – means to throw as many computing cycles as possible at a problem and to work on as much data pertinent to the problem as we can get. IBM’s Deep Blue, which beat Kasparov in a chess game, met Turing’s test of intelligence without even being close to a living chess player (master or beginner). Seemingly, Venter’s attempt at modifying life is also one based on brute force. While [Hacker \(1978, 2005\)](#) addressed anticipation (properly defined as *Vorwegnahme*, the German word that describes *antecipare*, the Latin word behind *anticipation*) in relation to trajectory evaluation – a frog catching prey is the classic example – the *Star Wars* effort was supported by brute force computation. *Star Wars* was not about anticipating an attack, rather integrating information (descriptions of attack circumstances) and providing a reaction based on the fact that the speed of even the fastest ballistic missiles, carrying nuclear warheads or other lethal charge, is limited, and lower than the speed of electrons. If a system can get information as close as possible to the launch pad, given the fact that there is a huge difference in speed between the missile’s movement and the propagation of information, it can generate predictive models and effectively target – like the frog catching an insect – the trajectory load. “Elementary, my dear Watson” – nothing more to add.

With all this in mind, the methodological question becomes: Is the non-algorithmic, or even the broader (ill-defined) non-computable, corresponding to a level of complexity beyond the threshold defining the living, equivalent to infinite computation? The characteristics of powerful computation of large amounts of data (which is the tendency in our days) and the characteristics of brute-force computation (on a larger scale) of small but appropriately selected data are different. Can we evaluate the difference between computing an infinite amount of data (all that we can get given the progress in sensor technology) and the infinite computation needed to process a small but relevant amount of data? Evidently, to know beforehand which data are significant is a matter of anticipation; and it might simply move the problem of computing anticipation from processing all available data to filtering what is significant.

In general, to represent life is to represent something that is in process, changing all the time. For better or worse, we can model/represent – ergo prevent – every oil spill that has already occurred, but not anticipate accurately the one that will happen next. We can similarly model/represent every terrorist attack, and every financial crisis, and every epileptic seizure that have already taken place, but not predict accurately new occurrences of the same. But since there are no laws that capture the uniqueness of extreme events, of disease, of art creation, of Shakespeare's writings, of scientific research, we cannot build a machine – similar to cars, rockets, AI-driven surgery, etc that will anticipate such things. The regular, patterned, and repetitive can be described as infinite representations. Therefore, machines can be conceived to effectively process such representations. The unique is the subject of idiographic knowledge, which is focused on the particular, not the general. Currently, more and more science is idiographic: visualizations are not abstractions equivalent to mathematical equations. To generalize from an individual's brain image to all brains is more a matter of faith than of science. Dynamic visualizations – i.e., “films” of certain processes – are even less so. They qualify rather as “histories” – which is the substance of biology – than as “theories.”

Anticipation, always expressed in action, is unique. Repetitive patterns, such as the frog's behavior in chasing a moving target (and in nature, this is not an exception), or mating behavior, do not result in laws, but in “chronicles” of successful and less successful actions. In ascertaining the action as expression of anticipatory processes, we implicitly ascertain realization at the pragmatic level, but not syntactic or even semantic performance, as is the case with machines. A computation “aware” of its state is an intentional procedure. Such a teleological computation, if we could conceive of it, would have a sense of purpose. It would also be an adaptive computation. Not unlike an artist, who never knows when the goal has been reached (for that matter, every work of art is open, i.e., unfinished), such a computation could also be seen as open-ended. Anticipatory computing (cf. [Nadin 2010b](#)), as a subject, will benefit from Rosen's many elaborations on the subject, in the sense of terminological coherence.

Once we reach the notion of complexity at **which causality itself is no longer reducible to determinism**, and the condition of the living integrates past, present, and future, a new form of adaptive behavior and of finality (*purposiveness*) emerges. In

the dimension of complexity, anticipatory processes become possible, although only as non-deterministic processes (after all, anticipation is often wrong (cf. [Dennett 1991](#))).

Life is process (to recall Whitehead, among others), more precisely, non-deterministic process. This makes the role of the physician, and of the economist and the politician, for that matter, so difficult. There is so much science that such endeavors are built upon. But there is always a whole lot of art that makes them successful. Therefore, in addressing causality in respect to the living (a person's health, the state of the economy, the politics associated with managing extreme events), we need to consider past and present (cause-effect, and the associated reaction), both well defined, in conjunction with creativity, i.e., a possible future realization, ill defined, ambiguous. When we have to account for higher complexity – the threshold beyond which reaction alone can no longer explain the dynamics – the anticipatory component must be integrated in our understanding. In logic ([Kleene 1950](#)) an impredicative definition is one in which the definition of an entity depends on some of the properties of the entities described. The definition of life is an example of *impredicativity*; that is, it is characterized by complexity, which in turn is understood as a threshold for the living. If this strikes you as circular, that's because it is. Impredicative definitions are circular. [Kerckel \(2007\)](#) noticed that ambiguity is an observable signature of complexity. He goes on to connect this to the issue of prediction: "Ambiguity of complexity shows that the 'unpredictable' behaviors of complex systems are not random, but are causally determined in a way that we (having no largest model of them) cannot completely predict." These words describe anticipatory dynamics.

Laws of Nature vs. Expressiveness. Generality vs. Singularity

At some moment in the research upon which these Prolegomena are based, I contacted several scientists known to have studied Rosen's work, some who even worked with him. In general, regardless of whether they agreed with his work or were critical of it, the attitude has been one of sincere respect. As radically new as Rosen's perspective was, it was based on a broad intellectual foundation. The fact that he answered questions that nobody else seems to have posed does not make those questions less relevant. Indeed, researchers and scholars from all fields of endeavor and who are captive to an accepted framework tend to function only within the premises they have accepted. Rosen questioned the premises. This was radical. An answer from one of the persons I contacted is telling: I have my reservations about LI, but I fully agree with AS [paraphrase mine].

How can this be? This question is prompted by the subject of the Prolegomena: What speaks in favor of an inquiry into anticipatory processes? In simpler terms: Will the time come when we go to a store and buy not only an iPhone or an iPad, but also an iAnticipator? It is easy (and comes close to demagoguery) to consider the most recent crises – financial meltdown, slowdown in the economy,

ash clouds from volcanoes, oil spills in the Gulf of Mexico – and exclaim: We need anticipation! The French speak about *le grand soir* (the great evening) – that would be the “evening” when we will know everything about what will happen and either avoid it (somehow), or at least prepare for it. That would mean the miraculous cure, the gene switch that we only have to put in the right position in order to cure cancer; no more stock market crashes, no more disastrous oil spills, etc. etc. Animated by an optimism that is implicit in every scientific endeavor, many talented and very serious scholars are trying to make such a *grand soir* happen. They believe in determinism. If it happens, there must be a cause behind it. The advent of the information revolution made membership in the holy dogma of reductionist determinism easier. These scholars and researchers are willing to concede that the short sequence of cause-and-effect might be an oversimplification, and then, since they have access to information processing, chase after longer and longer sequences. All kinds of systems, using computation as their underlying procedure or not, are conceived for a variety of reasons: to help prevent terrorist attacks, stock market crashes, pandemics, etc. Other systems are deployed for automating tasks involving a great deal of uncertainty: from nuclear power plant monitoring to autonomous airplane piloting, or even autonomous automobile driving. Finally, others go so far as to create new realities, where functioning according to such deterministic rules is implicit in their making. The clock of the remote past serves as the “universal” blueprint; the hardware was conceived as a sequence of cause-and-effect mechanics.

Synthetic life – yet another example of a knowledge domain rapidly ascertaining itself – is exactly what I am referring to. But where *Fabrication of Life*, which is part of the subtitle of Rosen’s LI, I interpret as “how life fabricates its own life,” synthetic life is the attempt – deserving our entire respect as a scientific goal – to create life from non-life on the basis of a cause-and-effect understanding of the living. All these attempts are based on the assumption that laws similar to those of physics (if not the same laws) can be formulated and applied in order to construct anticipatory machines or living machines with the same degree of success that humankind has already had in building cars, airplanes, rockets, computers, self-assembling nano-machines, and the like.

My own position on the matter is very clear: laws of physics or quantum mechanics fully apply to the physical aspects of the living, but not to its specific dynamics. Complexity is not the result of additive physical processes. Very recently, Kauffman and Longo came out with similar conclusions, in line with Rosen’s view of the living: No law entails the evolution of the biosphere ([Kauffman and Longo 2011](#)). [Kauffman \(2011\)](#) dedicated a lecture to “The end of physics as a world view.” Neither Kauffman nor Longo referred to anticipation; but sooner or later they will have to integrate it in their arguments leading to their sweeping assertions.

Readers of *AS* will understand that this book, even in its revised version, is not a *How to* manual, but rather a *Why?* inquiry into what makes anticipation a characteristic of the living. Could Rosen have put together an “Anticipatory Systems for Dummies” in the vein of all those “4 Dummies” books written with the aim of making us all better mechanics, plumbers, graphic designers, gardeners, users of word-processing programs, etc? First and foremost, that would not have been a task

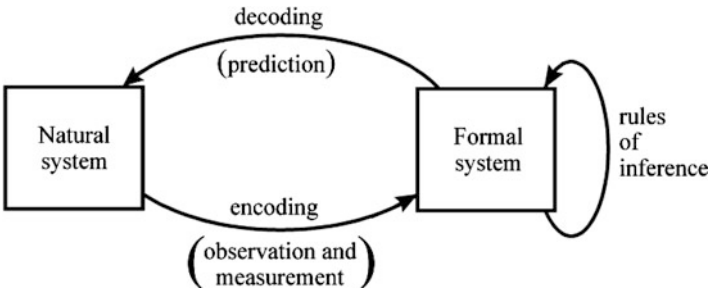


Fig. 4 Initial modeling relation

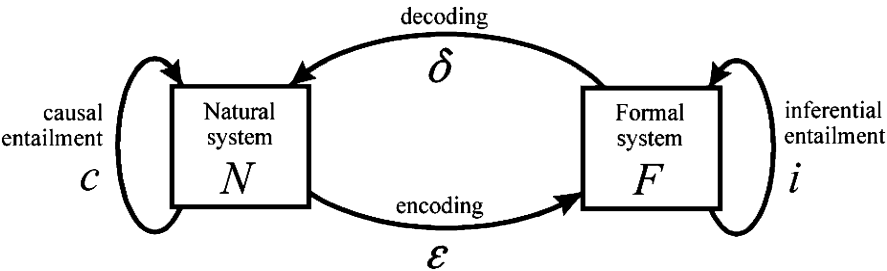


Fig. 5 Revised modeling relation

for a passionate intellectual who was fully committed to his science. Moreover, neither what defines the living nor what it means to measure, and even less what anticipation is are subjects for the do-it-yourself obsession of our time.

For something to be replicated, we need the understanding of what is needed for that entity to exist in the first place. Rosen expressed this kind of understanding in the modeling relation. This is, in a very concentrated form his epistemology (Fig. 4).

In AS, the natural system is represented by a formal system: knowledge is derived as inference and eventually guides practical activities. Rosen explains: Knowledge acquisition translates as an attempt to encode natural systems (i.e., represent them based on a semiotic code) into formal descriptions (equations, diagrams, etc.). Operating on representations, we can derive inferences or even theorems (statements that can be finally proven). Such operations eventually result in statements about how the natural system might behave. In fact, a modeling relation is, *in nuce*, a “theory of prediction.” For it to become a “theory of change,” (cf. Nadin 2010d), it would have to account for how the natural system behaves over time. Causality – causal entailment, in Rosen’s original thought – is the answer to the *Why?* question definitory of relational biology. In LI, causality is expressed as causal entailment and “rules of inference” are seen as the causal entailment encoded into inferential entailment, which completes the modeling relation diagram (Fig. 5).

Anticipation itself has the condition of entailment. It is clear that there is anticipation in the natural system; but this does not translate, in a one-to-one

relation, into the formal system, and even less into the ability to replicate it. In his words:

We seek to encode natural systems into formal ones [such that] the inferences or theorems we can elicit within such formal systems become predictions about the natural systems we have encoded into them (p. 74 of the original edition).

If we associate Rosen's clear statement with Einstein's (1921) observation, "In so far as the propositions of mathematics are certain, they do not apply to reality; and in so far as they apply to reality, they are not certain," we realize that only charlatans can promise to deliver a known future event. What is possible, however, is the design of systems that allow us to consider under which circumstances our descriptions of reality will have to be endowed with adaptive properties (no matter how primitive or limited in scope).

Earlier in the Prolegomena, the point was made that hyperboles do not help us better understand Rosen's original contributions. Beyond the hyperbole in "Biology's Einstein" lies an argument we need to consider: Rosen advanced a coherent view of the living within a broad understanding of what is traditionally called Nature. If in respect to the physical, the formal system allows us to infer laws, on the basis of which machines are built, then in respect to the living, we can at best describe successions, and further relations between the events or phenomena succeeding each other. Windelband (1894), who advanced the distinction between *noematic* sciences (focused on descriptions in the forms of laws) and the *idiographic* sciences (focused on descriptions of sequences, in the form of Gestalt), would have mentioned Rosen's view as illustrative of both (if he had not been his precursor by almost a century). Rosen specifically spells out (LI, p. 58) that science describes the world ("that is in some sense orderly enough to manifest relations or laws"), but that it also "says something about ourselves." The "orderliness" of the world (ambience) "can be matched by, or put into correspondence with, some equivalent orderliness within the self." This is what makes science, and also the scientist, possible.

What speaks in favor of an inquiry into anticipatory processes is the need to ascertain the complementarity of noematic and idiographic knowledge. Every process of anticipation, definitory of the natural system, involves knowledge as an expression of accumulated experience, but also of art as an expression of the creativity implicit in the living. In a very telling private communication (addressing Segel, who reviewed "Organisms as causal systems which are not mechanisms: an essay into the nature of complexity"), Rosen (1985c) uses a very intuitive image: "The point of view I have taken over the past 25 years is that the way we look at systems is no different than the way we look at each other . . . dynamic interactions between the (systems) are cognate to our own observing process." His intention in explaining his view to a colleague who "did not get it" (as we would say today) was very simple: "I'm writing the above only so that we may understand each other better." This second edition of AS is a delayed "letter" to a new generation of scientists who are now in a better position than Rosen's own generation to entertain provocative ideas that justify current interest (and hope) in anticipation studies.

Final Note

Rosen wanted to write a book on complexity. Many scholars and researchers wish he did. It was not to be; but it suggests to the scientific community that progress in understanding the relation between complexity and anticipation requires that we define the situation at which anticipation becomes possible – and probably necessary.

Science is not about doctrine – even less about faithfulness. Rosen’s work should therefore not be seen in a light different from his own: It is yet another hypothesis – probably one whose time has finally come. In publishing a second edition of *AS*, those involved in the process did not intend to suggest a return to the initial book, rather to stimulate further dialog and more probing scientific investigation. Judith Rosen has her own well-defined identity; she is also passionate about her father’s legacy and understands why dialog is important. She never hesitated in providing answers to questions I had or volunteering details I would not have access to. George Klir is dedicated to this book, and even more to the pursuit of academic dialog of integrity. A.H. Louie, who has also supported the effort of seeing a second edition of *AS* published, disseminates Rosen’s thoughts in his own distinguished publications. His teacher would have been proud of him. I have benefited from their competence and wisdom, and I admire their knowledge and appreciation of Rosen’s ideas. I also benefited from contacts with Peter Cariani, Roy Chrisley, Winfried Hacker, G. Hoffman, S. Kercel, Dobilas Kirvelis, Andres Kurismaa, Loet Leydesdorff, Helmut Loeckenhoff, Alexander Makarenko, E. Minch, H.H. Pattee, Dean Radin, Marion Wittstock (who made me aware of the work of the Dresden group), and Olaf Wolkenhauer, and want to express my gratitude for their help. In particular, my respect to Lotfi Zadeh for some very useful conversations, and to Otthein Herzog for making possible my research at the Information Technology Center (TZI, Bremen University, 2009–2010). Lutz Dickmann was a patient sounding board for many of my hypotheses. He challenged me in the spirit of his own research for his Ph.D. To what extent my research of anticipation enriched the work of everyone I interacted with (in the project Anticipation-Based Intelligent Multiagent Societies for Logistics) remains to be seen. A Fellow award at the Hanse Institute for Advanced Study (Delmenhorst, Germany) allows me to focus on the relation between anticipation and its representation. A first concrete result is the study dedicated to the expression of anticipation in innovation ([Nadin 2012c](#)) and the interaction I had with a distinguished musician (Tibor Szász, Freiburg) regarding anticipation expressed in musical creativity (in particular Enescu and Bartók). But this was also a time for putting the finishing touches on these Prolegomena. Publication of the second edition of *AS* does not do any favor to Rosen – his work stands on its own – but to everyone who is authentically interested in the subject. My prediction is that the book will stimulate more attempts to integrate the anticipatory perspective in our understanding of the world.

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Chapter 1

Preliminaries

1.1 General Introduction

The original germinal ideas of which this volume is an outgrowth were developed in 1972, when the author was in residence as a Visiting Fellow at the Center for the Study of Democratic Institutions in Santa Barbara, California. Before entering on the more formal exposition, it might be helpful to describe the curious circumstances in which these ideas were generated.

The Center was a unique institution in many ways, as was its founder and dominating spirit, Robert M. Hutchins.¹ Like Mr. Hutchins, it resisted pigeonholing and easy classification. Indeed, as with the Tao, anything one might say about either was certain to be wrong. Despite this, it may be helpful to try to characterize some of the ambience of the place, and of the remarkable man who created it.

The Center's spirit and *modus operandi* revolved around the concept of the Dialog. The Dialog was indispensable in Hutchins' thought, because he believed it to be the instrument through which an intellectual community is created. He felt that "the reason why an intellectual community is necessary is that it offers the only hope of grasping the whole". "The whole", for him, was nothing less than discovering the means and ends of human society: "The real questions to which we seek answers are, what should I do, what should we do, why should we do these things? What are the purposes of human life and of organized society?" The operative word here is "ought"; without a conception of "ought" there could be no guide to politics, which, as he often said, quoting Aristotle, "is architectonic". That is to say, he felt that politics, in the broadest sense, is ultimately the most important thing in the world. Thus for Hutchins the Dialog and politics were inseparable from one another.

For Hutchins, the intellectual community was both means and end. He said,

The common good of every community belongs to every member of it. The community makes him better because he belongs to it. **In political terms the common good is usually defined as peace, order, freedom and justice.** These are indispensable to any person, and no person could obtain any one of them in the absence of the community. An intellectual community is one in which everybody does better intellectual work because he belongs to a community of intellectual workers. As I have already intimated, an intellectual community cannot be formed of people who cannot or will not think, who will not think about anything

in which the other members of the community are interested. Work that does not require intellectual effort and workers that will not engage in a common intellectual effort have no place in [the intellectual community].

He viewed the Dialog as a continuation of what he called “the great conversation”. In his view,

The great conversation began with the Greeks, the Hebrews, the Hindus and the Chinese, and has continued to the present day. It is a conversation that deals – perhaps more extensively than it deals with anything else – with morals and religion. The questions of the nature and existence of God, the nature and destiny of man, and the organization and purpose of human society are the recurring themes of the great conversation. . .

More specifically, regarding the Dialog at the Center, he said,

Its members talk about what ought to be done. They come to the conference table as citizens, and their talk is about the common good. . . It does not take positions about what ought to be done. It asserts only that the issues it is discussing deserve the attention of citizens. The Center tries to think about the things it believes its fellow citizens ought to be thinking about.

The Dialog was institutionalized at the Center. Almost every working day, at 11:00 a.m., the resident staff would assemble around the large green table to discuss a pre-circulated paper prepared by one of us, or by an invited visitor. At least once a month, and usually more often, a large-scale conference on a specific topic, organized by one or another of the resident Senior Fellows, and attended by the best in that field, would be held. Every word of these sessions was recorded, and often found its way into the Center’s extensive publication program, through which the Dialog was disseminated to a wider public.

It might be wondered why a natural scientist such as myself was invited to spend a year at an institution of this kind, and even more, why the invitation was accepted. On the face of it, the Center’s preoccupations were far removed from natural science. There were no natural scientists among the Center’s staff of Senior Fellows, although several were numbered among the Center’s Associates and Consultants; the resident population, as well as most of the invited visitors, consisted primarily of political scientists, journalists, philosophers, economists, historians, and a full spectrum of other intellectuals. Indeed, Mr. Hutchins himself, originally trained in the Law and preoccupied primarily with the role of education in society, was widely regarded as contemptuous of science and of scientists. Immediately on assuming the presidency of the University of Chicago, for instance, he became embroiled in a fulminating controversy on curricular reform, in which many of the faculty regarded his position as anti-scientific, mystical and authoritarian. At an important conference on Science and Ethics, he said, “Long experience as a university president has taught me that professors are generally a little worse than other people, and scientists are a little worse than other professors”.

However, this kind of sentiment was merely an expression of the well-known Hutchins irony. His basic position had been clearly stated as early as 1931:

Science is not the collection of facts or the accumulation of data. A discipline does not become scientific merely because its professors have acquired a great deal of information. Facts do not arrange themselves. Facts do not solve problems. I do not wish to be

misunderstood. We must get the facts. We must get them all. . . But at the same time we must raise the question whether facts alone will settle our difficulties for us. And we must raise the question whether. . . the accumulation and distribution of facts is likely to lead us through the mazes of a world whose complications have been produced by the facts we have discovered.

Elsewhere, he said,

The gadgeteers and data collectors, masquerading as scientists, have threatened to become the supreme chieftains of the scholarly world.

As the Renaissance could accuse the Middle Ages of being rich in principles and poor in facts, we are now entitled to enquire whether we are not rich in facts and poor in principles.

Rational thought is the only basis of education and research. Whether we know it or not, it has been responsible for our scientific success; its absence has been responsible for our bewilderment. . . Facts are the core of an anti-intellectual curriculum.

The scholars in a university which is trying to grapple with fundamentals will, I suggest, devote themselves first of all to a rational analysis of the principles of each subject matter. They will seek to establish general propositions under which the facts they gather may be subsumed. I repeat, they would not cease to gather facts, but they would know what facts to look for, what they wanted them for, and what to do with them after they got them.

To such sentiments, one could only say Amen. In my view, Hutchins was here articulating the essence of science, as I understand it.

However, I had more specific intellectual reasons for accepting an invitation to spend a year at the Center, as, I think, the Center had for inviting me to do so. It may be helpful to describe them here. My professional activities have been concerned with the theory of biological systems, roughly motivated by trying to discover what it is about certain natural systems that makes us recognize them as organisms, and characterize them as being alive. It is precisely on this recognition that biology as an autonomous science depends, and it is a significant fact that it has never been formalized. As will be abundantly seen in the ensuing pages, I am persuaded that our recognition of the living state rests on the perception of homologies between the behaviors exhibited by organisms, homologies which are absent in non-living systems. The physical structures of organisms play only a minor and secondary role in this; the only requirement which physical structure must fulfill is that it allows the characteristic behaviors themselves to be manifested. Indeed, if this were not so, it would be impossible to understand how a class of systems as utterly diverse in physical structure as that which comprises biological organisms could be recognized as a unity at all. The study of biological organization from this point of view was pioneered by my Major Professor in my days as a graduate student at the University of Chicago, Nicolas Rashevsky (who, through no coincidence, idolized Robert Hutchins). Rashevsky called this study "relational biology", and we will have much more to say about it below.

The relational approach to organisms is in many ways antithetical to the more familiar analytic experimental approach which has culminated in biochemistry and molecular biology. Particularly in these latter areas, the very first step in any analytic investigation is to destroy the characteristic biological organization possessed by the system under study, leaving a purely physical system to be investigated by standard

physical means of fractionation, purification, etc. The essential premise underlying this procedure is that a sufficiently elaborate characterization of structural detail will automatically lead to a functional understanding of behaviors in the intact organism. That this has not yet come to pass is, according to this view, only an indication that more of the same is still needed, and does not indicate a fault in principle. The relational approach, on the other hand, treats as primary that which is discarded first by physico-chemical analysis; i.e. the organization and function of the original system. In relational biology, it is the structural, physical detail of specific systems which is discarded, to be recaptured later in terms of *realizations* of the relational properties held in common by large classes of organisms, if not universally throughout the biosphere. Thus it is perhaps not surprising that the relational approach seems grotesque to analytic biologists, all of whose tools are geared precisely to the recognition of structural details.

In any case, one of the novel consequences of the relational picture is the following: that many (if not all) of the relational properties of organisms can be realized in contexts which are not normally regarded as biological. For instance, they might be realized in chemical contexts which, from a biological standpoint, would be regarded as exotic; this is why relational biology has a bearing on the possibility of extraterrestrial life which is inaccessible to purely empirical approaches. Or they might be realized in technological contexts, which are the province of a presently ill-defined area between engineering and biology often called Bionics. Or, what is more germane to the present discussion, they may be realized in the context of human activities, in the form of social, political and economic systems which determine the character of our social life.

The exploration of this last possibility was, I think, the motivation behind the extending of an invitation to me to visit the Center, as it was my primary motivation for accepting that invitation. The invitation itself was extended through John Wilkinson, one of the Center's Senior Fellows, whose interest in the then novel ideas embodied in the structuralism of Levi-Strauss clearly paralleled my own concern with relational approaches in biology.

It is plain, on the face of it, that many tantalizing parallels exist between the processes characteristic of biological organisms and those manifested by social structures or societies. These parallels remain, despite a number of ill-fated attempts to directly extrapolate particular biological principles into the human realm, as embodied, for example, in Social Darwinism. Probably their most direct expression is found in the old concept of society as a "super-organism"; that the individuals comprising a society are related to one another as are the constituent cells of a multicellular organism. This idea was explored in greatest detail by the zoologists Alfred Emerson and Thomas Park, who studied insect societies, and who carefully established the striking degree of homology or convergence between social and biological organizations. (Coincidentally, both Emerson and Park were professors of zoology at the University of Chicago).

What would it mean if common modes of organization could be demonstrated between social and biological structures? It seemed to me that, in addition to the obvious conceptual advantages in being able to effectively relate apparently

distinct disciplines, there were a number of most important practical consequences. For instance, our investigation of biological organisms places us almost always in the position of an external observer, attempting to characterize the infinitely rich properties of life entirely from watching their effects without any direct perception of underlying causal structures. For instance, we may watch a cell in a developing organism differentiate, migrate, and ultimately die. We can perceive the roles played by these activities in the generation and maintenance of the total organism. But we cannot directly perceive the causal chains responsible for these various activities, and for the cell's transition or switching from one to another. Without such a knowledge of causal chains, we likewise cannot understand the mechanisms by which the individual behaviors of billions of such cells are integrated into the coherent, adaptive behavior of the single organism which these cells comprise.

On the other hand, we are ourselves all *members* of social structures and organizations. We are thus direct participants in the generation and maintenance of these structures, and not external observers; indeed it is hard for us to conceive what an external observer of our society as a whole would be like. As participants, we know the forces responsible for such improbable aggregations as football games, parades on the Fourth of July, and rush hours in large cities. But how would an external observer account for them?

It is plain that a participant or constituent of such an organization must perceive and respond to signals of which an external observer cannot possibly be aware. Conversely, the external observer can perceive global patterns of behavior which a participant cannot even imagine. Certainly, if we wish to understand the infinitely subtle and intricate processes by which biological organisms maintain and adapt themselves, we need information of both types. Within the purely biological realm, we seem eternally locked into the position of an external observer. But if there were some way to effectively relate biological processes to social ones; if, more specifically, both biological and social behaviors constituted alternate realizations of a common relational scheme, it might become possible to utilize our social experience as a participant to obtain otherwise inaccessible biological insights. Indeed, this capacity for transferring data and information from a system in which it is easy to obtain to a similar system in which it is hard to obtain is a unique characteristic of the relational approach. This was my basic hope; that I as a theoretical biologist could learn something new about the nature of organisms by judiciously exploiting the cognate properties of social systems.

The other side of that coin was equally obvious; that by exploiting biological experience, obtained from the standpoint of an external observer, we could likewise develop entirely new insights into the properties of our social systems. At that time, however, my detailed knowledge of the human sciences was essentially nil; to explore the possibilities raised above would require what appeared to me to be a major educational effort, and one which at first sight seemed far removed from my own major interests and capabilities.

It was at this point that I perceived the benefits of the community of scholars which Robert Hutchins had created. At the Center I could explore such ideas, while at the same time it was possible for me to learn in the most painless possible fashion

how the political scientist, the anthropologist, the historian, and the economist each viewed his own field and its relation to others. In short, the Center seemed to provide me with both the opportunity and the means to explore this virgin territory between biology and society, and to determine whether it was barren or fertile. I thus almost in spite of myself found that I was fulfilling an exhortation of Rashevsky, who had told me years earlier that I would not be a true mathematical biologist until I had concerned myself (as he had) with problems of social organization. At the time, I had dismissed these remarks of Rashevsky with a shrug; but I later discovered (as did many others who tried to shrug Rashevsky off) that he had been right all along.

Thus, I expected to reap a great deal of benefit from my association with the Center. But, as stressed above, the Center was an intellectual community, and to participate in it, I was expected to contribute to the problems with which the other members of that community were concerned. Initially, it appeared that I would have no tangible contribution to make to such problems as the constitutionalization of the oceans, the role of the presidency, or the press as an institution. Gradually, however, I perceived the common thread running through these issues and the others under intense discussion at the Center, and it was a thread which I might have guessed earlier. As I have noted, Hutchins' great question was: what should we do now? To one degree or another, that was also what the economists, the political scientists, the urban planners, and all the others wanted to know. However different the contexts in which these questions were posed, *they were all alike in their fundamental concern with the making of policy, the associated notions of forecasting the future and planning for it.* What was sought, in each of these diverse areas, was in effect a technology of decision-making. But underlying any technology there must be a substratum of basic principles; a science, a theory. What was the theory underlying a technology of policy generation?

This was the basic questions I posed for myself. It was a question with which I could feel comfortable, and through which I felt I could make an effective, if indirect, contribution to the basic concerns of the Center. Moreover, it was a question with which I myself had extensive experience, though not in these contexts. For the forecasting of the future is perhaps the basic business of theoretical science; in science it is called *prediction*. The vehicle for prediction must, to one degree or another, comprise a *model* for the system under consideration. And the making of models of complex phenomena, as well as the assessment of their meaning and significance, had been my major professional activity for the preceding fifteen years. In some very real sense, then, the Center was entirely concerned with the construction and deployment of predictive models, and with the use of these predictive models to regulate and control the behaviors of the systems being modeled. Therefore, the basic theory which must underlie the technologies of policy making in all these diverse disciplines is the theory of modeling; the theory of the relation between a system and a model of that system.

This in itself was a pleasing insight. And it led to some immediate consequences which were also pleasing. For instance: why did one need to make policies in the first place? It was clear that the major purpose of policy-making, of planning, was to eliminate or control *conflict*. Indeed, in one form or another, much attention at

the Center was devoted to instances of conflict, whether it be between individuals or institutions; that was what the Law, for example, was all about. In each specific case, it appeared that the roots of conflict lay not so much in any particular objective situation, but rather in the fact that differing *models* of that situation had been adopted by the different parties to the conflict; consequently, different predictions about that situation were made by these parties, and incompatible courses of action adopted thereby. Therefore, a general theory of policy making (or, as I would argue, a general theory of modeling) would have as a corollary a theory of conflict, and hopefully of conflict resolution.

I proceeded by attempting to integrate these thoughts with my overall program, which as I noted above was to establish homologies between modes of social and biological organization. Accordingly, I cast about for possible biological instances of control of behavior through the utilization of predictive models. To my astonishment, I found them everywhere, at all levels of biological organization. Before going further, it may be helpful to consider a few of these examples.

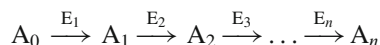
At the highest level, it is of course clear that a prominent if not overwhelming part of our own everyday behavior is based on the tacit employment of predictive models. To take a transparent example: if I am walking in the woods, and I see a bear appear on the path ahead of me, I will immediately tend to vacate the premises. Why? I would argue: because I can *foresee* a variety of unpleasant consequences arising from failing to do so. The stimulus for my action is not *just* the sight of the bear, but rather the output of the model through which I predict the consequences of direct interaction with the bear. I thus change my *present* course of action, in accordance with my model's prediction. Or, to put it another way, my present behavior is not simply *reactive*, but rather is *anticipatory*.

Similar examples of anticipatory behavior at the human level can be multiplied without end, and may seem fairly trivial. Perhaps more surprising is the manifestation of similar anticipatory behavior at lower levels, where there is no question of learning or of consciousness. For instance, many primitive organisms are negatively phototropic; they move towards darkness. Now darkness in itself has no physiological significance; in itself it is biologically neutral. However, darkness can be *correlated* with characteristics which are not physiologically neutral; e.g. with moisture, or with the absence of sighted predators. The relation between darkness and such positive features comprises a *model* through which the organism predicts that by moving towards darkness, it *will* gain an advantage. Of course this is not a conscious decision on the organism's part; the organism has no real option, because the model is, in effect, "wired-in". But the fact remains that a negatively phototropic organism changes state in the present in accord with a prediction about the future, made on the basis of a model which associates darkness (a neutral characteristic in itself) with some quality which favors survival.

Another example of such a "wired-in" model may be found in the wintering behavior of deciduous trees. The shedding of leaves and other physiological changes which occur in the autumn are clearly an adaptation to winter conditions. What is the cue for such behavior? It so happens that the cue is not the ambient temperature, but rather is day length. In other words, the tree possesses a model,

which *anticipates* low temperature on the basis of a shortening day, regardless of what the present ambient temperature may be. Once again, the adaptive behavior arises because of a wired-in predictive model which associates a shortening day (which in itself is physiologically neutral) with a *future* drop in temperature (which is not physiologically neutral). In retrospect, given the vagaries of weather, we can see that the employment of such a model, rather than a direct temperature response, is the clever thing to do.

A final example, this one at the molecular level, illustrates the same theme. Let us consider a biosynthetic pathway, which we may represent abstractly in the form



in which each metabolite A_i is the substrate for the enzyme E_{i+1} . A common characteristic of such pathways is a *forward activation* step, as for example where the initial substrate A_0 activates the enzyme E_n (i.e. increases its reaction rate). Thus, a sudden increase in the amount of A_0 in the environment will result in a corresponding increase in the activity of E_n . It is clear that the ambient concentration of A_0 serves as a *predictor*, which in effect “tells” the enzyme E_n that there *will be* a subsequent increase in the concentration A_{n-1} of its substrate, and thereby pre-adapts the pathway so that it will be competent to deal with it. The forward activation step thus embodies a model, which relates the present concentration of A_0 to a subsequent concentration of A_{n-1} , and thereby generates an obviously adaptive response of the entire pathway.

I remarked above that I was astonished to find this profusion of anticipatory behavior at all levels of biological organization. It is important here to understand why I found the situation astonishing, for it bears on the developments to be reported subsequently, and raises some crucial epistemological issues.

We have already seen, in the few examples presented above, that an anticipatory behavior is one in which a change of state in the present occurs as a function of some predicted future state, and that the agency through which the prediction is made must be, in the broadest sense, a model. I have also indicated that obvious examples of anticipatory behavior abound in the biosphere at all levels of organization, and that much (if not most) conscious human behavior is also of this character. It is further true that organic behaviors at all of these levels have been the subject of incessant scrutiny and theoretical attention for a long time. It might then be expected that such behavior would be well understood, and that there would indeed be an extensive body of theory and of practical experience which could be immediately applied to the problems of forecasting and policy-making which dominated the Center’s interests. But in fact, nothing could be further from the truth. The surprise was not primarily that there was no such body of theory and experience, but rather that almost no systematic efforts had been made in these directions; and moreover, almost no one recognized that such an effort was urgently required. In retrospect, the most surprising thing to me was that I myself had not previously recognized such a need, despite my overt concerns with modeling as a fundamental scientific

activity, and despite my explicit involvement with biological behavior extending over many years. Indeed, I might never have recognized this need, had it not been for the fortuitous chain of circumstances I have described above, which led me to think seriously about apparently alien problems of policy-making in a democratic society. Such are the powers of compartmentalization in the human mind.

In fact, the actual situation is somewhat worse than this. At its deepest level, the failure to recognize and understand the nature of anticipatory behavior has not simply been an oversight, but is the necessary consequence of the entire thrust of theoretical science since earliest times. For the basic cornerstone on which our entire scientific enterprise rests is the belief that events are not arbitrary, but obey definite laws which can be discovered. The search for such laws is an expression of our faith in causality. Above all, the development of theoretical physics, from Newton and Maxwell through the present, represents simultaneously the deepest expression and the most persuasive vindication of this faith. Even in quantum mechanics, where the discovery of the Uncertainty Principle of Heisenberg precipitated a deep re-appraisal of causality, there is no abandonment of the notion that microphysical events obey definite laws; the only real novelty is that the quantum laws describe the statistics of classes of events rather than individual elements of such classes.

The temporal laws of physics all take the form of differential equations, in which the rate of change of a physical quantity at any instant is expressed as a definite function of the values of other physical quantities at that instant. Thus, from a knowledge of the values of all the relevant quantities at some initial instant t_0 , the values of these quantities at the succeeding instant $t_0 + dt$ are determined. By iterating this process through an integration operation, the values of these quantities, and hence the entire behavior of the system under consideration, may be determined for all time. Carrying this picture to its logical conclusion, Laplace could say,

An intelligence knowing, at a given instant of time, all forces acting in nature, as well as the momentary position of all things of which the universe consists, would be able to comprehend the motions of the largest bodies of the world as well as the lightest atoms in one single formula. . . . To him nothing would be uncertain; both past and future would be present in his eyes.

This picture of causality and law, arising initially in physics, has been repeatedly generalized, modified and extended over the years, but the basic pattern remains identifiable throughout. And one fundamental feature of this picture has remained entirely intact; indeed itself elevated to the status of a natural law. That feature is the following: *in any law governing a natural system, it is forbidden to allow present change of state to depend upon future states*. Past states perhaps, in systems with “memory”; present state certainly; but *never* future states. It is perfectly clear from the above discussion why such a commandment is natural, and why its violation would appear tantamount to a denial of causality in the natural world.

A denial of causality thus appears as an attack on the ultimate basis on which science itself rests. This is also the reason why arguments from final causes have been excluded from science. In the Aristotelian parlance, a final cause is one which involves a purpose or goal; the explanation of system behavior in terms of final

causes is the province of *teleology*. As we shall see abundantly, the concept of an anticipatory system has nothing much to do with teleology. Nevertheless, the imperative to avoid even the remotest appearance of telic explanation in science is so strong that all modes of system analysis conventionally exclude the possibility of anticipatory behavior from the very outset.²

And yet, let us consider the behavior of a system which contains a predictive model, and which can utilize the predictions of its model to modify its present behavior. Let us suppose further that the model is a “good” model; that its predictions approximate future events with a sufficiently high degree of accuracy. It is clear that such a system will behave *as if it were* a true anticipatory system; i.e. a system in which present change of state does depend on future states. In the deepest sense, it is evident that this kind of system will not in fact violate our notions of causality in any way, nor need it involve any kind of teleology. But since we explicitly forbid present change of state to depend on future states, we will be driven to understand the behavior of such a system in a purely *reactive* mode; i.e. one in which present change of state depends only on present and past states.

This is indeed what has happened in attempting to come to grips theoretically and practically with biological behavior. Without exception (in my experience), all models and theories of biological systems are reactive in the above sense. As such, we have seen that they *necessarily* exclude all possibility of dealing directly with the properties of anticipatory behavior of the type we have been discussing.

How is it, then, that the ubiquity of anticipatory behaviors in biology could have been overlooked for so long? Should it not have been evident that the “reactive paradigm”, as we may call it, was grossly deficient in dealing with systems of this kind? To this question there are two answers. The first is that many scientists and philosophers have indeed repeatedly suggested that something fundamental may be missing if we adopt a purely reactive paradigm for consideration of biological phenomena. Unfortunately, these authors have generally been able only imperfectly to articulate their perception, couching it in terms as “will”, “Geist”, “élan”, “entelechy”, and others. This has made it easy to dismiss them as mystical, vitalistic, anthropomorphic, idealistic, or with similar unsavory epithets, and to confound them with teleology.

The other answer lies in the fact that the reactive paradigm is *universal*, in the following important sense. Given any mode of system behavior which can be described sufficiently accurately, *regardless of the manner in which it is generated*, there is a purely reactive system which exhibits precisely this behavior. In other words, any system behavior can be *simulated* by a purely reactive system. It thus might appear that this universality makes the reactive paradigm completely adequate for all scientific explanations, but this does not follow, and in fact is not the case. For instance, the Ptolemaic epicycles are also universal, in the sense that any planetary trajectory can be represented in terms of a sufficiently extensive family of them. The reason that the Copernican scheme was considered superior to the Ptolemaic lies not in the existence of trajectories which cannot be represented by the epicycles, but arises entirely from considerations of *parsimony*, as embodied for instance in Occam’s Razor. The universality of the epicycles is regarded as an extraneous

mathematical artifact irrelevant to the underlying physical situation, and it is for this reason that a representation of trajectories in terms of them can only be regarded as a simulation, and not as an explanation.

In fact, the universality of the reactive paradigm is not very different in character from the universality of the epicycles. Both modes of universality ultimately arise from the mathematical fact that any function can be approximated arbitrarily closely by functions canonically constructed out of a suitably chosen “basis set” whose members have a special form. Such a basis set may for instance comprise trigonometric functions, as in the familiar Fourier expansion; the polynomials $1, x, x^2, \dots$ form another familiar basis set. From this it follows that if any kind of system behavior can be described in functional terms, it can also be *generated* by a suitably constructed combination of systems which generate the elements of a basis set, and this entirely within a reactive mode. But it is clear that there is nothing unique about a system so constructed; we can do the same with *any* basis set. All these systems are different from one another, and may be likewise different from the initial system whose behavior we wanted to describe. It is in this sense that we can only speak of simulation, and not of explanation, of our system’s behavior in these terms.

Nevertheless, I believe that it is precisely the universality of the reactive paradigm which has played the crucial role in concealing the inadequacy of the paradigm for dealing with anticipatory systems.

Indeed, it is clear that if we are confronted with a system which contains a predictive model, and which uses the predictions of that model to generate its behavior, we cannot claim to understand the behavior unless the model itself is taken into account. Moreover, if we wish to construct such a system, we cannot do so entirely within the framework appropriate to the synthesis of purely reactive systems.

On these grounds, I was thus led to the conclusion that an entirely new approach was needed, in which the capability for anticipatory behavior was present from the outset. Such an approach would necessarily include, as its most important component, a comprehensive theory of models and of modeling. The purpose of the present volume in fact, is to develop the principles of such an approach, and to describe its relation to other realms of mathematical and scientific investigation.

With these and similar considerations in mind, I proceeded to prepare a number of working papers on anticipatory behavior, and the relation of this kind of behavior to the formulation and implementation of policy. Some of these papers were later published in the *International Journal of General Systems*.³ The first one I prepared was entitled, “Planning, Management, Policies, and Strategies: Four Fuzzy Concepts”, and it already contained the seeds of the entire approach I developed to deal with these matters. For this reason, and to indicate the context in which I was working at the Center, it may be helpful to cite some of the original material directly. The introductory section began as follows:

It is fair to say that the mood of those concerned with the problems of contemporary society is apocalyptic. It is widely felt that our social structure is in the midst of crises, certainly serious, and perhaps ultimate. It is further widely felt that the social crises we perceive have arisen primarily because of the anarchic, laissez-faire attitude taken in the past towards science, technology, economics and politics. The viewpoint of most of those who have

written on these subjects revolves around the theme that if we allow these anarchies to continue we are lost; indeed, one way to make a name nowadays is to prove, preferably with computer models, that an extrapolation of present practices will lead to imminent cataclysm. The alternative to anarchy is management; and management implies in turn the systematic implementation of specific plans, programs, policies and strategies. Thus it is no wonder that the circle of ideas centering around the concept of *planning* plays a dominant role in current thought.

However, it seems that the net effect of the current emphasis on planning has been simply to shift the anarchy we perceive in our social processes into our ideas about the management of these processes. If we consider, for example, the area of “economic development” of the underdeveloped countries (a topic which has been extensively considered by many august bodies), we find (a) that there is no clear idea of what constitutes “development”; (b) that the various definitions employed by those concerned with development are incompatible and contradictory; (c) that even among those who happen to share the same views as to the ends of development, there are similarly incompatible and contradictory views as to the means whereby the end can be attained. Yet in the name of developmental planning, an enormous amount of time, ink, money and even blood is in the process of being spilled. Surely no remedy can be expected if the cure and the disease are indistinguishable.

If it is the case that planning is as anarchic as the social developments it is intended to control, then we must ask whether there is, in some sense, a “plan for planning” or whether we face an infinite and futile anarchic regress. It may seem at first sight that by putting a question in this form we gain nothing. However, what we shall attempt to argue in the present paper is that, in fact, this kind of question is “well-posed” in a scientific sense: that it can be investigated in a rigorous fashion and its consequences explored. Moreover, we would like to argue that, in the process of investigating this question, some useful and potentially applicable insights into planning itself are obtainable.

After a brief review of the main technical concepts to be invoked (which was essential for the audience at the Center) I then proposed a specific context in which anticipatory behavior could be concretely discussed:

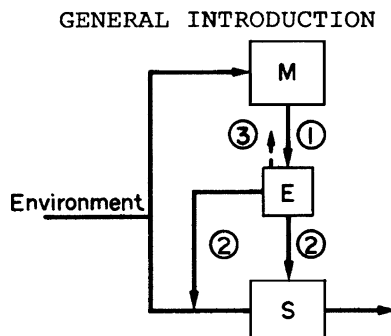
We are now ready to construct our model world, which will consist of a class of systems of definite structure, involving anticipation in an essential way, and in which the fuzzy terms associated with “planning” can be given a concrete meaning.

Let us suppose that we are given a system S , which shall be the system of interest, and which we shall call the *object system*. S may be an individual organism, or an ecosystem, or a social or economic system. For simplicity we shall suppose that S is an ordinary (i.e. non-anticipatory) dynamical system.

With S we shall associate another dynamical system M , which is in some sense a *model* of S . We require, however, that if the trajectories of S are parameterized by real time, then the corresponding trajectories of M are parameterized by a time variable which goes faster than real time. That is, if S and M are started out at time $t = 0$ in equivalent states, and if (real) time is allowed to run for a fixed interval T , then M will have proceeded further along its trajectory than S . In this way, the behavior of M *predicts* the behavior of S ; by looking at the state of M at time T , we get information about the state that S will be in at some time later than T .

We shall now allow M and S to be coupled; i.e. allow them to interact in specific ways. For the present, we shall restrict ourselves to ways in which M may affect S ; later we shall introduce another mode of coupling which will allow S to affect M (and which will amount to updating or improving the model system M on the basis of the activity of S). We shall for the present suppose simply that the system M is equipped with a set E of *effectors*, which

Fig. 1.1



allow it to operate either on S itself, or on the environmental inputs to S, in such a way as to change the dynamical properties of S. We thus have a situation of the type diagrammed in Fig. 1.1.

If we put this entire system into a single box, that box will appear to us to be an adaptive system in which prospective future behaviors determine present changes of state. It would be an anticipatory system in the strict sense if M were a *perfect* model of S (and if the environment were constant or periodic). Since in general M is not a perfect model, for reasons to be discussed in Chap. 5 below, we shall call the behavior of such systems *quasi-anticipatory*.

We have said that “M sees” into the future of S, because the trajectories of M are parameterized faster than those of S. How is this information to be used to modify the properties of S through the effector system E? There are many ways in which this can be formalized, but the simplest seems to be the following. Let us imagine the state space of S (and hence of M) to be partitioned into regions corresponding to “desirable” and “undesirable” states. As long as the trajectory in M remains in a “desirable” region, no action is taken by M through the effectors E. As soon as the M-trajectory moves into an “undesirable” region (and hence, by inference, we may expect the S-trajectory to move into the corresponding region at some later time, calculable from a knowledge of how the M- and S-trajectories are parameterized) the effector system is activated to change the dynamics of S in such a way as to keep the S-trajectory out of the “undesirable” region.

From this simple picture, a variety of insights into the nature of “planning”, “management”, “policies”, etc., can already be extracted.

The structure depicted in Fig. 1.1 possesses properties which relate directly to the generation and implementation of plans; I then proceeded to sketch these properties:

A. Choice of M.

The first essential ingredient in the planning process in these systems involves the choice of the model system M. There are many technical matters involved in choosing M, which will be discussed in more detail in Chap. 5 below. We wish to point out here that the choice of M involves paradigmatic aspects as well, which color all future aspects of the “planning” process. One simple example of this may suffice. Let us suppose that S is a simple early capitalist economic system. If we adopt a model system which postulates a large set of small independent entrepreneurs, approximately equivalent in productive capability and governed by “market forces”, we find that the system S is essentially stable; coalitions are unfavored and any technical innovations will rapidly spread to all competitors. On the other hand, if we adopt a model system M in which there are positive feedback loops, then we will see the same situation as *unstable*, much as an emulsion of oil and water is unstable. That is, initially small local accretions of

capital will tend to be amplified, and the initially homogeneous population of many small entrepreneurs will ultimately be replaced by a few enormous cartels. This, in a highly oversimplified way, seems to represent the difference between laissez-faire capitalism and Marxian socialism, proceeding from two different model systems of the same initial economic system S , and hence predicting two entirely different futures for S .

B. *Selection of the Effector System E .*

Once the model M has been chosen, the next step of the planning or management process for S is to determine how we are to modify the dynamics of S according to the information we obtain from M . The problem involves several stages. The first stage involves a selection of “steering” variables in S or in the environment of S , through which the dynamical properties of S can be modified. In general, several different kinds of choices can be made, on a variety of different grounds. In empirical terms, this choice will most often be made in terms of the properties of the model system M ; we will consider how M can be most effectively steered, and use the corresponding state variables of S (if possible) for the control of S . Thus again the initial choice of the model system M will again tend to play a major role in determining the specifics of the planning process.

C. *Design of the Effector System E .*

Once having chosen the control variables of S , we must now *design* a corresponding effector system. This is a technological kind of problem, governed by the nature of the control variables of S and their response characteristics. We may wish, for example, to employ only controls which are easily reversible.

D. *Programming of the Effector System E .*

The final aspect of the planning process involves the actual programming of the effector system; i.e. the specification of a dynamics on E which will convert the input information from M (i.e. information about the future state of S) into a specific modification of the dynamics of S . This transduction can be accomplished in many ways, and involves a mixture of “strategic” and “tactical” considerations.

E. *Identification of “Desirable” and “Undesirable” Regions.*

Ultimately the programming of the effectors E will depend heavily on the character of the regions we consider “desirable” and those we consider “undesirable”. This choice too is arbitrary, and is in fact independent of the model system M which we have chosen. It represents a kind of constraint added from the outside, and it enters into the planning process in an equally weighty fashion as does the model M and the effector system E .

F. *Updating the States of M .*

In Fig. 1.1 we have included a dotted arrow (labelled (3)) from the effector system back to the model. This is for the purpose of resetting the states of the model, according to the controls which have been exerted on the system S by the effector system. Unless we do this, the model system M becomes useless for predictions about S subsequent to the implementation of controls through E . Thus, the effector system E must be wired into M in a fashion equivalent to its wiring into S .

The enumeration (A)–(F) above seems to be a useful atomization of the planning process for the class of systems we have constructed. Within this class, then, we can proceed further and examine some of the consequences of planning, and in particular the ways in which planning can go wrong. We shall sketch these analyses in the subsequent sections.

The notion of “how planning could go wrong” was of course of primary interest to the Center; indeed, for months I had heard a succession of discouraging papers dealing with little else. It seemed to me that by elaborating on this theme I could establish a direct contact between my ruminations and the Center’s preoccupations. My preliminary discussion of these matters ended as follows:

We would like to conjecture further that, for any specific planning situation (involving an object system S , a model M , and suitably programmed effectors E), each of the ways in which planning can go wrong will lead to a particular kind of syndrome in the total system (just as the defect of any part of a sensory mechanism in an organism leads to a particular array of symptoms). It should therefore be possible, in principle, to develop a definite diagnostic procedure to “trouble-shoot” a system of this kind, by mimicking the procedures used in neurology and psychology. Indeed, it is amusing to think that such planning systems are capable of exhibiting syndromes (e.g. of “neurosis”) very much like (and indeed analogous to) those manifested by individual organisms.

Such considerations as these led naturally to the general problems connected with system error, malfunction or breakdown, which have always been hard to formulate, and are still poorly understood. Closest to the surface in this direction, especially in the human realm, were breakdowns arising from the incorporation of incorrect elements into the diagram shown in Fig. 1.1 above; faulty models, inappropriate choice of effectors, etc. I soon realized, however, that there was a more profound aspect of system breakdown, arising from the basic nature of the modeling process itself, and from the character of the system interactions required in the very act of imposing controls. These were initially considered under the heading of *side-effects*, borrowing a medical terminology describing unavoidable and usually unfortunate consequences of employing therapeutic agents (an area which of course represents yet another branch of control therapy). As I used the term, I meant it to connote unplanned and unforeseeable consequences on system behavior arising from the implementation of controls designed to accomplish other purposes; or, in a related context, the appearance of unpredicted behavior in a system built in accordance with a particular plan or blueprint. Thus the question was posed: are such side-effects a *necessary* consequence of control? Or is there room for hope that, with sufficient cleverness, the ideal of the *magic bullet*, the miraculous cure which specifically restores health with no other effect, can actually be attained?

Since this notion of side-effects is so important, let us consider some examples. Of the medical realm we need not speak extensively, except to note that almost every therapeutic agent, as well as most diagnostic agents, create them; sometimes spectacularly so, as in the thalidomide scandal of some years past. We are also familiar with ecological examples, in which man has unwittingly upset “the balance of nature” through injudicious introduction or elimination of species in a particular habitat; well-known instances of this are the introduction of rabbits to Australia, to give the gentlemen farmers something to hunt on the weekend; or the importation of the mongoose into Santo Domingo, in the belief that because the mongoose kills cobras it would also eliminate local poisonous snakes such as the fer-de-lance. Examples from technology also abound; for instance, we may cite the presence of unsuspected oscillatory modes in the Tacoma Bay Bridge, which ultimately caused it to collapse in a high wind; or the Ohio Turnpike, which was built without curves on the theory that curves are where accidents occur; this led to the discovery of road hypnosis. Norbert Wiener warned darkly of the possibility of similar disastrous side-effects in connection with the perils of relying on computers to implement policy. He analogized this situation to an invocation of magical aids as related in innumerable

legends and folk-tales; specifically, such stories as The Sorcerer's Apprentice, The Mill Which Ground Nothing But Salt, The Monkey's Paw, and The Midas Touch. And of course, many of the social and economic panaceas introduced in the past decades have not only generated such unfortunate side-effects, but have in the long run served to exacerbate the very problems they were intended to control.

The ubiquity of these examples, and the dearth of counter-examples, suggests that there is indeed something universal about such behavior, and that it might be important to discover what it is.

My first clumsy attempts to come to grips with this underlying principle, at this early stage, were as follows (the several references to an "earlier paper" in these excerpts refer to a paper written later than, but published before, the one being cited):

There is, however, a class of planning difficulties which do not arise from such obvious considerations, and which merit a fuller discussion. This class of difficulties has to do with the problem of *side effects*; as we shall see, these will generally arise, even if the model system is perfect and the effectors perfectly designed and programmed, because of inherent system-theoretic properties. Let us see how this comes about.

In a previous paper we enunciated a conjecture which I believe to have general validity: namely; that in carrying out any particular functional activity, a system *S* typically only uses a few of its degrees of freedom. This proposition has several crucial corollaries, of which we noted two in the preceding paper:

- (1) The same structure can be involved simultaneously in many different functional activities, and conversely.
- (2) The same functional activity can be carried out (or "realized") by many different kinds of structures.

We stressed in that paper how the fact that *all* of the state variables defining any particular system *S* are more or less strongly *linked* to one another via the equations of motion of the system, taken together with the fact that the many state variables not involved in a particular functional activity were free to interact with other systems in a non-functional or dysfunctional way, implied that any particular functional activity tends to be modified or lost over time. This, we feel, is a most important result, which bears directly on the "planning" process under discussion. The easiest way to see this is to draw another corollary from the fundamental proposition that only a few degrees of freedom of a system *S* are involved in any particular functional activity of *S*.

- (3) Any functional activity of a system *S* can be *modeled* by a system whose structure is simple compared to that of *S* (simply by neglecting the non-functional degrees of freedom of *S*). Indeed, it is largely because of this property that science is possible at all. Conversely,
- (4) No one model is capable of capturing the full potentialities of a system *S* for interactions with arbitrary systems.

The corollary (4) is true even of the best models, and it is this corollary which bears most directly on the problem of *side-effects*. Let us recall that *S* is by hypothesis a real system, whereas *M* is only a model of a particular functional activity of *S*. There are thus many degrees of freedom of *S* which are not modeled in *M*. Even if *M* is a good model, then, the capability for dealing with the non-functional degrees of freedom in *S* have necessarily been abstracted away. And these degrees of freedom, which continue to exist in *S*, are generally *linked* to the degrees of freedom of *S* which are modeled in *M*, through the overall equations of motion which govern *S*.

Now the planning process requires us to construct a real system *E*, which is to interact with *S* through a particular subset of the degrees of freedom of *S* (indeed, through a subset of those degrees of freedom of *S* which are modeled in *M*). But from our general proposition,

only a few of the degrees of freedom of E can be involved in this interaction. Thus both E and S have in general many “non-functional” degrees of freedom, through which other, non-modeled interactions can take place. Because of the linkage of all observables, the actual interaction between E and S specified in the planning process will in general be affected. Therefore, we find that the two following propositions are generally true: (a) An effector system E will in general have other effects on an object system S than those which are planned; (b) The planned modes of interaction between E and S will be modified by these effects. Both of these propositions describe the kind of thing we usually refer to as *side-effects*. As we see, such side-effects are unavoidable consequences of the general properties of systems and their interaction. They are by nature unpredictable, and are inherent in the planning process no matter how well that process is technically carried out. As we pointed out in our previous paper, there are a number of ways around this kind of difficulty, which we have partially characterized, but they are only applicable in special circumstances.

The basic principle struggling to emerge here is the following: the ultimate seat of the side-effects arising in anticipatory control, and indeed of the entire concept of error or malfunction in system theory as a whole, rests on the discrepancy between the behavior actually exhibited by a natural system, and the corresponding behavior predicted on the basis of a model of that system. For a model is necessarily an *abstraction*, in that degrees of freedom which are present in the system are absent in the model. In physical terms, the system is *open* to interactions through these degrees of freedom, while the model is necessarily *closed* to such interactions; the discrepancy between system behavior and model behavior is thus a manifestation of the difference between a closed system and an open one. This is one of the basic themes which we shall develop in detail in the subsequent chapters.

My initial paper on anticipatory systems concluded with several observations, which I hoped would be suggestive to my audience. The first was the following: that it was unlikely that side-effects could be removed by simply augmenting the underlying model, or by attempting to control each side-effect separately as it appeared. The reason for this is that both of these strategies face an incipient infinite regress, similar to that pointed out by Gödel in his demonstration of the existence of unprovable propositions within any consistent and sufficiently rich system of axioms. Oddly enough, the possibility of avoiding this infinite regress was not entirely foreclosed; this followed in a surprising way from some of my earliest work in relational biology, which was mentioned earlier:

There are many ramifications of the class of systems developed above, for the purpose of studying the planning process, which deserve somewhat fuller consideration than we have allowed. In this section we shall consider two of them: (a) how can we update and improve the model system M, and the effector system E, on the basis of information about the behavior of S itself and (b) how can we avoid a number of apparent infinite regresses which seem to be inherent in the planning process?

These two apparently separate questions are actually forms of the same question. We can see this as follows. If we are going to improve, say, the model system M, then we must do so by means of a set of effectors E'. These effectors E' must be controlled by information pertaining to the effect of M on S; i.e. by a model system M' of the system (S+M+E). In other words, we must construct for the purpose of updating and improving M a system which looks exactly like Fig. 1.1 except that we replace M by M', E by E', and S by S+M+E. But then we may ask how we can update M'; in this way we see an incipient infinite regress.

There is another infinite regress inherent in the discussion given of side-effects in the preceding section. We have seen that the interaction of the effectors E with the object system S typically give rise to effects in S unpredictable in principle from the model system M . However, these effects too, by the basic principle that only a few degrees of freedom of S and E are utilized in such interactions, are capable of being modeled. That is, we can in principle construct a new model system M_1 of the interaction between S and E , which describes interactions not describable in M . If these interactions are unfavorable, we can construct a new set of effectors, say E_1 , which will steer the system S away from these side-effects. But just as with E , the system S will typically interact with E_1 in ways which are in principle not comprehensible within the models M or M_1 ; these will require another model M_2 and corresponding new effectors E_2 . In this way we see another incipient infinite regress forming. Indeed, this last infinite regress is highly reminiscent of the “technological imperative” which we were warned against by Ellul¹ and many others. Thus the question arises; can such infinite regresses be avoided?

These kinds of questions are well-posed, and can be investigated in system-theoretic terms. We have considered² questions like these in a very different connection; namely, under what circumstances is it possible to add a new functional activity to a biological organization like a cell? It turns out that one cannot simply add an arbitrary function and still preserve the organization; we must typically keep adding functions without limit. But under certain circumstances, the process does indeed terminate; the new function is included (though not *just* the new function in general) and the overall organization is manifested in the enlarged system. On the basis of these considerations, I would conjecture that (a) it is possible in principle to avoid the infinite regresses just alluded to in the planning process, and in particular to find ways of updating the model M and the effectors E ; (b) *not every way of initiating and implementing a planning process allows us to avoid the infinite regress*. The first conjecture is optimistic; there are ways of avoiding this form of the “technological imperative”. The second can be quite pessimistic in reference to our actual society. For if we have in fact embarked on a path for which the infinite regresses cannot be avoided, then we are in serious trouble. Avoiding the infinite regresses means that developmental processes will stop, and that a stable steady-state condition can be reached. Once embarked on a path for which the infinite regresses cannot be avoided, no stable steady-state condition is possible. I do not know which is the case in our own present circumstances, but it should at least be possible to find out.

I hope that the above few remarks on the planning process will provide food for thought for those more competent to investigate such problems than I am.

The theoretical principle underlying this analysis of failure in anticipatory control systems is not wholly negative. In fact, we shall argue later that it also underlies the phenomena of *emergence* which characterize evolutionary and developmental processes in biology. It may be helpful to cite one more excerpt of a paper originally prepared for the Center Dialog, which dealt with this aspect:

It may perhaps be worth noting at this point that the above phenomenon is responsible for many of the evolutionary properties exhibited by organisms, and many of the developmental characteristics of social organizations. Consider, for example, the problems involved in understanding, e.g. the evolution of a sensory mechanism such as an eye. The eye is a complicated physiological mechanism which conveys no advantage until it actually sees, and it cannot see until it is complicated. It is hard to imagine how one could even get started towards evolving such a structure, however valuable the end-result may be, and this was one of the major kinds of objections raised by Darwinian evolution. The response to this objection is essentially as follows: the proto-eye in its early stages was in fact not involved in the function of seeing, but rather was primarily involved in carrying out some

other functional activity, and it was on this other activity that selection could act. If we now suppose that this other activity involved photosensitivity in an initially accidental way (simply because the physical structure of the proto-eye happened to also be photosensitive), it is easy to imagine how selection pressure could successively improve the proto-eye, with its accidental sensory capacity, until actual seeing could begin, and so that selection could begin to act on the eye directly as an eye. When that happened, the original function of the eye was lost or absorbed into other structures, leaving the eye free to evolve exclusively as a sensory organ.

This "Principle of Function Change" is manifested even more clearly by the evolution of the lung as an organ of respiration. Many fish possess swim bladders, a bag of tissue filled with air, as an organ of equilibration. Being a bag of tissue, the swim bladder is vascularized (possesses blood vessels). When air and small blood vessels are in contact, there will necessarily be gas exchange between the blood and the air, and so a respiratory function is incipient in this structure, designed initially for equilibration. It is easy to imagine how successive increases in vascularization of this organ, especially in arid times, could be an advantage, and thus how selection could come to act on this structure as a lung. This Principle of Function Change is thus one of the cornerstones of evolution (and indeed of any kind of adaptive behavior), and it depends essentially on the fact that the same structure is capable of simultaneously manifesting a variety of functions.

Thus the basic problem of avoiding infinite regresses in anticipatory control systems could be reformulated as follows: can we design systems which are proof against a Principle of Function Change?

This was the circle of ideas which I was led to place on the table at the Center. The response elicited thereby could perhaps best be described as restrained, but encouraging. I received some comments to the effect that my approach was logical and mathematical, and hence fundamentally inapplicable to politics. In particular, there seemed to be no room for perversity, a major factor in human behavior. Indeed, I had often noted that almost the only way for man to prove that he is truly free is to deliberately do himself an injury; to deliberately act against his obvious best interests. But I did not feel that this sort of objection was insuperable. Mr. Hutchins himself made few direct comments, except at one point to remark that one of my conclusions was the most outrageous thing he had ever heard. I took this as a high compliment.

A number of the Senior Fellows did feel that these ideas bore directly on the major concerns of the Center. Rex Tugwell, on the basis of a long life of practical experience in government policy-making, and a commitment to constitutionalization, provided numerous specific instances of many of the points I tried to make in a general context, and always seemed to ask the right questions. Harvey Wheeler, a political scientist, was a most effective and sympathetic Devil's advocate. The most enthusiastic response, however, was made by John Wilkinson, whose long advocacy of modeling and simulation for the understanding and exploitation of the lessons on history involved the questions I raised in an essential way.

For my own part, I continue to believe that the properties of anticipatory systems raise new questions for the scientific enterprise of the most basic and fundamental kind. These questions have led me to reformulate and refocus all of my previous work in the foundations of theoretical biology, and in the relation of biology to the physical and human sciences. Indeed, there is no aspect of science which can

be untouched by so fundamental an activity as a reconsideration of the reactive paradigm itself. The results of this reconsideration, and its implications, are the basic subject-matter of the developments which follow.

Now that we have reviewed the genesis of the theoretical problems with which the present volume is concerned, we may turn to a consideration of the problems themselves. The first and most basic of them is simply this: what is a model? What is the nature of the relation between two systems which allows us to assert that one of them is a model for the other? The basic property of the modeling relation is that we can learn something about a system in which we are interested by studying a model of that system. The crucial point to be developed here is that *causality* in the system must be represented by *implication* in the model. Chapter 2 is devoted entirely to abstract considerations of how this is done, and how a given system may be *encoded* into another so as to establish a modeling relation. These ideas comprise the heart of the book.

Chapter 3 provides a concrete survey of modeling relations between systems of many diverse kinds, drawn from all corners of natural science and mathematics. These examples not only exemplify the abstract development, but throw new light on the inter-relationships generated among systems through the existence of relations between system models. This part culminates with a thorough discussion of relational models and metaphors, to which we have alluded several times above.

Chapter 4 is concerned with laying the basis for a theory of dynamical models, which are the essential features of anticipatory systems. The main point of this chapter is a comprehensive discussion of *time*. Chapter 5, which is in fact closely related, is devoted to a discussion of system reliability and system error, in terms of the deviation between the actual behavior of a system and the behavior predicted by some model of the system. The basic point of view we take here is the one proposed earlier; namely, that this deviation can be regarded as the difference in behavior between a system closed to certain environmental interactions, and the same system open to those interactions. We illustrate these ideas with a discussion of emergent novelty in biology, and its relation to the general notion of a *bifurcation*.

Finally, in Chap. 6 we pull these various threads together to obtain a general theory for feedforwards and anticipatory systems. Of particular importance here is an extensive discussion of anticipation and causality. In the concluding part, we consider the unique possibilities for global failures in feedforward systems, which do not arise from local failure in any subsystem. We suggest how this possibility leads to a new theoretical approach to the spanning of biological properties, and to organism senescence.

As noted previously, the subsequent exposition will draw heavily on our previous work concerned with modeling and system epistemology. In particular, the material in a previous monograph entitled *Fundamentals of Measurement and Representation of Natural Systems* will be extensively used. However, we have made every effort to keep the present treatment as self-contained as possible. Where this was not feasible, extensive references to the literature have been supplied.

In conclusion, I would like to refer the reader once again to Mr. Hutchins' epic view of the Great Conversation, resounding through all ages that have been and are to come. I hope that this volume can be regarded as a small contribution to it.

References and Notes

1. Robert Hutchins died in 1978. A biography of this extraordinary man, who was Dean of the Yale Law School at 24, and President of the University of Chicago before he was 30, still awaits writing. At present, one can best learn about Hutchins from his own voluminous writings, or from the tapes of the dialogs in which he participated at the Center; these last reveal his unique and charismatic personality most directly.

The Center for the Study of Democratic Institutions also deserves a biographer. The Center has undergone a radical transformation over the past few years; it still survives, in a vastly mutated form, as part of the University of California in Santa Barbara. Its former character, under Hutchins, can best be appreciated by looking at its lively little magazine, the *Center Reports*, which unfortunately is now extinct. Its sister publication, the *Center Magazine*, still appears (as of this writing) but was always far more traditionally academic in tone.

The quotations of Hutchins cited in the text were taken from articles he prepared for Center publication, and from various talks and addresses he presented over the years.

2. The role of teleology in biological theory has had a long and confused history. In some forms of teleology (e.g. Lamarck) the individual organism is regarded as the telic agent; in others (e.g. Bergson) the organism is an instrumentality of an agent which lies outside itself. Further, the status of telic explanation has become inextricably confounded with the endless and futile mechanist-vitalist controversy. A bibliography of references on causality and teleology, and the philosophical issues they raise, would run to hundreds of pages; we content ourselves with a few representative discussions, from several different points of view, which the interested reader may pursue further:

Bunge, M., *Causality*. Harvard University Press, Cambridge, Massachusetts (1959).

Driesch, H., *History and Theory of Vitalism*. MacMillan, London (1914).

Grene, M., *Approaches to Philosophical Biology*. Basic Books, New York (1968).

Grene, M. and Mendelsohn, E., *Topics in Philosophy of Biology*. Reidel, Boston (1976).

Mackie, J. L., *The Cement of the Universe: A Study of Causation*. Clarendon Press, Oxford (1947).

Nagel, E., *Teleology Revisited and Other Essays*. Columbia University Press, New York (1979).

Woodfield, A., *Teleology*. Cambridge University Press, Cambridge (1976).

Woodger, J. H., *Biological Principles*. Routledge & Kegan Paul, London (1967).

Wright, L., *Teleological Explanations*. University of California Press, Berkeley (1976).

Among more recent works involving purposiveness in biological systems, we may especially mention Agar, W. E., *A Contribution to the Theory of the Living Organism*. Melbourne (1943). A perhaps surprising emphasis on purposiveness also appears in the book of the eminent molecular biologist Jacques Monod (*Chance and Necessity*, Knopf, 1971), for which he coins a new word (“teleonomy”). We may also cite some of the papers of J. M. Burgers (e.g. “Causality and Anticipation”. *Science*, **189**, 194–98) which are based on the philosophical writings of A. N. Whitehead.

There is of course an enormous literature on causality in physics, much of it provoked by quantum mechanics and especially by the uncertainty principle. A brief but clear discussion, which is still valuable, may be found in Weyl, H., *Philosophy of Mathematics and the Natural Sciences*. Princeton University Press (1949). More to our present point: the idea that causality forbids the future from influencing the present is all-pervasive in physics; as for instance in the automatic exclusion of certain solutions of Maxwell’s Equations (“advanced potentials”) which refer to future time, while retaining other solutions (“retarded potentials”) which refer to past time; see e.g. Sommerfeld, A., *Electrodynamics*. Academic Press (1954) or Landau, L. D. and Lifshitz, E. M., *The Classical Theory of Fields*. Pergamon Press (1975). The alacrity with which this is done has worried a number of theoretical physicists over the years; perhaps the most interesting attempt to rehabilitate advanced potentials may be found in a little-known paper of Wheeler and Feynman (*Rev. Mod. Phys.*, **21**, 425–43 (1949)).

For a typical system-theoretic argument that causal systems cannot be anticipatory, see e.g. Windeknecht, T. G., *Math. Syst. Theory.*, 1, 279 (1967).

3. Rosen, R., 1974a. *Int. J. Gen. Syst.* 1, 6–66.

Rosen, R., 1974b. *ibid.* 1, 245–252.

Rosen, R., 1975. *ibid.* 2, 93–103.

1.2 The Reactive Paradigm: Its Basic Features

Before we begin our systematic development, it will be well to briefly review the basic features of the reactive paradigm itself. This will be important to us for several reasons. The first of these is, of course, the overwhelming historical importance of reactive concepts in the development of science up to this point. Indeed, as we have noted, systems which appear to violate the hypotheses on which the reactive paradigm is based are routinely excluded from the scientific realm. This leads us to the second reason for being familiar with the reactive paradigm: at the present time, this paradigm dominates the very concept of modeling; since the concept of a model is the central feature of anticipatory behavior, we need to understand fully how the modeling relation and reactive behavior are related. Third, our extensive familiarity

with reactive systems will serve us as a source of examples and motivation, and as an effective contrast to the behavior of anticipatory systems as we proceed with our development.

The treatment which follows will be necessarily sketchy, and will be concerned entirely with formal aspects. We will leave it to subsequent chapters to develop the relations which exist between such formalisms and the natural systems these formalisms are intended to represent. The reader will probably be familiar with much or all of this material, and in any case we will provide extensive references to the literature in lieu of detailed discussions. Our intention is to exhibit the conceptual inter-relationships between the underlying ideas, and not to go deeply into the technical details.

We begin by noting that the reactive paradigm has been embodied in two distinct but related formalisms. The first, which is most closely related to the physics of material systems (from which it originally arose) may be called the *state variable* approach. It is expressed most directly in the mathematical theory of dynamical systems, and various generalizations and modifications of that theory. The second approach, which arose primarily from concerns of engineering, and especially from control engineering, may be called the *input-output* approach. Thus, the first approach arose from problems of system analysis, while the second was concerned more directly with system synthesis or system design. It is not surprising to find numerous close relationships between these approaches, and we shall be concerned with developing some of them in the present chapter.

Let us begin by considering the state variable approach. As we mentioned previously, the temporal laws of particle mechanics provided the initial impetus for this approach, and provided the guide for all of its subsequent elaborations.¹ These temporal laws take the form of differential equations. These in turn were originally expressions of Newton's Second Law, which *defined* the force acting on a mass point to be the rate of change of momentum:

$$F = dp/dt$$

where momentum p is defined as the initial mass m of the point multiplied by the velocity v . In its turn, velocity is defined as rate of change of position or displacement from some origin of coordinates; hence we have

$$F = m \frac{d^2x}{dt^2}$$

In order to apply this formulation to any specific systems, we must have an independent characterization of the force, expressed in terms of the varying quantities; in the present simple case, the force must be expressed in terms of the displacement x from some reference, and the velocity v . For instance, to obtain a representation of simple harmonic motion, we can posit

$$F(x, v) = -kx$$

where k is a *constitutive parameter*, whose magnitude is independent of the displacement and the velocity. We thus obtain the equation of motion

$$m \frac{d^2x}{dt^2} = -kx$$

which is to be solved for the displacement x as an explicit function of time. As a second-order differential equation, its general solution will contain two arbitrary constants, which are evaluated in terms of the *initial values* of the displacement and the velocity.

The single second-order equation can be written as a pair of first-order equations:

$$\begin{aligned} \frac{dx}{dt} &= \frac{p}{m} \\ \frac{dp}{dt} &= -kx \end{aligned} \tag{1.1}$$

The first of these equations simply defines velocity (or momentum), while the second specifies the force. It is in this form that the dynamics of mass points is most susceptible of useful generalizations to other situations.

We may also note that, according to Newton's Laws, the displacements and momenta of the particles constituting a given system are sufficient to characterize that system completely. That is, every other quantity pertaining to the system is assumed to be an explicit function of the displacements and momenta. Knowing these displacements and momenta at an instant of time thus suffices to specify the *state* of the system completely at that instant, and hence the positions and their associated momenta are said to constitute a set of *state variables* for the system. As we have seen earlier, a knowledge of the forces acting on the system (which allows us to write down explicitly the appropriate equation of motion) together with a set of initial values of the state variables completely determines the values of the state variables, and hence of every other system quantity, for all time. It should be noted explicitly that there is nothing unique about a set of state variables; indeed much of theoretical mechanics is concerned with transformations to new sets of state variables, in which the equations of motion take on particularly simple forms.

The system of first-order equations (1.1) can be considerably generalized. Let us suppose that $x_1, \dots, x_n$ represent a family of physical magnitudes which characterize the states of *any* given system, in the same way that position and momentum characterize the states of a system of particles. For instance, the x_i may represent the concentrations of reactants in a chemical system, or the population sizes in an ecosystem. Let us further suppose that the rates at which these quantities are changing, at any instant of time, is a specific function of the instantaneous values of the quantities themselves. Then we may write a set of first-order differential equations of the form

$$\frac{dx_i}{dt} = f_i(x_1, \dots, x_n), \quad i = 1, \dots, n \tag{1.2}$$

which are obvious generalizations of the equations of motion (1.1) for a simple particulate system. The system (1.2) is a mathematical object, which constitutes a *dynamical system*.² Let us note explicitly how the hypothesis of causality is embodied here; rate of change depends *only* on present state.

The basic paradigm of the dynamical system as embodied in (1.2) comprises the basic nucleus of all form of mathematical analysis of natural systems. It has of course been extended and modified in many ways; these modifications alter the technical character of the resulting theory, without affecting the conceptual features which are the main objects of present attention. For instance, if we wish to consider spatially extended or distributed systems, we must introduce appropriate spatial magnitudes as independent variables, in addition to the single independent variable (time) appearing in (1.2); the resulting equations of motion then become partial differential equations (field equations) instead of ordinary differential equations. Likewise, in many situations it is reasonable to pass from the continuous time parameter in (1.2) to a discrete-valued time parameter; the resulting equations of motion are difference equations, leading to a theory of discrete dynamical systems. If the set of states which such a system can occupy is likewise discrete, we obtain in effect a theory of automata. If we wish to consider systems with “memory”, this can be done by introducing time lags into the equations of motion. In each of these situations we may construct analogs of the arguments to be developed below.

Dynamical systems of the form (1.2) play a predominant role in the modeling of real-world phenomena. The construction of a dynamical model of a real system involves the identification of a suitable set of state variables x_i , and the characterization of the forces imposed on the system in terms of them (and of the constitutive parameters, such as the quantities m and k in (1.1), which identify the system). The further investigation of the behavior of the system then becomes entirely a question of determining the *mathematical* properties of the equations of motion (1.2). The basic mathematical concept involved here is that of *stability*, which we will now proceed to describe.

In mathematical terms, a *solution* of the system (1.2) is an explicit expression of each of the state variables x_i as a function of time, $x_i(t)$, in such a way that these functions identically satisfy the equations of motion. Geometrically, each solution corresponds to a *curve*, or *trajectory*, lying in the manifold of all possible states of the system (this manifold comprises the *state space* of the system.) The fundamental existence and uniqueness theorems for dynamical systems of the form (1.2) assure us that, under very mild mathematical conditions on the functions f_i , through each state in the state space there passes *exactly one* trajectory of the system; this is the *unique trajectory property*. It embodies a very strong notion of causality for dynamical systems, for it asserts that only one history can give rise to a particular state, and only one future can follow from it. As such, it is a corollary of the basic hypothesis that present change of state depends only on present state.

We may note parenthetically that the unique trajectory property allows the entire theory to be reformulated in a manner which is often more convenient for specific applications. Let x denote an element of the state space, and let us suppose that our system is in this state at some initial instant $t = 0$. At some later time t , the

system will be in a state $x(t)$, which by the unique trajectory property is uniquely determined by $x = x(0)$. Hence we may define a transformation T_t by writing $x(0) \rightarrow x(t)$ for every state in the state space; the unique trajectory property guarantees that this transformation is one-to-one and onto. We can do this for every instant t . The resultant family $\{T_t\}$ of mappings (often called a *flow*) possesses a group property; i.e. it satisfies the following conditions: (a) $T_0 = \text{identity}$; (b) $T_{t_1+t_2} = T_{t_2} T_{t_1}$; (c) $T_{-t} = T_t^{-1}$.

Such a family of transformations is called a *one-parameter group*. It is clear that the study of such one-parameter groups of transformations allows an alternate formulation (of somewhat different generality) of the theory of dynamical systems which we have been developing. For further details, the reader is invited to consult the references.

Very often, we are not interested in, or we cannot analytically determine, the specific solutions of (1.2). But we do wish to know the *asymptotic* behavior of the system (i.e. what happens as time becomes infinite). And we usually also wish to know how the system will respond to perturbations; i.e. how a particular trajectory is related to nearby trajectories. These are the province of stability theory.

Let us consider initially the simplest possible trajectories, those consisting of single states. Analytically, these correspond to solutions of (1.2) of the form $x_i(t) = \text{constant}$. Such solutions represent the *steady states* of (1.2); intuitively, a system placed initially in such a state will never leave it, since the rates of change dx_i/dt of the state variables all vanish in that state. In principle, the steady states of (1.2) are found by solving the simultaneous system of algebraic equations

$$f_i(x_1, \dots, x_n) = 0, \quad i = 1, \dots, n.$$

The *stability* of such a steady state is basically determined by whether the nearby trajectories approach that steady state or not as time increases. This is a question which can usually be answered analytically; since it is of importance for the sequel, we shall briefly describe how this is done.

Let us suppose that $(x_1^*, \dots, x_n^*)$ is a steady state of (1.2). Under very mild mathematical conditions, universally satisfied in applications, the functions $f_i(x_1, \dots, x_n)$ occurring in (1.2) can be expanded in Taylor's series in a neighborhood of the steady state; such an expansion typically has the form

$$f_i(x_1^* + q_1, \dots, x_n^* + q_n) = \sum_{j=1}^n q_j \frac{\partial f_i}{\partial x_j} + \sum_{j,k} q_j q_k \frac{\partial^2 f_i}{\partial x_j \partial x_k} + \dots$$

The partial derivatives which appear here are evaluated at the steady state itself; hence they are simply numbers. Moreover, if the neighborhood we are considering is sufficiently small, the higher powers of the displacement from the steady state can be neglected. Thus, in the neighborhood, the original equations of motion (1.2) are closely approximated by another set of equations of the form

$$\frac{\partial u_i}{\partial t} = \sum_{j=1}^n a_{ij} u_j \quad (1.3)$$

where we have written $a_{ij} = \frac{\partial f_i}{\partial x_j}$, evaluated at the steady state of (1.2). The new dynamical system (1.3) is *linear*; the state variables appear here only to the first power. Linear dynamical systems are completely tractable analytically; in general their solutions are sums of exponentials

$$u_i(t) = \sum_{j=1}^n A_{ij} e^{\lambda_j t} \quad (1.4)$$

where A_{ij} are constants determined by initial conditions, and the λ_j are the *eigenvalues* of the system matrix (a_{ij}) . Typically these eigenvalues are complex numbers

$$\lambda_j = p_j + iq_j$$

and so the exponentials in (1.4) can be written

$$e^{\lambda_j t} = e^{p_j t} e^{iq_j t}$$

Now the imaginary factor $e^{iq_j t}$ is bounded in absolute value between ± 1 . Therefore the time behavior of each of the exponentials is determined entirely by the real factor $e^{p_j t}$, and more specifically by the sign of p_j ; if this sign is positive, the corresponding exponential will grow without limit; if negative, the exponential will decay to zero as time increases. Stated another way: if the p_j are all negative, any trajectory of (1.3) sufficiently near the origin will approach the origin as time increases; if all of the p_j are positive, any such trajectory will move away from the origin; if some are positive and others negative, the behavior of a specific trajectory will depend entirely on the initial conditions; i.e. on the constants A_{ij} in (1.4). In the first case, the steady state is said to be *asymptotically stable*; in the second case is *unstable*; in the third case the steady state is *conditionally stable*. Let us also note here, for further reference, that the *magnitudes* of the p_j determine the *rate* at which the trajectories of (1.3) are traversed. Thus, systems for which all the p_j are large and negative are “more stable” than those for which the p_j are small and negative, and conversely.

It can be shown that, of all the p_j are not zero, then the behavior of the trajectories of (1.3) near the origin are the same as the trajectories of (1.2) near the steady state $(x_1^*, \dots, x_n^*)$. If all the p_j are zero, then we obtain no information about the stability of the steady state from the approximating linear system (1.3). The theorem expressing this fact is an assertion about the *structural stability* of the linear system (1.3); i.e. the invariance of its stability properties to perturbations of the *form* of the equations of motion. This idea of structural stability will be one of our main preoccupations in the ensuing chapters.

The above analysis refers to the stability of steady states. But we can investigate the stability of arbitrary trajectories in the same way; indeed, it can be shown that the stability of any trajectory of (1.2) can be reduced to the stability of a steady state of an associated dynamical system. For details on these matters, which we repeat belong entirely to pure mathematics, we invite the reader to consult the references.

Since the stability of trajectories embodies the manner in which a system responds to perturbations, it is clear that the stability of a trajectory represents the *homeostasis* of a particular mode of system behavior. Conversely, the instability of a trajectory can often be interpreted as representing a capacity for *self-organization*, in that the initially small perturbations away from an unstable trajectory are amplified by the system. This is true *independent* of the specific form of the system dynamics; for this reason, stability arguments have come to play a crucial role as metaphors for developmental processes in general. We shall return to these matters in a more comprehensive way below.

Let us now return to the other manifestations of the reactive paradigm, which we termed the input-output approach.³

Let us begin with a consideration of the proverbial “black box”; a device whose inner mechanism is unknown, but on which we can impose certain “signals”, interchangeably called *inputs*, *forcings* or *stimuli*. These signals can generally be regarded as-function of time. In its turn, the black box is able to generate corresponding signals, variously called *outputs* or *responses*, which are also functions of time, and which one can directly observe. We desire to know how each particular output of the box is determined by the corresponding input which elicits it, and from this to determine, to the extent possible, what is actually inside the box. We note explicitly that a knowledge of the relation of inputs to outputs also allows us to *control* the behavior of the box; i.e. to elicit any desired output, by supplying it to the corresponding input.

Clearly the box itself acts as a *transducer*, which converts inputs to outputs. In mathematical terms, since both inputs and outputs are represented as functions of time, the activity of the box is correspondingly represented by some kind of operator, which maps input functions into output functions. It is this operator which we wish to characterize, and our only means for doing so are our observations of how the box responds to inputs supplied to it.

Once again, this kind of question can only be answered completely for a special class of systems, whose input-output relation is generated by a *linear* operator T . More specifically, if $\{y_i(t)\}$ represents a family of possible input functions to our box, and $u_i(t)$ represents the family of corresponding outputs, then we require that

$$T \left(\sum_i r_i y_i(t) \right) = \sum_i r_i u_i(t) \quad (1.5)$$

where the r_i are arbitrary numbers. Simply stated, a linear box maps arbitrary sums of inputs into sums of corresponding outputs. A box represented by such a linear operator is called a *linear box*, or linear input-output system. The reader will

note that this definition of linearity appears quite different from that employed in connection with the dynamical system (1.3) above.

More generally, we shall suppose that linearity holds for even the situation in which the index i , instead of running over a finite or countable set, runs over a continuum. In this case, the summations appearing in (1.5) are replaced by integrations over this continuum.

We can see intuitively that we can completely characterize the input-output relation of a linear system, by invoking the kind of universality we mentioned above. Specifically, we have already noted that any function $y(t)$ can be uniquely represented in terms of an appropriate basis set $\{\phi_i(t)\}$, as a series of the form

$$y(t) = \sum_i a_i \phi_i(t). \quad (1.6)$$

Thus, if we know the outputs of the box for just the functions in the basis set, linearity allows us to determine the output of the box for any input $y(t)$. More generally, the index i can be taken as running over a continuum; in this case, the representation (1.6) becomes an integral of the form

$$y(t) = \int_{\tau_0}^{\tau_1} a(\tau) \phi(t, \tau) d\tau \quad (1.7)$$

Once again, if we know the response of the box to the particular functions $\phi(t, \tau)$, we can determine the response of the box to *any* input $y(t)$.

Let us speak now informally for a moment. If we are willing to stray outside the domain of continuous functions, we can pick the functions $\phi(t, \tau)$ in (1.7) very cleverly; in fact, in such a way that they are all determined by a single one. That one is the familiar δ -function, introduced by Heaviside (although it was already known to Cauchy, Poisson and Hermite) and most notably exploited by Dirac. This function is defined by the properties

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (1.8)$$

$$\delta(t) = 0 \text{ for } t \neq 0.$$

Clearly $\delta(t)$ is a most peculiar function, viewed from any classical standpoint. In fact it is not a function at all, but rather a *generalized function* or *distribution* in the sense of Laurent Schwartz. Briefly, it can be represented as a limit of a Cauchy sequence of continuous functions as for example

$$\delta(t) = \lim_{n \rightarrow \infty} \frac{n}{\pi(n^2 t^2 + 1)}$$

or

$$\delta(t) = \lim_{n \rightarrow \infty} \frac{n}{\sqrt{\pi}} e^{-n^2 t^2}$$

in much the same way that a real number can be represented as the limit of a Cauchy sequence of rational numbers. Its basic property is that it defines a linear functional on spaces of functions of the type we are considering, through what is usually called the *sifting property*:

$$\int_{-\infty}^{\infty} y(t) \delta(t) dt = y(0) \quad (1.9)$$

for any function y defined in a neighborhood of the origin.

From $\delta(t)$, we can obtain a continuum of functions of the form $\delta(t - \tau)$, one for each real number τ . These functions, the translates of $\delta(t)$, also satisfy a sifting property:

$$\int_{-\infty}^{\infty} y(t) \delta(t - \tau) dt = y(\tau) \quad (1.10)$$

Comparing (1.10) and (1.7), we see that we can represent any input $y(t)$ to a linear system in an integral form, using the translates $\delta(t - \tau)$ of the δ -function as a basis set.

Thus, let us suppose that we can supply $\delta(t)$ as an input to our linear black box, and let us suppose that the resultant output or response is $K(t)$. Clearly, the response of the box to any translate $\delta(t - \tau)$ will be $K(t - \tau)$. But now any arbitrary input $y(t)$ to the box can be written in the form (1.10); hence the linearity of the box implies that the corresponding output is

$$u(t) = \int_{-\infty}^t K(t - \tau) y(\tau) d\tau. \quad (1.11)$$

The relation (1.11) is the fundamental one which characterizes our linear black box, in the sense that it specifies the output $u(t)$ of the box when the input $y(t)$ is given. The function $K(t - \tau)$ is essentially what is called the *transfer function* of the system, although for technical reasons, the transfer function of the engineer is the Laplace Transform of K . The basic reason for this is that the Laplace transform converts the integral (1.11), which is a *convolution integral*, into an ordinary product of functions; thus, if L formally denotes the Laplace transform operation, (1.11) becomes

$$Lu(t) = G(s) Ly(t) \quad (1.12)$$

where $G(s)$, the Laplace transform of $K(t)$, is given by

$$G(s) = \int_0^{\infty} K(t) e^{-st} dt$$

For fuller details, see the Bibliography.⁴

We can cast the relation (1.11), expressing the input-output relations of our linear box, into a more familiar form under certain general circumstances. Namely, if the required derivatives exist, it is

$$\frac{du}{dt} = \int_{-\infty}^t \frac{\partial}{\partial t} K(t - \tau) y(\tau) d\tau$$

or more generally, that

$$\sum_{i=0}^n a_i(t) \frac{\partial^i u}{\partial t^i} = \int_{-\infty}^t \sum_{i=0}^n a_i(t) \frac{\partial^i}{\partial t^i} K(t - \tau) y(\tau) d\tau \quad (1.13)$$

where the $a_i(t)$ are the arbitrary functions of t . Let us suppose that we can find functions $a_0(t)$, $a_1(t)$, ..., $a_n(t)$ such that

$$\sum_{i=0}^n a_i(t) \frac{\partial^i}{\partial t^i} K(t - \tau) = \delta(t - \tau).$$

Then by the sifting property, (1.13) becomes

$$\sum_{i=0}^n a_i(t) \frac{\partial^i u}{\partial t^i} = y(t) \quad (1.14)$$

That is, *the relation between input $y(t)$ and corresponding output $u(t)$ is such that these functions satisfy the linear n th-order differential equation (1.14)*. Stated another way, the differential operator

$$\sum_{i=0}^n a_i(t) \frac{\partial^i}{\partial t^i}$$

and the integral operator

$$\int_{-\infty}^t K(t - \tau) d\tau$$

are inverse of one another.

The relationship between (1.11) and (1.14) allows us to draw an important connection between the linear black box, characterized by the function $K(t - \tau)$, and the theory of linear dynamical systems we touched on earlier. For the single n^{th} -order linear differential equation (1.14) can be expressed as a set of n linear first-order equations: i.e. as a dynamical system. Specifically, if we introduce new variables $x_1, x_2, \dots, x_n$ by writing

$$x_i = d^{i-1}x/dt^{i-1}$$

we obtain the linear dynamical system

$$\begin{aligned} \frac{dx_1}{dt} &= x_2 \\ \frac{dx_2}{dt} &= x_3 \\ &\vdots \\ \frac{dx_{n-1}}{dt} &= x_n \\ \frac{dx_n}{dt} &= -\frac{1}{a_n(t)} \sum_{i=1}^{n-1} a_i(t) x_i - y(t) \end{aligned} \tag{1.15}$$

This dynamical system is, apart from technical details, of the form (1.3) which we have seen before. The details in which (1.15) differs from (1.3) are: (a) in (1.15) the coefficients $a_i(t)$ are in general functions of time, and not constant as they were in (1.3); (b) the system (1.15) is not homogeneous; the last equation involves an additive time-dependent function $y(t)$, the system input. But it is easy to generalize the theory of linear systems like (1.3) to the situation of non-homogeneous systems (i.e. non-zero input) and time-varying coefficients; the details can be found in almost any text on linear differential equations. It is also shown from this analysis the close relation which exists between the response function $K(t - \tau)$ of the linear box and the matrix $A(t)$ of the associated linear dynamical system (1.15); namely

$$K(t - \tau) = e^{(t-\tau)A} \tag{1.16}$$

There is a further consequence of the representation (1.15) which should be noted. Initially, we regarded our black box as opaque; by hypothesis we had no knowledge of the internal structure of the box, but we assumed that we could observe the output $u(t)$ generated by a given input $y(t)$. On the other hand, the variables x_i introduced in (1.15) can be regarded as constituting a set of variables for the *box itself*. Moreover, the transfer function allows us to determine, through the representation (1.14) and (1.15), the equations of motion governing these state variables. Thus to this extent a characterization of the input-output relation allows us to *open the box*, at least to the extent we have indicated.

It should also be noted that the integral (1.11), which characterizes the input-output relation of our box, expresses the reactive paradigm in the following form: the *value* of the output $u(t)$ at any instant is obtained as an integral over all present and past values of the input $y(t)$, each appropriately weighted by the response function $K(t - \tau)$. In all linear theories of this form, the possibility of future input values entering into the present value of the output is expressly forbidden. When we represent the same system in differential form, as in (1.14) or (1.15), we find this translates into the usual condition that rate of change of *state* depends only on present state.

It should also be emphasized that the transfer-function approach to input-output systems is restricted to *linear* systems. There is in general no analog of the transfer function to describe the relations between inputs and outputs in systems which are not linear.

Now let us, in effect, turn the entire argument of the past few pages around. We shall *begin* with a linear dynamical system of the form

$$\frac{dx_i}{dt} = \sum_{j=1}^n a_{ij}x_j, \quad i = 1, \dots, n. \quad (1.17)$$

in which we can allow the coefficients a_{ij} to be definite functions of time. Let us now suppose, by analogy with (1.15), that we allow the rates of change dx_i/dt of the state variables x_i to depend in a linear fashion on a family $y_i(t)$, $\dots$, $y_r(t)$ of functions of time, which may be chosen from a class of admissible functions; i.e. we write the equations of motion as

$$\frac{dx_i}{dt} = \sum_{j=1}^n a_{ij}x_j + \sum_{k=1}^r b_{ik}y_k \quad (1.18)$$

Here the coefficients (b_{ik}) are assumed fixed. The functions $y_i(t)$ are generally called *controls* for the system (1.18). We shall further assume that the object of imposing such controls is to manipulate the values of some quantity H associated with the system. Since by hypothesis $x_1, \dots, x_n$ are state variables, the quantity H must be a specific function $H(x_1, \dots, x_n)$ of these variables, and we assume it is a linear function:

$$H(x_1, \dots, x_n) = \sum_{i=1}^n \beta_i x_i. \quad (1.19)$$

This is the basic format for the theory of the control of linear systems, resolved here into two distinct problems: (a) to determine, from (1.18), the manner in which a set of control functions $y_i(t)$ modify an appropriate set of state variables; (b) to compute from this the behavior of the output function H in which we are interested.

With appropriate modifications, we can in effect run backwards the argument which led from (1.11) to (1.15) above. More specifically, we can write the family

(1.18) of n first-order differential equations as a single n -th order equation; this equation will in general involve the control functions $y_i(t)$ and their time derivatives. From this n^{th} -order equation, we can work backwards to write down a generalized form of the input-output relation (1.11), thereby obtaining the most general form of linear theory of input-output systems; it allows us to explicitly express the relation between multiple controls and the resultant outputs.

It must not be thought that there is a complete equivalence between the dynamical approach (1.18) and the transfer function approach based on (1.11). In fact, the set of state variables x_i which we obtain in passing from (1.11) to (1.15) need not completely characterize our black box; in effect, we can only detect through the transfer function such state variables of which the outputs we observe are functions, and which are themselves functions of the imposed inputs. All other state quantities characterizing the interior of the box are *invisible* to this approach. They can be made visible only by augmenting either the class of admissible inputs to the box, or by gaining access to a larger class of outputs of the box (or both). The extent to which we can characterize the interior of a linear box in terms of a definite set of inputs we can impose, and a definite set of outputs we can observe, is the subject of an elegant development by R. E. Kalman, under the heading of a general theory of *controllability and observability*.⁵ As we shall see, the Kalman theory illuminates many important epistemological questions concerning systems in general. For further details regarding this theory, we refer the reader to the material cited in the Bibliography.

We now wish to develop another important relationship between the theory of input-output systems and that of stability of dynamical systems. We can motivate this development in the following way. We have already mentioned that the stability of a steady state of a dynamical system, or indeed of an arbitrary trajectory, can be thought of as a manifestation of homeostasis. On the other hand, the theory of control was largely developed precisely to generate and maintain homeostasis within a technological context. We have already formulated the basic notions of control, as far as linear systems are concerned, in the relations (1.18) and (1.19) above. What we wish to do now is to explicitly develop the relationship between dynamical stability and control theory. It is important to do this, because stability is a property of *autonomous* systems (i.e. whose equations of motion do not involve time explicitly) while control theory seems to involve time-varying quantities in an essential way; we can see at a glance that the control equations (1.18) do not fit under the rubric of stability as we defined it. We shall ultimately find that the relation between stability and control to be developed embodies all of the essential conceptual features of the reactive paradigm, in a particularly transparent and suggestive way. Before we take up this relationship, there is one more important point to be made about the input-output approach, which we now proceed to sketch.

We have seen that the basic unit of the input-output approach is the “black box”, which receives inputs from its environment, and produces outputs to the environment. This situation can be represented pictorially as in Fig. 1.2. Here the y_i represent particular modalities of inputs presented to the box, and the u_j are correspondingly output modalities. In the case of a linear box, we have seen that the

Fig. 1.2

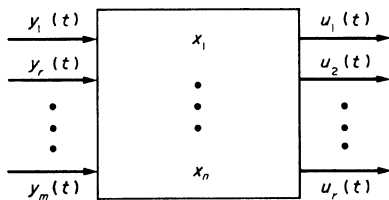
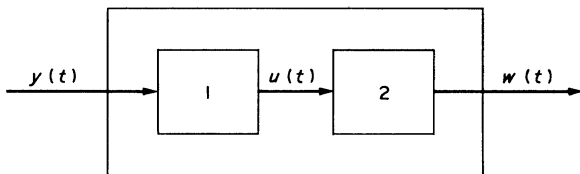


Fig. 1.3



box itself can be assigned a family of state variables $x_1, \dots, x_n$, whose temporal behavior depends on the inputs $y_i(t)$, and which in turn determine the properties of the outputs $u_j(t)$.

The reason that the representation of Fig. 1.2 is so suggestive is the following: we can imagine that the inputs $y_i(t)$ to the box are themselves outputs of other boxes; likewise, that the outputs $u_j(t)$ of the box are themselves inputs to other boxes. In other words, we can construct *networks* of such boxes; the response characteristics of such networks, considered as input-output systems in their own right, may then be expressed in a canonical way in terms of the properties of the constituent elements in the network, and the manner in which these are inter-related. Conversely, a given linear input-output system may be *analyzed* or resolved into a network of simpler systems. This crucial capacity for analysis and synthesis is not visible in a pure state-variable approach; although as we shall soon see, it is present.

As an illustration, let us consider the simplest possible networks. In Fig. 1.3, we see a system consisting of two linear boxes arranged in series; the output of the first is the input of the second.

As we have seen, each of the component boxes is represented by a response function, which we may denote by K_1, K_2 respectively. It turns out for our present purposes to be more convenient to work with the corresponding transfer functions G_1, G_2 , where, as noted above, $G_1 = LK_1$, $G_2 = LK_2$, and L is the Laplace transform operation. For it is then immediate to verify that

$$Lw(t) = G_2 G_1 Ly(t).$$

In other words, *the transfer function of a pair of linear boxes connected in series is the product of the transfer functions of the boxes.*

Likewise, we may consider the simple parallel network shown in Fig. 1.4:

Using the same notation as before, it is easy to see that

$$L(u_1(t) + u_2(t)) = (G_1 + G_2) Ly(t);$$

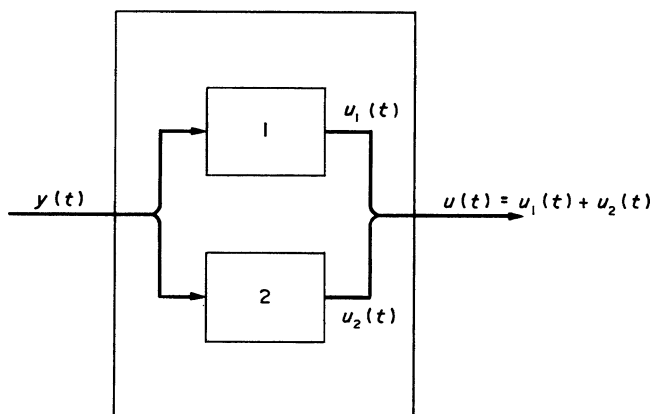
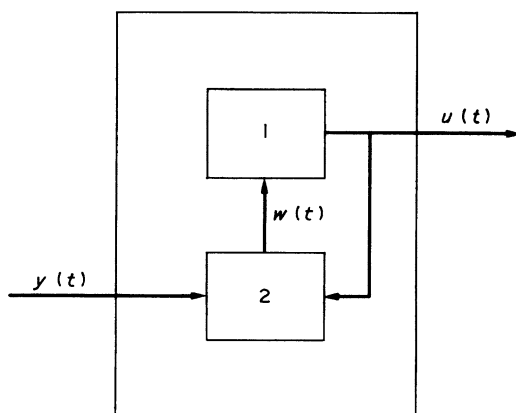


Fig. 1.4

Fig. 1.5

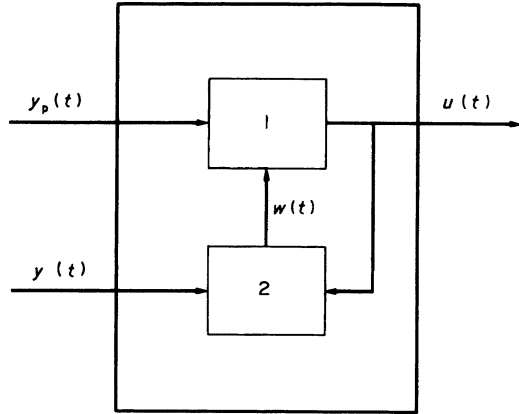


or in other words, that *the transfer function of a pair of linear boxes connected in parallel is the sum of the transfer functions of the individual boxes*. From these two simple results, we can readily see how the transfer function of any array of linear boxes connected in series and parallel can be computed, from a knowledge of the transfer functions of the component boxes and of the pattern of interconnections. Conversely, if the transfer function of a given system can be expressed in terms of sums and products of simpler functions, then the system itself can be effectively decomposed into a series-parallel network of simpler subsystems.

The most important kind of application of these ideas, for our purposes, arises when *the input signal to a box in a network is partially derived from its own output, either directly, or through the intermediary of other boxes*. Such a situation is called *feedback*, and is exemplified by the network shown in Fig. 1.5.

In control engineering, each aspect of this kind of network is given a special name, descriptive of the function of the network as a whole. These are as follows:

Fig. 1.6



The box labelled “1” is called the *controlled system*; The box labelled “2” is called the *controller*; The input $y(t)$ is called the *command signal*; $w(t)$ is called the *control signal*; The closed loop passing from the first box through the second and back to the first is the *feedback loop*.

Finally, if the entire system is to remain linear, it is clear that the total input to the second box, or controller, must be of the form $y(t) + u(t)$ or $y(t) - u(t)$. In the first case, we speak of *positive feedback*; in the latter case, of *negative feedback*. If the feedback is negative, then the total input $y(t) - u(t)$ to the controller is called the *error signal*. In general, a negative feedback tends to *oppose* the motion of the output $u(t)$, while a positive feedback tends to amplify it.

Let us for the moment restrict ourselves to the case of negative feedback. In this case, the diagram of Fig. 1.5 is the simplest instance of a *pure servomechanism*. Intuitively, the function of the controller here is to force the output signal $u(t)$ of the controlled system to follow or track the command signal $y(t)$. The controller’s input, or error signal, is determined by the discrepancy between $y(t)$ and $u(t)$; its output, the *command signal*, is designed to minimize this discrepancy. Finally, if G_1 , G_2 denote the transfer functions of the controlled system and controller respectively, it is not hard to show that the transfer function of the total system shown in Fig. 1.5 is

$$G = G_1 G_2 / (1 + G_1 G_2).$$

A variant of the network shown in Fig. 1.5 is given in Fig. 1.6.

Here we assume that the command signal is constant (such a constant command signal is often called a *set-point*). We now admit a new signal $y_p(t)$ directly into the controlled system; $y_p(t)$ represents ambient fluctuation or perturbation. If $y_p(t) = 0$, the output $u(t)$ of Fig. 1.6 will come to, and remain at, a constant value. A non-zero perturbing signal will tend to move the controlled system away from this value. The function of the feedback loop is to offset the effect of the perturbation, again by comparing the actual present value of the output with the set-point, and imposing a control signal whose magnitude is determined by the degree

of discrepancy between the two. For this reason, the network of Fig. 1.6 is often called a *pure regulator*, or *homeostat*.

We can now express the relationship between stability and control, which is most transparent in the context of feedback control. To fix ideas, let us consider the homeostat shown in Fig. 1.6 above. The purpose of imposing such a control is, of course, to maintain some output $u(t)$ of the controlled box at a constant level, in the face of externally imposed perturbations $y_p(t)$. It is precisely to accomplish this that we attach the controller in the fashion indicated.

Let us suppose that $x_1, \dots, x_n$ represent a set of state variables for the controlled system, so chosen that the autonomous equations of motion of that system, in the absence of external perturbation, or any signal from the controller, are in the form corresponding to (1.15) above; namely

$$\begin{aligned}\frac{dx_i}{dt} &= x_{i+1}, \quad i = 1, \dots, n-1 \\ \frac{dx_n}{dt} &= -a_1x_1 - a_2x_2 - \dots - a_nx_n\end{aligned}\tag{1.20}$$

We also arrange matters so that the state variable x_1 is also the system output in which we are interested; i.e. $x_1 = u$. It can readily be shown that these hypotheses involve no loss of generality.

Let us further suppose that the controller, considered in isolation, can be represented by a set of state variables $\xi_1, \dots, \xi_m$, where once again we can express the autonomous dynamics in the form

$$\begin{aligned}\frac{d\xi_i}{dt} &= \xi_{i+1}, \quad i = 1, \dots, m-1 \\ \frac{d\xi_m}{dt} &= b_1\xi_1 - b_2\xi_2 - \dots - b_m\xi_m\end{aligned}\tag{1.21}$$

and such that $\xi_1 = w(t)$; i.e. the command signal, or controller output, is the first of our state variables.

We now combine (1.20) and (1.21) in the fashion indicated in Fig. 1.6, remembering the relation (1.14). First, let us agree to represent the command signal, or set point, by a new formal state variable ξ_{m+1} of the controller. The equations (1.21) then become

$$\begin{aligned}\frac{d\xi_i}{dt} &= \xi_{i+1}, \quad i = 1, \dots, m-1 \\ \frac{d\xi_m}{dt} &= -b_1\xi_1 - b_2\xi_2 - \dots - b_m\xi_m + \xi_{m+1} \\ \frac{d\xi_{m+1}}{dt} &= 0\end{aligned}\tag{1.22}$$

Furthermore, the output $x_1(t)$ of the controlled signal is to be *fed back* into the controller. Thus, (1.22) finally becomes

$$\begin{aligned}\frac{d\xi_i}{dt} &= \xi_{i+1}, \quad i = 1, \dots, m-1 \\ \frac{d\xi_m}{dt} &= -b_1\xi_1 - b_2\xi_2 - \dots - b_m\xi_m + (\xi_{m+1} - x_1) \\ \frac{d\xi_{m+1}}{dt} &= 0\end{aligned}\tag{1.23}$$

Now we must represent the command signal ξ_1 as input to the controlled system. When this is done, (1.20) becomes

$$\begin{aligned}\frac{dx_i}{dt} &= x_{i+1}, \quad i = 1, \dots, n-1 \\ \frac{dx_n}{dt} &= -a_1x_1 - a_2x_2 - \dots - a_nx_n + \xi_1\end{aligned}\tag{1.24}$$

The equations (1.23) and (1.24) together form a single linear dynamical system, which represents the autonomous behavior of the entire homeostat shown in Fig. 1.6. We can see immediately that this system will have a unique steady state at which

$$x_1 = \xi_{m+1}$$

and all the other state variables are zero. This steady state will be stable if all the eigenvalues of the system matrix have negative real parts, and the degree of stability (i.e. the rate at which any particular trajectory approaches the steady state) will depend on the magnitudes of these real parts. In their turn, the eigenvalues themselves are determined entirely by the coefficients a_i and b_j , which are the constitutive parameters of the controlled system and the controller respectively. In particular, we can choose the b_j in such a way that, for a given system to be controlled, i.e. a given set of parameters a_i , all the eigenvalues of the total system have large negative real parts.

If we now impose an arbitrary perturbing signal $y_p(t)$ on the controlled system, subject only to the condition that $y_p(t)$ changes *slowly* compared to the rate at which the trajectories of the homeostat are approaching their steady state, it is intuitively clear that the homeostat will always remain close to its steady state. Thus, *the homeostatic property of the feedback control is directly identified with the stability property of the total system consisting of the controlled system and the controller.*

From this argument, we see that the *function* of a controller in a homeostat is to embed the controlled system in a larger system, in such a way that (a) the larger system possesses an asymptotically stable steady state, and (b) that steady state, when projected onto the states of the controlled system alone, possesses the desired property (here, the property that $x_1 = \xi_{m+1}$).

The above argument exemplifies the relation we sought between control theory and stability. Although we couched it in the language of linear feedback, the argument itself is perfectly general. Specifically, if we are given *any* dynamical system of the form (1.2), then in some neighborhood of any stable steady state, we have seen that the behavior of the system may be approximated by that of a linear system (1.3). In an approximate co-ordinate system (i.e. by a suitable choice of state variables) that approximating linear system may itself be decomposed into a part interpretable as a controlled system, and a part interpretable as a feedback controller. Of course this decomposition is not unique, but it does exhibit the essential equivalence between the concept of feedback control and the stability of steady states.

The same argument may be used to demonstrate the equivalence between positive feedback and instability; we leave the details to the reader.

It should be carefully noted that the linear formulation we have provided above is not adequate for the homeostats and servomechanisms which are found in biology. Basically, this is because biological feedback controls violate basic assumptions we have made: (a) that the autonomous dynamics of an input-output system are of the linear form (1.3), and (b) that the interaction between boxes is describable in the form (1.18), in which controls enter linearly into the autonomous dynamics. In connection with this latter point, it is found that interactions between biological systems, even if they could be regarded as linear, takes place through the modification of constitutive parameters; this is called *feedback through parameters*. Such control is analogous to a modification of the transfer functions through interaction; if we attempt to embody this type of control through arguments leading to the equations (1.23) and (1.24), the resulting system will necessarily be nonlinear. This comment, however, is primarily of a technical nature; it does not affect the essential relation between feedback and stability which we have developed above.

We may now conclude this brief overview of the basic aspects of the reactive paradigm with a few general remarks.

The first remark is concerned with the frankly reactive nature of feedback control. Indeed, the distinguishing feature of feedback control is the presence of an “error signal”, which is the difference between what the system output *should be* at a given instant (as embodied in the command signal) and what the output actually *is* at that instant. It is this error signal which is transduced by the controller into a control signal, whose purpose is to minimize the discrepancy between command signal and output. Clearly, the utilization of such an error signal means that the behavior of the controlled system *has already departed* from that which is desired *before* any control can be imposed on it. The exercise of feedback control is thus always corrective; always imposed as a reaction to a deviation which has already occurred. Nothing could be more indicative of the essential character of the reactive paradigm than this.

We can now graphically illustrate the fundamental distinction between a reactive and an anticipatory mode of control. We have just seen that all reactive behavior necessarily involves a response to circumstances which have already occurred. It

might be argued that by making a reactive system more stable, in the sense of increasing the magnitudes of the (negative) real parts of the system eigenvalues, or in other words by making the time constants of the system sufficiently short, this difficulty can for practical purposes be eliminated. Apart from the practical problems arising from attempting to do this, we can see that this is not so in principle by comparing two apparently similar homeostats; one arising in biology, and the other of a technological nature.

One of the best-studied biological homeostats is one involved in maintaining an optimal constancy of light falling on the retina of the vertebrate eye, the so-called *pupillary servomechanism*.⁶ Roughly speaking, in conditions in which there is a great deal of ambient light, the pupil contracts, and admits a smaller amount of light to the eye. Conversely, when the ambient light is dim, the pupil opens to admit more light. It has been established that the control system involved here is a true feedback system, whose output is represented by the actual amount of light falling on the retina. Thus, the sensor for the controller is at the retina, and the system *reacts to how much light has already been admitted to the eye*. The time constant for this servomechanism is not outstandingly small, but the system clearly functions well for almost all conditions the organism encounters.

Now let us consider the analogous problem of controlling the amount of light entering a camera to ensure optimal film exposure. Here again, the control element is a diaphragm, which must be opened when the ambient light is dim, and closed when the ambient light is bright. However, in this case, *we cannot in principle use a reactive mechanism at all*, no matter how small its time constant. For clearly, if the input to the controller is the light falling on the film, in analogy to the situation in the eye, then the film is already under- or over-exposed before any control can be instituted. In this case, the *only* effective way to control the diaphragm is through an anticipatory mode, and that is what in fact is done. Specifically, a light-meter is used to measure the *ambient* light. The reading of the light meter is then referred to a predictive model, which relates ambient light to the diaphragm opening necessary to admit the optimal amount of light to the camera. The diaphragm is then *pre-set* according to the prediction of the model. In this simple example we see all the contrasting features of feedforward and feedback; of anticipatory as against reactive modes of control.

The crucial distinction between anticipation and reaction manifests itself very strongly in what we may call the “style” of approach that each imposes on the study of system behavior. For instance, we have already discussed the anticipatory nature of simple organism behaviors, such as phototropisms. We argued that these tropisms embody predictive models, insofar as they correlate neutral ambient characteristics such as darkness with circumstances which are physiologically favorable. On the other hand, we may cite a passage from Norbert Wiener’s seminal book *Cybernetics*,⁷ dealing with negative phototropism in the flatworm:

... negative phototropism. . . seems to be controlled by the balance of the impulses from the two eyespots. This balance is fed back to the muscles of the trunk, turning the body away from the light, and, in combination with the general impulse to move forward, carries the animal into the darkest region accessible. It is interesting to note that a combination of a pair

of photo-cells, with appropriate amplifiers, a Wheatstone bridge for balancing their outputs, and further amplifiers controlling the input into the two motors of a twinscrew mechanism would give us a very adequate negatively phototropic control for a little boat.

There is in this discussion a total preoccupation with *mechanism*, and more tacitly, with the idea that reactive simulation provides a full understanding of the tropistic behavior. It can be seen from this example that the reactive paradigm necessarily restricts us from the outset to the *effector* aspects of behavior, and in itself leaves the adaptive aspects of that behavior, which arise precisely from its anticipatory features, untouched. In our little example of the pre-setting of a camera aperture as a function of ambient light, the analogous discussion to that cited above could only be concerned with the specific mechanism by which the diaphragm is actually moved, as a function of the predictions of the model.

Our final remark arises naturally from the preceding discussion, and involves the relation between homeostasis and adaptation in general. In the broadest terms, adaptation in biology refers to a modification of system structure or behavior, in such a way that the overall survival of the adapting system is enhanced. As such, adaptation connotes a dynamical process, in which changed circumstances are reflected in corresponding changes in the adapting organism. In these terms, adaptation itself can occur through either a reactive mode or an anticipatory mode; the former connotes responses to changes which have already occurred, and the latter connotes responses to changes which some model predicts will occur. Let us restrict attention for the moment to reactive adaptation.

In particular, let us consider the homeostat shown in Fig. 1.6; or better, a specific realization of such a homeostat, as for example the thermostatically controlled temperature of a room. Although of course “survival” is not an appropriate concept in this context, it is not too great an abuse of language to regard the thermostat as adapting to ambient fluctuations of the temperature of the room. We obtain two quite different pictures of the thermostat, depending upon which part of it we focus our attention upon. For instance, if we concentrate on the system output, we know that this remains approximately constant in the face of ambient fluctuations, and this reflects the homeostatic aspects of the system. On the other hand, we could concentrate upon, say, the dampers which control the amount of air entering the furnace. This part of the system is in more or less continual movement, and this movement reflects the adaptive aspect of the system. We must say, in general, that adaptation and homeostasis in this sense are always correlated; there is always some property of an adaptive system which is homeostatic in character, and conversely there is always some aspect of a homeostatic system which is adaptive in character. Whether we consider such a system as adaptive or homeostatic is determined entirely by whether we focus on the parts of the system which are changing, or on the parts which are staying fixed. The dualism between adaptation and homeostasis is in fact an essential feature of biological systems, to which we shall return many times as we proceed.

References and Notes

1. There are many excellent books dealing with classical mechanics, which is the wellspring of the reactive paradigm. Among more modern treatments, the following may be recommended:

Abraham, A. and Marsden, J. E., *Foundations of Mechanics*. Benjamin, Reading, Pennsylvania (1978).

Arnold, V. I., *Mathematical Methods of Classical Mechanics*. Springer-Verlag, New York (1978).

Santilli, R. M., *Foundations of Theoretical Mechanics*. Springer-Verlag, New York (1978).

Sudarshan, E. C. G. and Mukunda, N., *Classical Dynamics: A Modern Perspective*. Wiley, New York (1974).

2. The theory of dynamical systems have developed along a number of fronts, which overlap considerably but are not quite equivalent mathematically. The form closest to its origin in Newtonian mechanics is the theory of systems of first-order differential equations; an early and still excellent reference is

Birkhoff, G. D., *Dynamical Systems*. AMS, Providence (1927). A later but still classically oriented work is

Nemitskii, V. V. and Stepanov, A., *Qualitative Theory of Ordinary Differential Equations*. Princeton University Press (1960).

A more recent approach is to define a dynamical system as a vector field on a manifold. A good introduction, which nevertheless stays close to the classical roots, is

Hirsch, M. W. and Smale, S., *Differential Equations, Dynamical Systems and Linear Algebra*. Academic Press (1974).

Still another approach is to regard a dynamical system as a flow; a parameterized family of automorphisms on a topological space or a measure space. As an example of the former, we may mention

Gottschalk, W. H. and Hedlund, G. A., *Topological Dynamics*. AMS, Providence (1955).

for the latter, the reader may consult, e.g.

Ornstein, D., *Ergodic Theory, Randomness and Dynamical Systems*. Yale University Press (1971).

3. The input-output approach is characteristic of engineering rather than physics. The theory of such systems is almost entirely restricted to the situation in which the system in question is linear. On the mathematical side, this theory connects linear differential equations with integral equations (which is perhaps not surprising, if we recall that differentiation and integration are inverse operations). The application of this theory is broadly called *control theory*. The classic text on linear system theory is still

Zadeh, L. and Desoer, C. A., *Linear System Theory*. McGraw-Hill, New York (1963).

4. The mathematical theory underlying our discussion here is generally called harmonic analysis. A fairly typical exposition (which nowadays can be found in most treatments of real variable theory) is

Loomis, L. H., *Abstract Harmonic Analysis*. van Nostrand, New York (1955). Harmonic analysis is one of the most interesting and seminal areas in all of mathematics; for an incredibly rich historical survey of how these ideas ramify throughout mathematics, see Mackey, G. W., *Bull. Am. Math. Soc.* (New Series) 3, 543–698 (1980). A far more restricted development of those portions most familiar to engineers may still be found in

van der Pol, B. and Bremer, H., *Operational Calculus*. Cambridge University Press (1955).

5. See for instance

Kalman, R. E., Arbib, M. and Falb, P. L., *Topics in Mathematical System Theory*. McGraw-Hill, New York (1969).

6. A detailed discussion of the pupillary servomechanism, as well as many other examples of biological control, will be found in

Stark, L., *Neurological Control Systems*. Plenum, New York (1968).

7. Cybernetic control is in many ways the apotheosis of the reactive paradigm. The best reference is

Wiener, N., *Cybernetics*. MIT Press, Cambridge (1961).

Chapter 2

Natural and Formal Systems

2.1 The Concept of a Natural System

In the present chapter, we shall be concerned with specifying what we mean by a *natural system*. Roughly speaking, a natural system comprises some aspect of the external world which we wish to study. A stone, a star, the solar system, an organism, a beehive, an ecosystem, are typical examples of natural systems; but so, too, are automobiles, factories, cities, and the like. Thus, natural systems are what the sciences are about, and what technologies seek to fabricate and control. We use the adjective “natural” to distinguish these systems from the *formal systems* which we create to represent and model them; formal systems are the vehicles for drawing inferences from premises, and belong to mathematics rather than science. There are of course close relations between the two classes of systems, natural and formal, which we hope to develop in due course.

In coming to grips with the idea of a natural system, we must necessarily touch on some basic philosophical questions, on both an ontological and an epistemological character. This is unavoidable in any case, and must be confronted squarely at the outset of a work like the present one, because one’s tacit presuppositions in these areas determine the character of one’s science. It is true that many scientists find an explicit consideration of such matters irksome, just as many working mathematicians dislike discussions of foundations of mathematics. Nevertheless, it is well to recall a remark made by David Hawkins: “Philosophy may be ignored but not escaped; and those who most ignore least escape”. The discussion presented in this chapter is not intended to be exhaustive or comprehensive; it will, however, provide much of the basic conceptual underpinning for the detailed technical discussions of the subsequent chapters.¹

It is perhaps best to begin our discussion of natural systems with our fundamental awareness of sensory impressions, which we shall collectively call *percepts*. These percepts are the basic stuff of science. It is at least plausible for us to believe that we do not entirely create these percepts, but rather discover them, through our experience of them. If we do not create them, then their source must lie, at least

in part, outside of us, and thus we are led to the idea of an *external world*. We are further led to associate the sources of our percepts with definite *qualities* belonging to that external world; thence to identify such qualities with the percepts which they generate, and which we experience.

The first obvious step in our characterization of a natural system can then be stated: *a natural system is a member or element of the external world*. As such, then, it generates (or can generate, under suitable conditions) percepts in us, which we identify in turn with specific qualities, or properties, of the system itself.

The ability to generate percepts is not sufficient for us to say we have identified a natural system, although it is certainly a necessary part of such an identification. Intuitively, we would expect the concept of a system to involve some kind of inter-relation between the percepts it generates, and which then become identified with corresponding relationships between the external qualities which generated them.

The question of relations between percepts would be relatively simple if it were the case that such relations were themselves percepts. In that case, we could immediately extend our identification of percepts with qualities, so as to likewise identify perceptual relations with relations between qualities. However, we would not be justified in immediately taking such a step, because it is not clear that relations between percepts are themselves percepts or that they are discovered rather than created. This is a subtle and important point, which must be discussed in more detail.

Briefly, we believe that one of the primary functions of the mind is precisely to *organize* percepts. That is, the mind is not merely a passive receiver of perceptual images, but rather takes an active role in processing them and ultimately in responding to them through effector mechanisms. The *organization* of percepts means precisely the establishment of relations between them. But we then must admit that such relations reflect the properties of the active mind as much as they do the percepts which the mind organizes.

What does seem to be true, however, is the following: that the mind behaves *as if* a relation it establishes between percepts were itself a percept. Consequently, it behaves *as if* such a relation between percepts arises from a corresponding relation between qualities in the external world. Therefore it behaves as if such a relation between qualities in the external world were itself a quality, and as if the perception of this new quality consisted precisely of the relation it established between percepts. The basic point is simply that relations between percepts are, at least to some extent, creations of the mind which are then *imputed* to the external world. As such, they may be regarded as “working hypotheses”, or, to use a more direct word, *models* of how the external world is organized.

Now we have already argued that the function of the mind, in biological terms, is to act as a transducer between percepts and specific actions, mediated by the organism’s effector mechanisms. It is clear that these actions, and thus the models of the external world which give rise to them, are directly subject to natural selection. An organism which acts inappropriately, or in a maladaptive manner, will clearly not survive long. The very fact of our survival as organisms subject to natural selection

thus leads us to suspect that the mechanisms of the mind for organizing percepts (i.e. for establishing relations between percepts) must have some degree of correspondence with objective relations existing between qualities in the external world which is, after all, the agency through which selection mechanisms themselves operate. This type of argument from selection, while obviously not a proof (and equally obviously, hardly an argument from first principles), makes it at least plausible that (a) relations between perceived qualities exist in the external world, and (b) that such relations, too, can be discovered by the mind. The process by which such relations are discovered, however, is not of the same character as the discovery of the qualities themselves; rather, it arises through the *imputation* of mentally created relations between particular percepts to the qualities which generate these percepts.

This discussion provides the second necessary ingredient in our characterization of a natural system. Specifically, we shall say that *a natural system is a set of qualities, to which definite relations can be imputed*. As such, then, a natural system from the outset embodies a mental construct (i.e. a relation established by the mind between percepts) which comprises a hypothesis or model pertaining to the organization of the external world.

In what follows, we shall refer to a perceptible quality of a natural system as an *observable*. We shall call a relation obtaining between two or more observables belonging to a natural system a *linkage* between them. We take the viewpoint that the study of natural systems is precisely the specification of the observables belonging to such a system, and a characterization of the manner in which they are linked. Indeed, for us *observables are the fundamental units of natural systems, just as percepts are the fundamental units of experience*.

To proceed further, we need first to discuss in more detail how qualities, or observables, are defined in general. We have already stated that a quality is a property of the external world which is, or which can be made to be, directly perceptible to us. Let us elaborate a bit on this theme.

The first thing to recognize is that our sensory apparatus, through which qualities in the external world are transduced to percepts in us, must themselves be part of the external world. That is, they themselves are natural systems of a particular kind, or at least involve such natural systems in an essential way. It is more precise to say that it is a specific interaction between a natural system and our sensory apparatus which in fact generates a percept. In fact, it must ultimately be a *change* or *modification* in the sensory apparatus, arising from such an interaction, which actually does generate the percept.

Furthermore, we know that we can also interact with the external world through our own effector mechanisms; thus these effectors themselves must project into the external world. Such interactions cause changes in the external world itself, which can then be directly perceived.

From these assertions, it is a small but significant act of extrapolation to posit that, in general, natural systems are capable of interactions, and that these interactions are accompanied by changes or modifications in the interacting systems. We have seen that, in case one of the interacting systems is a part of our own sensory apparatus, the vehicle responsible for its modification is *by definition* a quality, or

observable, of the system with which that apparatus interacts. It is a corollary of these propositions that *if an interaction between any two natural systems S_1 , S_2 causes some change in S_2 , say, then the vehicle responsible for this change is an observable of S_1 , and conversely*. That is, we wish to define observables *in general* as the vehicles through which interactions between natural systems occur, and which are responsible for the ultimately perceptible changes in the interacting systems arising from the interactions.

The above ideas are in fact crucial for science, because they indicate how we can discover new qualities of observables of natural systems, besides the ones which are immediately perceptible to us. More specifically, since our own effectors are themselves natural systems, they can be employed to bring systems into interactions which were not so before. In this way we can actively perform *experiments*, rather than being restricted to passive observation. Suppose for example that we are interested in some specific natural system S_1 , some of whose qualities we can directly perceive. Suppose further that we bring S_1 into interaction with some other system S_2 . Suppose that as a result of this interaction there occurs a directly perceptible change in S_2 . We may then associate the change in S_2 with some specific quality or observable of S_1 , which may (or may not) be *different* from those observables of S_1 which were directly perceptible to us. In this situation, we are in effect employing the system S_2 as a transducer, which renders the corresponding observable of S_1 directly perceptible to us. As we shall see shortly, this situation is the prototype for the idea of a *meter*, which not only defines the observable through which S_1 interacts with it, but actually enables us to *measure* this observable. And it is also clear that such considerations directly generalize the manner in which our own sensory apparatus generates percepts.

Let us return again to the situation above, in which we have placed a system S_1 into interaction with another system S_2 . Let us suppose now, however, that the interaction results in a perceptible change in S_1 . We may then say that, as a result of the specific interaction with S_2 , a new *behavior* of S_1 has been generated. It is correct to say that this behavior is *forced* by the interaction with S_2 , or that it is an *output* to the input consisting of the observable of S_2 through which the interaction occurs.

Both of these situations are fundamental to experimental science, and are the prototypes of the two ways in which we acquire *data* regarding natural systems. We shall have much to say about both of them as we proceed.

The above ideas have one further crucial ramification, which we now proceed to describe. This ramification is concerned with associating *quantity* with qualities or observables, and is implicit in the preceding discussion. Let us recall that we have *defined* an observable of a natural system S_1 as a capability for specific interactions of S_1 with other systems, such as S_2 , and that such an interaction is recognized through a corresponding change in S_2 . Let us imagine now that the system S_2 has the following properties: (a) any change in S_2 arising from interaction with another system S_1 can be associated with the *same* quality or observable of S_1 ; and (b) prior to the establishment of interaction between S_2 and some other system S_1 , S_2 has been restored to an unchanged, fixed initial situation. The property (a) means that,

though S_2 can in principle interact with many different systems S_1 , the *agency* of S_1 responsible for a change in S_2 is a fixed observable; (b) means that the system S_2 is always the same before interaction is instituted.

Under these conditions, it may still happen that the particular perceptible change in S_2 arising from interaction with S_1 is different from that arising from interaction with another system S_1' . Since by hypothesis the *quality* responsible for the change in S_2 is in both cases the same, we can conclude that this quality may take on different *values*. In fact, we can *measure* the values of this quality by the differences between the changes arising in S_2 . By this means, we introduce a new *quantitative* aspect into the characterization of natural systems; we may say that *if a quality of such a system corresponds to an observable, a quantity corresponds to a specific value of an observable*.

The essence of quantitative measurement, or the determination of the magnitude or value of an observable, is thus referred to the change caused through interaction with a specific system like S_2 in the above discussion. The role played by S_2 is then, as noted before, that of a *meter* for the observable in question; under the conditions we have stated, the possible changes which can be induced in S_2 can be identified with the set of possible values of the observable which induces them. Such an identification introduces a crucial notion: that the change in a meter for an observable can *represent*, or replace, the specific value of the observable which generated it. In other words, the change in a meter serves to *label* the values of the associated observable. This will turn out to be the crucial concept underlying the development of the subsequent chapters, when we turn to the concept of formal systems, and the representation of natural systems in terms of formal ones.

There now remains one more basic notion to be introduced before we can proceed with a more comprehensive discussion of the linkage relations which can obtain between observables of a system. This is the notion of *time*, which is of course central to any discussion of change, and hence is a fundamental ingredient in any discussion of interaction of natural systems of the kind sketched above.² For the present, we shall restrict ourselves to the notion of time as quality; we shall return to the matter in more depth in Sect. 2.3 below, which is devoted to dynamics.

The notion of time is a difficult and complex one. We may recall the plaintive words of Saint Augustine, who in his *Confessions* remarks: "What then is time? if no one asks me, I know: if I wish to explain it to one that asketh, I know not."

In accord with the spirit of the discussion we have presented so far, we must begin with our direct perception of time, or what is commonly called the "sense" of time. This is a percept, in the sense in which we have employed the term above. But it is apparently not a simple percept; we can discern at least two quite distinct and apparently independent aspects of our time perception. These distinct aspects must be interpreted differently when, as with all percepts, we attempt to associate them with specific qualities pertaining to the external world. These aspects are: (a) the perception of co-temporality or *simultaneity*; and (b) the perception of *precedence*, involving the distinction between past and future. It should be noted at the outset that both of these aspects of time perception involve *relations* between percepts. It clearly makes no sense to apply either of them to a single percept. For this reason, it

does not seem appropriate to treat time directly as a *quality* or observable belonging directly to the external world. Indeed, we have interpreted all such qualities as potential capacities for interaction, as manifested especially in the moving of meters. It does not seem that either aspect of time perception is of this character.

On the other hand, we have the conviction that time can be “measured”. Thus, if time is not to be regarded as a quality or observable, we must understand the sense in which it can be measured, or resolved into specific time values (instants). We shall begin with a consideration of this question.

The crucial concept here will be that of a *label*; the employment of qualities of one system to represent qualities of another. We have seen, for instance, that the values of an observable can be represented, or labelled, by the specific changes induced in other systems, or meters. Now we have argued that one essential aspect of our time perception expresses the *simultaneity* of other percepts. If this aspect truly reflects a relation between percepts, it follows from our earlier discussion that this relation involves a creative mental act, or model, about the external world which generated the percepts. It is at least plausible to infer that *the specific creative act of the mind involved here is the segregation of percepts into mutually exclusive classes, in such a way that percepts assigned to the same class are regarded as simultaneous, and percepts assigned to different classes are not simultaneous*. In other words, the mind imposes an equivalence relation, not on any specific set of percepts, but on the set of *all possible percepts*; it is this act of extrapolation which characterizes the creative aspect of the sense of simultaneity. Now in general, whenever we are given an equivalence relation on a set of elements, we may select a single representative from each equivalence class, and regard it as a specific *label* for that entire class. As we shall see later, the search for a clever choice of such a label or representative dominates all theories of classification; in mathematics, for instance, we search for *canonical forms* to represent classes of equivalent objects. If this is the case, we can conclude that a “time meter”, or clock, is simply a natural system chosen in such a way that the particular percepts it generates serve as *labels* for the equivalence classes of simultaneous percepts to which they belong. Thus, a clock is not to be regarded as a meter at all, but rather as a generator of labels or representatives for these equivalence classes. In other words, a clock does not “measure” anything; it simply exemplifies the mentally constructed relation of simultaneity between percepts in a particularly convenient form. Its only relation to a true meter resides in its association of percepts with labels.

We may point out explicitly that there is no absolute or objective character to the classes of simultaneous percepts, or the corresponding classes of simultaneous qualities, which we have just described. Clearly, each of us creates these classes in his own way. We shall see later how we may use our basic time intuitions to construct an objective “common time”, which all observers can use, in place of the subjective and idiosyncratic “time sense” which we have been describing.

We turn now to the other aspect of our time perception; that embodied in the distinction between past and future. This aspect imposes a different kind of relation on the set of possible percepts than that embodied in the sense of simultaneity; it is compatible with it, but apparently not directly inferable from it.

To describe this relation, which essentially expresses the idea of predecessor and successor, it is simplest to use a mathematical notation. Let t denote a particular instant; as we have seen, t is itself a percept belonging to the class $C(t)$ of all possible simultaneous percepts, and which serves to *label* the class. Let t' be another instant, $C(t')$ that class of simultaneous instants labelled by it. If we perceive that t' is later than t , then for each percept $p(t)$ in $C(t)$ there is a uniquely determined percept $p(t')$ in $C(t')$ such that $p(t)$ and $p(t')$ are related in the same way that t and t' are related. We alternatively say that $p(t)$ is a *predecessor* of $p(t')$, or $p(t')$ is a *successor* of $p(t)$. Thus, the relations of precedence in time serve to relate percepts belonging to *different simultaneity classes*. We must apparently posit separately that these precedence relations are such that the set of instants (i.e. the set of labels for the simultaneity classes) thereby acquires the mathematical structure of a totally ordered set; in fact, we shall see later that the natural mathematical candidates to represent such a set of instants are the integers or all real numbers.

If we take the now familiar step of imputing relations between percepts to corresponding relations in the external world, the above considerations suffice to characterize time in that world. However, simultaneity in the external world relates qualities or observables (or more precisely, the values of observables) and the set of instants serves to label classes of simultaneously manifested qualities; likewise, the relation of precedence serves to relate the qualities belonging to different simultaneity classes, again in such a way that the set of instants becomes totally ordered. We once again emphasize that these two relations reflect apparently different aspects of our sense of time, and it is important not to confuse them.

Let us now return to our discussion of natural systems, following this extensive detour into the character of observables and the nature of time. We recall that we had defined a natural system to be a collection of observables satisfying certain relations. We can now be somewhat more specific about the nature of the relations, or linkages, which may obtain between the observables of a natural system. To do this, we are going to utilize in an essential way the structures we have discussed in connection with time. Namely, we have described the sets $C(t)$ of simultaneous percepts, and the way these sets are related through the successor-predecessor relation. If we are given a natural system consisting of a set of related observables or qualities, we can then naturally consider:

- (a) How these particular observables (or better, their specific values) are related as elements of a simultaneity class;
- (b) How these observables are related through the successor-predecessor relation.

The first of these will specify how the values assumed by particular observables in the system at an instant of time are related to the values assumed by other observables. The second will specify how the values assumed by particular observables at a given instant are related to the values assumed by these or other observables at other instants.

In an intuitive sense, we would not say that a set of observables actually constitutes a system unless both these kinds of linkages can be specified, at least in principle. A direct expression of these linkages would be equivalent to specifying

the *system laws*. For it is the function of system laws in science to tell us precisely how the values of the system observables at an instant of time are related, and how the values of these observables at a given time are related to their values at other times. As we shall see, the formal vehicles for expressing these laws are what we shall call *equations of state*. However, we shall argue that the precise formulation of such relations belongs not directly to the theory of natural systems, but depends in an essential way on the manner in which observables and their values are *represented* or *encoded* into corresponding formal systems. We will turn to such questions in the subsequent chapters of this section. For the present, it suffices to note that the existence of definite linkage relations of the kind we have described represent one embodiment of the notion of *causality*. Indeed, our initial insistence that definite linkage relations exist between the observables constituting a natural system restricts us from the outset to situations in which such a notion of causality is expressed. We shall later on see how this requirement can be loosened.

We wish to point out two other kinds of linkage relations which are implicit in the discussion of observables and measurement which was sketched above. However, whereas thus-far we have considered only linkages between the observables comprising a single system, the linkages we describe now relate observables belonging to *different* systems.

The first of these is implicit in the very concept of a meter. As we noted above, the essential function of a meter is to provide labels for the specific values of some observable, through the different changes arising in the meter when placed in interaction with a system containing the observable. This means precisely that the interaction between such a system and the meter establishes a relation, or linkage, between the observable being measured and some quality of the meter. Indeed, we shall see later that dynamical interactions provide the vehicles whereby linkages are established and lost in general; the case of the meter is a fundamental prototype for the modification of linkage relations. Thus in particular we can see how *the linkage relations themselves may be functions of time*. We note, for later reference, that the temporal variation of linkage relations arising from interactions will provide us with the vehicle for discussing phenomena of *emergence* and emergent novelty.

A final kind of linkage between observables belonging to different systems arises when, speaking roughly, two different natural systems comprising two different sets of observables, satisfy the same linkage relations. This shall provide us with the conceptual basis for the discussion of *system analogy*, and it is exemplified for example in the relation between a scale model and a prototype. We shall take up this kind of linkage further in our discussion of similarity.

Now let us pause for a moment and see where our discussion has led us so far. We have described the way percepts in us are associated with qualities in an external world. We have argued that such qualities, or observables, provide the vehicles for interaction between natural systems in that world. Such interactions can be exploited, through experiment, to make new qualities, initially imperceptible to us, visible. Moreover, such interactions provide us with a way of labelling the qualities of a system with changes in other systems; this is the essence of the concept of a meter. The importance of this concept lies not only in the fact that

it allows us to *define* observables, but actually allows us to *measure* them, in the manner we have described. The exploitation of these interactive capabilities provide us with *data* about the external world; the acquisition of such data is the basic task of experimental science.

We also described certain kinds of relations between observables, which we called *linkages*. We saw how the establishment of such relations involved a constructive or creative aspect of the mind, expressed in the organization of percepts. The concept of time plays a crucial role in this organization of percepts; first in establishing a relation of simultaneity, and then in establishing a relation of succession or precedence. These relations are then imputed to the external world, just as qualities or observables are imputed to percepts.

Much beyond this we cannot go without introducing a fundamental new idea. The basic reason lies in the fact that experiment (i.e. the acquisition of data) can only tell us about observables and their values; it cannot in principle tell us anything about the manner in which these observables are linked or related. This is because relations do not directly generate percepts; they do not directly move meters. As we have seen, such relations are posited or *imputed* to the external world, and this imputation involves an essentially creative act of the mind; one which belongs to *theoretical* science. But how do we know whether such imputations are correct? Observation and experiment cannot directly answer such a question; all that can be observed, in principle, are observables and their specific values.

This is why a new element is needed before we can proceed further. Put succinctly, this new element must enable us to make *predictions* about observables on the basis of *posited* linkages between them. There is absolutely no way to do this within the confines of natural systems alone. What we require, then, is a new universe, in which we can *create* systems. But in this universe, the systems we create will have only the properties we decide to endow them with; they will comprise only those observables, and satisfy only those linkages, which we assign to them. But the crucial ingredient of this universe must be a mechanism for making inferences, or predictions. This capacity will play the same role in our new universe that causality plays in the external world. But whereas in the external world we are confronted with the exigencies of time, and are forced to wait upon its effects, in our new universe we are free of time; the relation of inference to premise takes the place of the temporal relation between predecessor and successor.

We have, of course, created such a universe. In its broadest terms, that universe is mathematics. The objects in that universe are *formal systems*. Hence, before we can go further in the universe of natural systems, we must explore the one of formal systems. We shall begin this exploration in the next chapter.

Looking ahead a bit further still, the next obvious task is to relate the worlds of natural systems with that of the formal ones; to exploit the capability of drawing inferences in the latter to make predictions about the former. This will be the subject of the final section in this chapter.

References and Notes

1. The epistemological considerations developed herein are idiosyncratic, but I think not arbitrary. I am sure that none of the individual elements of this epistemology is new, but I think that their juxtaposition is new and (judging by what it implies) significant. I see no purpose in contrasting my approach to others; this would at best be tangential to my primary purpose. For a fuller treatment of how the epistemological ideas developed herein arose out of definite scientific investigations, the reader is referred to

Rosen, R., *Fundamentals of Measurement and the Theory of Natural Systems*. Elsevier, New York (1978).

2. For the purpose of the present discussion, it is sufficient to refer the reader to a general philosophical treatment of the problem of time, such as:
Whitrow, G., *The Natural Philosophy of Time*. Nelson, London (1980).
A more detailed discussion of how time is represented in different kinds of encodings will be found in the various Chap. 3 below, and the references thereto.

2.2 The Concept of a Formal System

The present chapter is devoted to developing the concept of a formal system. This development forces us to leave the external world of natural systems, with its observables which generate percepts and move meters. We must enter another world, populated entirely by mental constructs of particular kinds; this is the world of mathematics in the broadest sense.

This is not to say that the external world and the world of mathematics are entirely unrelated, or that our perception of natural systems plays no role in the construction of formal ones. As we shall see, there are many deep relations between the two worlds. Indeed, the main theme of the present book is the modeling relation, which rests precisely on linking the properties of natural systems with formal ones in a particular way. We wish merely to note here that mathematical objects, however they may initially be created or generated, possess an existence of their own, and that this existence is different in character from that of a natural one.

Furthermore, as we shall see, the world of mathematics really consists of two quite separate worlds, which are related to each other in very much the same way as the world of percepts is related to the external world. One of these mathematical worlds is populated by the familiar objects of mathematics: sets, groups, topological spaces, dynamical systems and the like; together with the mappings between them and the relations which they satisfy. The other world is populated with symbols and arrays of symbols. The fundamental relation between these two worlds is established by utilizing the symbols of the latter as *names* or *labels* for the entities in the former, and for regarding arrays of symbols in the latter as expressing *propositions* about the former. In this sense, the world of symbols becomes analogous to our previous world of percepts; while the world of mathematical objects becomes analogous to

an external world which elicits percepts. The primary difference between the two situations (and it is an essential one) is that the mathematical worlds are entirely constructed by the creative faculty of the mind.

Perhaps the best way to illustrate the ideas just enunciated is to begin to construct the two mathematical worlds in parallel. We will do this by establishing a few of the rudiments of what is commonly called the *theory of sets*. This has been universally adopted as the essential foundation, in terms of which all the rest of mathematics can be built. As our purpose here is illustrative rather than expository, we shall suppose that the reader already has some familiarity with the concepts involved.¹

Let us begin to populate our first world, which we shall denote by u . We will initially admit two kinds of entity into u , which we will call *elements* and *sets* respectively. We initially assume no relation of any kind between the entities called elements and those called sets.

Whenever we wish to speak of a particular *element*, we must characterize it by giving it a name; i.e. by labelling it with some symbol, say α , which serves to identify it, and distinguish it from others. This kind of labelling is very much like an act of perception; perceiving the particular element so labelled and its distinctness from others, which we labelled differently. Likewise, when we wish to speak of a particular *set* in u , we must provide it too with a name, or labelling symbol, say A .

Of course we must carefully distinguish the entities in and the symbols which we use to name these entities. The symbols themselves do not belong to u ; only the entities themselves have this property. Their names thus must belong to another world, established in parallel with u ; this world of names we shall call p .

As it stands, the world u is completely static and uninteresting. In order to proceed further, we must introduce some *relations* into it. We shall do this by admitting into our world a new kind of entity, neither a set nor an element. These new entities must also be labelled or named; let us agree to call such an entity (α, A) , where α is the name of an element of u and A is the name of a set. Intuitively, we are going to interpret the presence of such a pair (α, A) in u by saying that α is an *element of* A .

At this stage, we can note that the *character* of u now depends entirely on which pairs are admitted into it. Thus already at this early stage we confront the necessity of *choice*; and indeed the choice we make here will determine everything which follows. Retrospectively, we can see that populating our universe u with different pairs will in effect create different (indeed, radically different) set theories, and hence different mathematics. We will return to this point several times as we proceed.

In accord with long tradition, we will agree to denote the presence of (α, A) in u by the introduction of a new symbol into p . This symbol serves to connect the names of the element and set constituting the pair; specifically, we will write

$$\alpha \varepsilon A \tag{2.1}$$

The symbol ε thus names a *relation* obtaining between sets and elements in u , and the entire expression (2.1) is interpreted as naming the manifestation of the relation between a particular pair in u .

The expression (2.1) is our first example of a *proposition*. Such a proposition does not belong to u , but rather is the name of an *assertion* about u . Since this is a crucial point, we will spend a moment discussing it.

Once we have introduced the symbol ε into p , there is nothing to prevent us from taking the name of an arbitrary element of u , say β , and the name of an arbitrary set, say C , and producing the expression or proposition

$$\beta \varepsilon C \quad (2.2)$$

This too is a *proposition*, and in effect it *asserts* something about u ; namely, that (β, C) is a pair in u . But merely because the proposition (2.2) can be *expressed* in p , it does not follow that the relation it names actually *obtains* between β and C . Thus there are in general going to be *many more propositions in p than there are entities in u* .

If the particular proposition (2.2) happens to name a relation that actually obtains in u ; (i.e. is such that (β, C) actually is in u) then we shall say that the proposition (2.2) is *true*. Otherwise, we shall say that (2.2) is *false*. Thus, the truth or falsity of such a proposition depends entirely on the nature of u ; it cannot be determined from within the world p at all. In a sense, the characterization of (2.2) as a true proposition is an *empirical* question; we must look at u to determine if the particular pair (β, C) is in it. If not, then (2.2) is the name of a situation which does not exist in u ; a label which labels nothing. But such a label cannot be excluded from p on any syntactical grounds; once the symbol ε is allowed, so are *all* propositions of the form (2.2). This is why, even at this most elementary stage, the world p of propositions has already grown larger than the world u which originally generated it. If we restrict ourselves only to the *true* propositions, then the two worlds remain in parallel. But as we emphasize, there is no way to do this within p . Indeed, the propositions (2.2) which now populate p express all the choices which *could have been* made in constructing u ; they cannot tell us which ones actually *were* made.

It should also be stressed before proceeding that truth and falsity are attributes of *propositions*. It is therefore meaningless to speak of these attributes in connection with u ; they pertain entirely to p alone. Basically, a false proposition is one which looks like a name, but which is in fact not a name (of anything in u).

Let us now proceed a bit further. The next step is, in effect, to change the rules of the game in u . What we shall do is to *identify* a set A in u with the totality of elements which belong to it. In other words, the entity originally named A completely disappears from view, and the name itself is reassigned to a particular totality of elements. This procedure can be represented in p with the aid of some new symbols introduced for the purpose:

$$A \equiv \{\alpha : \alpha \varepsilon A\}. \quad (2.3)$$

In words, A now names a particular family of elements of u ; namely those for which the proposition $\alpha \varepsilon A$ (in the original sense) is true. The symbol " $\equiv$ " thus expresses a *synonymy* in p , and will be exclusively used for this purpose. The notation in

(2.3) also serves to assign names to sets of elements α , characterized through the expression of a particular relation or property which they satisfy in u .

This identification makes it clear why the initial population of u which pairs (α, A) is so important. For at this stage, with the identification of A with the totality of its elements, the character of all subsequent developments in mathematics is determined. Indeed, the “best” way to do this is far from a trivial matter, and in fact is not really known. We do possess a certain amount of retrospective experience bearing on this question, but it is mainly of a negative character. For instance, if we populate u too liberally with such pairs, then some of our later set-theoretic constructions will turn out to have paradoxical properties; i.e. certain objects will need to be simultaneously included in and excluded from our universe. The well-known paradoxes of early set theory can all be regarded as arising in this manner.

Modulo the identification (2.3), we can now introduce all of the familiar relations and operations of elementary set theory. At each stage, we need to introduce corresponding *names* of these relations and operations into p . As we do so, we find exactly the same situation as we encountered with the introduction of ε ; i.e. we are able thereby to form propositions in p which name no counterpart in u , and hence are to be considered *false*. In other words, as we build structure into u , we introduce a corresponding *syntax* into p , which generates a *calculus of propositions*.

To illustrate these last remarks, and incidentally to introduce some important terminology, let us consider the relation in u which expresses that a set A is a *subset* of a set B . By virtue of the identification (2.3) which specifies the sets of u in terms of their constituent elements, such a relation will devolve upon corresponding relations pertaining to the *elements* of A and B . The subset relation is said to obtain if u is such that, whenever a pair (α, A) is in u , the pair (α, B) is also; i.e. every element of A is also an element of B . Now to express this in p we need to relate the *propositions* $\alpha \varepsilon A$ and $\alpha \varepsilon B$. There is no way to do this with the machinery we have introduced into p so far; we need another *symbol* to express such a relation between propositions, just as we needed the new symbol ε to express a relation between names. This new symbol will be denoted by $\Rightarrow$, and will relate propositions in p , as in the expression

$$(\alpha \varepsilon A) \Rightarrow (\alpha \varepsilon B). \quad (2.4)$$

This expression is now a *new proposition*, which means: if $\alpha \varepsilon A$ is true, then $\alpha \varepsilon B$ is true. The subset relation can now be expressed in p , because if it obtains in u between a pair of sets A, B , the proposition (2.4) is true; conversely, if the proposition (2.4) is true for some A and B , we say that A is a subset of B . We still need a name for the subset relation itself; we thus write

$$A \subseteq B \equiv (\alpha \varepsilon A) \Rightarrow (\alpha \varepsilon B). \quad (2.5)$$

This name $A \subseteq B$ expresses now a relation between sets, instead of the oblique form (2.4) which relates propositions about elements. The relation named by (2.5) is called *inclusion*.

The implication relation $\Rightarrow$ introduced above is going to be the central feature of our discussion of formal systems. As we shall see, it will be our primary vehicle for expressing in $\mathbf{p}$ all of the properties pertaining in $\mathbf{u}$. Its basic characteristic is that *it never takes us out of the set of true propositions*; i.e. if p is any true proposition, and $p \Rightarrow q$ holds for some other proposition q , then q is also a true proposition. Stated another way, this implication relation between propositions enables us to express *linkages* between properties in $\mathbf{u}$; it will play the same role in the universe of formal systems that causality plays in the universe of natural ones.

Let us note in passing that the inclusion relation just introduced allows us to look at *synonymy* in a new light. For instance, if it should be the case that *both* of the inclusion relations

$$A \subseteq B, \quad B \subseteq A$$

hold in $\mathbf{u}$, then both of the sets in question consist of exactly the same elements. By virtue of (2.3), then, these sets are not different in $\mathbf{u}$, even though they bear two different *names*. Hence these names are synonyms for the *same* set in $\mathbf{u}$. We shall soon see that much of mathematics consists of establishing that different names, or more generally, different propositions in $\mathbf{p}$, are actually synonyms; in the sense that they name the same object or relation in $\mathbf{u}$. The vehicle for establishing such synonymy will consist of chains of *implications* relating propositions in $\mathbf{p}$; this is the essence of proof. We will take these matters up in more detail subsequently.

So far, we have considered how a set-theoretic relation (inclusion) is defined, and the manner in which it is accommodated in the world $\mathbf{p}$ of propositions, through the introduction of a corresponding relation between propositions. Let us now turn to the question of set-theoretic *operations*, using the specific example of the *union* of two sets.

In a sense, we can encompass a concept like the union of two sets under the rubric of set-theoretic relations, which we have already introduced. Intuitively, an operation like union defines a new set in a canonical fashion out of a pair of given sets. If C denotes this new set, we can equivalently define it in terms of the unique relation it bears to the sets from which it is built. That is, we can proceed by adding to our universe $\mathbf{u}$ a family of triples (A, B, C) satisfying the appropriate properties. For instance, we would informally require the set C in such a triple to be a subset of any set D of which A and B are both subsets.

However, it is more convenient (and more traditional) to proceed by specifying the new set C in terms of the specific elements of which it is composed, via (2.3). Intuitively, we wish an element α to belong to C in case α belongs to A , or α belongs to B , or belongs to both. To define C in this way requires us to articulate this idea as a *proposition* in $\mathbf{p}$. We have no proposition at our disposal for this purpose in $\mathbf{p}$ so far, but we can *build* one if we allow ourselves a new operation on *propositions*. Of course, this is the *conjunction operation*, which from any two propositions p, q in $\mathbf{p}$ allows us to produce a new proposition denoted $p \vee q$. In terms of this operation on propositions, we can now specify the set we seek:

$$C \equiv \{\alpha : (\alpha \in A) \vee (\alpha \in B)\}. \quad (2.6)$$

As before, we must give a specific name to the operation in u which constructs C for us out of A and B ; we do this by *defining*

$$C \equiv A \cup B.$$

In words, the proposition $(\alpha \in A) \vee (\alpha \in B)$ is true precisely when at least one of its constituent propositions $(\alpha \in A)$ or $(\alpha \in B)$ is true. The expression $A \cup B$ is now to be regarded as the *name* of the set in u defined by (2.6).

Let us pause now to note a fundamental distinction between the world u of mathematical objects and the world p of names and propositions pertaining to these objects. In a certain important sense, a set like $A \cup B$ already exists in u from the outset; we need only find a way of giving it an appropriate name, which exhibits explicitly its relation to other sets in u . But the corresponding propositions in p need to be *created*, or *generated*; a proposition like $p \vee q$ does not exist in p before we introduce the specific operation “ $\vee$ ”, and directly apply it to particular propositions in p . This is a crucial point, and marks the difference between those mathematicians who believe that the mathematical universe pre-exists, possessing properties which must be *discovered*, and those who believe that the mathematical universe must be *constructed* or *created*, and that this construction is an entirely arbitrary process. Ultimately, this difference resides in whether one conceives of the natural habitat for mathematics as u or p . We need not regard these alternatives as being mutually exclusive, though perhaps most mathematicians have regarded them as such; this fundamental distinction, resting as it does primarily on aesthetic grounds, is responsible for much of the acrimony and bitterness surrounding the foundations of mathematics. But in any case we can discern once again the similarity of the relation between u and p to the relation between qualities and percepts which we discussed in the preceding chapter. We shall return to this matter when we consider the question of axiomatics below.

Let us return now to a consideration of the set-theoretic operation of forming the union. Once it has been defined or named, we can discover that it possesses certain properties. for instance that it is associative. Associativity consists in the fact that certain *names* of sets are in fact synonyms; specifically, that the two names

$$(A \cup B) \cup C \quad \text{and} \quad A \cup (B \cup C)$$

both label the same set in u . The *proof* of synonymy consists of establishing certain *implications* in p ; namely, that

$$\alpha \in (A \cup B) \cup C \Rightarrow \alpha \in A \cup (B \cup C)$$

and

$$\alpha \in A \cup (B \cup C) \Rightarrow \alpha \in (A \cup B) \cup C$$

Thus, the sets named by $(A \cup B) \cup C$ and $A \cup (B \cup C)$ are each contained in the other; hence are identical. This identity is a *theorem*; an assertion which is true in u if the initial propositions

$$\alpha \varepsilon (A \cup B) \cup C$$

and

$$\alpha \varepsilon A \cup (B \cup C)$$

are true in $\mathbf{u}$. We leave it to the reader to fill in the details. The reader should also convince himself that the associativity of the set-theoretic operation $\cup$ *forces* the associativity of the operation $\vee$ on propositions in $\mathbf{p}$. The nature of this forcing is most illuminating in considering the relation between $\mathbf{u}$ and $\mathbf{p}$.

We can proceed in an analogous manner to define the other basic set-theoretic operations of intersection (denoted by $\cap$) and complementation (denoted by $\overline{}$); these in their turn force us to introduce new operations on propositions in $\mathbf{p}$ (namely disjunction, denoted by $\wedge$, and negation, denoted variously by $\neg$ or $\sim$). We will omit the details, which the reader can readily supply. These simple ideas already admit a rich calculus (or better, an algebra, which is usually associated with the name of the English mathematician Boole), in which one can prove many theorems of the character we have already described. For instance, we can establish the familiar de Morgan Laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

in $\mathbf{u}$, and their corresponding analogs in $\mathbf{p}$. The reader should establish the appropriate chains of implication in detail, and interpret their meaning in the light of the preceding discussion.

Before proceeding further, let us add a word about the intersection operation. Explicitly, in analogy with (2.6), the intersection of two sets A, B is defined as the set $A \cap B$ of elements which belong both to A and to B ; i.e. for which the propositions $(\alpha \varepsilon A)$ and $(\alpha \varepsilon B)$ are true. But clearly there need be no elements α in $\mathbf{u}$ for which $(\alpha \varepsilon A)$ and $(\alpha \varepsilon B)$ are both true. In this case we say that A and B are disjoint. By convention, and so that the intersection operation shall be universally defined for all pairs of sets A, B , we introduce a new concept; that of the *empty set*. The empty set intuitively possesses no elements, and is usually denoted by ϕ . Thus, if A and B are disjoint, we write

$$A \cap B = \phi.$$

The empty set is considered to be a perfectly respectable member of $\mathbf{u}$ (indeed, it can be regarded as generating all of $\mathbf{u}$, in a certain ironic sense)², and is likewise considered to be a subset of every set; i.e. $\phi \subseteq A$ for every set A .

We now possess all the machinery for defining in $\mathbf{u}$ all of the familiar structures of mathematics. These can all be regarded as arising from particular kinds of set-theoretic operations. The most important of these set-theoretic operations is the *cartesian product* operation; given any pair A, B of sets we can define a new set $A \times B$ in the following way:

$$A \times B = \{(\alpha, \beta) : \alpha \varepsilon A, \beta \varepsilon B\}.$$

An arbitrary subset $R \subseteq A \times B$ defines a *binary relation* between the elements of A and the elements of B . If R is such a relation, it is traditional to abbreviate the proposition $(\alpha, \beta) \in R$ by writing $\alpha R \beta$.

Binary relations formally lead us to some of the basic structures of mathematics, with which we shall be much concerned throughout the sequel. For instance, suppose that $R \subseteq A \times B$ is a binary relation. Suppose further that R possesses the following property

$$\alpha R \beta \quad \text{and} \quad \alpha R \gamma \Rightarrow \beta = \gamma.$$

A relation of this kind can thus be regarded as establishing a *mapping*, or *correspondence*, between the elements of A and the elements of B . Indeed, given a relation of this kind, we can express the proposition $\alpha R \beta$ by writing $\beta = f(\alpha)$, where f is the symbol expressing the mapping or correspondence. As we shall see, these mappings (also denoted by the symbol $f : A \rightarrow B$) introduce a dynamic element into $\mathcal{U}$; they are the vehicles through which different structures in $\mathcal{U}$ may be compared.

One of the basic properties of binary relations, which we shall exploit heavily in subsequent chapters, resides in the fact that they can be *composed*. Suppose that $R_1 \subseteq A \times B$, and that $R_2 \subseteq B \times C$, are two binary relations. We are going to define a new relation $R \subseteq A \times C$. To define it, we need to specify what elements of $A \times C$ are in R . We shall say that a pair (α, γ) is in R if and only if there is a β in B such that

$$(\alpha, \beta) \in R_1 \quad \text{and} \quad (\beta, \gamma) \in R_2.$$

We shall define this relation R to be the *composite* of R_1 and R_2 , and write $R = R_2 R_1$.

In case R_1 and R_2 are *mappings*, it is easy to see that this definition produces a relation which is itself a mapping. This generates the familiar idea of *composition of mappings*. Specifically, if $f : A \rightarrow B$, and $g : B \rightarrow C$ are mappings, then there is a uniquely determined mapping $h : A \rightarrow C$ defined in accordance with the above construction, and we *define* $h = gf$.

These considerations lead directly to another set-theoretic construction, which is as important as the cartesian product. Namely, given two sets A, B in $\mathcal{U}$, we can consider the set of *all* mappings from A to B . This is a new set, which we can denote by $H(A, B)$. This set $H(A, B)$ thus turns out to be intimately related to the cartesian product operation appropriately generalized, but we shall not develop that relation here.

If $A = B$, a binary relation $R \subseteq A \times A$ establishes a relation among the elements of A . By far the most important of these for our purposes are the *equivalence relations* on A . An equivalence relation $R \subseteq A \times A$ possesses the following properties:

- (a) $\alpha R \alpha$ for every α in A (reflexivity)
- (b) $(\alpha R \beta) \Rightarrow (\beta R \alpha)$ (symmetry)
- (c) $(\alpha R \beta) \text{ and } (\beta R \gamma) \Rightarrow (\alpha R \gamma)$ (transitivity).

Such equivalence relations will be of vital importance to us throughout our subsequent development. Intuitively, an equivalence relation is a generalization of equality (which is itself obviously an equivalence relation) which arises when we discard some distinguishing property which enabled us to recognize that particular elements were unequal; when the property is discarded, such elements now appear indistinguishable.

This central aspect of equivalence relations is manifested in their fundamental property, which we can formulate as follows: if R is an equivalence relation on a set A , and if $\alpha \in A$ is any element of A , then we can form the subset of A defined as follows:

$$[\alpha] = \{\alpha' : \alpha R \alpha'\}.$$

If β is another element of A , we can likewise form the subset

$$[\beta] = \{\beta' : \beta R \beta'\}.$$

It is then a *theorem* that either

$$[\alpha] = [\beta].$$

or

$$[\alpha] \cap [\beta] = \emptyset$$

The subset $[\alpha]$ defined above is called the *equivalence class* of α (under the relation R); the theorem just enunciated asserts that two such classes are either disjoint or identical. Hence the relation R can be regarded as partitioning the set A into a family of disjoint subsets, whose union is clearly all of A . The *set* of equivalence classes of A under an equivalence relation R is itself a set in u , denoted A/R , and often called the *quotient set* of A modulo R . It is easy to see that the equivalence relation R on A becomes simple equality on A/R .

There is a simple but profound relation between mappings and equivalence relations. Let $f : A \rightarrow B$ be a mapping. For each $\alpha \in A$, let us define

$$[\alpha] = \{\alpha' : f(\alpha') = f(\alpha)\}.$$

These sets $[\alpha]$ are all subsets of A , and it is easy to see that they are either disjoint or identical. They can thus be regarded as the equivalence classes of a corresponding equivalence relation R_f on A . Intuitively, this relation R_f specifies the extent to which the elements of A can be distinguished, or resolved, by f . Thus, if we regard the mapping f as associating with an element $\alpha \in A$ a name or label $f(\alpha)$ in B , the equivalence relation R_f specifies those distinct elements of A which are assigned the same name by f . Moreover, it is a fact that *any* equivalence relation may be regarded as arising in this fashion. In this sense, equivalence is a kind of inverse of synonymy; equivalence involves two *things* assigned the same *name*; synonymy involves two *names* assigned to the same *thing*.

After all this extensive preamble, we are now in a position to discuss some familiar formal system as mathematical objects. In the world u , such a formal system

can be regarded as a set together with some other structure; a set and a mapping; a set and a relation; a set and a family of subsets. As a simple example, we may consider a typical algebraic structure, such as a *group*.

A group may be regarded as a pair (G, τ) , where G is a set in $\mathcal{U}$, and $\tau : G \times G \rightarrow G$ is a definite mapping in $H(G \times G, G)$. This mapping τ is not arbitrary, but is supposed to satisfy the following conditions:

- (a) $\tau(g_1, \tau(g_2, g_3)) = \tau(\tau(g_1, g_2), g_3)$ (associativity)
- (b) $\tau(g, e) = \tau(e, g) = g$ for some distinguished element $e \in G$; the *unit element*
- (c) for every $g \in G$ there is a unique element $g^{-1} \in G$ such that $\tau(g, g^{-1}) = \tau(g^{-1}, g) = e$.

It is more traditional to write $\tau(g_1, g_2)$ as $g_1 \cdot g_2$, and consider τ as defining a *binary operation* on G . The properties (a) to (c) above are usually called the “group axioms”.

We can establish the framework for studying groups in $\mathcal{P}$ by mimicking the machinery which was established for sets, appropriately modified or constrained so as to respect the additional structure which distinguishes a group from a set. This machinery can be succinctly characterized under the headings (a) substructures, (b) quotient structures, (c) structure-preserving mappings. These are the *leitmotifs* which establish the framework in terms of which all kinds of mathematical structures are studied. For the particular case we are now considering, this *leitmotif* becomes: (a) sub-groups; (b) quotient groups; (c) group homomorphisms. Let us briefly discuss each of these in turn.

(a) Subgroups

As we have seen, a mathematical structure like a group is a set together with something else; in this case, a binary operation satisfying the group axioms. Thus a *subgroup* of a group G will intuitively first have to be a *subset* $H \subseteq G$ of the set of elements belonging to G . Further, this subset must bear a special relation to the binary operation, which is the other element of structure. It is clear what this relation should be: H should itself be a group under the same operation which makes G a group; and further should have the same unit element. A subset H with these properties is said to define a *subgroup* of G .

(b) Quotient Groups

To define quotient groups, we must first consider equivalence relations on G , considered as a set. But by virtue of the additional structure (the binary operation) we will only want to consider certain special equivalence relations, which in some sense are *compatible* with the additional structure. It is relatively clear how this compatibility is to be defined. Suppose that R is some equivalence relation on G . Let g_1, g_2 be elements of G . Suppose that $g_1 R g_1'$, and that $g_2 R g_2'$. In G we can form the *products* $g_1 \cdot g_2$ and $g_1' \cdot g_2'$. It is natural to require that *these two products are also equivalent under R* ; i.e. $(g_1 \cdot g_2) R (g_1' \cdot g_2')$. If we do this, then the quotient set G/R can itself be turned into a group, by *defining* the product of two equivalence classes $[g_1], [g_2]$ in G/R as

$$[g_1] \cdot [g_2] = [g_1 \cdot g_2].$$

That is: the product of equivalence classes in G/R is the equivalence class of the product in G . It is trivial to verify that this operation in G/R turns it into a new group; this new group is then called a *quotient group* of G .

(c) *Structure-Preserving Mappings*

Suppose that G and H are groups. Just as we did not wish to consider all equivalence relations on a group, but only those which were compatible with the additional structure, so we do not wish to consider all mappings $f: G \rightarrow H$, but only those which are also compatible with the structures on G and H . Just as before, this compatibility will consist in preserving the properties of the binary operations. Written succinctly, such compatibility means at least that

$$f(g_1 \cdot g_2) = f(g_1) \cdot f(g_2)$$

for any two elements g_1, g_2 in G . It should be noted that the binary operation appearing on the left side of this condition is the operation in G ; the operation on the right side is the operation in H . We will also require explicitly that f *preserve units*; i.e. if e_G, e_H are the unit elements of G and H respectively, then $f(e_G) = e_H$. From this it immediately follows that such a mapping f also *preserves inverses*: $f(g^{-1}) = (f(g))^{-1}$.

A mapping $f: G \rightarrow H$ which satisfies these compatibility conditions is called a *group homomorphism*. Such a homomorphism is an instrument through which the group structures on G and H can be *compared*.

On the basis of these simple ideas, we can construct a world u_G patterned after our original world u , except that instead of consisting of sets it consists of *groups*; instead of allowing arbitrary mappings, we allow only group homomorphisms; instead of allowing arbitrary equivalence relations, we allow only compatible equivalence relations. Such a world is now commonly called a *category*; what we have done in specifying u_G is to construct the *category of groups*.

The above line of argument is generic in mathematics; it can be applied to any kind of mathematical structure, and leads to the construction of a corresponding *category* of such structures. Thus, we can consider categories of rings, fields, algebras, topological spaces, etc. Our original world u can itself be considered as a category; the *category of sets*. One of our basic tasks, which we shall consider in the next chapter, is to see how different categories may themselves be compared; i.e. how different classes of mathematical structures may be related to one another. In abstract terms, this kind of study comprises the *theory of categories*³, and it will play a central conceptual role in all of our subsequent discussion.

In our discussion of formal systems so far, we have been concentrating on the world u . Let us now return to the world p of names and propositions, and see how such formal systems look in that world. As we shall see, the situation in that world is quite different. This is roughly because, as we pointed out before, we could essentially *discover* groups in u ; but they must be *created* in p .

Let us then begin to create a group; the procedure we use will be typical of the constructive or generative procedures which characterize doing mathematics in $\mathfrak{p}$. We are going to initiate our construction by taking an array of symbols $g, h, \dots$ from $\mathfrak{p}$, now considered in the abstract; i.e. without regard for anything external to $\mathfrak{p}$ which these symbols might name; we establish by fiat that these symbols shall be in our group. Now we must introduce a special rule which allows us to combine our symbols to obtain new ones analogous to the operations $\vee, \wedge$ etc., which allow us to combine propositions. We will call such a rule a *production rule*. The rule we use is a very simple one, called concatenation; if g, h are two of our symbols, then their concatenation gh will also be a symbol in our group-to-be. More generally, if we are given any finite number $g_1, \dots, g_n$ of such symbols, then the concatenation $g_1 \dots g_n$ is a symbol. In this way we build up finite *strings* of the symbols with which we started; such strings are conveniently called *words*. Most generally, if w_1, w_2 are words, then the concatenation $w_1 w_2$ is also decreed to be a word.

Now concatenation is obviously associative: $(w_1 w_2) w_3$ is the same string as $w_1 (w_2 w_3)$. Moreover, if we admit the *empty word* e into our system (where e is a string containing no symbols) then e plays the role of a unit element for concatenation; $w = ew = w$ for any word w . In this way, we have constructed in effect an algebraic system which has many grouplike properties; a binary operation (concatenation) which is associative and possesses a unit element. However, it is clear that there are no inverses in this kind of system, which is called the *free semigroup* generated by the symbols originally chosen (the significance of the word “free” will become clear in a moment).

To force the existence of inverses, we need to add them explicitly. Let us do this by taking another family of symbols $g', h', \dots$, one for each of the generating symbols initially chosen, but such that the two families of symbols are disjoint. Intuitively, we are going to force g' to behave like g^{-1} , h' like h^{-1} , and so on. We do this by adding the new primed generating symbols to the original ones, and once again constructing all finite strings by concatenation. But now on this larger set of strings, we are going to *force* a concept of synonymy. In particular, we are going to decree that the string gg' and the string $g'g$ be synonymous with the empty word e ; and likewise for all of the generating symbols. From this it is easy to see that, if $w = g_1 \dots g_n$ is an arbitrary string, and if we define

$$w' = g_n' \dots g_1',$$

then ww' and $w'w$ must also be synonymous with the empty word.

Intuitively, what this procedure does is to impose an equivalence relation R on our second set of strings; two such strings being called equivalent if one can be obtained from the other by the insertion or removal of substrings of the form ww' or $w'w$. It requires a bit of an argument to establish that the relation so introduced is indeed an equivalence relation, and that it is compatible with concatenation, but the details are not difficult and are left to the reader. Granting that this relation is a compatible equivalence, we can pass then to the quotient set modulo R . This quotient set, whose elements are *equivalence classes* of strings, has now all the properties of a group; it is called the *free group* generated by our original set of symbols.

The significance of the term “free” in these constructions is as follows: there is a sense in which the group we have constructed is the largest group which can be built from the generators we have employed. Stated another way: no *relation* is satisfied by the elements of this group, other than the one we have imposed to guarantee that they do indeed form a group.

If G is such a free group, then we can impose further (compatible) equivalence relations on G , and obtain new groups by forming the corresponding quotient groups G/R . It is not difficult to show that there is a sense in which *every* group can be regarded as a quotient group of a suitable free group, and in that sense, we have succeeded in constructing all groups in $\mathfrak{p}$. However, it is clear that the kind of constructive procedure we had to use to generate groups in $\mathfrak{p}$ makes a group look much different from the simple definitions which could be employed to discover the same groups in $\mathfrak{u}$. The treatment adopted in the above constructive argument is often called *the definition of a group through generators and relations*. To construct any specific group, we must first construct a free group, and then impose on it precisely the relation R which ensures that the quotient modulo R is in fact the desired group. It may be imagined that this is not always easy to do⁴.

We may note in passing that the free groups are in fact in $\mathfrak{u}$, but they are there characterized by the satisfaction of what appears to be a quite different property. For the reader's interest, we might sketch this property. Suppose that G is a group in $\mathfrak{u}$, and that A is simply a set. Let $f : A \rightarrow G$ be a mapping of A into G . Let H be any other group, and let $g : A \rightarrow H$ be any mapping. If in this case we can always find a group homomorphism $\phi : G \rightarrow H$ such that the composite mapping $\phi f : A \rightarrow H$ is the same as the chosen mapping $g : A \rightarrow H$, then G is called the free group generated by A .

In fact, the two definitions are equivalent, in the following sense. If we start with a set A in $\mathfrak{u}$, we can use the names of its elements as the symbols in $\mathfrak{p}$ from which to begin our construction. When we do this, there is a direct *interpretation* of our construction in $\mathfrak{u}$ at every stage, and in particular, the free group we generate names an object in $\mathfrak{u}$. This object can be shown to satisfy the condition just enunciated. In fact, this kind of argument is used to establish the *existence* of free groups. Indeed, by allowing our symbols to possess a definite interpretation in $\mathfrak{u}$ from the outset, we may pass freely back and forth between $\mathfrak{p}$ and $\mathfrak{u}$, exploiting both the constructive aspects of the former and the existential qualities of the latter. Mathematicians do this routinely. But it raises some deep questions, which we must now pause to examine.

These questions devolve once again on the relative roles played by $\mathfrak{u}$ and $\mathfrak{p}$ in the development of the mathematical universe. We have already touched on this above, and indeed, we have seen how differently the same mathematical object (a group) could look in the two realms. In general, we may say that those concerned with mathematics can be divided into two camps, or parties, which we may call the $\mathfrak{u}$ -ists and the $\mathfrak{p}$ -ists. The $\mathfrak{u}$ -ists minimize the role played by $\mathfrak{p}$ in mathematics; their basic position is roughly as follows: $\mathfrak{p}$ if we wish to establish some property of a mathematical system, we need merely look at that object in $\mathfrak{u}$ in

an appropriate way, and *verify* the property. According to this view, mathematics is akin to an experimental science and, as noted before, *u* is treated in the same way an experimental scientist treats the external world.

The *p*-ists have, however, cogent arguments on their side. For one thing, all of the machinery of proof; i.e. for the establishment of chains of inference, is in *p*. Moreover, we can build into *p* an abstract image of any imaginable mathematical object, as a family of propositions generated from an appropriate set of symbols and production rules, whether *u* is available for inspection or not. Hence, they argue, *u* can be entirely dispensed with, for it can be invented entirely within *p*.

The *u*-ists can counter with a very powerful argument. Namely, as long as at least some of the symbols and propositions in *p* are interpretable as names of objects in *u*, the concept of truth or falsity of the propositions in *p* is not only meaningful, it is determined. Within *p* alone, there is no vehicle for assigning truth and falsity. Thus, since as we have seen *p* is inevitably much bigger than *u*, we will never effectively be able to find an abstract image of *u* in *p*. Moreover, the fundamental weapon of *implication*, which as we have seen never takes us out of a set of true propositions, is of no value without a *prior* notion of truth and falsity.

The *p*-ists can counter such an argument to some extent, as follows. We do not care, they may say, about any notion of truth or falsity which is imported into *p* from outside. From within *p* itself, we may by fiat declare a certain set of propositions to be *true*, together with all the propositions which may be built from these according to the syntactical rules in *p*. All we need care about is that this choice be made in such a way that the set of propositions arising in this fashion is *consistent*; in other words, that for no proposition *p* is it the case that both *p* and $\neg p$ (the negation of *p*) belong to the set. And indeed, it may be argued that any such set of consistent propositions is an abstract image of *some* mathematical world *u*, and therefore *is* that world.

The *p*-ists have one more powerful argument, which is intended to rebut the idea that mathematical properties can be determined by inspection or verification in *u*. They point out that all mathematicians seek generality; i.e. to seek to establish the relation between mathematical properties in the widest possible context. Mathematicians seek to establish relations such as, “if *G* is *any* group, then some property *p* is true of *G*”. Or, stated another way, for *all* groups *G*, *P*(*G*) holds. In *p*, a proposition like this is represented through the introduction of the universal quantifier $\forall$, and would be expressed as

$$\forall G \ P(G)$$

Now the universal quantifier is closely related to the operation $\wedge$ of disjunction on propositions. For, if we had some way of enumerating groups, as say $G_1, G_2, G_3, \dots, G_n, \dots$, the above universal assertion could be written as

$$P(G_1) \wedge P(G_2) \wedge \dots \wedge P(G_n) \wedge \dots$$

To *verify* a proposition of this type requires an infinite number of verifications, which is impossible *in principle*. Thus, to the extent that a universal quantifier is meaningful in mathematics at all, it can only be meaningful within a limited constructive sense in a universe of totally abstract propositions.

The same argument may be given in connection with another kind of basic mathematical proposition, which asserts that for some particular property P , there *exists* an object A which satisfies the property. To enunciate such a proposition requires another quantifier, the *existential quantifier* $\exists$; in terms of it, the above assertion would be expressed as

$$\exists A \ P(A).$$

This existential quantifier is closely related to the conjunction operation; again, if all objects A could be enumerated in some fashion as $A_1, A_2, \dots, A_n, \dots$, the existential assertion would amount to the proposition

$$P(A_1) \vee P(A_2) \vee \dots \vee P(A_n) \vee \dots$$

Here again, to establish existence by verification requires an infinite number of separate acts of verification, which is impossible in principle. Hence existence also can only be given a very circumscribed formal meaning, which involves the actual *construction* of the object whose existence is asserted.

Once again the *u*-ists can rebut in a very powerful way. Quite apart from the utter aridity of the manipulation of symbols devoid of interpretation, which is the nature of the universe contemplated by the *p*-ists, they can point out that the validity of the entire *p*-ists program would depend on *effectively* being able to determine the consistency of families of propositions arbitrarily decreed in advance to be true. It is here that the name of Gödel⁵ first appears. To appreciate the full force of the argument we are now developing we must digress for a moment to consider the concept of *axiomatization*.

Axiomatization is an old idea, dating back to the codification of geometry by Euclid. The essence of the Euclidean development was to organize a vast amount of geometric knowledge by showing that each item could be established on the basis of a small number of specific geometric postulates, and a similarly small number of rules of inference or axioms. On this basis, it was believed for a long time that every true proposition in geometry could be established as a theorem; i.e. as a chain of implications ultimately going back to the postulates. Moreover, if some geometric assertion was not true, then its negation was a theorem.

Mathematicians came to believe that every branch of mathematics was of this character. Namely, they believed that the totality of true propositions about every branch of mathematics (say number theory) could all be established as theorems, on the basis of a small number of assumed postulates and the familiar rules of inference. It was this belief that in fact underlies the expectation that all of mathematics can be transferred from *u* to *p*; for the axiomatic method is entirely formal and abstract, and in principle utterly independent of interpretation. In this

way, an axiomatic *theory* in $\mathbf{p}$ proceeds by identifying some small (invariably finite) number of propositions or symbols as axioms, and entirely by the application of a similarly small number of syntactical production rules (rules of inference) proceeds to generate the implications or theorems of the system. As noted above, the only limitation imposed here is that the resulting set of theorems be *consistent*; i.e. that for no proposition p are both p and $\neg p$ theorems. The generation of theorems in such a system is entirely a *mechanical* process, in the sense that it can be done by a robot or machine; and indeed one approach to such systems proceeds entirely through such machines (Turing machines);⁶ this is the basis for the close relation between axiomatics and automata theory. But that is another story.

What Gödel showed in this connection was that this kind of procedure is essentially hopeless. If we have some body of propositions that we desire to declare true, and if this body of propositions is rich enough to be at all interesting in mathematical terms, then there will be no mechanical process for establishing all these propositions on the basis of any finite subset of them chosen as axioms; i.e. there will be true propositions which are unprovable from the axioms, however the axioms may be chosen. There is only one exception; the fatal one in which the body of propositions initially declared true is inconsistent. Moreover, it is generally impossible, given a system of axioms and a set of production rules, to determine whether or not an arbitrary proposition is in fact a theorem; i.e. can be reached from the axioms by chains of implications utilizing the given rules.

Thus the $\mathbf{u}$ -ists can say to the $\mathbf{p}$ -ists: you claim that the impossibility of carrying out infinitely many acts of verification means that mathematics must become a game of manipulating meaningless symbols according to arbitrary rules. You say that truth or falsity of propositions is likewise something which may be arbitrarily assigned. But you cannot effectively answer any of the crucial questions; not even whether a proposition is a theorem, nor whether a particular set of propositions is consistent. Surely, that is no basis for doing mathematics.

And in fact, this is where the matter still stands at present. The dispute is over what mathematics *ought* to be; a question which would gladden Mr. Hutchins' heart, but one which cannot apparently be answered from within mathematics itself; it can only be answered from within the mathematician.

We are going to make the best of this situation by ignoring it to the extent possible. In what follows, we shall freely move back and forth between the world $\mathbf{u}$ of formal mathematical systems and the world $\mathbf{p}$ of propositions relating properties of these systems as if it were legitimate to do so; unless the reader is a specialist in foundation problems, this is in fact what the reader has always done. We will freely make use of *interpretations* of our propositions; in fact, the generation of such interpretations will be the heart of the modeling relation. The above discussion is important in its own right, partly because it teaches us the essential difference between things and the names of things. But also it provides some essential background for the discussion of modeling itself; many of the purely mathematical struggles to provide meaning through interpretation of sets of propositions are in themselves important instances of the modeling relation, and will be discussed in some detail in Chap. 3 below.

We now turn to the next essential item of business; the establishment of relationships between the worlds of natural systems and of formal ones.

References and Notes

1. Georg Cantor, the creator of set theory, envisioned it as providing an unshakable foundation for all of mathematics, something on which all mathematicians could agree as a common basis for their discipline. He would have been astounded by what has actually happened; hardly two books about “set theory” are alike, and some are so different that it is hard to believe they purport to be about the same subject. We have tried to make this liveliness visible in the treatment we provide. Of the multitude of books about set theory, the one whose spirit seems closest to ours is:

van Dalen, D., Doets, H. C. and de Swart, H., *Sets: Naive, Axiomatic and Applied*. North-Holland (1978).

2. The empty set is perhaps the only mathematical object which survives Cartesian doubt; it must *necessarily* exist, even if nothing else does. Since it exists, we can form the class which contains the empty set as its single element; thus a one-element set also exists, and hence the number 1 can be defined. By iterating this procedure, we can construct a set with two elements, a set with three elements, etc., and mathematics can begin.
3. We shall discuss the Theory of Categories at greater length in Sect. 3.3 below; references are provided in the notes for that chapter.
4. In fact, it is generally impossible (in a definite, well-defined sense). At least, it is impossible in any sense that would be acceptable to a $\mathbf{p}$ -ist. The problem of determining effectively whether two words of even a finitely generated free group fall into the same equivalence class with respect to some relation is essentially the *word problem* for groups, and it can be shown to be unsolvable. See

Boone, W. W., Cammonito, F. B. and Lyndon, R. C., *Word Problems*. North-Holland (1973).

5. In a celebrated paper, Gödel showed that any axiom system strong enough to do arithmetic must contain undecidable propositions, unless the system is inconsistent to begin with. Gödel’s discovery devastated the so-called Hilbert Program, which sought to establish mathematical consistency entirely along $\mathbf{p}$ -ist lines. For a clear and simple exposition of Gödel’s argument and its significance, we may recommend

Nagel, E. and Newman, J. R., *Gödel’s Proof*. Routledge & Kegan Paul, London (1962).

6. The term “machine” is used in mathematics for anything that executes an algorithm; it does not refer to a physical device. Such “Mathematical machines” provide an effective way to navigate in the universe $\mathbf{p}$. The theory of such machines is known variously as recursive function theory and the theory of

automata. It also turns out to be intimately connected to the theory of the brain (cf. Sect. 3.5, Example 3B below) and cognate subjects. One way of showing that a class of problems (such as the word problems for finitely generated groups which was mentioned above) is unsolvable is to show that no machine can be programmed to solve the problem in a finite number of steps. General references bearing on these matters are:

Davis, M., *Computability and Unsolvability*. McGraw-Hill, New York (1958).

Minsky, M., *Computation: Finite and Infinite Machines*. Prentice-Hall, New York (1967).

2.3 Encodings Between Natural and Formal Systems

In the preceding two chapters, we have discussed the concepts of *natural systems*, which belong primarily to science, and *formal systems*, which belong to mathematics. We now turn to the fundamental question of establishing relations between the two classes of systems. The establishment of such relations is fundamental to the concept of a model, and indeed, to all of theoretical science. In the present chapter we shall discuss such relations in a general way, and consider a wealth of specific illustrative examples in Chap. 3 below.

In a sense, the difficulty and challenge in establishing such relations arises from the fact that the entities to be related are fundamentally different in kind. A natural system is essentially a bundle of linked qualities, or observables, coded or named by the specific percepts which they generate, and by the relations which the mind creates to organize them. As such, a natural system is *always incompletely known*; we continually learn about such a system, for instance by watching its effect on other systems with which it interacts, and attempting to include the observables rendered perceptible thereby into the scheme of linkages established previously. A formal system, on the other hand, is entirely a creation of the mind, possessing no properties beyond those which enter into its definition and their implications. We thus do not “learn” about a formal system, beyond establishing the consequences of our definitions through the application of conventional rules of inference, and sometimes by modifying or enlarging the initial definitions in particular ways. We have seen that even the study of formal systems is not free of strife and controversy; how much more strife can be expected when we attempt to relate this world to another of a fundamentally different character? And yet that is the task we now undertake; it is the basic task of relating *experiment* to *theory*. We shall proceed to develop a general framework in which formal and natural systems can be related, and then we shall discuss that framework in an informal way.

The essential step in establishing the relations we seek, and indeed the key to all that follows, lies in an exploitation of *synonymy*. We are going to force the name of a percept to be also the name of a formal entity; we are going to force the name of a linkage between percepts to also be the name of a relation between mathematical

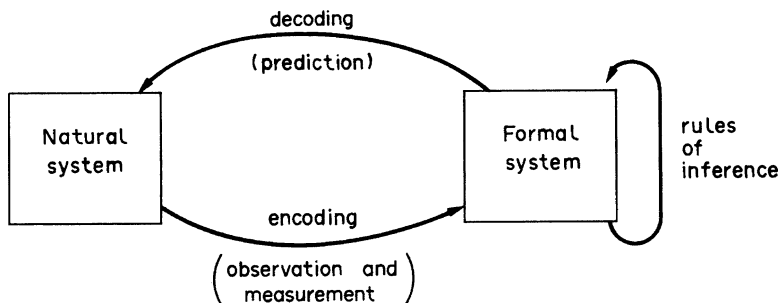


Fig. 2.1

entities; and most particularly, we are going to force the various temporal relations characteristic of causality in the natural world to be synonymous with the inferential structure which allows us to draw conclusions from premises in the mathematical world. We are going to try to do this in a way which is *consistent* between the two worlds; i.e. in such a way that the synonymies we establish do not lead us into contradictions between the properties of the formal system and those of the natural system we have forced the formal system to name. In short, we want our relations between formal and natural systems to be like the one Goethe postulated as between the genius and Nature: what the one promises, the other surely redeems.

Another way to characterize what we are trying to do here is the following: we seek to *encode* natural systems into formal ones in a way which is consistent, in the above sense. Via such an encoding, if we are successful, the inferences or theorems we can elicit within these formal systems become *predictions* about the natural systems we have encoded into them; consistency then means that these predictions will be *verified* in the natural world when appropriately *decoded* into linkage relations in that world. And as we shall see, once such a relation between natural and formal systems has been established, a host of other important relations will follow of themselves; relations which will allow us to speak precisely about analogy, similarity, metaphor, complexity, and a spectrum of similar concepts.

If we successfully accomplish the establishment of a relation of this kind between a particular natural system and some formal system, then we will obtain thereby a composite structure whose character is crudely indicated in Fig. 2.1 below:

In this figure, the arrows labelled “encoding” and “decoding” represent correspondences between the observables and linkages comprising the natural system and symbols or propositions belonging to the formal system. Linkages between these observables are also encoded into relations between the corresponding propositions in the formal system. As we noted earlier, the rules of inference of the formal system, by means of which we can establish new propositions of that system as implications, must be re-interpreted (or decoded) in the form of specific assertions pertaining to the observables and linkages of the natural system; these are the *predictions*. If the assertions *decoded* in this fashion are *verified* by observation; or what is the same thing, if the observed behavior of the natural system *encodes* into the same

propositions as those obtained from the inferential rules of the formal system, we shall say that (to that extent) the relation between the two systems which we have established, and which is diagrammed in Fig. 2.1, is a *modeling relation*. Under these circumstances, we shall also say that the formal system of Fig. 2.1, modulo the encoding and decoding rules in question, is a model of the natural system to which it is related by those rules.

Let us consider the situation we have described in somewhat more detail. We saw in the preceding chapter that rules of inference relating propositions in a formal system cannot be applied arbitrarily, but only to propositions of a definite form or character. If we are going to apply our rules of inference in accordance with the diagram of Fig. 2.1, then specific aspects of the natural system must code propositions to which they can be applied. In the broadest terms, the aspects so encoded can be called *states* of the natural system. A state of a natural system, then, is an aspect of it which encodes into a *hypothesis* in the associated formal system; i.e. into a proposition of that formal system which can be the basis for an inference. Speaking very informally, then, *a state embodies that information about a natural system which must be encoded in order for some kind of prediction about the system to be made*. It should be explicitly noted that, according to this definition, *the concept of state of a natural system is not meaningful apart from a specific encoding of it into some formal system*. For, as we are employing the term, a state comprises what needs to be known about a natural system in order for other features of that system to be determined. This in itself is somewhat a departure from ordinary usage, in which a state of a natural system is regarded as possessing an objective existence in the external world. It should be compared with our employment of the term “state” in Sect. 1.2 above; we will see subsequently, of course, that the two usages in fact coincide.

Let us now suppose that we have encoded a state of our natural system into a hypothesis of the associated formal system. What kind of inferences can we expect to draw from that hypothesis, and how will they look when they are decoded into explicit predictions?

We shall begin our discussion of this question by recalling our discussion of *time* and its relation to percepts. We pointed out that there were two different aspects of our time-sense, which we called simultaneity and precedence, which we impose upon percepts. We then *impute* similar properties to the qualities in the external world which generate these percepts. Accordingly, it is reasonable to organize assertions about natural systems (and in particular, the predictions generated from a model) in terms of temporal features.

We can then recognize three temporally different kinds of predictions: (a) predictions which are time-independent; (b) predictions which relate different qualities of our system at the same instant of time; and (c) predictions which relate qualities of our system at different instants of time. Predictions of the first type are those which are unrelated to the manner in which time itself is encoded into our formal system. Typically, they express relations or linkages between qualities which play an essential role in the recognition of the identity of the natural system with which we are dealing, and hence which *cannot* change, without our

perception that our original system has been replaced by a new or different system. To anticipate subsequent discussion somewhat, qualities of this type are exemplified by constitutive parameters; hence a time-independent prediction would typically take the form of a linkage between such constitutive parameters.

Almost by definition, all time-dependent qualities pertaining to our natural system can change without our perceiving that our system has been replaced by another. Hence, the two kinds of time-dependent predictions we might expect to make on the basis of a model will necessarily pertain to such qualities.

Predictions of the second type itemized above will generally be of the following form: if a state of our natural system at a given instant of time is specified, then some other quality pertaining to the system *at the same instant* may be inferred. That is, predictions of this type will *express linkages between system qualities which pertain to a specific instant of time*, and which are simultaneously expressed at that instant.

Predictions of the third type will generally be of the form: if the state of our natural system at a particular instant is specified, then some other quality pertaining to the system *at some other instant* may be inferred. Thus, predictions of this type *express linkages between qualities which pertain to different instants of time*. In particular, we may be interested in inferences which link *states* at a given instant to states at other instants. As we shall see, linkages of this type essentially constitute the dynamical laws of the system.

It will be noted that all three types of inferences, and the predictions about a natural system which may be decoded from them, express *linkage* relations between system qualities. The corresponding inferential rules in the formal system which decode into these linkage relations are what are usually called *system laws* of the natural system. It is crucial to note that, as was the case with the concept of state, the concept of system laws is meaningless apart from a specific encoding and decoding of our natural system into a formal one; the system laws belong to the formal system, and are in effect *imputed* to the associated natural system exactly by virtue of the encoding itself.

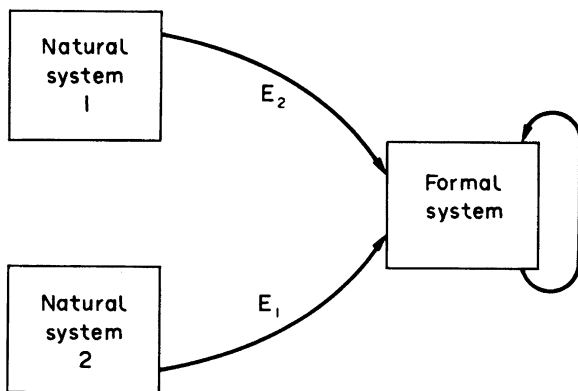
We shall see many specific examples of all of these ideas, in a broad spectrum of contexts, in Chap. 3 below; therefore we shall not pause to give examples here. However, the reader should at this point look back at the treatment of the reactive paradigm provided in Sect. 1.2 above, to gain an idea of just how much had to be tacitly assumed in writing down even the first line of that development.

We are now going to elaborate a number of important variations on the basic theme of the modeling relations embodied in Fig. 2.1 above. As we shall see, these variations represent the essential features of some of the most important (and hence perhaps, the most abused) concepts in theoretical science. They will all be of central importance to us in the developments of the subsequent chapters.

The first of these variations is represented in diagrammatic form in Fig. 2.2 below. In this situation, we have two natural systems, which we shall denote by N_1 , N_2 , encoded into the same formal system F , via particular encodings E_1 , E_2 , respectively.

The specific properties of the situation we have diagrammed depends intuitively on the degree of “overlap” in F between the set $E_1(N_1)$ of propositions which encode qualities of N_1 , and the set $E_2(N_2)$ of propositions which encode qualities

Fig. 2.2

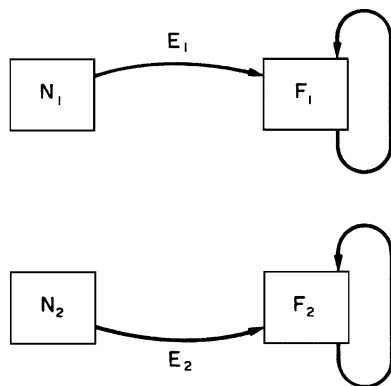


of N_2 . Clearly, if the *same* proposition of F simultaneously encodes a quality of N_1 and a quality of N_2 , we may thereby establish a relation between N_1 and N_2 . It is convenient for us to recognize three cases:

Case A1: $E_1(N_1) = E_2(N_2)$

In this case, every proposition of F which encodes a quality of N_1 also encodes a quality of N_2 , and conversely. Utilizing this fact, we may in effect construct a “dictionary”, which allows us to *name* every quality of N_1 by that quality of N_2 which encodes into the same proposition in F ; and conversely, every quality of N_2 can be named by a corresponding quality in N_1 . Since by definition of the modeling relation itself, the system laws imputed to N_1 and N_2 are also the same, this dictionary also allows us to decode the inferences in F into corresponding propositions about N_1 and N_2 respectively. In this case, we can say that *the natural systems N_1 and N_2 share a common model*. Alternatively, we shall say that N_1 and N_2 are *analogous systems, or analogs*.¹ As usual, we point out explicitly that this concept of analogy is only meaningful with respect to a particular choice of the encodings E_1, E_2 into a common formal system F . A moment’s reflection will reveal that our use of the term “analog” is a straightforward generalization of its employment in the term “analog computer”. It must also be emphasized that the relation of analogy is a relation between *natural systems*; it is a relation established precisely when these natural systems share a common encoding into the same formal system. In this sense, it is different in character from the modeling relation, which (as we have defined it) is a relation between a natural system and a formal system. Despite this, we shall often abuse language, and say that N_1 and N_2 are *models of each other*; this is a usage which is ubiquitous in the literature, and can really cause no confusion. It should be noted that our “dictionary” relating corresponding qualities of analogous systems behaves like a *linkage* between them; but the origin of such a linkage has its roots in the formal system F and in the properties of the encodings E_1, E_2 ; it does *not* reside in the natural systems N_1, N_2 themselves.

Fig. 2.3



Case A2: $E_1(N_1) \subset E_2(N_2)$

This case differs from the previous one in that, while every proposition of F which encodes a quality of N_1 also encodes a corresponding property of N_2 , the converse is not true. Thus if we establish a “dictionary” between the qualities of N_1 and N_2 put into correspondence by the respective encodings, we find that every quality of N_1 corresponds to some quality of N_2 , but not conversely. That is, there will be qualities of N_2 which do not correspond in this fashion to any qualities in N_1 .

Let N_2' denote those qualities of N_2 which can be put in correspondence with qualities of N_1 under these circumstances. If we restrict attention to just N_2' , then we see that N_1 and N_2' are analogs, in the sense defined above. If we say that N_2' defines a *subsystem* of N_2 , then the case we are discussing characterizes the situation in which N_1 and N_2 are not directly analogs, but in which N_2 contains a *subsystem analogous to N_1* . By abuse of language, we can say further that N_2 contains a *subsystem N_2' which is a model of N_1* .

Of course, if it should be the case that $E_2(N_2) \subset E_1(N_1)$, we may repeat the above argument, and say that N_1 possesses a subsystem N_1' which is analogous to, or a model of, N_2 .

Case A3: $E_1(N_1) \cap E_2(N_2) \neq \emptyset$

We assume here that neither of the sets $E_1(N_1)$, $E_2(N_2)$ of encoded propositions is contained in the other, but that their intersection is non-empty. In this case, an application of the above argument leads us to define subsystems N_1' of N_1 , and N_2' of N_2 , which can be put into correspondence by virtue of their coding into the same proposition of F . In this case, then, we may say that the natural systems N_1 , N_2 possess *subsystems* which are analogous, or which are models of each other.

There is of course a fourth possibility arising from the diagram in Fig. 2.3; namely $E_1(N_1) \cap E_2(N_2) = \emptyset$. But this possibility is uninteresting, in that the encodings E_1 , E_2 allow us to establish no relation between N_1 and N_2 .

Now let us consider a further variation on our basic theme, which generalizes the above discussion in a significant way. Let us consider the situation diagrammed in Fig. 2.3 below:

Here the natural systems N_1, N_2 encode into *different* formal systems F_1, F_2 respectively. We shall now attempt to establish relations between N_1 and N_2 on the basis of *mathematical* relations which may exist between the corresponding *formal* systems F_1, F_2 , respectively.

Here again, it will be convenient to recognize a number of distinct cases, which are parallel to those just considered.

Case B1: $E_1(N_1)$ and $E_2(N_2)$ are *isomorphic*

In this case, there is a structure-preserving mapping between the sets of encoded propositions, which is one-to-one and onto. Since we are dealing here with formal systems, this mapping is a definite mathematical object, and can be discussed as such. By “structure-preserving”, we mean here that this mapping can be extended in a unique way to all inferences in F_1, F_2 which can be obtained from $E_1(N_1), E_2(N_2)$ respectively.

Under these circumstances, we can once again construct a dictionary between the qualities of N_1 and those of N_2 , by putting qualities into correspondence whose encodings are images of one another under the isomorphism established between $E_1(N_1)$ and $E_2(N_2)$. This dictionary, and the fact that the system laws in both cases are also isomorphic, allows us to establish a relation between N_1 and N_2 which has all the properties of analogy defined previously. We shall call this relation an *extended analogy*, or, when no confusion is possible, simply an analogy, between the natural systems involved. It should be noted explicitly that the dictionary through which an extended analogy is established depends on the *choice* of isomorphism between $E_1(N_1)$ and $E_2(N_2)$.

Case B2: $E_1(N_1)$ is isomorphic to a subset of $E_2(N_2)$

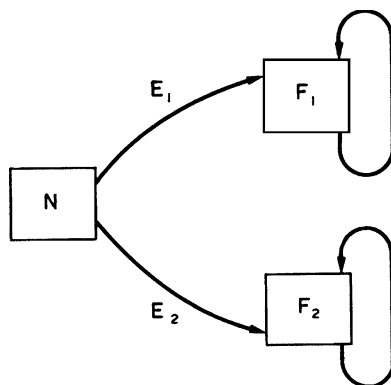
This case is, of course, analogous to Case A2 above. As in that case, we can establish our dictionary between N_1 and a subsystem N_2' of N_2 , and thereby establish a relation of extended analogy between N_1 and N_2' . It is clear that all aspects of our discussion of Case A2 also apply here. Likewise, we have

Case B3: A subset of $E_1(N_1)$ is isomorphic to a subset of $E_2(N_2)$

This case is the analog of Case A3 above, we can establish an extended analogy between subsystems N_1' of N_1 and N_2' of N_2 in the obvious fashion.

There is a final case to be considered before leaving the situation diagrammed in Fig. 2.3, which should be mentioned at this time, but about which we will not be able to be very precise until somewhat later. This is the circumstance in which the formal systems F_1, F_2 of that diagram satisfy some more general mathematical relation than that embodied in the existence of a structure-preserving mapping between them. As we shall see subsequently, such a relation will typically involve the satisfaction of some general criterion according to which formal systems may be classified; as for instance that F_1, F_2 are both “finite-dimensional” or “finitely generated” in some sense, but without any more detailed relationship which might be embodied in the existence of a structure-preserving mapping between them. In such a circumstance, we cannot establish a detailed dictionary relating qualities of the associated natural

Fig. 2.4



systems N_1, N_2 . However, we can utilize the fact that F_1 and F_2 are related in some more general sense, and hence are to be counted as alike in that sense, to *impute* a corresponding relation to N_1, N_2 themselves. The situation we are describing will provide the basis for saying that N_1 is a *metaphor* for N_2 , in the sense that *their models share some common general property*. Subsequently we shall see many examples of such metaphors, as for instance in the assertion that open systems are metaphors for biological developmental processes. This concept of metaphor will be important to us, but we will leave the more detailed discussion for subsequent chapters.

So far, we have considered situations in which different natural systems can be encoded into the same formal system, or into systems between which some mathematical relation can be established. We will now turn to the dual situation, in which the same natural system can be encoded into two (or more) formal systems; this situation is diagrammed in Fig. 2.4 below:

We will now take up the important question of the extent to which a mathematical relation can be established between the *formal* systems F_1 and F_2 , simply by virtue of the fact that the same natural system N can be encoded into both of them. Such a relation would essentially embody a *correspondence principle* between the encodings.

In a sense, the diagram of Fig. 2.4 can be regarded as a variant of that shown in Fig. 2.3, in which we put both N_1 and N_2 equal to N . Indeed, if we consider the simplest situation, in which exactly the same variates of N are encoded into F_1 and F_2 (via the encodings E_1 and E_2 respectively), then it is easy to see that we are indeed in the situation previously discussed, and in fact essentially in the Case B1. However, we shall ignore this relatively trivial situation, and instead consider what happens when *different qualities* of N are encoded into F_1 and F_2 . We are now in a quite different situation from that diagrammed in Fig. 2.3, for the following reason: we supposed that N_1 and N_2 represent *different* natural systems; this means that their respective qualities are generally unlinked. In Fig. 2.4, on the other hand, even though we may generally encode different qualities of N

into different formal systems F_1 , F_2 , these qualities are linked in N through the hypothesis that they pertain to the *same* system. However, the linkages between these differently encoded qualities are themselves encoded neither in F_1 nor in F_2 . Thus the question we are now posing is the following: *to what extent can these uncoded linkages manifest themselves in the form of purely mathematical relations between F_1 and F_2 ?* It will be seen that this is the inverse of the former cases, in which we *posited* a mathematical relation between F_1 and F_2 , and *imputed* it to the natural systems encoded into them. Stated otherwise: whereas we formerly sought a relation between natural systems through a mathematical relation between their encodings, we presently seek a mathematical relation between encodings, arising from uncoded linkages between families of differently encoded qualities.

There are again a number of cases which must be considered separately.

Case C1

Let us suppose that we can establish a mapping of F_1 , say, into F_2 . That is, every proposition of F_1 can be associated with a unique proposition of F_2 , but we do not suppose that the reverse is the case. Then every quality of N encoded into F_1 can thereby be associated with another quality of N , encoded into F_2 , but not conversely. There is thus an important sense in which the encoding F_1 is *redundant*; every quality encoded into F_1 can be recaptured from the encoding of apparently different qualities into F_2 . In this case, we would say intuitively that F_2 provides the more comprehensive formal description of N , and in fact that the encoding into F_1 can be *reduced* to the encoding into F_2 .

The fundamental importance of this situation lies in its relation to *reductionism* as a basic scientific principle.² Reductionism asserts essentially that there is a universal way of encoding an arbitrary natural system, such that *every other encoding* is reducible to it in the above sense. It must be stressed, and it is clear from the above discussion, that reductionism involves a *mathematical* relation between certain formal systems, into which natural systems are encoded. In order for reductionism in this sense to be established as a *valid* scientific principle, the following data must be given: (a) a recipe, or algorithm, for producing the universal encoding for any given natural system N ; (b) a *mathematical* proof that every other coding can be effectively mapped into the universal one. Needless to say, no arguments bearing on these matters have ever been forthcoming, and as we shall see abundantly, there are good reasons to suppose that the reductionistic principle is in fact false. Furthermore, in a purely practical sense, an attempt to obtain specific inferences about qualities of a given natural system N by detouring through a universal description would not be of much value in any case. Nevertheless, it is important to determine in what circumstances a given encoding E_1 (N) can be reduced to another one E_2 (N). For such a reduction implies important linkages between the qualities encoded thereby; linkages which as we noted above are themselves not explicitly encoded by either E_1 or E_2 .

Case C2

Here we shall suppose that a mapping can be established from a *subset* of F_1 to a *subset* of F_2 , in such a way that certain qualities of N encoded into $E_1(N_1)$ can be related to corresponding qualities encoded into $E_2(N_2)$. The mapping itself, of course, establishes a logical relation between the encodings, but one which does not hold universally. Thus in this case we may say that a *partial* reduction is possible, but one of limited applicability. As we shall see, when we leave the domain in which such a partial reduction is valid, the two encodings become logically independent; or what is the same thing, the linkage between the qualities in N which are encoded thereby is broken. Depending on the viewpoint which is taken, such a departure from the domain in which reduction is valid will appear as a *bifurcation*, or as an *emergence phenomenon*; i.e. the generation of an emergent novelty. We will develop explicit examples of these situations subsequently.

Case C3

In this case, no mathematical relation can be established between F_1 and F_2 . Thus, these encodings are inadequate in principle to represent any linkages in N between the families or qualities encoded into them. Into this class, we would argue, fall the various *complementarities* postulated by Bohr in connection with microphysical phenomena. Here again, it must be stressed that it is the mathematical character of the encodings which determines whether or not relations between them can be effectively established; in this sense, complementarity is entirely a property of formal systems, and not of natural ones.

One further aspect of the basic diagram of Fig. 2.4 may be mentioned briefly here. We are subsequently going to relate our capacity to produce independent encodings of a given natural system N with the *complexity*³ of N . Roughly speaking, the more such encodings we can produce, the more complex we will regard the system N . Thus, contrary to traditional views regarding system complexity, we do not treat complexity as a property of some particular encoding, and hence identifiable with a mathematical property of a formal system (such as dimensionality, number of generators, or the like). Nor is complexity entirely an objective property of N , in the sense of being itself a directly perceptible quality, which can be measured by a meter. Rather, complexity pertains at least as much to *us as observers* as it does to N ; it reflects *our ability to interact with* N in such a way as to make its qualities visible to us. Intuitively speaking, if N is such that we can interact with it in only a few ways, there will be correspondingly few distinct encodings we can make of the qualities which we perceive thereby, and N will appear to us as a *simple* system; if N is such that we can interact with it in many ways, we will be able to produce correspondingly many distinct encodings, and we will correspondingly regard N as complex. It must be recognized that we are speaking here entirely of complexity as an attribute of *natural* systems; the same word (complexity) may, and often is, used to describe some attribute of a *formal* system. But this involves quite a different concept, with which we are not presently concerned. We shall also return to these ideas in more detail in subsequent chapters.

Let us turn now to an important re-interpretation of the various situations we have discussed above. We have so far taken the viewpoint that the modeling relation exemplified in Fig. 2.1 involves the encoding of a given natural system N into an appropriate formal system F , through an identification of the names of qualities of N with symbols and propositions in F . That is, we have supposed that we *start* from a given natural system whose qualities have been directly perceived in some way, and that the encoding is established *from* N *to* F . There are many important situations, however, in which the same diagram is established from precisely the opposite point of view. Namely, suppose we want to *build*, or create, a natural system N , in such a way that N possesses qualities linked to one another in the same way as certain propositions in some formal system F . In other words, we want the system laws characterizing such a system N to represent, or *realize*, rules of inference in F . This is, of course, the problem faced by an engineer or an architect who must *design* a physical system to meet given specifications. He invariably proceeds by creating first a *formal system*, a blueprint, or plan, according to which his system N will actually be built. The relation between the system N so constructed, and the plan from which it is constructed, is precisely the modeling relation of Fig. 2.1, but now the relation is generated *in the opposite direction* from that considered above.⁴ From this we see that the analytic activities of a scientist seeking to understand a particular unknown system, and the synthetic activities of an engineer seeking to fabricate a system possessing particular qualities, are in an important sense inverses of one another. When we speak of the activities of the scientist, we shall regard the modeling relations so established in terms of the *encoding* of a natural system into a formal one; when we speak of the synthetic activities of the engineer, we shall say that the natural system N in question is a *realization* of the formal system F , which constitutes the plan. Using this latter terminology, then, we can re-interpret the various diagrams presented above; for instance, we can say that two natural systems N_1, N_2 are analogs precisely to the extent that they are *alternate realizations* of the same formal system F . Likewise, in the situation diagrammed in Fig. 2.4, we can say that the same natural system N can simultaneously realize two distinct formal systems; the complexity of N thus can be regarded as the number of distinct formal systems which we perceive that N can simultaneously realize. We shall consider these ideas also in more detail in the course of our subsequent exposition.

To conclude these introductory heuristic remarks regarding the modeling relation, let us return again to the situations diagrammed in Figs. 2.2 and 2.3 above. We saw in our discussion of those diagrams that the modeling relation is one established between a natural system N and a formal system F through a suitable encoding of qualities of N into symbols and propositions in F . We also saw that when two distinct natural systems N_1, N_2 can be appropriately encoded into the same formal system, it was meaningful to speak of a relation of *analogy* as being established between the natural systems N_1 and N_2 themselves. We wish to pursue a few corollaries of these ideas.

The main property of a modeling relation, which we shall exploit relentlessly in what follows, is this: once such a relation has been established (or posited) between a natural system N and a formal system F , *we can learn about* N *by*

studying F; a patently different system. Likewise, once an analogy relation has been established between two natural systems N_1 and N_2 , *we can learn about one of them by studying the other*. In this form, both modeling and analogy have played perhaps the fundamental role in both theoretical and experimental science, as well as a dominant role in every human being's daily life. However, many experimental scientists find it counter-intuitive, and therefore somehow outrageous, to claim to be learning about a particular natural system N_1 by studying a different system N_2 , or by studying some formal system F , which is of a fundamentally different character from N_1 . To them, it is self-evident that the only way to learn about N_1 is to study N_1 ; any other approach is at best misguided and at worst fraudulent. It is perhaps worthwhile, therefore, to discuss this attitude somewhat fully.

It is not without irony that the empiricist's aversion to models finds its nourishment in the same soil from which our concept of the modeling relation itself sprang. For they are both products of the development which culminated in Newton's *Principia*. In that epic work, whose grandeur and sweep created reverberations which still permeate so much of today's thought, it was argued that a few simple formal laws could account for the behavior of all systems of material particles, from galaxies to atoms. We have sketched a little of this development in Sect. 1.2 above. Newton also provided the technical tools required for these purposes, with his creation of the differential and integral calculus; i.e. he also provided the formal basis for encoding the qualities of particulate systems, and drawing conclusions or inferences pertaining to those systems from the system laws. Not only was the entire Newtonian edifice one grand model in itself, but it provided a recipe for establishing a model of *any* particulate system. To set this machinery in motion for any specific system, all that was needed was (a) a specification of the forces acting on the system (i.e. an encoding of the system laws), and (b) a knowledge of initial conditions.

The universality of the Newtonian scheme rests on the extent to which we can regard *any* natural system as being composed of material particles. If we believe that any system can in fact be resolved into such particles, then the Newtonian paradigm can always be invoked, and the corresponding machinery of encoding and prediction set in motion. The belief that this is always possible (and more than this, is always necessary and sufficient) is the motivating force behind the concept of reductionism, which was mentioned earlier.

Now the determination of system laws, and even more, the specification of initial conditions, which are the prerequisites for the application of the Newtonian paradigm, are not themselves conceptual or theoretical questions; they are empirical problems, to be settled by observation and measurement, carried out on the specific system of interest. Thus, hand in hand with the grandest of conceptual schemes we necessarily find the concomitant growth of empiricism and reductionism; two thankless offspring which incessantly threaten to devour that which gave them birth. For, once separated from the conceptual framework which originally endowed them with meaning and significance, they become in fact antithetical to that framework. Pursued in isolation, empiricism must come to claim (a) that measurement and observation are ends in themselves; (b) that measurement and observation pertain only to extremely limited and narrow families of interactions which we can generate

in the here-and-now; (c) and that, finally, whatever is not the result of measurement and observation in this narrow sense *does not exist*, and ergo cannot be part of science.

We can now begin to understand why the theoretical concept of a model, which has as its essential ingredient a freely posited formal system, is so profoundly alienating to those reared in an exclusively empirical tradition. This alienation can sometimes take bizarre and even sinister forms. Let us ponder, for example, the growth of what was called *Deutsche Physik*, which accompanied the rise of National Socialism in Germany not so very long ago.⁵ At root, this was simply a rebellion on the part of some experimental physicists against a threatening new theoretical framework for physics (especially as embodied in general relativity); the ugly racial overtones of this movement were simply an indication of how desperate its leaders had become in the face of such a challenge. They were reduced to arguing that “those who studied equations instead of nature” were traitors to their abstract ideal of science; that the true scientist could only be an experimenter who developed a kind of mystic rapport with nature through direct interaction with it. Similarly, the phenomenon now called Lysenkoism in the Soviet Union was associated with a phobia against theoretical developments in biology, especially in genetics.⁶ These are admittedly extreme situations, but they are in fact the logical consequences of equating science with empiricism (which in itself, it should be noted, involves a model).

Although perhaps most empirical scientists would dissociate themselves from such aberrations, they tend to share the disquiet which gave rise to them. They regard the basic features of the modeling relation, and the idea that we can learn about one system by studying another, as a reversion to magic; to the idea that one can divine properties of the world through symbols and talismans. Perhaps it must inevitably seem so to those who think that science is done when qualities are simply named (as meter readings) and tabulated. But in fact, that is the point at which science only begins.

From this kind of acrimony and strife, one might suppose that the modeling relation is something arcane and rare. We have already seen that this is not so; that indeed the modeling relation is a ubiquitous characteristic of everyday life as well as of science. But we have not yet indicated just how widespread it is. To do so, and at the same time to give specific instances of the situations sketched abstractly in our discussion so far, we will now turn to developing such instances, as they appear in many fields. We may in fact regard such instances as *existence proofs*, showing that the various situations we have diagrammed above are not vacuous. This will be the primary task of the chapters in Chap. 3 to follow.

References and Notes

1. Aside from the special case of “dynamical similarity”, which we shall consider at some length in Sect. 3.3 (especially Example 5), there has still been no systematic

study of the circumstances under which two different systems are analogous. Our treatment here is based on ideas first expounded in a paper of the author's: Rosen, R., *Bull. Math. Biophys.* 30, 481–492 (1968).

2. See also Sect. 3.6 below.
3. See Sect. 5.7 below.
4. The relation between model and realization is discussed more fully in Sect. 3.1.
5. For a good discussion of the philosophical issues involved here, see Beyerchen, A. D., *Scientists Under Hitler*. Yale University Press (1977).
6. The two definitive treatments of Lysenkoism are Medvedev, Zh. A., *The Rise and Fall of T. D. Lysenko*. Columbia University Press (1969).

Joravsky, D., *The Lysenko Affair*. Harvard University Press (1970).

These three books make fascinating reading, especially when read concurrently.

Chapter 3

The Modeling Relation

3.1 The Modeling Relation within Mathematics

We developed the modeling relation in the preceding chapter as a relation between a natural system and a formal system. This relation was established through an appropriate encoding of the qualities of the natural system into the propositions of the formal system, in such a way that the inferential structure of the latter correspond to the system laws of the former.

As it happens, an analogous relation to the modeling relation can be carried out entirely within the universe of formal systems. This situation is, for example, tacitly embodied on the diagram of Fig. 2.4 above, in which we sought to relate two formal systems F_1 , F_2 by virtue of the fact that they both encoded qualities pertaining to the same natural system N . The various cases discussed under that heading involved encodings between formal systems, and when accomplished, result in a diagram very similar in character to that displayed in Fig. 2.1. The primary difference between these two situations is, of course, that we must now consider encodings between two *formal* systems which preserve the *inferential* structures in the two systems, so that a theorem of the first corresponds to a theorem of the second.

More precisely, an encoding of one formal system into another can now itself be represented by a precise mathematical object, namely, a specific mapping $\Phi : F_1 \rightarrow F_2$. In the analogous situation establishing a modeling relation, the encoding of N into F could not be regarded as a mapping, because a natural system is not itself a formal object. But in the present case, we can now formulate the basic structure-preserving feature of the encodings we seek by saying precisely that, if p is a theorem in F_1 , then $\Phi(p)$ is a theorem in F_2 .

It is thus appropriate to begin our consideration of modeling relations by examining this analogous situation arising within mathematics itself. This will have the advantage that the relation of “system” to “model” becomes far more transparent; the objects in a formal system possess only the properties endowed to them through their definitions, and are not an incompletely specified family of qualities or observables. Moreover, very often in science we proceed by in effect

making a model of a model, through formal manipulations of a formal system F in which some natural system has already been encoded. Such formal manipulations lead precisely to a situation of the type we will be considering.

Our first examples will be drawn from geometry. Until the beginning of the last century, “geometry” meant the geometry of Euclid. As was mentioned previously, Euclid provided the earliest and most influential example of a mathematical theory as a system of axioms, to which we have alluded briefly in Sect. 2.2 above. All the examples to be considered now arose historically out of deeper investigations into the Euclidean system. In our first example, we shall consider the familiar arithmetization (or perhaps more accurately, algebraization) of geometry by Descartes, which was first published in 1637.

Example 1. Analytic Geometry

The axiom system provided by Euclid involved both general rules of inference, by which theorems could be proved, and specifically geometric postulates which, in effect, defined the properties of such geometric objects as points, lines, planes and the like. Since these axioms determine the entire geometry, it is necessary to formulate them explicitly, and then see how they can be encoded into another mathematical context.

The Euclidean axioms have been the subject of exhaustive study, perhaps most especially during the last half of the nineteenth century. Probably the most satisfactory reformulation of these axioms was that of David Hilbert, who in his monumental *Grundlagen der Geometrie* (first published in 1899) sought to place all geometric investigations into a common perspective. His reformulation involved eleven axioms, divided into five classes, which for plane geometry (to which we restrict ourselves here) may be set down as follows:

(a) *Axioms of Incidence*

1. Any pair of points lie on exactly one straight line.
2. At least two points lie on every line.
3. Given any straight line, there is a point not lying on that line.

(b) *Axioms of Order*

1. If three points lie on a line, one of them is between the other two.
2. Given any two points on a line, there is another point of the line lying between them.
3. A straight line divides the plane into two disjoint half-planes.

(c) *Axioms of Congruence*

1. An admissible motion of the plane carries lines onto lines.
2. Motions can be composed; i.e. two motions carried out successively produce the same result as a uniquely determined single motion, their composite.
3. Any motion is a composition of a translation, a rotation and a reflection about a line.

(d) *Axiom of Continuity*

Let L be a line. Let $P_1, P_2, \dots, P_n, \dots$ be any sequence of points such that P_{i+1} is between P_i and P_{i+2} for every i . If there is a point A of L such that for no i A is between P_i and P_{i+1} then there is a uniquely determined point B which also possesses this property, but if B' is any point for which B is between B' and A , then there is a P_i such that B' is between P_i and P_{i+1} .

(e) *Euclidean Axiom of Parallels*

Given a point P not lying on a line L , there is exactly one straight line through P which does not intersect L .

On the basis of these axioms, Hilbert showed that all of Euclidean geometry could be built up. Let us note that the various specific geometric terms, such as point, line, plane, between, etc., are considered as *undefined* (or more precisely, they possess only the properties endowed to them by the axioms).

We are now going to encode this system in another. This second system relies basically on the properties of the real number system, which can also themselves be developed axiomatically, but which we shall assume known (since our main purpose here is to establish the encoding, and not to do either geometry or arithmetic). To establish the encoding, we shall give the undefined terms in the above axioms new names, and then show briefly that the axioms, which define the properties of the undefined terms, are satisfied.

The new terminology we introduce is as follows:

point = ordered pair (a, b) of real numbers

plane = set of all ordered pairs of real numbers

line = set of all pairs (x, y) satisfying the condition $ax + by + c = 0$, where a, b, c are any three real numbers

From just this, it is a matter of straightforward verification that the Axioms of Incidence are satisfied.

Further, let $(a_1, b_1), (a_2, b_2), (a_3, b_3)$ be three points on a line.

We define the second to be *between* the other two if and only if the inequalities

$$a_1 < a_2 < a_3,$$

$$b_1 < b_2 < b_3$$

are satisfied. Then the Axioms of Order can be verified.

To verify the Axioms of congruence, we must define the admissible motions in terms of ordered pairs. We will do this by expressing a motion as a mapping of the set of all ordered pairs onto itself; it is thus sufficient to specify what the motion does to each particular pair (x, y) . If we define the motions as follows:

(a) *Translation*: $(x, y) \rightarrow (x + h, y + k)$ for some fixed numbers h, k .

(b) *Rotation*: $(x, y) \rightarrow (ax + by, -bx + ay)$, where a, b are fixed numbers such that $a^2 + b^2 = 1$.

(c) *Reflection*: $(x, y) \rightarrow (-x + 2a, -y + 2b)$ where (a, b) is a fixed point.

then it can be verified that the Axioms of Congruence are satisfied. The Axiom of Continuity is also verified directly from properties of real numbers.

Finally, if we are given a line, say in the form $y = mx + b$, and a point (ξ, η) not on this line, consider the line

$$y = mx + (\eta - m\xi).$$

It is immediately verified that this is a unique line which contains the point (ξ, η) and which has no point in common with the given line. Thus the Euclidean Axiom of Parallels is verified.

The Cartesian algebraization of Euclidean geometry which we have just described was one of the great syntheses of mathematics. It showed that every geometric proposition had an algebraic counterpart. It also showed how geometry itself needed to be extended, if the converse was to be true. We have become so accustomed to this synthesis that its properties have become commonplace, and its initial impact hard for us to imagine. But in fact, it was one of the supreme intellectual achievements; one which was indispensable for everything that followed. For instance, it is hard to imagine how Newton could have developed the calculus had not Descartes first established his encoding of geometry into algebra and arithmetic.

*Example 2. Non-Euclidean Geometry*¹

Let us preface our discussion with a few historical remarks. The encodings we are going to develop arose from consideration of the Euclidean Axiom of Parallels cited above. Already in ancient times, it was recognized that this axiom was of a different character than the others. Although no one doubted it was in some sense true, it seemed to lack the character of “self-evidence” which Euclid claimed for his axioms and postulates. In particular, its verification in any particular case involved a potentially infinite process; two given lines might have the property that, after being prolonged an arbitrarily long distance, they still had not met. But that fact could not establish whether or not those lines were in fact parallel. Thus it was felt in antiquity that the Axiom of Parallels should in fact be a theorem inferable from the other axioms. Thus were born the succession of attempts to “prove the Parallel Postulate”.

By the early part of the nineteenth century, it finally had to be admitted that these attempts had all failed. Most of these attempts were direct; they attempted to establish a chain of inferences from other Euclidean axioms to the Axiom of Parallels. But on close examination, all these direct attempts made tacit use of propositions equivalent to the Parallel Axiom itself, and hence were circular. A number of attempts were of a different character; they proceeded indirectly by assuming that the Parallel Axiom was false, and endeavored to derive a logical contradiction. The best known of these are associated with the names of Saccheri and Lambert. Finally, after a great deal of effort had been expended, the possibility gradually dawned that no logical contradiction could be derived in this way; that the negation of the Parallel Axiom led to a body of theorems which was logically

consistent. The first publication of this assertion was that of Lobachevsky in 1826, although Gauss and others had conceived of this possibility earlier. This was a sensational claim, for it had long been believed that the geometry of Euclid was the only “true” geometry. Moreover, the Cartesian identification of Euclidean geometry with algebra which we have just discussed was regarded as irrefutable evidence of at least the logical consistency of Euclidean geometry.

The body of theorems obtained by postulating the negation of the Parallel Axiom indeed seemed very peculiar. For instance, it was a theorem that no two triangles could be similar; or that there were no triangles of arbitrarily large area. However, there was no *logical* inconsistency visible in these theorems; i.e. no case of a theorem and its negation both being provable from these axioms. If this is so, then (a) more than one geometry is *logically* possible, and (b) it becomes an *empirical* question to decide which of the logically possible geometries actually pertains to *physical* space. This last assertion led, in a circuitous way, to the General Theory of Relativity. But nearly one hundred years earlier, Lobachevsky was already suggesting astronomical observations as a test of this question.

Now we have already seen in Sect. 2.2 above that the *consistency* of a body of propositions, such as the one which constitutes the geometry of Lobachevsky, cannot be established from the perusal of any finite sample, just as the parallelism or non-parallelism of a pair of lines cannot be established by any limited prolongation of them. Thus it could always be claimed that logical contradictions were inherent in the “non-Euclidean” geometries; they simply had not yet been discovered.

These hopes were laid to rest by the encodings we are now going to describe. Namely, it was shown that, via such encodings, it was possible to build a *model*² of the geometry of Lobachevsky completely within the confines of Euclidean geometry. These *models of non-Euclidean geometry* in fact had the following profound consequences: (a) they related the consistency of the non-Euclidean geometries to the consistency of Euclidean geometry, in such a way that the former could not be inconsistent without the latter at the same time being inconsistent; this was the first important example of a *relative* consistency proof; (b) they gave a concrete *meaning* to the “points”, “lines” and “planes” of non-Euclidean geometry; this was important because it was clear that these terms could not be synonymous with the traditional Euclidean usages; (c) at the same time, they suggested that mathematics could, and perhaps should, be done entirely in the abstract, devoid of *any* notion of meaning or interpretation; this paved the way for the position we described as p-ist in Sect. 2.2 above, and which we shall take up again shortly.

The first model we shall describe is that of Beltrami, who showed in 1868 that at least a portion of Lobachevsky’s non-Euclidean geometry could be encoded into the intrinsic geometry of a surface of constant negative curvature. The particular surface he used was the *pseudosphere*, which is a surface of revolution obtained by revolving a certain curve (called a tractrix) about its asymptote. In turn, the tractrix is that curve defined by the following property: its tangent at any point P intersects

the asymptote at a point Q such that the segment PQ has constant length. Let us establish then the following encoding:

point = point on the pseudosphere
 line = geodesic on the pseudosphere
 motion = geodesic-preserving map of pseudosphere onto itself

All of the other undefined terms of Lobatchevskian geometry are given their traditional Euclidean meanings, restricted to the surface of the pseudosphere.

Then to every theorem of Lobatchevskian geometry there corresponds a *fact* about the intrinsic geometry of the pseudosphere (or more accurately, to every such theorem which pertains to a finite part of the Lobatchevskian “plane”). What this encoding shows, then, is that the non-Euclidean geometry of Lobatchevsky can be regarded as an abstract version of the intrinsic geometry of the pseudosphere; a geometry which itself exists in an entirely Euclidean setting. More than this; as the curvature of the pseudosphere goes to zero, the propositions of the Lobatchevsky geometry approach their Euclidean counterpart in a certain well-defined sense; i.e. Euclidean geometry can itself be regarded as a limiting case of the Lobatchevskian.

A more comprehensive *model* of this kind was developed by Felix Klein in 1870; Klein showed how to encompass the infinite Lobatchevskian “plane” within the confines of Euclidean geometry. To develop Klein’s model, let us consider a circle in the ordinary Euclidean plane. Let us agree to establish the following dictionary:

point = interior point of this circle
 line = chord of this circle
 motion = mapping of the circle onto itself, in such a way that chords are carried onto chords.

As examples of such motions, we may consider the ordinary rotations of the circle; these are the Lobatchevskian rotations also. The translations are somewhat more complicated, and can best be described through reference to the Cartesian arithmetization of the Euclidean plane. If we take our circle to be the unit circle in the Euclidean plane, and establish a co-ordinate system in that plane with the center of the circle as origin, then the set of points interior to the circle are defined as those satisfying the inequality $x^2 + y^2 < 1$. Now we define the transformation

$$x \rightarrow \frac{x + a}{1 + ax}$$

$$y \rightarrow \frac{y\sqrt{1 - a^2}}{1 + ax}$$

where a is any real number such that $-1 < a < 1$. It is an easy exercise to verify that this transformation maps the interior of the unit circle onto itself, and leaves the circumference of that circle fixed. It is also readily verified that it maps chords onto chords.

Finally, reflections of this circle are just ordinary reflections in the Euclidean plane

Once again, it can be readily shown that all the axioms of the Lobatchevskian geometry are satisfied in this model. We may mention that this model captures the infinite Lobatchevskian plane because the definition of *distance* in the model is such that a segment of unit length (in the Lobatchevskian sense) is such that its Euclidean length shrinks to zero as the segment is translated towards the boundary of the circle. We shall omit the details, but they can be found in the references.

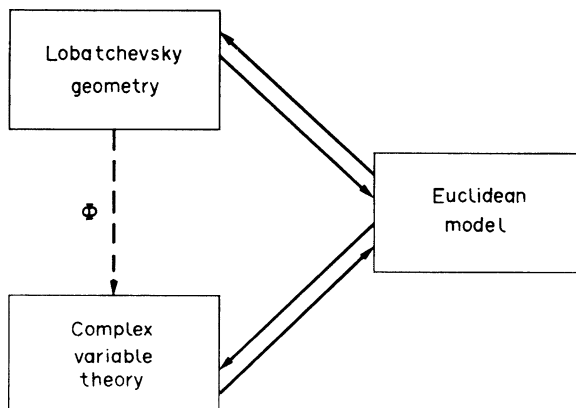
The Klein model shows explicitly that every theorem of Lobatchevskian geometry expresses, via the above encoding, a fact of Euclidean geometry within the chosen circle. In this case, the converse also happens to be true; namely, every fact of Euclidean geometry within the circle also corresponds to a theorem of Lobatchevskian geometry. In this way, we can use Euclidean theorems to “predict” Lobatchevskian theorems, thereby illustrating one of the fundamental properties of the modeling relation, here entirely within a mathematical context.

The fascinating story of Euclidean models of non-Euclidean geometry does not end here; we may describe one further ramification of it. In 1904, Poincaré modified Klein’s model, in the following way: instead of interpreting the Lobatchevskian lines as chords of a Euclidean circle, he interpreted them as arcs of circles intersecting the given one perpendicularly. Further, instead of interpreting a motion as a chord-preserving transformation of the circle onto itself, he required instead that a motion be an arbitrary *conformal* mapping (these are mappings which preserve Euclidean angles). Once again, he could verify that all the Lobatchevskian axioms are satisfied in this model. But it is well known that conformal mappings are also intimately connected with properties of analytic functions of a complex variable. Thus Poincaré could also encode important features of the theory of functions of a complex variable into the same mathematical object for which Lobatchevskian geometry was also encoded. In effect, then, he thus produced a diagram of the form which should be compared to Fig. 2.4 above. In this case, Poincaré was able to establish a partial correspondence, denoted by the dotted arrow in the above figure, by means of which he was able to re-interpret certain theorems of Lobatchevskian geometry as theorems about analytic functions. He was thereby able to establish important results in complex variable theory; results which apparently could not be easily obtained *in any other way*. We obviously cannot enter into the details here; nevertheless, the above story exhibits in a remarkably graphic way the power of modeling relations, here developed entirely within mathematics itself. We invite the reader to pursue the details independently.

Example 3. Formal Theories and Their Models^{2,3}

The experience with non-Euclidean geometries, some of which was described in the previous section, was but one of several profound shocks experienced by mathematics in the last part of the nineteenth century and the first part of the twentieth. Perhaps the worst of these shocks was the discovery of the paradoxes generated by apparently legitimate employment of set-theoretic constructions; these paradoxes precipitated the “foundation crisis” which is, in effect, still going on. As

Fig. 3.1



we saw earlier, many mathematicians felt that the only way to avoid such difficulties was to give up the idea that mathematics was “about” anything; i.e. to give up all idea of interpretation of mathematical ideas, and to consider mathematics itself as an entirely abstract game played with meaningless symbols manipulated according to definite syntactical rules. This, together with the Euclidean experience, led to an intense interest in formal theories, and thence to the concept of *models* for such theories. Since many of these ideas are direct generalizations of the preceding examples, and since they are for many purposes important in their own right, we will review them in more detail here.

Let us consider first what is meant by the term “formal”. Initially, any body of propositions in mathematics (and perhaps in all of science) is associated with a meaning or interpretation, which in general can be ultimately related back to corresponding percepts and relations between percepts. These meanings or interpretations enter tacitly into all arguments involving the propositions. The point is to get these tacit meanings, which affect the manner in which propositions are manipulated, themselves explicitly expressed as propositions. Once this is done, the meanings are no longer necessary, and can be forgotten. This process was perhaps best described by Kleene (1952):

This step (axiomatization) will not be finished until all the properties of the undefined or technical terms of the theory which matter for the deduction of theorems have been expressed by axioms. Then it should be possible to perform the deductions treating the technical terms as words in themselves without meaning. For to say that they have meanings necessary to the deductions of the theorems, other than what they derive from the axioms which govern them, amounts to saying that not all of their properties which matter for the deductions have been expressed by axioms. When the meanings of the technical terms are thus left out of account, we have arrived at the standpoint of formal axiomatics Since we have abstracted entirely from the content matter, leaving only the form, we say that the original theory has been *formalized*. In this structure, the theory is no longer a system of meaningful propositions, but one of sentences as sequences of words, which are in turn sequences of letters. We say by reference to the form alone which combinations of words are sentences, which sentences are axioms, and which sentences follow as immediate consequences from others.

This, of course, is the past p -ist position which we have considered previously.

Now a formal theory of the type contemplated here turns out to possess a well-defined anatomy, very much like the anatomy of an organism, which we now proceed to describe in a systematic way.

Specifically, any formal theory $\mathcal{J}$ must contain the following:

1. A class of elementary symbols called *variables*. This class is usually taken to be denumerably infinite, or at least indefinitely large.
2. Another class of elementary symbols called *constants*. Among this class, we discern two subclasses: (a) *Logical Constants*: These comprise the so-called sentential connectives ($\vee$, $\wedge$, $\sim$, $\rightarrow$), the quantifiers ($\forall$, $\exists$), and the identity symbol ($=$). (b) *Non-Logical Constants*: These comprise the names of predicates and operations available in the theory. Each predicate and operation is associated with a number, its *rank*; thus a unary operation is of rank one; a binary operation is of rank two, and so on; likewise, a one-place predicate is of rank one, a two-place predicate is of rank two, etc. There are also symbols for “individual constants” as required.

Initially, we allow any string of constants to be an *expression*. Certain expressions are called *terms*; the simplest of these are the symbols for the individual constants; a compound term is obtained by combining simpler terms by means of an operation symbol of the appropriate rank. Likewise, the simplest *formulas* are obtained by combining terms through a predicate of appropriate rank; compound formulas are built from simpler ones through the sentential connectives and the quantifiers.

If a formula contains a variable (x , say) then the variable is called *bound* if it is quantified over; otherwise it is called *free*. A formula with no free variables is usually called a *sentence*.

3. A set L of sentences called *logical axioms*.
4. A distinguished set U of operations called *operations of inference* or *production rules*. This set U is invariably taken as finite. When these rules are applied to sentences, they yield new sentences. A sentence is called *logically provable* or *logically valid* if it can be obtained from the sentences of L by arbitrary application of the rules in U .

Within this framework, further structure is usually required:

5. A set A of sentences, which are regarded as *non-logical axioms*. A sentence is called *derivable* if it can be obtained from the sentences in $A \cup L$ by applications of the rules of U . If A is empty, so that the sentence is derivable from L alone, we are back in the previous situation. Sometimes a sentence derivable from A is called *valid*, or *provable*, from A .

As we see, a theory $\mathcal{J}$ has a substantial amount of anatomy. Let us illustrate some of the anatomical units in the specific case of Euclidean geometry. A variable might be a symbol for “triangle”; a constant would be a symbol for a specific triangle. To say that two triangles are similar would express a two-place predicate. A logical axiom might be: “If equals are added to equals, the results are equal”. A non-logical axiom would be one of the specific geometric postulates; e.g. “Any pair of points

lie on exactly one straight line”, or the Parallel Axiom itself. Thus, if we replace the Parallel Axiom by its negation, we obtain a different theory, with a different set of valid or provable sentences.

In the last analysis, a formal theory is determined by its set of valid sentences. It must be stressed here, as before, that this concept of validity depends on arbitrary choices of L, A and U; there is no way from inside the theory of making these choices in a unique way.

Let us now turn to the concept of a *model* of a theory $\mathcal{J}$. Intuitively, the concept of a model is an attempt to re-introduce some notion of meaning or interpretation into the theory. Indeed, the choice of non-logical axioms in a theory $\mathcal{J}$ is generally made because we have some particular model in mind; the notion of validity in $\mathcal{J}$, which as we saw was arbitrary from within $\mathcal{J}$, can only be effectively made in this way.

In mathematics, we traditionally desire our models to themselves be mathematical objects, belonging to the universe U which was described in Sect. 2.2 above. We will proceed in two stages; the initial stage will be to construct a mathematical *realization* of a theory $\mathcal{J}$, and then we shall say which of these realizations are in fact *models*.

Accordingly, let U be a set in u ; we shall call U the *universe* of the realization. Let us now enumerate all the non-logical constants of our theory $\mathcal{J}$ in some fashion; say as a sequence

$$c_1, c_2, \dots, c_n, \dots$$

With each of these C_i , we are going to associate a structure involving U . Specifically, if C_i is an r -place predicate, we shall associate C_i with an r -ary relation on U ; i.e. with a subset $R_i \subseteq U \times U \times \dots \times U$, where there are r terms in this cartesian product. If C_i is an r -ary operation, we shall associate C_i with an r -ary operation on U ; i.e. with a mapping

$$\Phi_i : \underbrace{U \times U \dots \times U}_{r \text{ times}} \rightarrow U.$$

On this basis, we can in the obvious way translate sentences from the theory $\mathcal{J}$ into propositions about this system Σ of relations and operations on U . A sentence of $\mathcal{J}$ is said to be satisfied in Σ if it translates into a true proposition.

The system Σ of relations and operations on a set U is thus what we shall call a realization of $\mathcal{J}$. A realization becomes a *model* of $\mathcal{J}$ when the set of all valid sentences in $\mathcal{J}$ are satisfied in Σ . We remark once more that in practice, we generally choose a particular notion of validity in $\mathcal{J}$ because we want a certain mathematical system to be a model; formally, however, the above definition of model is always meaningful, however validity may be defined in $\mathcal{J}$.

In any case, it is clear that once a modeling relation between $\mathcal{J}$ and Σ has been established, it has all of the properties of the modeling relation displayed in Fig. 2.1 above. Specifically, any true proposition in Σ *decodes* into a valid proposition in $\mathcal{J}$, and hence we may make *predictions* about from the properties of its model Σ . It should be noted that we may arrange matters so that a model of a formal theory $\mathcal{J}$ is another formal theory $\mathcal{J}'$; in such a case we impart “meaning” to $\mathcal{J}$

through its encoding into the propositions of another formal theory. In this fashion, we come to some of the principal applications of the concept of a model in formal mathematics: the problem of consistency, which we have already mentioned several times in previous discussions.

In a formal theory, consistency means simply that no sentence and its negation can both be valid. It is a difficult, and indeed in some sense impossible, problem to establish the consistency of a formal theory which is at all mathematically interesting (e.g. such that the real number system can be a model). However, what the theory of models enables us to do is to establish *relative* consistency arguments: if Σ is a model of $\mathcal{J}$, then $\mathcal{J}$ can be consistent only if Σ is consistent. We have already seen specific examples of this kind of argument in our discussion of the models of Lobatchevskian geometry above. There is a very rich literature dealing with relative consistency arguments, which we urge the reader to consult for further examples.

Another principal application of the theory of models has been to a vexatious problem, already touched on above, which is generally called *decidability*. What is at issue here is the following problem: given an arbitrary sentence of a theory $\mathcal{J}$, is there some procedure for effectively deciding whether it is valid or not? In principle, all the valid expressions in such a theory can be effectively generated, one after another; but there are an infinity of them; hence the ability to *generate* them does not provide a means of *recognizing* one. For obviously, no matter how many valid sentences we have already generated, we have no way of knowing whether an arbitrary sentence not already generated ever will be generated. The fundamental distinction between the recognition problem and the generation problem is one of the vexing features of foundation studies and its offshoots; this includes such areas as automatic pattern recognition and cognate areas connected with the theory of the brain.

Let us give one example of a recognition problem which cannot be solved. We saw in Sect. 2.2 above that groups could be defined within the confines of a formal theory, by imposing compatible equivalence relations on a free group. The effect of such equivalence relations is to force certain words of the free group to be synonymous with each other. Let us ask the following deceptively simple question: given an arbitrary relation R , and an arbitrary pair of words w_1, w_2 in the free group, can we effectively decide whether these words are synonymous under R or not? This is the *word problem for groups*, and it can be shown that this problem is *unsolvable*; there is no recognition procedure which can answer this question for us in all cases in a finite number of steps.⁴

It will of course be noticed that the recognition problem is a generalization of the consistency problem; if we had a decision procedure for recognizing valid sentences, we could thereby establish consistency. Thus it is plausible that, just as we can establish relative consistency through the use of models, we can establish solutions to recognition problems in a theory $\mathcal{J}$ if we can solve them in a model Σ of $\mathcal{J}$. In fact, this idea leads to a very interesting theory in itself, concerned with *degrees* of unsolvability⁵; once again, a detailed discussion of such matters is not possible here, but we would encourage the reader to consult the literature.

Example 4. Equivalence Relations and Canonical Forms

We are now going to develop some examples which are going to be of the utmost importance to us in our subsequent work. We will begin with a consideration of a familiar problem of elementary linear algebra, which is based on the Cartesian arithmetization of Euclidean geometry described above.

Just as we identified the Cartesian plane as the set E_2 of all ordered pairs (a, b) of real numbers, so we can identify Euclidean n -dimensional space as the set E_n of all ordered n -tuples $(a_1, \dots, a_n)$ of real numbers. This is not just a set, but in fact inherits definite algebraic and geometric properties from the real numbers; indeed, Euclidean n -space is nothing but the cartesian product of the real numbers with itself n times.

For instance, if $\alpha = (a_1, \dots, a_n)$ and $\beta = (b_1, \dots, b_n)$ are two such n -tuples, we can form the sum $\alpha + \beta$, which is defined to be the n -tuple

$$(a_1 + b_1, a_2 + b_2, \dots, a_n + b_n).$$

This addition is associative, commutative, and in fact turns our space into a commutative group, with the n -tuple $(0, 0, \dots, 0)$ as unit. Moreover, if α is any n -tuple, and r is any real number, we can define the product $r\alpha$ as the n -tuple

$$(ra_1, ra_2, \dots, ra_n);$$

this *multiplication by scalars* distributes over addition; $r(\alpha + \beta) = r\alpha + r\beta$, and has all the other familiar properties.

Let us consider the n -tuple $e_1 = (1, 0, \dots, 0)$. The set of all scalar multiples of the form re_1 is a *line* in our space. Likewise, if e_i is the n -tuple with zeros in all but the i^{th} position, and 1 in that position, then the set of all multiples of the form re_i is another line in the space.

These lines play a distinguished role, because we can express every n -tuple α uniquely in the form

$$\alpha = a_1e_1 + a_2e_2 + \dots + a_ne_n. \quad (3.1)$$

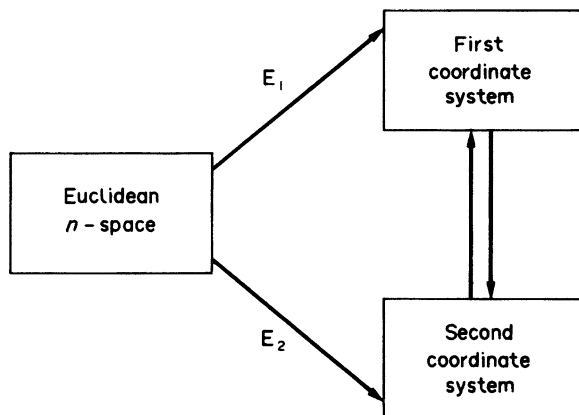
In fact, under this interpretation, the numbers a_i become the co-ordinates of α , referred to the lines defined by the e_i ; these lines are then the *co-ordinate axes*; the n -tuples $e_1, \dots, e_n$ form a *basis* for E_n . Thus, labelling a point in Euclidean n -space by an ordered n -tuple of numbers presupposes a specific choice of co-ordinate axes, or basis, in that space.

If we choose another set of co-ordinate axes, the points in Euclidean n -space would be labelled by different n -tuples. Thus we can ask the following sort of question: if we label the points of E_n in two different ways by choosing two different sets of co-ordinate axes, how are these labellings related to each other?

We can express this question in diagrammatic form, as indicated in Fig. 3.2 below:

In this diagram, each point P of Euclidean n -space is given two distinct encodings $E_1(P)$, $E_2(P)$ into the set of ordered n -tuples, by virtue of two different choices

Fig. 3.2



of co-ordinate axes. In this case, we would expect complete correspondences between these two encodings; these correspondences would amount to one-one transformation of the set E_n of ordered n -tuples onto itself. Of course we know that this is the case; all such encodings are thus *equivalent* to each other.

Let us look at this matter in more detail; in particular, let us see how these correspondences themselves can be expressed, with the aid of a Cartesian encoding. Let $e_1, \dots, e_n$ be as above, and let $e'_1, \dots, e'_n$ define the new co-ordinate axes. Now each of the e'_i can be referred back to the original axes; i.e. we can write

$$e'_i = p_{i1}e_1 + p_{i2}e_2 + \dots + p_{in}e_n. \quad (3.2)$$

Thus, expressing the new axes in terms of the old involves the n^2 numbers p_{ij} ; these numbers can be arranged in the familiar way into a *matrix* which we shall call P . The numbers in this matrix, which are the co-ordinates of the new basis elements e'_i in the original co-ordinate system, clearly also depend on that co-ordinate system.

This matrix P represents, in a sense, all we need to know about transforming co-ordinates, because once we know how the e_i transform to e'_i , we can express any n -tuple $\alpha = a_1e_1 + \dots + a_ne_n$ in the new co-ordinates.

Specifically, let us suppose we can write

$$\alpha = a_1e_1 + a_2e_2 + \dots + a_ne_n \quad (3.3)$$

in the original co-ordinate system. In the new co-ordinate system, we will have

$$\alpha = a'_1e'_1 + a'_2e'_2 + \dots + a'_ne'_n. \quad (3.4)$$

To find the a'_i , we need to express the e_i in terms of the e'_i ; this is done analogous to (3.2) by finding the numbers q_{ij} such that

$$e_i = q_{i1}e'_1 + q_{i2}e'_2 + \dots + q_{in}e'_n \quad (3.5)$$

The numbers q_{ij} are related to the p_{ij} as follows: the two matrices $P = (p_{ij})$ and $Q = (q_{ij})$ are *inverses* of each other with respect to matrix multiplication; i.e.

$$Q = P^{-1}.$$

In any case, if we now substitute (3.5) in (3.3), then collecting terms and comparing the resulting coefficients of the e_i' with (3.4), we find

$$a_i' = \sum_{j=1}^n a_{ij} a_j \quad (3.6)$$

So far, we have regarded such matrices as representing different encodings of the *same* point of E_n , referred to different choices of co-ordinate systems. However, we can equally well interpret them as *moving* the points of E_n , with the co-ordinate system held fixed. In that case a relation like (3.6) expresses the co-ordinates of the point to which a particular point with co-ordinates $(a_1, \dots, a_n)$ is moved. Under this interpretation, the matrices we have introduced represent *motions* of E_n . Thus, using the familiar matrix notation, we can write (3.6) as

$$\alpha' = Q\alpha = P^{-1}\alpha, \quad (3.7)$$

where P is the matrix which describes how the basis vectors e_i move under the transformation according to (3.2) above.

Thus, under this last interpretation, we see that the *motions* of E_n encode as matrices. But the entries of such matrices themselves depend upon the choice of co-ordinate system we have made in E_n ; if we choose a different co-ordinate system, we will obtain a different matrix, i.e. a different encoding, for the same motion of E_n . Thus, just as before, we may ask the following question: *under what circumstances do two different matrices represent encodings of the same motion?*

This question can be easily answered from (3.7). Suppose that A is a matrix, and that we can write

$$\alpha = A\beta \quad (3.8)$$

in some initial co-ordinate system $e_1, \dots, e_n$. If we then change co-ordinates to $e_1', \dots, e_n'$, then from (3.7) and (3.8) we can write

$$\alpha' = P^{-1}AP\beta' \quad (3.9)$$

where P is the appropriate matrix defining the co-ordinate transformation according to (3.2). This relation means that the matrix

$$P^{-1}AP$$

represents the same transformation in the new co-ordinates that the matrix A represented in the original co-ordinates.

Thus, if A and B are two matrices such that

$$B = P^{-1}AP \quad (3.10)$$

for some co-ordinate transformation P , they can be regarded as representing different encodings of the same motion, or transformation, of E_n . Under these circumstances, we say that A and B are *similar matrices*.

If we now look at the set of all square matrices of n^2 entries as itself a mathematical object, it is easy to see that (3.10) represents a binary relation between pairs of matrices A, B . In fact, this relation is an equivalence relation, as we might expect; reflexivity and symmetry are immediate, and transitivity follows from the ordinary properties of matrix multiplication (and the elementary theorem that the composition of the two co-ordinate transformations is again a co-ordinate transformation). Hence, the set of motions, or linear transformations, on E_n , is simply the set of all such square matrices, reduced modulo this equivalence relation.

Much of linear algebra is concerned intimately with this equivalence relation of similarity between matrices, and with the associated *decision problem* of deciding when two given matrices A, B are in fact similar. Since this circle of ideas will be vital to us in subsequent chapters, we will discuss it in some detail in this relatively familiar context.

Let us denote the set of all $n \times n$ square matrices by U_n , and let us denote the similarity relation by R . We wish to construct an effective *decision procedure* to decide when a pair of given matrices A, B in U_n are similar; i.e. fall into the same equivalence class under R .

One way to approach this problem involves seeking *invariants* of R . Let us suppose that $f : U_n \rightarrow \mathbb{R}$ is a mapping of matrices into real numbers. If this mapping f happens to be *constant on the equivalence classes of R* , then f is called an invariant of R . Specifically, if f is such an invariant, and if A, B are two matrices such that $f(A) \neq f(B)$, then A and B *cannot be similar*; i.e. the condition $f(A) = f(B)$ is a necessary condition for similarity. It is not in general a sufficient condition, unless f happens to take different values on *each* equivalence class. In that case, we find that our relation R on U_n is precisely the same relation as what we called R_f in Sect. 2.2 above; but in general, f is an invariant if every equivalence class of R_f is the *union* of classes of R (or, stated another way, if R *refines* R_f). We can thus solve the decision problem for equivalence of matrices if we can find enough invariants $f_1, f_2, \dots, f_m$ to satisfy the condition that $f_i(A) = f_i(B)$ *if and only if* A and B are similar. More explicitly, given any two matrices, we have only to compute the invariants $f_i(A), f_i(B)$. If these numbers are different for any $i = 1, \dots, m$, the matrices cannot be similar; if they are the same for each i , the matrices must be similar. In this case, $f_1, \dots, f_m$ is called a *complete set of invariants*.

We remark here, for later exploitation, that such invariants play a role analogous to the qualities or observables of natural systems; the machinery for their computation or observation are analogous to physical meters; a complete set of invariants is analogous to a set of state variables.

In the case of similarity of matrices, the finding of an appropriate set of invariants is intimately associated with the reduction of matrices to *canonical forms*. The question of finding canonical forms has two distinct aspects. Quite generally, if we are given any equivalence relation on a set, it is often helpful to be able to pick a specific representative from each equivalence class in a canonical way. A set of such representatives is then obviously in one-one correspondence with the quotient set modulo the given equivalence relation. In the present case, a set of such canonical representatives would itself be a set of matrices, and not a set of equivalence classes of matrices; this is a great convenience for many purposes. The second aspect is that if such a representative is chosen in a natural way, we can also solve the decision problem for two matrices A, B by reducing each of them to their canonical form; the matrices can be similar if and only if the canonical forms turn out to be identical.

The basic idea behind expressing a given matrix in a canonical form is, in effect, to find that co-ordinate system or basis of E_n in which that matrix is encoded in an especially simple fashion. To do this, of course, requires us to specify what we mean by “simple”. There are a variety of criteria which can be employed for this purpose; the best-known of these leads to the familiar *Jordan Canonical Form*. It so happens that, in the process of establishing this form, we simultaneously obtain a desired complete set of invariants. These invariants turn out to be the *eigenvalues* of a given matrix, together with their *multiplicities*. Once again, we shall omit the details, which can be found in any text on linear algebra.

The reader may now with some justice inquire what this discussion has to do with the modeling relation. The answer is this: we are going to take the point of view that *two similar matrices are models of each other*, in a sense now to be made precise.

Let A, B be matrices of the type under consideration. Then we have seen that each of them may be regarded as a motion; i.e. a transformation of E_n into itself. If α is an element of E_n , which we may think of as a possible *input* to the transformation, then $A\alpha$ is the corresponding *output*. (The reader should compare this usage with that employed in discussing linear input-output systems in Sect. 1.2 above.)⁶

Let us express this observation in a convenient diagrammatic form. If A and B are similar matrices, then we can find a co-ordinate transformation P such that (3.10) holds. If we regard A and B as transformations of E_n , then (3.10) may be re-expressed as a *diagram of transformations* of the form

$$\begin{array}{ccc}
 E_n & \xrightarrow{A} & E_n \\
 P \downarrow & & \downarrow P \\
 E_n & \xrightarrow{B} & E_n
 \end{array} \tag{3.11}$$

This diagram has the fundamental property that the composite mappings PA and BP are in fact the same mapping. A diagram with this property is called *commutative*. Thus, the relation of similarity between matrices can be expressed through the existence of a commutative diagram of the form (3.11).

Let us further look upon P as an *encoding* of the inputs to A into the inputs to B , and the inverse mapping P^{-1} as a *decoding* of the outputs of B into the outputs of A . The commutativity of the diagram then becomes precisely the statement that B is a model of A . Likewise, we can regard P^{-1} as an encoding of the inputs to B onto the inputs to A , and its inverse P as a decoding of the outputs of A onto the outputs of A . By the same argument, A is also a model of B ; hence in this case, the two matrices A, B are, as claimed, models of each other. Moreover, if we have a canonical form A^* for the equivalence class of similar matrices to which A belongs, then A^* is simultaneously a model for every matrix in that class.

Let us introduce some terminology which will be important to us subsequently. If α is regarded as an input to A , then we shall call $P(\alpha)$, which is its encoding into an input to B , as that input which *corresponds* to α . If $A(\alpha) = \alpha'$, then we shall say that $P(\alpha')$ likewise *corresponds* to α' . The commutativity of the diagram (3.11) then says precisely that A, B map corresponding elements onto corresponding elements. As we shall see, this kind of situation arises ubiquitously in the modeling of natural systems, under such guises as Laws of Dynamical Similarity or Laws of Corresponding States.

*Example 5. A Generalization: Structural Stability*⁷

The situation described in the preceding example can be vastly generalized. Some of the generalizations are among the deepest and most interesting areas of the contemporary mathematics. For instance: instead of looking at E_n , we may look at more general metric spaces (i.e. sets on which a distance function, and hence a metric topology, are defined). Thus, let X, Y be metric spaces. Let $H(X, Y)$ denote the set of continuous mappings or functions from X into Y . Let us say that two mappings f, g in $H(X, Y)$ are *equivalent* if we can find a pair of one-one mappings $\phi : X \rightarrow X, \psi : Y \rightarrow Y$ such that the diagram

$$\begin{array}{ccc}
 X & \xrightarrow{f} & Y \\
 \phi \downarrow & & \downarrow \psi \\
 X & \xrightarrow{g} & Y
 \end{array} \tag{3.12}$$

is commutative. It will be observed that this diagram (3.12) is an immediate generalization of (3.11), and the verification that the relation of equivalence between mappings in $H(X, Y)$ is indeed an equivalence relation on $H(X, Y)$ proceeds in exactly the same way.

Thus, $H(X, Y)$ is partitioned into equivalence classes. A major question that becomes the analog of the decision problem for matrices: given any particular mapping f in $H(X, Y)$, to which equivalence class does it belong? When are two mappings equivalent?

This question is rendered still more interesting by the following fact. The set $H(X, Y)$ of continuous mappings of X into Y is not *just* a set; there is a sense in which it inherits a further structure from the fact that X and Y are metric spaces. In particular, there are a variety of distinct and important ways in which $H(X, Y)$ can itself be made into a metric space; i.e. in which a distance function can be defined in $H(X, Y)$ itself.

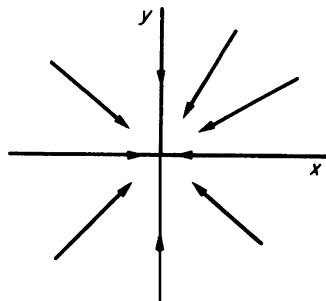
Thus if $f \in H(X, Y)$, we can talk about the mappings f' which are *close* to f , in the sense that the distance between f' and f is small. Indeed, we can regard f' as a “perturbation” of, or an approximation to, a nearby mapping f . The main feature of an approximation is this: we would expect that important mathematical properties of f are also manifested by sufficiently good approximations to f ; i.e. differing from f by a sufficiently small amount. In particular, we might expect that if f' is close to f in $H(X, Y)$, then f and f' are equivalent.

The fact is that this expectation turns out not to be in general true, no matter how the distance function on $H(X, Y)$ is defined. If it is true that, under a particular distance function, all nearby mappings f' are indeed equivalent to f , then f is called *stable*, or *structurally stable*. Some of the deepest questions of mathematics are associated with this notion of stability, which in turn reflects the interplay of the relation of equivalence on $H(X, Y)$ with a notion of closeness or approximation on $H(X, Y)$.

Speaking very roughly for a moment, the question of stability of mappings we have just raised boils down to this. We have already seen that the relation of equivalence involves a precise sense in which one mapping can be regarded as a model of another. Likewise, the notion of closeness or approximation also refers to a modeling situation; namely, to the idea that mappings which are sufficiently close to one another in a topological sense are also models of each other. The fact that there are mappings which are not stable thus means that *these two notions of model are essentially different from one another*. We can thus see emerging here a specific example of the situation diagrammed in Fig. 2.4 above, in which *we cannot establish a correspondence between two models of the same object simply from the fact that they are both models*. This is the situation we called Case C2 in the previous discussion, and we see that it arises here in a completely mathematical context. As we noted previously, such a situation is intimately connected with ideas of reductionism; in this case, we see that we cannot reduce the topology on $H(X, Y)$ to equivalence, nor vice versa; the two notions are in general logically independent of one another.

We also pointed out that this situation is intimately related to the notion of *bifurcation*. Indeed, let f be a mapping in $H(X, Y)$ which is not stable. Then however close we come to f , in the topology of $H(X, Y)$, there will be mappings of f' which are closer still to f and which belong to a different equivalence class. This is the same as saying that certain arbitrarily small perturbations of f will manifest essentially different properties than f does; this is precisely what

Fig. 3.3



bifurcation means. We also see clearly how a bifurcation represents a situation in which two different representations of the same object (here the mapping f) become logically independent of one another under these circumstances.

Let us describe here one specific context in which these ideas manifest themselves. Consider a simple linear dynamical system, of the form

$$\begin{aligned}\frac{dx}{dt} &= ax + by \\ \frac{dy}{dt} &= cx + dy\end{aligned}\tag{3.13}$$

We shall consider two situations: (a) the eigenvalues of the system matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

are real and negative; (b) the eigenvalues of the system matrix are pure imaginary.

In each case, there is a unique steady state, at the origin of the state space. Near the origin, in case (a), the trajectories are essentially as shown in Fig. 3.3: i.e. the origin is a stable node. The condition that the eigenvalues of A be real and negative may be expressed, in terms of the matrix coefficients, as a pair of inequalities:

$$\begin{aligned}(a + d)^2 &> 4(ad - bc) \\ (a + d) &< 0\end{aligned}$$

Suppose we now perturb the system, by replacing the coefficients in the matrix A by new values a' , b' , c' , d' , which are close to the original values. It is evident that, if this perturbation is sufficiently small, it will not affect the above inequalities; the perturbed system will have essentially the same dynamical behavior as the original one.

Now let us turn to case (b). In this case, which is essentially that of the undamped harmonic oscillator in one dimension, the trajectories will appear as shown in Fig. 3.4 below:

Fig. 3.4

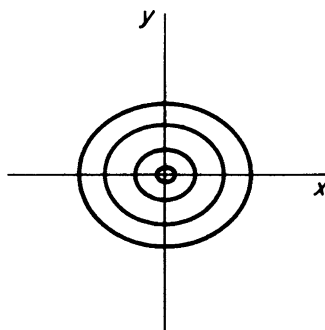
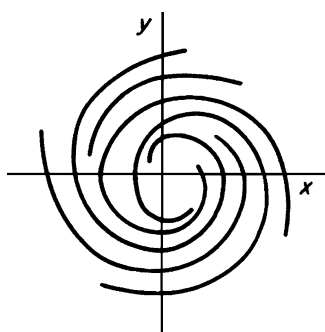


Fig. 3.5



The trajectories here are closed orbits; concentric ellipses; the origin is a *center*, and is neutrally stable. The condition that the eigenvalues of the system be pure imaginary are now

$$(a + d) = 0$$

$$(ad - bc) > 0.$$

The first of these conditions is an *equality*. Thus, if we now perturb this system as we did the preceding one, by replacing the coefficients a , b , c , d by new values close to the original ones, we see that in general, no matter how small a perturbation we impose, the equality

$$a' + d' = 0$$

will in general *not hold*. The eigenvalues of the new system will then not be pure imaginary, but will have some non-zero real part. This is sufficient to turn the trajectories of the system from the original closed, neutrally stable orbits into open spirals, which approach or diverge from the origin according to whether the sign of the real part of the eigenvalues is positive or negative; this behavior is shown in Fig. 3.5:

Thus in this case, an arbitrarily small perturbation can turn the origin from a neutrally stable center to a (stable or unstable) *focus*.

The first case we have discussed is an example of a structurally stable system. Let us recast its properties into the form exhibited in (3.12) above. As we saw earlier, we could regard a dynamics like (3.13) as generating a flow on the state space (in this case, on E_n); in turn, this flow could be represented by a one-parameter family of mappings $f_t : E_2 \rightarrow E_2$. Likewise, a perturbation of the system will generate a corresponding one-parameter family of mappings $f'_t : E_2 \rightarrow E_2$. The structural stability of the first system means that the character of the trajectories, although modified in a quantitative sense when we replace f_t by f'_t , remains qualitatively the same. In mathematical terms, this means that we can establish continuous one-one mappings of E_2 onto itself which *preserve the trajectories*. In general, then, if f_t is structurally stable, we can find continuous, one-one mappings ϕ, ψ such that the diagram

$$\begin{array}{ccc}
 E_2 & \xrightarrow{f_t} & E_2 \\
 \phi \downarrow & & \downarrow \psi \\
 E_2 & \xrightarrow{f'_t} & E_2
 \end{array} \tag{3.14}$$

commutes for every t . That is, the original and the perturbed systems are in this sense *models of each other*.

The second case is an example of a structurally unstable system. Indeed, it is clear from comparing Figs. 3.4 and 3.5 that we cannot hope to find continuous maps on E_2 which preserve trajectories.

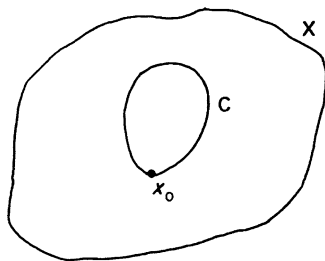
Thus, an arbitrarily small perturbation of a structurally unstable system can cause it to assume essentially new properties. In this case, there is a bifurcation of the following character: some perturbations of the neutrally stable system will result in a stable system; others will result in an unstable system. These two situations are not models of each other; nor is either of them a model for the original unperturbed situation.

Before concluding this section, we shall make one further comment about diagrams like (3.12) and (3.14). If in diagram (3.14) we regard f'_t as a perturbation of f_t , the commutativity of the diagram asserts precisely that *such a perturbation can be "annihilated" through suitable co-ordinate transformations ϕ, ψ* . That is, if such a perturbation is imposed, in the sense that the map f_t is replaced by a new map f'_t , then we can recapture the original situation simply by replacing all elements in the domain and range of f_t by their *corresponding* elements via the encodings ϕ, ψ . This too will prove to be an essential characteristic of modeling relations in general, with which we will meet constantly in subsequent chapters.

Example 6. Algebraic Topology and the Theory of Categories

In the preceding example, we briefly discussed the concept of the stability of mappings $f : X \rightarrow Y$ of a metric (topological) space X into a similar space

Fig. 3.6



Y . We saw that stability reflected the extent to which two mappings which are close to one another in a suitable topology of $H(X, Y)$ are also members of the same conjugacy class. We are now going to briefly describe a closely related circle of ideas, applied to another kind of classification problem.

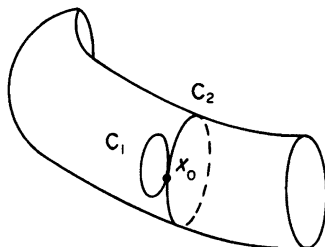
One of the basic problems of topology is again a decision problem: given two spaces X, Y , to decide whether they are *homeomorphic*. A homeomorphism $f : X \rightarrow Y$ is a mapping which is one-one and onto, and such that both f and its inverse mapping f^{-1} are continuous. Two homeomorphic spaces can be considered as abstractly the same; any topological property of one can be transformed into a corresponding property of the other. Thus, two homeomorphic spaces are *models of each other*; any topological property of one of them can be encoded (via a homeomorphism) into a corresponding property of the other, and conversely.

We can try to approach this decision problem for topological spaces in exactly the same way we approached the decision problem for similar matrices; we may attempt to construct suitable *topological invariants*. These invariants need not be numbers; they may be other mathematical objects $G(X)$ associated with a topological space X in such a way that if $G(X)$ and $G(Y)$ are different, then X and Y cannot be homeomorphic. In what follows, we shall consider a few of the simpler topological invariants of this type. We shall see (a) that such a topological invariant $G(X)$ is *itself a model of X* , in that it represents an encoding of properties of X into $G(X)$, and (b) that the general procedure by which a space X is associated with the invariant $G(X)$ leads us directly to the *theory of categories* which was mentioned earlier. In this sense, we shall see that the theory of categories can itself be regarded as a *general theory of modeling*, built up entirely within mathematics itself.

As we stated above, the topological invariants we shall describe are not themselves numbers, but rather are certain algebraic systems; in fact, they are groups. The basic idea behind the construction of these invariants goes back to Poincaré. We shall describe them in an informal fashion; full details can be found in any text on algebraic topology.⁸

Let us first describe the *fundamental group*. Suppose that X is a suitable topological space; e.g. a region of the plane, or the surface of a sphere, or the surface of a torus. Let us fix a point x_0 in X , and let us consider curves C of the type shown in Fig. 3.6 below:

Fig. 3.7



The curve C is just a closed curve which both initiates and terminates at our chosen point x_0 . We shall also provide such a curve C with an *orientation*, as follows: we can, starting at x_0 , traverse the curve either in a clockwise or in a counter-clockwise direction and return to x_0 ; if we choose the former, we shall say that the curve is given a positive orientation; if we choose the latter, we shall say that the curve is given a negative orientation.

Given two such curves, say C_1 and C_2 , we can obtain another such curve by traversing one of them from x_0 back to x_0 in a given orientation, and then traversing the other from x_0 back to x_0 . In this way, we can introduce a binary operation into the set of such curves. It is easy to see that under this binary operation, the set of curves becomes a group; the inverse of a curve C is the same curve traversed in the opposite orientation, and the unit element for the group is just the point x_0 itself.

Now we shall introduce a compatible equivalence relation on this group, in the following intuitive way. First, let us suppose that such a curve C can be continuously *deformed* down to the point x_0 ; i.e. that we can make C smaller and smaller in a continuous fashion, without ever leaving the space X . Intuitively, this means that the “interior” of the curve C is simply connected; or that C does not enclose a “hole”. We shall say that such a curve is “homotopic to zero”. For instance, if X is the interior of a circle, or is the surface of the sphere, then any such path C is homotopic to zero. On the other hand, if X is the surface of a torus, then the curve C_1 shown in Fig. 3.7 is homotopic to zero, but the curve C_2 is not:

if the composite curve $C_1 - C_2$ is homotopic to zero. It is easy to show that homotopy is an equivalence relation among such curves, and that it is compatible with the binary operation we have introduced. If $K(x_0)$ is the set of curves, and R is the homotopy relation, then $G(x_0) = K(x_0)/R$ is a group; the *fundamental group*, or *first homotopy group*, of the space X (it should be noted that a simple argument is required to show that this group does not in fact depend on the choice of x_0).

We might mention that the homotopy relation just introduced is really intimately connected with conjugacy of mappings. For we may regard any curve in X as a mapping f of the unit interval

$$I = \{r \mid 0 \leq r \leq 1\}.$$

into X . The curves originating and terminating at x_0 are precisely the ones for which $f(0) = f(1) = x_0$. Thus, homotopy of curves becomes an equivalence relation between certain mappings in the set of mappings $H(I, X)$. Moreover, we can represent the “deformation” of curves in X by means of curves in $H(I, X)$; two mappings f, g are joined by a continuous curve in $H(I, X)$ if there is a continuous mapping $\phi : I \rightarrow H(I, X)$ such that $\phi(0) = f$, $\phi(1) = g$; the existence of such a ϕ establishes the homotopy relation between f and g . It is not hard to show that this can be translated into a conjugacy diagram of the form

$$\begin{array}{ccc}
 I & \xrightarrow{f = \phi(0)} & X \\
 \alpha \downarrow & & \downarrow \beta \\
 I & \xrightarrow{g = \phi(1)} & X
 \end{array}$$

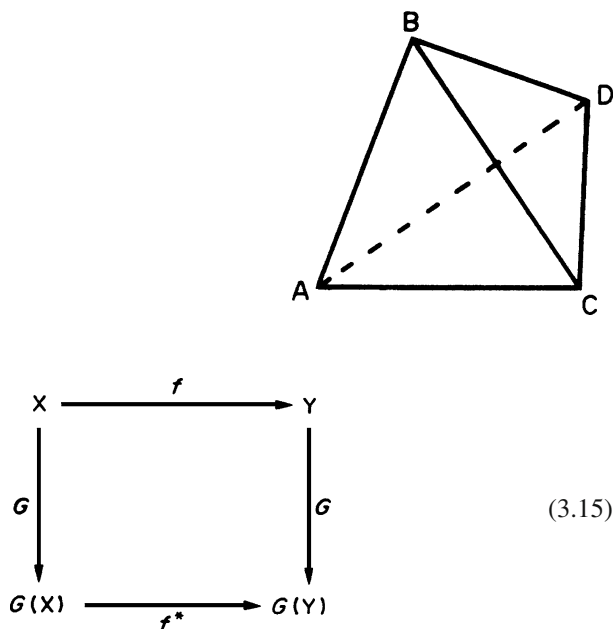
relating f and g . Thus, homotopy of curves can be regarded as a special case of conjugacy of mappings.

In any case, once we have shown that the fundamental group is independent of our choice of x_0 , it becomes in effect a function of the space X alone, which we may denote by $G(X)$. It can readily be shown that *homeomorphic spaces are associated with isomorphic fundamental groups*; thus if two spaces X, Y are such that $G(X)$ and $G(Y)$ are not isomorphic, it follows that X and Y cannot be homeomorphic. Thus $G(X)$ is a topological invariant; the isomorphism of fundamental groups is a necessary (but not sufficient) condition for X and Y to be homeomorphic.

As an obvious application, we can see immediately that the surface of a torus and the surface of a sphere cannot be homeomorphic. This is because on the surface of a sphere, every curve is homotopic to zero, while the torus admits curves which are not homotopic to zero, as we have seen.

One further crucial property of this construction must now be described. Suppose X and Y are topological spaces, and that $G(X), G(Y)$ are their associated fundamental groups. If we are given a continuous mapping $f : X \rightarrow Y$, then clearly f maps every curve in X into some corresponding curve in Y . It is again not hard to show that a continuous mapping preserves the homotopy relation between curves; if two curves C_1, C_2 are homotopic in X , then their images $f(C_1), f(C_2)$ are homotopic in Y . From this it follows that, associated with the continuous mapping f , there is a corresponding group homomorphism $f^* : G(X) \rightarrow G(Y)$. This fact may be represented by a diagram

Fig. 3.8



Here G is regarded as a correspondence (a *functor*) which associates with any space X its corresponding fundamental group. As we shall see in a moment, it is appropriate to write $f^* = G(f)$ as well. It must be carefully noted that the arrows labelled G in (3.15) are *not mappings* in the sense that f and f^* are mappings; G does not associate *elements* x in X with corresponding elements $G(x)$ in $G(X)$. Rather, G is a rule which shows how the *group* $G(X)$ is associated with the *space* X . Thus this diagram (3.15) has a different significance from, say, the conjugacy diagrams (3.11) and (3.12) above.

It should be noted that the construction we have just sketched can be vastly generalized, to yield a sequence of *higher homotopy groups*. The operations which associate a space X to these higher homotopy groups have the same properties which we have just observed for G ; in particular, they all yield diagrams like (3.15). We refer the reader to the literature for further details about these other topological invariants.

The study of what was traditionally called combinatorial topology yields other kinds of groups which are topological invariants; the *homology groups*, and their duals, the *cohomology groups*. We can only here give the flavor of the ideas leading to their construction. To take a very simple example, consider the tetrahedron $ABCD$ shown in Fig. 3.8 below:

This tetrahedron is bounded by four *triangles*; these in turn determine six distinct *edges*, and four *vertices*. These will be called the *elements* of the tetrahedron; the triangles are two-dimensional elements, and the vertices are zero-dimensional.

Let us consider one of the triangles, say ABC . We can give the triangle an *orientation* by agreeing to traverse its boundary in a given direction; say clockwise

or counter-clockwise. Two triangles are said to be *coherently oriented* if they are traversed in the same direction. It should be noticed that if two triangles with a common edge are coherently oriented, then the common edge is traversed in *opposite* directions in the two triangles. The reader should verify that the four triangles comprising the faces of our tetrahedron can be coherently oriented; it turns out that this property is itself a topological invariant.

We can also orient *edges*, in such a way that if, say, the edge AB is traversed from A to B in an orientation of ABC, then this orientation is inherited by the edge.

If we call one orientation positive (+) and the other negative (−), then we can designate the bounding faces of the tetrahedron as a *sum* of coherently oriented triangles:

$$ABCD = ABC + BDC + DBA + ACD. \quad (3.16)$$

Indeed, we can in this way talk about *any* linear combination of these triangles, where the coefficients are any positive and negative integers, or zero. Such a linear combination of two-dimensional elements is an example of a *2-chain*.

The particular 2-chain exhibited as (3.16) above is a special one; it represents the boundary of the tetrahedron. Now we can look at its boundary as a particular 1-chain; i.e. as a sum of one-dimensional elements, each with the proper orientation. If we write down this 1-chain explicitly, we see that it is *zero*. This illustrates the crucial fact that the *boundary of a boundary is always zero*.

Now a chain which is a boundary of another (and hence such that its own boundary is zero) is called a *cycle*.

The set of all *r*-chains of our tetrahedron (here $r = 0, 1, 2, 3$) is clearly an additive group under the binary operation $+$ exemplified in (3.16) above. It can readily be verified that the set of all cycles is a *subgroup*. Let us call two *r*-chains *homologous* if they differ by a cycle; a cycle itself is thus *homologous to zero*. This relation of homology is easily seen to be an equivalence relation compatible with the group operation; hence we can form the quotient group. This quotient group is the *r*-dimensional homology group. These homology groups are the prototypes of the topological invariants we seek.

In general, we can construct such homology groups for any topological space which we can dissect into triangles, tetrahedra and their higher-dimensional analogs, by generalizing the above discussion in the appropriate way. This approach to such spaces is the essence of combinatorial topology, and hence of algebraic topology.

If $K^r(X)$ is the *r*-dimensional homology group of a space *X*, then it is not hard to verify that a diagram of mappings essentially similar to (3.15) can be built (see (3.17)). As before, any continuous mapping $f : X \rightarrow Y$ induces a corresponding group homomorphism $f^* : K^r(X) \rightarrow K^r(Y)$. The homology groups are also topological invariants; if *X* and *Y* are homeomorphic, then their homology groups $K^r(X)$, $K^r(Y)$ are isomorphic as groups.

$$\begin{array}{ccc}
 X & \xrightarrow{f} & Y \\
 \downarrow \kappa^r & & \downarrow \kappa^r \\
 \kappa^r(X) & \xrightarrow{f^*} & \kappa^r(Y)
 \end{array} \tag{3.17}$$

As noted above, we are going to take the point of view that algebraic objects like $G(X)$ and $K^r(X)$ are *models* of the space X ; i.e. that *they encode in their algebraic properties particular features of the topological space X* . We are now going to describe how the relation between a space X and its algebraic models can be described in formal terms. This will lead us essentially into the theory of categories; in this theory, diagrams like (3.15) and (3.17) will play the fundamental role.

Let us recall that in Sect. 2.2 above, we briefly touched on the concept of a category.⁹ At that time, we regarded a category as the totality of mathematical structures belonging to a certain type which belong to a particular mathematical universe u , together with the structure-preserving mappings through which different structures in the category could be compared.

One category we might consider, then, would be the category of all topological spaces in u . This category would comprise two distinct elements of structure: (a) its *objects*; in this case, all the topological spaces in u ; (b) its *mappings*; in that, for every pair X, Y of topological spaces, we would associate the set $H_c(X, Y)$ of all *continuous* mappings from X to Y . Thus, the category itself becomes a mathematical object, bearing definite mathematical properties. Most of these revolve around the composition of mappings belonging to the category; for instance, if $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are continuous mappings, we know that there is a definite mapping $gf: X \rightarrow Z$ which is also in the category. These properties can be formulated in a completely abstract way (cf. the discussion of Example 3 above) to define an axiomatic approach to categories as mathematical objects in their own right.

Likewise, we could consider the category of all groups in u . The objects of this category would be individual groups; to each pair G_1, G_2 of groups we associate the set $H_g(G_1, G_2)$ of all group homomorphisms of G_1 into G_2 .

If such categories are themselves to be regarded as mathematical objects, bearing definite internal structures, we can study them in the same way we study any other mathematical objects; namely, by defining structure-preserving mappings between them. If a, b are categories, then, we need to define what we mean by such a structure-preserving mapping $F: a \rightarrow b$. Such a structure-preserving map between categories will be called a *functor*.

A category contains two items of structure: its objects, and the mappings which relate these objects. A functor F must be specified according to its behavior on both. Accordingly, let us define a functor F in two steps: (a) a functor F is a rule which associated, with each object A in a , an object $F(A)$ in b . (b) let $f: A \rightarrow B$ be a

mapping in $\mathbf{a}$. Then a functor F is also a rule which associated with f a mapping $F(f)$ in $\mathbf{b}$, such that

$$F(f) : F(A) \rightarrow F(B).$$

This rule F must respect composition of mappings, in the following sense: if $gf : A \rightarrow C$ is a composite mapping in $\mathbf{a}$, then $F(gf)$ is the same mapping as $F(g)F(f)$ in $\mathbf{b}$. Furthermore, if $F : A \rightarrow A$ is the *identity mapping* of A onto itself, then $F(f)$ is the identity mapping for $F(A)$.

We have essentially defined a particular kind of functor; a *covariant* functor of a single argument. We often need to consider other kinds of functors; e.g. *contravariant* functors. These are defined in the same way as are covariant functors, except that $F(f)$ is required to be a mapping of $F(B)$ into $F(A)$; i.e. contravariant functors reverse the arrows associated with mappings in $\mathbf{a}$.

Since this concept of a functor is so basic, we will give a few simple examples.

1. Let $\mathbf{a} = \mathbf{b} = \text{category of sets}$. If X is a set, let $F(X)$ be the set of all subsets of X . If $f : X \rightarrow Y$ is a mapping, let $F(f) : F(X) \rightarrow F(Y)$ be defined as follows: if S is a subset of X , let $F(f)(S) = f(S)$. It is immediate to verify that F is indeed a covariant functor of the category of sets into itself.
2. Let $\mathbf{a}, \mathbf{b}$ be as in the preceding example, and let us again take $F(X) = \text{set of all subsets of } X$. However, let us now define F on mappings in a different way. If $f : X \rightarrow Y$ is a mapping, we will define $F(f)$ as a mapping *from* $F(Y)$ *to* $F(X)$ according to the following requirement. If S is a subset of Y , then

$$F(f)(S) = f^{-1}(S).$$

Again, it is immediate to verify that F is a functor, but this time a *contravariant* functor.

3. Let us consider some examples of functors of more than one variable. Once again, let $\mathbf{a} = \mathbf{b} = \text{category of sets}$. If X, Y are sets, define

$$P(X, Y) = X \times Y.$$

That is, P associates with a pair of sets X, Y a new set, their cartesian product. To turn P into a functor, we must now describe what it does to mappings. Accordingly, let

$$f : X \rightarrow X', \quad g : Y \rightarrow Y'$$

be mappings. Then we must have

$$P(f, g) : P(X, Y) \rightarrow P(X', Y');$$

we do this by simply defining

$$P(f, g)(x, y) = (f(x), g(y))$$

for all elements $x \in X, y \in Y$. Again, it is immediate to verify that P is a functor under these conditions, covariant in both of its arguments.

4. A somewhat more involved, but equally important functor of two variables is one which enters into the very definition of a category. Specifically, if X, Y are sets, we can consider the new set $H(X, Y)$ of all (set-theoretic) mappings from X to Y as being part of a functor. We will now see how it can be extended to mappings. Namely, we shall define a new mapping $H(f, g)$, where $f : X' \rightarrow X$ and $g : Y \rightarrow Y'$ are given mappings. By definition, $H(f, g)$ must map $H(X, Y)$ into $H(X', Y')$; i.e. given any mapping $\theta : X \rightarrow Y$, we must specify a new mapping $H(f, g)(\theta) : X' \rightarrow Y'$. A moment's reflection reveals that if we define

$$H(f, g)(\theta) = g\theta f$$

then H becomes a functor, contravariant in its first argument, and covariant in its second argument.

From these few examples, it becomes clear that all of the important mathematical constructions are functorial in this sense. Indeed, it turns out that the functorial character of these constructions is the reason for their significance.

In terms of the concepts we have just introduced, the next step should be evident. Namely, the constructions which associate to a topological space X its fundamental group $G(X)$, or its r -dimensional homology group $K^r(X)$, are *functors of the category of topological spaces into the category of groups*. They associate topological spaces with groups, and continuous mappings with group homomorphisms, in such a way that they become covariant functors from spaces to groups. Indeed, it was primarily from a study of this relation that the concept of a category was extracted initially. The detailed study of these functors, from the standpoint of category theory, has grown into an exceedingly rich mathematical theory in its own right: *homological algebra*. We refer the reader to the references for further details.

Before we proceed to investigate the sense in which functors establish modeling relations, there is one more basic concept to be introduced. We have seen that a functor is a vehicle for comparing categories; we now wish to introduce a corresponding vehicle for comparing functors. In doing so, we will introduce the idea of a *natural transformation*. We shall preface our discussion with a few words of motivation.

We saw above that the cartesian product could be regarded as a functor of two variables; with each pair of sets X, Y , we could associate another set $P(X, Y) = X \times Y$. If we interchange the roles of X and Y , we thereby define another functor, which we can call $P'(X, Y) = Y \times X$. Strictly speaking, these two functors P and P' are different. Nevertheless, there is an important sense in which we do not wish to distinguish between $P(X, Y)$ and $P'(X, Y)$, nor between $P(f, g)$ and $P'(f, g)$. This sense clearly has to do with the fact that $P(X, Y)$ and $P(Y, X)$ are always *isomorphic* as sets; there is in fact a *distinguished* isomorphism

$$\phi : X \times Y \rightarrow Y \times X \tag{3.18}$$

defined by writing $\phi(x, y) = (y, x)$. This isomorphism is always present, for any pair of sets X, Y ; therefore it must have something to do with the *category*. In other words, it must be definable in categorical terms, rather than as an *ad hoc* mapping pertaining to individual sets X, Y . When we specify how $P(X, Y)$ and $P'(X, Y)$ are related in general, we will thereby have an example of a natural transformation (in fact, of a natural equivalence) which relates the two functors.

We have just pointed out that, given any pair of sets X, Y , there is a distinguished mapping $\phi(X, Y)$ defined by (3.18). The properties of this mapping can be expressed in diagrammatic form as follows:

$$\begin{array}{ccc}
 P(X, Y) & \xrightarrow{P(f, g)} & P(X', Y') \\
 \downarrow \phi(X, Y) & & \downarrow \phi(X, Y) \\
 P'(X, Y) & \xrightarrow{P'(f, g)} & P'(X', Y')
 \end{array} \quad (3.19)$$

The totality of these mappings $\{\phi(X, Y)\}$ comprises a general way of relating the two functors P, P' ; i.e. of translating or encoding the properties of P into the properties of P' in a systematic way. We shall call such a family of mappings a *natural transformation* between P and P' , and write $\phi : P \rightarrow P'$. In the present case, of course, the encoding can also be reversed, since each $\phi(X, Y)$ is itself an isomorphism, and thus can be inverted. In such a case, the natural transformation becomes a *natural equivalence*. Let us notice explicitly that for any definite pair of sets X, Y , the diagram (3.18) expresses a *conjugacy relation*.

In general, if $F, G : \mathbf{a} \rightarrow \mathbf{b}$ are (covariant) functors, we say that these functors are *naturally equivalent* if for every pair of objects A in $\mathbf{a}$, B in $\mathbf{b}$, we can find isomorphisms $\phi(A, B), \psi(A, B)$ in $\mathbf{b}$ such that the analog of (3.19), namely

$$\begin{array}{ccc}
 G(A) & \xrightarrow{G(f)} & G(B) \\
 \downarrow \phi(A, B) & & \downarrow \psi(A, B) \\
 F(A) & \xrightarrow{F(f)} & F(B)
 \end{array} \quad (3.20)$$

always commutes. The totality of these pairs of mappings constitutes the natural equivalence. If these mappings are not isomorphisms, the totality of these pairs constitutes a natural transformation of F to G .

Let us note one final feature of this concept of natural transformations. If $\mathbf{a}$ and $\mathbf{b}$ are categories, then we can consider the class of all functors from $\mathbf{a}$ to $\mathbf{b}$. To each

pair (F, G) of such functors, we can associate the set of all natural transformations from F to G . It is then easy to verify that the totality of these functors and natural transformations themselves form a category, the *functor category* $\mathbf{U}(a, b)$. Such functor categories will play an important role in subsequent discussions, for the following reason: if we consider $F(X)$ to be a model of X , and $G(X)$ to be another model of X , then a natural transformation between F and G represents a sense in which $G(X)$ is a model of $F(X)$; i.e. the extent to which we can extract a relation between $F(X)$ and $G(X)$ simply from the fact that they are both models of X . As a result, natural transformations have a profound bearing on such problems as reductionism, and conversely, on the possibility of bifurcation between the two encodings F, G of a system X . As we shall see later, such questions all devolve on the specific properties of a functor category. Thus, the above discussion should be considered in the light of the diagram shown in Fig. 2.4 in the preceding section, and the various cases pertaining to that diagram.

Having now described at some length a number of specific examples of encodings and modeling relations occurring entirely within mathematics, let us see what general concepts pertaining to modeling may be distilled from them. To do this, it may be helpful to review our examples and see in each case what role the modeling relation has played in understanding the system being modeled.

Our first example was that of the cartesian synthesis of Euclidean geometry and algebra. The encoding of geometry into algebra and arithmetic was not only a grand intellectual synthesis; it was of the utmost practical importance. For the availability of algebraic manipulative techniques made possible entirely new advances, which would have been difficult or impossible to contemplate in purely geometrical terms. In a certain sense, the inferential rules in the model system (algebra) are infinitely more transparent than the corresponding ones in the system being modeled (geometry). This fact indicates one of the basic features of a modeling relation: via an artful encoding, we can convert features difficult or impossible to extract from the original system into a new form which renders them visible, and amenable to study. Indeed, we may say that perhaps the bulk of modern geometry would have been literally *unthinkable* in the absence of algebraic encoding.

We also, however, see one important *caveat* to be borne in mind, which became especially significant in our discussion of similar matrices. Namely, the particular encoding we choose can introduce *artifacts*. Namely, any cartesian encoding carries with it a reference to a particular co-ordinate system, which of itself has no intrinsic geometric significance. When attempting to do geometry via a cartesian encoding, we must be careful to restrict ourselves to algebraic features which are, in some sense, *independent* of the system of co-ordinates which the encoding imposes on us. Indeed, this is why we must search for invariants and for canonical forms in the first place. The lesson here is the following: when encoding a system S_1 into another system S_2 , we must remember that not every property of S_2 is an encoding of some property of S_1 . In general, S_2 has properties of its own, unrelated to the encoding; we must recognize this fact, and draw our conclusions accordingly.

We see similar lessons emerging in the Euclidean models of non-Euclidean geometry. Here again, the encodings we discussed were of incalculable significance

for geometry; not only did they provide concrete interpretations of what had heretofore been purely formal structures, but they actually served to establish the logical consistency of these geometries relative to the Euclidean. And once again, we can see that such encodings made possible the discovery of many new theorems of non-Euclidean geometry, owing to the relatively much greater transparency of the properties of the model than those of their prototype. We repeat once again: a fundamental feature of the modeling relation is that it can convert relations obscure in the prototypic system into a tangible and graspable form. We will see this transductive feature of modeling relations again and again as we proceed.

Our discussion of similarity, conjugacy and stability brought to the fore another feature of the modeling relation; namely, its role in classification, discrimination and recognition among a set of related objects (matrices and mappings). Here the models were treated as *invariants*; we counted systems alike to the extent that they gave rise to identical models. Moreover, the models themselves could be constructed in a *canonical* fashion. Of particular interest here is the relation of our arguments to the notion of *metaphor*, which we mentioned earlier. In a certain sense, once we know how to construct a canonical form of *any* matrix, for instance, we know how to construct a canonical form for *every* matrix. All we need to know is that some mathematical structure is a matrix, or that it behaves “sufficiently like” a matrix, and we can instantly apply the battery of machinery which yields a canonical model. Indeed, it is precisely in this sense that we can regard any matrix as a metaphor for every other, regardless of its dimensionality or the character of its entries. Precisely this fact is exploited relentlessly in mathematics, as for example in the enormous theory of *representations*¹⁰ of arbitrary groups by matrices; a subject we might well have included among our examples of modeling relations in mathematics.

All of these features manifested themselves most clearly in our discussion of categories. In a sense, all the objects in a given category are metaphors for one another, in the sense that once we know how to encode any one of them into a model, we know how to encode every one of them. For instance, given *any* triangulable space, we can construct the corresponding homology groups. This indeed is the essence of a functor; the construction of a functor on a category simultaneously establishes (a) a sense in which all objects in the category are related to one another, and (b) thereby produces an explicit model of each of them. It must also be emphasized that the functorial relation between, say, a topological space and an algebraic model like a homology group is not a pointwise correspondence between them, but rather embodies the capturing of a certain *global* feature of the space in corresponding global algebraic properties of the group; it is not in principle a reductionistic kind of relation.

In a certain sense, whenever we have established a modeling relation between mathematical systems of differing characters, we have built at least a piece of a functor. Thus we emphasize once more the intimate relation between category theory as a mathematical discipline, and the general theory of modeling. We shall exploit this relation in various important ways as we proceed. Before we do this, however, we shall return once again to a consideration of natural systems, and see how they may be explicitly encoded into formal systems in fruitful ways.

References and Notes

1. A good historical treatment of the basic trends in Euclidean and non-Euclidean geometries may be found in

Greenberg, M. J., *Euclidean and Non-Euclidean Geometries*. Freeman (1974).

The various Euclidean models are succinctly described in

Coxeter, H. S. M., *Non-Euclidean Geometry*. University of Toronto Press.

A monumental and relatively recent study of axiomatic approaches to geometry is

Borsuk, K. and Smidiew, W., *Foundations of Geometry*. North-Holland (1960).

2. The reader may observe that it is perhaps a matter of taste whether we regard the constructions of Beltrami, Klein and Poincaré (and even the arithmetization of Descartes) as models or realizations. Indeed, the mathematical utilization of the term “model”, as we shall see it used throughout the present chapter, is often more closely akin to realization than it is to model. However, the terminology is so deeply entrenched that it is hopeless to change it now. The abstract distinction between the operation M of making a model, and the operation R of constructing a realization, may crudely be put as follows: $MRM = M$, but $RMR \neq R$ in general.

3. The “theory of models” has grown rapidly in the past several decades, and has had an important influence on logic, foundations, set theory and algebra. There are a number of good introductory texts, among which we may mention the following:

Smullyan, R. M., *Theory of Formal Systems*. Princeton University Press (1961).

Robinson, A., *Introduction to Model Theory*. North-Holland (1963).

Kopperman, R., *Model Theory and its Applications*. Allyn & Bacon (1972).

Chang, C. C., *Model Theory*. Elsevier (1973).

A general background text, still of great value for all of the foundational matters we have discussed, is

Kleene, S. C., *Introduction to Metamathematics*. van Nostrand (1952).

4. See Note 4 in Sect. 2.2 above.
5. Roughly, two unsolvable problems are said to be of the same degree if either implies the other. Thus, unsolvable problems come in many degrees. For a treatment of these matters, see
Schoenfield, J. R., *Degrees of Unsolvability*. North-Holland (1971).
6. The usage is in fact identical. This can be seen by writing the elements a_{ij} of a matrix A as $a(i, j)$, thus revealing the true status of the indices as independent variables of which the matrix elements are functions. Likewise, we will write the components x_i of a vector $\vec{x}$ as $x(i)$. Then the action of a matrix A on a vector $\vec{x}$ (the “input”) produces a new vector $\vec{y}$, (the “output”), whose i^{th} component can be written as

$$y(i) = \sum_{j=1}^u a(i, j) x(j).$$

If we now think of the variables i, j as ranging over continua instead of a discrete index set, the above sum goes over into an integral of precisely the form (1.2.11).

7. Stability of mappings is a subject of much current interest. In one sense, it is a straightforward generalization of the study of the following question: given a matrix A , under what conditions is every other matrix sufficiently close to A also similar to A ? If a matrix A satisfies these conditions, then it is *stable* to all sufficiently small perturbations. Stability of mappings underlies bifurcation theory, structural stability of dynamical systems, and catastrophe theory. A good reference is

Guillemin, V. and Golubitsky, M., *Stable Mappings and their Singularities*. Springer-Verlag (1973).

There are no texts devoted to the structural stability of dynamical systems *per se* (a subject which used to be known to engineers as sensitivity analysis, and to biologists as robustness). A good review of the basic approach may be found in Smale, S., *Bull. Am. Math. Soc.* **73**, 747–817 (1967).

8. Good general references for these important matters are

Wallace, A. H., *Algebraic Topology*. Benjamin (1970).

Switzer, R. M., *Algebraic Topology, Homotopy and Homology*. Springer-Verlag (1975).

Bourgin, D. G., *Modern Algebraic Topology*. MacMillan (1963).

For deeper discussions of homotopy, see for example

Hilton, P. J., *An Introduction to Homotopy Theory*. Cambridge University Press (1953).

Hu, S., *Homotopy Theory*. Academic Press (1959).

Gray, B., *Homotopy Theory*. Academic Press (1975).

For homology and cohomology, see

Hilton, P. J., *Homology Theory*. Cambridge University Press (1960).

MacLane, S., *Homology*. Springer-Verlag (1963).

Clearly, the techniques of algebraic topology grew out of geometry. However, it is interesting to note that the machinery characteristic of algebraic topology may be created and applied in purely algebraic situations. This creation of a kind of “fictitious geometry” in a purely algebraic setting is of the greatest interest, and is another application of the making of models of mathematical systems with mathematics itself. A typical reference in this area is

Weiss, E., *Cohomology of Groups*. Academic Press (1960).

It is also interesting to observe that classical homology theory was axiomatized by Eilenberg & Steenrod (*Foundations of Algebraic Topology*, Princeton 1952), and that the homology theory of algebraic systems, differential forms and the like can be regarded as *models* (or realizations) of these axioms.

9. As we have noted, category theory grew out of the construction of algebraic invariants of topological spaces. The paper which founded the subject was Eilenberg, S. and MacLane, S., *Trans. Am. Math. Soc.* **58**, 231–294 (1945). Initially it was very much skewed in these directions, under the guise of *homological algebra* (cf. Cartan, H. & Eilenberg, S., *Homological Algebra*, Princeton 1956), but it soon developed as an independent mathematical discipline, cutting obliquely across all others. As we have noted, we regard it as a general framework for the construction of models of mathematical objects. Some current references are:

Mitchell, B., *Theory of Categories*. Academic Press (1965).

MacLane, S., *Categories for the Working Mathematician*. Springer-Verlag (1971).

Arbib, M. and Manes, E. G., *Arrows, Structures and Functors*. Academic Press (1975).

10. A representation of an arbitrary group is a homomorphism of the group into the group of linear transformations acting on some linear space. Representation theory has an enormous literature, spurred in large part by its applications to physics. Good general references are

Dornhoff, L., *Group Representation Theory*. Marcel Dekker (1971).

Kirillov, A. A., *Elements of the Theory of Representations*. Springer-Verlag (1972).

The reader should also consult the review of Mackey referred to in Note 4 to Sect. 1.2 above.

3.2 Specific Encodings Between Natural and Formal Systems

In the preceding chapter, we reviewed a number of examples illustrating the manner in which formal systems of a given type could be encoded into formal systems of a different type. We could then utilize these encodings to bring the inferential apparatus characteristic of the latter system to bear, yielding entirely new theorems about the encoded systems; these are the analogs of *predictions* in natural science. Before we turn to explicit consideration of encodings of *natural* systems, we must review some general principles which underlie these encodings. This is the purpose of the present chapter.

We obviously cannot proceed to develop encodings of natural systems in the same terms as were employed when dealing with purely formal systems. We have seen in the preceding chapter that such encodings typically take the form of mappings or functors relating mathematical objects. But as we have stressed, natural systems are not formal objects; indeed, the main purpose of establishing a modeling relation is precisely to bring an inferential structure to bear on them; a structure which will mirror whatever laws govern the behavior of the natural system itself. Thus, we cannot hope to talk of mappings and encodings without some basic preliminary discussion.

As we have seen in Sect. 2.2 above, the fundamental unit of a natural system is what we called a quality, or an observable. We recall that the defining characteristic of an observable is that it represents an interactive capability, through which a given system may interact with another. In particular, an observable embodies either the capability of a natural system to move a meter, or through which the effect of a particular mode of interaction on the system itself may be assessed. For the moment, we shall concentrate on observables defined through their effects on the specific meters which measure them; we shall return to their own capacity to change as a result of interactions imposed on them shortly.

In any case, since an observable is the basic unit of a natural system, all problems pertaining to the encoding of natural systems in formal ones ultimately devolve upon the problem of how observables are to be encoded.

Before attacking this fundamental question, let us make two simple observations. The first is the following: *the same observable can be manifested by many distinct natural systems*. Indeed, insofar as an observable is operationally defined through the specific meters which it moves, we may say that any system capable of moving those meters manifests the *same* observable quality. The second observation is: *different systems manifesting the same observable, or even the same system considered at different instants of time, may exhibit the capacity of moving a meter to a different degree*. The totality of different ways in which such a meter may be moved, as a result of an interaction with a natural system manifesting the quality in question, can be considered as the set of possible *values* which that quality can assume. Following customary usage, we shall refer to this set of values as the *spectrum* of the observable.

Intuitively, the simplest kinds of meters, and those customarily employed in natural science, are those for which the values of the corresponding observable are expressed as *numbers*. Indeed, it is a basic postulate of physics that it is *sufficient* for all scientific purposes to restrict ourselves entirely to observables whose values can be represented in this way. For the moment, we shall limit ourselves to observables of this type. We leave open for the moment the question of whether numerical-valued observables are in fact adequate for all of science; we shall return to this question subsequently, when we consider the sense in which, for example, feature extraction can be regarded as measurement. For now, we simply note that (a) most of the well-known examples of encoding of natural systems into formal ones tacitly restrict themselves to numerical-valued observables, and (b) the employment of such observables allows us to make the most immediate contact between formal systems and natural ones, through the use of mathematical objects (numbers) as names for observable values.

Thus, for our present purposes, we suppose that an observable is an entity capable of assuming numerical values. To that extent, then, an observable possesses some of the features of a purely mathematical object; namely, a mapping into real numbers. However, it thus far lacks one of the basic properties of such a mapping; namely, we have as yet no idea about what the *domain* of such a mapping may be. An observable associates *something* with particular numbers; but until we specify in a formal sense

what that something may be, an observable is not yet a real-valued mapping. We shall now endeavor to do this, by enlarging on the notion of *state* of a natural system which was developed earlier.

As we developed the concept of state in Sect. 2.3 above, we saw that it in principle only had meaning with respect to a particular encoding; we shall in fact enlarge on this as we proceed. Basically, we regard a state as comprising those qualities which needed to be encoded into a specific formal system so as to provide a hypothesis to which the inferential machinery of that system could be brought to bear. We will now make a small but essential epistemological step; we shall *impute* this concept of state back to the natural system under consideration, *as a conceptual entity which exists apart from any specific encoding; indeed, as that part of the natural system which is encoded*. To distinguish this kind of state from the encoding-dependent usage which we employed previously, we shall endow it with a name of its own: an *abstract state*.

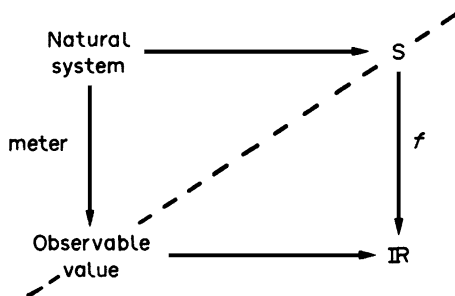
Speaking very roughly, an abstract state of a system is that on which the observables of the system assume their values. In this sense, the value of any observable on an abstract state is a way of naming that state; i.e. is already an encoding of the abstract state. In particular, if we restrict ourselves to observables which take their values in real numbers, the evaluation of an observable on an abstract state is an encoding of that state into a mathematical system; such an observable names a state through a corresponding number. As we shall see abundantly, these simple considerations in fact will lead to the most profound consequences.

Through the employment of this idea of abstract state, our concept of what constitutes a natural system becomes much enlarged. We began with the simple idea that a natural system comprises a set of qualities or observables. These could be defined through the effects produced in other systems (e.g., in meters) as a result of specific interactions; this in turn led to the idea that an observable could be regarded as manifesting a set of values, which comprise its spectrum. Now we have introduced the idea of an abstract state, which is the agency on which each particular observable assumes a definite value in its spectrum. Finally, we have seen that the specific value assumed by an observable on such an abstract state may be regarded as a name of that state.

We suggested above that observables possessed features of real-valued mappings, except that there was no way of specifying the *domain* of the observable. The concept of abstract state precisely fills this gap. For a natural system is now to be regarded not *only* as a family of observables, but also as a set of abstract states on which these observables are evaluated. Thus, to every natural system we shall associate a set of abstract states; we learn about these abstract states through specific acts of measurement; each such act results in the evaluation of some observable on a state, and at the same time names or encodes that state into real numbers in a particular way. We can express these considerations succinctly into a diagram (Fig. 3.9):

The dotted line bisecting this diagram separates those parts which pertain to the external world from those which pertain to the world of formal systems. The set *S* of abstract states sits partly in both worlds; in its definition it is imputed to the natural

Fig. 3.9



system itself, but as a set, it belongs to our mathematical universe u . The meter readings pertain also to the external world, but by definition they can be directly encoded into real numbers $\mathbb{R}$. The meter, which by definition serves to evaluate a specific observable on a particular state, is then itself encoded as a specific mathematical object; namely, as a real-valued function f defined on the set of abstract states of the natural system. Finally, since the meter *defines* the observable with which it is associated, we may regard the mapping f as *representing* that observable. Everything which follows will be seen to rest on these deceptively simple ideas.

Before proceeding further, let us note one crucial feature of the situation we have described. In a sense, an act of observation (i.e. an act of evaluating a specific observable on an abstract state) represents a description, or encoding, of that state. But any such act of observation clearly gives us only a *partial* description of the state. Thus any specific act of observation is at the same time an act of *abstraction*; an act of neglecting or forgetting all of the other ways in which that same state could be named; or what is the same thing, all of the other ways in which that same state could move other meters. It is usually considered a truism that abstraction is entirely a theoretical activity; but in fact, abstractions are imposed on us every time we make a measurement. Insofar as theoretical investigations must begin from the encodings provided by measurement, they are imposed on theory by the observer himself. Indeed, since one of the fundamental tasks of theory is to determine the relations between different descriptions or encodings, theory itself is one of the few ways that the shackles of abstraction imposed by measurement can be removed, or at least loosened. We shall return to this point several times as we proceed, for it is crucial to a proper understanding of the controversies revolving around the relative significance of “theory” and “experiment” in science.

Let us now examine some of the consequences of the picture we have drawn of an observable of a natural system encoded as a mapping $f : S \rightarrow \mathbb{R}$ from a set S of abstract states into the real numbers $\mathbb{R}$. Let us recall first that, given any mapping between sets, we can define an equivalence relation R_f on its domain, by saying that $s_1 R_f s_2$ if and only if $f(s_1) = f(s_2)$. In the present case, to say that two abstract states s_1, s_2 in S are so related means that both states produce the same effect on our meter, and are hence *indistinguishable* to it. They are thus both assigned the same name in $\mathbb{R}$; or, stated another way, the value assigned by f to such states is *degenerate*.

If we form the quotient set S/R_f , it is obvious that this set is in one-one correspondence with the spectrum $f(S)$ of S ; namely, if r is a number in $f(S)$, we associate with r the entire equivalence class $f^{-1}(r)$ assigned the label r by the observable f . But $f(S)$ is a set of *numbers* into which the abstract states have been encoded; by hypothesis, since we are at present considering only the single observable f , this encoding represents *all* the information available about the states in S . Thus it is reasonable to *identify* the spectrum $f(S)$ with S itself; indeed, $f(S)$ is the prototype of what is generally called a “state space” of our natural system. It must be carefully observed that the encoding of abstract states in S into real numbers returns us to our previous definition of “state” as defined in terms of a specific encoding.

Let us now generalise these ideas slightly. Suppose that we have at our disposal two observables of our system, encoded into respective mappings $f, g : S \rightarrow \mathbb{R}$. Each of them has a definite spectrum (namely $f(S)$ and $g(S)$ respectively) and these spectra are in one-one correspondence with the quotient sets $S/R_f, S/R_g$.

Let us suppose s_1, s_2 are two abstract states such that $f(s_1) = f(s_2)$. Then as we have seen, the observable f cannot distinguish or resolve these states, and assigns the same label to both of them. However, it may very well be the case that the observable g can resolve these states; $g(s_1) \neq g(s_2)$. Thus, as we would expect, we can obtain more “information” about S by employing a second observable g than we could with our original observable f alone.

Indeed, it is suggested that instead of naming a state s either with $f(s)$ or $g(s)$ alone, we name s by encoding it into the *pair* $(f(s), g(s))$. Let us pursue this idea, and consider some of its consequences.

First, in a certain sense, we are treating the pair of observables (f, g) as if they comprised a single observable. It is clear that we may regard the pair (f, g) as imposing a single equivalence relation R_{fg} on S , defined by writing $s_1 R_{fg} s_2$ if and only if $f(s_1) = f(s_2)$ and $g(s_1) = g(s_2)$; i.e. two abstract states are equivalent under R_{fg} if and only if neither f nor g can distinguish between them. It may be noted that the equivalence classes of this new relation R_{fg} are the totality of intersections of the equivalence classes of R_f with those of R_g ; symbolically, $R_{fg} = R_f \cap R_g$. Thus, once again, we may form the quotient set S/R_{fg} ; given an equivalence class in S/R_{fg} , we may uniquely associate it with the pair of numbers $(f(s), g(s))$, where s is any abstract state in that class.

It is clear that this procedure of labelling abstract states with pairs of numbers produces a mapping

$$\phi : S/R_{fg} \rightarrow f(S) \times g(S). \quad (3.21)$$

However, we must note carefully that this mapping ϕ is not, in general *onto* the cartesian product $f(S) \times g(S)$. In other words, given an arbitrary number $r_1 \in f(S)$, and an arbitrary number $r_2 \in g(S)$, *there need not exist an abstract state s such that $\phi(s) = (r_1, r_2)$* . Even though $f(S)$ is the spectrum of f , and $g(S)$ is the spectrum

of g , the cartesian product $f(S) \times g(S)$ is not generally the spectrum of ϕ ; all we can say is that it *contains* that spectrum.

It is easy to verify that $f(S) \times g(S)$ will coincide with the spectrum of ϕ if and only if every equivalence class in S/R_f intersects every equivalence class in S/R_g , and conversely. Intuitively, when this happens, we may say that a knowledge of $f(s)$ places *no restriction* on the possible values of $g(s)$. In all other cases, a knowledge of $f(s)$ generally does limit the possible values of $g(s)$; in the extreme situation, where every class in S/R_f is a union of classes in S/R_g , and hence can intersect only one such class, a knowledge of $f(s)$ *completely determines* the value of $g(s)$. We have seen this situation before; it is the case for which g is an *invariant* for the equivalence relation R_f .

Thus, if the mapping ϕ of (3.21) above is not onto $f(S) \times g(S)$, we may say that there is some *relation* obtaining between the values of the observables f and g on the set S . This kind of relation is our first indication of what we have previously called *linkage* between qualities or observables; we see it emerging here directly out of the formalism into which we encode the basic concept of an observable. In precise terms, a linkage relation involves the *degree of logical independence* between the labelling of abstract states by f and labelling them by g . We shall explore this in more detail shortly.

Before taking up this point, however, let us look once again at the cartesian product $f(S) \times g(S)$. We have seen so far that this set of pairs of numbers contains the spectrum of the mapping ϕ , but is not in general coincident with this spectrum. However, it is customary to treat $f(S) \times g(S)$ as if it were the spectrum of ϕ , and thus, just as before, to call it a “state space” of our natural system. Since we now utilize two observables f, g in our encoding, this “state space” is *two-dimensional*; we typically call a pair of numbers (r_1, r_2) in this product a “state” of our system. Once again, we point out explicitly that this use of the term “state” is dependent on a particular encoding, and must not be confused with the abstract states in S on which our observables are evaluated. Indeed, as we have seen, if there is a linkage between the observables f and g , not every pair (r_1, r_2) can be the name of an abstract state. When such a situation arises in applications, it is customary to say that the system is *constrained*, or that there is some *selection rule* operating to eliminate those pairs which do not label abstract states.

Let us formalize the construction of this “state space” in more detail. Given any set S , we can produce a mapping

$$\delta : S \rightarrow S \times S \quad (3.22)$$

by defining $\delta(s) = (s, s)$; this mapping δ is often called the *diagonal map*. Moreover, we have seen previously that the cartesian product operation is a *functor*; thus in particular, given any pair of mappings $f, g : S \rightarrow \mathbb{R}$, we can write

$$f \times g : S \times S \rightarrow \mathbb{R} \times \mathbb{R}. \quad (3.23)$$

Thus, the process by which we label abstract states through pairs of numbers can be represented, putting (3.22) and (3.23) together, by a diagram of mappings of the form

$$S \xrightarrow{\delta} S \times S \xrightarrow{f \times g} f(S) \times g(S) \quad (3.24)$$

and indeed, we see that the mapping ϕ in (3.21) is nothing but the composite $(f \times g)\delta$.

The diagram (3.24) can be considerably generalized. In particular, if we are given n observables $(f_1, \dots, f_n)$, each of which is represented by a mapping of S into $\mathbb{R}$, we can construct an n -dimensional “state space” by means of the diagram

$$S \xrightarrow{\delta} S^n \xrightarrow{f_1 \times \dots \times f_n} \prod_{i=1}^n f_i(S) \quad (3.25)$$

where we have written S^n as the n -fold cartesian product of S with itself, and δ is here the appropriate map of S into the n -fold product.

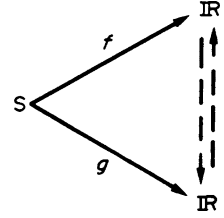
It is these spaces $\prod_{i=1}^n f_i(S)$ of n -tuples of numbers which constitute the “state spaces” that are the point of departure for scientific investigations of natural systems. It is perhaps not generally appreciated just how much conceptual work is involved before we can talk meaningfully about these spaces.

As with any encoding, we attempt to utilize the mathematical structure present in the formal system to learn about the properties of the natural system encoded into it. In the situation diagrammed in (3.25) above, these mathematical properties arise because we are dealing with cartesian products of sets of real numbers. The real numbers possess many rich mathematical structures of an algebraic, topological and measure-theoretic kind. Let us consider for a moment the topological properties, which arise from the ordinary Euclidean metric, or distance function. In particular, it is meaningful to inquire whether two n -tuples in such a cartesian product are close; i.e. whether the distance between them is small. If these n -tuples both name abstract states in S , *it is reasonable to impute the distance between the names of the states to the states themselves.* Thus, we would like to say that two abstract states s_1, s_2 are close in S if their images $(f_1 \times \dots \times f_n)\delta(s_1), (f_1 \times \dots \times f_n)\delta(s_2)$ are close in the ordinary Euclidean metric. But it cannot be too strongly emphasized that this kind of imputation is dependent on the encoding, and has no intrinsic significance in S ; if we choose other observables $(g_1, \dots, g_m) : S \rightarrow \mathbb{R}$, thereby creating another “state space” for S through the diagram

$$S \xrightarrow{\delta} S^m \xrightarrow{g_1 \times \dots \times g_m} \prod_{i=1}^m g_i(S) \quad (3.26)$$

we could be led to impute an entirely different metrical structure on S . Indeed, we could then ask the following kind of question: to what extent do the metrical structures arising from (3.25) and (3.26) coincide with S ; i.e. to what extent does

Fig. 3.10



a knowledge that two abstract states s_1, s_2 appear close in (3.25) imply that these same states appear close in (3.26)? This is a question intimately related to notions of *stability* which we encountered in the previous chapter, and provides the general context in which *bifurcations* can be discussed. However, as we shall now see, questions of this type devolve once again to notions of linkage, and thence back to the essence of the modeling relation itself. Let us then take up the question of linkage of observables in more detail.

Let us begin with the simplest case, in which we are given two observables $f, g : S \rightarrow \mathbb{R}$. Let us represent this situation by means of the diagram shown in Fig. 3.10 below. This diagram should remind the reader of that shown in Fig. 2.4 above, in which a single natural system (here represented by its set of abstract states) is encoded into two distinct formal systems; here the encodings are given by the observables f and g themselves, through their evaluations on abstract states. The linkage between these two observables is, as we have indicated, manifested by how much we learn about $g(s)$ when $f(s)$ is known, and conversely; thus the linkage between f and g involves the specification of relations between the encodings themselves, considered as formal systems, arising from the fact that the same abstract set of states is being encoded. Such relations are expressed in the diagram by the dotted arrows. Thus, if such relations exist, they measure the extent to which the two encodings are models of each other, in the sense of the preceding chapter.

Let us consider a few special cases, to fix ideas. The simplest possible situation is that for which $R_f = R_g$; i.e. every equivalence class in S/R_f is also an equivalence class for S/R_g . In this case, the value $f(s)$ of f on any abstract state s completely determines the g -value, and conversely. This means that there is actually an *isomorphism* Φ between $f(S)$ and $g(S)$, and we can write

$$\Phi(f(s)) = g(s) \quad (3.27)$$

for every abstract state $s \in S$.

Restated in slightly different language, if $R_f = R_g$, the $R_{fg} = R_f = R_g$. The image of S/R_{fg} in $f(S) \times g(S)$ is then simply a *curve* in the two-dimensional “state space”, whose equation is given by (3.27).

The next most elementary case is that in which f is an invariant for R_g ; i.e. each class in S/R_f is a union of classes in S/R_g . In this case, knowing the value $g(s)$ completely determines $f(s)$; but knowing $f(s)$ does not in general determine

$g(s)$. It is easy to see that, in this case, $R_{fg} = R_g$. We can thus establish a mapping $\Phi : g(S) \rightarrow f(S)$, but this mapping is not an isomorphism; it cannot be inverted. In this case, the encoding arising from f is “coarser” than that arising from g ; the f -encoding may be *reduced* to the g -encoding, but not conversely.

In this case, we shall say that f is *linked* to g . The linkage relation can again be represented geometrically in the two-dimensional “state space” $f(S) \times g(S)$ as a curve; i.e. as a relation

$$\Phi(f(S), g(S)) = 0 \quad (3.28)$$

in which $g(s)$ plays the role of independent variable; i.e. (3.28) can be solved as a single-valued function of $f(s)$, but not conversely.

In a still more general situation, where a given value $f(s)$ of f partially constrains the possible values of $g(s)$, a repetition of these arguments shows that the image of S/R_{fg} in $f(S) \times g(S)$ is no longer a curve, but some more complicated subset of $f(S) \times g(S)$. Nevertheless, we can still find a relation of the form (3.28); it can however not be solved in general for $f(s)$ as a single-valued function of $g(s)$, or conversely. Thus, there is in this case no mapping between the encodings in either direction; so that neither of them can be regarded as a model of the other.

Finally, in the case that the value of $f(s)$ places no restriction on the value of $g(s)$ and conversely (i.e. the case in which every class in S/R_f intersects every class in S/R_g and conversely) there is no relation of the form (3.28) at all; S/R_{fg} maps *onto* all of $f(S) \times g(S)$, and we say that the two observables f, g are *unlinked*.

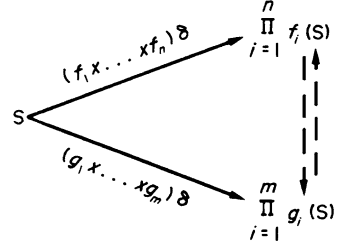
In general, we can say that a linkage between two observables f, g can be expressed as a relation of the form (3.28), which characterizes some locus in $f(S) \times g(S)$. Such a relation (3.28) will be called an *equation of state* for the system. An equation of state is thus seen to be a relation between encodings of a natural system, expressing the degree to which the codings are linked. *An equation of state is not an observable*, but expresses a relation between observables; as we shall see, such equations of state represent the *encoding of system laws*, imposed by a choice of encoding of individual observables in the fashion we have described.

In the light of the above discussion, let us consider the more general situation, in which we are given two *families* $(f_1, \dots, f_n), (g_1, \dots, g_m)$ of observables of our system S . With each of these families we can associate corresponding “state spaces”, which we can denote by $\prod_{i=1}^n f_i(S), \prod_{i=1}^m g_i(S)$ respectively. Using the notation of (3.25) and (3.26) we can construct the following analog of the diagram of Fig. 3.11, as shown overleaf. The arguments above can be repeated word for word in the new situation. Any linkages between the f_i and the g_i considered individually or in sets will be expressed by a more general equation of state, of the form

$$\Phi(f_1(s), \dots, f_n(s), g_1(s), \dots, g_m(s)) = 0 \quad (3.29)$$

which is the analog of (3.28). The character of Φ will express the manner in which the observables in these families are linked.

Fig. 3.11



Of special importance is the case in which $m = 1$ (i.e. our second family of observables consists of only a single member) and in which the relation (3.29) can be solved for this member; i.e. put into the form

$$g(s) = \Psi(f_1(s), \dots, f_n(s)). \quad (3.30)$$

In this case, the value of $g(s)$ is completely determined by knowing the values of the f_i on s , for any abstract state s . Stated another way the relation R_g is precisely the intersection of the relations R_{f_i} , $i = 1, \dots, n$. In this case, the linkage is expressed by saying that the observable g is a *function* of the observables f_i , and the equation of state (3.30) can be used to determine its value if the values of the f_i are known. This is perhaps the simplest instance of how we may utilize the encodings we have discussed, and the equations of state to which they give rise, to make specific *predictions*. We shall see many examples of this as we proceed.

So far, we have not made any mention of dynamics; i.e. of the manner in which the states of a system may change in time; nor of the manner in which time itself must be encoded into a formal system. We will take up these important matters in due course. For the moment, we will content ourselves with pointing out how more general equations of state may be written within the scope of the discussion we have presented above, and of which dynamical laws can be regarded as a special case. Let us return to the diagram (3.25) above, which specifies how we may encode abstract states in terms of a family $(f_1, \dots, f_n)$ of observables. A particular role is played in that diagram by the formal mapping δ , the diagonal mapping. Intuitively, we may think of δ as specifying an “instant of time” at which our natural system is in some abstract sense s , and it is at that instant that the observables f_i are evaluated. With this intuitive picture, it is clear that if we replace δ by some more general mapping of S into its n -fold cartesian product, and then carry out the same analysis as before, we obtain thereby *equations of state which relate the values of the observables f_i evaluated on different states of S* .

Thus, in particular, if we think of these different abstract states of S as pertaining to different instants of time, *such an equation of state represents a linkage relation between observables evaluated at different instants*. For instance, we might interpret an equation of state like (10) under these circumstances as a relation linking the value of an observable g at some instant to the values of the observables f_i evaluated at some other instant. In this fashion, such equations of state become the essential

vehicles for dynamical prediction. For the present, we content ourselves with these general remarks; we shall take up dynamical questions in more detail in a subsequent chapter.

Let us conclude this chapter with a few considerations bearing on the encoding of observables we have described. Let us first suppose that we have at our disposal a family $\mathcal{F} = (f_1, f_2, \dots, f_n)$ of observables. One question we may ask is whether we may find a subset $\mathcal{F}' \subset \mathcal{F}$ of these observables, such that every observable in $\mathcal{F} - \mathcal{F}'$ can be related to those in $\mathcal{F}'$ by an equation of state of the form (3.30). In that case, we can speak of $\mathcal{F}'$ as constituting a family of *state variables* for $\mathcal{F}$. In practice, we also try to pick such a set $\mathcal{F}'$ which is of minimal size, and for which the observables of $\mathcal{F}'$ are pairwise unlinked. The existence of such a subset $\mathcal{F}'$ is a purely *formal* question, and receives its precise formulation in terms of the corresponding equivalence relations involved. It is universally supposed in applications (a) that we can always find such a set of state variables; (b) that the number of observables in such a set is always the same; i.e. is an invariant of $\mathcal{F}$; and (c) if $\mathcal{F}'$ is a set of state variables for $\mathcal{F}$, it is also a set of state variables of a smaller set. It turns out, however, that none of these assumptions is generally valid; i.e. that there is nothing intrinsic about a set of state variables. We shall not dwell on these matters here; fuller details concerning problems of state variables, and indeed of all of the encodings we have discussed in the present chapter, may be found in our earlier monograph on measurement and system description.

We will conclude this chapter with a word about bifurcations. To fix ideas, let us consider again the simple diagram shown in Fig. 3.10 above, and let us suppose that some equation of state of the form (3.28) relates the two encodings via f, g respectively. We are going to inquire how the equation of state behaves with respect to the topologies of $f(S)$ and $g(S)$, and the manner in which they separately impute metric considerations to S . Basically, the question we ask is this: if we know that $f(s)$ and $f(s')$ are close in $f(S)$, can we conclude that $g(s)$ and $g(s')$ are likewise close in $g(S)$? Stated in slightly different language: if $f(s')$ is an *approximation* to $f(s)$, will $g(s')$ be an approximation to $g(s)$?

This question obviously devolves on the *continuity* of any relation between $f(s)$ and $g(s)$ established by the equation of state Φ . More specifically, suppose $r \in f(S)$. Let U be a small neighborhood of r . Now consider $f^{-1}(U)$; the totality of abstract states in S assigned a value in U by the observable f . Let us form $g(f^{-1}(U))$; this is a set of values in $g(S)$. The relation between U and $g(f^{-1}(U))$ is obviously another way of expressing the equation of state (3.28). Clearly, if $g(f^{-1}(U))$ is a small neighborhood in $g(S)$, and if $f(s) = r \in U$, then replacing s by another abstract state s' for which $f(s') \in U$ will keep $g(s')$ within a small neighborhood of $g(s)$. Under such circumstances, the equation of state (3.28) is a continuous relation within U , and we can say that any $r \in U$ is a *stable point* of f with respect to g . The totality of such stable points comprises an open subset of $f(U)$. The complement of the set of stable points will be called the *bifurcation set* of f with respect to g ; clearly, if a number r belongs to the bifurcation set, then the relation between any neighborhood U of it and the corresponding set $g(f^{-1}(U))$ established by the equation of state will not be a continuous one. In other words,

no matter how close $f(s')$ is to $f(s) = r$ in this case, we cannot conclude that $g(s')$ and $g(s)$ are close; i.e. no matter how closely we approximate to r in the f -encoding, we do not approximate correspondingly in the g -encoding. It will be seen that this notion of bifurcation is exactly the same as that described in the preceding chapter. We reiterate that in present circumstances, when we are dealing with encodings of observables of natural systems, *bifurcation depends on the continuity of the linkage relations between observables on a set of abstract states*. Thus, as we pointed out before, a bifurcation arises when the properties of distinct encodings become logically independent; the properties in question here are topological relations between respective sets of observable values.

We point out explicitly that so far, we have considered stability and bifurcation of the observable f with respect to the observable g ; in particular, the bifurcation set we defined was a subset of $f(S)$. We can clearly interchange the roles of f and g in the above discussion; we thereby obtain a dual notion of stability and bifurcation of g with respect to f . Thus, bifurcation sets come in pairs; in the present case, as a pair of subsets of $f(S)$ and $g(S)$ respectively.

The discussion of stability and bifurcation can clearly be generalized to equations of state of the form (3.29). We leave the details as an exercise for the reader.

We shall now turn to some specific examples of the encodings we have described above, beginning with those which characterize the study of physical systems. As we shall see, all of the deep problems in the study of such systems arise from the attempts to inter-relate distinct encodings; i.e. from attempts to establish equations of state of the form (3.30). It should now be evident that all of these problems involve at root the essence of the modeling relation itself.

References and Notes

The material presented in this chapter is an elaboration of some of the ideas originally developed in the author's earlier monograph, *Fundamentals of Measurement and the Representation of Natural Systems* (Elsevier, 1978). Further background and details may be found there.

3.3 Encodings of Physical Systems

We are now in a position to discuss the modeling of natural systems in detail. The present chapter will be concerned with the encoding of physical systems. As in Sect. 3.1 above, we will subdivide our development into a number of examples, each of which will be concerned with a different kind of encoding of a physical system into a formal or mathematical one. We shall emphasize in our discussion how these encodings are related formally to each other, or not, as appropriate. We will thus

construct an armory of specific examples of the modeling relation between natural and formal systems, which we shall draw upon heavily as we proceed.

*Example 1. Particle Mechanics*¹

Our first example will be concerned with the Newtonian approach to systems of material particles, which we already touched on briefly, in a superficial way, in Sect. 1.2 above. We will now bring to bear the basic discussion of the previous chapter, to see what is actually involved in the encoding of particulate systems.

The first order of business is to specify the fundamental observables to be encoded, and to define the corresponding “state space”. The first great contribution of Newtonian mechanics was to propose such a set of observables, or state variables, which provides the foundation for everything to follow.

In brief, Newton proposed that these fundamental observables could be taken as: (a) the (instantaneous) displacements of the particles in the system from some convenient reference, or origin of coordinates, and (b) the corresponding (instantaneous) values of the velocities of these particles. Thus, given a system composed of N particles, each abstract state of the system is represented by a set of $6N$ numbers; three co-ordinates of displacement for each particle, and three corresponding co-ordinates of velocity. This gives us an encoding of abstract states into the points of a $6N$ -dimensional Euclidean space, corresponding to 3.2.5 of the preceding chapter. This Euclidean space is the *phase space* of mechanics; the $3N$ -dimensional space comprising the displacement co-ordinates alone is generally called the *configuration space* of the system; the $3N$ -dimensional space of velocity co-ordinates is called the *velocity space*.

The next step in the Newtonian encoding is to postulate that *every other observable of a particulate system can be represented by a real-valued function on this phase space*. This innocent-sounding step is really of great profundity and significance, for it asserts that every other observable is totally linked to these state variables on the set of abstract states S . That is, if g is any observable of our system, then there is a linkage of the form (3.30) above, which expresses its value on any abstract state as a function of the values of the state variables on that state. We shall put aside for the moment the extent to which such a postulate can be expected to be universally satisfied; we shall simply treat it here as a definition of the scope of the Newtonian encoding; that is, the observables with which Newtonian mechanics deals are precisely those which are representable in this manner. The force of this postulate is that it removes the necessity of ever referring back to the set S of abstract states; we can confine ourselves from now on entirely to the phase space into which S is encoded by the state variables, and to the real-valued functions defined on this phase space.

We must now say a word about the manner in which *time* is encoded in the Newtonian scheme. Indeed, the very notion of velocity or rate, which enters into the choice of the basic state variables, presupposes some kind of temporal encoding. We have also tacitly utilized such an encoding when we used the term “instantaneous” in discussing these state variables. Basically, the Newtonian view of time involves two aspects: one of simultaneity, and one of temporal succession, as described in 2.1 above. For Newton, an *instant* of time is that at which the state variables must

be evaluated in order to provide a coding of abstract states into points in the phase space. It follows that a particulate system must be in exactly one such state at any instant of time; the determination of which state our system is in at some instant is equivalent to the *simultaneous* evaluation of all of the state variables at that instant. It should be noted that it is precisely here that the modern treatments of quantum theory depart from the Newtonian scheme; in quantum theory it is in fact *impossible* (according to the Heisenberg uncertainty principle) to simultaneously evaluate these variables; hence in quantum theory there is (a) no such thing as an instantaneous state, and (b) the encoding of time itself in the quantum-theoretic formalism is shrouded in the deepest obscurity.

Thus, for Newtonian mechanics, a point in the phase space represents an *instantaneous* state; it tells us what our system is like at an instant of time, and thereby tacitly specifies what is meant by an instant.²

The next step, basic to dynamics, is to specify what is meant by temporal succession. It is here that we find the famous identification of the set of instants with the real number continuum, and of absolute time flowing along this continuum of itself. In terms of the phase space of a mechanical system, this must mean the following: if at some instant t of time our system is represented by a point $\sigma(t)$ in the phase space, then as time flows of itself along the time continuum, the state $\sigma(t)$ itself will in general move in the phase space, tracing out thereby a curve, or *trajectory*, which is a continuous image of the time continuum. Each system trajectory, then, will be represented by a continuous mapping of the set $\mathbb{R}$ of instants into the phase space.

We have already pointed out that in Newtonian mechanics, we assume that *every* observable pertaining to a particulate system is represented by a function of the state variables. In particular, the *rates* at which the state variables themselves are changing in any particular state must be a function of the state alone (it will be noted that this basic hypothesis of the reactive paradigm, which was described at length in Chapter 1.2 above, is seen here simply as a consequence of the way observables are encoded in the Newtonian scheme). *Thus, if these rates are themselves observables, there must be equations of state of the form (3.30), which express the manner in which the rates of change of the state variables depend on the values of the state variables themselves.* It is this fact which leads directly to the entire dynamical apparatus of Newtonian physics.

Let us suppose that $x_1, \dots, x_n, v_1, \dots, v_n$ are respectively the instantaneous co-ordinates of displacement and velocity of a state of a particulate system. Let us denote their respective rates of change of that state in the traditional way by $dx_1/dt, \dots, dx_n/dt, dv_1/dt, \dots, dv_n/dt$. Then by (3.30) we must have $2n$ equations of state, of the form

$$\begin{aligned} \frac{dx_i}{dt} &= \phi_i(x_1, \dots, x_n, v_1, \dots, v_n) & i &= 1, \dots, n \\ \frac{dv_i}{dt} &= \psi_i(x_1, \dots, x_n, v_1, \dots, v_n) & i &= 1, \dots, n \end{aligned} \tag{3.31}$$

These equations of state play the predominant role in the theory; they are essentially the *equations of motion* of the system.

Before proceeding further, let us pause for a moment to look at the equations (3.31). Each of the functions ϕ_i, ψ_i appearing in these equations will typically involve, in addition to the state variables themselves, a number of numerical quantities *which are not themselves functions of the state variables*. For instance, if these functions are represented mathematically in terms of some basis set, we will find a corresponding family of coefficients. Such numerical quantities pertain to the specific character of the system with which we are dealing, for they determine the precise form of the linkage between the state variables and the rates at which these variables are changing. We cannot change them without in a precise sense changing the *identity* of the system with which we are dealing (i.e. we would thereby force a change in the way the observables of our system are linked to each other). These numerical quantities thus play the role of *structural* or *constitutive parameters* of our system, which were mentioned briefly in Sect. 2.3 above. We see here that such specific numerical parameters appear here automatically, by virtue of the basic hypothesis that rates of change of state variables in a state are linked in a specific way, via (3.31), to the values of the state variables themselves.

Of course, such constitutive parameters may be measured independently on our system; we shall see in a moment that they correspond to such quantities as inertial masses, elastic constants, coefficients of viscosity, and the like. Two points should be observed regarding them: (a) they appear as necessary consequences of the encoding of state variables and observables, and thus their appearance as a logical consequence of the encoding itself represents a specific *prediction* about the system; (b) since they are observable quantities, and not functions of the state variables, we see that our initial identification of observables as functions of the state variables is already proving to be too narrow. We shall see in a moment how to broaden our encoding of observables so as to accommodate them.

Let us now return to the incipient equations of motion of our system, embodied in (3.31) above. It was the second great step of Newton to give a specific interpretation to the functions ϕ_i, ψ_i in these equations of state. The specification of the first n of these functions is simple; we merely put

$$\phi_i(x_1, \dots, x_n, v_1, \dots, v_n) = v_i \quad (3.32)$$

In fact, these n equations are nothing but the *definition* of velocity as rate of change of displacement. The real innovation was Newton's identification of the functions ψ_i with the mechanical concept of *force*, and the essential identification of force with rate of change of velocity. Since the forces imposed on a system can generally be posited directly as functions of displacements alone (conservative forces) or as functions of displacement and velocity, the second n equations in (3.31) become not only non-circular, but in fact enable us to specifically encode *any* particulate system in the fashion we have indicated.

We thus can write our $2n$ equations of motion in the form

$$\begin{aligned}\frac{dx_i}{dt} &= v_i \\ \frac{dv_i}{dt} &= F(x_1, \dots, x_n, v_1, \dots, v_n)\end{aligned}\tag{3.33}$$

where F_i is the force acting on the particle for which dv_i/dt is a velocity co-ordinate. It is in this form that we can bring the full inferential power of mathematical analysis specifically to bear on the encoding of our system, and obtain the most far-reaching predictions. But before dealing with this question, let us look again at the constitutive parameters entering into the equations (3.33) by virtue of the general discussion given above.

Let us first observe that these parameters appear in the equations of motion precisely in the definition of the forces imposed on the system; i.e. in the functions F_i . If we suppose that there are r such parameters $q_1, \dots, q_r$, then these equations of motion can be written as

$$\begin{aligned}\frac{dx_i}{dt} &= v_i \\ \frac{dv_i}{dt} &= F_i(x_1, \dots, v_n, q_1, \dots, q_r)\end{aligned}\tag{3.34}$$

Since these parameters by definition cannot change without changing the very identity of the system, and since they are *unaffected* by the forces being imposed on the system, we can treat them as if they were additional state variables for our system, governed by dynamical equations of the form

$$\frac{dq_i}{dt} = 0 \quad i = 1, \dots, r\tag{3.35}$$

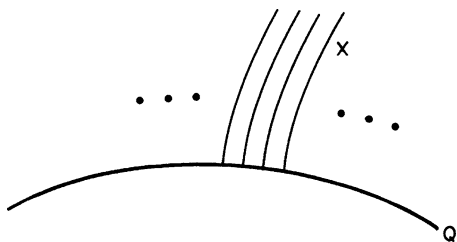
Indeed, we can even treat time as if it were a state variable τ , by writing

$$\frac{d\tau}{dt} = 1\tag{3.36}$$

Thus, with these assumptions, the constitutive parameters can be included directly into the essential Newtonian formalism. Indeed, utilizing (3.36), we can allow the forces F_i in (3.34) to contain time t explicitly.

The next point to make is that we can imagine *changing* the values of these constitutive parameters to new values q_i' , leaving every other aspect of our encoding unaffected. We would thereby in effect create an encoding of *another system*, *quantitatively* different from the original one, but with the same phase space, and with the same *qualitative* properties. Indeed, we can identify each such system obtained in this fashion with the corresponding r -tuple $(q_1, \dots, q_r)$ of structural parameters. The set of all possible r -tuples of constitutive parameters for these

Fig. 3.12



systems fills out a set Q , which is a subset of r -dimensional Euclidean space; this set Q may be called the *parameter space* for the systems so defined; each specific system is obtained by identifying a single element of this parameter space. Looked at in this light, the equations of motion (3.34) represent not a single system, but an entire class of systems, differing from one another only in the values assigned to their constitutive parameters.

If we denote the phase space shared by the systems in such a class by X , then the class itself is represented by the cartesian product $Q \times X$. We may note here, and we will enlarge upon this subsequently, that the class of systems so obtained has many of the properties of a mathematical object called a *fiber bundle*, with base space Q and fiber X . In diagrammatic form, the class can be represented as shown in Fig. 3.12 below:

where each point $q \in Q$ is associated with a copy of the phase space X , and the equations of state obtaining in the pair (q, X) are given by (3.34) and (3.35) with the appropriate numerical values of the q_i inserted into them. We shall have extensive recourse to this picture in some of the examples to be considered subsequently.

This, then, is the general picture of the Newtonian encoding of particulate systems. We do not encode a specific system as a general rule, but an entire class of qualitatively similar but quantitatively distinct systems, as we have just described. These classes are encoded into objects of the type shown in Fig. 3.12 above; objects with an exceedingly rich mathematical structure, which themselves form a category. This fact too will be exploited in the examples to be considered subsequently.

Let us return briefly to a specific system in such a class, specified by a definite pair (q, X) , and with the equations of motion (3.35). This is now a formal, mathematical object, and as we noted above, we can impose the full power of mathematical analysis on such objects. Let us briefly review some of its basic properties, and how these may be decoded into predictions pertaining to the corresponding natural system from which it was obtained.

Specifically, if we fix a q_0 in Q , then the equations of motion (3.36) become, in mathematical terms, a family of $2n$ simultaneous first-order differential equations. We have already discussed the properties of such a system of equations in Sect. 1.2 above; specifically, how we may obtain solutions of such a system in analytic form. In general, a solution comprises a family of $2n$ functions of the independent variable t :

$$\{x_1(t), \dots, x_n(t), v_1(t), \dots, v_n(t)\}$$

which satisfy the equations of motion identically. We have seen also (in Sect. 1.2) that, under very weak mathematical conditions on the forces F_i , a solution is uniquely determined by any point in the phase space through which it passes; this is the unique trajectory property in analytic form, when we note that a solution precisely defines a continuous mapping of the space of instants into the phase space. Thus, in principle, knowing a point in the phase space (i.e. knowing the values of the state variables x_i, v_i at a single instant of time), the equations of motion determine the entire trajectory on which the point lies.

In particular, the knowledge of such a set of initial conditions $\{x_1(t_0), \dots, x_n(t_0), v_1(t_0), \dots, v_n(t_0)\}$ allows us to draw *inferences* of the form

$$\{x_i(t_0), v_i(t_0)\} \rightarrow x_j(t) \quad (3.37)$$

and

$$\{x_i(t_0), v_i(t_0)\} \rightarrow v_j(t) \quad (3.38)$$

That is, knowing the values of the state variables at any instant t_0 allows us to *predict* the values of any state variable at any other instant. These predictions are simply expressions of linkage between the initial values of the state variables and their values at other instants, of the type which were mentioned in the preceding chapter; that is, they represent a linkage between observables evaluated on *different* abstract states of our physical system. Further, using the hypothesis that every relevant observable is directly linked to the state variables, we can use the inferences (3.37) and (3.38) to predict the value of *any* observable on the state of our system at *any* instant t , knowing the values assumed by the state variables on the state at some fixed instant t_0 .

Thus, we can make an enormous number of inferences regarding the temporal behavior of our physical system when we know: (a) the forces acting on our system; i.e. the way in which rates of change of the state variables are linked to the state variables themselves; (b) the values assumed by the state variables on any initial state. These inferences are then specific predictions about our physical system, which decode into the results of specific measurements or observations carried out on our physical system at particular instants. If such predictions are verified, then the mathematical system we have described must, by definition, be related to the physical system encoded into it by a modeling relation. That is, the linkages embodied into the equations of motion, together with the purely mathematical inferential structure which can be brought to bear on these equations, faithfully represent the laws governing the change of abstract states of that system in the external world. If the predictions we have made on this basis are found *not* to be realized when the appropriate observations are performed, and if the discrepancy between prediction and observation is not due to faulty observation, then we are faced with a number of obvious alternatives: (a) we have not selected an adequate set of state variables, so that the assumed linkage between rates of change of state variables and the variables themselves fails to encode essential qualities, or (b) our hypothesis about the form of the linkage, embodied in the specification of the

forces on the system, do not faithfully mirror the circumstances obtaining in the physical system, or (c) the entire encoding procedure we are using is fundamentally inapplicable to the given physical system, for one reason or another. We shall see examples of all of these circumstances as we proceed.

We can carry out this procedure for each system in our family of systems (q, X) , by inserting the appropriate values of the constitutive parameters q in Q into the equations of motion. In terms of the diagram of Fig. 3.12 above, we obtain thereby a fibering of each of the state spaces X into trajectories, determined by solutions of the equations of motion associated with the corresponding point of q . It then becomes an interesting question to relate the fiberings corresponding to different values q, q' of Q ; it will be seen that this question is a special case of structural stability, which was discussed in Example 5 of Sect. 3.1 above. To see this, let us recast the dynamical equations (3.34) above into the form of a one-parameter family of transformations of X , which was described in Sect. 1.2. Thus, to each q in Q , we will have a one-parameter family

$$T_t^q : X \rightarrow X. \quad (3.39)$$

If q' is another point in Q , it will be associated with a corresponding one-parameter family

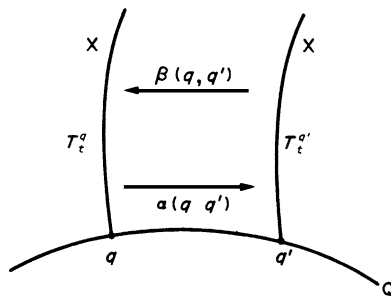
$$T_t^{q'} : X \rightarrow X. \quad (3.40)$$

Roughly speaking, if the effect of replacing q by q' in (3.39) can be annihilated by co-ordinate transformations of X ; i.e. if we can find mappings $\alpha = \alpha(q, q')$, $\beta = \beta(q, q')$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{T_t^q} & X \\ \alpha \downarrow & & \downarrow \beta \\ X & \xrightarrow{T_t^{q'}} & X \end{array} \quad (3.41)$$

is commutative for every t , then by definition the dynamics imposed by (3.34) is structurally stable to the perturbation $q \rightarrow q'$. In this case, we may say that the particular system (q, X) is itself a *model* of the different system (q', X) . This kind of diagram, which we have already seen in abstract terms in Fig. 2.3 above, will be the basis for our discussion of similarity in Example 5 below. The reader should also note for future reference the relation between the situation diagrammed in (3.41) and the notion of a natural transformation of functors. Briefly, what we have shown is the following: to the extent that a modification of constitutive parameters in Q

Fig. 3.13



can be annihilated by co-ordinate transformations s on X , to that extent are the two physical systems themselves related to one another by a modeling relation; and to that extent is a model for one of them also a model for any other. The ramifications of this situation are most profound, and reach into every corner of both theoretical and practical science.

Since the situation we have just described is so important, let us recast (3.41) into a diagram analogous to that of Fig. 3.12:

This diagram allows us to enlarge somewhat on our previous remark that $Q \times X$ itself possesses the structure of a fiber bundle. On each “fiber” X there is imposed a group of transformations; namely the corresponding dynamics T_t^q ; the notion of structural stability allows us to consider mappings of the entire space $Q \times S$ which (a) leaves Q fixed, and (b) preserves dynamics on the fibers. Later, we shall consider a broader class of mappings on spaces $Q \times X$ which do not leave Q fixed; these are intimately connected with what the physicist calls *renormalization*. We shall take up these matters once again in the appropriate sections below.

Example 2. Continuum Mechanics³

We have seen in the preceding example how to encode a natural system consisting of a discrete set of mass points (particles) into a mathematical formalism, and how that formalism leads in turn to specific predictions about the system so encoded. This kind of encoding represents one of the two possible logical alternatives we may entertain about the ultimate nature of an extended physical system; namely, that it is comprised of such particles. The assertion that every material system is composed of particles forms the basis for the atomic theory; thus, to the extent that the atomic theory holds universally, the concepts of Newtonian mechanics (or some appropriate generalization, such as quantum theory) can be regarded as providing an encoding of every physical system.

The other logical alternative we may entertain regarding the structure of extended material systems is that, in some sense, they are continua. That is, they may be indefinitely subdivided, in such a way that any portion of the system, however small it may be, retains all of the qualities of the entire system. The pursuit of this logical alternative leads to a quite different mechanical theory, which is generally called continuum mechanics. As we shall see, the character of the encoding required for a continuum is in many ways radically different from the one we have just described. Furthermore, the point of departure for this encoding is logically incompatible with

particle mechanics, as we have just seen. In the present section, we shall briefly explore the nature of the encodings appropriate to continua, and then we shall describe the manner in which these two incompatible encodings may be forced to co-exist as alternate descriptions of the same extended material system. We will thus be exploring a special case of the situation diagrammed in Fig. 2.4 above; and as we shall see, much of classical physics is concerned precisely with the confrontation of these incompatible pictures, neither of which, by its very nature, can be reduced to the other.

The first item of business is to decide on the encoding of an *observable* of a continuous extended system. As this is far from a trivial matter, we shall discuss it in some detail.

It is reasonable to suppose initially that a quality or observable pertaining to a continuously extended system involves two distinct aspects: (a) an aspect involving the actual manner in which the system is extended in space, and (b) an aspect independent of the extension. For by the hypothesis of infinite divisibility, any portion of such a system, however small it may be, retains the characters of the system as a whole; this would be impossible if all characters were dependent on the way each particular system is extended. In the limit, then, each *point* of the system must manifest these characters, independent of all the others.

Moreover, we must relate the encoding of observables to the encoding of time. We retain the idea that instants of time are encoded as real numbers, and thus that the set of instants is the real number system $\mathbb{R}$. We encode instants in a manner similar to that employed for particulate systems; we assume that *both* aspects of observables of continua (i.e. the aspect dependent on extension and the aspect independent of it) are what are evaluated at specific instants.

We shall suppose that, at any instant t_0 of time, our system occupies a region $U(t_0)$ of Euclidean three-dimensional space, E_3 . Thus, we assume that any point of $U(t_0)$ can be identified by a triple (u_1, u_2, u_3) of real numbers, where the point is referred to some convenient system of co-ordinates in E_3 . Thus, any quality pertaining entirely to the spatial extension of the system can simply be regarded as a real-valued function of $U(t_0)$; as examples, we may cite the observables used to describe the motions of rigid bodies in space.

We now turn to the representation of observables which are independent of spatial extension. We will proceed analogously to the example of particle mechanics, by assigning to any *point* of an extended system a set S of abstract states. An observable will then, as usual, be encoded into a real-valued function $f : S \rightarrow \mathbb{R}$ on this set of abstract states. If we assume as before that any point of the system is in exactly one abstract state $s(t_0) \in S$ at any instant of time t_0 , then the value of f at the instant t_0 is just $f(s(t_0))$.

Let us now put these two aspects together. If at an instant t_0 our extended system occupies a definite region $U(t_0)$, then each point of this region is assigned a copy of the set S of abstract states. Moreover, we suppose that in each such copy of S , there is exactly one distinguished state, on which observables independent of extension may be evaluated at that instant. Thus, the extended system as a whole is represented by a set of 4-tuples of the form

$$(s(t_0), u_1, u_2, u_3);$$

i.e. is a definite element of $S \times U(t_0)$. Thus, if $f : S \rightarrow \mathbb{R}$ is an observable independent of extension, we may encode this observable into a 4-tuple of real numbers.

$$(f(s(t_0)), u_1, u_2, u_3);$$

i.e. into the cartesian product $f(S) \times U(t_0)$. Thus, an observable of S defines, at any instant of time, a *scalar field* of $U(t_0)$. Observables of this character are the ones predominant in studies of elastic solids, and of fluids of various types. We shall call the number $f(s(t_0))$ appearing at the first co-ordinate of such a 4-tuple the *value* of f at the point (u_1, u_2, u_3) of our extended system at time t_0 . In this way, we can ultimately encode such observables as realvalued functions of the four real variables t, u_1, u_2, u_3 ; this is analogous to, but quite different from, our encoding of observables of particulate systems.

The effect of this kind of encoding is to convert observables f which are independent of spatial extension into new observables of a particular extended system; these new observables can be regarded as *densities*. We invariably suppose that these densities, considered as functions of four real variables, are continuous and smooth in time and space. Thus, they have rates of change in time (velocities) and rates of change in space (gradients). If we assume that these rates of change are *themselves observables* of the same character, we have as before the possibility of establishing linkages between them; such linkages will be the analogs of the equations of motion for a particulate system, and will represent the encoding of system laws into mathematical objects, which are now *partial* differential equations.

In general, the encoding of such linkages does not proceed, as it did in particle mechanics, by attempting to find a set of “state variables” in terms of which any observable can be expressed, and then establishing equations of motion for the state variables. Rather, if we are interested in any particular observable f on an extended system, we attempt to find a family of observables $g_1, \dots, g_r$ such that f is linked to the g_i , and the equations of motion for each g_i depend only on the other g ’s. Thus the g_i play the role of state variables for the observable f in which we are interested, but they need not play the same role for another observable f' .

The actual expression of such linkages depends heavily on the local geometry of Euclidean space. For instance, if f is an observable of an extended system, we can talk about the rate at which f is changing in any direction $\vec{r} = (r_1, r_2, r_3)$ at some instant t_0 and some point (u_1, u_2, u_3) ; this is identified with the scalar product

$$r_1 \frac{\partial f}{\partial u_1} + r_2 \frac{\partial f}{\partial u_2} + r_3 \frac{\partial f}{\partial u_3} = \vec{r} \cdot \nabla f.$$

We can seek to establish a linkage between these rates, or *flows* of the quality f , with its velocity $\frac{\partial f}{\partial t}$; such a linkage may for instance take the familiar form

$$\frac{\partial f}{\partial t} = -\nabla^2 f + Q(f, g_1, \dots, g_r) \quad (3.42)$$

Such a linkage may be interpreted as follows: the rate at which a quality f is changing at an instant t , at a point (u_1, u_2, u_3) of our extended system, is given by the rate at which it flows from the point together with the rate at which it is produced or consumed at the point. Typically these rates themselves depend on other such observables g_i and their flows; to encode the system laws into a mathematical form from which inferences may be drawn, it is necessary to specify these observables as well, in the manner we have indicated. We then obtain a system of simultaneous partial differential equations, in terms of which we may make predictions about the observable in which we are interested.

Just as in the case of particle mechanics, linkage relations like (3.42) typically involve numerical parameters which characterize the *identity* of the system; these then play the role of constitutive parameters. Likewise, the form of the linkages arising in the description of any particular system may be determined by subsidiary relations among the observables involved, which are generally called *constitutive relations*; these play the role of constraints in particle mechanics, and again refer to the specific nature of the system with which we are actually dealing.

We cannot in the present limited space go too deeply into the detailed features of continuum mechanics; for these the reader may consult any of the excellent texts on e.g. elasticity or on hydrodynamics. We have said enough, however, to point out the fundamental differences between it and particle mechanics, not only in terms of the basic hypothesis of the infinite divisibility of extended matter, but in terms of the mathematical form of the symbolism into which observables and linkages of extended systems are encoded.

We now turn to the question of how the irreconcilable encodings of extended material systems we have described can be related to each other.⁴ As we have already noted above, the atomic theory of matter postulates precisely that matter is not infinitely divisible, but in fact consists of discrete particulate units. However, it also follows that these discrete units are exceedingly small compared to the units of extension typically used in describing bulk matter; hence in any bulk volume there will be an exceedingly large number of them. The basic idea, then, is to use this fact to “approximate” to a continuum by the exploitation of *average values* of observables pertaining to such a family of particles; at the same time, we must “discretize” the continuum by breaking it into regions which are (a) large enough so that the average process is meaningful (i.e. large enough to contain many particles) but (b) small enough so that they can be treated as infinitesimals. In this way, we obtain a distortion of both theories; it is these distortions which may then be related. The basic question is whether, in the process of formal distortion, we may lose the encodings themselves.

Let us look at the matter in more detail. Let us suppose that U is a region of an extended system, containing a number N of particles at some instant t_0 . Then this particulate system can be encoded as described previously, and in particular, the displacements and velocities of the particles form a set of state variables. We can then think of representing the entire system of particles by a single “mean particle”, encoded into a single 6-tuple $(\bar{u}_1, \bar{u}_2, \bar{u}_3, \bar{v}_1, \bar{v}_2, \bar{v}_3)$, where each $\bar{u}_i, \bar{v}_i$ is the *average* of the corresponding co-ordinates for the individual particles. We can then further

evaluate any observable f of the particulate system on this “mean particle” to obtain an average value $\bar{f}$ of f in the region, by writing

$$\bar{f} = f(\bar{u}_1, \bar{u}_2, \bar{u}_3, \bar{v}_1, \bar{v}_2, \bar{v}_3). \quad (3.43)$$

We see that, if we could suppose our region shrunk to a point in such a way that the number of particles within the region was always large enough to make these averages meaningful, they would converge to what we earlier called *densities*. However, we clearly cannot do this; therefore, the mean observables we have described can only be regarded as formal *approximations* to such densities.

Conversely, let us suppose that we are given a continuum. Let (u_1, u_2, u_3) be a point of this continuum, and let V be a small neighborhood of this point. If now f represents a density (i.e. an observable of the continuum) then we can eliminate the pointwise variation of f in V by a process of integration; i.e. we can form the quality

$$\bar{\bar{f}} = \frac{1}{||V||} \int_V f \, du_1 du_2 du_3. \quad (3.44)$$

where $||V||$ is the volume of the neighborhood V . This is a kind of averaging in the *continuum*, which approximates to f at the given point (u_1, u_2, u_3) in the sense that $\bar{\bar{f}}$ converges to f as V shrinks to (u_1, u_2, u_3) .

What we need to do is to identify these two kinds of averages. But we must note that the approximations involved themselves require incompatible hypotheses. The first approximation requires a large number of particles, and hence a large region over which the averages are formed. The second requires, on the other hand, a small region. If we suppose that these incompatible requirements can be satisfied, so that we can identify the two kinds of averages; i.e. write

$$\bar{f} = \bar{\bar{f}}, \quad (3.45)$$

then the two approaches become *equivalent*. In particular, we can treat the densities of continuum mechanics as if they were averages of appropriate observables of an underlying particulate system. The linkages of the continuum thus become expressions of how these *average* values are related. Further, the predictions made on the basis of equations of state on the continuum become exact predictions about the *average* behavior of this particulate system.

However, it is clear that no such formal compromise between the particulate and continuous pictures can be valid. In particular, if we insist on large volumes, so that the averages of the particulate system are meaningful, we lose the properties on which the encoding of the continuum are based. Conversely, if we insist on small volumes, we lose the meaningfulness of the averages. In either case, there will necessarily be a discrepancy between the predictions obtainable through such a compromise and the behavior to be expected in observation of the system itself.

Stated another way: if the particulate or continuum encoding of a natural system is a model of that system, then the compromise we have described above cannot be; conversely, if the compromise is a model, then neither the particulate or continuum encodings of that system can be a model.

We can attempt formally to correct for this discrepancy by adding an *ad hoc* term to the equations of motion, which represents the *fluctuations* about the average value arising from the necessity for averaging over small regions. In a sense, then, these fluctuations encode *nothing* about the natural system; they are entirely mathematical artifacts arising from an attempt to reconcile two fundamentally incompatible encodings of the same system. Indeed, the entire theory of *heat*, which is precisely the study of how the fluctuations must behave in order for the compromise to work at all, is in this sense an artifact. It arises entirely from the desire to use equations of motion appropriate to continua in dealing with systems assumed in fact to be particulate. To this extent, theories of heat, fluctuation and randomness which play such a predominant role in thermodynamics are simply the necessary formal consequences of utilizing incompatible encodings simultaneously. We shall see a further example of this kind of artifact when we take up the relation of particle mechanics to thermodynamics in our next example, and will meet with this circle of ideas in a more general context when we take up the concept of *error*.

There is one further important point to be noticed in regard to equations of motion like (3.42) above, which express linkages within an encoding of a continuously extended system. In the light of (3.43), (3.44) and (3.45), such an equation must represent the behavior of an *average* quantity, which may be interpreted *either* as an average of a particulate quality, *or* as the average of a continuous quantity. If in (3.44) we consider a quantity f which is independent of position, then there is a sense in which the average f becomes identical with the value of f . In such a case, the term in (3.42) representing flow drops out; the partial derivative $\partial f / \partial t$ becomes the total derivative $d f / dt$, and (3.42) itself becomes an *ordinary* differential equation of first order. If the observables g_i in (3.42) are also of this character, then the equations of motion for the entire family become a set of simultaneous first-order differential equations in f and the g_i ; i.e. a dynamical system.

This is indeed the origin of the dynamical systems described in Sect. 1.2 above, which are used as models for all kinds of rate processes in chemistry and biology. In Sect. 1.2, we considered them as analogs of the equations of motion of particle mechanics, but we see here that they are actually of quite a different character. The dependent variables in such a dynamical system must be interpreted as *averages* which are independent of position; the equations themselves describe the time course of these average quantities, and the linkages they embody have a physical meaning entirely different from those of particle mechanics (where they embody Newton's Second Law). Moreover, these average quantities *must* be supplemented by *ad hoc* fluctuations; in certain cases, as in the diffusion-reaction systems to be considered in the next chapter, the fluctuations become in fact the dominant factor in interpreting the system's behavior. Thus, despite the formal similarity between dynamical systems arising as encodings of particulate systems and other kinds of dynamical

models, the nature of the encodings through which they respectively represent temporal phenomena is essentially different, and it is important to bear this in mind.

*Example 3. Thermodynamics, Particle Mechanics and Statistical Mechanics*⁵

We shall begin our considerations by briefly describing the nature of thermodynamics as a branch of physical science. It is distinguished from other branches of physics by its central concern with the concept of *heat*, and with the interconversion of heat into mechanical work.

The simplest kind of system with which thermodynamics deals is a homogeneous fluid, such as a gas. A system of this kind is considered as a continuum, and thus falls within the province of the encoding of continua described in the previous example. There is thus a certain overlap between thermodynamics and the dynamics of continua. However, the former is distinguished both by its preoccupation with heat and temperature, and by its *essential restriction to conditions of thermal and mechanical equilibrium*.

A thermodynamic system is, as are all the physical systems we have considered thus far, encoded in mathematical terms through the identification of the observables with which it deals into real-valued functions. Like particle mechanics, certain of these observables are distinguished as state variables, and all others are considered to be functions of the state variables in the manner we have already described. For a homogeneous gas, only three state variables are necessary: the volume V occupied by the system, the pressure P , and the temperature T . In this fashion, the abstract states of a thermodynamic system are encoded into a Euclidean three-dimensional space in the familiar manner. However, the essential restriction to equilibrium conditions means that in general, these three state variables are not independent, but are connected by an *equation of state* of the form

$$F(P, V, T) = 0. \quad (3.46)$$

Indeed, this is the origin of the terminology for “equations of state” as expressions of linkage which we have used above. A thermodynamic system is completely characterized by its equation of state, which can be interpreted geometrically as specifying a locus, or surface, in the thermodynamic space of states.

For instance, (3.46) may be taken of the form

$$PV = rT \quad (3.47)$$

which is generally called the *Ideal Gas Law*. The significance of the parameter r appearing in (3.47) depends on our choice of units for temperature and volume. If the choice of units is made arbitrarily, then r is a *constitutive parameter* of the kind we have seen before; its specific value is determined by the intrinsic character of the gas we are describing. However, if we measure temperature on the absolute scale, and we measure volume in *mols* (which means that we must use a different unit of volume for each gas) then r becomes a universal constant, *independent* of the nature

of the gas; this is because we have in effect incorporated the constitutive parameter into the unit of volume.

The study of ideal gases in thermodynamic terms involves inferences about the effect of infinitesimal changes in the state variables on the equilibrium surface (3.47), and the effects of such changes on other specific thermodynamic state variables, such as energy, work, enthalpy, entropy, and the like. It turns out that all of thermodynamics can be built up on the basis of a small number of formal laws, limiting the form of the equations of state which link the basic thermodynamic observables to the state variables. The First Law of Thermodynamics expresses in effect the conservation of energy, by relating the energy to heat absorbed by the system, and work done by the system. The Second Law of Thermodynamics involves the concept of entropy, and basically asserts that in any transition to equilibrium in an isolated system, the entropy always increases. We assume that the details are well known; they can be found in any of the standard texts on thermodynamics.

The phenomenological development of thermodynamics which we have briefly sketched was essentially completed before the end of the last century; the ideal gas laws had been postulated a century before that. However, during all this time, the atomic theory had been extensively developed and applied to all parts of physics. As we have seen, according to this theory, any physical system was to be encoded according to the Newtonian scheme described in Example 1 above. In particular, a thermodynamic system such as a homogeneous gas was accordingly to be regarded not as a structureless, infinitely divisible continuum, but as a family of a large number of particles. Moreover, the universality of the Newtonian encoding required that *every* physical quantity be representable as a real-valued function on an appropriately encoded space of states. The question thus arose: what was the relation between the thermodynamic state variables and the underlying atomic picture? In short: how could thermodynamics be *reduced* to mechanics? The task here is thus a special case of the situation diagrammed in Fig. 2.4 above; to relate two encodings of the same physical system (e.g. a homogeneous gas) in such a way that the thermodynamic state variables become expressible in terms of the purely mechanical ones.

From the consideration of the preceding example, the strategy to be employed should be fairly clear: namely, to relate the thermodynamic state variables to certain averages of the underlying mechanical encoding, in such a way that not only the averages but also the associated fluctuations can be expressed directly in mechanical terms. The machinery for accomplishing this, being a relation between two *mathematical* systems, is itself of a purely mathematical character, and has come to be called *statistical mechanics*. In fact, the statistical approach to thermodynamics grew historically alongside the more phenomenological approach, although from the formal and logical point of view, the two are quite distinct; in principle either of them can be developed in total ignorance of the other, or indeed of any physical interpretation whatever.

The basic idea of statistical mechanics involves the transfer of attention from the temporal behavior of individual states (i.e. from trajectories) to the temporal behavior of *sets* of states. The original motivation for doing this was the following:

it was known from the time of Avogadro that macroscopic volumes of gases contain enormous numbers of particles; according to Avogadro's Law, one mol of a gas must contain of the order of 10^{24} particles. Thus, the appropriate phase space for such a system would be a Euclidean space of approximately 6×10^{24} dimensions; a complete knowledge of a state of the gas in Newtonian terms would require the specification of the same number of initial conditions. This was recognized as at least a technical difficulty by even the most fervent disciples of the atomic theory. Thus, the question became: what can we say about a gas from this Newtonian picture when our knowledge about it is more modest; e.g. if we know simply that our gas is initially in a state for which the energy lies within a certain range?

In general, constraints of this form specify regions of the phase space, in which initial states compatible with them may lie. Any state in such a region is a candidate for the initial state of our gas. Each such state will trace out a unique trajectory in time, and accordingly, the initial *region* will correspondingly appear to flow in the state space. If U_0 is our initial region in the phase space X , then any point $u(0)$ in U_0 will move to a point $u(t)$ in X after a time t has elapsed, and thus the region U_0 will move to a new region U_t . What can we say about U_t from an initial specification of U_0 , knowing only the equations of motion of the system?

Briefly, the answer to this question is given by *Liouville's Theorem*: if the total energy of the system is conserved on trajectories, then the flow we have just described in the phase space is like that of an incompressible fluid. Thus, if our initial region U_0 has a certain volume $\mu(U_0)$, and if after a time t this initial region has moved to the region U_t , then $\mu(U_t) = \mu(U_0)$; or more succinctly,

$$\frac{d}{dt} \mu(U_t) = 0$$

This theorem, together with the conservation of energy (of which it is a corollary) forms one of the great cornerstones of statistical mechanics.

The other great cornerstone is a concept called *ergodicity*. This may be explained as follows. If energy is conserved, then an exact knowledge of the energy H_0 of our system constrains all possible trajectories to lie on the hypersurface $H = H_0$ in the phase space X . The "ergodic hypothesis" of Gibbs essentially states that a *typical* trajectory (in a precise measure-theoretic sense) wanders through this hypersurface in such a way that it spends roughly the same amount of time in any neighborhood of any point in that hypersurface. Thus, in a rough sense, watching a single typical trajectory for a sufficiently long time gives the same result as a single random sampling of the entire hypersurface. In other words, if we average any mechanical quantity along a single trajectory in time, we obtain the same answer as if we averaged the same quantity over the entire energy hypersurface in which the trajectory lies; or in more technical language, time averages (on single trajectories) may be replaced by phase averages (over such hypersurfaces).

The totality of points in X consistent with some partial knowledge about where an initial state lies is an example of what is called an *ensemble*. An ensemble is just a set of points in the phase space X . However, it may equally well be regarded as

an encoding of a *set* of samples of our gas, each initially in some different state. In other words: our uncertainty as to exactly where the initial state of our sample of gas lies in X is equivalent to an ensemble of different samples, each of which may be perfectly known.

Let then $E \subset X$ denote such an ensemble. Let us *suppose* that there is some observable $\rho : X \rightarrow \mathbb{R}$ which can be interpreted as the *density* with which the points in E are distributed in X ; it will be observed that this notion of density in X is analogous to the one considered in the previous example, except now we are dealing with density in a phase space. If $U \subseteq X$ is now any subset, we can define a new volume for U , which we will denote by $m_\rho(U)$, by writing

$$m_\rho(U) = \int_U \rho dt$$

It is then a corollary of Liouville's Theorem that

$$\frac{dm_\rho(U_t)}{dt} = 0 \quad \text{iff} \quad [\rho, H] = 0$$

where H is the energy observable (the Hamiltonian) and $[\rho, H]$ is the Poisson Bracket, defined by

$$\sum_{i=1}^n \left[\frac{\partial \rho}{\partial x_i} \cdot \frac{\partial H}{\partial v_i} - \frac{\partial \rho}{\partial v_i} \cdot \frac{\partial H}{\partial x_i} \right]$$

The ensemble E is called *stationary* if $[\rho, H] = 0$; i.e. the density with which the points of E are distributed in X is independent of time.

It follows immediately from this argument and from the hypothesis of ergodicity that an ensemble is stationary only if its density ρ is constant on surfaces of constant energy; i.e. is a function of energy alone.

To bring this machinery to bear on thermodynamic questions, we need to find some way of talking about *temperature*. Intuitively, temperature is a measure of energy flow between systems brought into interaction or contact with each other; when two systems have the same temperature, then there is no net energy flow between them. Thus, to talk about temperature, we need to find a way of talking about stationary ensembles for interacting systems of this type.

We can make contact with our previous discussion in the following way. Suppose that X is the phase space considered before, but let us now suppose that it can be written as a cartesian product

$$X = \prod_{\alpha=1}^k X_\alpha$$

where each X_α is itself the phase space of some *subsystem*, which we shall also refer to as X_α . We shall suppose that the decomposition is made in such a way that, if $(x_1^\alpha, \dots, v_m^\alpha)$ are the state variables for X_α , the equations of motion of X can be written in the form

$$\begin{aligned}\frac{dx_i^\alpha}{dt} &= v_i^\alpha \\ \frac{dv_i^\alpha}{dt} &= F_i^\alpha(x_1^\alpha, \dots, v_m^\alpha) + \phi_i\end{aligned}$$

where each function ϕ_i is assumed (a) to be a function of the state variables of X not in X_α , and (b) ϕ_i is small compared to F_i ; i.e. the interaction between the subsystems X_α is small.

By hypothesis, energy is conserved in the entire system X , but in general it is not conserved in the subsystems X_α . That is, we can regard the X_α as exchanging energy with one another. If we assume that our knowledge of X is represented by a *stationary* distribution ρ , we can ask such questions as: what fraction of the time would we expect X_α to be in a state with energy less than some given value E_0 ?

The answer to this question can be obtained from the equations of motion, Liouville's Theorem, and the ergodic hypothesis, with the aid of much mathematical manipulation and approximation. It involves an important function, called the *partition function*, defined as

$$Z \equiv \int_{-\infty}^{\infty} e^{-\beta E} \mu'(E) dE$$

where E is a given energy value, and $\mu(E)$ is the volume of the set of all states in X_α with energy less than E . The parameter β depends on the way in which X_α is coupled to the other subsystems, and on their dynamics, and is thus intimately related to what we would call the *temperature* of X_α ; in fact, it turns out that

$$\beta = 1/kT$$

where k is a universal constant (the Boltzmann Constant) and T is temperature in degrees Kelvin. We then proceed to *identify* the two statements:

- (a) X_α has temperature T ;
- (b) The fraction of time during which X_α is in a state with energy less than E_0 is given by

$$\frac{1}{Z} \int_{-\infty}^{E_0} e^{-E_0/kT} \mu'(E) dE.$$

When this is done, we have a way of building a concept of temperature, the fundamental thermodynamic variable, into the Newtonian picture, in terms of an energy *distribution* in X_α . We have thus carried out the essential step in relating the thermodynamic variables of state to an underlying particulate picture; i.e. we have built a mapping from the Newtonian encoding to the thermodynamic encoding of the same system. This is what the physicist means by asserting that thermodynamics can be *reduced* to mechanics.

IF must be stressed that the character of this reduction is *entirely mathematical*, and rests on the assumption of a number of important mathematical properties

which must be possessed by the equations of motion of the particulate system. If a particulate system does not admit an encoding with these mathematical properties, the reduction we have described will not work. It is this fact which, more than any other, has limited the scope of statistical mechanics. We shall see this in more detail in the next chapter.

*Example 4. The Mechano-Optical Analogy*⁶

In the previous example, we have described an instance of *reduction* in physics; i.e. a specific example of the situation diagrammed in Fig. 2.4 above. We now wish to consider an example of the dual situation, diagrammed as Fig. 2.2. In this case, two distinct kinds of physical systems are related through the fact that they admit a common encoding; i.e. they stand in a modeling relation to the *same* formal system. To use the terminology we developed before, these two distinct kinds of physical systems are then analogs of one another, and we can learn about one of them by studying the analogous properties of the other. It will be noted that this relation of analogy is a relation between natural systems, and not between encodings or formal systems; thus it is necessarily of a completely different character from reduction (which is a relation between encodings).

Let us begin with a number of mathematically equivalent reformulations of the Newtonian encoding of a particulate system. We will exhibit these reformulations in a special simple example, but in fact these ideas are perfectly general. To that end, let us consider once more the one-dimensional undamped harmonic oscillator, whose equations of motion were given in (1.1) above. We note that in these equations we have replaced the velocity v by the momentum $mv = p$, where m is the mass (a constitutive parameter). The trajectories of this system were shown in (3.4) above; they are a family of concentric ellipses, whose equations in the phase plane are

$$p^2/2m + kx^2/2 = \text{constant}.$$

The observable $H(x, p)$ defined by

$$H(x, p) = p^2/2m + kx^2/2 \quad (3.48)$$

plays a predominant role in all of our subsequent considerations. It is called the *Hamiltonian* of our system, and its value on a state (x, p) is simply the *total energy* of the state. We see that it is a sum of two terms:

$$H(x, p) = T(p) + U(x), \quad (3.49)$$

the first of which depends only on velocity or momentum, and the second of which depends only on displacement. Accordingly, $T(p)$ is called *kinetic energy*, and $U(x)$ is called *potential energy*. We note that the potential energy $U(x)$ is related to the force imposed on the system by

$$F = -kx = -\frac{\partial U}{\partial x}$$

A force related to a potential in this way is called *conservative*.

It will also be observed that the equations of motion of our system are completely determined by the Hamiltonian, for we can verify that

$$\begin{cases} \frac{dx}{dt} = \frac{\partial H}{\partial p} \\ \frac{dp}{dt} = -\frac{\partial H}{\partial x} \end{cases} \quad (3.50)$$

That is, the time derivatives of the state variables are precisely the partial derivatives of H as functions of phase. The equations (3.50) are the *Hamiltonian form* of the equations of motion.

If we differentiate the first of these equations with respect to time and substitute into the second, remembering that $p = m dx/dt$, we find

$$\frac{d}{dt} \left(\frac{\partial H}{\partial p} \right) + \frac{\partial H}{\partial x} = 0$$

If we now introduce a new observable

$$L(x, p) = T(p) - U(x)$$

we can rewrite this last expression in terms of L as

$$\left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial p} \right) = 0. \quad (3.51)$$

This observable L is called the *Lagrangian* of the system, and (3.51) is called the *Lagrangian form* of the equations of motion. It can be regarded as a single second-order equation in the displacement x ; in fact, expanding the total derivative in (3.51), we have explicitly

$$\left(\frac{\partial L}{\partial x} - \frac{\partial^2 L}{\partial t \partial p} \right) - \left(\frac{\partial^2 L}{\partial x \partial p} \right) \cdot \frac{dx}{dt} - \left(\frac{1}{m} \frac{\partial^2 L}{\partial p^2} \right) \cdot \frac{d^2 x}{dt^2} = 0$$

It will be noted that all of our arguments can be run backward; the original equations of motion are completely equivalent to both the Hamiltonian and Lagrangian forms.

All of our arguments apply to general conservative systems. If we know the total energy $H(x_1, \dots, x_n, p_1, \dots, p_n)$ then the equations of motion of the system can be put into the Hamiltonian form

$$\begin{cases} \frac{dx_i}{dt} = \frac{\partial H}{\partial p_i} \\ \frac{dp_i}{dt} = -\frac{\partial H}{\partial x_i} \end{cases} \quad i = 1, \dots, n$$

or into the Lagrangian form

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial p_i} \right) = 0 \quad i = 1, \dots, n$$

and all of these are completely equivalent.

Now the Lagrangian form (3.51) of the equations of motion arise in another class of problems in mathematics; namely, in that part of mathematics called the *calculus of variations*. The basic problem here is the following: to find that curve $x = x(t)$ which minimizes (or maximizes) some function of the form

$$J(x) = \int_{t_0}^{t_1} F(t, x(t), x'(t)) dt \quad (3.52)$$

For instance, we might wish to find that curve $x(t)$ passing between two points $(t_0, x(t_0)), (t_1, x(t_1))$ in the plane such that the arc length

$$\int_{t_0}^{t_1} \sqrt{1 + \left(\frac{dx}{dt} \right)^2} dt$$

is minimal.

It is shown in the calculus of variations that a necessary condition for a curve $x = x(t)$ to extremize a function of the form (3.52) is precisely that $x(t)$ satisfy the differential equation

$$\frac{\partial F}{\partial x} - \frac{d}{dt} \left(\frac{\partial F}{\partial x'} \right) = 0$$

This equation is called the Euler equation associated with the variational problem (3.52). On comparing (3.53) and (3.51), we see that *the Lagrangian form of the equations of motion is the Euler equation of the variational problem*

$$A(x, p) = \int_{t_0}^{t_1} L(x, p) dt = 0 \quad (3.53)$$

We can interpret this situation as follows. In mechanics, x is a co-ordinate of configuration; the totality of displacement co-ordinates define the configuration space of our system. If we ask the question: of all paths in the configuration space between an initial configuration $x(t_0)$ and a later configuration $x(t_1)$, which one will

actually be traced out by the system? We find that this path must extremize the quantity $A(x, p)$ defined by (3.53). This quantity is called the mechanical *action*, and the property we have just mentioned (i.e. that of all possible paths joining two configurations in configuration space, the one actually followed by the system extremizes action) constitutes essentially the *Principle of Least Action*.

The discovery that all of Newtonian mechanics could be developed on the basis of a global minimum principle of this type was a very exciting one in its time, for it was in accord with the idea that nature behaved in a way which was, in some sense, *optimal*. It was thus very much in accord with the eighteenth century idea that the actual world of nature was the “best of all possible worlds”. It also showed that under some circumstances, the differential equations of motion, which strictly speaking express linkages between a given state and infinitesimally nearby states, could be equivalent to global principles pertaining to arbitrarily extended time intervals.

It was W. R. Hamilton who first formulated the Principle of Least Action in definitive form, in the course of the most far-reaching studies on the foundations of mechanics. He also noticed that this principle was formally identical with an independent principle postulated by Fermat, from which all of geometric optics could be derived. This was Fermat’s Principle of Least Time, which we now proceed to describe.

If we suppose that a ray of light is passing through a point x of some optical medium, in a direction $\vec{r}$, then the velocity with which it moves in general depends on both x and x' ; i.e. $v = v(x, x')$. If the light ray moves between two points of the medium along some curve $x = x(s)$ (where s is some parameter), then the time of transit for the ray between the two points along that curve is just

$$\tau = \int_{t_0}^{t_1} \frac{x'}{v(x, x')} ds = \int_{S_0}^{S_1} G(x, x') ds \quad (3.54)$$

Fermat’s Principle of Least Time states that, of all curves connecting two points in an optical medium, light will *actually* move along that curve which minimizes (3.54); i.e. for which the *time of transit* is least.

On the basis of (3.54), all the facts of geometric optics can be expressed; just as, on the basis of (3.53), all the facts of particle mechanics can be expressed. Hamilton now noticed that a *dictionary* could be established between mechanics and optics, by analogizing action with time of transit, and the Lagrangian L with refractive index G . In this way, by exploiting constructions in optics such as Huyghens’ Principle, and translating them via his dictionary into mechanical terms, he was led to many deep insights into mechanics itself. For instance, he was able to derive a partial differential equation (the Hamilton-Jacobi equation) for the spreading of “wave fronts” of constant action in configuration space, which ultimately led directly to the Schrödinger equation of quantum mechanics. In fact, this synthesis of optics and mechanics was of such power and beauty that one of the finest textbooks on the subject (Lanczos) prefaces the discussion of the Hamilton-Jacobi equation with the following words from Exodus: “Put off thy shoes from off thy feet, for the place whereon thou standest is holy ground”.

We stress again that there is in this discussion no question of a *reduction* of optics to mechanics, or vice versa. Rather, by exploiting a purely mathematical relation between the manner in which optical and mechanical systems are independently encoded, a dictionary can be established between the *systems themselves*, in such a way that the behavior of each system is a model for the behavior of the other. Such analogies have had a most fruitful exploitation in theoretical physics, through their suggestion of how to formulate equations of motion (or their equivalent action principles) in particular physical domains, assuming an analogy between such a domain and another for which action principles are known. As far as we are aware, the extent and significance of the employment of such analogies as a basic methodological tool in theoretical physics has never been systematically studied. But by any standard, its significance has been at least as great as that of any reductionistic scheme.

Example 5. Similarity⁷

We are now going to consider an extension of the relation of analogy discussed in the preceding example, so as to provide important illustrations of the situation diagrammed in Fig. 2.3 above. In so doing, we will throw new light on the discussion focused around the diagram (3.32) above, which was originally proposed in the context of encoding of particulate systems.

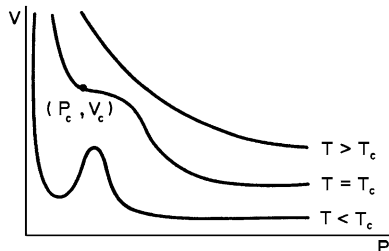
We shall begin with an example drawn from classical thermodynamics. It had long been known that the behavior of real gases was only approximately given by the ideal gas law (3.47), with the approximations between the predictions of this law and experimental observation becoming poorer as the temperature was lowered and the pressure raised. Indeed, it was experimentally known that, past certain critical values of pressure and temperature, a *phase transition* would occur; the original gaseous phase would be replaced by a liquid phase, possessing quite different physical properties. Of even the possibility of such a phase transition, there is no hint in the ideal gas law. Thus, the ideal gas law cannot be a universally valid encoding of the equilibrium behavior of real gases.

By utilizing considerations based on atomic theory, a more general equation of state was proposed by van der Waals in 1873. This equation of state is of the form

$$(P + a/V^2) (V - b) = rT. \quad (3.55)$$

It will be noticed that this new equation of state involves two new constitutive parameters a , b , and that it reduces to the ideal gas law (3.47) when $a = b = 0$. The parameters a , b are associated with the particulate constitution of real gases. Thus, the parameter a embodies the fact that the particles in such a gas will attract each other; hence the pressure P actually experienced by a gas will be larger than the value P' obtained from a manometer or other measuring instrument. Likewise, the parameter b embodies the fact that the particles of a gas themselves occupy a volume, which must be subtracted from the measured volume V' . Indeed, we may think of (3.55) as an admission that the instruments which we use to evaluate the state variables P , V are in fact measuring *different* quantities P' , V' , related to the ones we want by

Fig. 3.14



$$P' = P - a/V'^2,$$

$$V' = V + b;$$

and when we take this into account, the ideal gas law in fact still holds. This is an important point which should be kept in mind for our later discussion of renormalization.

The van der Waals equation (3.55) describes the equilibrium behavior of a family of “non-ideal” gases. Indeed, to every triplet of values (a, b, r) , there corresponds in principle such a gas. Let us describe all these gases in a common framework, utilizing ideas which we have seen before. Namely, let us denote by Q the totality of all triples (a, b, r) . To each point of Q , we associate a copy of the state space E_3 ; i.e. the space of all triples (P, V, T) . In each such space there is a distinguished surface; the surface of steady states of the gas given by the van der Waals equation (3.55), in which the appropriate values (a, b, r) are inserted. Thus, our family of gases is put into the form diagrammed in Fig. 3.12 above, with $Q = E_3$.

We are now going to argue that a transition $q \rightarrow q'$ in Q can be annihilated by co-ordinate transformations of the associated (P, V, T) spaces. In other words, we are going to show that all the different gases obeying the equation of state (3.55) are *similar* to each other. To see this, we shall digress for a moment to discuss some general properties of (3.55).

We said earlier that the ideal gas law (3.47) did not admit phase transitions. This can be seen as follows: if we fix any temperature $T = T_0$, then (3.47) becomes the equation of a hyperbola in the (P, V) -plane; this curve is called an *isotherm* for obvious reasons. Clearly all these isotherms are hyperbolic, and each one behaves like any of the others. If we do the same thing with the van der Waals equation, however, we find a situation as diagrammed in Fig. 3.14 below:

Heuristically, because of the fact that the volume V enters (3.55) in a cubic fashion, there will be three qualitatively different kinds of isotherms, depending on the number of real roots possessed by (3.55) as a function of V . For large T , the isotherms are of the same hyperbolic character as we found for the ideal gas law. For very small T , the isotherms will no longer be monotonic functions, but will possess a unique maximum and a unique minimum. Separating these two classes of isotherms there will be a *critical* one; for some definite temperature $T = T_c$ the

corresponding isotherm will have a point of inflection. We have labelled this unique point of inflection by (P_C, V_C) in Fig. 3.14.

This very special point, whose co-ordinates in the state space are (P_C, V_C, T_C) , is called the *critical point* of the system. We can easily evaluate its co-ordinates, using the facts that (a) the critical point is an equilibrium state of the gas, and hence satisfies the van der Waals equation (3.55); (b) the critical point causes the derivative dV/dP to vanish; (c) the critical point is a point of inflection, thus causing the second derivative d^2V/dP^2 to vanish. We can solve the resulting three equations for the three co-ordinates of the critical point; we find

$$\begin{aligned} P_C &= \frac{a}{27b^2} \\ V_C &= 3b \\ T_C &= \frac{8a}{27rb} \end{aligned} \tag{3.56}$$

Let us note that the co-ordinates of the critical point in the state space depend *only* on the constitutive parameters (a, b, r) . No other point satisfying (3.55) has this property.

Thus, if we make a transition $q \rightarrow q'$ in Q (i.e. if we replace the constitutive parameters (a, b, r) by new values (a', b', r') we clearly move the critical point from its original location to a new one. If we want to annihilate such a transition, we need to find a co-ordinate transformation of the state space which preserves critical points, and also maintains the manner in which arbitrary states are related to the critical point. Let us consider the transformation $\gamma(q, q') : E_3 \rightarrow E_3$ defined by

$$\gamma(P, V, T) = (P', V', T')$$

where

$$\begin{cases} P' = \frac{P'_c}{P_c} P \\ V' = \frac{V'_c}{V_c} V \\ T' = \frac{T'_c}{T_c} T \end{cases} \tag{3.57}$$

and (P_C, V_C, T_C) , (P'_C, V'_C, T'_C) are the critical points corresponding to the values $q = (a, b, r)$, $q' = (a', b', r')$ in Q respectively. Then it is immediately verified that this transformation indeed annihilates the transition $q \rightarrow q'$ made initially. Indeed, the transformation (3.57) embodies what is usually called the *Law of Corresponding*

States for gases satisfying (3.55); it is a precise assertion of the sense in which two such gases are similar to each other.

Let us reformulate these considerations in a slightly different fashion. If we fix a set of constitutive parameters (a, b, r) , the van der Waals equation may be regarded as a rule for associating to each pair (P, T) a corresponding value of V . Let us denote the space of pairs (P, T) by X , and the space of values of V by Y . Then the van der Waals equation (3.55) can be represented as a mapping

$$\Phi_{abr} : X \rightarrow Y. \quad (3.58)$$

Here we have used the notation Φ_{abr} to indicate that the mapping (3.58) depends on the specific values of the constitutive parameters.

If we take a different set of constitutive parameters (a', b', r') , we obtain a different mapping

$$\Phi_{a'b'r'} : X \rightarrow Y \quad (3.59)$$

The Law of Corresponding States then says precisely that there exist co-ordinate transformations $\alpha : X \rightarrow X, \beta : Y \rightarrow Y$, which make the diagram

$$\begin{array}{ccc} X & \xrightarrow{\Phi_{abr}} & Y \\ \alpha \downarrow & & \downarrow \beta \\ X & \xrightarrow{\Phi_{a'b'r'}} & Y \end{array} \quad (3.60)$$

commutative; i.e. which annihilate the transition from unprimed to primed constitutive parameters. These transformations α, β depend only on the initial and terminal values of these parameters, and are given in the obvious way from (3.57). Stated another way, the above construction shows that all the mappings Φ_{abr} are *conjugate*.

We will now point out a fundamental property of such conjugacy classes of equations of state, which generalizes a remark made earlier about ideal gases. We noticed, in connection with (3.47) above, that the parameter r appearing therein could be interpreted either as a constitutive parameter characteristic of the gas, or as a universal constant independent of the gas. In this last case, we saw that it was necessary to choose a unit of volume specifically related to the gas being measured (the mol), so that in effect the constitutive parameter disappears from the equation into the units in which its terms are measured.

Exactly the same thing is true for the van der Waals equation. This can be seen in two related ways. First, let us suppose that we wish to choose our units so that *any* gas satisfying this equation has its critical point located at $(1, 1, 1)$ in the state space. Clearly, the choice of unit will then depend on the gas. From (3.56), it is easy to see that the specific values of a, b, r which assign unit values to the critical volume, temperature and pressure are given by

$$a = 3; \quad b = 1/3; \quad r = 8/3;$$

and substituting these values into the van der Waals equation itself, we find

$$(P + 3/V^2)(3V - 1) = 8T. \quad (3.61)$$

With this choice of units, *any* gas satisfying the van der Waals equation will also satisfy the equation (3.61), in which *no constitutive parameters appear*. As before, these parameters are now hidden in the units in which the constitutive parameters, and hence also state variables, are measured.

Let us look at what we have just done in a slightly different way. Given any gas satisfying the van der Waals equation, let us introduce new state variables (π , v , τ) by writing

$$\pi = \frac{P}{P_c}; \quad v = \frac{V}{V_c}; \quad \tau = \frac{T}{T_c} \quad (3.62)$$

These new variables have the property that they are *dimensionless*; i.e. they are pure numbers, independent of the units in which pressure, volume and temperature are measured. Once again, we see that these dimensionless variables depend on the gas we are dealing with; the constitutive parameters (a , b , r) appear explicitly in them through their determination of the critical point of the gas. It is evident that if we substitute these variables into the van der Waals equation, we obtain precisely

$$\left(\pi + \frac{3}{v^2} \right) (3v - 1) = 8\tau \quad (3.63)$$

which is identical with (3.61) in form. For this reason, (3.63) is often called the *dimensionless form* of the van der Waals equation.

It will now be apparent that the conjugacy expressed in the diagram (3.60) above is equivalent to the existence of a single “dimensionless form” into which all the equations of state of the conjugacy class can be transformed; and that the co-ordinate transformations α , β in that diagram are precisely the ones which express this fact. In effect, what we have done is to pick a particular *canonical* gas out of the conjugacy class (namely, the one for which the critical point has co-ordinates (1, 1, 1)) and refer every other element of the class to it. These remarks should be compared with our discussion of similar matrices and canonical forms, in Example 4 of Sect. 3.1.

Using the notation that the dimensionless form (3.61) or (3.63) is a canonical form of equation of state for an entire conjugacy class, we can reformulate the Law of Corresponding States in the following way. If (P , V , T) and (P' , V' , T') are respectively states of gases (a , b , r) and (a' , b' , r'), then the states are corresponding if and only if they are both mapped onto the *same* state of the canonical gas by the appropriate transformations (3.57). Or, stated somewhat differently: if (π , v , τ) is a state of the canonical system, and if (a , b , r) specify any gas, there is a unique state (P , V , T) of that gas which maps onto (π , v , τ) under (3.57). Any two such states, for any

two gases, are corresponding states, and conversely. Under this formulation, we see that we annihilate any change of constitutive parameters by in effect transforming the initial gas and the perturbed one to canonical form.

Let us agree in general to say that *two natural systems are similar if they admit encodings into equations of state which are conjugate*. This is a perfectly general definition of similarity among natural systems, and plays the same role in a class of natural systems as the modeling relation does between natural and formal systems. It should also be compared with the notion of natural equivalence between functors (taking a natural system as an object in a category, and their equations of state as their images under distinct functors).

These ideas find many important applications in engineering, because they underlie the notion of a “scale model”. In general, let us suppose that we can encode some natural system into an equation of state analogous to (3.58) above; i.e. as a mapping

$$\Phi_{a_1 \dots a_r} : X \rightarrow Y$$

where $a_1, \dots, a_r$ are constitutive parameters. As before, let us denote by Q the space of r -tuples $(a_1, \dots, a_r)$; then to each q in Q , we have a mapping Φ_q between copies of X and Y . In all the mappings Φ_q are conjugate, we can essentially repeat the discussion given above for the van der Waals equation word for word; we can identify a canonical form of the equation of state which is *dimensionless* in the above sense. There is a Law of Corresponding States arising just as before; if q, q' are distinct elements of Q , then a pair (x, y) of the first system corresponds to a pair (x', y') of the second if and only if they map onto the *same* state (ϵ, η) of the canonical system.

To see how these ideas are applied, let us suppose for example that we are interested in the hydrodynamic properties of some airplane. We can in principle write down an equation of state, expressing the physics of the situation, and attempt to draw inferences from it directly, in the fashion we have described in previous examples in this chapter. But we can also note that, by varying the constitutive parameters, we may be able to construct a different system, obeying the same equation of state, and which is thus *similar* to the case we are interested in. This second system is called a *scale model*. By *observing* the hydrodynamic behavior of the scale model, and by utilizing the transformations which establish similarity, we can thereby directly determine the hydrodynamic properties of the *corresponding states* of the actual airplane. Such ideas serve to bring many otherwise intractable problems within reach; although we shall describe a number of other important examples of this character as we proceed, we cannot enter into full details here. We must once again urge the reader to consult the relevant literature.

Example 6. Symmetry and Invariance

In our discussion of similarity of matrices, we pointed out that the entire question arises from the cartesian encoding of geometry into arithmetic, through the fact that such an encoding distinguishes a particular system of co-ordinates. In this system, the n -tuples of cartesian n -space become the co-ordinates of specific points in the geometric space. If we choose co-ordinates differently, the labelling of points by n -tuples changes.

We may argue that exactly the same situation obtains in the encoding of any natural system into a formal system; particularly into one arising from numerical evaluations of observables through specific meters. Such encodings are basic to our construction of phase spaces, which in turn provide the foundation for the inferential structure from which predictions are made, and which are supposed to mirror the laws (linkages) obtaining in the natural system itself. Such encodings are open to precisely the same problem that we found in the cartesian encoding of geometry; if we change the encoding through a change in meters, or more generally through any shift in reference frame, we will thereby change all the linkages.

In physics, as in geometry, we feel that such different encodings must in some precise sense be *equivalent*. In particular, it must be the case that predictions about a *given* natural system, made on the basis of two encodings differing by only a “change of co-ordinates”, must be the same, in some precise sense. This is expressed in physics through the proposition that system laws, or better, the “laws of nature”, must be *independent of the observer*, and hence of any special way of encoding or co-ordinatizing a physical system. Stated another way, the linkages obtained from any particular encoding must be *invariant* to transformations of co-ordinates. Thus, if we have two distinct encodings related by such a transformation, we must be able to (a) express the co-ordinates of either encoding in terms of the other, and (b) show that any linkage in either encoding purporting to express system laws is invariant to the transformation relating the co-ordinates.

This kind of situation has in fact permeated all of our previous discussions of conjugacy and similarity. For let us suppose that, by virtue of some specific encoding of a natural system, we obtain an equation of state of the form

$$\Phi(q, x, y) = 0 \quad (3.64)$$

where q denotes an r -tuple of constitutive parameters, and x, y denote vectors of state variables and observables respectively. As we have seen, this kind of encoding presupposes a distinguished co-ordinate system in each of the three spaces Q, X, Y whose elements enter into the equation of state (3.64).

Let us suppose we perform a general co-ordinate transformation on the space $Q \times X \times Y$ into which our system is encoded; i.e. we introduce a new co-ordinate of the form

$$\begin{cases} q' = q'(q, x, y) \\ x' = x'(q, x, y) \\ y' = y'(q, x, y) \end{cases} \quad (3.65)$$

If this transformation can arise from a change of observer, or a change of meter, then the invariance of natural laws to choice of co-ordinate system means that the linkage (3.64) must be preserved; i.e. that

$$\Phi(q', x', y') = 0 \quad (3.66)$$

and hence that the linkage (expressed by the relation Φ) must be invariant to the transformation (3.65).

If we now suppose that (3.64) can be expressed as a family of mappings

$$\Phi_q : X \rightarrow Y \quad (3.67)$$

and that all the mappings Φ_q are conjugate for each choice of $q \in Q$, then we have seen that (3.67) is indeed invariant to transformations of the form

$$\begin{cases} q \rightarrow q' \\ \alpha(q, q') : X \rightarrow X \\ \beta(q, q') : Y \rightarrow Y \end{cases} \quad (3.68)$$

For the van der Waals equation, the invariance of the equation of state to the transformation (3.68) is simply the conjugacy diagram (30) above. Therefore, if we reinterpret the case of the van der Waals equation to refer to a *single* gas, encoded relative to different observers who measure (a, b, r) and (P, V, T) in different scales related to one another by (3.68), our previous discussion of similarity becomes precisely a discussion of invariance under coordinate transformations of particular types.

It is not hard to verify that the totality of transformations of the type (3.68) form a *group* of transformations acting on $Q \times X \times Y$, and that the conjugacy of all the mappings Φ_q means precisely that the equation of state (3.67) is invariant to every transformation of this group. We can now proceed just as well in the opposite *direction*: given a group G of transformations on the space $Q \times X \times Y$; under what circumstances will an equation of state of the form (3.66) or (3.67) be invariant to every transformation in the group? If we have independent reasons for believing that the transformations of such a group G are in fact simple changes of co-ordinatization or observer in the encoding of a natural system, then only those equations of state invariant to G can be candidates for expression of linkage relations of that system.

This was in fact the procedure underlying the development of the special theory of relativity.⁸ In the study of mechanical systems, it had long been known that Newtonian equations of motion were invariant to the transformations of the Galilean group; these transformations related observers moving relative to one another at constant velocities. However, Einstein observed that the equations of electro-dynamics (the field equations of Maxwell) were not invariant under these transformations, but rather under those of a different group (the Lorentz group). Invoking the invariance of *mechanical* laws under change of observer, and utilizing the Lorentz group instead of the Galilean, Einstein was led to a profound reformulation of all of mechanics, which by now is well known.

Similar ideas of invariance have come to be of decisive importance in quantum mechanics.⁹

We wish to draw the reader's attention to several features of this kind of situation.

First, that symmetry and invariance are all variations on the basic ideas of conjugacy and similarity. They refer initially to mathematical relations between *encodings*, and are then imputed back to the natural systems which give rise to the encodings.

Second, we must point out that the often rather mysterious ideas collectively called *renormalization*¹⁰ can be looked upon as arising from considerations of invariance under transformation. To see this, let us compare the general transformation (3.65) with the special transformations (3.68), under which the equation of state (3.67) is invariant. As we saw, this invariance is simply an expression of conjugacy of the mappings Φ_q . But suppose that (3.68) is invariant to a larger class of transformations of $Q \times X \times Y$ than those given by (3.68). Under such a larger class, the new constitutive parameters q' will, according to (3.65), now depend on the original state variables in X and Y ; likewise, the new state variables x' , y' will depend on the original constitutive parameters in Q . In other words, the new constitutive parameters cannot be defined in terms of the original ones; nor can the new state variables be defined entirely in terms of the original ones. We thus retain the *form* of the original equation of state, now referred to a set $Q' \times X' \times Y'$, but we can no longer identify this product directly with $Q \times X \times Y$ in a simple fashion. As we noted above, in connection with the van der Waals equation, this is tantamount to passing to a completely *new encoding* of our natural system, and the introduction of a completely new set of observable quantities. We cannot go deeply into the details of specific renormalization procedures in physics, but the reader can observe from what has already been said its essential theoretical significance.

Third, the idea that system laws are invariant to the observer refers only to the case in which *the observers are basically measuring the same set of observables*. Specifically, the observables measured by any particular observer must be totally linked to those measured by any other observer. *Indeed, this fact is what makes transformations between the encodings generated by these different observers possible at all.* But we have seen that different encodings of the *same* system may involve unlinked observables; this was the case in our discussion of discrete versus continuous encodings, which constituted the first two examples of this chapter. In that case, we cannot expect these encodings to be related by any kind of transformation; they are fundamentally irreducible one to the other. We have already alluded to this possibility in Sect. 3.2 above, and we suggested that it plays the essential role in our perception of *complexity* of natural systems; this is a point we will develop in detail subsequently. Indeed, as we shall see, there is a sense in which biology is possible only *because* there exist intrinsically inequivalent observers, whose encodings cannot be transformed into one another. Nevertheless, within their sphere of applicability, symmetry and invariance arguments are of fundamental importance; they may roughly be regarded as expressing a sense in which different *observers* are models of each other.

References and Notes

1. The references cited in Note 1 to Sect. 1.2 above are appropriate here also.
2. See especially the discussion in Sect. 4.2 below.

3. Continuum mechanics comprises, among other things, the traditional approaches to hydrodynamics, aerodynamics, elasticity, plasticity and the like. In its mathematical form, it can be regarded as transmuting into the theory of fields (e.g. electrodynamics and electromagnetism). Some good general references on the mechanics of continuous media are:

Truesdell, C. A., *Elements of Continuum Mechanics*. Springer-Verlag (1966).

Eringen, A. C., *Mechanics of Continua*. Wiley (1966).

Scipio, L. A., *Principles of Continua with Applications*. Wiley (1966).

For a treatment of fluid mechanics from the same general point of view, see Aris, R., *Vectors, Tensors and the Basic Equations of Fluid Mechanics*. Academic Press (1962).

For a cognate treatment of elastic solids, see

Brillouin, L., *Tensors in Mechanics and Elasticity*. Academic Press (1964).

The volumes of Elasticity and Fluid Mechanics in the Landau-Lifschits series are worth consulting. A treatment of fields may be found in

Artley, J., *Fields and Configurations*. Holt, Reinhart & Winston (1964).

4. In the main, the relationship between particle and continuum mechanics has tacitly been treated as a complementarity (in the sense of Bohr), like the wave-particle duality of quantum mechanics. From the purely mechanical point of view, the problem is not urgent; the atomic theory tells us that matter is ultimately particulate, but we may use continuum methods when dealing with bulk phenomena, whose scale of length is large compared to the interatomic distances. The problem becomes unavoidable when we deal with the effects of fields on matter. At this point, the usual approach is to quantize the fields, which in effect buries the epistemological issues in an opaque formalism. Our main point here is not with the applicability of continuum or particulate encodings of specific situations, but rather to point out that the encodings are distinct and mutually irreducible.
5. Each of these subjects has an enormous literature. Thermodynamics is a subject which can be, and has been, treated as a logically closed entity, independent of the rest of physics (indeed, for an axiomatic treatment, see Giles, R., *Mathematical Foundations of Thermodynamics*, Pergamon Press (1964). The classical texts, such as those of Planck (*Treatise on Thermodynamics*, 1867, reprinted Dover 1965) and Sommerfeld (*Thermodynamics and Statistical Mechanics*, Academic Press 1956) are perhaps still the best. Of the many standard references on statistical mechanics, one of which the author is particularly fond is
 Khinchine, A. I., *Statistical Mechanics*. Dover (1949).
 We shall discuss the formalism of statistical mechanics more deeply in Sect. 4.3 below.
6. The two best current references for the relation between geometric optics and the action principles of mechanics are:

Lanczos, C., *The Variational Principles of Mechanics*. University of Toronto Press (1966).

Yourgrau, W. and Mandelstam, S., *Variational Principles in Dynamics and Quantum Theory*. Saunders (1968).

It is interesting to note further that the very same formalism is exploited nowadays in yet another context; that of the theory of optimal control. The two basic approaches to optimal control theory, namely the Maximum Principles of Pontryagin, and the Dynamic Programming of Bellman, are analogous to the Euler-Lagrange equations and the Hamilton-Jacobi equations of mechanics respectively. A remarkably clear treatment of all these matters may be found in Gelfand, I. A. and Fomin, S. V., *Calculus of Variations*. Wiley.

See also the chapter on optimal control in the volume by Kalman, Arbib and Falb mentioned in Note 5 to Sect. 1.2 above. See also Hestenes, M. R., *Calculus of Variations and Optimal Control Theory*. Wiley (1966).

7. The treatment of similarity given here was first developed in a paper of the author's (*Bull. Math. Biophys.* 40 (1978), 549–580). Further general references on the subject may be found in that paper. Particularly see the articles of Stahl referred to therein for incredibly complete bibliographies.
8. The special theory of relativity asserted that the “laws of nature” were the same for all observers moving relative to each other with constant velocities. The general theory asserted the same for all observers, regardless of their relative states of motion. A good modern treatment of these matters may be found in

Adler, R., Bazin, M. and Schiffer, M., *Introduction to General Relativity*. McGraw-Hill (1975).

Some of the older books on relativity are full of interesting insights and suggestions. We may recommend particularly the book of H. Weyl (*Space, Time and Matter*, Dover 1948).

9. At issue here are the groups of “symmetries” under which the Schrödinger equations of quantum mechanics are invariant. The classic references in this area are

Weyl, H., *Group Theory and Quantum Mechanics*. Methuen (1923).

Wigner, E., *Group Theory and its Application to the Quantum Mechanics of Atomic Spectra*. Academic Press (1959).

The theory of group representations, which was mentioned in Note 10, Sect. 3.1 above, plays a crucial role in these developments. The review of Mackey, to which we have referred several times, provides a panoramic overview of these developments, as well as references to more detailed treatments.

10. In a certain sense, renormalization is an admission that we measure the wrong things. As we have already noted, the passage from the ideal gas law to the van der Waals equation is a kind of renormalization. Ideas of this kind became decisive in physics with the study of the interaction of particles and fields in quantum physics; the results were plagued by divergences and other conceptual difficulties. A similar state of affairs emerged more recently in the study of phase transitions and other critical phenomena. For a treatment of these matters the reader may consult

Amit, P. J., *Field Theory, the Renormalization Group and Critical Phenomena*. McGraw-Hill (1978).

See also the relevant articles in the series *Phase Transitions and Critical Phenomena* (C. Domb and M. S. Greene, eds., Academic Press), especially volumes 1 and 6.

3.4 Encodings of Biological Systems: Preliminary Remarks

In the preceding chapter, we explored some of the ways in which physical systems could be encoded into formal ones. The present chapter is concerned with encodings of biological systems. The point of departure for all physical encodings was seen to reside in the fundamental concept of an *observable*. In phenomenological terms, an observable represented a capacity for interaction (manifested by the ability to move a meter) and in formal terms it could be regarded as a mapping from abstract states to real numbers. Correspondingly, we must begin our treatment of biological encodings with a discussion of observables of biological systems, and the nature of biological observations generally, before turning to specific examples.

Let us consider initially the most primitive of biological concepts; the recognition that a particular system is in fact an *organism*. The very word connotes that it is possible for us to discriminate in some fashion between systems which are organisms and systems which are not. The basis for such a discrimination must lie in some quality, or qualities, which are possessed by organisms but not by other systems, and in the fact that we can directly recognize these qualities.

Now we have seen that qualities in general, as percepts pertaining to the external world, are to be identified with definite observables of the systems which manifest them. As we have repeatedly noted, the diagnostic of a quality is precisely its ability to be recognized under appropriate circumstances; that is, its ability to move another system in a definite way. Insofar as the quality of being an organism is directly perceptible to us, we are ourselves playing the role of a measuring system for that quality; our very perception of an organism means that the quality in question is capable of moving us, or imposing dynamics on us, in a unique way. We cannot say precisely that we ourselves constitute *meters* for the numerical expression of this quality; nevertheless, the process whereby we recognize that a particular system is an organism is exactly analogous to the kinds of things we call measurement in dealing with physical systems.

This quality through which we can directly discriminate between organisms and non-organisms, which embodies the basic perception underlying the very existence of biology as an autonomous natural science, is in fact typical of the class of qualities with which biology must deal. There are many ways in which such qualities behave like the observables of the physicist, but there are clearly many fundamental differences as well. The similarities allow us to bring to bear on biological problems most of the machinery we have developed above for dealing with observables of natural systems; in particular, the basic concepts of abstract states and of linkage.

On the other hand, the differences between them endow biology with its unique properties, and make the encoding of biological systems much different from that encountered in our previous examples. Of particular interest and importance are the cases in which we wish to simultaneously encode biological qualities and physical observables, and to establish linkages between them; as we shall see, the whole question of reductionism in biology devolves onto the existence of such encodings. Indeed, the basic (and as yet unanswered) question as to whether or not we can define the quality of being an organism in purely physical terms amounts precisely to *linking* the quality to an appropriate family of physical observables through the analog of an equation of state; i.e. of finding a family of physical observables which will serve as a complete set of invariants for the quality of being an organism.

Just as we directly perceive the quality of being an organism, so too do we directly perceive other similar qualities which in organisms are linked to it. Such terms as irritability, adaptation, growth, development, metabolism, reproduction, evolution, and indeed most of the basic vocabulary of biology, refer precisely to such qualities. Once again, we can say that our direct perception of such qualities means that they share the basic attributes of numerical observables; but the differences between them require a substantial modification of the encoding procedure we have used heretofore. Likewise, our encoding of the linkage relation itself must also be extended, in such a manner as to encompass not only such qualities, but those represented by the numerical observables with which we have so far been exclusively concerned.

Let us try to characterize further these basic biological qualities, and how they can be related to qualities which can be encoded as real-valued functions. Indeed, any organism with which we are confronted must manifest qualities of *both* types; it must manifest biological qualities precisely because we can recognize it as an organism, and it must manifest numerical qualities because it is above all a natural system. We must use these latter qualities to discriminate between organisms, just as we use them to discriminate between other kinds of natural systems. We note that this is the basis of *taxonomy*, which rests on the fact that the quality of being an organism can be manifested by an unlimited number of different kinds of natural systems, whose numerical observables (and their linkages) can be vastly different from system to system. Thus, taxonomic concepts provide a good starting-point for our characterization of biological qualities, and the manner in which they are linked to each other and to the more familiar numerical qualities. There is indeed a sense in which it is correct to say that *the quality of being an organism is represented by an observable which takes its values in the taxa of the taxonomist*. We shall enlarge on this point as we proceed.

Traditionally, a taxonomic unit, or taxon, is defined entirely through the numerical-valued observables which any organism must manifest as a natural system. Hence a specific organism is *assigned* to a taxon if it manifests the defining observables, and if their values are linked to each other in the appropriate way. Thus, to the extent that belonging to a particular taxon is a biological quality, we see that this quality is, by definition, itself linked to the numerical observables through which the taxon itself is specified.

We might use this fact to try to relate *any* biological quality of an organism to the real-valued observables which it manifests, in the following way. We could formally identify a taxon with the class of all organisms which belong to it, in much the same way as a cardinal number is defined in mathematics as the class of all sets of that cardinality. However, such a procedure is not generally valid here, for the following reason. Insofar as a taxon is defined *entirely* through linkages between numerical observables, it pertains to all natural systems, and not just to those we recognize as organisms. For instance, if we define the taxon “leopard” by *any* set of observables and linkages, however extensive, we can imagine constructing a completely artificial system which exhibits precisely those observables and linkages. Such an artificial system is clearly not an organism, but it must belong to the taxon because it satisfies the defining criteria. This indeed is precisely the difficulty faced in trying to *identify* the quality of being an organism with a set of numerical observables and linkages, which is basically the attempt to “define life”. We could stipulate at the outset that our definition of a taxon is to be restricted to the class of organisms, but this simply begs the crucial question.

The basic problem exemplified in our discussion so far is in fact one which we have seen before. It involves the fundamental distinction made earlier between *recognition* and *generation* which we considered in Sects. 2.2 and 3.1. We saw, for instance, that for any system of axioms rich enough to be mathematically interesting, there must always exist propositions we can *recognize as true*, but which we cannot effectively *generate* from the axioms by means of the available production rules; this is the content of Gödel’s theorem. Translating this situation to the terminology we have been using, Gödel’s theorem asserts that the *quality* of being a theorem is one which can be recognized directly, but for which no specific linkage can be effectively constructed between this quality and our axioms.

We find similar situations in physics. The *quality* we call temperature, for instance, is directly perceptible by us, but cannot be directly linked to microscopic qualities measured in the conventional units of mass, length and time alone. To measure it, we need to construct a new kind of meter, and a special set of units (degrees); what we showed in Example 4 of the preceding chapter was that the quality of temperature was *consistent* with mechanical observables, but not directly derivable from those observables through specific linkage relations. Indeed, we saw that temperature is a property of an *ensemble* of mechanical system, and not of an individual one. Thus, temperature provides a specific, physical example of a quality which can be directly recognized, but not generated in a purely mechanical context. In a different physical realm, the basic qualities through which the electromagnetic field is defined can likewise be directly recognized, but not measured in terms of the same meters appropriate to mechanical systems; we must again invent new meters, and new units, to characterize these qualities. Thus it would not be an unprecedented circumstance for the qualities we recognize as pertaining to organisms to be of this character as well. Above all, it implies nothing mystical, vitalistic or unphysical about these qualities, just as there is nothing mystical or unphysical about such qualities as temperature and charge. At root, as we have emphasized, such a situation is merely an expression of the fact that we can recognize more qualities than we can generate.

In sum, there are three viewpoints which may be adopted regarding the relation of biological qualities of an organism to those which we can represent by numerical-valued observables. These are:

1. All biological observables are directly linked to an *appropriate* set of underlying purely physical observables. This is the reductionist position, and typically it is sought to link biological qualities directly to the character of specific particles (molecules) of which an organism is composed. In the words of one of the devoted postulants of this point of view: “The point of faith is this: make the polypeptide sequences at the right time and in the right amounts, and the organization will take care of itself”. Although superficially plausible, and apparently directly related to observation, this position is in fact the least tenable alternative, for reasons which we have discussed in detail elsewhere.
2. Let us consider some specific biological quality, such as motility. This is a quality manifested by organisms, but it is not itself co-extensive with the quality of being an organism; in particular, there are many non-organic systems which are motile. Any non-organic motile system can be entirely represented by encodings of numerical observables, in the fashion we have already considered in detail. In such a system, we may expect that the quality of motility will be directly linked to an appropriate family of such observables. The class of such motile systems, then, can be regarded as consisting of *metaphors* for organic motility, in the sense we used that term in Sect. 2.3. In particular, we may hope to understand the specific quality of motility *in organisms* by considering them as elements of the larger class of motile systems, and thereby formally relating their motility to appropriate numerical observables in the fashion characteristic of that class. It is this procedure which underlies most of the approaches taken to a theoretical understanding of individual organic qualities; most of the examples we shall consider below are precisely of this character.

The essence of this approach is to obtain insight into a *particular* quality, like motility, by embedding a given system manifesting the quality into a *class* of systems bearing a metaphorical relation to it, and for which linkages can effectively be established between this quality and others. In physics, for example, once we recognize some particular system as a gas, we can apply it to the full machinery of thermodynamics, simply because we know how this machinery can be applied to other gases. As we shall see, this exploitation of metaphoric relations provides us with a great deal of “free information” (to use a term of Eddington’s) which we can bring to bear on *any* particular system simply from a knowledge that the system belongs to a certain class. In slightly different language: by embedding a given system into a class in this fashion, we create a *context*, which in itself is often sufficient to specify the form of the basic linkage relations pertaining to our system. It is by exploitation of such context, for example, that a paleontologist is able to reconstruct an entire anatomy from a single bone; utilizing the context created by the comparative anatomy of extant organisms as a source of information. Such considerations of metaphor, or context, also underlie many empirical studies of biological behaviors, such as the

employment of inorganic films to elucidate properties of biological membranes, or the use of catalytic surfaces to help understand enzymes, or the employment of rats and other organisms to help understand disease processes and therapies in humans. In the limit, metaphor converges to *analogy*, and the usages we have described become identical with the employment of a scale model to determine the exact behavior of all systems similar to it.

We have seen how we can employ the relation of metaphor to throw light on how some specific biological quality, like motility, is linked to other observables. We can in principle do the same for *any* particular biological quality, we determine a class of systems manifesting it, in which this quality is specifically linked to others. Now let us imagine that we form the *intersection* of all these classes. A system in this intersection will by definition exhibit *all* of the basic biological qualities, and we will know in principle how to link each such quality, considered individually, to other observables. This intersection is not empty, because the organisms themselves must lie in it. The question is whether this intersection contains *only* the organisms. The conviction that this is the case underlies what used to be called *biomimesis*; the attempt to construct natural systems exhibiting the basic biological qualities. The idea was that a natural system manifesting enough of the basic biological qualities would itself be an organism. It is important to recognize, though, that this approach to biomimesis through an exploitation of relations of metaphors does not itself imply reductionism; the biomimetic hypothesis can be true without reductionism being true.

3. The third approach strives to deal with biological qualities and their linkages directly, without relating them initially to numerical observables of any specific kind of natural system. This is the approach characteristic of *relational biology*. We can then recapture individual systems, and the observables and linkages which characterize them, through a process of *realization*. The relational approach is thus analogous to the study, say, of transformation groups by regarding them initially as abstract groups. Any particular transformation group can then be recaptured through a process of representation; i.e. by a “structure-preserving mapping” which maintains the abstract linkages while endowing the abstract elements with specific properties consistent with these linkages. We shall consider this relational approach, which is the most radical of those we have described, in more detail below.

3.5 Specific Encodings of Biological Systems

In the present chapter, we turn to some specific examples of encoding of biological systems, with special reference to the way in which characteristically biological qualities are related to the numerical observables manifested by all natural systems. For this reason, we shall not consider reductionistic approaches in detail, nor any of the encodings belonging entirely to *biophysics*. The essence of such encodings is precisely that they treat biological systems exclusively in terms of their numerical

observables. Thus, for example, we shall ignore the large literature on the flow of blood as a hydrodynamic problem, pertaining to the flow of viscous fluid in a family of elastic vessels. For our purposes, this literature is part of hydrodynamics, and not of biology. It is of course true that such studies are often of vital practical importance, but they raise no question of principle beyond what we have already considered in Sect. 3.2 above; the biological origins of such studies and their application to cardiovascular problems are essentially irrelevant to the encodings themselves, and the manner in which inferences are drawn from them. We shall concentrate instead on examples of encodings which exhibit a basic metaphorical aspect, and on relational encodings. It is only in these cases that particular biological qualities play a dominant role.

*Example 1. The Mass-Action Metaphor*¹

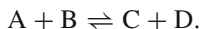
We have seen, in Example 1 of Sect. 3.3 above, that one way in which to encode dynamical processes is to specify the manner in which the rates of change of observable quantities depends on their instantaneous values. Indeed, we stressed earlier (cf. Sect. 1.2) that such a specification is fundamental to the reactive paradigm itself. In mechanics, it is Newton's Second Law which allows us to do this, by linking the concept of force explicitly to the rate of change of velocity or momentum. Thus, if we are going to encode any kind of dynamical process in a natural system, or class of such systems, we need to find the appropriate analog of Newton's Second Law; i.e. we need to postulate the linkage relations which express instantaneous rates of change of observables as a function of their instantaneous values on states.

Let us suppose that the system under consideration consists of a family of interacting units of various types. A natural set of observables to employ for encoding such a system comprises the instantaneous sizes of the populations of the "species" of units involved in our system. More generally, if our units are arranged in space in some fashion, we may regard these observables as *concentrations* or *densities*. Following the usual procedure, we shall suppose that these observables have spectra which are *continua* of real numbers. That is, if f is such an observable, and if there is an abstract state s such that $f(s) = r$, then there is another abstract state s' for which $f(s') = r + \epsilon$, where ϵ is arbitrarily small. It will be noted that this presupposition involves the compromise described in Example 2 of Sect. 3.3 above; our population of units is in fact discrete and finite, while the representation of a concentration or density in the above fashion presupposes the properties of the continuum. Thus, we must regard our observables f as *average* quantities, to which the discrete and continuous pictures can only be approximations.

Let us suppose that the units of our system may be inter-converted as a result of their interactions, according to definite rules. How are we to link the rates of change of our observables to their instantaneous values in such a situation? The general postulation whereby we may accomplish this is the *Law of Mass Action*, which thereby plays exactly the same role here as Newton's Second Law did for mechanical systems. This law states that the instantaneous rate of change of any concentration or density is proportional to the product of the concentrations of the

interacting units involved. If more than one type of interaction among the units can contribute to the rate of change of concentration, then this rate of change is a sum of such products.

The Law of Mass Action was originally proposed to account for the rates at which chemical reactions proceed. A chemical system can be regarded as a family of interacting populations of chemical species (molecules); a particular reaction in the system is a mechanism whereby specific kinds of units are converted into others. Thus, for example, suppose that a particular reaction can be expressed in the familiar chemical shorthand as



Suppose we are interested in the rate at which the concentration $[A]$ of the chemical species A is changing. According to the Mass Action Law, we note that there are two independent mechanisms contributing to this rate of change: (a) the “forward reaction”, through which A is consumed, and (b) the “backward reaction”, through which it is produced. We can then write our linkage explicitly as

$$d[A]/dt = -k_1[A][B] + k_2[C][D],$$

where k_1 , k_2 are constants (rate constants), and the choice of sign of the summands is determined by whether the particular interaction is producing reactant or consuming it. We can write down similar rate equations for the other reactants; we obtain thereby a dynamical system, which can be treated as a formal encoding, or *model*, of the dynamical process in which we are interested. Thus, the Law of Mass Action is the linchpin in the study of chemical kinetics, just as Newton’s Second Law is in mechanics.

It should be carefully noted that there is nothing in the discussion so far which specifically singles out chemical systems; everything we have said so far pertains equally well to *any* natural system of interacting populations of units, in which units may be interconverted as a result of their interactions. Chemical systems indeed represent one large subclass of such systems, in which we also know that the Mass Action Law generates faithful dynamical models. We can then think of using the subclass of chemical systems as *metaphors* for other systems of interacting populations, even though on physical grounds these other systems may be far removed from chemistry. If this is done, we may extend the Mass Action Law from its initial restricted domain in chemistry, and postulate it as a universal mechanism for linking rates of change to population size, valid throughout the entire class.

It was the fundamental insight of Lotka and Volterra to carry out this procedure in detail in an ecological context. Here the interacting units are not chemical reactants but biological individuals. As a result of such interactions, the population sizes comprising such an ecological system will change, just as the concentrations of the reactants will change in a chemical system. The nature of this change in ecological population sizes is particularly transparent in the special case of predator-prey interactions. Thus, with Lotka and Volterra, let us consider the simplest possible

such system, consisting of a single species x (the prey species) and a single species y (the predator species). We suppose that three “reactions” occur in such a system:

- (a) $x \rightarrow x$; i.e. there is an autocatalytic first-order increase in the prey species;
- (b) $x + y \rightarrow y$; i.e. there is a conversion of prey to predator as a result of their interaction;
- (c) $y \rightarrow 0$; i.e. there is a first-order loss of predator, which can be regarded as a decay to an inactive or non-reactive form.

Applying the Law of Mass Action under these circumstances, we find the following rate equations:

$$\begin{aligned} dx/dt &= k_1 x - k_2 xy \\ dy/dt &= k_2 xy - k_3 y \end{aligned} \tag{3.69}$$

where k_1 , k_2 , k_3 are appropriate rate constants. These are the prototypes of the general Lotka-Volterra rate equations for describing the dynamics of interacting populations in an ecosystem.

The procedures we have just sketched shows in detail how the “chemical metaphor” leads directly to a system of rate equations in a non-chemical system, but which nevertheless shares certain *basic features* with chemical systems. It should be noticed that the specific features on which the metaphor is based (namely, that both chemical systems and ecological systems involve the interactions of different kinds of populations, through which interconversions of these populations occur) are *themselves observable properties* and as such must represent *qualities* manifested in common by both classes of systems. The metaphor can then be regarded in the following way: it is a linkage between this shared quality and the formal property of encodings embodied in the Mass Action Law. The Mass Action Law itself expresses a specific linkage between rates of change of certain observables and their instantaneous values; thus *the metaphor serves to identify this kind of linkage universally with the quality which assigns a particular natural system to our class*. This is the essential conceptual step involved in the employment of a metaphor; we shall see it again and again as we proceed.

It should also be noted that, as with all relations of metaphor and analogy between natural systems, there is no question of a *reduction* of ecology to chemistry, or vice versa. The relation of metaphor between natural systems involves a mathematical relation *between encodings*, which is then regarded as embodying a quality manifested by the systems themselves. In the present case, the mathematical relation between encodings involves the presence of a certain kind of linkage between rates of change of observables and their instantaneous values; we identify this with a *quality* manifested by the systems themselves, but *which is not itself directly encoded* (or indeed encodable) in terms of numerical observables evaluated on abstract states of an individual system.

Once we have obtained the rate equations (3.69), the investigation of their formal properties leads us to a number of other relations between the ecosystems they describe and other classes of natural systems which we have already considered.

We shall here note only one. It can readily be shown that the rate equations (3.69) can be converted, through a simple mathematical change of variables, into a form in which *all the conditions are satisfied which allow the formalism of statistical mechanics (cf. Example 3 of Sect. 3.3 above) to be applied to them*. This was already noted by Volterra, but has been most systematically exploited by E. H. Kerner.² By exploiting this formal property of the encoding, we can construct an exact analog for ecosystems of the procedure by which we can pass from the microscopic Newtonian dynamics to a macroscopic thermodynamics. We can thus establish a relation of *analogy* between ecosystems and thermodynamics systems, which is formally of the same character as the mechano-optical analogy relating optics and mechanics in physics (cf. Example 4 of Sect. 3.3 above). This “ecosystem-thermodynamics analogy” is extremely suggestive in many ways. Unfortunately, its specific usefulness as an empirical tool has been thus far limited by the fact that we have, as yet, no empirical analogs of the *meters* (such as thermometers and manometers) which play the crucial role in physical thermodynamics. Stated another way: our present perception of ecosystems is currently restricted to the microscopic level specified by the dynamical equations (3.69). We cannot, as yet, perceive them directly in macroscopic or thermodynamic ways. Therefore, we have no direct insight into the *significance* of the thermodynamic analogy for the behavior of ecosystems. But exactly the same would be true if, in physics, we could not perceive a gas except through the mechanical state variables describing its individual particles. The thermodynamic analogy makes it clear that there must be a “macroscopic” level in terms of which ecosystems can interact; we must await the development of specific empirical instruments to make their qualities directly perceptible to us.

Example 2. Morphogenetic Metaphors

The phenomena of morphogenesis and pattern generation in development are among the most picturesque and striking in all of biology. Until recently, at any rate, there appeared to be nothing within the compass of our experience with inorganic systems which was remotely like them. Experience with the phenomena of embryology led experimentalists like Driesch to despair *in principle* of finding physical explanations (i.e. linkages between physical quantities and those characteristic of developmental phenomena) and thus to a mystical vitalism. In a similar vein, but from the opposite direction, the physicist Maxwell proposed that living units were “demons” who could violate the Second Law of Thermodynamics.³

What are the basic characteristics of morphogenetic phenomena, which has so bemused successive generations of scientists up to the present time? Let us list a few of them, initially in informal language; we shall then see to what extent each of them can be captured through metaphorical means.

The first, and most obvious, characteristic of morphogenesis is the apparently magical emergence of successive novelties of structure and function, which in simplest terms can be regarded as the growth of heterogeneity. Quite apart from the simple growth in size (which is in itself remarkable), development proceeds from a relatively structureless and homogeneous zygote to a complete functional organism

of almost unfathomable complexity in a very short time. The progression from relative homogeneity to limitless heterogeneity, which is contrary to all physical intuition, proceeds in an irreversible fashion. Furthermore, the gross features of the developmental process itself are startlingly similar even in organisms of the utmost diversity.

The second basic feature of morphogenesis which we shall note here is its stubborn stability against external interference. It is in fact very difficult to experimentally interfere with the process, in any but trivial ways, without killing the organism. Let us give a few examples of this stability: (a) It was shown by Driesch that if the two blastomeres arising from the first division of a fertilized frog egg, each of which normally gives rise to half the cells of an adult frog, are separated from each other, then each blastomere will develop into a complete frog (it was these experiments, more than any other, which convinced Driesch that no mechanistic explanation for developmental phenomena was possible in principle). (b) It was shown by E. Newton Harvey that a fertilized sea urchin egg could be centrifuged for hours at several thousand g, completely disrupting all spatial relations between cellular structures; nevertheless, such a cell would still give rise to a normal developmental sequence. (c) It was shown by Holtfreter, Moscona and many others that if the cells from an embryonic organ were randomized, they would spontaneously "sort out" to restore at least an approximation to the original histology of the organ. (d) If a limb bud or other rudiment from one embryo is grafted onto a second, it will continue its development as if it were still in its original site.

These stability phenomena, and many others which could be mentioned, collectively came to be called *equifinality*⁴; the tendency of a developing organism to attain the same final configuration, despite external interference of the type we have mentioned.

Finally, developmental phenomena are under exceedingly strict control both in space and in time. Part of this control is exerted through direct genetic mechanisms; other parts are epigenetic. Examples of this kind of control are shown by the experiments of Spemann on organizer regions; these produce specific signals (evocators) which evoke specific developmental responses in competent populations of target cells. Both the capacity to organize, and the competence to be organized, are sharply limited in time; hence developmental events of this character must be kept synchronous, and this synchrony, of course, is evidence of the rigor of developmental control.

Let us now turn to the question of how such phenomena may be understood. As we said before, our approach will be of a metaphorical character; we shall attempt to embed each of the behaviors we have described in a wider context, in which it may be linked to more conventional numerical-valued observables.

2A. *Metaphors for the emergence of heterogeneity*

Let us consider how we may approach the first of the phenomena we described: the emergence of heterogeneity from an initially homogeneous state. We shall

restrict our attention here to the simplest form of this problem; the spontaneous generation of a polarity or axis of symmetry in a system which originally does not possess one.

The crucial insight underlying the approach we shall develop was first articulated by N. Rashevsky.⁵ He recognized that a *spontaneous* passage from a homogeneous to an inhomogeneous situation, in any context, could only occur under circumstances which rendered the homogeneous state *unstable*, in a mathematical sense. He proceeded to show how such a situation could arise, within a plausible physical context.

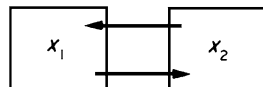
Heuristically, Rashevsky's argument was as follows: Let us imagine a membrane-bounded chemical system, which to a first approximation may itself be regarded as a metaphor for a biological cell. Suppose that some reactant is produced at various sites distributed in this system, and which then proceeds to diffuse away from the sites. Suppose further that the system contains particles which inhibit the activity of these sites. Rashevsky had shown earlier (in the course of his investigation of a metaphor for cytokinesis, or cell division) that flows of reactants down diffusion gradients could exert mechanical forces on suspended particles. In the present situation, these forces would drag the inhibitory particles to regions of low reactant concentrations. Since the particles by hypothesis inhibit particle sources, the effect of the flow is to magnify any gradients which exist; this much is clear intuitively.

Rashevsky then showed that if the initial distribution of particles and sources were spherically symmetric, the spherical symmetry would be preserved by such a flow. However, this condition of spherical symmetry is *unstable*; if any hemisphere of the system should, by chance, come to have a higher concentration of sources or inhibitory particles than the other hemisphere, the effect of the flow would be to *amplify*, or magnify, this initial asymmetry. Eventually, all the inhibitory particles would come to lie in the hemisphere which initially contained more of them, and all the sites at which reactant is produced would come to lie in the opposite hemisphere. Moreover, this new situation is stable; as Rashevsky pointed out, no matter how the contents of the system are stirred or divided, the same kind of final configuration would be produced (cf. the experiments of Driesch and Harvey referred to above). Intuitively, a system of this kind spontaneously establishes a polarity, in which two opposite hemispheres come to possess completely different properties.

The elementary discussion we have given above exemplifies the basic features of the metaphors we shall consider. These features are: (1) a coupling of chemical reactions and diffusion in a spatially extended system; (2) the rendering of the homogeneous state *unstable* through such a coupling. (3) In such circumstances, the apparently spontaneous transition of the system from homogeneity to inhomogeneity.

Thus, the setting for our metaphors will comprise the class of encodings of diffusion-reaction systems. (We shall, as usual, call the *encodings* of a diffusion-reaction system also by that name.) It should be observed that (a) we have seen informally that this class of systems can indeed exhibit behaviors characteristic of

Fig. 3.15



those seen in developing systems; (b) the operant dynamic features of the class, namely chemical reactions and diffusion, are universally manifested in developing systems.

The basic points which Rashevsky enunciated were independently rediscovered a decade later by the English mathematician Turing,⁶ who proposed them in a far simpler context. To fix ideas, let us consider the simplest possible Turing system, which is shown in Fig. 3.15 below:

We envision here two identical “cells”, each of which is completely specified by the concentration of a single reactant (Turing used the term “morphogen”); the amount of morphogen in the first cell at an instant t will be denoted by $x_1(t)$; the amount of morphogen in the second cell at the instant t will be denoted by $x_2(t)$. We suppose that morphogen can flow between the cells by ordinary Fickian diffusion; and that morphogen is produced at some constant rate S , and is lost by a first-order process, which is the same for both cells. Thus, according to Mass Action, we can encode these postulations into a dynamical system of the form

$$\begin{cases} dx_1/dt = -ax_1 + D(x_1 - x_2) + S. \\ dx_2/dt = -ax_2 + D(x_2 - x_1) + S. \end{cases} \quad (3.70)$$

where a is a rate constant for first-order decay or morphogen, and D is a diffusion constant. This appears to be a completely linear system; it is not in fact a linear system, because the morphogen concentrations x_1, x_2 must remain non-negative. However, in the interior of the first quadrant of the “state space” of the system, it does behave linearly. Let us look at this behavior.

First, we observe that the system has a steady state given by

$$x_1^* = x_2^* = S/a. \quad (3.71)$$

At this steady state, the concentration of morphogen is equal in the two cells; i.e. the steady state is homogeneous.

Next, we inquire into the stability of this steady state. We know that the stability is determined by the eigenvalues of the system (3.70); i.e. by the eigenvalues of the system matrix

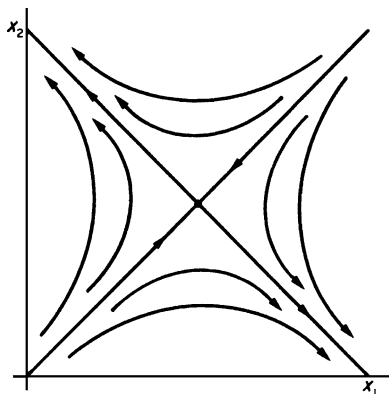
$$\begin{pmatrix} D - a & -D \\ -D & D - a \end{pmatrix}$$

It is easy to verify that the two eigenvalues of this matrix are given by

$$\lambda_1 = -a; \quad \lambda_2 = 2D - a.$$

The first of these eigenvalues is always negative. The second is negative as long as the diffusion constant D between the two cells is not too large; in more precise

Fig. 3.16



terms, it is negative as long as $D < a/2$. In such a case, the steady state (3.71) is stable. However, if $D > a/2$, the eigenvalues λ_2 becomes positive; the steady state (3.71) becomes a *saddle point*, and the trajectories of (3.70) are as indicated in Fig. 3.16 below:

In this case, as in the Rashevshy systems mentioned earlier, any deviation from homogeneity becomes successively amplified because of the manner in which reaction and diffusion are coupled in (3.70). Since by hypothesis morphogen concentrations cannot go negative, the dynamics drives us to a situation in which all the morphogen is either in the first cell or in the second cell, depending on the character of the initial deviation from homogeneity. Thus, here again a polarity is spontaneously established.

The situation we have just described can clearly be enormously generalized. For instance, we can consider any finite number N of cells, arranged in an arbitrary way in space.⁷ We can suppose that instead of a single morphogen we may have any finite number of n of morphogens, interacting with each other according to any arbitrary reaction scheme. Thus, if u_{ij} represents the amount of the i^{th} morphogen in the j^{th} cell, we obtain analogous to (3.70) a system of Nn rate equations of the form

$$du_{ij}/dt = f_i(u_{1j}, \dots, u_{nj}) + \sum D_{ijk}(x_{ij} - x_{ik}) \quad (3.72)$$

where the summation is taken over all cells k adjacent to the j^{th} .

In this scheme, the second index j plays the role of a position co-ordinate. We can pass to a continuum by allowing this index to range over a continuum in an appropriate fashion; we then find that the equations (3.72) become partial differential equations of the form

$$\begin{aligned} \frac{\partial u_i}{\partial t} &= f_i(u_1, \dots, u_n) + D_i \nabla^2 u_i, \\ i &= 1, \dots, n. \end{aligned} \quad (3.73)$$

which we have seen before (cf. (3.42) above).

It may be noted that the rate equations (3.70) or (3.72) or (3.73) always come equipped with the fluctuations required to manifest any instability of the homogeneous states. This is because, as we have repeatedly pointed out, the quantities whose rates of change are described thereby are *average* quantities, and thus they necessarily deviate in a random way from the observables they encode. These random deviations are precisely the fluctuations needed initially to move the system away from a steady state; thus, in this picture, the generation of inhomogeneity is indeed totally intrinsic.

Rate equations of this character have been intensively studied over the past decade or so, under the rubric of “symmetry-breaking” or “dissipative structures”, especially by I. Prigogine⁸ and his associates. A primary motivation for this study has been the recognition, motivated in large part by the phenomena of biological development which we have been describing, that classical thermodynamics is not adequate to deal with essential physical situations. We have already noted (cf. Example 3 of Sect. 3.3 above) that thermodynamics is essentially restricted to considerations of equilibrium in closed, isolated systems. In other words, we cannot hope to effectively encode into a thermodynamic context any kind of physical phenomenon which involves far-from-equilibrium behavior in an open system. Thus, from this point of view, *a main purpose of the reaction-diffusion metaphor is to suggest general procedures for encoding physical situations falling outside the scope of thermodynamics*. In fact, it is correct to say that biology is here providing a metaphor for physics. We shall return to this view in subsequent chapters, when we discuss the general properties of open systems and their relation to modeling.

Let us note one further feature of the simple formal system (3.70), which has provided the point of departure for another related field of active research. As we saw, the constitutive parameter D occurring in (3.70) plays a crucial role in determining the dynamical behavior of that system. When D is small, there is only one steady state for the system, corresponding to a homogeneous situation, and it is stable. As D is increased through the critical value $D = a/2$, this single steady state *splits*, or *bifurcates*, into (in effect) a pair of new stable steady states, while the original homogeneous state becomes unstable. Such bifurcation phenomena had long been known in purely physical contexts; an essentially similar behavior is shown by a vertical column on which a compressive axial force is imposed. For small values of the force, the vertical position is stable; past a critical value, the vertical position becomes unstable, and new steady states (corresponding to the buckled column) appear. A parameter like D in the system (3.70), or like the axial force imposed on a vertical column, is generally called a *bifurcation parameter*.

We have already discussed the general notion of bifurcation in Sect. 2.3 above, and in Example 5 in Sect. 3.1. We here see the bifurcation phenomenon appearing in yet another context, which it is instructive to relate to our earlier discussions. As we noted above, a bifurcation is generally to be regarded as manifesting a logical independence, or loss of linkage, between two distinct modes of system encoding. In the case of the systems (3.70), we can encode these systems either in terms of the bifurcation parameter D , or in terms of the stability of the homogeneous state. For small D , these two encodings are equivalent; two systems characterized

by similar values of D will also exhibit similar stability properties. But in *any* neighborhood of the bifurcation point $D = a/2$, we can find systems which are arbitrarily close in terms of their D -values, and at the same time qualitatively different in their stability properties. Thus, in any neighborhood of the bifurcation point, we cannot generally annihilate an arbitrary small change of D by means of a structure-preserving co-ordinate transformation on the state space; the two encodings have become completely independent of one another.

The upshot of these observations is the following. We have been considering reaction-diffusion systems as metaphors for developmental processes characterized by a spontaneous transition from homogeneity to heterogeneity. We have seen that, in some sense, the *reason* that reaction-diffusion systems exhibit such behavior resides in their capability to exhibit bifurcations. Thus the reaction-diffusion systems themselves can be regarded as metaphors for a wider class of bifurcating systems; or stated otherwise, *any system which exhibits a bifurcation can be a metaphor for these developmental processes*. This approach to morphogenesis has been intensively exploited in recent years, especially by Rene Thom, under the rubric of the *Theory of Catastrophes*.⁹ The interested reader should consult the literature for further insight into this kind of approach.

Let us conclude our discussion of reaction-diffusion metaphors by observing that their power lies precisely in the fact that *any formal reaction-diffusion system, like (3.73), is to be regarded as a possible encoding of some natural system*. Any natural system, of which a system like (3.73) is an encoding, must then exhibit the characteristic spontaneous transition from homogeneity to heterogeneity. The class of systems (3.73) is thus identified as the class of *potential models* for such behavior, because any system in the class can, in principle, stand in the modeling relation to some natural system. Conversely, insofar as developing biological systems are encodable as diffusion-reaction systems, we tacitly expect that such a system, when encoded, will yield a formal system in the class (3.73), and thus, that any behavior exhibited by virtue of belonging to that class will *a fortiori* pertain to natural systems encodable into the class. This is the basic characteristic feature of a metaphorical approach to biological behavior in general.

2B. Phase transitions and phase separations as morphogenetic metaphors

We saw, in our consideration of reaction-diffusion systems, that the basic property through which they generate inhomogeneity is the presence of a bifurcation parameter. Below some critical value of this parameter, a homogeneous situation is stable; above the critical value, the homogeneous state becomes unstable; new (necessarily inhomogeneous) stable states appear, and the system will spontaneously tend to one of them. Consequently, we argued that any system with a bifurcation parameter of this type could be regarded metaphorically as an encoding of a morphogenetic process.

Many such situations are known to the physicist. A large class of them fall under the general heading of *phase transitions*. We have already considered certain

phenomenological aspects of phase transitions, from a thermodynamic point of view, in our discussion of Example 5 of Sect. 3.3 above. In that discussion, we tacitly treated temperature as a bifurcation parameter, in the context of the van der Waals equation as an encoding of equilibrium states of non-ideal gases. If we look at Fig. 3.14, we can actually see the bifurcation. Specifically, above the critical temperature $T = T_c$ all isotherms are monotonically decreasing. At the critical temperature T_c , a point of inflection (the critical point itself) appears on the corresponding isotherm. Below the critical temperature, this point of inflection splits, or bifurcates, into a minimum and a maximum. As we noted, it is this bifurcation which corresponds to the phase transition.

We can, of course, treat the same phenomenon from the point of view of statistical mechanics. A favorite device employed by physicists for this purpose, is a metaphor generally called the *Ising Model*.¹⁰ In fact, the Ising model is a class of formal systems which can exhibit phase transitions, just as the diffusion-reaction systems are a class of formal systems which can pass from homogeneity to inhomogeneity. As we shall see in a moment, the two classes are in fact closely related.

It is worth while to spend a moment in briefly describing the Ising model. In its simplest form, we imagine a family of units arrayed in space in some regular fashion; e.g. on the vertices of a square lattice. At any instant of time, we suppose that a particular element can be in one of two possible states. We suppose further (a) that the elements can interact, in such a way that at any instant an element tends to be in a state favored by its nearest neighbors in the lattice; (b) the elements may be perturbed by an external influence (such as temperature, or an imposed field). Thus, interactions favor homogeneity of state throughout the lattice; the external perturbation favors inhomogeneity.

Under these circumstances, we can expect that there will be a critical value of the external perturbing influence, below which homogeneity is favored in the system as a whole, and above which heterogeneity is favored. Treating the external perturbation as a bifurcation parameter, we thus expect that at this critical value there will be an abrupt transition between the disordered situation (in which the states of adjacent elements are uncorrelated) to the ordered situation (in which adjacent elements will be in the same state). This is the Ising model version of the phase transition; this intuitive behavior is borne out by detailed statistical-mechanical considerations.

It is clear that the “Ising model” is not one system, but rather a large class of systems manifesting similar features. We can generalize it in many ways; we can consider more general spatial distributions of our elements, or we can consider that our elements have any number of possible states; we can postulate any type of non-nearest-neighbor interaction between the elements. Thus the Ising model is a *metaphor* for phase transitions, and not a “model” in our sense.

The main point at which we are driving can now be stated. We have interpreted the two states of our lattice elements as alternate states in which a definite element can be found at an instant of time. Thus, the transition of an element from one state to another can be looked upon as a *differentiation* of the element. But there are at least two other interpretations which can be made of the same situation:

- (a) Let us suppose that instead of a population of identical elements, each of which may be in one of two alternate states, we envision a population of two distinct kinds of elements, each of which possesses only a single state. In that case, a change of state must be interpreted as a *replacement* of an element of one type by an element of the other type. That is, we view a change of state in these circumstances as a *movement* of one type of element away from a particular lattice position, and a corresponding *movement* of the other type of element into that lattice position.

What is a phase transition under these circumstances? The “ordered” phase, in which elements tend to be in the same state as their nearest neighbors, will represent a situation in which all elements of one type are clustered together in one part of the lattice, and all elements of the other type are clustered together elsewhere in the lattice. The “disordered” phase will represent a situation in which the elements of both types are maximally intermingled. Thus, a “phase transition” in this situation will actually be a *phase separation*, like the breaking of an oil-water emulsion. We notice that the *formalism* describing this situation is exactly as it was before; only the *interpretation* has changed.

- (b) Let us suppose that our population consists only of one kind of element. The two states which can be assigned to a particular lattice position will now be interpreted as the *presence* of an element at that position, or the *absence* of an element. Thus, a change of state in these circumstances can be regarded either as the *birth* of an element, or the *death* of one. This situation is called the “lattice gas” by the physicist; the “disordered” situation corresponds to a uniform distribution of elements and “holes”; the “ordered” situation to a precipitation or accumulation of elements, with simultaneous generation of a “vacuum”. Here again, the basic situation is exactly as it was before; only the *interpretation* has changed.

Let us now return to the problem of morphogenesis, after this brief detour into the Ising model. The biologist knows that there are three basic kinds of processes underlying all specific morphogenetic or developmental phenomena.¹¹ These are:

1. *Differentiation*; the cells of a developing system become progressively different from one another, in their chemical constitution and their physiological properties.
2. *Morphogenetic movement*; the cells of a developing system systematically change their relative positions, as for instance in the phenomena of “sorting out” to which we briefly alluded above.
3. *Differential birth and death*; the various cell populations of a developing system change their relative sizes through modulations of their rates of multiplication and death.

It will be observed that these are exactly the three processes manifested metaphorically in the Ising model, arising from the different interpretations of state transitions of individual lattice elements. Thus, the Ising model provides us with metaphors for all of these basic morphogenetic mechanisms.

Moreover, the reaction-diffusion systems we considered above are themselves special cases of suitable Ising models. For at the molecular level, a chemical reaction is simply the disappearance (“death”) of elements of reactant, with the concomitant appearance (“birth”) of elements of reaction product. Likewise, diffusion is simply the *movement* of a diffusing element from one lattice position to another. Thus, the reaction-diffusion systems can be obtained from the Ising models by a suitable superposition of movement with birth-and-death. The phase transition in such a system is simply the bifurcation we saw before.

Thus, we now have at our disposal a new and larger class of metaphors for developmental processes. This is, as always, a class of *formal* systems, each member of which is a presumptive encoding of some specific physical or biological situation. Moreover, it is a class of metaphors which serves to relate the morphogenetic processes characteristic of developing biological systems to homologous phenomena characteristic of inorganic systems. It thus serves to exhibit a continuity between the behaviors of the organic and inorganic realm, which seemed initially so disparate. It should be carefully noted, as always, that this conceptual continuity has nothing to do with a *reduction* of developmental phenomena to physics; it rather asserts that developmental phenomena may admit encodings falling into the same class of formal systems as do certain phenomena of inorganic nature.

2C. *Metaphoric aspects of genetic control*

We noted at the outset of this discussion that developmental phenomena are under careful control, and that at least some portion of this control must be of a *genetic* character. We wish now to consider the meaning of this kind of assertion, within the context of the metaphors for morphogenesis presented so far.

At the simplest level, the genome of an organism is to be regarded as determining the *species* of biological system with which we are dealing. Thus, almost by definition, any physiological, morphological or developmental characteristic of an organism which is *species-specific* must ultimately be referred to the genome. In particular, if a developmental metaphor of the type we have been discussing is to be fully meaningful biologically, we must be able to identify the sense in which its specific properties are genome-determined. At the same time, we note that the very concept of a genome seems utterly alien to inorganic phenomena, which are also represented by this same class of metaphors. We shall thus begin our considerations of genetic control in morphogenesis by showing that this is not so; incipient in even the simplest example of a morphogenetic metaphor there is already present a genetic component which we can extract.

Let us take as an example a simple inorganic natural system; e.g. a gas. There is clearly a sense in which such a gas possesses a “species”; thus we can tell, for example, that chlorine and nitrogen are *different* gases, or that two samples of nitrogen belong to the same “species”. A moment’s reflection will reveal that the qualities on which our perception of “species” depends reside in the *constitutive parameters* which enter into the equation of state describing *all* species of gas. To say that we have replaced one species of gas by another means that we have, at bottom, made a change in one or more of these constitutive parameters.

If we take, say, the van der Waals equation (3.55) as the equation of state governing the various species of gas under consideration, we see that there are three such constitutive parameters, which we denoted by (a, b, r) . Thus, within the confines of the encoding of our gas into this formalism, it is reasonable to refer to the specific values of these parameters as determining the *genome* of the gas. It must, of course, be carefully noted that the property of being a constitutive parameter belongs to a *formal* system; but we will as usual *impute* this property to the natural system encoded into it.

In the example we are considering, the three constitutive parameters (a, b, r) are related to the equilibrium values of the state variables (P, V, T) by an equation of state of the form

$$\Phi(P, V, T, a, b, r) = 0.$$

Here two of the state variables (say P and T) can be fixed externally; they can thus be regarded as properties pertaining to the *environment* with which the gas is equilibrating. The equilibrium volume V is then determined by the equation of state. Moreover, when P and T are fixed, the specific *value* of the equilibrium volume under these conditions is determined entirely by the genome (a, b, r) . It is not too great an abuse of language to refer to the volume so determined as the *phenotype* of the gas, determined by the genome (a, b, r) and the environment (P, T) .

In general, if the genome (a, b, r) is fixed, then the equation of state defines a specific mapping Φ_{abr} from the space X of all environments (i.e. all (P, T) pairs) to the space Y of all phenotypes (all values of V). That is, the equation of state can be expressed as a relation of the form

$$\Phi_{abr} : X \rightarrow Y$$

which precisely determines what the corresponding phenotype will be under a given set of environmental conditions.

Exactly the same kind of considerations can be applied to any linkage between observables which involves constitutive parameters. The equation of state which expresses this linkage can thus always be interpreted as specifying phenotype as a function of genome and environment. Thus, insofar as “genome” can be identified with “species-determining”, we can always specify a genome, for *any* encodings of linkages, in *any* natural system. Moreover, we can do so in a way which allows us to bring to bear the entire formal apparatus of *similarity*. As we have seen, similarity governs the relations which exist between systems with different genomes which obey the same equation of state. This observation in itself discloses an exceedingly rich and fertile area of study, but one which we shall not pursue in detail here; our present point is simply to show that the concept of a genome, in the limited sense of “species-determining”, is already present in any encoding; and in particular, is available in the morphogenetic metaphors we have already discussed.

However, the term “genome” means much more to a biologist than simply “species-determining”. Several decades of intensive research into the molecular basis of cellular activities have led to the conclusion that the ultimate role of the

genome (in a cell) is to modulate the rates of the reactions which take place in the cell, by determining the kinds and amounts of the specific catalysts (enzymes) which catalyze these reactions. Furthermore, we can see from the facts of differentiation of cells in development that, even at this elementary level, the relation between the genome of a cell and the reactions which occur in the cell is not a simple one. For we must suppose that all cells in a developing organism possess the same genome; yet cell differentiation means precisely that *cells can possess the same genome and nevertheless be completely different from one another in chemical, morphological and physiological properties*. The question then becomes: how can this be?

Briefly, the answer to this question lies in the concept of *gene expression*. According to this viewpoint, we must imagine that any individual gene may at any instant of time be found in one of a number of *alternate states*. For simplicity, we may suppose that there are two states available to any gene, which we may designate simply as “on” and “off”. Thus it is not merely a question of which genes are present in the genome; it is a question of which genes are *expressed* (“on”) and which are not at specific instants; it is a question of the *time course* of gene expression. To a biologist, then, the term “species” means not simply genome, but refers to a temporal pattern of genome expression.

Now insofar as a temporal pattern of genome expression is a “species” characteristic, *it must itself be determined by the genome*. We are thus immediately led to the idea that the change of state of any individual gene at an instant must be determined by antecedent patterns of expression of the genome as a whole, and of course by whatever environmental influences may contribute directly to such a change of state. We are thus led to a situation not very different from that of the Ising model discussed above; a family of interacting elements, in which the change of state of any individual element depends on the states of its “neighbors”, and on the character of the environment. In the present circumstances, the term “neighbor” is not necessarily to be understood in terms of geographic proximity, but in terms of a capability for direct interaction between elements; the two ideas are identified in the Ising model, but are in fact distinct. The picture of the genome itself which is thus emerging can be called a *genetic network*. Thus, a genetic network is a formal system representing an interacting population of two-state elements, and as such comprises a metaphor for the genome itself.

We thus see emerging two distinct metaphors for genetic control. The first one, which is shared by all natural systems, identifies the genome with appropriate constitutive parameters appearing in an equation of state. In biology this viewpoint is adequate for many purposes, especially those involving anatomical or taxonomic considerations; indeed, it was in this kind of context that the very concept of the gene was originally proposed. On the other hand, at a more microscopic level, we find a different picture: the idea of a genome as a network of interacting multistate elements, whose characteristic features involve temporal patterns of expression. This second kind of metaphor is apparently unique to biological systems; it is hard to imagine an analog of this kind of encoding for, say, a gas obeying the van der Waals equation. Thus, we may say that biological systems appear to admit a wide variety of different encodings of their genomes, while non-biological systems do not.

How are we to relate the encoding of a biological genome in terms of constitutive parameters at a morphological level, and in terms of a genetic network at a microscopic level? We are now in the situation diagrammed by Fig. 2.4 above, in which the *same* system admits these two distinct encodings. The question is: does there exist a mapping between the encodings, which allows us to *reduce* one of them to the other? Or is there a more general mathematical relation between them? Or perhaps there may be no relation of a formal character between them at all.

At the present time, these questions simply cannot be answered. All that can be said is that, *as yet*, no such relation between these pictures of the genome can be exhibited. It is, of course, an article of faith of molecular biology that the picture of the genetic network is in essence the fundamental one, to which every other encoding of genetic activity can be effectively reduced. For the moment, however, faith is the only vehicle for asserting such a conclusion; all the other possibilities we have mentioned remain unexcluded.

Example 3. Metaphors for the Central Nervous System

Of all the organs of man, the brain is perhaps the one which is the subject of greatest interest. It is the seat of all the qualities we perceive as peculiarly human, as well as perhaps the major biological organ of control. As such, it has been the subject of modeling and metaphor from very early times. Of greatest interest to the investigator is the manner in which the brain generates and propagates the electrochemical signals which have come to be regarded as the essential features of its activity. In the present section, we will devote ourselves to two distinct modes of formally encoding the phenomena of propagation. We shall then explore some of the ramifications of these encodings, both for an understanding of the brain itself, and to illuminate some of the general principles we have been developing.

We must preface our description of the encodings themselves with a brief description of the biological presuppositions on which they are based. Each of these presuppositions is ultimately founded on experimental observation, but the general form in which we shall present them involves a substantial amount of theory as well. That is, they already involve a considerable amount of tacit metaphorical encoding. For our purposes, it is sufficient to treat them as we would treat geometric postulates; the same metaphorical machinery can be brought to bear on any similar system of presuppositions. Since our major interest at this point is in the metaphors, it thus suffices to regard these particular presuppositions merely as a convenient point of departure.

The presuppositions we shall employ are then the following:

1. The functional unit of conduction and propagation may be identified with an individual nerve cell or neuron. A neuron is initially an *anatomical* unit, which we now endow with specific functional capability.
2. Neurons are anatomically interconnected in a definite spatial pattern. Hence, propagation and conduction phenomena can be thought of as restricted to this pattern of interconnected neurons, which we shall call a *neural network*.

3. Individual neurons are *all-or-none* elements. That is, we may think of them as being in one of two possible states at an instant of time, which we may call “on” or “off”.
4. The state of a neuron at an instant depends on its own state, and on the states of the neurons afferent to it in the network.
5. Neurons are *threshold elements*. In order to make a state transition from “off” to “on”, the magnitude of excitation reaching a given neuron from those afferent to it in the network must exceed some critical value, or threshold.
6. The excitation reaching a neuron from another neuron afferent to it may facilitate the transition from “off” to “on”, or it may inhibit this transition. In the former case, we speak of an *excitatory* interaction; in the latter case, of an *inhibitory* interaction.
7. Neurons are *refractory*; when they have made the transition from “off” to “on” at some instant, they cannot make this transition again for a characteristic time (*refractory period*). During this period, we may regard the threshold of the neuron as infinite.

This is clearly a substantial set of properties, which we assume at the outset. Most of them are subject to extensive qualification, but we repeat that we are using them primarily for illustrative purposes. We also repeat that most of them already involve a substantial amount of tacit encoding of experimental observation into a quasi-formal system. The metaphors we now proceed to describe complete this process of encoding in different ways.

3A. *Two-factor metaphors*¹²

The properties itemized above allow us to regard a neuron as a kind of *meter* for the excitation impinging upon it. That is, each excitation is capable of inducing a dynamics in the neuron, in a manner linked to the magnitude of the excitation. The case of the neuron is a particularly simple one, since we allow the neuron only two states; thus, the spectrum of the neuron (considered as a meter) consists only of two possible values, which we may take as 0 and 1.

The neuron can thus be considered as a “black box” (cf. Sect. 1.2 above), which relates input signals (magnitudes of excitation) to its responses or outputs. Indeed, observation of an individual biological neuron typically takes the form of cataloguing its responses to different magnitudes of input or excitation.

The two-factor theories of excitation now to be described represent metaphors for what is inside such a “black box”. They are thus analogous to the treatment we have given of linear boxes in Sect. 1.2; they are an attempt to carry out the equivalent of a state-variable analysis for individual neurons. The treatment we describe was pioneered by N. Rashevsky; several other equivalent formulations are possible.

We shall suppose that the internal state of a neuron can be characterized by a pair of observables, or state variables, which we shall call x and y . We shall suppose that the response of the neuron is a function of these two state variables; $Y = Y(x, y)$. We shall further suppose that these state variables possess the character of chemical concentrations, as we have described extensively above. We shall now

impose a linkage between the rates of change of these state variables, and an afferent excitation $I = I(t)$, considered as a function of time. We shall suppose that both of these quantities increase at a rate proportional to the instantaneous magnitude of the excitation, and at the same time are subject to first-order decay. Thus, by Mass Action, we can write these linkages as

$$\begin{cases} dx/dt = -ax + bI \\ dy/dt = -cy + dI \end{cases} \quad (3.74)$$

Now we shall *define* an observable $Y = Y(x, y)$ by the relation

$$\begin{aligned} Y(x, y) &= 1 \quad \text{iff } x \geq \theta y, \\ &= 0 \quad \text{otherwise.} \end{aligned} \quad (3.75)$$

The parameter θ appearing in this definition is to be identified with the *threshold* of the neuron. Intuitively, if $Y(x, y) = 0$, we shall say that the neuron is in the “off” state; if $Y(x, y) = 1$, we shall say that the neuron is “on”. Thus, we see that a state transition from “off” to “on” in the neuron requires that the state variable x exceeds the value of y by an amount exceeding threshold. Thus, it is natural to regard x as an *excitatory factor*, and y as an *inhibitory factor*. This is the origin of the terminology, “two-factor” for the metaphor we are describing.

Now let us take the next indicated step. Let us suppose that we have some finite number N of two-factor elements, arranged in space in some fashion. We shall imagine that these elements are organized into a network, by specifying for each individual element which others are afferent to it. We shall regard the output $Y_i(x_i, y_i)$ of the i^{th} element as playing the role of an input to each element efferent to it. Thus, the entire network can be described by a system of $2N$ rate equations of the form

$$\begin{cases} \frac{dx_i}{dt} = -a_i x_i + \sum_j \lambda_j Y_j(x_j, y_j) + b_i I_i(t) \\ \frac{dy_i}{dt} = -c_i y_i + \sum_j \mu_j Y_j(x_j, y_j) + d_i I_i(t) \end{cases} \quad (3.76)$$

where the summations are taken over all elements afferent to the i^{th} , and the coefficients λ_i, μ_i can for simplicity be taken to be ± 1 , depending on whether the interaction is excitatory or inhibitory. A formal system of the form (3.76) will be called a *two-factor network*; these rate equations then describe the manner in which excitation flows in such a network.

The two-factor networks thus comprise a class of metaphors for the behavior of real networks of neurons, considered as encoded into a certain family of highly nonlinear dynamical systems. We shall have more to say about this two-factor metaphor presently.

3B. McCulloch-Pitts networks¹³

We shall now consider a completely different kind of encoding of the *same* basic biological situation. The basic element in this encoding is a formal object, which we shall call the *McCulloch-Pitts formal neuron*.

The crucial distinction between the McCulloch-Pitts metaphor and the two-factor systems just described involves the manner in which time is encoded in the two approaches. In the latter systems, time is encoded as a one-dimensional continuum, which we conventionally take as the real numbers $\mathbb{R}$. In the former, on the other hand, time is assumed quantized into a set of discrete instants, which we may take as the set of integers $\mathbb{Z}$. The rationale behind this drastic step lies in the refractory property of neurons (cf. property (3.75) above). The idea is that we need only consider the state of a neuron when it is *capable* of being excited, and that if we use the length of the refractory period as a time unit, we thus need only consider the neuron's state at integral multiples of the refractory period. This seems like a considerable idealization, but in fact it can be formally shown that the class of McCulloch-Pitts systems, which by definition satisfy this property, is essentially identical with the apparently far broader class of discrete-time systems which are not synchronous in this sense. We shall have more to say about this situation in a moment, as it reflects a most important property of the McCulloch-Pitts metaphors; namely their *universality*.

Let us now proceed to define the McCulloch-Pitts neuron, which we emphasize again is a *formal* object. This neuron will consist of the following data:

1. A number $e_1, \dots, e_m$ of afferent *input lines*, which will be called *excitatory*. At any instant of time, each of these lines may be in one of two *states*, which we will designate as +1 ("active") or 0 ("inactive").
2. A number $i_1, \dots, i_n$ of afferent input lines, which will be called *inhibitory*. At any instant of time, each of these lines may be in one of two states, which we will designate -1 ("active") and 0 ("inactive").
3. A non-negative number θ ("threshold").
4. A pair of states for the neuron itself, which will again be taken as +1 ("active") and 0 ("inactive").
5. If $s(t)$ denotes the state of our neuron at the instant t , then this state is linked to the above data at the *preceding* instant by the following rule:

$$\begin{aligned}
 s(t) = +1 \quad & \text{iff} \left[\sum_{k=1}^m e_k(t-1) + \sum_{k=1}^n i_k(t-1) \right] \geq \theta; \\
 & = 0 \quad \text{otherwise.}
 \end{aligned}
 \tag{3.77}$$

That is, the formal neuron is active at time t if and only if the number of active excitatory lines at $(t-1)$ exceeds the number of active inhibitory lines at $(t-1)$ by at least the threshold θ .

We can now organize such formal neurons into networks in the obvious fashion; we regard the state of an individual neuron as its output, and we allow this output to flow to all other neurons efferent to it along appropriate input lines.

It is clear that if we know the connectivity of such a network, and any initial pattern of activity, we can determine the subsequent patterns of activity for every subsequent instant.

At this point, we may pause to notice the vast difference wrought by the apparently purely technical passage from continuous time to discrete time. If we compare the linkage (3.77), determining the state-transition rule of a McCulloch-Pitts neuron with its two-factor analog (3.74) and (3.75), we can see immediately that the former is entirely an algebraic object, while the latter is an analytic object. Thus, the inferential machinery which can be brought to bear on these encodings of neural phenomena is vastly different. This difference is reflected in the kinds of conclusions which can be drawn within these metaphorical situations, which we shall notice as we proceed.

It was already stressed by McCulloch and Pitts that the class of networks of formal neurons is closely related to the kind of mathematical universe which we called $\mathfrak{p}$ in Sect. 2.1. It will be recalled that $\mathfrak{p}$ was a universe of *names* of objects, relations and operations arising in another mathematical universe $\mathfrak{u}$. We saw that as we proceeded to define various structures in $\mathfrak{u}$, we simultaneously created a calculus of *propositions* in $\mathfrak{p}$. The relation between $\mathfrak{u}$ and $\mathfrak{p}$ revolved around the fundamental fact that it was essentially through interpreting propositions in $\mathfrak{p}$ as assertions about $\mathfrak{u}$ that we could assign definite *truth values* to these propositions. We saw that it was essential to have a means of assigning truth values in order to be able to develop any kind of mathematical theory in $\mathfrak{p}$ at all.

Now let us suppose that p is any proposition in $\mathfrak{p}$; any proposition whatsoever. Let us *force* P to be synonymous with the following: the neuron N_i in a particular neural net is active at the instant t_0 ; i.e. p now *means* that $N_i(t_0) = 1$. Then the activity of this neuron automatically *assigns* a truth value (1 or 0) to the proposition p .

If we make this kind of forcing in a judicious fashion, we find that we can construct neural networks which will represent any proposition in $\mathfrak{p}$. For instance, suppose that p, q are arbitrary propositions in $\mathfrak{p}$. How would we represent the proposition $p \wedge q$?

We can do this with a network of three formal neurons; one (call it N_p) to represent p , one (N_q) to represent q , and one ($N_{p \wedge q}$) to represent $p \wedge q$. We need only arrange matters so that

$$\begin{aligned} N_{p \wedge q}(t) &= 1 \quad \text{iff } N_p(t-1) = 1 \quad \text{and} \quad N_q(t-1) = 1 \\ &= 0 \quad \text{otherwise.} \end{aligned}$$

Clearly, the neuron $N_{p \wedge q}$ must have two (excitatory) input lines, and a threshold of exactly 2. The neurons N_p and N_q must be afferent to $N_{p \wedge q}$, each connected along one of the input lines to $N_{p \wedge q}$. We see that *we have thus represented the proposition $p \wedge q$ by forcing its truth or falsity to be determined by the activity of $N_{p \wedge q}$.*

By extending this idea in the obvious way, we can in fact represent any proposition which can be constructed from a finite number of others through application of the basic logical operations. Thus, in a sense, the class of networks of formal neurons is *co-extensive* with the calculus of propositions in $\mathbf{p}$. It is this enormously suggestive fact, arising as it does in a class of metaphors for the brain, which has made the neural network idea so powerful an influence in the theory of the central nervous system. For 'it' asserts that, *if any behavior whatsoever can be described in propositional terms, there is a neural net which will exhibit precisely this behavior*. In other words, the neural networks form a universal class for realizing all propositions of this type.

This universality result, which was already enunciated by McCulloch and Pitts, was exceedingly influential in its day. For it showed that the complexities of the brain could be approached in a constructive fashion through the metaphor of the neural network. Thus, it was a kind of existence proof arguing the possibility, and even the feasibility of understanding brain function in neural terms. Only later did it come to be recognized that the very universality of the class of McCulloch-Pitts networks raised difficulties of its own. We have already touched on this kind of difficulty when we considered the universality of the reactive paradigm (to which the neural nets in fact belong), at the very outset, in Sect. 1.1. Namely, the fact that any behavior can be *represented* in terms of a universal class of this type does not at all imply that the behavior is in fact *generated* within the class. Thus, the class of neural nets may *simulate* any behavior which the brain can exhibit, but we cannot conclude anything from the fact of such a simulation about how the behavior was generated.

Despite this, we cannot overemphasize the historical importance of the McCulloch-Pitts metaphor, and its intimate relation with the logical machinery, on which all of mathematical theory is based. Indeed, this circle of ideas closes upon itself; if mathematics is a creation of the human brain, and if the human brain is representable by the machinery of mathematics, then the intrinsic limitations of mathematics (e.g. as exemplified by Gödel's theorem) are *a fortiori* limitations of the brain. Such ideas have been the source of some profound developments in both brain theory and the foundations of mathematics, as well as of some lively controversy. We may refer the reader to the literature for fuller discussions of these developments.

Let us note here, for further reference, that the logical correlates of the McCulloch-Pitts metaphor seem unreachable from the continuous-time universe of the two-factor metaphor. We content ourselves here with the remark that this unreachability can in fact be fruitfully circumvented.

Example 4. Relations Between the Morphogenetic and Brain-Theoretic Metaphors

We shall now pause to point out some extremely interesting and important formal relations which exist between the various metaphors we have described, for morphogenesis on the one hand, and for the central nervous system on the other. Since these relations pertain to formal encodings of different classes of natural systems, they involve corresponding relations of *analogy* between the natural systems themselves;

cf. the diagram Fig. 2.3 above, and our discussion of the mechano-optical analogy in Sect. 3.3.

Let us first direct our attention to the relation between the two-factor metaphor for the central nervous system, and the reaction-diffusion metaphor for morphogenesis. In particular, let us compare the basic dynamical representation (3.74)–(3.75) of the former with the simple diffusion-reaction system (3.70). On inspection, we see that these dynamical equations (which we recall can be regarded as equations of state; i.e. as linkage relations between states and their rates of change) are essentially identical; *the only difference between them is in the form of interaction assumed between the respective units*. Specifically, in the diffusion-reaction systems, the interaction was simple Fickian diffusion; in the two-factor systems, the interaction is through the observable Y . Thus, we can in effect transform a diffusion-reaction picture to a two-factor picture by modifying the postulated form of the interactions allowed between our units.

On the basis of this simple observation, we can see that the biologically quite disparate problems of chemical differentiation in population of cells, and the propagation of excitation in a network of neurons, are in fact *similar*. Indeed, we may regard the temporal properties of excitations in neural nets as a form of morphogenesis; or better, we can say that both morphogenesis in cell populations and excitation in neural networks are alternate concrete realizations of a class of abstract processes for *pattern generation*.¹⁴

A similar relationship exists between the McCulloch-Pitts networks and the genetic networks which were discussed in connection with gene expression. Indeed, the “genes” in such a network were considered exactly as two-state elements. A very influential treatment of genetic networks from this point of view is tacit in the familiar *operon hypothesis*, which was put forth in 1960 by the French microbiologists Jacob and Monod,¹⁵ on the basis of their experimental work on adaptation in bacteria. It is instructive to spend a moment reviewing this work.

Jacob and Monod postulated a separation of the genome of a cell into two parts: (a) the “structural genes”, which are identified with segments of DNA actually coding for a specific catalytic protein or enzyme, and (b) DNA regions called “operators”, which contain specific binding sites for particular ambient metabolites. The complex consisting of one or more structural genes with the operator which controls them is called an *operon*. In the simplest version, the binding of a metabolite to an operator may render its associated structural genes unavailable for transcription, and thus *repress* these structural genes; in the terminology we used above, such repressed genes are simply not expressed, or are in the “off” state. Other metabolites, which may combine with the repressors or otherwise render them inactive, will clearly release the repression, or *induce* the expression of the structural genes; these will then be in the “on” state. If we regard the small repressor and inducer metabolites as themselves arising from reactions catalyzed by enzymes coded by other structural genes, we see that a population of such operons becomes a network; the “wiring” of the network in this case is determined entirely by the specificities of the operator binding sites. Moreover, the interaction between inducer and repressor is clearly identical with what we called “excitatory and inhibitory

input lines” in our discussion of the McCulloch-Pitts neurons. We can thus take over our discussion of the McCulloch-Pitts networks word for word to the case of the operons; thus *the metaphor we developed for neural nets becomes simultaneously a metaphor for operon networks*. Stated otherwise, neural nets and operon nets are analogs of each other; they are alternate realizations of the same formal class of abstract mathematical objects.¹⁶

This striking fact was not noticed by Jacob and Monod; however, some of the examples they proposed of simple operon networks, which illustrated how such networks exhibit differentiation in terms of gene expression, are absolutely identical with neural networks proposed 30 years earlier by N. Rashevsky and H.D. Landahl, to illustrate how phenomena like learning and discrimination could arise from differential neural excitation. This neural-genetic analogy, which has so far been surprisingly unexploited, possesses many remarkable properties, which the reader is invited to explore; our purpose here is merely to point it out as a specific illustration of our general treatment of modeling relations, analogies and metaphors. It should be regarded as a biological analog (this word is chosen purposely) of the mechano-optical analogy in physics, which was discussed previously.

One further ramification of this circle of ideas may be mentioned. Precisely the same class of systems into which we may encode neural and genetic networks also serves to encode an important class of non-biological systems; namely, the class of *digital computers*. This fact was noted almost from the outset, by McCulloch and Pitts themselves, and by many others; probably the most extensive early exploitation of this fact was by the mathematician von Neumann. As always, the fact that two distinct kinds of natural systems possess a common encoding means that the systems themselves are *analogs* of each other; thus from the beginning, it was recognized that we could in principle construct artificial systems which would carry out brain-like activities. Indeed, an entire field (“artificial intelligence”)¹⁷ has grown up to exploit this analogy, and thereby simultaneously obtain insight into the design of “intelligent” machines, and into the activity of the brain. The close relation between such devices and numerical computation, coupled with the universality to which we have already alluded, are also responsible for the efficacy of digital machines in solving dynamical problems. Indeed, there is a precise sense in which a digital computer, programmed to solve a specific dynamical problem, is analogous to any natural system described by those dynamics. This is a very rich and fruitful circle of ideas, which we can unfortunately only hint at here.¹⁸ However, it should be kept in mind as we proceed with our discussion.

So far, we have considered relations between encodings of distinct classes of biological phenomena; namely those arising from development, and those arising from neural activity. Let us now turn briefly to the relations which exist between *different* encodings of the *same* class of phenomena. As an illustration, let us consider what, if any, relation exists between the continuous encoding of neural phenomena as two-factor networks, and the discrete encodings of the same phenomena as McCulloch-Pitts networks of formal neurons.

We have already pointed out the radical mathematical differences which exist between the two kinds of encoding. Any kind of relation we could establish between

these kinds of encodings would be tantamount to constructing a continuous version of a logical system, or of a formal theory (in the sense of Example 3 of Sect. 3.1), or alternatively, a discrete version of an arbitrary dynamical system. This has long been a tantalizing dream within mathematics itself; the realms of the continuum and of the discrete have always resisted all attempts at unification. However, in the present circumstances, we have a new idea to bring to bear; namely, that specific discrete and continuous systems can serve as encodings for the *same* natural system. If the qualities encoded separately into the two realms are at all linked in the natural system itself, then this linkage will precisely specify a formal relation (like an equation of state) between the two encodings. This is an idea which we shall pursue extensively in subsequent chapters.

Example 5. A Relational Metaphor: The Rashevsky Principle of Biotopological Mapping

In this section we shall consider the first of two metaphors of a quite different character from those which we have already seen. We have so far considered certain qualities associated with biological activities, as for example those manifested by developing systems, or the organized activity of the brain. The thrust of the metaphors we developed for these qualities was to establish how they might be linked to the kinds of numerical valued observables which comprise the basis of models of inorganic systems. We have not, as yet, attempted to link such biological qualities to each other. The exploration of such linkages of biological qualities, without the necessary intervention of numerical-valued observables at all, is the province of relational biology; a term coined by N. Rashevsky in 1954. Our first illustration of a relational metaphor will be that initially developed by Rashevsky, and which he termed the “principle of biotopological mapping”,¹⁹ for reasons soon to become apparent.

Before turning to the metaphor itself, let us begin with a few words regarding the basic motivation for Rashevsky’s approach. From what has already been said in this chapter, the reader will recognize that Rashevsky had been the outstanding pioneer in seeking to understand the properties of biological systems by relating them to physics. Thus, for example, he was the first to propose that cellular properties could be understood by linking them directly to the ubiquitous phenomena of chemical reactions and diffusion; he was among the few who understood that the brain could be productively approached as a network of excitable elements; he also carried out early and outstanding work in a multitude of other biological areas from this point of view. However, despite the fertility and success of the viewpoint which he was the first to consistently espouse, and to which he contributed so much, he came to feel a growing dissatisfaction with the entire approach, primarily because of what seemed to elude capture in this way. It is instructive to consider his own words in this connection:

... There is no successful mathematical theory which would treat the *integrated* activities of the organism as a whole. It is important to know how pressure waves are reflected in blood vessels. It is important to know that diffusion drag may produce cell division. It is important to have a mathematical theory of complicated neural networks. But nothing so far

in those theories indicates that the proper functioning of arteries and veins is essential for the normal course of intercellular processes. . . And yet this integrated activity of the organism is probably the most essential manifestation of life. So far as the theories mentioned above are concerned, we may just as well treat, in fact *do* treat, the effects of diffusion drag forces as a peculiar diffusion problem in a rather specialized physical system, and we do treat the problems of circulation as special hydrodynamic problems. The fundamental manifestation of life mentioned above drop out from all our theories. . . .²⁰

Thus, Rashevsky's development of relational biology was an attempt to circumvent this decisive shortcoming by introducing the essential integrative aspects characteristic of biology from the very outset, and by postulating that, whatever else may occur at the physical level, these aspects must remain invariant.

Thus, Rashevsky was forced to deal from the outset with characteristic biological qualities, and with the manner in which they are linked. The qualities with which he chose to deal could initially only be characterized crudely and verbally, through such terms as ingestion, digestion, absorption, assimilation, respiration, locomotion, reproduction, and the like. He hoped, and expected, that more precise characterizations would become possible through the mathematical form of the formalism into which they were encoded. This formalism was to stress the manner in which these biological qualities are linked; linkages which must be preserved invariant throughout the entire class of biological organisms. To a large extent, this is indeed true, as we shall see.

Now these biological qualities are directly recognizable by us, as we noted earlier. Thus, they must formally correspond to the kinds of observables with which we have been dealing all along, except that they need not take their values in numbers. The whole essence of the relational approach is to treat these qualities exactly like the observables which characterize all other qualities of natural systems. Let us see what this means in detail.

Suppose that v denotes a biological quality, which we can recognize directly. Like any observable, then, v must be encoded into a mapping of some kind. Our numerical-valued observables, we recall, were encoded as mappings $f : S \rightarrow \mathbb{R}$ from a set S of *abstract states* to a spectrum of numerical meter readings. Likewise, v must be encoded as a mapping $v : \Omega \rightarrow M$, where Ω plays the role of the abstract states on which the observable is defined, and M is a convenient set of *values*. What intuitively are the abstract states on which a biological quality is defined? It is clearly something like the "class of all organisms"; an element $\omega \in \Omega$ must be regarded as an "abstract organism", on which the quality v assumes a certain value $v(\omega)$.

Let us suppose that we are given some arbitrary natural system N , and a family of numerical observable qualities $\mathcal{F} = \{f_1, \dots, f_n\}$ defined on the abstract states S of N . The main task of science, as we have abundantly seen above, is to establish the linkage relations which exist between the observables f_i on the states of N .

In general, as we have seen, we cannot expect there to be a single linkage relation involving all these observables; rather, we can find a variety of subsidiary relations linking various *subsets* of these observables. Thus, if $\{f_{i_1}, \dots, f_{i_r}\}$ is a subset of these observables, we may be able to find an equation of state of the form

$$\Phi_i(f_{i_1}, \dots, f_{i_r}) = 0. \quad (3.78)$$

Let us suppose that $\Phi_1, \dots, \Phi_k$ are such equations of state linking the observables in $\mathcal{F}$. More precisely, each such equation of state asserts that, for any abstract state $s \in S$, we have

$$\Phi_i(f_{i_1}(s), \dots, f_{i_r}(s)) = 0 \quad i = 1, \dots, k \quad (3.79)$$

for an appropriate subset $\{f_{i_1}, \dots, f_{i_r}\}$ of $\mathcal{F}$. This is the essential content of the discussion provided in Sect. 3.2 above, which the reader should now consult once more.

The equations of state (3.78) or (3.79) which hold among the family of observables $\mathcal{F} = \{f_1, \dots, f_n\}$ introduce relations on $\mathcal{F}$ itself. For instance, if a subset of these observables is linked through an equation of state, we must say that this subset constitutes a *linkage group*. Let us say that two observables f_i, f_j are *related* if there is some such linkage group to which they both belong. This introduces a *binary relation* into $\mathcal{F}$, which in fact turns $\mathcal{F}$ into a *graph*. Indeed, *this graph is characteristic of the natural system N on which the observables are defined, and may in fact be regarded as a new encoding of N itself*.

Suppose that we replace N by a new natural system N', on whose abstract states precisely the same set of observables $\mathcal{F}$ is defined. We can repeat exactly the same procedure on N'. There is no reason to expect in general that the new linkage groups into which $\mathcal{F}$ is now decomposed will be the same as before; hence the graph $G(N')$ we obtain in this way will be different from $G(N)$.

Now let us see what these ideas amount to when translated into the terminology of biological qualities. Let $V = \{v_1, \dots, v_n\}$ be a set of such qualities, defined on the abstract set Ω . Just as before, we will expect these qualities to be linked on Ω in definite ways, through the analogs of equations of state of the form (3.78) or (3.79), and hence that V itself will fall into a family of linkage groups. Thus, once again, we can define a binary relation, or graph, $G(\Omega)$, characteristic of Ω . Moreover, given any particular $\omega \in \Omega$, we can consider the corresponding graph of *values*, which we can denote by $G(\omega)$.

Considerations of this type are the essential feature of Rashevsky's construction. However, there is one further idea which must be introduced before we can actually articulate that construction, and which we can motivate in the following way. Consider once again the class Ω of all "abstract organisms"; i.e. the class on which the biological qualities v_i are defined. *Each $\omega \in \Omega$ can itself be regarded as being a natural system*, and hence can be identified with a set S of abstract states on which ordinary numerical observables are defined. These are precisely the observables to which Rashevsky referred in the quotation above, on which specific theories of biological activity had heretofore been based. We can see now that any such specific theory *must necessarily* miss the integrative characteristics residing in the biological qualities, *because these latter are defined on Ω , and not on any individual S*. Indeed, it was Rashevsky's essential insight to recognize that, if we would capture the missing integrative aspects, it is necessary to move the discussion from an individual S to a *class* like Ω .

Rashevsky recognized that the *differences* we perceive between organisms reside in observables defined on individual abstract state sets S , but that the *commonalities* shared by all organisms reside in the qualities defined on Ω (and in the manner in which they are linked). Thus in his initial treatment, he allowed room for both kinds of qualities. Specifically, he supposed given a set $V = \{v_1, \dots, v_n\}$ of qualities defined on Ω , and another set $\mathcal{F} = \{f_1, \dots, f_m\}$ of ordinary observables, defined on each individual $\omega \in \Omega$. He then proceeded to form the graphs $G(\omega)$ appropriate to both sets of qualities. His *Principle of Biotopeological Mapping* was an assertion about how any two such graphs $G(\omega_1)$, $G(\omega_2)$ must be related. The essence of the principle was, of course, that the portion of these graphs coming from the biological qualities V must be maintained invariant; i.e. any two such graphs could be homomorphically related in such a way that there was always a certain invariant subgraph. This invariant subgraph is substantially what Rashevsky called the *primordial graph*, and on which he based his entire approach to relational biology.

Let us now observe that *any family of homomorphic graphs of this type could represent an encoding of biological qualities*. Thus, any such family of graphs could comprise what Rashevsky called an *abstract biology*. It follows that any proposition true for every abstract biology must *a fortiori* apply to biology itself. He established a number of such propositions, which are of the greatest interest. For example, he could show that under rather general conditions, the invariant primordial graph could be effectively determined from *any representative* of the class of graphs homomorphic to it. In effect, he showed that any biology is in principle completely determined by any organism in it. This is a startling kind of result, but one which has been entirely neglected by biologists. Indeed, it has ironically enough been dismissed as entirely alien to the spirit of biology, whereas the entire purpose of the relational approach was to capture in definite terms that which is intrinsically biological. We have seen that the relational approach, initially perceived as utterly abstract and remote from “experiment”, is in fact based on exactly the same considerations which underlie any other kind of observation. It is thus no more remote, and no more abstract, than any other empirically-based theoretical development. In this regard, its only distinguishing characteristics are its universal significance, and the unusual insight required to perceive it initially.

It should be remarked that Rashevsky’s initial treatment was superficially rather different from that presented above. However, we would as usual urge the reader to consult the original work for fuller details.

*Example 6. Another Relational Metaphor: The (M, R)-Systems*²¹

The (M, R)-systems represent another class of relational metaphors, this time developed within the more restricted confines of the biology of cells.

The point of departure for these ideas lies once again in this fact; that although the biologists can recognize many different kinds of cells, which may differ from one another more or less radically in physical terms, there are certain invariant commonalities which allow them all to be recognized as cells. As before, we can discern that these commonalities must be expressed in terms of biological qualities,

and that these in turn must be defined on a class of systems, rather than on single ones. Once these are characterized and the linkages between them determined, we will have at our disposal a relational characterization of all cells. From such a characterization, we will be able to understand what cellular properties follow from cellularity *per se*, and which devolve upon the specific physical features of particular realizations.

Two basic biological qualities play the fundamental role in characterizing our class of systems. The first of these may be crudely summed up in the single word *metabolism*. However much individual specific cells differ from one another in chemical constitution, they are alike in that they are open systems, capable of ingesting and processing environmental materials, and converting them into new forms. The second of these qualities may roughly be called *repair*. It has to do with the utilization of processed materials to reconstitute the specific machinery by which metabolic processing occurs. Speaking very crudely, metabolism corresponds to cellular activities which are collectively called *cytoplasmic*, while repair corresponds to cellular activities generally called *nuclear* or genetic. We shall call a system capable of manifesting both these activities, linked to one another in a specific way, an (M, R)-system.

To proceed further, we must have some way of formally encoding the crude ideas of metabolism and repair just enunciated into a more formal language. We shall regard a unit of metabolic activity, termed a *metabolic component*, essentially as a black box which transduces inputs into outputs. The inputs are drawn from the environment of the component, and the outputs also become part of the environment. Now we have seen before (cf. Sect. 1.2, for instance) that any such transducer can be regarded as a mapping $f : A \rightarrow B$ from an appropriate space A of inputs into a space of outputs. Thus our first step is to suppose that every metabolic component can be encoded into such a mapping.

As usual, whenever we are confronted with a family of transducers of this kind, we can arrange them into *networks*, through identifying the ranges (outputs) of certain of them with the domains (inputs) of others. In the present case, we can see that a network of metabolic components will comprise a finite family of mappings, related by the fact that the ranges of some of them are the domains of others, or are embedded in these domains as factors of larger cartesian products. Thus, for our purposes, a *metabolic system* M will comprise such a family of mappings. Conversely, any such family, as a possible encoding of a metabolic system, will be thought of as a metaphor for the metabolic activity of cells.

Let us now turn to the repair aspect. We wish to encode a repair element as a mapping, just as a metabolic element was so encoded. Let us see what such a mapping will have to be like. If $f : A \rightarrow B$ is a metabolic element in a metabolic system M , then f belongs to a *family* of mappings which we earlier (cf. Example 6, Sect. 3.1) denoted by $H(A, B)$. Any process which generates copies of f must, if represented by a mapping Φ_f , have its range in $H(A, B)$. Thus, it is suggested that we do the following: with *every* mapping f of a metabolic system M , we associate a corresponding repair element Φ_f . This element Φ_f will be a mapping into the set $H(A, B)$ to which f belongs.

What, now, is the domain of such a repair element Φ_f ? This domain must be related to the metabolic activity of the entire system M . We will thus suppose that the domain of each Φ_f must be a cartesian product of the ranges of mappings in M itself; the inputs to Φ_f represent outputs of M as a whole. A system composed of mappings f , and mappings Φ_f related to them in this fashion, is an (M, R) -system.

These (M, R) -systems possess many remarkable properties. Let us consider a few of them. One of the most obvious is intimately connected with the repair aspect of such systems. Let us suppose that by some means we remove a metabolic component f from the system. The associated repair component Φ_f , however, remains present and, if it receives its inputs, can continue to produce copies of f . Thus, the damage initially done to the system is in fact repairable. However, in order for this to be possible, it must be the case that the inputs to Φ_f do not depend on f ; i.e. that the removal of f from the system should not affect the inputs to Φ_f or any of their necessary antecedents. If this is the case, then as we have seen, Φ_f will repair the damage. Let us call a component f with this property *re-establishable*; a component without this property will be *non-reestablishable*. Re-establishability of a component f thus depends entirely on (a) the manner in which the metabolic components are inter-connected, and (b) on the domains of the repair components Φ_f .

It can easily be shown that *every (M, R) -system must contain at least one non-re-establishable component*. That is, there must be at least one way of damaging such a system which cannot be repaired by the system. Moreover, we can show that there is an inverse relation between the number of such non-re-establishable components and their importance for the functioning of the system as a whole. For instance, if there should be exactly one non-re-establishable component, then the removal of that component will cause the entire system to fail. In other words, although such a system can repair almost all such injuries, the single one which it cannot repair is lethal to every component.

Let us consider another kind of result which is unique to the theory of (M, R) -systems. We have argued that the repair components perform functions usually regarded as nuclear, or genetic, in cells. One of the decisive features of genetic activity in real cells pertains to the *replication* of the genetic material. Thus it would be most important to show that, already within the (M, R) -system formalism as we have defined it, and without the need for making any further *ad hoc* assumptions, a mechanism for such replication was already present. This is what we shall now proceed to show; namely, that already within our formalism, there exists machinery which can *replicate the repair components*. Let us see how this comes about.

Quite generally, if X and Y are arbitrary sets, we can define for each element x in X a mapping

$$\hat{x} : H(X, Y) \rightarrow Y$$

by writing

$$\hat{x}(f) = f(x).$$

This mapping is often called the *evaluation map* of x . We thus obtain an embedding $x \rightarrow \hat{x}$ of X into the set of mappings

$$H(H(X, Y), Y).$$

Let us suppose that this evaluation mapping x is *invertible*. Then clearly

$$\hat{x}^{-1} \in H(Y, H(X, Y)).$$

When will $\hat{x}$ be invertible? It is clearly sufficient that

$$\hat{x}(f_1) = \hat{x}(f_2) \quad \text{implies} \quad f_1 = f_2.$$

But by definition of the evaluation map, this means that

$$f_1(x) = f_2(x) \quad \text{implies} \quad f_1 = f_2.$$

That is, *the invertibility of $\hat{x}$ is a condition on $H(A, B)$* ; this latter set must be such that any two maps f_1 and f_2 which agree on x agree everywhere. (We may observe that this condition is reminiscent of the unique trajectory property in dynamical systems.)

Now in particular, let us put

$$X = A; \quad Y = H(A, B).$$

We thus obtain, for any b in B for which $\hat{b}^{-1}$ exists, a mapping

$$\beta_b : H(A, B) \rightarrow H(B, H(A, B)).$$

where we have put $\beta_b = \hat{b}^{-1}$. It is now easy to see that this β_b plays the role of a replication for the simplest (M, R) -system

$$A \xrightarrow{f} B \xrightarrow{\Phi_r} H(A, B). \quad (3.80)$$

By pursuing this kind of argument in more general settings, we obtain conditions under which replication maps exist for the repair components of any arbitrary (M, R) -system. To our knowledge, this kind of result is unique; there is no other situation in which replicative capacity can be made to *follow* from repair activity without the intervention of *ad hoc* assumptions. Thus it appears that this replicability is a *relational result*; i.e. independent of any particular physical mechanism or realization.

There are two other noteworthy features of the above construction. The first is that replication is not an *obligatory* feature of repair, but depends on the invertibility of a certain mapping. This in turn depends on the character of the entire set $H(A, B)$

with which we are dealing. It may be expected that this condition will not usually be satisfied, and hence that most (M, R) -systems cannot replicate. Thus, as we would expect, replication is a relatively rare and unusual situation.

The second noteworthy aspect of our construction arises from the relation of the replication map (when it exists) to the other mappings in an (M, R) -system. This can be seen by extending the diagram of the simplest (M, R) -system, shown in (3.80), in the following way:

$$A \xrightarrow{f} B \xrightarrow{\Phi_f} H(A, B) \xrightarrow{\beta_b} H(B, H(A, B)). \quad (3.81)$$

Let us now notice that the first three maps in (3.81) constitute our original (M, R) -system, in which f represents the metabolic component, and Φ_f represents the repair component. But let us now consider the last three maps in the diagram (3.81). It is easy to see that *these three maps themselves constitute an (M, R) -system*; but now one in which the original repair component Φ_f plays the role of metabolic component, and the original replication map β_b plays the role of the repair component. From this we see the curious fact that *there is nothing intrinsic about the biological qualities of metabolism, repair and replication*; our perception of them depends on the total system in which they are embedded. In fact, we can imagine the diagram (3.80) extended indefinitely on both sides, with any successive triplet of mappings being an (M, R) -system, and in which any map could be either a metabolic component, a repair component or a replication map, depending on which triplet was selected as primary. This too is a most remarkable result, of an entirely relational character.

These few results should suffice to give the flavor of the (M, R) -systems as cellular metaphors. It should be noted that, in formal terms, we may form (M, R) -systems in any category of sets and mappings (cf. Example 6, Sect. 3.1 above). Indeed, it can be shown that the totality of (M, R) -systems which can be formed in any category is itself a category in a natural way, and can serve as an index of the structure of the original category. Moreover, the property of being an (M, R) -system is preserved invariant by functors between categories. This fact has some interesting consequences which we cannot go into here; we refer the reader to the original paper for fuller details.

One final feature of the (M, R) -systems may be mentioned. As we have seen, such systems may be formed from any category of sets and mappings. In particular, we may think of the mappings appearing in an (M, R) -system as representing equations of state, or linkage relations, of some natural system. Now we saw in Example 2 above that one natural encoding of a biological genome is in terms of a set of structural or constitutive parameters appearing in such an equation of state. If we do this, then the output of the associated repair mapping can be thought of as precisely such a set of constitutive parameters; we obtain thereby a way of interpreting these repair components, which in fact represent nuclear activity in cells, in terms of the more familiar genetic concepts arising from morphological and biochemical considerations. Once again, we cannot pursue this interesting circle of ideas here,

but perhaps the reader can observe from the outline presented above how rich such relational considerations can be.

References and Notes

1. The Law of Mass Action is at the heart of chemical kinetics. Roughly speaking, it allows us to convert a hypothesized chemical reaction mechanism into a system of rate equations, describing the rate of change of reactant concentrations in time. Although mass action had been known for a long time, the modern form of the law was derived from statistical mechanical considerations by Guldberg & Waage in 1867; their argument was based on the calculation of collision frequencies of the reactant molecules. It seems to have first been noticed by Alfred Lotka (*Principles of Physical Biology*, Williams and Wilkins 1925) that mass action was in principle applicable to any system of populations of interacting units. He applied this idea widely to problems in population biology. Similar ideas were developed at about the same time by the Italian mathematician Vito Volterra (*Lecons sur la Theorie Mathematique de la Lutte pour la Vie*, Gauthier-Villars, 1931.) Between them, Lotka and Volterra laid the foundation for all theoretical studies of the dynamics of populations e.g. mathematical ecology.
2. For the original papers, see Kerner, E. H., *Bull. Math. Biophys.* 19, 121–146 (1957); *ibid.* 21, 217–255 (1959); *ibid.* 23, 141–158 (1961). For a discussion of this and other attempts to exploit statistical-mechanical ideas in biology, see Rosen, R., *Dynamical System Theory in Biology*. Wiley (1970).
3. For an interesting discussion of the Maxwell Demon and its history, see Brillouin, L., *Science and Information Theory*, Academic Press (1963). Brillouin, as had others (notably Szilard) before him, attempted to circumvent the possibility of such demons by showing that in a closed system, a demon would have to pay for the information it extracted by becoming increasingly disordered. This kind of argument was one of the main justifications of identifying the formal notion of “information” with the negative of physical entropy.
4. Equifinality in developing systems was advanced by Driesch (cf. Note 2 to Sect. 1.1 above) as his “first proof of vitalism”. In particular, Driesch pointed out that the characteristic feature of equifinality, the independence of the end state from initial conditions, was totally at variance with the equilibria studied at that time in physics and chemistry. It seems to have first been noticed by von Bertalanffy that such independence is in fact characteristic of stable steady states of open systems. This observation constituted major conceptual advance; it not only posed a counter-example to the arguments of Driesch, but also (a) revealed the inadequacy of contemporary physics (based on closed systems) for biological problems, and (b) opened a rich new field of investigation. A good discussion may be found in von Bertalanffy, L., *Problems of Life*, Harper (1960).

5. Rashevsky, N., *Bull. Math. Biophys.* 1, 15–25 (1940).
6. Turing, A. M., *Philos. Trans. R. Soc. B.* 237, 5–72 (1952).
7. See e.g. Othmer, H. G. and Scriven, L. E., *J. Theoret. Biol.* 32, 507–537 (1971).
8. Prigogine was an early pioneer in the study of what may be called the thermodynamics of open systems. He and his collaborators have been remarkably prolific over the years, especially since belatedly recognizing the conceptual significance of the Rashevsky-Turing metaphor (cf. Notes 5 and 6 above). The most recent extended development of these ideas may be found in two books: Glansdorff, P. and Prigogine, I., *Thermodynamic Theory of Structure, Stability and Fluctuations*, Wiley (1971), and Nicolis, G. and Prigogine, I., *Self-Organization in Nonequilibrium Systems*, Wiley (1977). For a further discussion of these ideas, see Sect. 4.3 below. See also the author's review of the book by Nicolis and Prigogine, *Int. J. Gen. Syst.* 4, 266–269 (1978).
9. The original book of Thom (*Stabilité Structurelle et Morphogenese*, Benjamin 1972) still makes the most provocative reading. A recent book by Poston, T. and Stewart, I., (*Catastrophe Theory and its Applications*, Pitman 1978) provides a remarkably clear survey of the mathematical ideas and many of its areas of application. See also Zeeman, E. C., *Catastrophe Theory: Selected Papers*. Addison-Wesley (1977).
10. For a description of the Ising models in their traditional setting, see McCoy, B. M. and Wu, T. T., *The Two-Dimensional Ising Model*. Harvard University Press (1973). See also the relevant articles in the series *Phase Transitions and Critical Phenomena* edited by Domb and Green cited in Note 10 to Sect. 3.3 above.
11. The analysis of morphogenetic phenomena in biology into these three basic mechanisms was apparently first suggested by J. T. Bonner (*Morphogenesis: An Essay on Development*. Atheneum 1963).
12. For a good discussion of the various forms of the two-factor theories of excitation and conduction, their history and their biological implications, see Rashevsky, N., *Mathematical Biophysics*. Dover (1960).
13. The original paper on neural networks is McCulloch, W. S. and Pitts, W., *Bull. Math. Biophys.* 5, 115–134 (1943). For a survey of applications to the theory of the brain, see Rashevsky's book (cf. Note 12 above). A lively introduction to the relation of neural net theory to automata theory and logic as well as to biology is Arbib, M., *Brains, Machines and Mathematics*. McGraw-Hill (1964).
14. See Rosen, R., *Bull. Math. Biophys.* 30, 493–499 (1968).
15. Monod, J. and Jacob, F., *Cold Spring Harbor Symp. Quant. Biol.* XXVI, 389–401 (1961).
16. This fact appears to have been first pointed out by M. Sugita, (*J. Theoret. Biol.* 1, 413–430 1961).
17. The term “artificial intelligence” means different things to different individuals. The basic disagreement is over the interpretation of “intelligence”. For instance, an early and acrimonious dispute in the field concerned whether chess-playing programs were or were not paradigmatic for intelligence. Several representative surveys of the field are:

- Bannerji, R. B., *Theory of Problem Solving*. Elsevier (1969).
- Nillson, N. J., *Problem-Solving Methods in Artificial Intelligence*. McGraw-Hill (1971).
- Slagle, J., *Artificial Intelligence*. McGraw-Hill (1971).
- Schank, R. and Colby, K. M., (eds.) *Computer Models of Thought and Language*. Freeman (1973).
- Simon, H. A. and Siklossy, L., (eds.) *Representation and Meaning*. Prentice-Hall (1972).
18. The question here is the relation between discrete-time and continuous-time encodings of the same class of phenomena. The problem is not so much the establishment of a single over-arching mathematical formalism which encompasses both; this is relatively easy (cf. for instance Arbib, M. A., *Automatica* 3, 161–189 1963). The problem is rather whether a relation of correspondence is established between the encodings themselves, along the lines of Fig. 2.4 above. If the answer is affirmative, it would provide a relation between the mathematical concept of a “machine” and the physical concept of a mechanical device; it would provide a physical analog to the mathematical concept of unsolvability (cf. Note 6 to Chapter 2.2 above); and among other things would reveal whether unsolvability in the mathematical sense places a constraint on *physical* laws. If the answer is negative, it would then be interesting to discover exactly why; i.e. why two encodings which seem so similar in spirit should be so unlinked.
 19. Rashevsky, N., *Bull. Math. Biophys.* 16, 317–348 (1954). This work was pursued through a long series of papers, published mainly in the Bulletin of Mathematical Biophysics between 1955 and 1965. See also the posthumously published monograph *Organismic Sets*, published in 1972 by J. M. Richards Laboratory.
 20. Rashevsky, N., *Mathematical Biophysics*. Dover (1960).
 21. For a succinct review of the basic ideas, see Chap. 4: *Some Relational Cell Models: The (M, R)-Systems*, in Volume 2 of *Foundations of Mathematical Biology* (R. Rosen, ed.) Academic Press 1973).

3.6 Models, Metaphors and Abstractions

In the present chapter, we are going to draw some general conclusions about modeling relations from the specific examples we have developed. We shall begin with a discussion about the concept of a metaphor as an embodiment of some general quality, and the relation which exists between such qualities and those embodied by numerical observables.

Let us begin by recalling that a metaphor was defined (cf. Sect. 2.3 above) as a class of encodings (or possible encodings) which share some property. Since encodings are *formal* systems, the properties which they can share are mathematical properties. As always, the mathematical relation manifested by the encodings is

then *imputed* to the natural systems so encoded. Let us then consider a few typical mathematical properties which may be so shared, and see how the properties themselves may be expressed in terms of the formal systems which exhibit them. We may for example consider the following:

1. An integer is *prime* if it has no divisors other than itself and unity.
2. A set X is *finite* if there is no 1–1 mapping of X onto a proper subset of itself.
3. A group is *simple* if it contains no non-trivial proper subgroup.
4. A topological space is *connected* if it cannot be expressed as a union of two subsets whose closures are disjoint.
5. A dynamical system

$$dx_i/dt = f_i(x_1, \dots, x_n)$$

is *closed* if $\sum_{i=1}^n f_i = 0$.

6. A mapping $f : X \rightarrow Y$ between two topological spaces is *stable* if every nearby map is conjugate to f .
7. A mathematical theory J is *consistent* if no proposition and its negation are both theorems.

Here the terms *prime*, *finite*, *simple*, *connected*, etc. name mathematical qualities, which may or may not be shared by mathematical objects to which they are applicable.

Given any such mathematical property or quality, we can imagine the totality of all mathematical structures which share or exhibit the property. Thus, we can consider the set of all prime numbers; the class of all finite sets; the class of all simple groups; etc. Along with these mathematical structures typically come appropriate families of structure-preserving mappings. Thus, each mathematical property defines a *category* containing all the structures which manifest the property.

Let us consider such a category $\mathcal{C}$. Using the appropriate concept of isomorphism between objects in $\mathcal{C}$, we can partition $\mathcal{C}$ into equivalence classes; two objects X, Y in $\mathcal{C}$ fall into the same equivalence class if and only if there is an isomorphism between them. Intuitively, these equivalence classes represent the number of different *ways* in which the mathematical property defining the category can be manifested. In other words, the equivalence classes arising in this fashion can be regarded as defining the *spectrum of values* of the quality in question. Further, the class to which a particular object X belongs can be regarded as a *label* or *name* for X ; it precisely specifies the value our quality assumes on the object X .

It will be noted that the situation we are describing is exactly analogous to the manner in which we treated numerical-valued observables, as devices for naming or labelling the abstract states on which they were defined. Indeed, suppose that $f : S \rightarrow \mathbb{R}$ is such an observable. We saw in Sect. 3.1 above that the effect of such an observable is to partition S into equivalence classes; two elements s, s' in S fall into the same class if and only if $f(s) = f(s')$; i.e. if and only if the observable f assigns to s and s' the same label in $\mathbb{R}$. We also saw that the set $f(S)$ of such labels was in 1-1 correspondence with the set S/R_f of equivalence classes. Thus, if we look upon

f as embodying some property or quality of the elements of S , the *values* which this property can assume are identical with the classes of elements which assume these values.

Thus we can see, what we have repeatedly emphasized, that general qualities can be regarded exactly like observables; in each case there is an equivalence relation imposed on a class of objects bearing the quality, and the set of equivalence classes is the spectrum or set of values of the quality. The only difference lies in the interpretation given to the domains and ranges of the qualities.

In general, given an object X in such a category $\mathcal{C}$, how can we tell what value of the defining quality is assumed on X ? It is evident that this is a general form of the *classification problem*, which we saw repeatedly in Sect. 3.1. As noted at that time, one general way of solving the classification problem arises from the concept of a *canonical form*. Namely, suppose that we possess a means for extracting out of each equivalence class some specific representative. We can solve the classification problem for X by determining to which of these representatives X is isomorphic. In a certain sense, this approach to the classification problem involves a kind of “template matching”, in which we use the canonical forms as a set of “templates” to which any particular object X is to be compared. We again note that this procedure is exactly analogous to the situation for numerical-valued observables; we classify an unknown state s in S by matching its “meter reading” $f(s)$ to some set of standard readings, and assign s to the same class as the state which gave the standard matched by $f(s)$.

Thus, in most general terms, what we have called a *metaphor* is simply a category with an equivalence relation on its class of objects. The only restriction placed on this equivalence relation is that it be preserved by the isomorphism of the category, and thus may be thought of as arising from the isomorphisms. Any two objects in the category are thus *metaphorically related*; they both manifest the quality through which the category is defined. They assume the same *value* of the quality if and only if they fall into the same equivalence class; i.e. they are isomorphic as objects of $\mathcal{C}$. We stress again that these notions are exactly the same as those arising in the case of qualities encoded into numerical-valued observables; indeed, the latter can be regarded simply as a special case of the former, if we consider categories whose only mappings are the identity maps as sets of abstract states.

Let us now extend these ideas, by imputation, to natural systems. Suppose that a natural system N can be encoded in principle into one of the objects in a category $\mathcal{C}$ of the type we have been describing. There is thus a precise sense in which any object in that category is a metaphor for N , since it shares the quality which assigns the encoding of N to the category. It is in this sense that an arbitrary diffusion-reaction system was regarded above as a metaphor for morphogenetic processes; or that an arbitrary neural network could be regarded as a metaphor for the brain. Thus, we can learn something about N even without knowing how to encode it explicitly; in particular, anything true for the objects in the category in general will *a fortiori* be true for the encoding of N , and hence, by imputation, for N itself.

We will now turn to the more specialized circle of ideas involving modeling relations. From the outset, these should be regarded as special cases of the metaphors

we have just described. We recall that two formal systems are metaphorically related if they share some general mathematical property, as we have described. In the case of a model, this relation can be more precisely specified: it involves the existence of a *homomorphism* of one of them into the other; i.e. a dictionary through which the properties of one of them can be translated into corresponding properties of the other. We are now going to discuss the ramifications of this most important situation by introducing some new terminology, which will be most useful to us later. In the process, we shall see some important new relations emerging between natural and formal systems, which belong to the general area of *system analysis*. This new terminology will center around the concept of abstraction; a term which we have not used heretofore. It will also involve the concomitant concept of a *subsystem*¹ of a natural system.

Let us return for a moment to the fundamental concept of an individual observable. We recall that observables represent potential interactive capacities, made manifest through the moving of corresponding meters. The meter readings are labels for the values assumed by the observable on particular abstract states, and we have seen how these sets of labels are the point of departure for all formal encodings.

Let us suppose we are given only a single meter, which may interact with the states of some natural system N. As far as the meter is concerned, the *only* quality of N visible to it is the one embodied in the observable it measures. As far as that meter is concerned, no other qualities of N exist; and indeed N behaves for that meter as if it possessed no other qualities. Instead of seeing the full range of interactive capabilities which N can manifest, then, the meter sees only one.

It is in this sense that we shall say that *any observation procedure, applied to a natural system N, generates an abstraction*. For instead of telling us about N, the observation procedure can by definition tell us only about a single quality of N; it must necessarily *forget*, or neglect, all the other qualities which N may manifest. It allows us to see only a *projection* of N along a single interactive dimension. This kind of forgetting, or neglect, of existent qualities is the essence of abstraction.

In popular parlance, “abstraction” has a pejorative connotation. The antonym of “abstract” is “concrete”; thus in the popular view (which is shared by many empirical scientists) abstraction is a property of theory, and one engaged in the direct observation of nature cannot be accused of performing abstractions. Above all, one who observes nature directly must always be solidly anchored in concrete reality.

We see, however, that the facts are quite otherwise. The observing procedure is the very essence of abstraction. Indeed, no theory, and no understanding, is possible when only a single mode of observation (i.e. a single meter) is available. With only a single meter, there cannot be any science at all; science can only begin when there are at least two meters available, which give rise to two descriptions which may be compared with one another.

If an abstraction involves a loss or neglect of properties present in the external world, then clearly the qualities which are in fact captured in the abstraction can only refer to some restricted part of the available qualities of any natural system N. We are going to call this part a *subsystem* of N. Thus, for us, the concepts of abstraction

and of subsystem go hand in hand; a subsystem is that part of N which a restricted family of observation procedures allow us to see, and conversely, an abstraction can never capture more than a subsystem.

It follows therefore that all the modes of system representation which we have been considering can at best describe *subsystems* of a given natural system N . Thus, any modeling relation which we establish between N and any formal system, insofar as it proceeds from an encoding of a limited family of observables, really pertains only to such a subsystem. Different encodings of the same system N must then generally pertain to different subsystems. Conversely, it is reasonable to say that if two *different* natural systems N_1 , N_2 admit an encoding into the same formal system, that N_1 and N_2 admit a *common subsystem* (of which N_1 , N_2 constitute *alternate realizations*).

It will be recalled that these were the two prototypic situations discussed in a preliminary way in Sect. 2.3 above. They are the basic situations of science, which as we have noted, can only begin when we have two representations or encodings to compare. If the two encodings we are comparing both pertain to the same natural system N , then our science is directed towards ascertaining under what circumstances these encodings are related, and under what circumstances they are not. As we have seen, the capacity to establish a homomorphism between such distinct encodings of the same system underlies the idea of *reduction* of one encoding to the other; in that case, our two subsystems effectively collapse to a single one. If the two encodings bifurcate from one another, then they each capture separate aspects of the reality of N ; neither of them can be reduced to the other. Conversely, if our two encodings refer to different systems, then the extent to which we can establish homomorphisms between them is the extent to which our two natural systems share a common subsystem captured by the encodings; this determines the extent to which our natural systems can be regarded as *analogs* of each other.

Now the whole point of making models; i.e. of encoding natural systems into formal ones, is to enable us to make specific *predictions* (particularly temporal or dynamical predictions) about natural systems, utilizing the inferential structure of the model as an image of the processes occurring in the natural system itself. To what extent can we hope to do this, if the best a model can capture is a subsystem?

In general, if two natural systems N_1 , N_2 interact, they must do so because observables in each of them evoke dynamical changes in the other. Indeed, the prototype of a general interaction between two natural systems is the image of two meters looking at each other. In general, such an interaction will not involve *all* the observables of either system; rather, there will generally be two *subsystems* $M_1 \subset N_1$, $M_2 \subset N_2$ which already contain all the *relevant* qualities. To this extent, we could in fact replace N_1 by its subsystem M_1 , and N_2 by its subsystem M_2 ; the interaction between these subsystems would be exactly the same as the interaction between the original systems. In other words, the replacement of our original systems by *these* subsystems would have no observable effect on the interaction; it would be *invisible* to that interaction.

Thus, if we had an encoding of each of these systems into formal systems, or models, we would have captured all of the relevant interactive capabilities of both systems. Even though we are as always dealing with abstractions, the abstractions in this case have only neglected interactive capabilities irrelevant for the interaction in which we are interested. Consequently, any predictions we make regarding this interaction will be verified (at least to the extent that our encodings are faithful). The conclusion to be drawn from this argument is simply this: to say that any encoding is necessarily an abstraction is not in itself a reproach; indeed, we have seen that we can deal with nothing but abstractions. What we must do is see to it that the qualities retained in our abstraction are precisely the relevant qualities actually manifested in an interaction in which we are interested. The diagnostic for this is, as mentioned above, that we could replace each of the interacting systems by the subsystems we actually encode, with no effect on the interaction itself.

In such a case, it is correct to say that the subsystems M_i are themselves models of the larger systems N_i to which they belong. Thus, any formal system into which the M_i are encoded will also be encodings of N_i .

In the preceding discussion, we have considered subsystems as units of interaction between natural systems. We seek to understand such an interaction by identifying the relevant observables of the interacting system, where an observable is specified by a special kind of interaction between the system and a meter. The extraction and characterization of subsystems in terms of individual observables, and the employment of such characterizations to encode particular interactions, is the essence of *system analysis*.

Now we must note that there is another, quite different way in which the term “subsystem” has been employed in science. All of our discussion so far has been framed in terms of observables which are measured by the dynamics they *impose* on the other natural systems (i.e., on meters). On the other hand, *we can ourselves impose dynamics on natural systems*. We could think of developing a parallel discussion of system analysis by considering the dynamics which can be *imposed upon* a natural system, and using these as a point of departure. Let us briefly see where such a discussion leads.

We have argued at great length elsewhere¹ that the general result of imposing dynamics upon a natural system N is a modification of the linkage relations which originally related the observables of N . In particular, linkage relations can be abolished by means of imposed dynamics; i.e. two observables initially related through an equation of state on N may become unlinked. The abolition of a linkage relation of this type may be interpreted as a “partition” of our original system N into independent parts. For instance, if our original system N is a mixture of alcohol and water, we can by raising the temperature cause the alcohol to distill off, resulting in a spatial separation of the original mixture. This spatial separation uncorrelates the observables characteristic of the water from those of the alcohol, whereas in the mixture definite linkages existed between them.

Families of observables which have been unlinked by an imposed dynamics will be called *fractions* of the original system; the dynamics itself will be called a *fractionation*.² It is very common in science to identify such fractions, produced

by a dynamics imposed on a system, with what we have earlier called subsystems. It is this identification of “fraction” with “subsystem” which we wish now to briefly discuss. We shall argue that, *although any fraction may (with some qualification) be regarded as a subsystem, not every subsystem is a fraction*. Thus, the concepts of fraction and subsystem are not co-extensive. It follows further that the capacity of a natural system *N* to induce dynamics on other systems need not be, and indeed is generally not, identifiable with the capacity of *N* to be resolved into fractions by dynamics imposed on it. As we shall see, the mistaken notion that (subsystem = fraction) is the main conceptual underpinning for *reductionism* as a general philosophy of systems analysis. The other, equally important, side of the coin, is: the existence of subsystems which are not fractions vastly enlarges the scope of system analysis in non-reductionistic directions. In these directions it is the concepts of modeling and metaphor which necessarily play the dominant role.

To show that the concepts of subsystem and fraction are not co-extensive, it suffices to exhibit a single natural system *N* for which we can characterize a subsystem which is not a fraction. Of the many examples which could be given, let us consider a simple biological one: namely, an enzyme molecule. The *subsystem* will be taken to be the *active site*. This subsystem can be characterized directly in terms of its interactive capabilities, and indeed encoded thereby into formal systems in a variety of ways. It is thus a well-defined subsystem of the enzyme molecule. But we can conceive no fractionation which will unlink precisely those observables defining the active site from all the other observables. That is, we can imagine no dynamics, imposed on the molecule in which the site is embedded, which will separate the molecule into a fraction which is site, and a fraction which is “everything else”. Moreover, it should be observed that the *same* site may be embedded in distinct enzyme molecules, thus providing a concrete example of how different systems (i.e. different molecules) can share a common subsystem (or alternatively, how the same subsystem can be alternately realized in different ways).

Precisely this point was piquantly (if tacitly) argued long ago by the organic chemist R. Willstätter, who would never accept the identification of enzyme with protein. He admitted, of course, that enzymic activity was *associated* with protein, but he felt that the protein constituted an unavoidable contaminant of the enzyme. Insofar as we may identify enzymic activity with an active site, Willstätter was entirely correct; the active site is *not* the molecule manifesting the site, but the site cannot be separated from the molecule by any imaginable dynamics. This is not a quibble over semantics; it is a point of the deepest significance for the understanding of natural systems.

In a completely different context, the same point was made by the physicist Eddington, in considering the interactions of elementary particles. He was forced to identify the units of interaction (what we have termed subsystems) as possessing exactly the same physical reality as “real” particles, even though they could not be fractionated out of such particles. It is instructive to cite his words on the subject:

The word “particle” survives in modern physics, but very little of its classical meaning remains. A particle can now best be defined as *the conceptual carrier of a set of variates*. . .

We shall freely invent particles to carry the sets of variates which our form of analysis groups together. The provision of a carrier is not so much a necessity of thought as a necessity of language. It might seem desirable to distinguish the “mathematical fictions” from the “actual particles”, but it is difficult to find any logical basis for doing so. “Discovering” a particle means observing certain effects which are accepted as proof of its existence, but it seems to be a matter of fashion or convention that one sort of effect rather than another is accepted as critical for this purpose.”³

Having seen that the concepts of fraction and subsystem are not coextensive, we may further observe that it requires quite a non-trivial argument to accept that fractions are in fact subsystems. Indeed, we may recall that linkage relations between observables are defined in terms of the intersections of the equivalence classes which the observables define on a common set of abstract states. To say that a linkage is changed or broken means, in effect, that we must change the set of abstract states on which the system is defined. In order to fractionate a system, then, we must make radical changes in its very definition, in such a way as to include new observables, belonging to the system imposing the fractionating dynamics; for fuller details, we refer the reader to an extensive discussion of this point in our earlier monograph on measurement. Thus, the most we can actually conclude about a specific fraction is that it *contains* a subsystem of the original system, but it is generally not identical with such a subsystem.

Now let us briefly discuss the relation between fractionation and reductionism. As we have defined the term, reduction refers to a specific relation between encodings F_1 , F_2 of the same natural system N . More precisely, we say that we can reduce F_2 to F_1 if we can construct a monomorphism $\theta : F_2 \rightarrow F_1$, which allows us to express every property of F_2 in terms of properties of F_1 . A specific example of reduction was given in Example 3 of Sect. 3.3 above, where we utilized the machinery of statistical mechanics to reduce a thermodynamic encoding to a dynamical one.

We now observe that the atomic theory itself (at least in its simplest form) tacitly involves fractionations in an essential way. In asserting the particulate nature of every material system, we are inherently granted the capability of (at least in principle) fractionating any specific constituent particle from the system without changing its nature. Indeed, the totality of fractions of such a material system consists precisely of all subsets of its particles. In this sense, the *analysis* of a material system *means* the characterization of the particles of which it is composed. Once this is properly done, the forces which these particles impose on each other can be determined, and the equations of motion of the system follow from Newton’s Second Law. In such a way, as we have seen, we indeed obtain an encoding of our original material system.

The basic hypothesis of reductionism is, of course, that *every other encoding of our system may be reduced to this one*. In sum, then, reductionism as a philosophy of system analysis posits the following: (a) There is a “universal” way of encoding any material system, so that every other encoding is reducible to the universal one. (b) This universal encoding can be canonically determined from an appropriate series of *fractions*, which can be obtained from any material system. (c) These

fractions constitute the individual particles of which the system is composed, or sets of such particles. These fractions may in turn be isolated by imposing well-defined dynamics on the system.

Thus, reductionism is an *algorithm*, or recipe, for the analysis of any material system. We merely apply a known spectrum of fractionating dynamics to any material system, appropriately characterize the resulting fractions, and from these, a universal encoding is obtained, to which any other encoding can be reduced.

Many forms of reductionism have been proposed, especially in biology. Indeed, the well-known *cell theory*, originally propounded in the nineteenth century, is a reductionistic theory; it asserts that every quality of an organism can effectively be reduced to the properties of individual cells, or groups of cells. Here the individual cells are the basic fractions. Molecular biology is at root a refinement of this theory, insofar as it asserts that every property of a cell can be effectively reduced to the properties of its constituent particles (molecules); here the molecules constitute the fractions.

Unfortunately, the lack of correspondence between fractions and subsystems destroys the universality of the reductionist algorithm; an encoding of a subsystem which is not itself a fraction cannot be reduced to an encoding of fractions alone. This is a mathematical or formal fact, which no ingenuity of argument can change. Moreover, as we have seen, fractions need not in general even be subsystems; they need only contain subsystems. Thus, the extent to which a description of a fraction isolated from a natural system is even a description of the system itself is a question which cannot be settled in the abstract; it will depend on the details of the fractionating dynamics in each particular case.

On the other hand, we have argued above that the subsystems of a natural system, as interactive units, possess exactly the same physical reality as the system itself, or as any fraction extracted from the system. The characterization of such subsystems, and their expression as linked sets of observables, allows us many modes of system analysis which are intrinsically non-reductionistic. Such characterizations are always encoded in terms of models (or more generally, in terms of metaphors), from which specific predictions about interactions can be made. Thus the concept of a model, as a mathematical representation of a subsystem, comes to play a central conceptual role in system analysis. In this approach, there is no longer any overarching universal encoding, to which all others can be reduced. Rather, we have a family of potential interactive or analytic units, whose encodings may be compared and related in the manner we have been describing.

To sum up: the reductionistic algorithm may tell us many things about a natural system, but it cannot tell us all about the system. To find out more about the system than the reductionistic recipe can tell us, the concepts of models and metaphors are our only recourse. As noted above, they vastly extend the arsenal of analytic weapons we may bring to bear on the study of natural systems. Indeed, the thrust of everything we have done in the present section, sketchy as it may have been, exhibits the power and fertility of this approach. Specifically, for most of the basic biological qualities we have discussed in the preceding chapter, there is in fact no other way.

References and Notes

1. For fuller details regarding these ideas, consult Rosen, R., *Fundamentals of Measurement and the Representation of Natural Systems*. Elsevier (1978).
2. The idea that system analysis is synonymous with fractionation (i.e. dissection) finds its ultimate roots in Newtonian mechanics; we have seen that the Newtonian paradigm allows us to make complete predictions about any system of particles if we know (a) what the particles are like individually and (b) the forces they exert on each other. Somewhat independently came the rise of Baconian empiricism, which asserted essentially that the way to understand complex things is to take them apart into simple things. Such ideas are invariably found at the basis of most empirical sciences; the resolution of “mixtures” (complex things) into “pure substances” or “pure phases” (simple things). The postulation that the only acceptable mode of system analysis is to be found in a resolution of the system into such fractions is the essence of reductionism.

It may be noted that fractionation is entirely an operation on state or configuration. It is an operation which is intended to simplify state description. In the case of a dynamical system, however, each state has associated with it a tangent vector. An operation (fractionation) geared to simplify state description may, and generally will, do terrible things to the tangent vectors. This, in a nutshell, is the main reason why reductionism fails as a universal analytical scheme for natural systems.

Chapter 4

The Encodings of Time

4.1 Time and Dynamics: Introductory Remarks

In the preceding sections, we have seen many examples of modeling relations, which have spanned a wide range of physical and biological contexts. Most of the models and metaphors we have introduced have involved dynamical processes, and the manner in which the inferential structure of a formal encoding represents the dynamical properties of the natural system so encoded. This is indeed the main thrust of the entire book; to determine how we may employ formal models of natural systems to make temporal predictions about the systems themselves, and ultimately to utilize such predictions to modify the systems' present behavior. In order to accomplish this, we must investigate modeling relations involving dynamical systems in more detail than we have heretofore done.

The crucial concept in dynamics, of course, is time. In all of the dynamical models we have considered, the essential step involves the establishment of linkages between instantaneous states and their rates of change. We have pointed out in Sect. 2.1 above that the concept of time involves two distinct aspects: an aspect of *simultaneity* and an aspect of *temporal succession*. Both of these are intimately involved in the encoding of dynamics. The very definition of an instantaneous state depends on simultaneity, while the definition of rate involves temporal succession. Thus, if we are to have a thorough understanding of the nature of dynamical encoding in general, and the nature of temporal predictions about a natural system based on a dynamical model, we must explicitly consider the encoding of time into our models. Specifically, we must clarify the relations which exist between time, instantaneous state, and instantaneous rate of change, and the properties of the linkage relations which can exist between them. This will be the main task of the present section.

If we look back over the examples we have developed above, it will become apparent that we have tacitly treated the concepts of time and rate in several quite different ways. At the very simplest level, we have employed two distinct encodings for the basic set of instants. In most of our physical and biological examples, we

encoded the set of instants as the continuum $\mathbb{R}$ of real numbers. Indeed, in these examples, we made extensive and fundamental use of the mathematical properties of $\mathbb{R}$, especially the topological (metric) properties. The algebraic properties of $\mathbb{R}$ also enter in a fundamental way, at several levels; the fact that $\mathbb{R}$ is an additive group is basic to our treatment of dynamical systems as flows (cf. for instance Example 3 of Sect. 3.3 above). At an even deeper level, the mathematical concept of a derivative makes essential use of all the algebraic properties of $\mathbb{R}$; the encoding of rates of change of observables as derivatives thus ultimately devolves directly upon the algebraic properties of the set of instants. Nevertheless, we also saw (cf. Example 3 of Sect. 3.5) that important kinds of dynamical encodings could be obtained when the set of instants was encoded as the discrete set $\mathbb{Z}$ of integers. Since $\mathbb{Z}$ is an additive group, we can retain the idea of a (discrete) flow, but we lose all of the metric and topological properties; this was seen to have profound effects on every aspect of encoding.

Even in the continuous-time encodings, the role of time differs greatly in the various examples we discussed. In the encoding of particulate systems (cf. Example 1 of Sect. 3.1) time simply plays the role of a parameter which indexes states along trajectories. On the other hand, in thermodynamics, the restriction of our encodings to equilibrium or near equilibrium situations implies an entirely different treatment of time, arising from the fact that thermodynamic systems can only approach equilibrium and not depart from it. This fact, as we shall see below, imposes an “arrow” on temporal aspects of thermodynamic phenomena which is absent from mechanics. This “arrow” is formally manifested by the Second Law of Thermodynamics, which basically asserts that certain kinds of linkages cannot arise in thermodynamic encodings, and is intimately related to probabilistic considerations which have no counterpart in mechanics.

The discrete-time encodings force on us a fundamental dissociation between the concepts of *time* and *rate*, which are of course basic to continuous-time encodings. Thus, discrete time stresses the *sequential* aspects of the set of instants. The idea of time as sequence introduces yet another important aspect into our thoughts about time, which we must develop carefully. We have already seen, (cf. Example 3B, Sect. 3.5) that discrete-time systems are closely related to logic and to mathematical inference; indeed, there is a profound sense in which the successive states of a discrete-time system can be regarded as *theorems*, each arising from the preceding state as axiom, and the dynamics itself as production rule or rule of inference. This idea of the temporal evolution of a state as a *logical* process, independent of any notion of rate, represents still another aspect of time; one which will have many important ramifications for us as we proceed. Moreover, as we shall see, it has many important connections with probabilistic ideas, which are also rate-independent.

All of these different aspects of time capture different essential qualities of our time-sense, just as different observables capture different qualities of natural systems. Each of them allows a different kind of dynamical encoding to be built upon it. As always, it then becomes important to *compare* these kinds of dynamical encodings; i.e. to establish relations between them. The establishment of such relations tells us, in turn, how the different qualities of our time-sense are themselves

related, and perhaps more important, the senses in which they are distinct and irreducible one to another. From such an understanding, we will arrive at a clearer idea of what a temporal prediction about a natural system actually entails.

Thus, in Sect. 4.2 below, we shall take up the question of “Newtonian time”; i.e. the role of time in the encoding of particulate systems. In Sect. 4.3, we shall consider thermodynamic or statistical time, with its fundamental relation to irreversibility. In Sect. 4.4, we shall extend these considerations to probabilistic time in general, and see how the laws of irreversible thermodynamics can be regarded as embodiments of such results as the Law of Large Numbers. In Sect. 4.5, we return to the notion of continuous dynamics; we shall see that despite the formal similarities which exist between Newtonian encodings and those arising in more general dynamical contexts, the notions of time in the two classes of systems are completely different from one another. In Sect. 4.6, we shall turn to another and quite distinct aspect of time; namely discrete or logical time, and the relation between time and sequence. In Sects. 4.7 and 4.8, we shall discuss the related notions of similarity in dynamical systems, and the inter-relationships between time and age.

4.2 Time in Newtonian Dynamics

Let us suppose that we are given a system of particles in space, whose displacements from some origin of co-ordinates at any instant can be represented as an n -tuple of numbers $x_1, \dots, x_n$. In the terminology of Example 1 of Sect. 3.3 above, this n -tuple of numbers encodes a *configuration* of the system, and the totality X of all such n -tuples is the *configuration space*.

Let us further suppose that the forces imposed on the system depend only on the configurations, and moreover in such a way that the force F_i imposed on the i^{th} configuration variable x_i can be encoded as

$$F_i = - \frac{\partial U(x_1, \dots, x_n)}{\partial x_i} \quad (4.1)$$

for some definite function U defined on the configuration space X . The function $U(x_1, \dots, x_n)$ is called the *potential*; forces F_i satisfying (4.1) for some potential are called *conservative*.

The theory of systems of particles under the action of conservative forces is the cornerstone of Newtonian mechanics. For such a system, it is shown in any textbook on classical mechanics that there is a function $H = H(x_1, \dots, x_n, p_1, \dots, p_n)$ defined on the *phase space* of the system, which has the following properties:

$$(a) \quad H = T(p_1, \dots, p_n) + U(x_1, \dots, x_n). \quad (4.2)$$

That is, H is the sum of two terms, the first of which depends only on the momenta of the particles, and the second of which is our potential, which depends only on the displacements.

- (b) The equations of motion of our system can be expressed entirely in terms of H ; explicitly, they can be put into the form

$$dx_i/dt = \frac{\partial H}{\partial p_i} \quad dp_i/dt = -\frac{\partial H}{\partial x_i} \quad (4.3)$$

The function H is called the *Hamiltonian function*. It can be interpreted as the total energy of our system. The summand $T(p)$ in (4.2), which depends only on momentum, is accordingly called the *kinetic energy*, while U is the *potential energy*. The equations of motion in the form (4.3) represent the *Hamiltonian form* of these equations; they are entirely equivalent to Newton's Second Law. Thus from (4.3), we can see that the Hamiltonian function determines the entire dynamics.

The Hamiltonian function, and the Hamiltonian form of the equations of motion, are remarkable in many ways. For our present purposes, we wish to stress the following: that the equations (4.3) serve to relate the *temporal* rates of change of the state variables to the gradient of a certain function H in *phase space*. These latter partial derivatives themselves *contain no explicit mention of time*. In fact, we shall take the viewpoint that the Hamiltonian equations (4.3) implicitly *define* the time differential dt , and hence all temporal rates of change, in terms of the quantity H and the differentials dx_i , dp_i , all of which pertain entirely to the phase space. That is, instead of taking the customary viewpoint (in which time is primary, rates are specified as time derivatives, and Newton's Second Law relates rates of change to forces), we shall *begin* with a Hamiltonian on a phase space, and work backwards towards the equations (4.3) as a *definition* of time.

Let us then completely forget about Newtonian dynamics, and about Newton's Second Law. Let us retain only the following: (a) a family of state variables $(x_1, \dots, x_n, p_1, \dots, p_n)$, where we may think of the x_i as variables of configuration, but the p_i are simply other observables; in particular, we do not think of them as velocities or momenta; (b) there is a function $H = H(x_1, \dots, x_n, p_1, \dots, p_n)$ defined on the space $X \times P$ of all n -tuples $(x_1, \dots, x_n, p_1, \dots, p_n)$.

We will interpret the function H as imposing a *linkage relation*, or *equation of state*, on the states of our system, in the following way: the only states we shall consider are those satisfying the relation

$$H(x_1, \dots, x_n, p_1, \dots, p_n) = C \quad (4.4)$$

where C is a fixed constant. There is in this picture no dynamics visible; no forces, and no explicit mention of time.

Let us suppose that our system is initially in some state $(x_1^0, \dots, p_n^0)$, and that through some external intervention each of the values of the state variables is changed slightly. Using the familiar abuse of language, we shall say that the

new state arising from the intervention is $(x_1^0 + dx_1, \dots, p_n^0 + dp_n)$. If this new “perturbed” state is again to be a state of our system, it must satisfy (4.4). On the other hand, H itself can be expanded in a Taylor series about $(x_1^0, \dots, p_n^0)$, to give

$$\begin{aligned} H(x_1^0 + dx_1, \dots, p_n^0 + dp_n) &= H(x_1^0, \dots, p_n^0) \\ &+ \sum_{i=1}^n \left(\frac{\partial H}{\partial x_i} dx_i + \frac{\partial H}{\partial p_i} dp_i \right) \\ &+ \text{higher order terms} \end{aligned} \quad (4.5)$$

If we now invoke (4.4), and neglect the higher-order terms, we see that the following relation must be satisfied by the perturbations dx_i, dp_i :

$$\sum_{i=1}^n \left[\frac{\partial H}{\partial x_i} dx_i + \frac{\partial H}{\partial p_i} dp_i \right] = 0 \quad (4.6)$$

Here the partial derivatives are, of course, evaluated at the initial unperturbed state $(x_1^0, \dots, p_n^0)$, and are thus simply numbers. We can thus think of (4.6) as a *linkage relation* which must be satisfied by the perturbations, imposed on them by the linkage relation (4.4) governing the state variables.

We stress again that these perturbations dx_i, dp_i make no reference to any notion of time or rate. The relation (4.6) merely expresses a condition which must be satisfied by these perturbations (i.e. a linkage imposed on them) arising from the hypothesis that the perturbation of a state satisfying (4.4) once again gives rise to such a state. We have as yet no notion of “how long it takes” to go from the initial state to the perturbed one; such a question is as yet not even meaningful.

Now the relation (4.6) obtains between perturbations, or differentials of the state variables themselves. Let us suppose that we can find an analogous relation which involves only observables. That is, suppose the observables

$$M_i = M_i(x_1, \dots, p_n)$$

$$N_i = N_i(x_1, \dots, p_n)$$

are such that the relation

$$\sum_{i=1}^n \left[M_i \frac{\partial H}{\partial x_i} + N_i \frac{\partial H}{\partial p_i} \right] = 0 \quad (4.7)$$

is identically satisfied. What relation can be established between the observables M_i, N_i and the corresponding perturbations dx_i, dp_i ?

To put the matter simply, let us now *define* a differential quantity dt , as that quantity that simultaneously satisfies all of the relations

$$\begin{aligned}
 dx_i &= M_i dt, \\
 dp_i &= N_i dt.
 \end{aligned}
 \quad i = 1, \dots, n \quad (4.8)$$

In such a quantity dt can be defined, it will be called the *time differential* arising from the linkages (4.4) and (4.7).

Now we observe that we have natural candidates for the observables M_i , N_i . In particular, let us write

$$\begin{aligned}
 M_i &= \frac{\partial H}{\partial p_i} \\
 N_i &= -\frac{\partial H}{\partial x_i}
 \end{aligned}
 \quad i = 1, \dots, n \quad (4.9)$$

With this choice, (4.7) is always identically satisfied. Thus these observables define a time differential, according to (4.8). On the other hand, we can rewrite (4.8) as

$$\begin{aligned}
 dx_i/dt &= M_i \\
 dp_i/dt &= N_i
 \end{aligned}
 \quad i = 1, \dots, n, \quad (4.10)$$

thus explicitly defining the *rates of change* of the state variables themselves, with respect to the appropriate time differential. With the specific choice (4.9), these rates of change are precisely Hamilton's equations.

It is important to observe explicitly that *there is a different time differential dt arising from each choice of observables M_i , N_i satisfying (4.7)*. Each such choice gives rise to a system of dynamical equations satisfying the equation of state (4.4). The specific choice of the time differential for which the dynamical equations (4.10) become Hamilton's equations is, in a sense, arbitrary; once made, however, it serves to *define time* consistently. Indeed, arguing backward from (4.9) to Newton's Second Law, we see that the forces imposed on our system are precisely those for which the time differential dt gives precisely the rates at which the states change as a function of force.

It is also important to observe that, if we carry out the same construction as above for a *different* conservative system, we will obtain a new time differential dt , defining time consistently *for that new system*. *There is a priori no guarantee that the time differentials so defined for different conservative systems are in fact the same*. Indeed, the entire problem of calibration arising from the use of particular conservative systems as clocks (e.g. harmonic oscillators) indicates that the time differentials arising from different systems are actually different, and a relatively elaborate system of mappings is required to transform consistently from the time variable of one system to the time variable of any other. We shall return to this question in Sect. 4.7 below; we merely call attention to the problem here, because it arises from the very outset, and it is especially transparent in this general context.

Let us note that we have so far only formally encoded the time differential dt into the formalism. We have yet to relate this differential, defined by (4.8) above, with the usual encoding of specific instants of time into real numbers. To do this requires a mathematical process of *integration*. If this is done, the usual properties of the definite integral translate the time differential into a set of instants identical with $\mathbb{R}$, and which is an additive group into the bargain. This, it must be stressed, is a purely mathematical (i.e. formal) procedure relating the differential of time (which is all that can be defined from a Hamiltonian) to a global set of instants.

The procedure we have used above is, of course, closely related to the traditional procedure of *virtual displacements*. Essentially, what we have done in passing from (4.6) to (4.9) is to introduce a time differential in such a way that a particular set of these virtual displacements becomes identical with the actual infinitesimal displacements along a trajectory in the state space. This trajectory is simply that (unique) trajectory of (4.10) on which our initial state lies. The global equation of state (4.4) is satisfied by every such trajectory. By repeating the procedure for every value of the constant C in (4.4), the equations (4.10) become the equations of motion describing every such system, and the corresponding trajectories fill out the whole of the state space in the usual fashion.

The procedure we have used is somewhat reminiscent of the concept of *ergodicity*, as it was defined in Example 3 of Sect. 3.3 above. This, it will be recalled, was one of the cornerstones of statistical mechanics, and it required that we be able to replace a notion of time (i.e. a sampling of states along a single trajectory) by a “phase average” (i.e. a sampling of points satisfying the equation of state (4.4)). Our procedure for defining a time differential dt is, in a sense, an inverse of ergodicity; it involves the replacement of operations in the phase space (namely, a perturbation of a point of the space in such a way that the equation of state continues to hold) by a new operation which involves time. We shall return to this important relation to ergodicity in the next chapter.

We already noted in Example 1 of Sect. 3.3 that the quantity t , whose differential we have defined by (4.8), can itself be treated as a variable of configuration. In general, each variable of configuration in a mechanical system is accompanied by a corresponding variable of momentum. Thus we can ask what would be the observable playing this role, as the “momentum” associated with t . The particular choice of M_i , N_i embodied in (4.9) allow us to answer this question in a simple way; the momentum associated with the time t defined by (4.8) and (4.9) is simply $-H$. Indeed, we can treat the mechanics of a conservative system with n co-ordinates of configuration as a problem in a phase space of $2n + 2$ dimensions, by considering time as a new configuration variable and $-H$ as its associated “momentum” in a completely symmetrical way. Once again, we omit the details; they can be found in any standard text on Newtonian mechanics.¹

It is interesting to consider the relation between the procedure we have outlined for consistently defining a time differential dt , and the principle of causality. As we have already seen, one expression of causality appears in the “unique trajectory property”, which says that any state of a system can give rise to only one future behavior, and can arise from only one past, as long as the forces on the system

remain the same. In a certain sense, this causality principle is already implicit in the equation of state (4.6) relating the perturbations which may be imposed on a state satisfying (4.4). Indeed, a relation like (4.6) may be looked upon as itself expressing a kind of conservation law among the perturbations. Causality itself may be regarded as expressing such conservation conditions; indeed, the dynamical equations (4.10) say precisely that, in any state, *only one* infinitesimal perturbation is allowed; namely the one whose magnitude and direction are given by (4.8). These ideas relating causality and time will also be important to us later, especially in Sect. 4.5 below.

We turn now to one of the most important properties of the time differential dt which we have introduced. Namely, the dynamical equations (4.10) satisfy the property that they remain invariant to the replacement of dt by $-dt$; i.e. *these dynamical equations are symmetrical in the time we have defined*. Thus, for a conservative mechanical system, the distinction between future and past is entirely conventional; there is no way within such a system to attach an objective meaning to the “direction” in which time is flowing. In addition, the principle of causality holds equally well, whichever direction is (arbitrarily) chosen to represent the positive flow of time.

It has long been felt that this aspect of *time reversibility* in conservative mechanical systems is at variance with our own intuitive perception of the flow of time. Indeed, very few dynamical phenomena in our experience are actually symmetric to the flow of time; we perceive the sharpest possible asymmetry between past and future. And yet, to the extent that the mechanics of conservative systems is itself a formalization of physical experience, and to the extent that any system can be regarded as composed of particles moving in potential fields, time reversibility appears as an inherent feature of the world. This situation has been correctly perceived as paradoxical. One immediate conclusion we can draw, which is nonetheless important, is the following: *the encoding of time in conservative mechanics is not the only encoding possible*. In particular, it fails to capture the basic quality we perceive in the distinction between past and future. Just from these simple remarks, we see that time is *complex*, in the sense that it allows (and indeed, requires) more than one encoding. Different encodings of time may then be compared with each other, just as different encodings of a natural system may be. Indeed, the characterization of such different encodings, and the relations which exist between them, is the main purpose of the present section. We may go so far as to say that many (though not all) of the problems traditionally associated with time arise from a failure to recognize that time is in fact complex, and that its different qualities *require* more than one kind of encoding.

Traditionally, there are two main kinds of suggestion available for resolving the paradox of time reversibility in a perceptibly irreversible world. These are:

- (a) The dynamics of conservative systems is indeed fundamental, in some ultimate microscopic sense. Hence the irreversibilities we perceive in ordinary experience arise at another (macroscopic) level. Just as the quality of temperature appears to emerge when we pass from microscopic mechanics to

macroscopic thermodynamics, so too does the apparent irreversibility of macroscopic experience. Like temperature, irreversibility depends on the fact that macroscopic experience involves statistical averages over large number of microscopic events. That is, irreversibility at the macroscopic level is connected with ensembles on which we can only deal with probabilities.

- (b) The reversibility of time inherent in the mechanics of conservative systems shows that this formalism is not sufficiently general to encode all natural phenomena. That is, even at the most microscopic level, we must generalize the formalism of conservative Newtonian dynamics to include a broader class of systems for which time reversibility does not hold. The apparent time reversibility of Newtonian dynamics is thus essentially an artifact arising from the adoption of too narrow a formal framework for physical encodings. In a properly general framework, irreversibility will be manifested at the most fundamental microscopic levels, independent of considerations of probabilistic averaging at a macroscopic level.

It should be noted that these two alternatives are *not mutually exclusive*.

We shall consider the first of these possibilities in detail in the next chapter. For the moment, we merely note that in developing it, we are basically confronted with the necessity for developing a new encoding of time *de novo*, this time on the basis of statistical arguments. Such an encoding will necessarily be vastly different from the one we have developed above.

The possibility (b) can itself be studied within the confines of Newtonian dynamics. We can successively weaken the assumptions on which the encoding developed above is based. For instance, we can allow the potential U to depend not only on configurations but also on momentum, and even on time itself; in the most general case, it can readily be shown that the appropriate expression of (4.1) becomes

$$F_i = -\frac{\partial U}{\partial x_i} + \frac{d}{dt} \left(\frac{\partial U}{\partial p_i} \right) \quad (4.11)$$

Such systems are in general no longer conservative. But the question immediately arises; what is the time differential dt appearing in this expression? It no longer has the meaning attributed to it by the argument we have given above; that argument is in fact *equivalent to* the conservation of total energy (this is a corollary to a famous theorem due originally to E. Noether).² Thus, if we are to widen the scope of our encodings in this fashion, *the time differential, and time itself, must be redefined*.

Exactly the same is true if we give up the idea of a potential entirely, and allow forces which do not satisfy (4.1) or (4.11) for any function U . This is the case with so-called *dissipative systems*, of which the simplest example is the damped harmonic oscillator

$$\begin{cases} \frac{dx}{dt} = \frac{p}{m} \\ \frac{dp}{dt} = -kx - \beta p \end{cases} \quad (4.12)$$

We can ask again: what is the meaning of the time differential dt , in terms of which the defining rates are written? Is it the same as the time differential which can be defined as above in the case $\beta = 0$? Or must it be given a different interpretation? We shall consider such questions more deeply in Sect. 4.4 below.

Considerations of this kind are essential if we are to attach any definite meaning to temporal predictions arising from dynamical encodings of natural systems. It is well to stress again that such considerations are not concerned with the question of what time “really” is, but the manner in which its several perceptible qualities are captured within different kinds of encodings. Thus in a real sense our considerations are entirely *formal*; they take place entirely within mathematics. However, we shall impute these properties to time itself, just as we impute topological qualities to sets of abstract states through the observables defined on them. In particular, of course, we wish to impute a definite meaning to such assertions as: “if a system is in state $s(t_0)$ at an instant t_0 , it is in some definite state $s(t_1)$ at another instant t_1 ”. The fact that time itself is *complex* is what makes it difficult to do this in a universal fashion, valid for all encodings. But unless it is done, we have no way of converting an assertion of the kind given above into an actual temporal prediction about a specific natural system.

References and Notes

1. See for example any of the references cited in Note 1 to Sect. 1.2 above. The assertion here is that energy and time are conjugate variables; this fact takes on profound formal significance (though in different ways) in both relativity and in quantum theory.
2. This theorem establishes a remarkable connection between one-parameter groups of transformations (“symmetries”) in phase space under which Lagrange’s equations (3.3.21) are invariant, and corresponding conserved quantities; i.e. observables which are constant on the trajectories. Indeed, the theorem shows that there is a reciprocal relation between such conserved quantities (like total energy in conservative systems) and corresponding *symmetries* of the equations of motion. It is very easy to show that the conserved quantity arising from symmetry to time reversal $t \rightarrow -t$ is the total energy H ; in general, if some dynamical variable is subject to symmetry in the above sense, the conjugate variable is conserved. For a comprehensive treatment of these matters and their physical implications, see e.g.

Lopes, J. L., *Lectures on Symmetries*. Gordon & Breach (1969).

4.3 Time in Thermodynamics and Statistical Analysis

We briefly discussed the character of thermodynamics and statistical mechanics in Example 3 of Sect. 3.3 above. In the present chapter, we will consider these matters in more detail, with special reference to the role of time.¹

We have already noted that classical thermodynamics is restricted to essentially the situations of *equilibrium*. There are several reasons why this is so, which it is worth while explaining in some detail. In the first place, we have already seen that a thermodynamic description of e.g. a mole of gas requires only three state variables (namely pressure, volume and temperature), while a mechanical description of that same system requires on the order of 10^{25} state variables. This tremendous compression comes at a cost, and that cost is the restriction to equilibrium. When a macroscopic system is in equilibrium with its surroundings, there is a precise sense in which its state variables become functions of the environment with which the system has equilibrated. If this environment can be characterized in terms of a small number of quantities of state, then so can the system itself, no matter how many state variables might be necessary to describe the system in other circumstances. Thus, if pressure and temperature are regarded as parameters imposed by the environment, at equilibrium there will be an equation of state describing how every observable of the system is linked to them. Experience indicates that the only such observable we need to consider is volume, and the resultant linkage takes the form of the ideal gas law, or the van der Waals equation, or some corresponding state equation. In non-equilibrium situations, it is clear that not every state variable of the thermodynamic system under consideration will be uniquely determined by the properties of the environment, and hence any encoding of the system will necessarily be much larger; indeed, we will be essentially in the circumstances described in Example 2 of Sect. 3.3.

There is another, even more serious reason for the restriction of classical thermodynamics to equilibrium situations. This resides in the fact that *the fundamental thermodynamic observables of pressure and temperature are only meaningful at equilibrium*. Like all other observables, their values on (thermodynamic) abstract states are determined by the interaction of these states with meters. In the case of temperature, for example, the appropriate meter is a thermometer. However, the essence of a thermometer requires that we must allow our system to *equilibrate* with the thermometer before we can say that the temperature of the system is encoded into the reading of the thermometer. Hence in particular, it is very difficult to attach a meaning to a temperature measurement performed on a system which is not itself in equilibrium. Thus, the other crucial factor, restricting classical thermodynamics to equilibrium is simply that the observables with which it deals may simply not be defined in other circumstances.

Clearly, if thermodynamics were entirely restricted to states of equilibrium, there would be no need for a concept of time, and no thermodynamic sense in which such a concept could be introduced. However, even within the classical formulations of thermodynamics, there is some degree of flexibility, and with this flexibility the first tacit temporal notions come creeping in. If we imagine a thermodynamic

system in equilibrium with some environment, we have seen that the thermodynamic quantities of state associated with the system are all functions of the environment. Let us now imagine the environment changes “slowly”; i.e. slowly enough so that we may imagine *the system is always in equilibrium with it*. In this way we can discuss the incremental changes (usually written as *differentials*) arising in our system as a result of such environmental changes, and introduce the concepts of work, energy, etc. which are the main subject-matter of classical thermodynamics. However, the restriction that the system be always in equilibrium with its environment (or in other words, that these changes are, in a certain sense, “infinitely slow”) restricts us entirely to *reversible* changes in our system; i.e. to the cycles characterizing classical thermodynamic analysis. Such situations are called “quasi-static”, and clearly there is no way to talk about temporal rates of change under these circumstances.

On the other hand, to the extent that thermodynamics is concerned with phenomena of heat, there must be some way within it to capture the basic fact of experience, that heat *flows* from warm bodies to colder bodies, and not the other way around. In other words, the flow of heat represents an *irreversible* situation, and any manner of encoding this aspect of the flow of heat must necessarily carry with it, at least tacitly, another notion of *time*. For the kind of irreversibility manifested by the flow of heat is intimately related with our intuitive ideas of temporal succession; a state of a two-body system in which there is a temperature difference between the bodies must *antedate*, or temporally *precede*, a state of the same system in which the temperatures of the two bodies are equal. Thus, any formulation of the notion of irreversibility tacitly partitions the class of thermodynamic states in to *earlier* and *later*, in which later states can be “reached” from earlier ones, but not vice versa. Hence we see that irreversibility carries with it an essential temporal component, but that this temporal component must be of a totally different character from the time of classical mechanics.

In fact, the fundamental thermodynamic concept of *entropy* can be regarded precisely as a partitioning of a set of states into *earlier* and *later* in this sense. Entropy can roughly be regarded as a real-valued function of thermodynamic state (at least, of thermodynamic state near enough to equilibrium so that we may imagine the thermodynamic variables to be well defined, and still be expressible as definite functions of the environment), defined in such a way that “later” states are characterized by higher entropy values than “earlier” ones. The thermodynamic state of maximal entropy is thus the state of thermodynamic equilibrium itself.

Thus it can be seen that the concept of entropy takes us out of the framework of classical thermodynamics, insofar as it characterizes situations which depart from thermodynamic equilibrium. On the other hand, the Second Law of Thermodynamics (which essentially was stated in the preceding paragraph) is traditionally considered an integral part of thermodynamics itself. It essentially involves the characterization of systems *approaching* thermodynamic equilibrium, but not at equilibrium. The *temporal* sense in which such a system is “approaching” equilibrium is characterized only insofar as the Second Law embodies the distinction between “earlier” and “later”.

The detailed study of states of thermodynamic systems near enough to equilibrium to allow thermodynamic characterization, but not at equilibrium, is the province of “the thermodynamics of irreversible processes” or “nonequilibrium thermodynamics”.² Let us briefly sketch how this kind of approach works. Let us suppose that we can characterize a system near equilibrium by a family $u_1, \dots, u_n$ of thermodynamic variables of state. In more detail, if s is an *abstract state* of thermodynamic equilibrium, then $u_i(s) = u_i^*$ are well defined numbers associated with s by unambiguous measuring processes. If we apply these measuring processes to another state and obtain readings $u_i(s')$ which are close to u_i^* , then we say that s' is a state *near equilibrium*, and we characterize s' by the numbers $u_i(s')$. We do this *even though*, as we have seen, the meters applicable to equilibrium states are strictly not applicable to any other states. By ignoring this last fact, we obtain an encoding of abstract states s' which by imputation are “near” the equilibrium state s . Nonequilibrium thermodynamics works entirely with such neighborhoods, encoded into Euclidean n -dimensional space in the usual way.

On such a neighborhood of the equilibrium state, which of course is itself encoded into the n -tuple $(u_1^*, \dots, u_n^*)$, we further suppose that entropy has been defined, i.e. as a real-valued function S defined on a neighborhood U of $(u_1^*, \dots, u_n^*)$. By hypothesis, it assumes a maximum at equilibrium. Thus, by the mathematical theorem known as the Morse Lemma, there is a neighborhood $V \subseteq U$ of $(u_1^*, \dots, u_n^*)$ in which S can be written as

$$S(u_1, \dots, u_n) = \sum_{i,k=1}^n a_{ik} u_i u_k \quad (4.13)$$

where we now interpret the quantities u_i in (4.13) as the *deviations* from their equilibrium values u_i^* . That is, in the neighborhood V of equilibrium, the entropy is essentially a quadratic function of the deviations from equilibrium. The coefficients a_{ik} appearing in (4.13) are the second derivatives

$$a_{ik} = \frac{\partial^2 S}{\partial u_i \partial u_k}$$

evaluated at the equilibrium $(u_1^*, \dots, u_n^*)$.

Intuitively, our system is *approaching* equilibrium; thus we must express the manner in which the quantities u_i are changing in the neighborhood V . This is the essential dynamical step characteristic of irreversible thermodynamics. We shall relate an incremental change du_i of the i th thermodynamic state variable to the *Hamiltonian time differential* defined in the preceding chapter by writing

$$du_i = J_i dt \quad (4.14)$$

where $J_i = J_i(u_1, \dots, u_n)$ is called the *flow*, or *flux*, of u_i .

Let us pause to comment on this definition. It will be seen that the Hamiltonian time differential is rather dragged in by the heels; it does not arise intrinsically, as it did before. If we use it as indicated in (4.14), we have a way of talking about the incremental changes du_i of deviations from equilibrium in a common temporal framework. We recall that in the preceding chapter we *defined* the time differential dt in terms of increments in (mechanical) state variables, relating increments to states by virtue of an equation of state. Here we have no such equation of state, and we must use an extrinsic time differential dt to relate the increments du_i . This is a vast difference from the preceding situation, and it has some profound practical and conceptual consequences, as we shall see. For the moment, we simply point out that (a) the choice of time differential dt determines the nature of the flows J_i ; changing this differential changes the form of the flows; (b) the choice of the Hamiltonian time differential is not conceptually compatible with the irreversibility expressed by the flow itself.

To construct the analogs of the equations of motion of a mechanical system from (4.14), we need, as usual, an analog of Newton's Second Law. This is obtained by *postulating* that the "forces" X_i responsible for the flows can be expressed in terms of the entropy (4.13) through the relations

$$x_i = \frac{\partial S}{\partial u_i} = - \sum_{k=1}^n a_{ik} u_k \quad (4.15)$$

and that the resulting equations of motion are of the form

$$J_i = \sum_{k=1}^n L_{ik} X_k \quad (4.16)$$

Thus, these equations are guaranteed to move the system encoded by (4.16) in the direction of increasing entropy; this is built in by the manner in which the "thermodynamic forces" X_i are defined.

In mathematical terms, the rate equations (4.16) are so constructed that the function S is a Lyapunov function for the system in the neighborhood V . The equations of motion (4.16) are, in a sense, the simplest of this kind which can be postulated; they assert that the flows J_i are linearly related to the forces X_i . The coefficients L_{ik} which serve to "couple" the flows to the forces are phenomenological coefficients, which obey the celebrated *Onsager reciprocal relations* $L_{ik} = L_{ki}$.

The Onsager relations are in a sense the backbone of irreversible thermodynamics. Onsager's own proof of these relations shows that they arise as a *consequence* of the introduction of the reversible Hamiltonian time differential dt . Indeed, if we were to choose any other time differential in defining the flows J_i , the reciprocal relations would not hold in general.

We may note that one of the basic quantities in irreversible thermodynamics is the "rate of entropy production"

$$dS/dt = \sum_{i=1}^n J_i X_i = - \sum_{i,k=1}^n a_{ik} u_k du_i/dt$$

which again depends crucially on the adoption of the Hamiltonian time differential.

It will be seen from this brief outline how restricted the domain of applicability of irreversible thermodynamics really is, and how contradictory are the assumptions on which it is based. On the other hand, the kinds of predictions which are obtained from the equations of motion (4.16) in specific situations are verified to perhaps a surprising extent. These equations (4.16) express how the thermodynamic (entropic) concepts of “earlier” and “later” are to be explicitly translated into a *reversible* Hamiltonian time frame.

Let us now see how the above thermodynamic considerations look when considered from the viewpoint of statistical mechanics, based on a Hamiltonian picture at the microscopic or molecular level.³ We recall that the statistical-mechanical approach is based on the concept of an *ensemble*. The kinds of ensembles which are important for thermodynamics are, of course, the stationary ensembles (cf. Example 3 of Sect. 3.3 above). Among these, we single out for special consideration the *canonical ensembles*, which characterize a system in equilibrium with a “temperature bath” at a fixed temperature T . These represent systems at thermodynamic equilibrium. More explicitly, it is shown in any standard textbook on statistical mechanics that the canonical ensemble for a system of energy E and at a given temperature T is given by the Maxwell-Boltzmann distribution:

$$\rho = \frac{e^{-E/kT}}{\int_{-\infty}^{\infty} e^{-E/kT} u'(E) dE}$$

and thus, the probability of finding our system with an energy less than some fixed value E_0 is given by

$$\frac{\int_{-\infty}^{E_0} \mu'(E) e^{-E/kT} dE}{\int_{-\infty}^{\infty} \mu'(E) e^{-E/kT} dE}$$

(where the terminology is the same as that used in the discussion of Example 3 of Sect. 3.3 above).

Thus, in the statistical mechanical treatment, the situation of thermodynamic equilibrium is represented by a canonical ensemble. Likewise, measurements of thermodynamic variables are associated with averages over such canonical ensembles. We note that these averages can be regarded, by quasi-ergodicity, either as phase averages or as time averages.

A canonical ensemble is uniquely determined by the underlying Hamiltonian function, and by the temperature T . In turn, at equilibrium, the Hamiltonian can be regarded as depending only on the volume available to the system. Any change in such a system can thus only arise from a change in temperature T , or from a change in Hamiltonian (i.e. in the volume available to the system), or both. This

argument shows in another way why a thermodynamic system at equilibrium can be characterized by such an exceedingly small number of state variables.

Now let us turn to the statistical-mechanical view of entropy. Since the Second Law of Thermodynamics is essentially concerned with non-equilibrium situations, it deals essentially with non-canonical ensembles. It should be noted that if an ensemble is initially non-canonical, it must remain non-canonical forever (this is a consequence of Liouville's Theorem). The concept of entropy in statistical-mechanical terms is basically a means of characterizing the deviation between a given ensemble and a canonical one. As with entropy, this deviation from canonicity is defined through a function of distributions $F(\rho)$ which assumes a maximum when ρ is the canonical distribution. Standard statistical-mechanical arguments again show that such a function $F(\rho)$ can be written as

$$F(\rho) = - \int \rho \ln \rho \, dx_1 \dots dp_n \quad (4.17)$$

where the integral is extended over the entire phase space.

To see how this function F can be related to the entropy, let us consider two initially separate (i.e. non-interacting) systems, with phase spaces P_1 , P_2 respectively. Initially, each system is represented by a canonical distribution in its respective phase space. When the two systems are brought into interaction, the phase space P of the composite system is simply the cartesian product $P_1 \times P_2$ of the phase spaces of the original systems. The resulting distribution in P is not canonical, and by Liouville's Theorem will never become so (as long as P is isolated). However, we may compare it to the canonical distribution in P , using the function F . In fact, it is easy to show that

$$F(\rho) = F_1(\rho_1) + F_2(\rho_2)$$

where ρ_1 , ρ_2 are the original canonical distributions in P_1 , P_2 respectively, and ρ is the resulting distribution in P . But by definition of F , we must have

$$F(\rho) \leq F(\rho_C)$$

where ρ_C is the canonical distribution in P uniquely determined by the average energy and temperature of ρ . This is essentially the Second Law, in statistical mechanical terms.

Finally, if we put this composite system, characterized by a distribution ρ with a definite average energy and temperature, in contact with an external "temperature bath" at the same temperature, the distribution will by definition ultimately become canonical. In the process, the entropy will necessarily increase. The detailed arguments regarding how the distribution changes from its initial noncanonical character to the final canonical distribution are the analog of (4.16) above.

It should be noted that the quantity $F(\rho)$ defined by (4.17) above becomes identical with thermodynamic entropy when ρ is the canonical distribution. For this distribution is determined entirely by the Hamiltonian (i.e. by the volume) and by

the temperature, and thus $F(\rho_c)$ can be looked upon as a thermodynamic variable of state. If ρ is not canonical, the measure of its deviation from canonicity may be regarded as the statistical mechanical analog of S as defined by (4.13) above; but all of the difficulties associated with interpreting S as a thermodynamic quantity away from equilibrium remain in this situation as well.

Now let us return to the role of time in this statistical-mechanical treatment. As we saw in the preceding chapter, the Hamiltonian time differential dt is well defined in the phase spaces underlying the statistical-mechanical arguments. With respect to this time differential, the times required for time averages over ensembles to be meaningful (i.e. to have small variance) are exceedingly long. To interpret such averages as thermodynamic quantities thus requires, from the Hamiltonian point of view, correspondingly long times. Likewise, the time required for a non-canonical ensemble placed in contact with a temperature bath to become canonical is in general also exceedingly long compared to dt . On the other hand, it is precisely this same time differential which is employed in establishing the fundamental equations of motion of irreversible thermodynamics; we cannot in fact dispense with this time differential without also time giving up the reciprocal relations, and with them, the entire apparatus we developed earlier. Thus, the encoding of time into irreversible thermodynamic processes seems to possess fundamentally paradoxical properties. These paradoxical properties are enhanced rather than diminished by the arguments of statistical mechanics, because in these arguments the same time differential dt must necessarily play two distinct and incompatible roles, at the microscopic and macroscopic levels. In this peculiar situation, perhaps the most paradoxical result of all is that the dynamical encodings of statistical mechanics lead to predictions which are experimentally verified.

It may perhaps be pertinent to point out in this connection that the definition of a time differential by a Hamiltonian bears on the *total system* P which the Hamiltonian describes. If we consider the case of a system composed of weakly coupled subsystems, which is the standard situation in statistical mechanics, we could not establish a consistent time frame by considering any of the subsystems in isolation, for these subsystems are not energetically closed in P ; i.e. not conservative, while to separate them from P changes their Hamiltonian. Thus, the establishment of a consistent time frame is a kind of *holistic property* of a conservative system. This remark should be considered in the light of our discussion of fractionation and reductionism given in Sect. 3.6 above.

Now let us briefly turn to a consideration of the probabilistic aspects of time in thermodynamics and statistical mechanics. The best way to do this is through the formulation of the Maxwell-Boltzmann distribution, and to the explication of the concept of entropy, which was originally given by Boltzmann himself (the procedure we have used previously was due to Gibbs). The Boltzmann argument depends essentially on probabilistic considerations, and in its simplest form arises from a combinatoric problem: in how many ways can N identical particles be distributed among $M < N$ boxes? Each such distribution is characterized by M numbers $N_1, \dots, N_M$, where N_i is the number of particles in the i th box; hence $N_1 + N_2 + \dots + N_M = N$. It will be noted that these numbers N_i are discrete analogs

of the continuous density ρ which we used before to characterize distributions in phase space. Thus we will denote a distribution $(N_1, \dots, N_M)$ of particles in our boxes by a similar symbol ρ .

It is easy to show that, given a particular distribution $\rho = (N_1, \dots, N_M)$, we can realize this distribution in

$$P(\rho) = \frac{N!}{N_1! N_2! \dots N_M!} \quad (4.18)$$

different ways; that is, we can assign N particles to M boxes in such a way that the i th box receives N_i particles in $P(\rho)$ different ways. We shall call $P(\rho)$ the *probability* of the distribution ρ . (Strictly speaking, $P(\rho)$ is only proportional to the actual probability, but following customary terminology we shall omit the normalizing factor of proportionality).

We can now ask: what is the distribution ρ_m of greatest probability; i.e. for which $P(\rho)$ is maximal? If N and the numbers N_i are large enough, we can approximate to the factorials in (4.18) by using Stirling's relations

$$N! = N^N; \quad N_i! \cong N_i^{N_i}.$$

Taking logarithms, we find that

$$\ln P(\rho) = N \ln N - \sum_{i=1}^M N_i \ln N_i \quad (4.19)$$

which will clearly be maximal when $\sum_{i=1}^M N_i \ln N_i$ is minimal. It is easy to see that this last sum is minimal when all the N_i are equal; i.e. when the density of particles in each box is uniform. The quantity $\ln P(\rho)$ is essentially what Boltzmann identified with the entropy $S(\rho)$. Indeed, the resemblance of (4.19) to (4.17) is already manifest.

Boltzmann's actual derivation of the Maxwell-Boltzmann distribution consists of adapting the above arguments to an appropriate phase space. Phase space is partitioned into a family of small regions, which are analogs of the boxes considered above. In phase space, each fixed region is associated with a certain average energy; thus the distribution of states among such regions is associated with a conservation of average energy instead of conservation of number of particles. The details of the argument, which leads to a discrete approximation to the Maxwell-Boltzmann distribution, can be found in the standard textbooks of statistical mechanics, and will be omitted here.

We see that Boltzmann's ideas identify the state of thermodynamic equilibrium (i.e. the state of maximal entropy) with the state which is "most probable". More precisely, a thermodynamic state corresponds to a distribution in some appropriate phase space; the thermodynamic state of maximal entropy is thus identified with the distribution of highest probability. In this sense, the transition of a system to thermodynamic equilibrium, which as we have seen automatically classifies such

states as “earlier” and “later”, can be expressed entirely in probabilistic terms; the *earlier* states are those corresponding to distributions which are less probable than the *later* states.

Another commonly employed terminology for these circumstances is the following: if we identify the distributions of high probability as *disordered*, and those of low probability as *ordered*, then the transition to thermodynamic equilibrium involves a passage from order to disorder. Thus, the Second Law itself is often formulated in the following way: that thermodynamic systems pass from ordered states to disordered states, and never the other way. In this language, the distinction between earlier and later becomes identical with the distinction between order and disorder. It is for this reason that, for instance, the phenomena of biological development (cf. Example 2 of Sect. 3.5) seemed so puzzling to physicists; they seem to involve a spontaneous transition from disorder to order, with a consequent reversal of the apparent sense of the arrow of time. To this point of view we must make the following remarks: (a) the systems encountered in developmental biology are *open systems*, to which the Second Law of Thermodynamics is not directly applicable; (b) the identification of “disorder” with distributions of high probability depends entirely on how probabilities are defined.

This last point is of fundamental importance. It deserves an independent discussion, which will be the substance of the next chapter.

References and Notes

1. Some interesting discussions of the matters to be considered here, as well as the relation of the properties of “thermodynamic” time to problems of causality and determinism, may be found in

Stuart, E. B., Bramard, A. J., and Gal-Or, B. (eds.), *A Critical Review of Thermodynamics*. Mono, Baltimore (1970).

2. Some classic standard references to the field of “irreversible thermodynamics” are

Prigogine, I., *Introduction to the Thermodynamics of Irreversible Processes*. Wiley (1967).

de Groot, S. R. and Mazur, P., *Non-Equilibrium Thermodynamics*. North-Holland (1963).

Gyarmati, I., *Non-Equilibrium Thermodynamics*. Springer-Verlag (1969).

3. See the references in Note 5 to Sect. 3.3 above.

4.4 Probabilistic Time

The present chapter represents somewhat of a digression from the main line of argument, in that it concerns some purely formal considerations regarding probability. However, it is so closely related to the notions of thermodynamics and

statistical mechanics described in the preceding chapter that it is natural to place it here. Moreover, it represents an interesting exercise in the general notions of encoding of observables which were developed in Sect. 4.2 above, and has had far-reaching implications for the general study of natural systems.

Let us return to the prototypic Boltzmann problem of throwing particles into M boxes. If $M = 2$, the problem is essentially one of coin-tossing, with the first box marked “heads” and the second box marked “tails”; if $M = 6$, it is that of rolling a die, etc. The first assumption we shall make about the problem is one of *ergodicity*. Namely, we shall assume that it is immaterial whether we throw N particles into our boxes at the same instant, or throw a single particle N times. For simplicity, we shall suppose that we are doing the latter; i.e. that our elementary operation is to throw a single particle into one of a family of M boxes.

In such a situation, each toss can result in one of M possible outcomes, which we shall designate as $u_1, \dots, u_M$. The symbol u_i thus represents the outcome that our particle has landed in the i th box. Let the set of possible outcomes be U . Thus, $U = \{u_1, \dots, u_M\}$.

Let us now suppose that we cast our particle N times. Each time we cast our particle, we obtain one of our M possible outcomes. Hence in this way we generate a *sequence*, or word, formed from the elements of U . The totality of all such sequences will be denoted by S_N .

In more formal terms, any element σ in S_N can be regarded as a mapping

$$\sigma : I_N \rightarrow U$$

where $I_N = \{1, 2, \dots, N\}$ is the set of the first N integers. In what follows, we are going to regard these sets I_N as sets of “instants”, and interpret $\sigma(k)$ as the outcome of the k th toss of our particle. In this way, we identify S_N with $H(I_N, U)$.

Now we are going to construct an encoding of the sequences $\sigma \in S_N$, with the aid of some new observables defined on S_N . Specifically, for each $i = 1, 2, \dots, M$, define

$$\lambda_i(\sigma) = \begin{array}{l} \text{the number of times the} \\ \text{event } u_i \text{ occurs in } \sigma. \end{array}$$

That is, $\lambda_i(\sigma)$ is the number of times our particle has landed in the i th box after our N tosses are completed. Since clearly we must always have

$$\sum_{i=1}^M \lambda_i(\sigma) = N,$$

it is convenient to normalize these observables, by writing

$$\mu_i(\sigma) = \frac{\lambda_i(\sigma)}{N}, \quad i = 1, \dots, M.$$

Thus, to each sequence $\sigma \in S_N$, we can associate an M -tuple of numbers

$$(\mu_1(\sigma), \dots, \mu_M(\sigma)),$$

such that

$$0 \leq \mu_i(\sigma) \leq 1 \quad \text{and} \quad \sum_{i=1}^M \mu_i(\sigma) = 1.$$

In this way, we can encode *any* S_N into a common M -dimensional Euclidean space E_M .

Now we know (cf. Sect. 3.2 above) that any such encoding partitions S_N into a family of equivalence classes; specifically, two sequences σ, σ' fall into the same equivalence class if they both give rise to the same M -tuple; i.e. if

$$\mu_i(\sigma) = \mu_i(\sigma'),$$

for every $i = 1, \dots, M$. Denote the resulting set of equivalence classes by S_N/R .

We shall now finally define a mapping $F : S_N/R \rightarrow \mathbb{R}$ by writing

$$F[\sigma] = \sum_{i=1}^M \mu_i(\sigma) \ln u_i(\sigma) \quad (4.20)$$

It is easy to see that this function F takes its maximum value on that class in S_N/R containing the *most elements*; indeed, F is a kind of measure of the *sizes* of the equivalence classes. We may call $F[\sigma]$ the *entropy* of the sequence σ , by analogy with what we have done before.

Now let us consider this entire situation as a function of N , the number of throws or trials. In particular, we want to know what happens to the numbers $\mu_i(\sigma)$ as N grows very large. The answer to this question is essentially the Law of Large Numbers¹; this asserts first of all that the $\mu_i(\sigma)$ converge to definite numbers; i.e.

$$\lim_{N \rightarrow \infty} \mu_i(\sigma) = p_i.$$

Furthermore, the numbers p_i possess the property that they maximize F ; i.e.

$$p_i = 1/M. \quad (4.21)$$

Let us see what this assertion means, especially in terms of the way in which sequences σ are encoded into E_M .

Intuitively, as we successively cast our particle at the boxes, the corresponding sequence σ of outcomes will grow in length. Regardless of the length of such a sequence, it will always be encodable as an M -tuple of numbers; i.e. as a point in E_M . Each new cast of the particle will change the numbers μ_i slightly, so that the representative point in E_M will appear to move in E_M as a function of “time”

N. Consequently, a growing sequence σ will appear to trace out a trajectory in E_M (although since “time” is here discrete, the resulting trajectory will consist of discrete points of E_M). Thus the growth of a sequence σ by successive casts of our particle will be represented in E_M by something very much like a dynamics on E_M . The Law of Large Numbers, in the form we have sketched it here, asserts that this dynamics possesses a unique, asymptotically stable steady state; namely

$$(1/M, 1/M, \dots, 1/M) \quad (4.22)$$

to which all (or more precisely, “almost all”) of these trajectories converge as “time” (i.e. the number of successive casts of the particle) becomes infinite.

Stated otherwise, the entropy of every growing sequence tends to increase as a function of N, so that these sequences become progressively more alike in terms of the observables M_i , the longer they grow. We can see also that the character of this dynamics on E_M is *irreversible*; the trend of any sequence is *towards* the condition of maximal entropy, and never away from it.

The numbers $p_i = 1/M$ may be heuristically identified with the *sizes* of the boxes at which we cast our particles. We have tacitly constructed the situation so that these boxes are of equal size; it is easy to see how to modify the construction when the boxes are of unequal size (so that the particle cast at the boxes is more likely to land in a larger box than in a smaller one). In that case, the Law of Large Numbers asserts that the $\mu_i(\sigma)$ will converge to numbers p_i proportional to the sizes of the boxes. Thus, these numbers p_i can be identified with the *a priori probabilities* of the outcomes u_i of U.

Now let us look more closely at the “probabilistic time” which is inherent in the above construction. We have already noted that (a) this probabilistic time is discrete, and (b) that it is irreversible. We may further observe that this kind of time carries with it no notion of *rate*, at least at the level of the growing sequences σ . However, for large N (i.e. close to the steady state (4.22)), we may attach an approximate sense to the expressions

$$d \mu_i(\sigma)/dN \quad (4.23)$$

in E_M . Indeed, if we now re-interpret the quantities $\mu_i(\sigma)$ as deviations from their steady-state values (4.22), we will obtain precisely the same equations (4.16) as we found earlier as the basic equations of irreversible thermodynamics, with the function F defined above playing the role of the entropy S.

From considerations of this kind, we see that there is a sense in which the fundamental equations of motion of irreversible thermodynamics are simply re-interpretations of purely probabilistic assertions, and in particular are reformulations of the Law of Large Numbers. If this is so, we see that the probabilistic “time differential” dN is not related to Hamiltonian time directly at all, but rather refers to the improvement of our averages by increasing the length of σ through the addition of another sample. In statistical mechanical terms, the addition of more samples means taking (Hamiltonian) time averages over longer pieces of trajectory; hence the probabilistic time considered above is monotonically related to Hamiltonian

time, but this is as far as the relation between them extends. For in our probabilistic considerations, the “time differential” dN is entirely *rate independent*, as we have seen.

Let us now briefly consider how the concepts of order and disorder appear from a purely probabilistic point of view. If we make the identification of disorder with high entropy, we see that the characterization of disorder depends entirely on the dynamics we impose on E_M ; i.e. on the specific relation we establish between the derivatives (4.23) and the points of E_M at which they are evaluated. That is, *disorder depends on dynamics*. A disordered state is one to which a dynamical system autonomously tends; or alternatively, a state which requires work to keep the system out of. *There is no way to characterize order or disorder in terms of states alone*; thus, to the extent that entropy is *simply* a state function on E_M , unrelated to any dynamics on E_M , an assertion that the state of maximal entropy is the state of maximal disorder is entirely meaningless. It is only when entropy possesses a very special relation to an imposed dynamics (e.g., when it is a Lyapunov function for the dynamics) that such an identification of entropy with disorder is meaningful. In all the examples we have considered in the past two chapters, what we have done is to define entropy *first*, and then *impose* dynamics for which entropy is a Lyapunov function. But if a dynamics is given *first*, there is no reason to expect any *a priori* notion of entropy, such as that embodied in (4.20) above, to have any meaningful relation to notions of order and disorder arising from that dynamics. What we are asserting, then, is the following: that order and disorder are *dynamical* qualities, and not *thermodynamic* qualities. One cannot define disorder in a dynamical vacuum, and this is what classical thermodynamics essentially is. Consequently, except in very special situations, the notions of entropy and disorder are not co-extensive; entropy is always defined purely as a state function, independent of any imposed dynamics, while disorder can only be meaningful in a specific dynamical context.

Precisely the same arguments hold for cognate fields, such as Information Theory.² Information Theory is basically an attempt to associate numbers with words in a language, in such a way as to equate “information” with improbability of a word (considered as a sequence of letters from an alphabet analogous to the set U above). The limitations of this approach, which is often described as equating “information” with “negative entropy” are well known; they revolve around the fact that “information” is identified with purely syntactical aspects, totally divorced from semantics. This is simply another way of saying that entropy, whether negative or positive, is independent of dynamics. However, these fascinating matters are tangential to our main considerations, and we cannot explore them further here; we refer the reader to the literature for further details. We shall, however, return to the general probabilistic ideas developed above when we discuss *complexity* and *error* in Sects. 5.6 and 5.7 below.

The difficulties associated with attempting to define order and disorder in purely probabilistic terms, which have been described above, turn out to be closely related to the problem of reductionism (cf. Sect. 3.6 above). This was first clearly pointed

out by the physicist Walter Elsasser.³ To Elsasser, reductionism is embodied by the expression of all macroscopic quantities (as for instance those encoded into biological descriptions or organisms) as averages over some appropriate phase space, in the manner we have described. By ergodicity, these averages may be regarded as time averages along individual trajectories; this means that “almost all” trajectories become statistically identical *if* we wait long enough. It is this mysterious phrase “almost all” which Elsasser stresses. In mathematical terms, this phrase refers essentially to what happens in regions of zero volume; such regions are always *necessarily* neglected in any formal probabilistic argument. Nevertheless, sets of “zero volume” can be appreciable in other mathematical senses. For instance, the set of rational numbers is of “zero volume” in the set of all real numbers, and hence “almost all” real numbers are irrational. On the other hand, the rational numbers are actually topologically *dense* in the real numbers; between any two rational numbers, however close together they may be, there is another one. Nevertheless, in any probabilistic argument, or more generally, in any averaging or integration process over the real numbers, what happens on the rational numbers is necessarily irrelevant.

Elsasser pointed out that the phase space of any organism, considered as a purely physical system, is in general of very high dimension. He argued further that the states in this phase space which are compatible with life will generally tend to be *sparsely distributed* in such a phase space; specifically, they will form a set of zero volume. Hence “almost all” states, and thus “almost all” trajectories in this phase space, will be incompatible with life. Furthermore, any attempt to form averages over the entire phase space will necessarily discard the biologically relevant states. Consequently, Elsasser argued that insofar as physics must deal entirely with such averages at the macroscopic level, biology is *in principle* irreducible to physics. It further follows that the laws governing the behavior of biological systems are not inferable from physical laws although they are compatible with them. This is a very powerful argument. In general, insofar as we must identify “order” with the properties of sets of small volume, and insofar as averaging processes neglect what happens on such sets, general statistical laws of physics cannot in principle be applied to ordered situations. In slightly different language, the assertion is that encodings of ordered situations cannot be reduced to encodings pertaining to disordered ones.

We can see this very strongly if we return to our earlier example of the relation between the rational numbers and the reals. The fact that the rationals are a set of zero volume may be interpreted, in gross probabilistic terms, as asserting that it is “infinitely improbable” for a number taken at random from the real numbers to be a rational number. On the other hand, if we ask someone to “pick a number”, the number picked will almost certainly be rational. The reason for this is that human beings are strongly biased towards the rational numbers, and we cannot explain this bias, nor understand our own relationship to the real number system, by means of *a priori* probabilistic or measure-theoretic considerations.

References and Notes

1. The “Law of Large Numbers” is the name given to a class of related results, originating in probability theory, and asserting essentially that as the number of trials increases, the corresponding sequences of “average results” (i.e. outcomes averaged over all the trials that have already taken place) converges. This result and its ramifications are discussed in almost all standard texts on probability theory. A concise but comprehensive treatment may be found in

Revesz, P., *The Law of Large Numbers*. Academic Press (1968). It is not surprising that there is a close relationship between the Law of Large Numbers and the notion of ergodicity as it was originally formulated in statistical mechanics. The relation is described e.g. in the review of Mackey to which we have had occasion to refer several times before (cf. Note 4 to Sect. 1.2).

2. On its mathematical side, Information Theory is closely related to the matters we have been discussing. A good treatment, which makes all sides of these relations visible, is

Billingsley, P., *Ergodic Theory and Information*. Wiley (1965).

3. Elsasser’s original arguments may be found in

Elsasser, W., *The Physical Foundations of Biology*. Pergamon Press (1958).

See also Elsasser, W., *The Chief Abstractions of Biology*. Elsevier (1975).

4.5 Time in General Dynamical Systems

In the present chapter, we shall be concerned with the role of time as it appears in the general dynamical encodings of natural systems into models and metaphors. We have seen many examples of such dynamical encodings in the preceding chapters. What we wish to do here is to point out that the encodings of time into these systems are as different from Hamiltonian and thermodynamic time as these two are from each other.

Let us in general suppose that $x_1, \dots, x_n$ represent the (actual or metaphorical) encoding of observable quantities, in such a way that every abstract state of the system is encoded into a point of a manifold M in Euclidean n -dimensional space. We recall that such an encoding already tacitly involves a definite temporal aspect; namely, that of simultaneity. Indeed, the numbers $(x_1, \dots, x_n)$ which identify a point of M are to be interpreted as the values of the observables x_i , evaluated simultaneously on some abstract state s , and thus serve to characterize s at some *particular instant* of time.

We have also already seen that a set of rate equations on M , of the form

$$dx_i/dt = f_i(x_1, \dots, x_n), \quad i = 1, \dots, n \quad (4.24)$$

embody several further hypotheses about our system. For one thing, they assert that the rate of change dx_i/dt of each state variable x_i is itself an observable, and in fact is precisely the observable f_i . Thus each of the rate equations in (4.24) express a linkage between the rates of change of an observable at an instant of time and the values at that instant of the state variables themselves. Further, in mathematical terms, the rate equations (4.24) serve to attach a definite velocity vector to each point of the manifold M into which the abstract states of our system are encoded; that is, they define a vector field on M . Thus, these equations specify on each point of M a unique *direction* in M associated with that point. Finally, we observe that the mathematical operation of *integrating* the equations (4.24) results in principle in the explicit expression of each state variable as a definite function of the time parameter t , and thus links a definite state of our system to each time instant t and every initial state $(x_1(t_0), \dots, x_n(t_0))$.

We have also seen that, under very general conditions, the rate equations (4.24) manifest the basic properties of *causality*. These are expressed in the unique trajectory property, which says that any point of our manifold M uniquely determines the entire trajectory passing through it; hence only one future arises from each state, and only one past can give rise to each state, as long as the equations of motion (4.24) remain valid.

It will be observed that, in this general setting, there is no concept of force or potential visible, of the kind which was dominant in our consideration of the encodings of particle mechanics; in their place we have simply a postulated linkage between instantaneous rate of change of state variables and the instantaneous values of the state variables themselves.

Now in the employment of rate equations of the form (4.24) as encodings of dynamical processes in natural systems, the usual point of view is that, roughly speaking, the differential increments dx_i of the state variables are known, and the time differential dt is known, and the observable $f_i(x_1, \dots, x_n)$ is their ratio. In what follows, we are going to take a different point of view, which is roughly this: the principles of causality, together with the differential increments dx_i of the state variables, essentially serve to specify the observables f_i ; these together with the dx_i in fact *define* the time differential dt . The time differential dt which arises from such an argument in general is quite different from the Hamiltonian time differentials which were defined in Sect. 4.2, and in fact are generally different for different dynamical systems of the form (4.24). Thus, each dynamic encoding requires an encoding of time into a time differential unique to it. We shall consider some of the consequences of this curious fact at the conclusion of the present chapter.

Let us then proceed to the details. We shall initially suppose only that we are given a manifold M , whose points may be expressed as n -tuples of the form $(x_1, \dots, x_n)$ as before. Since M is a manifold, we can give a definite and unambiguous meaning to the differentials $dx_1, \dots, dx_n$ at each point of M . This much is available to us simply from the fact that M is a manifold.

We are now going to impose on this situation a notion of causality. Intuitively, if the manifold M encodes the abstract states of some natural system, the simplest form of causality we can impose is to require that *the differentials dx_i are not*

all independent, but satisfy some linkage relations, depending on the particular point of M to which they refer. Moreover, we should require that *the totality of linkage relations satisfied by these differentials should suffice to determine them uniquely*. That is, to each point $(x_1, \dots, x_n)$ of M , there will be attached a *unique* set $(dx_1, \dots, dx_n)$ of differential increments, defined by these linkage relations. Clearly, to define these differential increments $dx_1, \dots, dx_n$ uniquely will require n independent linkage relations. We shall now suppose that these linkage relations are of the simplest possible form; namely, that they are *linear* in the dx_i . Let us then write these relations as

$$\sum_{j=1}^n a_{ij} dx_j = \omega_i \quad i = 1, \dots, n \quad (4.25)$$

where the numbers a_{ij} depend on the state $(x_1, \dots, x_n)$ to which the differentials dx_i are attached. The existence of such relations (4.25), then, arise entirely from the weakest possible notions of causality in the system whose states are encoded by the points of M , as stated above. It is a consequence of the stronger causality assertion that we may assume the determinant $|a_{ij}| \neq 0$.

Now we shall make our first real hypothesis. We shall suppose that each of the ω_i in (4.25) can be written in the form

$$\omega_i = df_i \quad (4.26)$$

where $f_i = f_i(x_1, \dots, x_n)$ is an observable; i.e. a real-valued function on the manifold M .

If (4.26) holds, then we can interpret the linkage relations (4.25) as each specifying an *accounting*, or a *conservation condition*, relating to the differential increments dx_i attached to a state. Each equation of (4.25) says that these dx_i are not entirely arbitrary; some linear combination of them is the differential increment of a particular observable f_i . The totality of these relations (4.25) then define the dx_i uniquely.

Another consequence of (4.26) which may be noted is that the numbers a_{ij} in (4.25) are immediately interpretable, as

$$a_{ij} = \frac{\partial f_i}{\partial x_j}$$

where the partial derivatives are evaluated at the point $(x_1, \dots, x_n)$ to which the differential increments dx_i are attached.

The final step in defining our differential time increment is now clear. We shall suppose that we can write

$$dx_i = f_i dt \quad (4.27)$$

for each $i = 1, \dots, n$. These relations serve simultaneously to define the time differentials dt , and to allow us to encode the dynamical behavior of our

system into the form (4.24). We stress again the order of ideas here: (a) the differential increments dx_i are related by causality by conservation conditions of the form (4.25), which suffice to determine them uniquely; (b) we *posit* that these conservation conditions define, in mathematical terms, exact differentials; (c) these in turn define global observables f_i , $i = 1, \dots, n$; and finally (d) we define the time differential by requiring it to satisfy the relations (4.27) for each i . The procedure we have used is strongly reminiscent of that employed in Sect. 4.2 for defining the Hamiltonian time differential; however in that case we *started* from a global linkage (the Hamiltonian of the system) and used it to derive conservation conditions on the differentials of the state variables analogous to (4.25) above. In the present case, we obtain these relations directly from a hypothesis of causality, without postulating any global linkage (equation of state) at all.

Now let us consider some of the consequences of the encoding of time we have developed. The first observation to make is that in general the dynamical time differential dt defined by (4.27) is *not reversible*. That is, the equations of motion (4.24) to which it gives rise is not invariant to a replacement of dt by $-dt$. Irreversibility is thus built into such a system from the outset, without any notion of equilibrium or any kind of probabilistic or entropic consideration. In this context, irreversibility is exhibited entirely as a *dynamical* phenomenon, and not as a thermodynamic one.

The next important observation to make is that every dynamical system of the form (4.24) in general defines its own time differential dt , and that *the differentials arising from different dynamical systems are in general different from one another*. This can be seen immediately from the defining relations (4.27). Indeed, if we take two different dynamical systems (4.24) defined on the same manifold M , the differential increments dx_i are here the same for both systems, but the conservation relations (4.25) (i.e. the numbers a_{ij}), and hence the functions f_i , will in general be quite different from one another. Since we are *defining* the time differential dt for the two systems in terms of a common set of differentials dx_i and different sets of functions f_i , there is no reason to expect that the resulting time differential dt will be the same for the two systems. Furthermore, there is no reason to expect that any such time differential will be related to a Hamiltonian time differential of the kind defined in Sect. 4.2 above. This fact raises some exceedingly deep questions pertaining to the temporal relations which exist between different dynamical systems, and in particular to the employment of a common set of clocks (which invariably involve a Hamiltonian time differential) to “keep time” for all dynamical processes. In effect, we have shown that each dynamical process (4.24) defines its own *intrinsic* time differential; the question we are posing is this: how are different intrinsic times, arising from different dynamical processes, to be related to each other? More specifically: how are these different intrinsic times to be related to a common “clock time”, which in general is different from all of them? Clearly, it is hard to make meaningful dynamical predictions from an encoding of the form (4.24) unless this question is answered.

To approach this kind of question, let us consider again the basic rate equations (4.24), this time taking the point of view that the parameter t (and its differential dt)

appearing therein is simply an arbitrary parameter; e.g. we could take it to be arc length along trajectories. In that case, we can multiply each of these rate equations by the same non-vanishing function $a(x_1, \dots, x_n)$ without changing the trajectories or the stability properties of the system; all we do is change the rates at which the trajectories are traversed, *relative to some fixed time scale*. In fact, this procedure may be interpreted as replacing the intrinsic time differential dt defined by (4.27) by a new time differential of the form $a(x_1, \dots, x_n)dt$.

Indeed, when we write down a set of rate equations encoding some natural dynamical system on the basis of a general law, like the Law of Mass Action (cf. Sect. 3.5 above), the time differential arising in these equations is only defined up to such an arbitrary function $a(x_1, \dots, x_n)$. There is no way within such a system to characterize this function, and hence no way of predicting from knowledge of trajectories the state in which we will find the system at a particular instant. The intrinsic time scale defined by (4.27) for these rate equations will clearly be the one for which the scale factor $a(x_1, \dots, x_n)$ becomes identically unity in M , but we have no guarantee (and indeed it will be false in general) that the time tacitly encoded into the Law of Mass Action is the intrinsic time to be associated with the natural system itself. Thus, we must regard all of our dynamical models as containing an arbitrary scale factor, which cannot be determined by employing any kind of intrinsic argument pertaining to the model itself.

We attempt to take account of this arbitrary scale factor by choosing the scale so that the intrinsic time differential dt defined by (4.27) is related to a Hamiltonian time differential dt_H by a relation of the form

$$dt = a(x_1, \dots, x_n)dt_H. \quad (4.28)$$

If this is done, the scale factor becomes absorbed into the constitutive parameters (rate constants) appearing in the equations of motion, and the time differential appearing in the resulting rate equations becomes interpretable directly in terms of “clock time”. But it must be recognized that this procedure involves a profound distinction between the constitutive rate constants which are measured, and the intrinsic rates at which our natural system is changing. This distinction is far from trivial; for instance, only for a particular choice of scale factor can the Lotka-Volterra equations of population dynamics be converted into a Hamiltonian form, for which Liouville’s Theorem holds. In other words, our ability to apply the machinery of statistical mechanics to the Lotka-Volterra equations (which are, it will be recalled, based precisely on Mass Action) depends entirely on making a proper choice of scale factor relating intrinsic time to Hamiltonian time.

We can conclude then that *the encoding of time as a general dynamical parameter cannot be effectively made within a limited dynamical context*. Rather, we must in each case arbitrarily select a scale factor $a(x_1, \dots, x_n)$ which will convert intrinsic time to Hamiltonian time via (4.28); only in this way can we use clocks to define a “common time” in terms of which dynamical predictions can be made. Even this is not the end of the matter, for we have already point out that Hamiltonian time differentials can be different from one another. Thus,

if we change from one Hamiltonian time differential to another, the resulting relations (4.28) force these Hamiltonian differentials to be related in a particular way if (4.28) is to remain invariant to this change. A change of this kind can be looked upon as a change of *observer*, and thus we impinge on the kinds of relativistic considerations which are intrinsic to any discussion of synchronization. However, we shall not pursue this matter further here; our main purpose is merely to show what is involved in attempting to consistently define temporal encodings and rates in such a way that meaningful temporal predictions can be obtained. As we have seen, any such dynamical encoding can only define what we have called an intrinsic time, related to a common “clock time” through an arbitrary scale factor, which cannot be specified by dynamical considerations entirely within the system.

We shall now briefly discuss the interpretation of the linkage relations (4.25), which as we have seen embody simply a hypothesis of causality. We have seen that, if a relation (4.26) also holds, we can interpret the numerical coefficients a_{ij} in (4.25) as

$$a_{ij} = \frac{\partial f_i}{\partial x_j} = \frac{\partial}{\partial x_j} \frac{dx_i}{dt} \quad (4.29)$$

Let us now introduce the following terminology: we shall say that the state variable x_j *activates* the state variable x_i in the state $(x_1, \dots, x_n)$ if the quantity a_{ij} in (4.29) is positive, and that it *inhibits* x_i if this quantity is negative. Intuitively, if x_j activates x_i , then an increase in x_j increases the *rate* at which x_i changes (or alternatively, a decrease in x_j decreases the rate at which x_i changes). Likewise, if x_j inhibits x_i , then an increase in x_j decreases the rate at which x_i changes (and a decrease in x_j increases the rate at which x_i changes). This terminology of activation and inhibition is in fact meaningful for any dynamical system of the form (4.24); it is interesting to see how far we may characterize a system of rate equations on the manifold M through a specification of functions $a_{ij} = a_{ij}(x_1, \dots, x_n)$ which determine an *activation-inhibition pattern* among the state variables $x_1, \dots, x_n$. As we see, such a pattern arises automatically from the expression of conservation conditions (4.25) among the differential increments dx_i at a state, as a direct consequence of causality.¹

In this terminology, we may now re-interpret the conservation relations

$$df_i = \sum_{j=1}^n a_{ij} dx_j \quad (4.30)$$

in the following way: *the differential change df_i (i.e. the differential change in the rate of change of the i th state variable x_i) is the sum of the activations and inhibitions imposed on x_i by the other state variables, each of which is weighted by the differential change of the corresponding state variable.* Thus we can interpret the sum $\sum_{j=1}^n a_{ij} dx_j$ as the *net excitation* received by the state variable x_i in the given state. Likewise, we may interpret the resulting increment df_i as the *response* to this net excitation.

Using this language, we in effect construct a *network* of excitable elements, very much like a neural net (cf. Example 3B of Sect. 3.5 above).

In order for such an interpretation to be valid for an arbitrary family of functions $a_{ij}(x_1, \dots, x_n)$ it is necessary and sufficient that the net activations $\sum_{j=1}^n a_{ij} dx_j$ which we have defined should be exact differentials. This is, of course, exactly the requirement we earlier stipulated in (4.26) above, and explicitly embodied in (4.30). Furthermore it is only in this case that we are able to define an intrinsic time differential dt , through which the concept of rate itself becomes meaningful.

If the manifold M is simply connected, it is easy to write down necessary and sufficient conditions on the functions $a_{ij}(x_1, \dots, x_n)$ which turn the net activations into exact differentials. These conditions involve only the partial derivatives of the functions a_{ij} with respect to the state variables x_i in the manifold M . For instance, if $n = 3$, these conditions become

$$\frac{\partial a_{i1}}{\partial x_3} = \frac{\partial a_{i3}}{\partial x_1}; \quad \frac{\partial a_{i1}}{\partial x_2} = \frac{\partial a_{i2}}{\partial x_1}; \quad \frac{\partial a_{i2}}{\partial x_3} = \frac{\partial a_{i3}}{\partial x_2},$$

for each $i = 1, 2, 3$. Thus we can see that the kinds of functions a_{ij} which can appear in the relations (4.25) above, which we noted are merely an embodiment of causality on M , must be severely restricted if a consistent notion of intrinsic time (and hence a corresponding set of rate equations of the form (4.24)) is to be defined at all. Thus we see emerging a close relation between time and causality, which is of course at the heart of using dynamical models for predictive purposes. We also draw attention, for subsequent reference, to the close formal relationship between general rate equations of the form (4.24) and the excitable networks which we considered earlier.

References and Notes

1. The ideas embodied in this discussion go back to Lotka, but the first explicit connection between the coefficients a_{ij} and biological function seems to have been made by Higgins (*Ind. Eng. Chem.* 59, 18–62 1977) in the context of developing a model for oscillations in sequences of enzyme-catalyzed reactions. A more detailed discussion of activation-inhibition patterns, and their relation to dynamical systems, may be found in Rosen, R., *Bull. Math. Biol.* 41, 427–446 (1979). Somewhat similar ideas can also be found in the dynamical analysis of ecosystems (e.g. Levins, R., *Lecture Notes in Biomathematics* 18, 152–199 1977).

4.6 Time and Sequence: Logical Aspects of Time

In the present chapter, we wish to explore yet another aspect of temporal encoding, different in character from those we have considered so far. This will concern the *logical* character of time. More specifically, we wish to develop the sense in which a dynamical prediction of the form “if a system is initially in state $x(0)$, then at time t it will be in state $x(t)$ ” can be regarded as a *theorem*, derived from $x(0)$ as axiom, with the aid of production rules (rules of inference) arising from the equations of motion. We thus wish to treat a dynamical system as a kind of *formal theory* in the sense of Example 3 of Sect. 3.1 above.

Let us begin by returning to the situation described in that earlier example, in which a formal theory was regarded as a family of axioms and production rules. The axioms themselves could be regarded as words $w_1, \dots, w_n$ written in some alphabet, and the production rules were a set of operations $\mathcal{J} = \{T_1, \dots, T_m\}$. An operation T_i in $\mathcal{J}$ can be regarded as a mapping defined on r -tuples of words, and associating with each r -tuple in its domain a definite word in its range. Thus, in algebraic terms, each T_i is an r -ary operation on the set of all words, where of course r depends on the operation.

Let A_0 denote the set of axioms; $A_0 = \{w_1, \dots, w_n\}$. We are now going to define a set of words A_1 , which are the *immediate consequents* of the axioms in A_0 under the production rules in $\mathcal{J}$. Briefly, A_1 consists of all the words that can be produced by applying the production rules in $\mathcal{J}$ to all possible r -tuples of axioms for which they are defined. That is, given any word w in A_1 , we can find a production rule T_k in $\mathcal{J}$, and axioms $w_{i_1}, \dots, w_{i_r}$ in A_0 , such that

$$w = T_k(w_{i_1}, \dots, w_{i_r}). \quad (4.31)$$

Thus, A_1 is a well-defined set of words; clearly, every $w \in A_1$ is a *theorem* of our system, and the *proof* w consists precisely in exhibiting the appropriate production rule T_k and r -tuple of axioms which yield w as a consequent.

Now let us go one step further. We shall define a set of words A_2 , which consist of all words produced by applying an operation in $\mathcal{J}$ to an r -tuple consisting of axioms in A_0 , or theorems in A_1 . Once again, A_2 is a well-defined set of words, and any word $w \in A_2$ is a *theorem*. The *proof* of such a theorem now consists of several stages. We must exhibit: (a) a production rule T_k such that (4.31) is satisfied, and the operands w_{i_j} belong either to A_0 or A_1 ; (b) for each operand w_{i_j} in A_1 , we must exhibit an explicit proof; an expression of the form (4.31) all of whose operands lie in A_0 .

Thus, to prove that a word w is a theorem in A_2 requires a *sequence* of elementary steps. These steps produce words in A_1 from the axioms; the final step produces a word in A_2 . Each step requires the specification of a production rule, and also of the r -tuple on which that rule is to act. The sequence itself constitutes the proof.

We can obviously iterate this procedure. For instance, we can define the set A_3 of words satisfying (4.31), such that each operand w_{i_j} belongs either to A_0 , A_1 or

A_2 . Once again, each such word is a *theorem*, whose proof consists of exhibiting a sequence of elementary steps of the kind we have described. We stress again that each of these elementary steps requires us to specify a particular production rule in $\mathcal{J}$, and also to specify the particular r -tuple of words on which this rule is to act. Likewise, we can generate the sets of words $A_4, A_5, \dots, A_N, \dots$ for every integer N .

We are now going to argue that *the set of indices for the A_i comprise a temporal encoding*. That is, these indices may be regarded as a (discrete) set of *instants* for a kind of logical time scale. To say that a word w belongs to A_N says that it requires N “instants” in this time scale for the word to be produced from the axioms. Hence the axioms themselves play the role of “initial conditions”; they specify the situation at the initial instant $N = 0$. Let us see how this can be done.

Suppose w is a word in A_N . Then by definition, there is a production rule T_k in $\mathcal{J}$, and words $w_{i_1}, \dots, w_{i_r}$ belonging to the sets $A_0, A_1, \dots, A_{N-1}$ such that (4.31) is satisfied. We shall think of w as being produced at the instant N . Now some of the operands of T_k will lie in A_{N-1} . We will think of these as being produced *simultaneously* at the instant $N - 1$, by means of expressions like (4.31), in which the operands now lie in $A_0, \dots, A_{N-2}$. Thus, at each instant of our time scale, we will produce all of the words which will be required at the next instant. Carrying this procedure back to the axioms themselves, we can regard them as words produced at the 0^{th} instant.

In this way, a proof can be regarded as a sequence of successive steps, indexed by the logical time we have defined. At each such step, a set of words is produced from the words generated at *earlier* steps, and which themselves will be utilized to produce new words at *later* steps. Thus, the procedure we are envisioning is a *recursive* one, in which each new step depends on the results of the preceding ones. We could write down explicitly how such a recursive procedure works, but the preceding discussion should be sufficient to illustrate how it works; we shall leave more precise formal statements to an exercise for the reader.

It may help fix ideas to pause at this point and give some concrete realizations. A simple and familiar one comprises a game¹ like chess. In chess, the axioms are embodied in the initial position; the production rules are the legal moves of the game. The logical time which we have introduced indexes the successive moves played by the players; the position which is arrived at following any particular move is the analog of the theorem, inferred from the starting position (the axioms) by means of successive applications of production rules. Of course, we can take any legal initial configuration of pieces on the chessboard as axioms; this is the format for the familiar chess problems, which require the establishment of theorems of the form “white to play and mate in two moves”.

Another most important embodiment of the considerations developed above may be found in the McCulloch-Pitts neural networks, described in Example 3 of Sect. 3.5 above. Instead of a chessboard, we now have a definite array of interconnected neurons. The production rules specify which neurons shall be excited at a particular instant, as a function of those neurons excited at a preceding instant. The instants themselves are, of course, arbitrary. The pattern of excitation of the

network at an instant N can thus be regarded as a theorem, derived from an initial pattern of excitation (the pattern at $N = 0$) through the successive application of the production rules. Thus we see, in another way, how the neural nets can be regarded as realizations of logical systems; this time as realizations of formal theories, in which the temporal structure of the network is directly interpretable into the logical or sequential time we have introduced above.

Before going further, let us itemize the main properties of the logical time frame we have constructed. First, we see that this time is *discrete*. Second, it is entirely independent of any physical notion of *rate*. Thus, the logical structure of a game of chess is entirely independent of the rates at which the individual moves are made, considered in any physical sense. The only thing which matters is the *order* in which the moves are made. Third, this logical time is *irreversible*. It will be noted that these properties were also manifested by the probabilistic time introduced in Sect. 4.4 above, and are at this point quite different from the properties of Hamiltonian or dynamical time.

Now the relation between logical time and neural networks on the one hand, and between neural networks and dynamical systems on the other, provides a motivation for supposing that dynamical time and logical time may be related to one another. If such a relation could be established, we could simultaneously view dynamical systems as very special kinds of formal theories, and formal theories themselves as generalizations of dynamics. The remainder of the present chapter will be devoted to exploring how this can be done.²

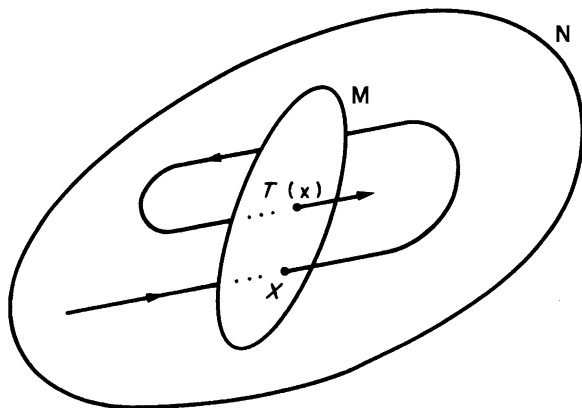
Let M be any set, and let $T : M \rightarrow M$ be any automorphism of M ; i.e. a 1-1 mapping of M onto itself. Since any automorphism may be composed with itself, we can form the automorphisms $T^2, T^3, \dots, T^n, \dots$. If we define the identity automorphism I to be T^0 , the set of all these automorphisms is a group G_T , where we have $T^m T^n = T^{m+n}$ for any integers m, n . This is the cyclic subgroup of all automorphisms $a(M)$ of M , generated by T . We are going to think of T as a *production rule* on M ; if $x \in M$ is regarded as an axiom, then $T(x)$ is the *theorem* obtained by applying T to the axiom.

With this terminology, the exponents of the powers of T in G_T become interpretable as *instants of logical time*. Thus, if $x(m) = T^m x(0)$, we can regard $x(m)$ as the theorem (in this case, the unique theorem) inferable from the axiom $x(0)$ in m steps of logical time. In this situation, we can characterize the theorems inferable from the axiom $x(0)$ very simply; they are the *orbit* of G_T on which $x(0)$ lies. More precisely, x is a theorem inferable from $x(0)$ if there is an instant m of logical time such that $x = T^m x(0)$.

In particular, if we take M to be a *manifold*, the specification of a group G_T converts the manifold into a *discrete dynamical system*. In this case, we have a manifold full of choices for our initial axiom, but once having chosen an axiom, the set of theorems is completely determined. Moreover, each theorem is indexed by an instant of logical time, which recursively relates the theorem to the axiom.

This situation is, of course, closely related to the specification of a continuous dynamical system, as a one-parameter group $\{T_t\}$ of automorphisms on a manifold M .

Fig. 4.1



Here the index t refers to a corresponding *dynamical time*, such as is embodied in the dynamical time differential dt defined in the preceding chapter. Indeed, if we are given a continuous dynamical system $\{T_t\}$ in this form, we can define many discrete dynamical systems from it; for any real number r , we can consider the cyclic subgroup generated by T_r (i.e. the subgroup whose elements are $T_r^n = T_{nr}$ for each integer n), and this will define a discrete dynamical system on M . In this case, we would have an explicit interpretation (i.e. a *realization*) of the *logical* time in the discrete system, in terms of the *dynamical* time t of the continuous system. Precisely this method is used in obtaining numerical solutions of systems of differential equations by means of digital computers.

Unfortunately, this procedure of relating logical time in a discrete system to dynamical time in a continuous one only can work if the dynamical time is defined *first*, and a logical time is specified by choosing some arbitrary unit time step $t = r$. We cannot go backwards, starting from a given logical time (i.e. a discrete group G_T) and embed it into a continuous one-parameter group. Moreover, there are many other possible relationships which can exist between discrete-time and continuous-time dynamical systems. For instance, suppose that M is embedded as a submanifold in a larger manifold N , and suppose that a one-parameter family $\{T_t\}$ of transformations is defined on the larger manifold N . Given a particular trajectory in N , we may define a mapping of M onto itself as indicated in Fig. 4.1 below:

Here the image $T(x)$ of a point x in M is defined as the point at which the trajectory through x next intersects M . Under appropriate circumstances, this mapping $T : M \rightarrow M$ defines a discrete dynamical system on M , but there is no continuous dynamical system on M of which G_T is a cyclic subgroup. It will be noted that the instants of logical time for the discrete system G_T in this case are not even a subgroup of the set of instants of dynamical time on N in general. The idea we have just sketched is often called the *suspension* of such a discrete system on M into a continuous one on a bigger manifold N , an idea originally due to Poincaré.³

Nevertheless, there is a sense in which we can interpret the dynamical time parameter t as a logical index for the theorems $x(t)$ implied by an axiom $x(0)$. It

will be noted that in the case of a discrete dynamical system, the entire dynamics is determined by a generator T ; an automorphism of M which can be applied to any point of M chosen as initial axiom. The logical time index m then refers to the number of times this generator must be applied to an axiom to yield a particular theorem. Thus there is an identification of this generator with the production rule of a formal system. If such ideas could be applied to the continuous case, we would need some concept of *infinitesimal generator*, from which any automorphism T_t could be obtained by an integration process.

In the continuous case, the role of such an infinitesimal production rule is played by the tangent vector. More precisely, if M is a manifold, we can turn M into a dynamical system precisely by specifying a tangent vector $\tau(x)$ for every point x in M ; i.e., we define a vector field on M . As we saw in the preceding chapter, this is sufficient to impose conservation conditions on the differential increments attached to x , and hence to define the concept of dynamical time on M . From the present point of view, however, the tangent vector $\tau(x)$ can also be regarded locally as an infinitesimal production rule, specifying an infinitesimal change of state $x \rightarrow x + dx$ associated with the differential dt of dynamical time. Of course this production rule is in general different from point to point in M ; indeed, the dynamical context is such that to each point x of M , a unique infinitesimal production rule $\tau(x)$ is also specified.⁴ By means of an integration process, a trajectory on M can be looked upon as a *continuum* of theorems, indexed by the instants of dynamical time; thus we can force dynamical time and logical time to coincide.

In this way, a dynamics can be viewed as a very special kind of formal theory. It is special in that, given any initial point M as axiom, there is exactly one production rule which can be applied to it (namely, the rule specified by the tangent vector to that point). On the other hand, this viewpoint leads to a significant generalization of formal theories; namely, that the theorems no longer form a discrete set, but are indexed by continua. The advantage of such a generalization is that discrete, rate-independent logical time coincides in this situation with rate-dependent dynamical time. Indeed, it is only under these circumstances that a relation between the two kinds of time can be established at all.

References and Notes

1. Indeed, any kind of game thus constitutes a formal system. However, games manifest additional structure, because the point of playing a game is to win. This means that there are certain distinguished propositions ("wins") in the system which a player attempts to establish, and others ("losses") which he wishes to avoid establishing. The theory of games is thus dominated by the idea of a *strategy*, the establishment of chains of propositions which culminate in "wins". Ordinarily, such strategies are generated through considerations of utility. The mathematical theory of games was originally developed in a classic book: von Neumann, J. and Morgenstern, O., *Theory of Games and Economic Behavior*. Princeton University Press (1944).

It is interesting to observe that, just as the rules of the game comprise a formal system, the concept of utility which generates strategies relates game theory to control and optimal control theory; this point is taken up further in Note 4 below. Further, since the generation of “good” strategies ordinarily requires skill (i.e. intelligence) on the part of the players, it is not surprising to find a good part of the literature in Artificial Intelligence (cf. Note 17 to Sect. 3.5 above) to be concerned with game-playing. Thus, game theory is an exceedingly lively area. It also impinges directly on our present subject of anticipatory behavior, since a strategy may obviously be regarded as a model of the environment with which a player interacts.

2. See Note 18 to Sect. 3.5 above.
3. A good recent reference to Poincaré mappings, and to an approach to discrete dynamical systems based on them, is

Gumowski, I. and Mira, C., *Recurrence and Discrete Dynamical Systems*. Springer-Verlag (1980).

4. If we regard the tangent vector to a state, not as fixed by the equations of motion, but as freely selectable from some submanifold of the tangent space to each state, we find ourselves in the framework of *control theory*, rather than dynamical system theory. At the same time, such a *control system* becomes nearly analogous to the discrete logical systems which we considered initially. A choice of tangent vector (control) at each point then produces a trajectory; different choices will produce correspondingly different trajectories. If we then superimpose on this picture a notion of utility (or, in this context, of *cost*) the result is the theory of optimal control. The notion of utility can also be regarded as turning a control system into a *game*, and indeed one in which logical time and continuous time coincide. Such games are commonly called *differential games*, and the theory of differential games may be regarded as a generalization of ordinary optimal control theory. A good general reference to differential games is

Friedman, A., *Differential Games*. Wiley (1971).

The flavor of the relation between game theory and optimal control theory may be indicated by a sampling of the papers in the following:

Roxin, E. O., Liu, P. T. and Steinberg, R. L., *Differential Games and Control Theory*. Dekker (1974).

4.7 Similarity and Time

In the present chapter, we shall be concerned with exploring how the concepts of similarity and invariance, as we developed them in Examples 5 and 6 of Sect. 3.3, can be extended to dynamic and temporal situations.

The province of similarity studies, it will be recalled, is the study of those transformations of the arguments in an equation of state which leave that equation invariant. Thus, for example, in our study of the van der Waals equation (3.55),

we found that this equation was invariant to: (a) an arbitrary replacement of the constitutive parameters (a, b, r) by any other values (a', b', r'). (b) a simultaneous replacement of each state (P, V, T) by the *corresponding* state (P', V', T'), where the values (P', V', T') are given by (3.57).

This invariance of the van der Waals equation was expressed precisely by the commutativity of the diagram of mappings displayed in (3.60). Conversely, all invariance properties of an equation of state can be expressed by such a conjugacy diagram; this fact was summed up in the relations (3.60).

Now in general, the introduction of dynamical considerations into our encodings of natural systems serves to introduce new equations of state into these encodings. Initially, of course, we have the equations of motion themselves, which as we have seen, express linkages between instantaneous rates of change of observables and the instantaneous values of the observables. The integrated form of these equations of motion are likewise linkages, between the values of observables evaluated at some initial instant, and the values of these observables at a later instant. Thus we may expect that considerations of similarity will be applicable to dynamics also. The question now is: *what transformations of the arguments appearing in dynamical equations of state leave those equations invariant?*

We have already seen a very special case of this kind of question in Sect. 4.2 above. Our assertion regarding the reversibility of Hamiltonian time is precisely an assertion of similarity; the Newtonian equations of motion remain invariant when an instant t is replaced by its negative $-t$, and all other arguments remain fixed. Conversely, the non-reversibility of the other kinds of time we have discussed means that the equations of motion which define them do not remain invariant under such a replacement. The remarks to follow should be looked upon as generalizations of these simple results, and those which we have developed previously.

To fix ideas, we will begin with a study of a particular example, which we have seen before; namely, the undamped harmonic oscillator. The equations of motion of this system were presented in (1.1) above. These equations of motion are, as usual, to be regarded as linkage relations, or equations of state, relating rates of change of the state variables to the instantaneous values of these variables. The integrated form of these equations of motion can, of course, easily be written down:

$$x(t) = x(0) \cos \sqrt{\frac{k}{m}}t + \frac{p(0)}{m} \sqrt{\frac{m}{k}} \sin \sqrt{\frac{k}{m}}t \quad (4.32)$$

There is a similar equation for $p(t)$, but it can be obtained from (4.32) by differentiation. The equation of state (4.32) should be considered as a linkage between an initial state ($x(0), p(0)$), an instant t , and the corresponding value $x(t)$ of the displacement at that instant.

The equation (4.32) can be regarded as a function of five arguments:

$$\Phi(x, t, m, k, x(0), p(0)) = 0. \quad (4.33)$$

To study the similarity properties of the oscillator, we must as always find those transformations

$$\begin{cases} x \rightarrow x' \\ t \rightarrow t' \\ m \rightarrow m' \\ x(0) \rightarrow x'(0) \\ p(0) \rightarrow p'(0) \end{cases} \quad (4.34)$$

which leave (4.33) or (4.32) invariant. We shall proceed in general just as we did with the van der Waals equation. It will be recalled from Example 3 of Sect. 3.3 that we extracted a triple (a, b, r) of arguments of the equation of state; this triple consists of the constitutive parameters whose values determine the *species* of gas with which we are dealing. These parameters we regarded as capable of arbitrary variation; i.e. we placed no restriction on a transformation

$$(a, b, r) \rightarrow (a', b', r') \quad (4.35)$$

of these arguments. We then found that the only transformations of the remaining arguments of the van der Waals equation

$$(P, V, T) \rightarrow (P', V', T') \quad (4.36)$$

which left the equation itself invariant were those given by the Law of Corresponding States. We recall that this law is expressed as follows: a state (P, V, T) of a gas with structural parameters (a, b, r) is *similar*, or *corresponding*, to a state (P', V', T') of a gas with structural parameters (a', b', r') if and only if

$$P/P_C = P'/P'_C; \quad V/V_C = V'/V'_C; \quad T/T_C = T'/T'_C \quad (4.37)$$

(where the subscripts c refer to the critical point, determined by the values of the appropriate structural parameters alone).

We shall now do exactly the same with the equation of state (4.32) or (4.33), which now involves time instants t as values of one of the arguments. If we agree to allow *arbitrary* transformations of the form

$$(m, k, x(0)) \rightarrow (m', k', x'(0)), \quad (4.38)$$

we ask what are the transformations of the remaining two arguments

$$\begin{cases} x \rightarrow x' \\ t \rightarrow t' \end{cases} \quad (4.39)$$

which leave the equation of state (4.32) or (4.33) invariant. It is easy to show that these transformations are given by

$$\begin{aligned} x' &= \frac{x'(0)}{x(0)}x \\ t' &= \sqrt{\frac{km'}{k'm}}t \end{aligned} \tag{4.40}$$

From the first of the equations in (4.40), it is immediate upon differentiation that we must also have

$$p' = \frac{x(0)}{x'(0)}p. \tag{4.41}$$

The relations (4.40) and (4.41) are the exact analogs of the Law of Corresponding States for the van der Waals equation; just as in that case, we can use these relations to write the equations of state (4.32) or (4.33) in a *dimensionless* form, in which no constitutive parameters appear.

The equations (4.38), (4.40) and (4.41) express the full similarity properties of the class of one-dimensional frictionless harmonic oscillators. These relations may be summarized in the following general form: *two oscillators satisfying (4.38), (4.40) and (4.41) are in corresponding states at corresponding instants.*

Let us look more closely at the second relation in (4.40) above, which tells us which instants are corresponding instants. We notice that the time scales appropriate to the two oscillators are related by a factor depending on the masses and stiffness coefficients of the oscillators; hence any transformation (4.38) for which $m \neq m'$ or $k \neq k'$ *necessitates* a change of time scale. Conversely, any transformation (4.38) which leaves these constitutive parameters fixed, and only changes the initial displacement $x(0)$, does not affect the time scale. We also note that the scale factor connecting the two time scales involves a *radical*, with its attendant ambiguity of sign; this is simply a manifestation of the fact that a simple replacement of t by $-t$ in (4.32) or (4.33) already leaves these relations invariant. The results just obtained thus reassert in another way our conclusions about the Hamiltonian time; that it is reversible, and that different Hamiltonian systems (such as two oscillators differing by a transformation of the form (4.38)) generate different intrinsic times.

In fact, the arguments we have given above are perfectly general, and can be applied to any dynamical equations of state. Indeed, it will be recognized that the argument we have given is nothing other than the establishment of the structural stability of the class of systems satisfying the equations of state to arbitrary perturbations of the form (4.38); the transformations (4.39) and (4.40) of the remaining arguments are those which annihilate the imposed perturbation (4.38). In general, we simply find an arbitrary maximal set of arguments of an equation of state which serve to parameterize such a structurally stable family; in both of the cases we considered (namely the van der Waals equation and the harmonic oscillator) it turned out that three such quantities sufficed.

We repeat the essential temporal feature of such arguments: *if time is not used as one of the parameters subject to arbitrary perturbation, then the effect of such a perturbation is to necessitate a change in time scale if the equation of state is to be*

preserved invariant. Under such a change in time scale, *similar systems will be in corresponding states at corresponding instants.*

We will conclude this discussion of similarity in dynamical systems by considering the sense in which the two dynamical systems

$$dx_i/dt = f_i(x_1, \dots, x_n), \quad i = 1, \dots, n \quad (4.42)$$

and

$$dx_i/dt = a(x_1, \dots, x_n)f_i(x_1, \dots, x_n), \quad i = 1, \dots, n \quad (4.43)$$

where $a(x_1, \dots, x_n)$ is any arbitrary non-vanishing function of the state variables, are similar to each other. It is clear that these two systems possess the same trajectories and the same stability properties; thus the only possible difference between them can reside in the rates at which their trajectories are traversed, relative to some fixed external time scale.

Once again, it is simplest to proceed by examining first a special case. Since we have already considered the harmonic oscillator at some length, let us return to it once more for this purpose. Let us simplify still further, and consider the case in which the arbitrary function $a(x, p)$ is put equal to some constant C . Thus our question is: in what sense is the new system

$$\begin{cases} \frac{dx}{dt} = \frac{Cp}{m} \\ \frac{dp}{dt} = -Ckx \end{cases} \quad (4.44)$$

similar to the original oscillator (for which $C = 1$)?

We observe that the transition from the original oscillator to (4.44) is obtained by making the transformation

$$(m, k) \rightarrow (m/C, kC). \quad (4.45)$$

We now look at our general similarity rules (4.38), (4.40), (4.41) for the harmonic oscillator; we find that the state variables (x, p) are unaffected by this transformation, but we must make a *time* transformation $t \rightarrow t'$; specifically, we must have

$$t' = Ct. \quad (4.46)$$

Now we observe that exactly the same argument holds if C is not constant, but is any non-vanishing function of state, $C = a(x, p)$. Thus, even in this general case, we find that

$$t' = a(x, p)t$$

which is indeed just what we would expect from our general considerations regarding dynamical time. We see then that the relation between a general dynamical

system (4.42) and its derived systems (4.43) is precisely one of similarity, and in particular, is entirely concerned with establishing a correspondence between their intrinsic time scales which leaves the equation of state of (4.42) invariant.

4.8 Time and Age

In the preceding chapters, we have abundantly seen that the quality we perceive as time is complex. It admits a multitude of different kinds of encoding, which differ vastly from one another. Thus, we have considered reversible Hamiltonian time; irreversible dynamical time; thermodynamic time; probabilistic time; and sequential or logical time. Each of these capture some particular aspects of our time sense, at least as these aspects are manifested in particular kinds of situations. While we saw that certain formal relations could be established between these various kinds of time, none of them could be reduced to any of the others; nor does there appear to exist any more comprehensive encoding of time to which all of the kinds we have discussed can be reduced. This, as we shall see later, is indeed the essence of complexity.

Moreover, we saw that within each class of temporal encoding, many distinct kinds of time were possible. Thus, there are many kinds of Hamiltonian time; many kinds of dynamical time, etc. Each such time was intrinsically generated by the particular kind of dynamics in which it arose. As we indicated, these times cannot strictly be reduced one to the other, but they can be *compared*, by means of more or less complicated *scale factors*. As we saw in the preceding chapter, this is most readily accomplished within a class of similar systems, all of which obey a common equation of state in which the appropriate intrinsic time enters as one of the arguments.

In the present chapter, we wish to illustrate further the complexity of temporal encodings by considering the distinction between *time* (or duration) and *age*. We shall show that these two concepts are not co-extensive; in the process of establishing the differences between them, we shall be able to gain additional insights into the nature of temporal encodings of dynamical processes.

Let us begin with consideration of a simple example. Consider a first-order decay process, such as the decay of a radioactive material. Under conventional hypotheses, a natural system of this kind can be encoded formally as a single first-order differential equation of the form

$$dx/dt = -\lambda x. \quad (4.47)$$

Here the variable x denotes the amount of radioactive material present at some instant t of clock time; the coefficient λ is a constitutive parameter, representing the (constant) fraction of material which decays in a single unit of clock time.

As usual, the equation (4.47) may be interpreted as a linkage relation, or equation of state, relating the rate of decay at an instant to the amount of radioactive material

present at that instant. By an integration process, we can convert this linkage to another one, of the form

$$x(t) = x(0)e^{-\lambda t} \quad (4.48)$$

which is of the general form

$$\Phi(x, x(0), \lambda, t) = 0 \quad (4.49)$$

analogous to (4.33) above. We stress explicitly that the linkage (4.48) or (4.49) relates the amount $x(0)$ of material initially present to that present at any *subsequent* instant of clock time.

We shall now treat (4.48) or (4.49) according to the general considerations of similarity developed in the preceding chapter. In particular, we shall ask the following question: if we make an *arbitrary* transformation

$$\begin{cases} \lambda \rightarrow \lambda', \\ x(0) \rightarrow x'(0), \end{cases} \quad (4.50)$$

what transformations must we make of the remaining arguments in order to keep the equation of state (4.48) invariant?

It is easy to see that the required transformations are:

$$\begin{cases} x \rightarrow x' = \frac{x'(0)}{x(0)}x, \\ t \rightarrow t' = \frac{\lambda'}{\lambda}t. \end{cases} \quad (4.51)$$

If as before we say that two states x, x' are *corresponding* if they satisfy the first relation in (4.51) above, and two instants t, t' are *corresponding* if they satisfy the second relation, we once again see that the equation of state (4.49) defines a two-parameter similarity class (parameterized by the pair $(\lambda, x(0))$), and the transformation rules (4.50), (4.51) express the Law of Corresponding States for this class; any two systems in the class are in corresponding states at corresponding instants if these rules are satisfied.

Indeed, if we introduce new (dimensionless) variables by writing

$$\xi = \frac{x}{x(0)}, \quad \tau = \lambda t \quad (4.52)$$

then the *entire class* can be represented by a single dimensionless equation of state of the form

$$\frac{d\xi}{d\tau} = \xi \quad (4.53)$$

in which no arbitrary parameters are visible; as usual, these parameters have been absorbed into the scales into which the remaining quantities are measured.

This last assertion means the following with respect to the time variable. If we measure time in units proportional to the rate of decay (e.g., in *half-lives*), then all systems obeying (4.48) *decay at the same rate*. On the other hand, if we persist in using a common clock time, simultaneously imposed on all the systems obeying (4.48), then different systems will necessarily appear to decay at different rates. If the constitutive parameters λ , λ' are quite different for two such systems, one of them will thus appear to “*survive*” *for a greater duration*; but the systems will both survive for exactly the same number of half-lives. What we now wish to assert is the following: *duration as measured in clock time is not an appropriate measure of age*; the appropriate measure for age in this case is the dimensionless quantity τ defined in (4.52). The Law of Corresponding States in this case thus asserts that *two systems satisfying (4.48) are of the same age if and only if they are in corresponding states at any instant in the dimensionless quantities*.

We are now going to assert that the distinction between elapsed time (or duration) according to some fixed scale, and any natural concept of *age*, possesses the qualities illustrated by the above simple example. That is, age is a property shared by certain states belonging to a *class* of similar dynamical systems. Two different systems in such a class (i.e. characterized by different values of constitutive parameters and/or initial conditions) are of the same age if and only if they are related by a similarity transformation; i.e. they are in corresponding states at corresponding instants of clock time.

The relation between elapsed time, age and similarity is a crucial one for many purposes in biology. Thus, there is an obvious sense in which a two-year-old human is “younger” than a two-year-old rat, even though the absolute durations (measured in clock time) through which their respective lives have extended are the same. However, to explicitly write down the sense in which such an interspecific comparison is meaningful is a far from trivial matter. What we propose here is the following: that to the extent that two natural systems can be compared in terms of age, they must be encoded into a common similarity class of dynamical systems. The difference between systems within the class is represented by a difference in the specific values attached to particular constitutive parameters and/or initial conditions. The ages of the two systems, at any instant of clock time, are then expressed in terms of those transformations of state and time which preserve the equation of state governing the class invariant. Thus, to the extent that it is meaningful at all to compare the ages of a rat and a human, we might say that a two-year-old human is the same *age* as a two-week-old rat, expressed in terms of duration. Age, in other words, is seen to be a reflection of a more comprehensive similarity relation obtaining within a class of similar dynamical systems; this similarity relation involves not only elapsed time (duration), but also the instantaneous states of the systems involved.

Thus we can see that, insofar as *age* represents a quality pertaining to the states of a natural system, it is related only in a complex way to the encoding of other qualities, and to the encoding of time.

The above considerations reflect only one of the various ways in which the concepts of time and age can be juxtaposed. Other such juxtapositions arise, for

example, from attempts to relate the kind of discrete, logical time described in Sect. 4.6 above to a continuous dynamical time. For instance, in discussions of “clonal aging” in cell cultures, the “age” of a culture is expressed in discrete time, as the number of *doublings* undergone by the culture.¹ Normal cells under these conditions exhibit a definite limit (the Hayflick Limit) to the number of doublings they can exhibit, which is then identified with their *life span*. This *logical* measure of age is unrelated to any concept of *duration*, as measured in some external continuous clock time. Thus, for example, a culture may be indefinitely arrested at any level of doubling (e.g. by immersing it in liquid nitrogen); when restored to normal temperature conditions, the cells proceed to complete precisely the expected number of doublings required to reach the Hayflick Limit, as determined by their logical age when arrested. Thus, the measure of age in terms of discrete doublings is *rate independent*, in the same sense as a sequence of moves in (say) a game of chess is rate-independent. This situation represents another sense, distinct from those subsumed under similarity, in which the concepts of duration and age are seen to be independent.

Aside from the examples described above, many other situations in which the notions of elapsed time and the quality of age are not coextensive can be imagined. For instance, we may express the age of a system in terms of a probabilistic time (cf. Sect. 4.4 above), which is again distinct from a continuous Hamiltonian or dynamical time. All of these considerations point once again to the complexity associated with temporal qualities and their various encodings, and to the fact that there is no apparent way to describe temporal qualities in a single comprehensive fashion.

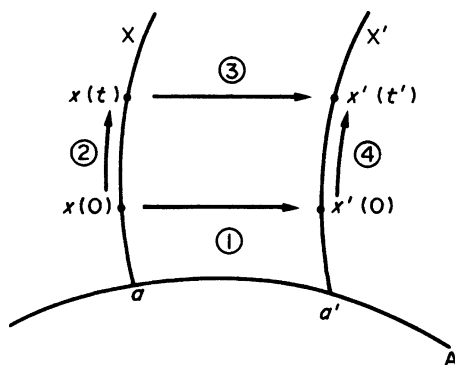
We shall conclude this section with some further remarks on similarity, time and age. Namely, we shall use the illustrations developed in the last several chapters as examples of *modeling relations* between natural systems, as these were defined in Chaps. 4.2 and 4.3 above. To do this, it is most convenient to refer to a refined version of Fig. 2.2 above; namely, to the diagram

Here, as before, we let A represent an appropriate space of constitutive parameters, which identify individual systems within a similarity class. To each point a in A , we associate a copy of a common state space X . In each of these state spaces, a specific dynamics is imposed, such that any initial state $x(0)$ is carried to a uniquely determined state $x(t)$ after some time interval t has elapsed. In the figure, two such points a, a' are indicated.

As we have seen, the linkages imposed on each state space X by the dynamics determines how the state $x(0)$ at some initial instant is linked to the state $x(t)$ at a future instant. Thus, *the dynamics determines how a present state can be regarded as a model of a future state of the same system*. This relation between present state and future state is represented in the figure by the arrows labelled (2) and (4).

On the other hand, the similarity relation imposed on the entire class indicates how each specific system in the class can be regarded as a model of any other. In particular, we have seen that the dynamical form of the similarity relations determine correspondences between (a) the states of systems belonging to different parameter values a, a' ; (b) the time scales used to parameterize the dynamics in

Fig. 4.2



these different systems. Thus, given any state of a given system in the class, there is a *corresponding* state in every other system; moreover, the time scales between the systems are so related that *any pair of systems are found in corresponding states at corresponding instants*.

These relations are represented in the figure by the arrows labelled (1) and (3). According to these relations, each state of a particular system in the class is also a model of all of its corresponding states; i.e. is a model of state belonging to different systems.

By putting these two different kinds of modeling relations together, we can employ dynamical properties of *any* system in our class to obtain information about *every other system*. For instance, by symbolically composing the arrows (2) and (3) in Fig. 4.2, we can see explicitly how the initial state $x(0)$ of one system can be regarded as a model of the state $x'(t')$ of a different system in the class, at a different instant. Hence, it follows that the combination of dynamics with similarity provides an exceedingly powerful inferential tool for the study of such classes of systems.

References and Notes

1. The term “doubling” is used here as a kind of average replicative potential for a species of differential cell type in culture. More technically, a few cells are initially plated on a culture dish; they will multiply until they cover the surface of the available medium in a monolayer (“normal” cells will only grow in a single layer on the medium, and they will stop multiplying when no more free surface is available to them, through a poorly understood mechanism called contact inhibition). At this point, some of the cells are replated on a fresh surface, and are once again grown to confluency. The major discovery of Leonard Hayflick is that this process cannot be continued indefinitely; normal cells can only be propagated for a relatively small number of such “doublings”, depending on the species from which the cells are taken, the particular cell type involved, and

the nature of the donor. This Hayflick Limit varies (roughly) directly with the longevity of the donor species, and inversely with the age of the individual donor; thus the Hayflick phenomenon is often called “clonal aging”, and plays a major role in the search for relatively simple models for whole-organism senescence. For a more detailed review, and further references, see for instance Rosen, R., *Int. Rev. Cyt.* 54, 161–191 (1978).

Chapter 5

Open Systems and the Modeling Relation

5.1 General Introduction

In the previous sections of this volume, we have developed in great detail the properties of the modeling relation, illustrated with a profusion of examples drawn from many scientific disciplines. The preceding section was specifically devoted to a consideration of dynamical modeling. Dynamical models are henceforward going to be our central concern, because they are the vehicles through which temporal predictions are made. Our most immediate item of business, to be taken up in the present section, is to combine our general insights regarding modeling relations with the specific properties of dynamical systems. The aim will be to obtain an integrated theory of dynamical models, which will play the central role in our treatment of anticipatory systems.

Let us begin by reviewing some of the salient features of the modeling relation, as we have previously developed them. We have seen that a model is essentially an encoding of qualities, or observables, of a natural system into formal mathematical objects. Just as the system qualities are interrelated in the original system, so these mathematical objects are related through corresponding linkage relations, or equations of state. It is then precisely the correspondence between the inferential structure of the model and the interrelation of qualities in the original natural system which allows us to utilize models for predictive purposes.

We have also seen that a model is by its very nature an abstraction, in the sense that any encoding must necessarily ignore or neglect qualities which are present in the original natural system. To that extent, a model represents a *subsystem* of the original system, rather than the system itself; much of our previous work was concerned with determining how, and whether, relationships between such subsystems could be established. In any case, the fact that a model is an abstraction means that the predictive capabilities of the model are limited to circumstances in which those interactive qualities of the system not present in the model are not

significant. Stated another way: the model behaves as the real system would behave if it possessed *only* those interactive capabilities which are explicitly encoded into the model.

Thus the construction of a model intrinsically presumes an absence of linkage between the behaviors we wish to study and those qualities not encoded into the model itself. Our main present concern will be to study the effects of such a presumption, in the special case of dynamical models. We shall approach this study by bringing it within the purview of the general theory we have already developed. Specifically, we have already seen how the properties of distinct encodings of the same system may be compared. This comparison tells us, of course, how the subsystems represented by the models are related to each other in the natural system itself. We shall choose the two encodings to be compared as follows: one of them will be a dynamical model we wish to study, and the other will comprise that model together with some further encoded interactive capabilities. The comparison between the two will tell us the extent to which predictions made from the original model can remain valid in the presence of uncoded qualities.

This kind of question is of great importance within the sciences themselves. Let us take a specific example; that of particle mechanics, which we have considered at great length above. The most important systems of classical mechanics are frank idealizations, in that they deliberately neglect or ignore qualities known to be present in the world. Thus, the most important and deeply studied mechanical systems are the conservative ones, in which the forces acting on the system are derived from time-independent potentials, and in which total energy is conserved. Such an idealization necessarily neglects all dissipative or frictional aspects, which we know to be essential and irreducible features of real natural systems. And yet, the dynamical predictions of conservative particle mechanics have played perhaps the central rôle in our attempts to understand the properties of real systems.

Thus, the kind of question with which we shall be involved concerns, as a special case, the relation between dissipative and non-dissipative (conservative) systems. Indeed, this fact will dictate the nature of the general approach we shall take to the relation between a model (or abstraction) and the system being modeled. For a conservative system is one in which we have neglected all dissipative or frictional interactions between the system and its environment. It is a system which is, in an important sense, *closed* to such interactions. The distinction between a conservative system, such as an undamped oscillator, and its dissipative counterparts is thus essentially a distinction between a closed system and an open one; between a system uncoupled to its environment, and the same system in direct interaction with a definite modality of that environment.

Thus, speaking very roughly, we are going to develop a kind of proportionality relation, of the form:

$$\frac{\text{model}}{\text{system}} = \frac{\text{closed system}}{\text{open system}}.$$

We shall begin by studying the right-hand side of this proportionality in a dynamical context, by comparing the dynamical properties of a system closed to environmental interactions with the corresponding properties of the *same* system when open to such

interactions. The principal result in this connection, to be established in Sect. 5.4 below, is the following: *the discrepancy between a dynamical behavior in a closed system and the corresponding behavior in an open system is an increasing function of time*. The sense in which “time” is to be understood in the above assertion will be made clear in the course of the argument itself.

We shall then see how this result can be interpreted as embodying a proposition relating the predictions of a dynamical model to the actual behavior of the system being modeled. In this context, the result will become: *the discrepancy between the dynamical behavior predicted from a model, and the actual behavior of the system being modeled, is an increasing function of time*. This is in fact the main conclusion of the present section. Because of its profound significance for many parts of system theory, and the crucial role it plays in understanding the properties of anticipatory systems, we shall examine it from a number of different points of view. For instance, we shall argue that the concept of *error*, as a presumed quality of system behavior, arises precisely from the discrepancy between the actual behavior of a natural system and the behavior *predicted* on the basis of some dynamical model of the system. This in turn will provide us a basis on which to rigorously discuss the relation between error and system complexity. In subsequent sections we shall draw important corollaries pertaining to the reliability of anticipatory systems, and their physiological, behavioral and evolutionary implications.

We shall also stress the profound distinction between this result and the Second Law of Thermodynamics. It will be shown that the increasing discrepancy between corresponding behaviors in closed and open systems has nothing whatever to do with entropy and disorder, or with approach to equilibrium in isolated systems, or with any considerations of probability or stochastics. Yet it embodies in a rigorous form the same kind of result for which the Second Law is often (wrongly) invoked, and hence it is essential to be clear about the distinction between them.

5.2 Open, Closed and Compensated Systems

In many branches of science, such as thermodynamics, we can classify the systems studied therein into a number of distinct classes, depending upon the way they interact with the environment. Thus, in thermodynamics, we can recognize a class of systems we call *isolated*, which exchange neither heat nor matter nor any other form of energy with their environment; much of the classical theory of thermodynamics is restricted to this class of isolated systems. Likewise, a system is called *closed* if it can exchange heat, but not matter, with its environment. A system which can exchange both energy and matter with its environment is generally called an *open system*.

Let us note the two crucial features of such a classification. In the first place, *we require above all an absolute distinction between system and environment*; between what is “inside” and what is “outside”. Indeed, without such a distinction,

the categories of closed, isolated, open, etc., are meaningless. Secondly, having made such a distinction, *we need to stipulate the kinds of linkage relations we allow between qualities which are internal to the system, and qualities which are assigned to the environment*. Thus, in an isolated thermodynamic system, the system qualities are presumed *unlinked* to any environmental quality; in a closed thermodynamic system, thermodynamic qualities of the system are linked to environmental temperature, but to no other environmental qualities.

Thus, the first essential order of business is to examine the following question: to what extent can we obtain an *intrinsic* characterization of what is system and what is environment? Very roughly, we seek a way of determining, from a consideration of the system qualities themselves, whether there is anything outside the system. For, if we cannot infer by intrinsic means that any environmental qualities actually exist, this is equivalent to saying that the system is in fact isolated (in the thermodynamic sense). It is clear that if such a distinction between system and environment can be meaningfully made at all, it must be represented in terms of the linkage relations which characterize the system.

In most branches of system theory, the distinction between system and environment is simply presumed to be given *a priori*, and not further discussed. However, there is one field in which there is indeed an absolute distinction between what is system and what is environment; that field, not surprisingly, is classical Newtonian mechanics. Accordingly, let us consider how an intrinsic system-environment distinction arises in mechanics, and then proceed to see how we may formulate this distinction in more general situations.

The simplest possible mechanical system consists of a single particle. As usual, we suppose that this particle can be characterized at any instant by its displacement x from some reference position, and by its velocity v . We shall also suppose the particle endowed with a single constitutive parameter; its (inertial) mass m . Hence at any instant the particle possesses a momentum $p = mv$.

The situation in which such a system is totally uncoupled or unlinked to any environmental quality is explicitly expressed in Newton's *First Law*: such a particle is either at rest, or in uniform motion in a straight line with constant velocity.¹

Thus, if a particle is *not* moving in a straight line, and/or if its velocity is not constant, then it is *by definition* experiencing some interaction with its environment. This is expressed by saying that there is a *force* (or more precisely, an external force) acting on it. The extent of the deviation of the particle from uniform motion can in fact be taken as a *measure* of the external force. This deviation is characterized by the instantaneous value of the particle's *acceleration* a ; thus the substance of Newton's *Second Law* is to identify the environmentally imposed force with the acceleration, weighted by the particle's inertial mass: $F = ma$.

Particle mechanics thus begins by explicitly characterizing what a non-interacting system is; namely, it is a system in which acceleration is absent. This characterization is intrinsic, and in Newtonian mechanics it has an absolute character.

Let us now consider a slightly more complex mechanical system; one consisting of several particles. We suppose that these particles can exert forces on each other,

but such forces are entirely *internal* to the system itself. How can we determine whether such a system is also interacting with its environment; i.e. whether there are *external* forces imposed upon it?

Here again, classical mechanics provides an absolute, intrinsic prescription. We simply write down all the forces acting on each particle of the system, including both those arising from the other particles, and those arising externally. We now invoke Newton's *Third Law*, which states that if any particle of the system exerts a force on a second particle of the system, then the second exerts an equal and opposite force on the first. Thus, when we add up all the forces imposed on all the particles, *the internal forces will sum to zero*. Consequently, if the sum of all the forces on all the particles of the system is identically zero, there are by definition no external forces acting on the system; the system is isolated. Conversely, if the system is isolated, the sum of all the forces will vanish identically. It is easy to show that the diagnostic that our system be isolated from external forces is that its center of mass should not accelerate.

Thus, the Three Laws of Newton can be precisely regarded as specifying when a mechanical system is isolated. As we noted earlier, an isolated system is one for which we cannot infer the existence of anything outside the system by looking at the system itself. These isolated systems thus provide the fundamental standards by which environmental interactions (i.e. impressed external forces) are recognized. It is precisely the deviation of a system from the behavior of an isolated system which allows us to infer that something exists outside the system, and which we can in fact take as the *measure* of that external quality. This situation is indeed analogous to that of a *meter*, which cannot provide meaningful measurements until a zero-point is established through isolating it initially from the quality it is intended to measure. Here, the behavior of a mechanical system unacted upon by external forces is the zero-point; that behavior is precisely what is characterized by Newton's Laws. This is in fact an important general point, which should be kept in mind as we proceed: a system which is open can be regarded as a meter for those environmental qualities with which it is in interaction, when its behavior is referred to that manifested in the *absence* of such an interaction.

Let us now recast these remarks in terms of explicit linkage relations. Suppose we have a system of N particles, with displacements $x_1, \dots, x_N$, and velocities $v_1, \dots, v_N$. Let us suppose further that the only forces acting on any particle are those arising from other particles, and that Newton's Third Law is satisfied. We shall now write down the equations of motion of the system.

According to the Second Law, the equation of motion of the k th particle in the system is

$$m_k \frac{d^2 x_k}{dt^2} = \sum_{i=1}^N F_{ik} \quad (5.1)$$

where m_k is the inertial mass of the k th particle, and F_{ik} is the force imposed on that particle by the i th particle of the system. We suppose that the value of this force F_{ik} depends only on the distance between the i th and k th particles at an instant. We

allow the differential of time in this case to be arbitrary; it will “cancel out” in the considerations to follow. Indeed, if we multiply both sides of the equation (5.1) by the velocity dx_k/dt , we may rewrite the equation as

$$m_k v_k dv_k = \left(\sum_{i=1}^n F_{ik} \right) dx_k. \quad (5.2)$$

There is a kind of linkage which we have seen extensively in the preceding section; specifically, it links a differential velocity increment dv_k with a differential displacement increment dx_k , when the system is in some state $(x_1, \dots, x_N, v_1, \dots, v_n)$. We have such a linkage relation (5.2) for each particle of the system; i.e. for each $k = 1, \dots, N$.

The family of linkage relations (5.2) provide a local *accounting*, which tells us how an increment of displacement is to be converted into an increment of velocity at any state of our system. We have seen earlier that such linkages arise as simple expressions of causality; they must exist if a given initial state is to uniquely determine the state after a differential time increment has elapsed. The specific *form* of the linkages is equivalent to the equations of motion (for a suitable definition of time increment, as we saw abundantly in Sect. 4.3 above), and in the present case is determined entirely by the forces imposed on the system.

Thus, if our mechanical system is isolated from external forces, the local accounting by which differential displacement increments are converted to differential velocity increments at any state must be given by the linkages (5.2), in which $F_{ik} = -F_{ki}$ by Newton’s Third Law. If we think of these differential increments dv_k , dx_k as representing *flows* of the state variables, then this accounting tells us how any flow dx_k is compensated by a corresponding flow dv_k in any state, and conversely. Thus, *according to these linkages (5.2), every flow in the system is compensated; there are no uncompensated flows.*

What happens if we now impose some external force on our system? Obviously, the original equations of motion (5.1) must now be replaced by new equations of the form

$$m_k \frac{d^2 x_k}{dt^2} = F_k + \sum_{i=1}^N F_{ik} \quad (5.3)$$

(where F_k denotes the external force acting on the k th particle, and is assumed to be a function of displacements alone). Likewise, the original accounting rules (5.2) are no longer valid; they must be replaced by rules of the form

$$m_k v_k dv_k = \left(F_k + \sum_{i=1}^N F_{ik} \right) dx_k \quad (5.4)$$

If we now compare (5.4) with (5.2), we see the effect of imposing the external force F_k : *when such a force is present, the original accounting rules (5.2) give rise*

to an uncompensated flow. In particular, we find discrepancies between the actual increments dx_k arising from a given increment dv_k in a state, and those which our accounting (5.2) tells us compensate for that increment. It is in this sense that the imposition of an external force appears to generate uncompensated flows.

Thus it is that the imposition of a new interaction between the system and its environment (i.e. the imposition of an external force) causes our initial accounting rules to break down. As we see, the diagnostic of this breakdown is the appearance of uncompensated flows; differential increments of state variables which cannot be “paid for” by decrements elsewhere internal to the system. On this basis, *the appearance of such an uncompensated flow implies the existence of something outside the system*, which acts as a source or sink for the uncompensated flows. The effect of passing from (5.2) to (5.4) is to “internalize” such sources and sinks, by establishing a *new accounting* in which all the flows are again compensated.

Let us note two features of this situation, before we reformulate its basic features in general dynamical terms. The first point is the following: when we impose an external force on a system of particles, the expression of that force as a conversion factor between a differential increment of displacement and a differential increment of velocity typically requires us to introduce new quantities into the equations of motion, which play the role of new constitutive parameters. Thus for instance, if we convert a single free particle into a harmonic oscillator by introducing an external force of the form $F = -kx$, the parameter k expresses precisely how a unit increment dx of displacement in a state (x, v) is to be converted to an increment of velocity; the analog of (5.4) for this system is

$$kx \, dx + mv \, dv = 0. \quad (5.5)$$

This quantity k pertains to the *external* force imposed on a free particle. But when the quantity is “internalized” to give the new accounting relation (5.5), in which all flows are now compensated, we can regard k as an *internal* constitutive parameter of our new system, playing the same kind of role as does the inertial mass m . Thus, in a sense, the number of constitutive parameters appearing in accounting relations like (5.5) (and hence in the equations of motion to which these relations give rise) indicate how many external forces have been internalized in passing from an absolutely free (i.e. non-interacting or isolated) system to the one under consideration. We shall return to this point in a moment.

The second feature we wish to discuss is the relation between a particular kind of accounting, like (5.2) or (5.4), in which every flow is compensated, and the formulation of global conservation laws. Intuitively speaking, if there is no interaction between a system and its environment, then there can in particular be no flow or exchange of qualities between them. Specifically, there can be no sources or sinks for system variables in the external environment. Thus, we would expect some kind of conservation law to be applicable to the system, which would take the form of a global linkage relation involving those system qualities which are

sources or sinks for each other. The differential accounting relation (5.2) or (5.4), in which all flows are compensated, provide in fact a local expression of this kind of expectation.

The passage from local accounting relations to a global conserved quantity is formally equivalent to the integration of a differential form. In general, a local accounting like (5.2) or (5.4) can be expressed as

$$\sum_{i=1}^N M_i dx_i + \sum_{i=1}^N N_i dv_i = 0 \quad (5.6)$$

where the M_i , N_i are functions of the state to which the differential increments dx_i , dv_i are referred. If this differential form (5.6) is exact, then there is a single function $K = K(x_1, \dots, v_n)$ such that

$$dK = \sum_{i=1}^N M_i dx_i + \sum_{i=1}^N N_i dv_i; \quad (5.7)$$

i.e. such that

$$M_i = \frac{\partial K}{\partial x_i}; \quad N_i = \frac{\partial K}{\partial v_i}$$

In that case, it follows that this observable K is a conserved quantity in a global sense, and we may say that there is no “flow” of this observable between the system and its environment. In mechanics, of course, such an observable is generally available; it is the Hamiltonian or total energy of the system. To say that energy is conserved in a system like (5.3) means precisely that the effects of all forces have been internalized into the accounting relations (5.4); i.e. that there are no uncompensated flows in the local accounting. However, not every local accounting can give rise to a global conserved quantity; even if all flows are locally compensated, there need be no observable of the system which is conserved. This is the case when the system is “dissipative”; we shall return to this situation in more detail subsequently.

Let us now return to the main line of our argument, and consider the extent to which the concepts we have introduced can be formulated in general terms. The heart of the argument was to compare the properties of a completely isolated system with its properties when allowed to interact with its environment; i.e. when an externally arising force is imposed on it. The non-interacting situation is characterized by the presence of local accounting relations, in which every flow is compensated. Relative to this accounting relation, the imposition of an external force creates non-compensated flows. The nature of these non-compensated flows then leads to a characterization of the external force responsible for them, and thereby to a new accounting relation, describing the properties of the interaction, with respect to which all flows are again compensated.

The crux of this situation will be seen to rest upon the following data:

1. A system in which all flows are compensated, to serve as a reference;
2. The same system, which is now placed in interaction with its environment in a particular way;
3. The generation of uncompensated flows, as measured against the reference system.

In our previous discussion, we have taken as reference system one which is initially totally isolated. But we can now see that this restriction is quite unnecessary; we may take as our initial reference *any system in which all flows are compensated*. In some absolute sense, a system of this kind may generally be regarded to be interacting with its environment, but the assumption of compensation means that these interactions have all been internalized. Hence, such a system can be regarded as isolated from any other environmental interaction, aside from the internalized ones. It can thus be utilized as a reference system, against which we may characterize the effects of new environmental interactions. We need to have a name for such a reference system, which need not be isolated in an absolute sense, but can be regarded as isolated from the standpoint of a particular modality of environmental interaction. In line with the above arguments, we propose to call such a system a *compensated system*. A compensated system is thus one for which local accounting rules are given; these rules embody the internalization of all sources and sinks for flows of the state variables. In thermodynamic terms, a compensated system could only be described as “partially open”. The environment of such a system comprises every quality which has not been internalized, and by definition, the system is totally isolated from that environment.

The study of such compensated systems, and the relations which can exist between them, will be the main object of the remainder of the present section.

References and Notes

1. This kind of discussion always presupposes a single (fixed) observer. Otherwise, relativistic considerations immediately come into play; e.g. an accelerated observer will immediately impute a force (i.e. a particle-environment interaction) to account for the motion he observes.

5.3 Compensation and Decompensation

In accord with the ideas developed in the preceding chapter, let us suppose we are given a compensated system S° , which will play the role of a reference system. As we saw, S° need not be isolated from its environment in any absolute sense; but the hypothesis that it is compensated means precisely that those interactions which it undergoes with its environment have already been internalized into its local accounting relations.

Let us express these ideas formally. Suppose that the states of S° can be represented by means of the values assumed by a set $\{x_1, \dots, x_n\}$ of state variables, and that the states so represented form a manifold in Euclidean n -dimensional space. Then the hypothesis that S° is compensated means that there is a linkage relation of the form

$$\sum_{i=1}^n M_i dx_i = 0 \quad (5.8)$$

which relates the increments dx_i attached to a state to the state itself. The coefficients M_i are, of course, functions of state $(x_1, \dots, x_n)$. However, they will in general also contain constitutive parameters $\alpha_1, \dots, \alpha_r$ which arise from the internalization of whatever environmental interactions to which S° is exposed. Thus, we may write $M_i = M_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_r)$.

Now let us suppose that an additional external force is imposed on S° ; i.e. a new interaction modality is established between S° and its environment. As we have seen in the preceding chapter, the effect of this force will be to produce flows in the increments dx_i which are uncompensated from the standpoint of the accounting (5.8). Let us suppose that the uncompensated portion of the i th flow can be expressed in the form

$$N_i dx_i \quad (5.9)$$

where each N_i is a function of state, and in general also involves some new constitutive parameters $\alpha_{r+1}, \dots, \alpha_{r+s}$. The existence of these functions N_i , in terms of which the uncompensated portion of the flows can be expressed, is a consequence of the general hypothesis of causality which we have employed repeatedly before.

We can now *internalize* these uncompensated flows (5.9), by combining them with (5.8). Namely, each total flow in the new situation can be regarded as a sum of two terms: (a) a term arising from the original accounting (5.8), and (b) a term comprising exactly that part of the flow which is uncompensated according to that accounting. Thus, the linkage relation appropriate to describe the effect of the new force imposed on S° is no longer given by (5.8), but must be written as

$$\sum_{i=1}^n (M_i + N_i) dx_i = 0. \quad (5.10)$$

The coefficients of the differential form (5.10) are functions of the state variables, the original set of constitutive parameters $\alpha_1, \dots, \alpha_r$ describing the system when isolated from the new external force, and the additional constitutive parameters $\alpha_{r+1}, \dots, \alpha_{r+s}$ which enter into the expression (5.9) of the uncompensated flows arising from that new force. The new accounting relations (5.10) thus describe a new system S ; S is itself a compensated system, but it clearly has quite different properties from the isolated system S° from which it arose. However, let us note explicitly that (5.10) is a *generalization of (5.8)*; it *reduces exactly to (5.8)* when $N_i = 0$ for each i ; i.e. when S is again isolated from the new external force.

It is of great importance that this entire argument can be run backwards. Namely, suppose that S is now a compensated system, whose local accounting is expressed by the linkage relation

$$\sum_{i=1}^n M_i dx_i = 0 \quad (5.11)$$

where $M_i = M_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_r)$. Let us now suppose that we isolate this system from one of the environmental forces imposed upon it, and suppose that this isolation is equivalent to placing some subset of the constitutive parameters equal to zero. In this way, we obtain a new compensated system S° , describable by a linkage of the form

$$\sum_{i=1}^n M_i^\circ dx_i = 0. \quad (5.12)$$

where the M_i° are obtained from the corresponding functions M_i by setting the appropriate constitutive parameters equal to zero. Then S and S° are related as in the previous discussion; from the standpoint of S° , S appears to exhibit uncompensated flows given precisely by

$$(M_i - M_i^\circ) dx_i, \quad i = 1, \dots, n.$$

These relations will be of great importance to us when we come to compare the dynamical behaviors arising in two such systems S and S° .

Finally, let us see explicitly what happens to conserved quantities when we pass between systems related like S and S° . Thus, let us return to the expression (5.8) above, which we will suppose is a perfect differential; i.e.

$$\sum_{i=1}^n M_i dx_i = dH^\circ = 0$$

where H° is some observable of the form $H^\circ(x_1, \dots, x_n, \alpha_1, \dots, \alpha_r)$. Then we know that H° is conserved; there is no flow of H° between the system S° and its environment.

If S° is now exposed to a new external force, so that the new linkage between states and their differential increments is given by (5.10) rather than (5.8), we can still write (5.10) in the form

$$dH^\circ + \sum_{i=1}^n N_i dx_i = 0. \quad (5.13)$$

By definition, the differential form appearing in (5.13) cannot vanish; i.e.

$$\sum_{i=1}^n N_i dx_i \neq 0.$$

Thus, H° cannot remain conserved in the larger system S . However, it can happen that we can write

$$\sum_{i=1}^n N_i dx_i = dH$$

for some observable $H = H(x_1, \dots, x_n, \alpha_{r+1}, \dots, \alpha_{r+s})$, which depends on the original state variables and some new constitutive parameters. In that case, it is clear that the function $K = H^\circ + H$ is conserved on S , although neither H° nor H is separately conserved. In the special case in which H° is a Hamiltonian for S° , and the N_i are mechanical forces arising from a potential, so that

$$N_i = -\frac{\partial U}{\partial x_i}$$

for some potential function U , then H is the additional energy arising from the applied force, and $K = H^\circ + H$ is the Hamiltonian for the larger system S .

Just as before, this argument can be run backwards. Thus, if K is conserved on the larger system S , and we obtain S° from S by setting the new constitutive parameters equal to zero, there will be a conserved quantity H° on S° if and only if the resulting differential form (5.8) is exact.

5.4 The Main Theorem

We are now ready to develop some crucial results which relate the temporal behavior of a system which is closed to some modality of environmental interaction to that of the same system when open to that interaction.

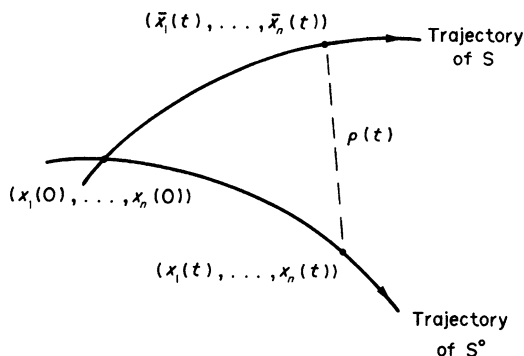
Let us begin by restating our terminology. We shall start with a given system S° ; this will be our closed system, or reference. We assume that it is encoded into a family $x_1, \dots, x_n$ of state variables, and that the totality of states thus encoded fill a state space which is a manifold $X \subseteq E_n$. We suppose also that the system is further determined by a family $\alpha_1, \dots, \alpha_r$ of constitutive parameters, which remain unaffected by any change of state which occurs in the system. Finally, we suppose that the temporal behavior of S° is governed by a family of equations of motion, of the form

$$dx_i/dt = f_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_r) \quad (5.14)$$

for $i = 1, \dots, n$. These equations of motion are to be interpreted according to the discussion given in Sects. 5.2 and 5.3 above.

We shall be comparing the behaviors of S° with those of a new system S , which intuitively is obtained by allowing S° to be open to a new modality of environmental interaction. More specifically, we shall assume: (a) that the states of S are encoded into exactly the same state space X into which the states of S° were encoded; (b) that

Fig. 5.1



in addition to the original structural parameters $\alpha_1, \dots, \alpha_r$ of S° , we must adjoin an additional number $\alpha_{r+1}, \dots, \alpha_{r+s}$ of new parameters; (c) that the equations of motion of S can be represented in the form

$$dx_i/dt = g_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_{r+s}) \quad (5.15)$$

for $i = 1, \dots, n$. Finally, we shall express the idea that S° and S are the *same* system, albeit exposed to different environmental circumstances, by requiring that

$$g_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_r, 0, \dots, 0) = f_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_r).$$

That is, by putting new constitutive parameters $\alpha_{r+1}, \dots, \alpha_{r+s}$ equal to zero in S, we obtain precisely the system S° .

Now let us suppose that at some initial instant of time t_0 , both of our systems S° and S are in the same state, namely $(x_1(0), \dots, x_n(0))$. Assuming that the unique trajectory property holds for both systems, this common initial state will determine a single trajectory in X representing the temporal behavior of S° , and another single trajectory in X representing the temporal behavior of S. This situation is diagrammed in Fig. 5.1 below:

After some time interval t has elapsed, the system S° will be in a state $(x_1(t), \dots, x_n(t))$ on its trajectory, while the system S will be in another state $(\bar{x}_1(t), \dots, \bar{x}_n(t))$, as indicated in the figure. The time scale t is taken to be a common time valid for both systems, as can be chosen by selecting appropriate scale factors according to the discussion in Sect. 4.3 above; we henceforth assume that the dynamical equations (5.14) and (5.15) already incorporate these scale factors.

It is clear, from the way in which the systems (5.14) and (5.15) were defined, that the respective tangent vectors to the initial common state $(x_1(0), \dots, x_n(0))$ *cannot be the same*. Hence the two trajectories determined in S° and S respectively by this common initial state must diverge from each other, at least initially. This divergence is represented explicitly in Fig. 5.1 by the Euclidean distance $\rho(t) = [\sum (x_i(t) - \bar{x}_i(t))^2]^{\frac{1}{2}}$. Our main interest will be to study the behavior of this divergence

as a function of time. Intuitively, this divergence can be regarded as the “dissipation” introduced into S° by opening it to a new modality of environmental interaction, this dissipation is a direct consequence of the creation of uncompensated flows.

We have already seen that the dissipation, or divergence, between trajectories of S° and S determined by a common initial state is initially positive. We shall now argue that it is also terminally positive; i.e. that

$$\lim_{t \rightarrow \infty} \rho(t) > 0.$$

We shall do this by showing that the transition from S° to S can be regarded as inducing a transformation $\alpha : X \rightarrow X$ of the common state space of S° and S . The effect of this transformation is to move the states in X , and in particular, to move the limiting states (or more generally, the limiting sets) to which trajectories of S° and S tend as $t \rightarrow \infty$. This kind of result is, of course, expected on intuitive grounds; if we recall our analogy between an open system S and a meter, we will observe that the essence of a meter is to manifest a different limiting behavior when a measurement is being performed than when the meter is isolated.

Let us then suppose that our system S° satisfies the general accounting relation

$$\sum_{i=1}^n M_i(x_1, \dots, x_n) dx_i = 0 \quad (5.16)$$

in accord with (5.8) above. We shall further suppose that the open system S satisfies the accounting relation

$$\sum_{i=1}^n (M_i + N_i) d\bar{x}_i = 0, \quad (5.17)$$

where we will denote the state variables for S by dashed quantities $\bar{x}_i$ to distinguish them from those pertaining to S° . Thus, in particular, when discussing S , we shall write $M_i = M_i(\bar{x}_1, \dots, \bar{x}_n)$, $N_i = N_i(\bar{x}_1, \dots, \bar{x}_n)$.

We next observe that, according to the discussion of Sect. 4.5 above, the equations of motion (5.14) and (5.15), describing S° and S respectively, express the relations between differential increments of the state variables at a state and a differential increment of time. Thus, for S° , we have

$$dx_i = f_i dt, \quad i = 1, \dots, n; \quad (5.18)$$

and for S we have

$$d\bar{x}_i = g_i dt, \quad i = 1, \dots, n. \quad (5.19)$$

According to Sect. 4.5, we can take these differential time increments to be the same by incorporating an appropriate scale factor $a(\bar{x}_1, \dots, \bar{x}_n)$ into the functions g_i ; for simplicity, we shall assume this done, so that the trajectories of S° and S are traversed in a common time.

Combining (5.18) and (5.19) with (5.16) and (5.17), we obtain global linkage relations for S° and S : explicitly, we have

$$\sum_{i=1}^n M_i f_i = 0 \quad (5.20)$$

for S° , and the corresponding relation

$$\sum_{i=1}^n (M_i + N_i) g_i = 0 \quad (5.21)$$

for S . These relations are the basic ones we shall need to complete our argument.

Let us look initially at the relation (5.20). We observe that this relation has the form of an orthogonality condition; it basically asserts that two n -vectors, namely

$$\vec{f} = (f_1, \dots, f_n)$$

and

$$\vec{M} = (M_1, \dots, M_n)$$

are orthogonal to each other in some suitable vector space of functions. Likewise, the relation (5.21) asserts that the two vectors

$$\vec{g} = (g_1, \dots, g_n)$$

and

$$\vec{M} + \vec{N} = (M_1 + N_1, \dots, M_n + N_n)$$

are orthogonal in this *same* space.

Now the replacement of $\vec{M}$ by $\vec{M} + \vec{N}$ in this vector space must involve a *rotation* of the space. If we think of this replacement as a *perturbation* of $\vec{M}$, the result of the perturbation cannot simply be to multiply the components of $\vec{M}$ by a constant factor. For, by our preceding arguments, the effect of multiplying the M_i by such a constant factor would be simply to change the time scale according to which the trajectories of S° are traversed, while leaving the trajectories of S° invariant. On the other hand, we know that the trajectories of S° and S are different, as was indicated in Fig. 5.1 above. From this it follows that the transition from $\vec{M}$ to $\vec{M} + \vec{N}$ must involve a rotation, as asserted.

One consequence of this rotation of $\vec{M}$ is that $\vec{M} + \vec{N}$ will no longer be orthogonal to $\vec{f}$. To restore orthogonality, we must correspondingly rotate $\vec{f}$; in effect, the vector $\vec{g}$ can be regarded as arising by *applying the same rotation to $\vec{f}$ as was applied to $\vec{M}$* . Indeed, the relation (5.21) embodies simply the statement that such a rotation of both $\vec{M}$ and $\vec{f}$ preserves orthogonality.

Now it is a general theorem that a rotation in a space of functions defined on a space X arises from a suitable transformation $\alpha : X \rightarrow X$ of that underlying space. More specifically, if U denotes such a rotation, and if $\vec{h}$ is a function of X , we can write

$$(U\vec{h})(x) = \vec{h}(\alpha x) \quad (5.22)$$

for any x in X . Intuitively, this transformation α is the one which relates the points on the trajectories of S° to those of S , as for instance illustrated in Fig. 5.1 above. In particular, it readily follows that the limiting behaviors of these trajectories will also be moved in general by α . Stated another way, the *attractors* to which trajectories of S° and S , initially determined by a common state, will move in X are different, and hence the distance between them must be positive.

Now let us obtain an estimate for the discrepancy between the state $(x_1(t), \dots, x_n(t))$ of S° and the corresponding state $(\bar{x}_1(t), \dots, \bar{x}_n(t))$ of S at a given instant t , given that the two systems were initially in the same state. As we noted above, this discrepancy is measured by the Euclidean distance

$$\rho(t) = \sum_{i=1}^n [(x_i(t) - \bar{x}_i(t))^2]^{\frac{1}{2}} \quad (5.23)$$

between these states in X . Using this relation, we can also obtain an expression for the rate at which this discrepancy is increasing at any time. In fact, by differentiating (5.23), and considering the quantities $x_i(t)$, $\bar{x}_i(t)$ as independent variables, we find

$$\rho \frac{d\rho}{dt} = \sum_{i=1}^n (x_i - \bar{x}_i) \left(\frac{dx_i}{dt} - \frac{d\bar{x}_i}{dt} \right) \quad (5.24)$$

In fact, however, these quantities are not independent; they are connected by the transformation α defined above. That is, given that the two trajectories of S° and S respectively are determined by a common initial state in X , we can generally express the dashed quantities, pertaining to S , as explicit functions of the undashed quantities, pertaining to S° , evaluated at the same instant. Thus we can write

$$\bar{x}_i(t) = \bar{x}_i(x_1(t), \dots, x_n(t)). \quad (5.25)$$

Using these relations, the expression (5.24) can be considered as a function of the n variables $x_1(t), \dots, x_n(t)$ alone, instead of as a function of $2n$ variables. Thus, from a knowledge of the state of S° at a given instant, and the transformation rules (5.25), we can compute the discrepancy between this state and the corresponding state of S determined by the same initial state as S° at $t = 0$.

As we noted earlier, the discrepancy $\rho(t)$ between states of S° and S which we have defined is a measure of the *dissipation* introduced by opening S° to a new modality of environmental interaction. It is thus of interest to express this dissipation explicitly in terms of the differential flows of the state variables at a state. From (5.24) and (5.25) this can be readily done. In fact, we can write

$$\rho d\rho = \sum_{i=1}^n (x_i - \bar{x}_i(x_1, \dots, x_n)) \left(dx_i - \sum_{j=1}^n \frac{\partial \bar{x}_j}{\partial x_j} dx_j \right) \quad (5.26)$$

$$= \sum_{i=1}^n K_i(x_1, \dots, x_n) dx_i \quad (5.27)$$

where the functions $K_i(x_1, \dots, x_n)$ can be written explicitly from (5.26). These functions K_i depend both on the instantaneous values of the state variables of S° and S , and of their instantaneous rates of change. On the other hand, the expression (5.27) also clearly represents the uncompensated flows arising at a state of S° when S° is opened to the new modality of environmental interaction. From this, we can see that these functions K_i are essentially the same as the functions N_i introduced in Sect. 5.3 above, and allow an alternative expression of them, measured now in terms of dissipation arising from the interaction.

Finally, let us note that in order to compute $\rho(t)$ explicitly, it is necessary and sufficient that both S° and S be known; i.e. that the functions f_i in (5.14) and the functions g_i in (5.15) be given. From this data, we can in particular compute the instant at which the deviation between S° and S becomes greater than some pre-assigned threshold quantity θ . As we shall see in subsequent chapters, this kind of “critical instant” has a profound significance for the relation between a system S and a dynamical encoding of it.

The results of the present chapter can be summed up succinctly in the following way: *no system S can be indistinguishable from a subsystem S° of itself*. There must always exist a set of environmental circumstances in which a discrepancy between the behavior of S , and the corresponding behavior of the subsystem S° in those circumstances, will appear.

5.5 Models as Closed Systems

We are now in a position to reconsider the modeling relation, in the light of the results just obtained. In particular, we are going to explore the proposition that the relation between a natural system and a dynamical model of it is the same as that existing between a system closed to some modality of environmental interaction and the same system which is open to such an interaction. In each case, there will be a discrepancy between the behavior of the system involved; between corresponding trajectories of the closed and open system, and between the predictions of the model and the behavior being modeled. And in each case, this discrepancy arises because of a tacit process of *abstraction*; the neglect or omission of interactions present in one of the systems (the open one) and necessarily absent in the other.

Let us begin by re-stating the modeling relation between a natural system and a formal one. We recall that a model is generated by an encoding of observables. It

is necessarily an abstraction, because any observation or evaluation of observables on states is already an abstraction (cf. Sect. 3.2 above). Indeed, we may recognize that there are two distinct kinds of abstractions involved in such an encoding: (a) a neglect of observable quantities pertaining to the system itself, and (b) a neglect of environmental quantities, which are capable of causing change of state even in that subsystem retained in the encoding. Thus, as we have stated above, any such encoding serves tacitly to *unlink* the encoded observables from those other qualities of both system and environment which are not encoded.

Using the language introduced in the preceding chapters of the present section, let us designate that subsystem of a natural system S which has been encoded into a formal model by S° . We shall assume that the inferential structure present in the formal model is in fact a faithful representation of the behavior of S° , when S is in fact closed to all interactions not explicitly incorporated into the model. These interactions include both those arising from the remainder of the system S , and those which come from outside of S . We repeat that this formally assumes that the behavior of S° is *unlinked* to any quantity not explicitly encoded; i.e. an arbitrary change can be made in any such quality without affecting any behavior of S° .

It will now be seen that the relation we have posited between S° and S is precisely the one which has been discussed at length above. It is this fact which makes it possible for us to discuss the relation between a system and a model in the same terms that we used to discuss the relation between a closed system and an open one. In particular, we want to translate the growth of discrepancies between behavior of S° and S into the deviation between the predictions of a dynamical model and the actual behavior of the system being modeled.

Let us restate these hypotheses, using the formal language developed in Sect. 3.3 above. We assume that we have a natural system, whose set of abstract states we will designate by S . We have also a family of observables $x_1, \dots, x_n$, which may be evaluated on these abstract states. In particular, if $s_0 \in S$ is the abstract state of our system at some particular instant of time, we encode this state into a corresponding n -tuple of numbers:

$$s_0 \rightarrow (x_1(s_0), \dots, x_n(s_0)).$$

The totality of n -tuples arising in this way we suppose to fill out a manifold X in Euclidean n -dimensional space; this is the state space of the system arising from the particular encoding we have chosen. The entire situation can be represented by a diagram of the form

$$S \xrightarrow{\delta} S^n \xrightarrow{(x_1, \dots, x_n)} X$$

which we have seen before.

We now suppose that the state of our natural system S is changing in time. We further suppose, for the moment, that this change of state in S can be faithfully represented in formal terms by a dynamics imposed on the state space X . At this level of formalization, it is sufficient to represent the dynamics by a one-parameter family of transformations on X :

$$T_t : X \rightarrow X.$$

Explicitly, when we say that the change of state in S is *faithfully* represented by the dynamics $\{T_t\}$, we mean the following: given any initial state s_0 in S , we evaluate the observables $x_1, \dots, x_n$ on s_0 to obtain a definite encoding $(x_1(s_0), \dots, x_n(s_0))$ of s_0 in X . We then apply any transformation T_t in our dynamics to this encoded state. We obtain thereby a uniquely determined point of X , of the form

$$T_t(x_1(s_0), \dots, x_n(s_0))$$

which we may denote by $(x_1(t), \dots, x_n(t))$. On the other hand, the state of our natural system is also changing in S . Suppose that at the particular instant t considered above, the system is in a definite state $s(t)$. Let us now evaluate our observables $x_1, \dots, x_n$ on this state $s(t)$. This clearly gives us another n -tuple $(x_1(s(t)), \dots, x_n(s(t)))$ in X . We assert that the two n -tuples obtained in this fashion *are the same* for each instant t ; i.e. that

$$x_i(t) = x_i(s(t))$$

for each $i = 1, \dots, n$. In other words, the *prediction* about the state of our natural system at time t , given an initial encoded state in X , is always exactly verified. Thus, the inferential structure of our encoding (i.e the dynamics $\{T_t\}$) is a precise expression of the dynamical processes occurring in S ; every *theorem* in our formal system (cf. Sect. 4.6 above) is verified by direct observation in S .

As usual, we can represent this kind of situation by means of a commutative diagram:

$$\begin{array}{ccc}
 S & \xrightarrow{(x_1, \dots, x_n)\delta} & X \\
 \Theta_t \downarrow & & \downarrow T_t \\
 S & \xrightarrow{(x_1, \dots, x_n)\delta} & X
 \end{array} \tag{5.28}$$

Here we have denoted by Θ_t the actual dynamical changes occurring in the abstract set of states S . The commutativity of the diagram means precisely that for each instant t , the mappings Θ_t and T_t are *models* of each other; we can translate between them precisely by means of the encoding mapping $(x_1, \dots, x_n)\delta$. In more precise mathematical language, Θ_t and T_t are *conjugate*. We note explicitly that we will have a separate diagram of the form (5.28) for each instant of time t ; i.e. (5.28) represents a one-parameter family of diagrams.

Now we observe that what we have in fact captured through the above encoding is a specific subsystem S° of S . The hypothesis that the dynamics $\{T_t\}$ on X precisely captures the change of state in S means that this subsystem S° is unlinked to the remainder of the system, or to any mode of environmental interaction besides those responsible for the changes of state Θ_t in S° . Under these circumstances, our model for S° is also a model for all of S .

Now let us assume that at some initial instant (which we may take as $t = 0$) these hypotheses fail to be satisfied. That is, at $t = 0$ some linkage relation is established between S° and uncoded observables in S , and/or between S° and a new environmental modality. As we have seen, this means that the system S° is no longer closed; it also means that the diagrams (5.28) will no longer be exactly commutative.

However, there is a sense in which the diagrams will remain *approximately* commutative, in the following sense. Let $\varepsilon > 0$ be some small positive quantity. Then for any time t , we may compute for any initial state $s_0 \in S$ the discrepancy in X arising from the two paths in the diagram (5.28); i.e. the distance between

$$T_t(x_1, \dots, x_n)\delta(s_0)$$

and

$$((x_1, \dots, x_n)\delta) \Theta_t(s_0).$$

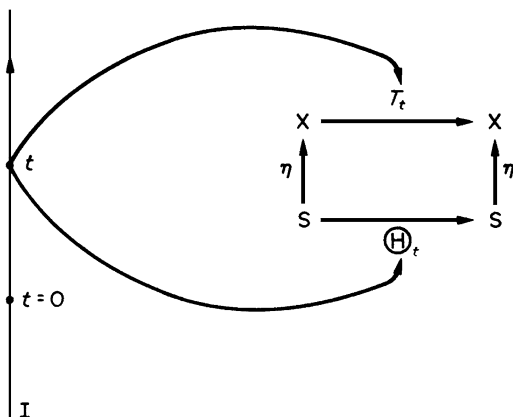
We may say that the diagram *approximately commutes* for this time instant t if this discrepancy is smaller than ε .

The results of the preceding chapter can now be invoked. In general, there will be some finite *critical instant* $t = t_c$ for which *the diagram no longer approximately commutes*. At this critical instant, the prediction obtained by applying our computative model T_{t_c} to the encoding of our initial state s_0 differs from the encoding of the actual state $\Theta_{t_c}(s_0)$ by more than the allowed amount ε . Thus, at the critical instant, we lose even the approximate conjugacy of the mappings Θ_t and T_t . In other words: for all times t such that $0 < t < t_c$, we retain an approximate conjugacy between our original model T_t on X and the actual dynamics Θ_t on S . For times greater than the critical time, this approximate conjugacy fails.

The actual value of the critical instant t_c obviously depends on (a) the specific character of the dynamics $\{T_t\}$ on X , and (b) the manner in which the subsystem S° has become linked to other non-encoded quantities. It is clear that this latter is in principle *not predictable* solely from a knowledge of S° as a closed system. It can only be predicted if we know the new linkages established between S° and its environment. In the discussion of the preceding chapter, this knowledge was assumed; hence in that case it was possible to explicitly predict the value of t_c for any given initial state s_0 and any particular value of the measure of closeness ε . Thus, from the standpoint of S° alone, we cannot in principle predict that the modeling relation will fail, nor is it possible to predict when such a failure will take place.

It is important to cast these considerations into the language of *bifurcations*. As we saw in Example 5 of Sect. 3.1 above, bifurcation and the failure of a conjugacy relation are essentially synonymous; hence it should not be surprising to find that the opening of a system like S° should involve a bifurcation in an essential way. We stressed earlier that a bifurcation always involves the appearance of a logical independence between two descriptions; or what is the same thing, the failure of a specific linkage relation between the descriptions to be maintained. Of course, that

Fig. 5.2



is exactly what is happening in the present case; a description based on a closed system S° is becoming unlinked to the behavior of the same system when opened to further interactions. In this case, it is the *time* which is acting as the bifurcation parameter; at a definite critical instant $t = t_c$, the discrepancy between the behavior of the two systems exceeds a preassigned threshold.

In greater detail: let I denote the set of instants in which time is measured in the original system S_0 . Then in particular, the elements of I constitute the set of parameters for both of the one-parameter families of transformations $\{T_t\}$, $\{\Theta_t\}$ which represent dynamical changes of state in X and S respectively. In what follows, we are going to treat the mappings $t \rightarrow T_t$, $t \rightarrow \Theta_t$, which associate instants with transformations, as *observables of I* . That is, we shall regard each transformation $T_t : X \rightarrow X$ as a *description of t* , and each transformation $\Theta_t : S \rightarrow S$ as another description of the same time instant t . These observables thus defined on I are not numerical-valued; rather, their values are themselves mappings of definite types.

If we regard T_t and Θ_t as separate descriptions of the particular instant $t \in I$ with which they are associated, the question arises (as always) as to how these descriptions may be compared. We always compare mappings or transformations by seeing if we can construct a commutative diagram which allows us to translate between their actions; if such a diagram can be constructed, the mappings are equivalent (conjugate). In the present case, the diagram we need to construct is already embodied in (5.28) above. It must be explicitly noted that we cannot translate between the action of T_t on X and Θ_t on S in an arbitrary way; rather, we are restricted to the encoding maps $(x_1, \dots, x_n)\delta$, by virtue of the way in which X and S are related.

Let us denote the situation we are describing diagrammatically (Fig. 5.2):

In this diagram, we have written $\eta = (x_1, \dots, x_n)\delta$ as the encoding map relating X and S . If we think of t as moving in I , the maps T_t and Θ_t are correspondingly changing, but the encoding map η must remain fixed.

Let us now agree that the two descriptions T_t , Θ_t of an instant t in I are *equivalent* if the diagram (5.28) approximately commutes, in the sense defined above. In particular, the descriptions will surely approximately commute when $t = 0$; intuitively, this means that S and S° are in the same state. As we now let t increase in I , we have seen that there will be a growing discrepancy between the two arms of the diagram (5.28), but the diagram will still approximately commute. As t continues to increase, we will ultimately come to the critical value $t = t_c$, at which the property of approximate commutativity is lost. At this point, the two descriptions T_{t_c} , Θ_{t_c} of the time instant t_c are no longer equivalent; we can no longer translate between these descriptions. This, of course, is the essence of a bifurcation.

It is most interesting to note that the bifurcation parameter in this situation is most conveniently taken to be the set I of time instants. This fact points out in yet another way the crucial role which time plays in dynamical encodings. Indeed, throughout the above discussion, we treated I precisely in the same terms as we treated S ; namely, as a set of “abstract time states”, labelled or encoded by different families of dynamical processes. The fact that these labellings can bifurcate with respect to one another is simply a way of stating that time is complex; a conclusion we reached on other grounds in Sect. 5.4 above.

Another crucial aspect of the above discussion may now be noted. When the critical time t_c is exceeded, we have seen that we can no longer directly translate between the description T_t of S° and the actual dynamics in S . We can hope to restore the commutativity (or approximate commutativity) of the diagram (5.28) in a variety of different ways. The simplest is to “recalibrate”; namely, to restore the coincidence between the state of X at a time t and the actual state of S encoded into X at that time. Such a recalibration essentially puts the clock back to zero, and gives us a further time interval of approximate commutativity before a new critical time is reached. By repeating this process often enough, even a faulty model can retain approximate commutativity indefinitely.

Often, however, the continual recalibration of a model is impracticable, especially since the critical time cannot be predicted within the context of the model itself. The next simplest procedure for retaining approximate commutativity of the diagram (5.28) is to replace the dynamics $\{T_t\}$ on X by a new dynamics, while retaining all other features of the encoding (in particular, retaining the original encoding map η). Essentially, this can be done by recompensating the flows in X ; i.e. introducing new accounting relations between the differential increments of the state variables at each point of X . This is the most obvious alternative, and is of course the one most generally adopted.

However, there are other alternatives. For instance, we may note that the loss of translation between the original dynamics $\{T_t\}$ on X and the actual dynamics $\{\Theta_t\}$ on S when the critical time t_c is exceeded arises *relative to the retention of the particular encoding map η which we are using*. It will be recalled that we have kept this encoding map fixed throughout the preceding discussion, as indeed we must if we are to preserve the given relation between S and X . However, our discussion does not preclude the possibility of restoring the approximate commutativity of the diagram (5.28) when t_c is exceeded, *providing we are willing to change the encoding*

map. To make a change in the encoding map means, of course, that we are changing our description of the abstract states in S more or less radically. In the simplest case, we may retain the manifold X , but modify the manner in which the n -tuples in X are attached to abstract states. This confronts us with an abstract version of what the physicist generally calls *renormalization*; a situation we have encountered before (cf. Sect. 3.3) in what the reader will recognize to be an essentially similar context. A more radical change of encoding is to give up the manifold X entirely, and recode the abstract states of S into a new manifold Y , using different observables of S . These new observables may be more or less distinct from those originally used to define X , and their introduction is a manifestation of the phenomenon of *emergence*, or *emergent novelty*. It will be noted that what is “emerging” here is not an intrinsic property of S , but rather the failure of an encoding to model S over time. This is an essential point, which we will consider in detail in the following chapter.

In any case, we see that the existence of a critical time makes it necessary to replace a given description or encoding of S by another, which may differ more or less radically from the original one. At root, this necessity arises entirely because a model is an abstraction, tacitly supposing the system being modeled is closed to non-encoded interactions. From the failure of this tacit supposition, all of our conclusions immediately follow.

5.6 The Concept of Error

In the present chapter, we are going to recast the results we have obtained into another form, which will make manifest its intimate connection with the important concepts of *error* and system *complexity*. In the process, we shall see that the results obtained for continuous-time dynamics also hold for the other kinds of time-encoding discussed in Sect. 5.4. The procedure we shall develop also illuminates many of the general results on encoding obtained in that section, on which the entire concept of the modeling relation is based.

Let us fix ideas, as usual, with a few purely formal or mathematical considerations, which are of a particularly simple and transparent character. Let $\mathbb{R}$ be the set of real numbers. We are going to treat $\mathbb{R}$ like a set of abstract states, and put ourselves in the position of an observer who has access to the elements of $\mathbb{R}$ only through a limited family of “measuring instruments” at his disposal. At the same time, we shall retain the godlike perspective arising from our knowledge of $\mathbb{R}$ and its elements as mathematical objects, so that at every step we will be able to evaluate the limited results of our observations from a more comprehensive view of what is actually happening.

We know, as mathematicians, that every real number $r \in \mathbb{R}$ can be uniquely represented by a decimal expansion, which is generally infinite and non-repeating. On the other hand, as empirical observers of the elements of $\mathbb{R}$, we can only characterize these elements through some finite family of “observables” $f_1, \dots, f_n$. We shall suppose that these observables have the following interpretation: $f_1(r)$ is

the first (non-zero) coefficient in the decimal expansion of r ; $f_2(r)$ is the second coefficient in the decimal expansion of r , and so on; thus in general, $f_i(r)$ is the i th coefficient in the decimal expansion of r . We suppose also that the observer can discriminate the decimal point; i.e. the numbers 213 and 2.13 are not counted as alike by the observer.

To see how $\mathbb{R}$ looks to such an observer, we may immediately invoke the general discussion of Sect. 3.2 above. Specifically, the observables $f_1, \dots, f_n$ which we have defined collectively impose an equivalence relation on $\mathbb{R}$; two real numbers r_1, r_2 fall into the same equivalence class under this relation if and only if we have

$$f_i(r_1) = f_i(r_2)$$

for $i = 1, \dots, n$. Thus, our observer does not see all of $\mathbb{R}$; to him, two real numbers are indistinguishable if and only if their first n decimal coefficients coincide. From a mathematical point of view, our observer must replace any given real number r by a (generally different) number $r^{(n)}$; the first n decimal coefficients of $r^{(n)}$ coincide with those of r , but all the remaining decimal coefficients of $r^{(n)}$ are equal to zero. In other words, the decimal expansion of $r^{(n)}$ is a truncated initial segment of the expansion of r , and any two real numbers with the same truncated initial segment are counted as identical by our observer.

There are an infinitude of such truncated initial segments $r^{(n)}$, but from a mathematical point of view they comprise a *discrete subset* of $\mathbb{R}$. Moreover, each equivalence class defined by the observables f_i contains an uncountable set of real numbers, all of which are indistinguishable to our observer, both from each other, and from the single $r^{(n)}$ which represents the entire equivalence class. However, the mathematician can see that there is a *discrepancy* between any number r in such a class and the truncated initial segment $r^{(n)}$ which represents the class; a discrepancy which is of course invisible to our observer.

This invisible discrepancy can, however, be made visible to our observer under appropriate circumstances. In particular, we can allow two arbitrary real numbers r_1, r_2 to *interact*; say by forming their product $r_1 r_2$ in $\mathbb{R}$. Now our observer will see this interaction as occurring between the two corresponding truncated initial segments $r_1^{(n)}$ and $r_2^{(n)}$. Moreover, he can only see the first n coefficients of the product $r_1 r_2$; i.e. he can only see

$$f_1(r_1 r_2), f_2(r_1 r_2), \dots, f_n(r_1 r_2),$$

which for him will specify the *result* of the interaction. Thus, for him, the multiplication operation will be represented in the form

$$r_1^{(n)} \times r_2^{(n)} = (r_1 r_2)^{(n)}.$$

From the more general mathematical perspective, we can recognize that *the operation of multiplication in $\mathbb{R}$ is not compatible with the observer's equivalence*

relation. That is, if r_1 and r_1' are equivalent to our observer (i.e. have the same truncated initial segment), and r_2 and r_2' are equivalent, it need not be true that

$$(r_1 r_2)^{(n)} = (r_1' r_2')^{(n)}.$$

That is, multiplication in $\mathbb{R}$ can *split* the observer's equivalence classes. This splitting arises, of course, because the decimal expansion of a product may involve decimal coefficients of the factors beyond the ones visible to our observer; such coefficients may vary widely within each of the observer's equivalence classes, and are precisely the source of (invisible) variability within those classes.

Let us now place ourselves in the position of our observer. Suppose he repeats the "experiment" of multiplying two numbers r_1, r_2 a large number of times. Since he can discriminate numbers only up to a truncated initial segment, his repetition of the experiment will in fact involve a sampling of the equivalence classes of the numbers being multiplied. Thus, what appear to him to be the *same* numbers are actually distributed over classes of numbers. In general, he will conclude that "most of the time" the multiplication determines a unique product; i.e. the equivalence class to which he assigns the products he sees will be the same. Occasionally, however, he will have to assign the resulting product to a different equivalence class.

The visible splitting of the observer's equivalence classes by multiplication represents precisely what we called a *bifurcation*. In this case, the bifurcation is between the observables $f_1, \dots, f_n$ and those invisible decimal coefficients actually involved in the multiplication.

How will our observer describe such a bifurcation when it occurs? To him, it will appear that a fixed interaction between two definite numbers does not always give rise to the same result. In general, he will be led to suppose that there is *some source of variability, or noise, or error in the interaction*.¹ From our more general mathematical perspective, we can see that there is indeed a source of variability, but it lies in the incompleteness of the observer's description of his numbers, and not at all in their interaction. On the other hand, since our observer is unaware that his description is incomplete, he will conclude that his inability to uniquely classify a product indicates a (stochastic) source of error *in the interaction itself*.

Let us formulate the situation even more sharply. Suppose that our observer possesses a calculating machine, which can multiply real numbers exactly. To our observer, who can see only the truncated initial segments of the numbers he supplies to the machine as input, and who likewise classifies the output in terms of such a segment, it will appear in the above circumstances as if the *machine* is making errors. It must be stressed that the *error* involved here appears to the observer as the assignment of a particular product to the "wrong" equivalence class; it is a failure of classification. This error arises from an invisible discrepancy between the numbers actually being multiplied and those the observer believes are being multiplied, but *this discrepancy is not itself the error* as seen by the observer.

From our more general mathematical perspective, it is clear that the errors of classification seen by the observer are a form of round-off error, or overflow,

arising from a replacement of the actual numbers being multiplied by abstract approximations to them. Since actual multiplication splits the equivalence classes defined by these approximations, the errors of classification are simply bifurcations between the qualities retained in the approximation, and those actually involved in forming a product. Such a bifurcation makes the initially invisible variability appear in the form of errors of classification, which are now patent to the observer. As we have noted, the observer will tend to regard these errors of classification as stochastic in nature; he can actually develop a detailed statistical theory of such errors. However, we can see that such a theory will merely represent the manner in which he is sampling his equivalence classes as he repeats what (to him) is the experiment of multiplying a pair of fixed numbers.

In a sense, the above discussion shows how errors of classification arise when there are *hidden variables*²; i.e. degrees of freedom invisible to an external observer, but manifesting themselves in a particular interaction. The errors arise because the hidden variables can bifurcate from the ones available to the observer. All of this is clearly seen in the example we have presented.

In yet another picturesque mathematical terminology, the replacement of a real number r by an initial segment $r^{(n)}$ is a passage from actual numbers to *germs* of numbers. This terminology is drawn from an exactly analogous situation, in which the local behavior of a function at a point is referred to the equivalence class of functions whose Taylor expansions at the point coincide up to the first n terms. Here again, these equivalence classes (called *germs* of functions) can be split by mathematical operations involving higher terms; indeed, it is of the greatest interest to note that the classical theory of bifurcations can be developed entirely from this point of view (although to do so here would take us much too far afield).³

For the most direct comparison of the situation we are presently envisioning with what we have done in previous chapters, let us suppose that a (discrete) dynamics is now imposed on $\mathbb{R}$, say by defining $T^n(r) = r^n$. Here, the index n plays the role of a discrete encoding of time; cf. Sect. 4.6 above. It is clear that such a dynamics on $\mathbb{R}$ will in general also split our observer's equivalence classes; by a repetition of the above argument, he will one again be led to conclude that there is a source of error in the dynamics.

We shall now relate the situation we have just described to the developments of the preceding chapters. As we have seen, our observer is necessarily placed in the position of replacing a general real number r , with its infinitude of decimal coefficients, by an abstraction generated by his limited set of measuring instruments (i.e. by a truncated initial segment of these coefficients). These decimal coefficients correspond to the degrees of freedom, or interactive capabilities, of the abstract states of a natural system. The abstraction retains only a small fraction of these. The abstractions thus correspond to what we earlier called a *closed* system S° . In any interaction in which the neglected coefficients do not enter significantly, the abstraction approximates sufficiently closely to what it represents so that the behaviors in S° and in the full system S remain indistinguishable to our observer. But in any interaction for which the neglected coefficients become significant, there will be an observable discrepancy, or bifurcation, between the behavior of

the abstractions in S° and the actual behavior observed in S . This bifurcation is ultimately quite visible to our observer, but as we have seen, it will tend to be ascribed to *error in the system*,⁴ and not to his description.

Indeed, it is curious and poignant, and not without deep epistemological significance, that our observer of the real number system, armed as he is with only a limited capacity for discerning the qualities of numbers, would regard the *mathematical* view of that system as the abstraction, and his own description as the reality. If our mathematician tried to convince the observer that the real numbers involved unobserved qualities which, when properly formulated, made numerical operations always associative, commutative, distributive, etc., he would be dismissed by the observer as a metaphysician; an inventor of abstractions totally divorced from observable reality. The relation between the mathematician and the observer which we have just sketched is in fact itself a model for the unfortunate chasm which often separates theoretical from empirical sciences. For the moment, we shall merely note that this chasm exists, and that its roots are visible even in this purely formal discussion; its properties will become a major preoccupation in the sections to follow.

Let us return to the notion of error which we have introduced above. We see that it arises because of a discrepancy between the behaviors exhibited by actual real numbers, and the behaviors of corresponding truncated initial segments, or number germs. It thus represents a discrepancy between objects which are closed to certain interactive qualities, and objects which are not closed to those qualities. We see also that the character of this discrepancy is in principle unpredictable from the standpoint of the number germs themselves.

We shall now argue that *the empirical concept of error always involves such a discrepancy between objects open to certain interaction and abstract representations which are closed to those interactions*. In other words, error represents a deviation between how a given system actually behaves, and the manner in which it *would* behave if it were not open. This concept of error thus provides another language in which to discuss the relation between natural systems and their models, which is of the greatest importance.

In empirical terms, the concept of error arises already at the most basic level; namely, in the operation of those meters or measuring instruments through which we formalize the notion of an observable. It is universally recognized that any measuring instrument is subject to “noise” or variability; or, stated another way, that every measuring instrument has a finite *resolving power*, below which it cannot effectively discriminate.

To inquire further into the nature of this finite resolving power, let us return to our consideration of the real numbers $\mathbb{R}$, as seen from the standpoint of a mathematician, and from that of an empirical observer who can only see number germs. Suppose that our observer constructs a new measuring instrument, of the type sketched in Fig. 5.3 below:

From the mathematical standpoint, this instrument multiplies any number r to be observed with a fixed number a , and then specifies the i th decimal coefficient of this product. It thus defines an *observable* g_a of $\mathbb{R}$, defined by

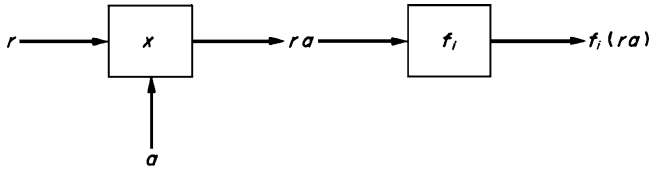


Fig. 5.3

$$g_a(r) = f_i(ra)$$

as seen in abstract mathematical terms. The structure diagrammed in Fig. 5.3 above is thus a *meter* for g_a , which can be applied to any real number r . It thus *classifies* real numbers r in the usual way: r_1 and r_2 are alike if $g_a(r_1) = g_a(r_2)$.

Let us inquire into the resolving power of this meter, as it will appear to our empirical observer. Of course, he can only specify the number r to be observed to a limited number of decimal coefficients; i.e. he cannot discriminate real numbers which differ beyond the n th coefficient. It is easy to see that if r is large compared to a , our observer's inability to discriminate numbers belonging to the same germ will have no effect on $f_i(ra)$. In mathematical terms, the (invisible) variability in r will in these circumstances not affect the observed value of $g_a(r)$, and the operation of the meter for g_a will appear absolutely precise. However, as the input r becomes sufficiently small with respect to a , a *dispersion* will appear in the output of the meter, which will become more and more pronounced as r becomes smaller with respect to a . At some point, depending on the magnitude of a , the discriminating capability of the meter will be completely lost, and we would have to say that the *resolving power* of the meter has been exceeded.

What is happening here? To see this, let us actually perform a simple multiplication; say $534 \times 27 = 14418$. In terms of decimal expansions, we have

$$534 = 5 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$$

and

$$27 = 2 \times 10^1 + 7 \times 10^0.$$

We obtain the decimal expansion of the product by multiplying these two decimal expansions. Let us look at this in some detail.

There will be a term in the product expansion of the form

$$\begin{aligned} (4 \times 10^0) \times (7 \times 10^0) &= 28 \times 10^0 \\ &= 2 \times 10^1 + 8 \times 10^0. \end{aligned}$$

Clearly there will be no other contributions to the coefficient of 10^0 coming from other terms of the product expansion. Thus the last term 8×10^0 has a coefficient which depends only on the last coefficients (4 and 7) of the factors. However, these

last coefficients also produce a contribution (namely 2×10^1) to the coefficient of 10^1 in the product.

The other contributions to the coefficient of 10^1 in the product come from the terms

$$(3 \times 10^1) \times (7 \times 10^0) = 2 \times 10^2 + 1 \times 10^1$$

and

$$(4 \times 10^0) \times (2 \times 10^1) = 8 \times 10^1.$$

These two terms combine to produce a net contribution to the product of

$$2 \times 10^2 + 9 \times 10^1$$

and hence the total coefficient of 10^1 in the product is obtained by combining the results of the three terms we have considered; it is

$$\begin{aligned} 9 \times 10^1 + 2 \times 10^1 &= 11 \times 10^1 \\ &= 1 \times 10^2 + 1 \times 10^1. \end{aligned}$$

No other terms can contribute anything further to the coefficient of 10^1 in the product. However, the terms we have already considered do make a contribution (3×10^2) to the next higher coefficient (that of 10^2) in the product.

Thus we see: (a) the coefficient of 10^0 in the product depends *only* on the coefficients of 10^0 in the factors; (b) the coefficient of 10^1 in the product depends *only* on the coefficients of 10^1 and those of 10^0 in the factors, but on all of these coefficients in general.

Pursuing this kind of argument, it follows that the coefficient of 10^p in a product depends only on the coefficients of 10^q in the factors, for $q < p$, but it generally depends on *all* of these coefficients. We also see that for large numbers, the *significance* of the lowest-power coefficients tends to diminish.

Let us now return to our observable $g_a(r)$. This involves forming a product rx_a , and extracting the i th decimal coefficient of the product. From the above argument, we can see that if r is large compared to a , the i th coefficient of rx_a will in general depend on the coefficients of 10^j in both r and a , where $j < i$. But the *effects* of coefficients of *both* numbers for which $j \ll i$ on this coefficient will generally be negligible; this negligibility will be sufficient to abolish the variability arising from working with number germs rather than with the numbers themselves. On the other hand, if r is small compared to a , the effect of multiplying r by a will be to *magnify* the contributions of lower-power coefficients of both r and a to the i th coefficient. At some definite point, this magnification will make the variability visible, and beyond this, we will ultimately come to see *only* the variability.

If we thus think of the infinitude of decimal coefficients of a real number $r \in \mathbb{R}$ as *degrees of freedom* of r , we see that the observable $g_a(r)$ ultimately depends on all these degrees of freedom. Replacing r by its “germ”, which is what our observer is forced to do, makes most of these degrees of freedom invisible to the observer;

in fact it tacitly puts them equal to zero. They become visible only when a number enters into an interaction in which they are important; e.g. in the formation of the product ra , where r is small compared to a . The variability, or noise, or error we have noted in the meter displayed in Fig. 5.3 above thus arises because these degrees of freedom are present in the numbers actually involved, but absent in the number germs by which the observer replaces them. The meter is thus telling us the net effect of *all* the interacting degrees of freedom present in both the number being observed and in the meter itself (i.e. in the number a), and *not simply those present in the germs of those numbers*. That is, the variables which are “hidden” in this case belong as much to a as they do to r .

To see that these apparently formal considerations actually do pertain to our intuitive concept of error in natural systems, let us consider a few specific contexts in which error plays an essential role.

1. Information theory

The basic problem with which Information Theory is concerned is the transmission of “messages” through a noisy channel. Let us recognize that such a “message” must always be carried by some physical vehicle. Such a vehicle may be regarded as an equivalence class of abstract states of some natural system S , corresponding to our number germs, and like them, defined through the values assumed on the states of S by some finite family F of observables. On the other hand, there are many other observables defined on the states of S , totally unrelated to those through which these states are classified as message vehicles. Likewise, the channel through which the message vehicles must pass are states of some natural system S' . Some of the observables of S' pertain to the channel’s capability to transmit these message vehicles, but there are in general many other observables of S' , unrelated to this capability.

To say that such a channel is noisy means precisely that discrepancies can arise between message vehicles observed to enter the channel and those observed to leave it; here observation means the evaluation of observables in F . Such a discrepancy, as usual, is referred to as an *error*, and it is localized *in the channel*; in such circumstances we say that the channel is noisy. However, through a line of reasoning exactly analogous to that carried out above, we can equally well suppose that the characterization of a message vehicle by *only* those observables in F , pertaining to message recognition, is equivalent to replacing a real number by a number germ, or to replacing a state open to certain interactions by another state closed to those interactions. Thus, the *physical* channel can interact with the *physical* message vehicles through observables entirely distinct from those in F on which the transmission of a message through the channel depends, and which are of course the only ones retained when the channel is regarded purely as a message transmission system. As we have seen, such an interaction can introduce a discrepancy between what the channel *actually* does to a message vehicle and what we *expect* the channel to do on the basis of our abstractions. This is exactly the noise, which we interpret as *error*.

Indeed, in a precise sense, the effect of a communication channel on a message vehicle can be regarded as an *observation* of the channel by the message vehicle; the message vehicle itself becomes analogous to a meter. The observable measured by this meter is, of course, distinct from those relating to the transmission of messages, and hence in general bifurcates from them. As we have repeatedly pointed out, it is such bifurcations which are interpreted as noise or error; their presence indicates that the systems involved are not closed to interactions which we have tacitly postulated to be absent.

Since the character of the “errors” arising in a channel are contingent upon observables distinct from those which characterize the messages, there is as usual no way to predict anything about the character of these “errors” from our descriptions of the messages themselves. The only way to gain access to these “errors” is to pass to a new description, which explicitly involves the observables and interactions on which they depend. In Information Theory, and indeed, in most treatments of noise and error, this is done in a roundabout fashion; namely, by superimposing on our original family of observables some kind of *probability distribution*. In this way, we treat the channel as a *stochastic* entity. The function of such a probability distribution is, of course, to take account of the observables not retained in our initial abstraction. However, this is not in general a good way to proceed; from our more comprehensive point of view, we see that the appearance of error is not a stochastic phenomenon at all.⁵ Nevertheless, under many circumstances, the imposition of a stochastic element upon a closed system description can provide a useful model of the corresponding open system, and it does represent a general strategy for dealing with such open systems. We shall have more to say about this in later chapters.

2. “Errors” in genetic coding

One of the high points of molecular biology has been the identification of the “symbol vehicles” for primary genetic material with strands of DNA, and the specific “information” carried by such a strand with the order of nucleotide bases linearly arranged along such a strand. The genetic material itself must already be involved in at least two distinct kinds of information-transmission processes: (a) a *replicative* process, whereby genetic information is transmitted to descendants, and (b) a *decoding* process, comprising a transcription step (into RNA) and a translation step (whereby an initial DNA sequence is converted into a corresponding linear sequence of amino acids along a corresponding polypeptide chain).

We can obviously think of replication and decoding as comprising *channels* through which message vehicles are passed. Just as in the situation considered above, discrepancies can arise between a message vehicle entering a channel, and the corresponding vehicle leaving the channel; as always, such discrepancies are regarded as *errors*. Thus, a discrepancy between a DNA sequence entering a replication and a daughter sequence emerging from the replication comprises the kind of error called a *mutation*; a discrepancy between a DNA strand entering a decoding and (say) the polypeptide it codes for represents a *mistranscription* or a *mistranslation*.

Clearly, the only observables of a polynucleotide strand, or of a polypeptide chain, which are involved in the transmission of genetic information are those which determine sequence. The only operations retained to describe the transmission of this information are those which preserve sequence. Thus as far as coding is concerned, genetic transmission is closed to all observables and interactions which do not characterize sequence. On the other hand, it is clear that the vehicles which carry information in this case are chemical substances; hence there are many modes of interaction between these substances and their environment which can modify the perceived sequence. The occurrence of such an interaction (e.g., between a DNA strand and a chemical mutagen, or between such a strand and ambient radiation) may thus result in an error; i.e. a deviation between the initial sequence and the subsequent sequence. Once again, the source of these “errors” resides precisely in those observables not involved in determining sequence, and hence abstracted out in any description of information transmission. Consequently, as before, we must either pass to a more comprehensive description of the actual information vehicles, or else attempt to supplement our description with a stochastic element; a probability distribution geared to take account of the missing observables.

3. *Errors in the central nervous system*

An exactly parallel discussion to those given above may be developed for the central nervous system. This should be clear already, on the basis of the many formal relations we have established in preceding sections between the central nervous system and genetic systems, computation, information transmission and logic. However, it is well to sketch the basis for this discussion independently.

As we saw abundantly in Example 3 of Sect. 3.5 above, the basic theoretical tool for investigating the central nervous system is the neural network; and the basic analytical unit for describing the properties of such networks is the individual neuron. Let us therefore begin by considering such an individual neuron, say for simplicity the formal neuron of McCulloch and Pitts; an analogous discussion can be provided for any other kind of formal unit.

As we saw earlier, the basic descriptive elements in the formal neuron are the states of excitation of the input lines to the neuron at specific instants of discrete time. The neuron itself can be regarded as a *meter*, defining an observable on the set of instantaneous states of the input lines through the relation (3.77). This observable is defined through the state of excitation of the neuron’s output line at the succeeding instant, and it also depends on the threshold of the neuron.

Let us note the similarity between this situation and that diagrammed in Fig. 5.3 above. In these terms, the neuron (as indeed any meter) can be regarded as a classifier of the set of states on which it is defined. Thus, as before, an *error* in the behavior of such a classifier can only be recognized in terms of a discrepancy between the class assigned to a particular input by the neuron, and the class to which that input “should” be assigned according to (3.77). That is, the error manifests a bifurcation between the classification actually made and that embodied in the abstract description (3.77).

As usual, we recognize that our description (3.77) is that of a system *closed* to interactions to which the neuron being described is in fact *open*. Thus, the ingredients of that description are like the number germs we considered earlier. If this is the case, then as before, the actual interactions on which the output of a neuron depends may generally bifurcate from those retained in our description. The result will be, from our point of view, a misclassification of particular inputs, which we will ascribe to an *error* of the neuron. Of course, the character of such an error will be unpredictable from our description, which by definition has abstracted away the basis for it. We will thus conclude that it is the neuron which is erroneous or noisy, and we will be led to superimpose a probability distribution on our description (3.77) to take account of its errors.

Ideas of this character have some interesting consequences when elevated to the level of neural networks. In particular, let us suppose that we have a population of such apparently fallible neurons, organized in some network. The behavior of the network as a whole will then be described by utilizing the description (3.77) as applied to each individual neuron in the network. As we saw earlier, such a network can be regarded as realizing some *proposition* in a logical system. If a neuron in the network appears to make an error, then the network need no longer realize that proposition; i.e. the *entire network* can thus be said to be in error. In other words, an error of *the network* is defined in exactly the same terms as is an error in an individual neuron; and indeed, the latter is the diagnostic of the former. In an analogous situation, we would recognize an entire mathematical proof as erroneous if any step of the proof is erroneous.

Thus in a general sense, the *reliability* of a neural network by definition devolves upon the errors made by its constituent neurons. This has had profound implications for theories of the brain. For it is observed that the brain as a whole appears to function “reliably” over prolonged periods of time (e.g. of the order of a century), despite the fact that its neurons randomly die at a significant rate, and even under conditions which greatly modify the properties of its constituent neurons (e.g. intoxication, anesthesia, sleep). Thus the question can be posed: how can the reliability of a neural network be greater than the reliability of its constituent neurons?⁶

This kind of question has received much attention over the past years, not only in the context of neural networks, but also in cognate areas of communication and computation. The general answer is: through the artful employment of redundancy, any statistically rooted variability in the individual elements of a network can be effectively neutralized or eliminated. However, it should be noted that this answer involves a tacit equivocation involving the term “reliability”; in particular, our heuristic argument requires us to identify the overall *functional* reliability of the brain with *freedom from error* of its constituent units. As we shall see later, there are no grounds for making such identifications in general; they involve different descriptions of the systems involved and these descriptions need not themselves be linked.

In a similar fashion, we can elevate the concept of *genetic error*, as outlined in the previous example, to the level of networks of interacting gene products; i.e. to the physiological behavior of cells and organisms. One such approach is embodied in the “error catastrophe” hypothesis of L. Orgel.⁷ Briefly, Orgel’s argument is as follows: The translation step of protein synthesis itself requires protein (the “synthetase” enzymes). Any error arising in these synthetases will introduce errors of sequence into *all* proteins subsequently synthesized, including the synthetases themselves. Thus, the subsequent generation of synthetase molecules will tend to be more erroneous than the initial one. This subsequent generation of synthetases will thus introduce still further errors of sequence into *all* proteins they synthesize, including again the synthetases themselves. Thus in effect a positive feedback loop exists, which serves to magnify the initial errors through successive rounds of protein synthesis. As a result of this positive feedback loop, it is concluded that all cell proteins become increasingly dysfunctional, until the entire cell becomes inviable. Further, as cells become inviable, so too must any organism composed of these cells. Hence, the “error catastrophe” itself escalates into a theory of mortality and senescence.

Here again, there is a tacit identification between reliability at some functional level and the occurrence of “errors” at a different level. That such an identification is not justified in general is shown in this case from a more detailed dynamical investigation of the “error catastrophe” itself. Such investigations show that the positive feedback loop envisioned by Orgel can only arise under physiologically unreasonable circumstances; thus there is no necessary relation here between errors in constituent units and reliability of larger systems composed of these units.⁸ The same is doubtless true in the central nervous system. Indeed, we shall take quite a different approach to reliability in the following section.

Now let us draw together the main features illustrated by the examples we have considered, and relate them to the results of the preceding chapters. We can state these features succinctly as follows:

1. The basis for error lies in the fact that any description of a system is an abstraction. As such, it cannot characterize the states of a system beyond assigning them to equivalence classes, generated by a limited set of observing instruments or meters. Hence there is an uncontrollable variability present in states which the observer classifies as identical.
2. This variability is not yet error. It *becomes* error when the states of the system enter into interactions which split the observer’s equivalence classes. This splitting comprises a bifurcation between the observables employed in the classification, and those actually involved in the interaction.
3. The bifurcation or splitting of the observer’s equivalence classes appears to him as a *mis-classification*, and will be interpreted by the observer as an *error* in the system being described.
4. Thus, error in the system appears to the observer as a discrepancy between the actual behavior of a system, and the behavior expected of the system, each determined by the observer on the basis of the descriptions generated by his observing instruments.

5. Any bifurcation can be regarded as a logical independence of the descriptions which are being compared. Hence in particular, the nature and character of the errors detected by the observer are unpredictable in principle from his descriptions.
6. In order to deal with the errors he detects, the observer must supplement his description, in order to take account of the observables involved in the interaction, but absent from his description. One general way of attempting to do this is to superimpose an appropriate probability distribution on his description; i.e. to treat error as a stochastic process intrinsic to the system.
7. In an interaction, the observables of *both* of the systems involved in the interaction must be taken into account. Hence in particular, error is in fact not intrinsic to a system considered in isolation. In fact, it is better regarded as a meter for environmental observables.

On the basis of these conclusions, we can now better appreciate the relation between the phenomenon of *error*, and the discussion of the previous chapters. In essence, the distinction between an abstract description of a system and the system itself is analogous to that between a system closed to environmental interactions and the same system opened to those interactions. As we saw in Sect. 5.4 above, there will always be a discrepancy between the behavior of a closed system and the same system opened to a family of environmental interactions; in a dynamical context, this discrepancy will grow with time. Such a discrepancy, as we have observed, is the underlying source of variability on which the perception of errors depends, but by itself it does not constitute error. Error, properly so called, arises when this underlying variability is sufficient to cause a bifurcation to appear under appropriate conditions of interaction. More specifically, two states which appear to be equivalent (or approximately so) under the observer's description are in fact sufficiently different in their other degrees of freedom to give widely different results when exposed to an interaction involving those other degrees of freedom. Hence, in a sense, error and bifurcation are equivalent terms; they both indicate the inadequacy of a particular mode of system description, and imply the existence of others. When such a bifurcation occurs, the original modeling relation which existed between the system and its description now ceases to hold. As we have seen, such a failure of a modeling relation can also be interpreted in terms of *emergence*; here too, a given description or model of a system must be replaced by another.

A most important conclusion can be drawn from these considerations, which underlies much of the distinction between biology and physics. Perhaps the most unassailable principle of theoretical physics asserts that *the laws of nature must be the same for all observers* (cf. Example 6, Sect. 3.5 above). By this is meant that these laws must be invariant to the position or state of motion of any observer. But the principle requires that the observers in question should otherwise be identical; i.e. equipped with precisely the same meters. If the observers themselves are not identical; i.e. if they are inequivalent, or equipped with different sets of meters, there is no reason to expect that their descriptions of the universe will be the same, and hence that we can transform from any such description to any other. In such a case, the observers' descriptions of the universe will *bifurcate* from each other (which is

only another way of saying that their descriptions will be logically independent; i.e. not related by any transformation rule or linkage). In an important sense, biology *depends* in an essential way on the proliferation of inequivalent observers; it can indeed be regarded as *nothing other than* the study of populations of inequivalent observers and their interactions. These remarks lend a profound significance to the view that biology is the science of mutability; i.e. the science of error.

References and Notes

1. The “error” we are considering here is well known to computer scientists and numerical analysts as overflow, or round-off error. It is discussed at length (though not from our point of view) in every standard text on numerical analysis, as for example Phillips, G. M. and Taylor, P. J., *Theory and Applications of Numerical Analysis*. Academic Press (1973).
There is also an interesting discussion of “computer arithmetic” in the article by A. S. Householder, included in the collection:
Sarty, T. L. (ed.), *Lectures on Modern Mathematics*. Wiley (1963).
2. There is an extensive and contradictory literature on hidden variables in quantum theory. The essentially statistical features of quantum theory have bothered many physicists (notably Einstein) and from the beginning the question was raised as to whether the quantum-mechanical description of nature was incomplete. If so, a situation obviously analogous to the one we are discussing arises. For a good general discussion see, e.g. d’Espagnat, B., *Conceptual Foundations of Quantum Mechanics*. Benjamin (1976).
Belinfante, F. J., *A Survey of Hidden-Variable Theories*. Pergamon (1973).
3. See for example Bröcker, T., *Differentiable Germs and Catastrophes*. Cambridge University Press (1976).
4. As we are employing the term, “error” refers to the discrepancy in behavior between a system and a model of that system. In the traditional canonization of scientific method, we are supposed to choose the observed behavior of a system as the standard, and refer the predictions of a model to it. If a discrepancy is found, the resultant error is to be imputed to the model; i.e. we would say that we have an erroneous model, and seek a better encoding. In the present circumstances, however, it is the model which is chosen as the standard, and the actual system behavior is referred to it; any discrepancy is then imputed to the system, and we say that the system is erroneous.
5. The viewpoint we are propounding here has certain analogies to, for instance, the calculation of absolute rate constants for chemical reactions in terms of the frequencies of molecular collisions. In order for two molecules to interact chemically to produce a new molecule, it is not enough that they collide; they must collide in a way adequate to pass over a potential barrier separating reactant

configurations from product configurations. Whether this occurs or not depends upon a variety of circumstances; among them temperature, (formally, all these factors enter into the concept of chemical potential). Likewise, when a symbol is passing through a communication channel, an interaction of the symbol vehicle with the channel depends on a variety of circumstances characteristic of the vehicle and the channel. The coding theorems of information theory, which state that by suitable encoding and decoding the probability of “error” (i.e. interaction of symbol vehicles with channel) may be made arbitrarily small are analogous to the assertion that by suitably lowering the temperature, a reaction rate can be made arbitrarily small. It would be interesting to develop the relation between coding and “channel temperature”, but so far this has not been done.

6. See for instance, von Neumann, J., in *Automata Studies* (Shannon, C. E. and McCarthy, J., eds.) Princeton University Press, 43–98 (1952).
7. See e.g. Orgel, L., *PNAS* 49, 517–519 (1963); Orgel, L., *Nature* 243, 441–443 (1973).
8. Goel, N. and Ycas, M., *J. Theor. Biol.* 55, 245–282 (1975); Goel, N. and Ycas, M., *J. Math. Biol.* 3, 121–132 (1976).

5.7 Error and Complexity

There has always been a close relationship between the notion of error and the important notion of complexity. The relation between the two can be summed up in the proposition that “simple systems do not make errors”. Thus, it is meaningless to speak of a system of mechanical particles making an “error”.¹ The capacity to behave erroneously thus seems to be a function of complexity; hence it is natural to consider in greater detail the relation between the two notions, in the light of the discussion of the preceding chapter. In particular, we want to consider the following question: to what extent is complexity a *quality* of a system? That is: to what extent can complexity be regarded as an observable, characterizing an intrinsic property of systems?

To the ancients, complexity was epitomized by the movements of celestial bodies. The hold which Newtonian mechanics continues to exert on the human mind rests in large part on its ability to cope effectively with such complexity, on the basis of a small number of indisputably simple and universal laws. If celestial mechanics continues to pose formidable problems, those problems are no longer of a conceptual, but rather of a computational character; these are of a quite different order. The rationalist point of view towards any problem, however complex it may initially appear, is colored by the experience of mechanics; surely any natural phenomenon, however complex, can be resolved by exploiting the same techniques which resolved the mysteries of the skies.

On the other hand, our views regarding the nature of complexity have tended to remain as richly varied as the concept itself. There have always been scientists uneasy with the view that all complexity could be reduced or explained by means of simple laws, and not all of them can be dismissed as “vitalists” or “holists”.

For one thing, we have all heard it said that complex systems are those which are “counter-intuitive”, meaning that their behaviors are different from what “common-sense” (i.e. extrapolation from simple rules) suggests that it ought to be. John von Neumann² argued that there was a kind of “threshold” of complexity, below which systems behaved with their traditional regularities, but above which entirely new kinds of phenomena emerge, such as self-reproduction, evolution, and free will, which are *sui generis*, and which can have no counterpart in systems of lesser complexity.

A corollary of the idea that complexity possesses such a threshold is that it is a quality of systems which can in fact be measured, or at least computed on the basis of other measurements. Thus, a great deal of literature on system complexity exists; this literature stresses such things as the number of components or variables involved in a particular description of the system, or the richness of the inter-connection of these components, or the length of the shortest algorithm required to construct the system.³ In all of these approaches, complexity is not only viewed as intrinsic to a system, but even is referred to a single particular mode of description of that system.

In what follows, we are going to take a quite different approach. Namely, *we are going to define a system to be complex to the extent that we can observe it in non-equivalent ways*. As we have seen extensively, each mode of observation of a system involves the evaluation of certain of the observables of the system on its abstract states. Each of these in turn gives rise to a particular kind of representation, encoding or description of the system. Thus, for us, a system will be complex to the extent that it admits non-equivalent encodings; encodings which cannot be transformed or reduced to one another. Stated yet another way: since each encoding describes a subsystem of the given system, a system is complex to the extent that we can discern many distinct subsystems of it.

This approach to complexity is novel in several ways. For one thing, it requires that complexity is not an intrinsic property of a system nor of a system description. Rather, it arises from the number of ways in which we are able to interact with the system. Thus, complexity is a function not only of the system’s interactive capabilities, but of our own. A moment’s reflection will reveal, however, that this situation is not as strange as it may appear at first sight; and indeed, is in accord with our intuitive notions regarding complexity. For instance, most of us would regard a stone, say, as a simple system; an organism, on the other hand, is clearly complex. Why do we believe this? Clearly, this intuition rests on the fact that we typically interact with a stone in only a few ways, while we can interact with an organism in many ways. As we multiply the number of ways with which we interact with a stone, its complexity appears to grow; to a geologist, who interacts with a stone in many distinct ways, it can appear infinitely complex. Conversely, as we circumscribe the number of ways we interact with an organism, its complexity accordingly appears to diminish. Thus in this intuitive sense, our characterization of complexity is a reasonable one.

Such considerations lead naturally to the ideas involved in the *analysis* of complex systems. One such strategy for system analysis is that of reductionism, which we have discussed at some length above. The idea here is to resolve a given

system into a spectrum of subsystems, and to reconstruct the properties of the entire system from those of the subsystems into which it has been resolved. We have already discussed some of the ramifications of this approach in Sect. 3.6, and we need not repeat it here. We merely point out once again that the multiplicity of descriptions which provide the very definition of complexity precludes any one class of subsystems, or any one mode of system analysis, from being universally valid over all system properties. Indeed, the lack of such a universally valid strategy for system analysis is but an alternate definition of complexity.

We have related complexity to the idea of many non-equivalent subsystems; i.e. modes of description or encoding which cannot be transformed or reduced to one another. However, we have also seen that such non-reducible, non-conjugate encodings must therefore bifurcate from one another. It is exactly the appearance of such bifurcations which we showed in the preceding chapter to be the essence of error. Therefore according to our definition of complexity, we recapture the intuitive result that a complex system is one in which errors can occur; furthermore, the more complex a system, the greater the number and kinds of errors which will appear, and the greater the deviation between the behavior of the system as a whole and any abstract (i.e. closed) subsystem of itself. Indeed, it is precisely this feature which is responsible for the “counter-intuitive” behavior of complex systems which we noted earlier; we can see that such “counter-intuitive” behavior is simply another way of talking about the capacity for making errors. Once again, this feature can in fact be employed to provide another alternate definition of complexity itself.

References and Notes

1. In the computation of planetary orbits on the basis of Newtonian mechanics, for example, two major discrepancies arose between prediction and observation. These involved the planet Uranus, and the advance of perihelion of the planet Mercury. In the former case, an assumption that there was an additional force arising from an as yet unobserved planet culminated in the discovery of a new planet (Neptune). In the latter case, it turned out that the entire apparatus of Newtonian physics had to be replaced by a new encoding (general relativity).
2. See Note 6, Sect. 5.6. See also the article by von Neumann in *Cerebral Mechanisms in Behavior* (L. A. Jeffress, ed.; Wiley (1951) pp. 1–41), and the volume *Theory of Self-Reproducing Automata*, completed from von Neumann’s notes by Arthur Burks (University of Illinois Press, 1966).
3. Indeed, there is an entire new field devoted to questions of “computational complexity”, which is, roughly speaking, a finite analog of the theory of degrees of unsolvability (cf. Note 5 to Sect. 3.1). The complexity of an algorithm is measured by the number of instants $f(n)$ of logical time which the algorithm requires to process an input string of length n . Alternately, instead of such a “temporal” measure of complexity, we could substitute a “spatial” one, by asking how the amount of energy storage increases with the length of the input.

The basic question is: can we place upper and lower bounds on the complexities of algorithms required to solve particular problems, or classes of problems? The interested reader may consult:

Karp, R. M., *Network* 5, 45–68 (1975).

Traub, J. F. (ed.), *Algorithms and Complexity*. Academic Press (1976).

Garey, M. R., *Computers and Intractability*. Freeman (1979).

Ferranti, J. and Rackoff, C. W., *Complexity of Logical Theories*. Springer-Verlag (1979).

It may be noted explicitly here that one can approach probabilistic notions in the same way; for instance, the question of how “random” a sequence is may be related to the complexity of the algorithm which generates its successive terms.

4. See Rosen, R., *J. Theor. Biol.* 63, 19–31 (1976); also *Int. J. Gen. Syst.* 3, 227–232 (1977).

5.8 Order and Disorder

In this section, we shall return to the basic arguments developed in Sects. 5.4 and 5.5 above. We saw there that if we take initially identical states of a closed system S° and the same system opened to a new modality of interaction with its environment, a growing discrepancy will appear between the behaviors in S° and S . We have already discussed this discrepancy from the standpoint of error and complexity. We now wish to consider this discrepancy from another standpoint, namely that of *order and disorder*. In the process, we shall find that an interesting new light is thrown on associated thermodynamic concepts, especially the concept of entropy.

We have already briefly discussed the concepts of order and disorder in Sects. 4.3 and 4.4 above. We saw there that thermodynamics in essence provides a standard of disorder for a system, and that standard is *absolute*. Namely, the state of absolute disorder for any thermodynamic system S is the state of thermodynamic equilibrium which the system approaches *when it is closed and isolated*. This state is by definition the standard of absolute disorder; it is absolute in the sense that no matter how the system S was open, and no matter what interactions in which it participated, it will always reach the same state of thermodynamic equilibrium when the system is closed and isolated from these interactions. The closed, isolated system thus plays the same role for thermodynamics as the system of particles with non-accelerating center of mass does for Newtonian mechanics; it establishes an intrinsic standard of non-interaction.

We also saw that this standard of absolute disorder for a system S admitted two quite independent characterizations: (a) The state of maximal disorder for a system S is the state on which *entropy is maximal*. Entropy in this case can be regarded as a pure state function; it can in principle be evaluated on any state of S , irrespective of any dynamics or any interactions in which S is involved. (b) The state of maximal disorder of a system S is also the state to which S will autonomously tend when S is closed and isolated.

The former characterization of disorder, in terms of the concept of entropy is thus a static concept, while the latter pertains to a particular dynamics on S . By evaluating entropy, we could in principle find the maximally disordered state of S without any knowledge of its dynamical behavior. On the other hand, from (b), we could also find the state of maximal disorder by isolating the system S and watching the autonomous dynamics, without any knowledge of entropy at all. The relation between these two independent characterizations of disorder is, of course, the following: *the entropy function is also a Lyapunov function for the dynamics arising when S is isolated from all interactions*. Thus, in the particular case under consideration, these two independent ways of defining disorder coincide.

If the system S is not completely closed to environmental interactions, the two notions no longer coincide. In particular, the dynamics imposed on S through such interactions will be different from that arising when S is completely isolated, and thus whatever attractor states exist for this dynamics will be different from the state of thermodynamic equilibrium. Thus, referred to that state of thermodynamic equilibrium as an absolute standard of disorder, such attractor states necessarily appear ordered. Indeed, this is the standard tacitly applied (e.g. by Prigogine and his collaborators) by identifying the departure from thermodynamic equilibrium in an open system with *morphogenesis* (cf. Sect. 3.5 above). They dignify this departure with such names as *symmetry-breaking* and *dissipative* structures.

On the other hand, according to the criterion (b), the attractors of S as an open system are themselves states of maximal disorder, *relative now to the dynamics imposed on S* . For when such a dynamics is given the attractor states are the ones to which the states of S autonomously tend, and those for which work must be done on the system to keep it away from. Thus, according to (b), the state of thermodynamic equilibrium is now itself to be considered as an *ordered* state, relative to the new dynamics. In particular, the entropy considered in (a) is no longer a Lyapunov function for the dynamics, and indeed need have no special relation to the dynamics at all.

Thus, we see that in effect the two notions of disorder, which coincide when S is closed and isolated, fail to coincide under other circumstances. Indeed, we may correctly say that these two characterizations of disorder *bifurcate* from one another.

Clearly, each of these bifurcating definitions possesses some valuable features, which we would like to retain. The characterization of disorder in terms of a state function (e.g. entropy) has the advantage of being absolute; disorder can be characterized by a pure observable universally applicable independent of any imposed dynamics. This absolute characterization has the important property that it allows us to compare the states of *different* systems with respect to order and disorder. On the other hand, we have attained this absolute standard at the cost of singling out, in an arbitrary way, the properties of a closed and isolated system; there is no good *a priori* reason to do this. Moreover, it is often of little interest what the behavior of an open system S *would* be if that system were to be closed and isolated; it thus seems entirely reasonable to expect that any concept of disorder in a system should take some account of its actual dynamics, even if this should

mean that we must relativize the concept, and even if we must give up the idea of comparing disorder in systems with different dynamics.

In what follows, we shall attempt to resolve this contradictory state of affairs by entirely reformulating the concepts of order and disorder. This will be done in such a way that the desirable features of both of the characterizations described above are

$$\begin{cases} \Phi_1(f_{11}, \dots, f_{1r_1}) = 0 \\ \Phi_2(f_{21}, \dots, f_{2r_2}) = 0 \\ \vdots \\ \Phi_k(f_{k1}, \dots, f_{kr_k}) = 0 \end{cases} \quad (5.29)$$

Then in the particular encoding of S into E^n which we employed above, the set of linkages (5.29) will define a subset $\Omega_P \subset E^n$; this is the subset on which the property P is satisfied. Thus, Ω_P is precisely the set of *ordered states* relative to the property P (and of course relative to the encoding we are using). This set Ω_P is fixed once and for all by the linkages (5.29) independent of any dynamics imposed on S (and hence on E^n). Therefore, regardless of any imposed dynamics, we can say whether or not a given state is ordered with respect to P , and if not, we can in principle say how disordered it is.

It will be noted that we successively identify the idea of a property P with a family of linkages, and thence with a specific subset of S (or E^n) which satisfies these linkages. The subset Ω_P we have defined is thus completely determined by its characteristic function, which is an observable of S ; specifically, it is the observable $\chi_P : E^n \rightarrow \mathbb{R}$ which takes the value unity for states in Ω_P and the value zero for all other states. Superimposed on this *absolute* characterization of the property, there is a *metric* aspect; we specify the *degree* to which a state fails to possess the property P by the distance of that state from Ω_P in E^n . The incorporation of this metric aspect brings us very close to the notion of a *fuzzy set*, as originally designed by Lotfi Zadeh.¹ The reader may find it of interest to develop in detail the relation between the abstract concept of a fuzzy set and the picture of disorder relative to a particular property P which we have developed; we shall not pursue the matter further here.

The circle of ideas we have developed above allows us now to make explicit contact between the developments of Sects. 5.4 and 5.5 above, and the concepts of order and disorder. In particular, we are going to view the deviation $\rho(t)$, which specifies the discrepancy between trajectories determined by the same initial state in an open system S and a closed system S° , in terms of the loss of a property P . In other words, *we are going to treat this deviation as a disordering of S° .*

In order to do this, we must specify that property P which is satisfied by the given trajectory in S° , but which fails to be satisfied by the corresponding trajectory in S . But this is very easy. We shall take the linkages which specify the property P to be merely those linkages in E^n which determine the trajectory of S° in question. If we do this, retained (and indeed, so that the thermodynamic situation is recaptured as a

special case). Moreover, this reformulation will automatically relate to the modeling considerations inherent in our results of Sects. 5.4 and 5.5 above.

Intuitively, what we shall do is the following. We shall begin by relativizing the concepts of order and disorder, by always referring them to some property P . However, this property P will be an absolute property, in the sense that whether it obtains or not does not pertain to whether a particular system is open or closed. We will call a state of any such system an *ordered state* relative to P if it exhibits the property; *disordered* relative to P if it does not exhibit the property. Thus, the set of states of a system which exhibit the property is the set of ordered states; the complement of this set is the set of disordered states. For each disordered state, we may in fact measure the *extent* to which it is disordered in terms of its distance from the set of ordered states in the state space. Thus, our definition of order will be relative, in that it pertains only to a specific property P , specified in advance; it will be absolute, in that the state of any system can be characterized, independent of any dynamics on the system, as ordered or disordered relative to the property in question.

The first step in making these heuristic considerations precise is specifying what we shall mean by a property P of a system. For us, such a property shall always be characterized in terms of specific *linkage relations* between observables of our systems. We recall that a linkage relation is of the form

$$\Phi(f_1, \dots, f_n) = 0 \quad (5.30)$$

where the $f_i : S \rightarrow \mathbb{R}$ are observables of our system. More exactly, we can define the set of abstract states of S on which such a linkage relation holds. i.e. for which

$$\Phi(f_1(s), \dots, f_r(s)) = 0.$$

The totality of these abstract states define a definite subset of any encoding of S ; in particular, they will define such a subset in the encoding $S \rightarrow E^n$ arising from the state variables $x_1, \dots, x_n : S \rightarrow \mathbb{R}$ which we employed in Sects. 5.4 and 5.5 above.

Thus, for us, a property P will be a set of linkage relations between observables of a system S , of the form (5.30). For each system S on which these observables are defined, there will be a subset of states (perhaps empty) which satisfy the property, and as we have seen, this subset will be imaged in any specific encoding of S . These imaged points in an encoding will be called *the ordered states of S* relative to the property P . Clearly, this definition of order is independent of any dynamics, or indeed of any other property which might be possessed by such a system S ; it thus has the same kind of absolute character as that manifested by the entropy in thermodynamics. On the other hand, it is relative in that it is defined only with respect to a definite property P ; i.e. a definite family of linkage relations between observables of S .

In particular, let us suppose that a property P is defined through a *family* of linkage relations of the form then the corresponding set Ω_P of ordered states with

respect to the property becomes merely the trajectory itself. Alternatively: the trajectory in question is specified by the linkages which define the dynamics in S° (i.e. the linkages between the values of the state variables evaluated at a state and the instantaneous rates of change of these state variables) supplemented by the linkages which specify the initial state. These uniquely determine the trajectory, and hence again the property P . Under these circumstances, it is now immediate that the discrepancy between trajectories of S° and S , specified by the same initial state, represents a disordering of S° , and in fact a disordering which grows in time.

But now let us further observe that the set of formal linkages which defines a property P can also be regarded as a *model* of P . Indeed, the set of linkages between states and their instantaneous rates of change in E^n which characterize the system S° are precisely what we have called a model of S° . Thus, the disordering of S° which we have demonstrated above, and which represents the discrepancy between the behavior of S° and a more open system S , can be interpreted precisely as the discrepancy between the actual behavior of the system S and that of a model of a subsystem S° . Since S° is itself regarded as a model of S , we may conclude that *the concept of disorder always refers to a discrepancy between the behavior of a system and the behavior of a model of that system*. Stated another way: order represents a concordance of behavior between a system and a model, in that the linkages which characterize the model are also satisfied by the system; disorder represents the extent to which those linkages fail to be satisfied. Hence, disorder is always to be understood relative to some particular description, or model, or property; it is thus perfectly possible for a system to be disordering with respect to one property, but not with respect to another. It is precisely this fact, which has long been intuitively sensed but not clearly articulated, which is responsible for much of the controversy which has arisen from the concepts of order and disorder.²

Let us now make some concluding observations about the circle of ideas which we have developed above. The first is to observe that, in our treatment of the deviation $\rho(t)$ between corresponding trajectories of S° and S , given in Sects. 5.4 and 5.5 above, there is apparently a certain symmetry between the roles played by S° and S . Namely, this discrepancy is the same, whether we regard it as arising from opening the system S° to environmental interaction at $t = 0$, or whether we regard it as arising from closing the system S to those interactions at that instant. In the former case, it appears that we obtain a disordering of the closed system S° by opening it; in the latter case, we obtain a disordering of the open system S by closing it. This is indeed the case; but we must recognize carefully that *the properties P which define order and disorder in the two situations are different*. Thus, if we opened a closed system S° , the linkages defining the property P (and hence which constitute the *model* against which order and disorder are characterized) are those pertaining to S° . On the other hand, if we close an open system S , these linkages are those pertaining to S . The discrepancy between the behaviors of the two systems is the same in both cases, and represents a disordering in our sense; but the property P against which order and disorder are recognized is determined by which of the two systems we take as the reference.

Finally, let us return briefly to the situation in thermodynamics, and the absolute characterization of disorder as the state of thermodynamic equilibrium of a system completely closed and isolated. In an intuitive sense, the state of thermodynamic equilibrium (as seen from a microscopic perspective, such as the Maxwell-Boltzmann distribution) is a state with no properties; it is characterized by total homogeneity, isotropy and stationarity in the underlying phase space. Thus, any departure from thermodynamic equilibrium will appear as the establishment of linkages between the underlying state variables, through which the departure from homogeneity, isotropy and stationarity can be described. These linkages correspond to *properties*, which are by definition lost as the system approaches thermodynamic equilibrium. With respect to any such property, then, thermodynamic equilibrium is indeed a state of maximal disorder. This is indeed, we believe, the intuitive sense underlying the identification of equilibrium with disorder; the loss of all properties involving departures from absolute uniformity. On the other hand, if we consider instead those properties which characterize absolute uniformity, and express these in terms of linkages, then the state of thermodynamic equilibrium becomes a state of maximal *order*. Thus, we see how important it is to specify in advance the property P with respect to which order and disorder are to be referred.

We will finally note, for future reference, that the disordering which arises from opening a system is ultimately to be identified with such things as “wear-and-tear”, and the other kinds of deleterious effects which appear in the operation of real systems; it is also closely related to the “side-effects” about which we spoke earlier (cf. Sect. 1.1 above). All these things, as well as the “errors” discussed in the preceding chapters, are simply ways of expressing the growth of discrepancies between a real system S, open to environmental interactions, and a model of S, which represents a subsystem closed to such interaction.

References and Notes

1. A fuzzy set was originally defined by Zadeh (*Information and Control* 8 (1965), 338–353) as one whose characteristic function takes values in the whole unit interval, rather than the two-point set $\{0, 1\}$. Since then, Zadeh and others have extended the idea enormously, into the theory of mappings, relations, systems, languages, etc. It has been Zadeh’s contention that the preponderance of circumstances with which men deal are inherently imprecise; hence models built out of ordinary precise (i.e. non-fuzzy) mathematical objects are inadequate to deal with them. There is now a burgeoning literature in this area; for an overview, see e.g.
Zadeh L. et al., (eds.) *Fuzzy Sets and Their Applications to Cognitive and Decision Processes*. Academic Press (1975).
2. Relevant here is the overly facile identification of “order” with “information” which characterized so many early attempts to apply Shannon’s results on coding (cf. Sect. 5.6 above; also Note 3 to Sect. 3.5). Perhaps the responsibility for

this ultimately rests with E. Schrödinger, who in his little book *What is Life?* (Cambridge University Press 1944) suggested that organisms avoid disordering (i.e. increase in entropy) by feeding on order; i.e. on *negative* entropy. From this it was a simple step from negentropy to order to information. Many valiant attempts were made, e.g. to calculate the “information content” of a DNA sequence, or even an entire organism (cf. for instance the volume *Information Theory in Biology* (edited by H. P. Yockey, R. L. Platzman and H. Quastler, (Pergamon, 1958); these essentially have come to naught, for reasons we have considered at great length above.

5.9 The Stability of Modeling Relations

The first four sections of the present volume were devoted to establishing the nature of the modeling relation, and exhibiting this relation in a profusion of particular cases. As we saw abundantly, a modeling relation can be regarded as an encoding of a natural system into a formal one, in such a way that the inferential structure of the formal system allows us to mirror the actual behavior in the system being modeled. The modeling relation thus establishes a kind of conjugacy between the system and the model; this enables us to employ the inferential structure in the model to make specific predictions about the system being modeled.

However, the present section has been devoted to exploring what must in effect be regarded as the failure of modeling relations, particularly in the context of dynamical models. We have argued that, since any model can only encode a fraction of the degrees of freedom, or interactive capabilities, of the system being modeled, it can only represent a closed subsystem of that system. Thus, the relation between a system and a model is the same as the relation between an open system and the same system when closed to particular environmental interactions. The result is a discrepancy which grows with time; a discrepancy between the inferential structure (and hence the predictions) of the model, and the actual behavior of the system to which these predictions are to be compared.

We have seen at length above how this growing discrepancy can be interpreted in terms of error, dissipation, disordering, and other similar phenomena, all of which represent a growing failure of the modeling relation; i.e. a growing loss of conjugacy between the actual temporal behavior of the system itself, and the predictions of that behavior made on the basis of a model.

Thus, it becomes necessary to reconcile the two basic propositions with which we have been concerned; namely: (a) modeling relations based on abstract encodings, can be established between natural and formal systems, and indeed provide essential tools of science; and (b) the properties of the modeling relation, particularly of dynamical models, tend to be lost in the course of time; i.e. the modeling relation initially established ultimately fails. This forces us to formulate problems relating to *the stability of the modeling relation itself*. In the process, we shall learn some

important new things about the modeling relation, which must ultimately reflect some very deep properties of our world, and our perceptions of it.

Let us begin to formulate the notion of the stability of a modeling relation, in the light of these remarks. For simplicity, we shall stay within the context of dynamical models, and in particular, we shall continue to regard the relation between system and model as the relation between an open system S and the same system S° closed to some modality of interactions.

Thus, we shall suppose that S° and S are encoded into a common set of states, which we shall assume to be a manifold X . Let us further suppose that the dynamics on S° can be encoded into a one-parameter family of automorphisms

$$T_t^a : X \rightarrow X,$$

where a denotes an appropriate set of constitutive parameters, and that the dynamics on S can be encoded into another one-parameter family of automorphisms

$$T_t^{a'} : X \rightarrow X$$

where a' is a different set of constitutive parameters. Then for each instant t , we can construct a diagram of the form

$$\begin{array}{ccc} X & \xrightarrow{T_t^a} & X \\ i_X \downarrow & & \downarrow \gamma(t) \\ X & \xrightarrow{T_t^{a'}} & X \end{array} \quad (5.31)$$

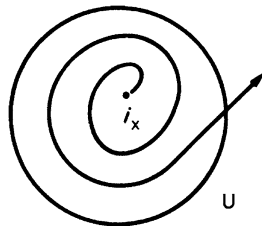
where i_X is the identity map on X , and $\gamma(t)$ is the map defined by

$$\gamma(t)(T_t^a(x_0)) = T_t^{a'}(x_0)$$

for each x_0 in X . Clearly, if the two dynamics were identical, then $\gamma(t) = i_X$ for every t . However, as we have seen, the two dynamics will generally not be identical; thus $\gamma(t)$ in effect measures the discrepancy between the *predicted* state $T_t^a(x_0)$ and the *actual* state $T_t^{a'}(x_0)$, when the system S and the model S° are started on the same initial state x_0 . It is clear that the map $\gamma(t) : X \rightarrow X$ is well defined, because T_t^a and $T_t^{a'}$ are automorphisms of X .

We can thus refer the discrepancy between the system S and the model S° under these circumstances to the discrepancy between $\gamma(t)$ and the identity map i_X . Namely, if $\gamma(t)$ is “close” to i_X in some appropriate metric, then the discrepancy at time t between the predicted behavior and the actual behavior will be small. Up

Fig. 5.4



to this measure of smallness, the predictions based on S° will approximate to the actual behavior of S , and we can say that the modeling relation between S° and S continues to hold. Of course, the *magnitude* of the map $\gamma(t)$ depends upon t ; thus, as t varies, $\gamma(t)$ will trace out some trajectory in the space $H(X, X)$ of mappings of X onto itself, and as we have seen, will ultimately leave any small neighborhood of the identity i_X in $H(X, X)$. As long as this trajectory $\{\gamma(t)\}$ remains in such a neighborhood, however, we may say that the modeling relation between S° and S holds to a *particular level of approximation*.

Let us look at this situation in another way. Let us suppose that the set $H(X, X)$ is given some appropriate metric. In this metric, consider the sphere U of radius $\varepsilon > 0$ about the identity map i_X . At time $t = 0$, we have by definition $\gamma(0) = i_X$. As t increases from 0, the map $\gamma(t)$ will trace out a trajectory or curve in $H(X, X)$, which we will suppose to be continuous in the metric on $H(X, X)$. As long as this trajectory stays within U , the discrepancy between $T_t^a(x_0)$ and $T_t^{a'}(x_0)$ will also be small in X , for all x_0 in X ; i.e. the predictions about S based on the model S° will be approximately verified. In other words, the diagram

$$\begin{array}{ccc}
 X & \xrightarrow{T_t^a} & X \\
 i_X \downarrow & & \downarrow i_X \\
 X & \xrightarrow{T_t^{a'}} & X
 \end{array} \tag{5.32}$$

which embodies the hypothesized modeling relation between S and S° , will *approximately* commute; i.e. will be insensitive to a replacement of i_X by $\gamma(t)$, for all values of t such that $\gamma(t)$ lies in U . We shall say that the modeling relation (5.32) is *stable* for all such values of t .

In a sense, the above ideas serve to define the stability of our modeling relation between S° and S in terms of the stability of the identity map i_X in $H(X, X)$ with respect to the dynamics

$$i_X \longrightarrow \gamma(t).$$

Diagrammatically, in $H(X, X)$, we have the situation shown in Fig. 5.4:

The situation we are describing must be carefully distinguished from that of the structural stability of the dynamics T_t^a . The above considerations are meaningful whether this dynamics is structurally stable or not; indeed, the question regarding the stability of the modeling relation (in which *time* plays the role of the “bifurcation parameter”) has little to do with the structural stability of T_t^a (in which the constitutive parameters a are the bifurcation parameters). However, there is an important sense in which the two concepts are related. We recall that if the dynamics T_t^a is structurally stable to the replacement of a by new constitutive parameters a' , then there are mappings $\alpha, \beta : X \rightarrow X$ such that the diagram

$$\begin{array}{ccc}
 X & \xrightarrow{T_t^a} & X \\
 \alpha \downarrow & & \downarrow \beta \\
 X & \xrightarrow{T_t^{a'}} & X
 \end{array} \quad (5.33)$$

commutes for all times t ; the mappings α and β depend only on the initial values and the perturbed values of the constitutive parameters (i.e. only on a and a'). This relation between the dynamics is one of *conjugacy*, and its interpretation is quite different from the situation diagrammed in (5.32) above. For in (5.32), we keep track of the states of S° and S through the *same* family of state variables for both systems; in (5.33) we must pass to *new* families of state variables for the system S , related to those describing the states of S° by the mappings α and β . However, let us observe that we can *reduce* (5.33) to the form (5.32) by rewriting it in the form

$$\begin{array}{ccc}
 X & \xrightarrow{T_t^a} & X \\
 i_X \downarrow & & \downarrow i_X \\
 X & \xrightarrow{\beta T_t^{a'} \alpha} & X
 \end{array} \quad (5.34)$$

In this form (5.34), we retain the same state variable for describing S° and S , but we must rewrite the dynamics on S ; i.e. we replace the original dynamics $T_t^{a'}$ by a new dynamics $\beta T_t^{a'} \alpha$, which directly incorporates the passage to new state variables into the dynamics itself.

The relation between the diagrams (5.33) and (5.34) allows us to make two general comments, each of which is important:

- (a) We have seen that the replacement of a closed system S° by an open system S (i.e. a perturbation of the constitutive parameters a to a different set a') changes

the dynamics, when both of them are viewed with respect to the same set of state variables. However, if the respective dynamics T_t^a , $T_t^{a'}$ are conjugate (i.e. if S° is structurally stable to this perturbation) then the discrepancy between them can be annihilated if we are willing to pass to a new set of state variables to describe S . That is, we can *renormalize* either our state descriptions (as in the diagram (5.33)) or the dynamics itself (as in the diagram (5.34)). The effect of this renormalization, from the standpoint of the mapping $\gamma(t)$, is basically to keep forcing it to coincide with i_X in the renormalized coordinates. We thus force i_X to be stable in $H(X, X)$, and hence we thereby stabilize the entire modeling relation (5.31) for all times t . We stress, however, that the price to be paid for this is the requirement to pass to a new encoding of S , different from that for S° . And obviously, if the dynamics T_t^a is not structurally stable to the perturbation $a \mapsto a'$, then no renormalization of this kind can stabilize the modeling relation for all times.

- (b) We recall that a conjugacy diagram of the form (5.33) already expresses a modeling relation between a pair of dynamics imposed on a manifold X . In this case, we can view both of these dynamics as pertaining to closed subsystems S° , $S^{\circ'}$ of *different* systems S , S' respectively. Invoking our previous results, the behaviors of S and S' will in fact be different from those encoded into S° and $S^{\circ'}$, and thus *the conjugacy relation between them will itself cease to hold under these circumstances*. However, by passing from the diagram (5.33) to the diagram (5.34), we can formulate the growing deviation from conjugacy in terms of the stability of the diagram (5.34). In this way, we can generalize our considerations of stability from its initial setting, and make it apply to *any conjugacy relation whatsoever*. We thus have an effective means of talking about the stability of *any* modeling relation between systems, and not merely of the stability of the dynamics of a single system S° when the system is opened to environmental interactions.

The further development of these propositions raises numerous interesting technical problems, which are of great importance for the epistemology of the modeling relation, but would take us too far afield to consider here. Their specific import for our main argument is that they draw attention to the *domain of stability* of a given modeling relation. More specifically, attention is drawn to a definite span of time over which a modeling relation like (5.31) or (5.33) remains stable. The very definition of stability means that, within this time span, the predictions of the model (i.e. the behavior of a closed system S°) gives us at least approximately correct information about the corresponding behavior of the system being modeled (i.e. the system S° opened to environmental interactions). When this time span is exceeded, we may say that the behavior of the model, and of the system being modeled, have become so different that a *bifurcation* has occurred between them; in that case, the model becomes useless for predictive purposes, and must be replaced by a different model in one way or another.

To talk in more detail about the span of time over which a modeling relation remains stable, it is helpful to reformulate the problem of stability in the other

equivalent languages we have developed. For the stability of a modeling relation means that, for practical predictive purposes, *we may replace an open system S by a certain closed subsystem S° ; or that we may replace a complex system by a simple one*. By definition, this replacement is *without significant visible effect*, modulo the encoding we are employing, for as long as the modeling relation remains stable. Our empirical experience with modeling relations, as exemplified by the many specific illustrations we have given in the preceding sections, indicate that the time interval over which stability persists can be substantial, even if it cannot be indefinitely long in general. Thus, although the behaviors of system and model will ultimately bifurcate, *they will in fact remain similar until bifurcation occurs*.

This fact of experience can be stated in another way. Until bifurcation occurs, a *complex* system may be replaced by a simpler system (the model) with no visible consequences, at least with respect to a particular encoding. During this period, for as long as the stability of the modeling relation persists, the complex system is in fact behaving *as if it were* a simple system. This means that as far as the modeled behavior is concerned, *only a few degrees of freedom, or interactive capabilities, of the complex system are actually involved in that behavior*. We may in fact elevate this assertion to a general principle governing the interaction of complex natural systems; namely, *given any specific interaction between such systems, there is a definite time interval in which this interaction involves only a few degrees of freedom of the interacting systems*. This means precisely that during such a time interval, the complex systems may be replaced by simpler (abstract) subsystems, or by models, without visible effect on that interaction.

The principle we have enunciated is perhaps the reason why science is possible at all. Indeed, for periods of time which are substantial to us, behaviors of complex systems do indeed appear as if they involved simple systems. These simple systems are themselves models for the complex systems they may replace, and further can be encoded into formal systems in a direct fashion. If it were not so, the world would appear as a total chaos, from which no laws, rules or principles could be extracted. For by definition, the formulation of scientific laws depends precisely on simple observable regularities which relate events, and such regularities are characteristic of simple systems. Indeed, if our measurement procedures could not isolate *single* degrees of freedom (i.e. single observables) the first step of the encodings on which science is based could never be made.

However, as we have seen, we cannot *indefinitely* expect a complex system to behave like a simple one, nor an open system to behave like a closed one. Ultimately, the two behaviors will bifurcate from one another. At such a time, we must replace our model by another; i.e. by another *simple* system, which encodes the degrees of freedom then dominating the interaction. From our point of view, as we have stated above, it will appear that *new properties of the system have emerged*; indeed, the phenomena of emergence so prominent in biology and elsewhere are nothing but the bifurcations between the behavior of a complex system and a simple model of it. In general, such a bifurcation represents the failure of linkage relations which previously existed, and the establishment of new ones; it is the encoding of these which lead us to the new model with which our original model must be replaced.

To these considerations, we must append a few general remarks. The first is this: all our considerations were based on the restricting of attention to a single behavior of our complex system. Any such behavior may be treated in the fashion we have indicated above, but different behaviors of the same system already require different encodings; i.e. different models. There need be no relation between the encodings appropriate to different behaviors of the same system, as we have abundantly seen. Thus, each modeling relation we establish, pertaining each to a distinct system behavior, will have its own domain of stability. We cannot therefore generalize about the stability of arbitrary system behaviors from the properties of a single one.

The second general remark is this: by definition, we cannot know the domain of stability of a particular modeling relation *from within the context of that modeling relation itself*. The bifurcation between system behavior and model behavior which we have described is itself an emergent novelty from the standpoint of the simple model; the basis for it has by definition been abstracted away in the very act of formulating the model. In order to *predict* this domain of stability, we need in fact another description; i.e. another model, with respect to which the original model will bifurcate. Indeed, this was the procedure we used when we initiated our discussion, and why we chose to do so in the context of closed and open systems. All we can say in general is that the domain of stability of a given modeling relation is *finite*; but we cannot in general predict the magnitude of this domain except through the agency of another model.

Chapter 6

Anticipatory Systems

6.1 General Introduction

At the very outset of this work, in Sect. 1.1 above, we tentatively defined the concept of an anticipatory system; a system containing a predictive model of itself and/or of its environment, which allows it to change state at an instant in accord with the model's predictions pertaining to a later instant. Armed as we now are with a clearer idea of what a model is, we may now return to this concept, and attempt to obtain a deeper understanding of some of the principal properties of such systems. It is well to open our discussion with a recapitulation of the main features of the modeling relation itself, which is by definition the heart of an anticipatory system.

We have seen that the essence of a model is a relation between a natural system S and some suitable formal system M . The modeling relation itself is essentially a linkage between behaviors in S and inferences drawn in M become predictions about the behavior of S . This linkage is itself expressed in terms of a commutative diagram of mappings of the kind we have seen abundantly in the preceding section, and which in mathematical terms expresses a *conjugacy* between the properties of S , and the properties of M which are linked to it.

In the last analysis, a modeling relation of this kind is established through an *encoding* of qualities or observables of S into formal or mathematical objects in M . These mathematical objects serve as labels or symbol vehicles which symbolically represent the qualities of S with which they are associated. The very definition of a system implies that such encoded qualities are *linked* to one another in definite ways; such linkages are represented in M by relations satisfied by the objects which represent the qualities. These relations are called *equations of state* for the system. The rules of inference in M allow us to draw conclusions from these relations, which as we have seen then become predictions about behaviors of S .

We have also seen that a relation of analogy is established between two natural systems S_1 , S_2 which can be encoded into the same formal system M . To the extent that an analogy exists between two such systems, each of them can be regarded as

a model for the other. Analogous systems can thus also be regarded as *alternate realizations* of a common formal system M .

We found that any encoding of a natural system S into a formal system M involved an act of abstraction, in which non-encoded qualities are necessarily ignored. Such an act of abstraction is not peculiar to theory; indeed, any act of observation (on which any kind of encoding must be based) is already an abstraction in this sense. What is captured by such an abstraction is thus not all of S , but rather a subsystem. In this sense, analogous systems are those which share a common subsystem. Almost by definition, however, any behavior of S which involves qualities not encoded into a particular model M is in principle unpredictable from M .

The relationship between M and S can thus be regarded as expressing what the behavior of S *would* be if S were closed to interactions involving non-encoded qualities. The actual relation of M to S is thus seen to be that between a system open to modalities of environmental interaction, and the same system isolated from those modalities. In the context of dynamical models, we saw that in general there will be a discrepancy between the behaviors of S and the corresponding behavior predicted by M , and that this discrepancy will generally grow in time. The character of this discrepancy is, of course, unpredictable from within M , since the basis for it has been abstracted away in the process of the encoding from which M was generated. It is the growth of such discrepancies which is responsible for phenomena associated with error and “disorder”, and also with the emergence of novel behaviors (as judged from the standpoint of M) in complex systems. The root of these discrepancies lies in the appearance of bifurcations between corresponding behaviors of M and S over time, and was summed up in the principle: no natural system S is indistinguishable from a closed subsystem of itself.

Now let us see what is necessary in order to extend this discussion of the modeling relation, in such a way that it can be brought to bear on the characterization and study of anticipatory systems. Let us first observe that, according to our characterization of analogy between natural systems, two natural systems S_1 , S_2 are analogous precisely when they may be encoded into a common formal system M ; i.e. precisely when we can construct a diagram of the form for suitable encodings E_1 , E_2 .

$$\begin{array}{ccc} S_1 & \xrightarrow{E_1} & \\ & \searrow & \\ S_2 & \xrightarrow{E_2} & \Rightarrow M \end{array} \quad (6.1)$$

As we have seen, this means precisely that the natural systems S_1 , S_2 possess a common closed subsystem. Thus, it is certainly true that S_1 contains a model of S_2 , and that S_2 contains a model of S_1 ; indeed, in formal terms, the relation $E_2^{-1}E_1$ can be regarded as an encoding of the states of S_1 into states of S_2 ; likewise, the relation $E_1^{-1}E_2$ represents an encoding of the states of S_2 into the states of S_1 . Thus, the relation of analogy always gives rise to an essential feature of anticipatory systems; namely, that such a system possesses a model of another system.

However, the mere fact that a model of S_1 can be found in S_2 , while clearly necessary if S_2 is to be an anticipatory system, is far from sufficient to make it so. As we recall, the decisive feature of an anticipatory system is that, in effect, it employs its model to make predictions about S_1 , and more important, change of state occurs in S_2 at an instant as a function of its predictions. Let us see, in a rough and heuristic way, how this decisive feature may be incorporated into our discussion; details and examples will be presented in subsequent chapters of this section.

First, it is trivially clear that those aspects of S_2 which comprise or embody the model of S_1 must not exhaust all of S_2 . That is, there must be qualities of S_2 which are not related to the encoding E_2 displayed in (1) above. In terms of that diagram, we may say more formally that $E_2^{-1}E_1(S_1) \neq S_2$.

The question then arises: how are the qualities of S_2 which are encoded into M related to those which are not so encoded? To answer this, let us suppose that $f : S_2 \rightarrow \mathbb{R}$ is an observable of S_2 not encoded into M , and let us consider what would happen if there were a linkage of M to f . Then by definition, given an arbitrary m in M , and an arbitrary number r in $f(S_2)$, there need be no state of $s \in S_2$ such that $E_2(s) = m$ and $f(s) = r$. Stated another way, fixing a value $r = f(s)$ will make some elements of M *inaccessible* if E_2 is linked to f . This follows immediately from the fact that in these circumstances the mapping

$$f \times E_2 : S_2 \rightarrow \mathbb{R} \times M$$

is into and not onto. Intuitively, this means that the *full model M of S_1 is not accessible from states of S_2 encoded in this fashion*. Thus, if every observable f of S_2 were linked to those comprising M , it would follow that M could not in fact be a model of S_1 , which contradicts our initial hypothesis.

Thus, the set of all observables of S_2 unlinked to those in M is not empty. It follows from this that we can in fact construct an encoding of S_2 which is precisely of the form

$$S_2 \rightarrow M \times X \tag{6.2}$$

where X is a manifold, co-ordinatized by observables f such that M is unlinked to f . That is, the mapping (6.2) is *onto*. The existence of an encoding of the form (6.2), in which there is an orthogonality between the subsystem of S_2 comprising the model, and other observables of S_2 to which the model is unlinked, is the second necessary condition which must be satisfied if S_2 is to be an anticipatory system.

A third condition follows now from the fact that the model M serves to influence present change of state in S_2 . This can be interpreted as follows: although the observables defining M are unlinked to those defining X in (6.2) above, a definite linkage exists between the observables of M and the *rates of change* of the observables of X . Thus, the behavior of S_2 , as viewed through the encoding into X alone, will depend on M as well as on X . In a certain sense, the encoding of X alone defines thereby a closed subsystem of S_2 , whose properties describe *how S_2 would behave if there were no model*. Indeed, we may think of X as a closed system, and $X \times M$ as a corresponding open system; we may then apply the

results of Sects. 5.4 and 5.5 directly to this situation. As we saw, there will be a growing discrepancy between the behaviors of X and XxM ; in the present case, this discrepancy represents precisely *the effect of the model* M on the behavior of S_2 .

This then represents the third necessary condition which is required if S_2 is to represent an anticipatory system: namely, the state of the model M must modify the properties of other observables of S_2 (namely, those involved in the encoding of S_2 into the manifold X). Indeed, we can regard the total encoding (6.2) of S_2 as follows: the elements of M represent particular values of constitutive parameters of S_2 ; to each $m \in M$ there is a copy of the manifold X , and the behavior in any such manifold depends on the value of M with which it is associated. The difference in behavior between two such copies of X is precisely the discrepancy we encountered before. We may ourselves anticipate future developments by calling the change in behavior as seen in X , arising from the particular element of the model M involved, an *adaptation* of X (or more accurately, in the case of a predictive model, a pre-adaptation).

Next, if we suppose that S_2 is modifying its present state (i.e. its behavior as seen through the encoding of S_2 into X) on the basis of a model, there would be no objective significance to this modification unless S_2 were interacting (or at least potentially interacting) with S_1 itself. For M is a model of S_1 by hypothesis; clearly, to attach significance to the conditions we have developed, it is necessary that the modification arising in S_2 have something to do with S_1 . We will take the point of view that *the above described change in S_2 , arising from the predictive model of S_1 , must be manifested in a corresponding change in some actual or potential interaction between S_1 and S_2* . In other words, there must be some discrepancy between the interaction between S_1 and S_2 which actually occurs, and the interaction which would have occurred had the model not been present. This represents a fourth condition required to turn S_2 into an anticipatory system.

Finally, we must now introduce some dynamical considerations, to capture the idea that M is a *predictive* model. To do this, we must recall some properties of temporal encodings of dynamics, as they were described in Sect. 4.5 above. Let us suppose that $T_t : S_1 \rightarrow S_1$ is an abstract dynamics on S_1 . If M is to be a dynamical model of this abstract dynamics, then there must exist a dynamics $\bar{T}_{x(t)} : M \rightarrow M$ such that the diagram

$$\begin{array}{ccc}
 S_1 & \xrightarrow{T_t} & S_1 \\
 E_1 \downarrow & & \downarrow E_1 \\
 M & \xrightarrow{\bar{T}_{x(t)}} & M
 \end{array} \tag{6.3}$$

commutes. In this dynamics, the instant $x(t)$ is that which *corresponds* to the instant t ; it will be recalled that, in general, the time scales for the two dynamics will not be the same, and that a conversion factor (here denoted by x) is necessary between them.

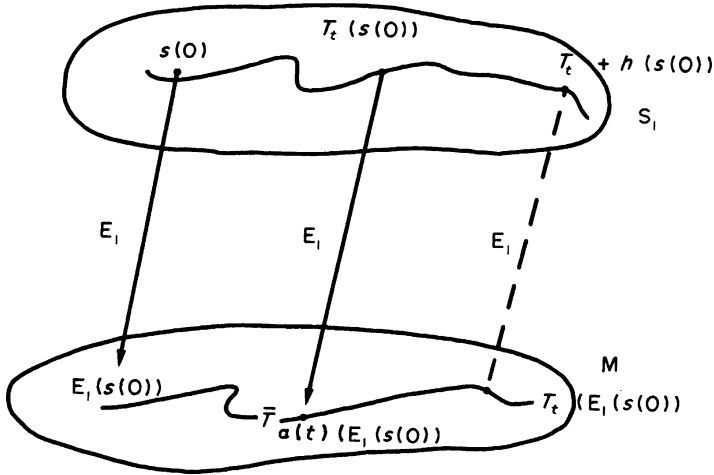


Fig. 6.1

If we ignore the conversion factor, and attempt to use the time scale of the dynamics T_t on S_1 as an *absolute* scale, then T_t and $\bar{T}_t$ are in general not conjugate. This is because with respect to any absolute time scale, the trajectories in S_1 and the corresponding trajectories in M are traversed at different rates. The temporal relations between such corresponding trajectories are illustrated diagrammatically in Fig. 6.1.

In this figure, we have supposed that $\alpha(t) < t$. Accordingly, suppose that the system S_1 is initially in state $s(0)$, and the model M is in the corresponding state $E_1(s(0))$. Then after a time t has elapsed for S_1 , this system will be in state $T_t(s(0))$. If we use the *same* scale for M , then M will at this instant be in state $\bar{T}_t(E_1(s(0)))$. As we can see from the figure, these two states *do not correspond*; in fact, M is now in a state *for which the corresponding state will not be reached in S_1 until further time has elapsed*. In other words, if we look at the state of the model M at time t , then this state of M actually encodes a state which S_1 will not reach until time $t + h$, where all times are referred to S_1 . Intuitively, with respect to this time scale, the trajectories in M are traced out faster than are the corresponding trajectories in S_1 ; thus the state of M at an instant actually refers to the state of S_1 at a later instant. It is in this sense that we can regard M as a predictive model; the present state of M (i.e. its state at an instant t) is actually an encoding of a future state of S_1 , when referred to the same time scale. In this scale, by looking at the *present* state of M , we obtain information pertaining to a *future* state of S_1 .

Thus we arrive at our fifth and final condition for S_2 to be an anticipatory system: the model of S_1 which is encoded into S_2 must be a predictive model, in the sense we have just defined.

These conditions may be succinctly summed up as follows: An anticipatory system S_2 is one which contains a model of a system S_1 with which it interacts.

This model is a predictive model; its *present* states provide information about *future* states of S_1 . Further, the present state of the model causes a change of state in other subsystems of S_2 ; these subsystems are (a) involved in the interaction of S_2 with S_1 , and (b) they do not affect (i.e. are unlinked to) the model of S_1 . In general, we can regard the change of state in S_2 arising from the model as an *adaptation*, or pre-adaptation, of S_2 relative to its interaction with S_1 . The conditions we have enunciated above establish precisely these properties.

Now let us focus our attention on the specific effect of a predictive model on an interaction. What we shall do is, as before, compare the actual interaction which occurs in an anticipatory system with the interaction which *would have occurred* if there were no predictive model; i.e. if the subsystem M of S_2 were absent. At the present level of generality, we can do this only in a correspondingly general way. Our approach will be to employ a modified version of the arguments used in a different connection in Sects. 5.4 and 5.5 above; namely, we shall argue that *there is some property P of S_2 which is satisfied when S_2 is anticipatory, and which is not satisfied when S_2 is non-anticipatory.*

Before taking up the specific argument required to establish this result, let us spend a moment assessing its significance. If such a property P can be identified, then we must say that the *role of the predictive model is to stabilize this property*; or, to put it another way, that the anticipatory behavior serves as a homeostat for the property P. To use another language: the predictive model *prevents a disordering* of S_2 , with respect to the property P in question, which by virtue of its interaction with S_1 would otherwise occur. Indeed, it is *only* with respect to the stabilization of a property P that we can refer to the anticipatory change of state, occurring in S_2 in accordance with the state of its predictive model, as an adaptive change, or more precisely as a *pre-adaptation*. Seen in this light, the anticipatory change of state in S_2 , determined by the state of the predictive model S_1 in S_2 , serves precisely to modulate the interaction between these two systems in such a way that the property P is retained. We shall have a great deal more to say about these notions of adaptation and pre-adaptation as we proceed. We shall also have occasion to consider the following kind of converse question: *given* a property P which we wish to stabilize in S_2 , can we convert S_2 into anticipatory system in such a way that the given property becomes precisely the one stabilized in the above sense? As will be seen, this is the central problem in what might be called the theory of predictive control.

With these words of motivation, let us return to the problem of finding a property P which is stabilized by the predictive model of S_1 in S_2 . But this is now very easy; indeed, we shall simply exploit the character of the discussion already given in Sect. 5.8 above. Let us recall that the presence of the model M causes the interaction between S_1 and S_2 to be *different* from what it would have been, had the model not been present, and had it not been capable of pre-adapting the states of S_2 . Moreover, we recall that the effect of any interaction between systems is to establish new linkages between particular observables of these systems. As we saw in Sect. 5.8, the specification of a linkage is equivalent to the characterization of an associated property P. Accordingly, we shall merely identify the property P in question with

the linkages established through the interaction between S_1 and S_2 which actually occurs as a consequence of the predictive model M ; by its very definition, this property is stabilized by the presence of the model, and will be lost if the model is not present. Hence there is always at least one property P which is stabilized by the presence of a predictive model in an anticipatory system. And if there is one such property, there will always be many others; namely, those which are in an obvious sense *linked* to P . These facts will be most important to us as we proceed.

It is now instructive to compare the conditions developed above to characterize a feedforward system with the heuristic discussion of Chap. 1 above, and in particular, with the diagram of such a system illustrated in Fig. 1.1. It will be seen that all the features of this diagram are in fact embodied in the conditions we have posited. What we have called the anticipatory system S_2 in the above discussion comprises the three boxes labelled respectively M , E and S in Fig. 1.1; M represents that encoding of S_2 which comprises the model, S represents that encoding of S_2 which we called X , and E denotes the linkage between M and the dynamics in X through which a state of M establishes an adaptation, or pre-adaptation in X (and what we earlier called the *effectors* of the system). What we called the environment in Fig. 1.1 is to be identified with the system S_1 . Indeed, the only difference between the two discussions is the following: in Fig. 1.1, we tacitly supposed that M , E and S were three *different* systems, with M controlling S through the effectors E ; on the other hand, in the above discussion, M and S represented different encodings of the *same* system, and E corresponds to a linkage between the observables coded into M and the rates of change of those involved in the interaction with S_1 . We thus see that our initial heuristic discussion of Sect. 1.1 was in fact well founded in terms of the more comprehensive conceptual treatment of modeling relations which we have developed.

The properties of anticipatory systems will be central to all of our subsequent discussion. Our aim is to develop at least the rudiments of a theory of anticipatory systems, whereby the behavior of such systems can be characterized. Most important, we wish to contrast the behavior of anticipatory systems with those of the reactive systems which have been our main preoccupation thus far, and thereby lay the foundations for what may be called an *anticipatory paradigm* for the treatment of such biological phenomena as adaptation, learning, evolution and other basic organic behaviors. Such a paradigm provides an important alternative to the pure *reactive paradigm*, on which all approaches to such behaviors have heretofore been based. Furthermore, we wish to develop at least some of the implications of such an anticipatory paradigm for human systems, through its bearing on questions of social organization, and its implications for social technologies involving planning and forecasting. These will be our major preoccupations for the remainder of the present volume.

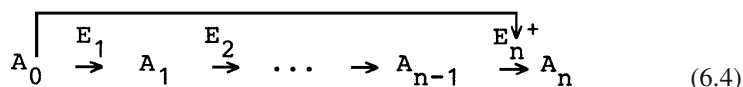
Let us then outline the character of the discussion to follow. In Sect. 6.2, we shall consider in detail perhaps the simplest example of an anticipatory system. This is an example which was already briefly introduced in Sect. 1.1, and involves a forward activation step in a biosynthetic pathway. In particular, we shall see how the general considerations regarding anticipatory systems which were developed above appear

in a concrete situation. In Sect. 6.3 we shall once again generalize from this example, with an emphasis on the property P which is stabilized in such a system, and the manner in which this property can be related to functional considerations bearing on fitness and survival. Section 6.4 will examine the implications of the stability of modeling relations which was developed in Sect. 5.9, for the long term behavior of anticipatory systems. In particular, we shall show that such systems can manifest novel modes of “system failure”, characterized by the loss of properties (linkages) in the system as a whole, without anything which could be identified as a *local failure* occurring in any subsystem.

As an application of these ideas, we shall consider their bearing on the senescence of organisms and human cultures. In Sect. 6.5, we shall return to the general problem of adaptation, and develop the ideas of selection and evolution in the context of anticipatory systems. In Sect. 6.6, we shall turn to the general question of the interaction between two systems which are both anticipatory; this will lead us into problems bearing on *conflict* and *conflict resolution* in the human realm. Finally, in Sect. 6.7, we shall consider what may be called the *ontogenesis* of anticipatory systems, and the general phenomenon of *learning*.

6.2 An Example: Forward Activation

In the present chapter, we shall be concerned with developing in some detail the properties of a particular example which we have seen before; namely, the biosynthetic pathway

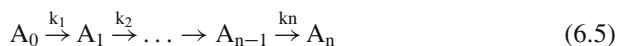


with a forward activation step, as indicated. In such a pathway, the initial substrate A_0 also serves as an activator of the enzyme E_n ; thus, the rate at which E_n catalyzes its reaction is a function of how much initial substrate A_0 is present. We have already indicated that such a pathway constitutes a simple example of an anticipatory system; we shall here consider its properties, not only as illustrations of the concepts developed above, but as motivation for subsequent theoretical developments.

Let us begin by reviewing the sense in which the pathway (6.4) actually constitutes an anticipatory system. We recall that, by virtue of the forward activation step, the concentration $A_0(t)$ of the initial substrate A_0 , at any instant t , serves also as a *predictor* of the concentration $A_{n-1}(t+h)$ of the substrate of E_n at a later instant $t+h$. Therefore, by modulating the rate of the enzyme E_n , the initial substrate A_0 serves to adapt, or better, to pre-adapt, this enzyme so that it will be competent to process its own substrate A_{n-1} at that future time.

We may now formulate these heuristic remarks about the system (6.4) in precise mathematical terms. We begin by observing that the function of the pre-adaptation of enzyme E_n is obviously to prevent the accumulation of the intermediate A_{n-1} in the face of ambient fluctuations of the initial substrate A_0 . We shall regard this feature of the forward activation step as precisely being the property P of the system which is in fact stabilized by the pathway. This represents a biologically plausible situation, and it illustrates the manner in which such properties can be immediately related to the biological concept of function.

As a first approximation, we can regard the pathway (6.4) as a simple series of first-order chemical reactions; i.e. as a system of the form



in which k_i denotes the apparent rate of the i th reaction in the pathway. We shall treat all of these reaction rates k_i as constant, except for the last one k_n ; this last rate shall be taken as a definite function $k_n(A_0)$ of the initial substrate concentration. The explicit functional dependence of k_n on A_0 will thus embody the forward activation step of the system (6.4) in the system (6.5).

With these hypotheses, the rate equations for the system (6.5) can be immediately written down; they are:

$$\begin{aligned} dA_i/dt &= k_i A_{i-1} - k_{i+1} A_i, \quad i = 1, \dots, n-1 \\ dA_n/dt &= k_n A_{n-1} \end{aligned} \quad (6.6)$$

where we suppose that $A_0 = A_0(t)$ can be taken as an arbitrary function of time.

Let us consider first the specific rate equation governing the behavior of the intermediate A_{n-1} ; namely

$$dA_{n-1}/dt = k_{n-1} A_{n-2} - k_n A_{n-1}. \quad (6.7)$$

The hypothesis that A_{n-1} is not to accumulate can be embodied in the requirement that dA_{n-1}/dt shall be non-positive; for simplicity, we shall take the stronger condition that in fact dA_{n-1}/dt shall always be zero, independent of the level of the initial substrate A_0 . Thus, from (6.7), we require that the condition

$$k_{n-1} A_{n-2} = k_n A_{n-1} \quad (6.8)$$

shall hold. We attempt to achieve this condition by making k_n an explicit function of A_0 ; the *form* of this function will embody the model implicit in the forward activation loop.

Now we observe that the system of rate equations (6.6) is a linear system, of the type we considered in Sect. 1.2 above; in fact, these equations are precisely of the form (1.3). According to (1.7), we can write the concentrations of the reactants A_{n-2} , A_{n-1} appearing in (6.8) at any instant of time t , as explicit functions of A_0 :

$$\begin{aligned}
 A_{n-2}(t) &= \int_0^t K_1(t-\tau)A_0(\tau)d\tau; \\
 A_{n-1}(t) &= \int_0^t K_2(t-\tau)A_0(\tau)d\tau.
 \end{aligned}
 \tag{6.9}$$

Here K_1 and K_2 are determined entirely by the appropriate rate constants k_i appearing in the equations (6.6); their precise form is not important for us here, but the reader may find it a useful exercise to compute them according to the discussion given in Sect. 1.2. The main thing to note is that the values of these reactants at any instant t is determined by the value of A_0 at earlier instants; or, stated another way, that the value of A_0 at a given instant determines the values of the reactants A_{n-2} and A_{n-1} at later instants according to the reaction scheme (6.6).

We now express the control condition 6.2.5 in terms of the explicit relations (6.9). We find that (6.8) will be satisfied if we put

$$k_n = k_{n-1} \frac{\int_0^t K_1(t-\tau)A_0(\tau)d\tau}{\int_0^t K_2(t-\tau)A_0(\tau)d\tau}. \tag{6.10}$$

That is: the rate k_n of the final step of (6.5), at any instant of time t , is determined by the value of the initial substrate A_0 at a *prior* instant $\tau = t - h$. Equivalently, the value of A_0 at a particular instant τ determines k_n at a *subsequent* instant $t = \tau + h$.

Thus, assuming that the behavior of the intermediates A_{n-2} , A_{n-1} is in fact governed by the rate equations (6.6), a specification for the forward activation step in accordance with (6.10) will guarantee that the control condition (6.8) is satisfied; i.e. will assure that the whole pathway satisfies the property P . Under these circumstances, we can see the exact sense in which the concentration of the initial substrate A_0 at an instant serves to *pre-adapt* the pathway, so as to stabilize this property. The pre-adaptation itself is what we earlier called the *competence* of E_n to maintain the property P ; indeed, we should stress that terms like adaptation and competence are only meaningful relative to some property P which has been previously specified.

Let us now reconsider this specific example in terms of the general discussion of the preceding chapter. In particular, using the terminology of Sect. 6.1, let us put

$$\begin{aligned}
 A_{n-1} &= S_1, \\
 A_0 &= M, \\
 E_n &= X, \\
 (A_0) \times (E_n) &= M \times X = S_2.
 \end{aligned}
 \tag{6.11}$$

That is: the anticipatory system, which is called S_2 in the preceding chapter, consists of two parts: (a) the part comprising the model M , which in the present case is A_0 , and (b) the part comprising the interactive observables X , which is the present case

is the enzyme E_n . The system S_1 , with which S_2 interacts, and which M models, is in this case the substrate A_{n-1} of E_n . As before, the model M is not linked to the observables in X (i.e. the value of A_0 is not dependent upon the state of the enzyme E_n), but A_0 does determine the reaction rate of the enzyme; e.g. by inducing a conformational change in enzyme molecules which modifies their rates in accordance with (6.10). With respect to the property P , it is this conformational change, or rather, its manifestation as a change in rate, which is precisely the crucial pre-adaptation step. It is now immediate to verify that, with these identifications, the pathway (6.4) does indeed satisfy all the conditions of an anticipatory system which were developed in the preceding chapter.

It is worthwhile to look more closely at the roles of the initial substrate A_0 in this pathway. Clearly, A_0 plays a double role here: it is on one hand a *substrate* for the entire pathway, and on the other hand it is a *predictor* for subsequent values of the specific substrate A_{n-1} of the enzyme E_n which it activates. As a predictor, *its role is essentially a symbolic one*; it represents, or encodes, or labels, a *future* property of A_{n-1} . In this role, A_0 serves essentially as a *linguistic* object; it is a symbol, or more precisely, a symbol vehicle, for the value of A_{n-1} to which it is linked by (6.9) above. In a precise sense, A_0 in its capacity as activator of E_n is a *hormone* ("messenger"). This terminology is provocative in the present context, but it is accurate, and is intended to be suggestive.

Let us also note that the homeostasis maintained in the pathway (6.4) is obtained entirely through the modeling relation between A_0 and A_{n-1} (i.e. entirely through symbolic processes) by virtue of the relation (6.10) which links the prediction of the model to the actual rate at which E_n operates. That is, the homeostasis is maintained entirely through pre-adaptation generated entirely on the basis of a *predicted* value for A_{n-1} . In particular, there is no *feedback* in the pathway, and no mechanism available for the system to "see" the value of the quantity which is in fact controlled, or indeed any feature beyond that embodied in the predictive model relating A_0 to A_{n-1} .

Let us make one further observation. The discussion so far has been couched in terms of a property P , which can be defined entirely in terms of the pathway (6.4) considered as an independent system. But in fact, such a pathway can be regarded as a single component in a larger biochemical network, which itself comprises a functional unit of an intact cell. Thus, the property P of the pathway can be directly related to properties P' of the successively larger systems with which it interacts, and ultimately to properties pertaining to the *viability* of the cell itself. Conversely, we may say that, at the level of the intact cell, there will be properties pertaining to its viability which are directly linked to the property P in the pathway (6.4). This kind of "cascading" of properties, by means of which properties of subsystems can be linked to properties of larger systems not directly visible from within the subsystem, will become most important to us shortly.

We shall now develop a further crucial property of the pathway (6.4) which is a direct consequence of its anticipatory character. To do this, let us return to the relation (6.8) above which will be seen to have two distinct but related meanings. On the one hand (6.8) expresses the actual behavior of the pathway, as seen from

the standpoint of its chemical kinetics, *as a consequence of the hypotheses we have made about the reactions in the pathway*. On the other hand, this same linkage is built, via the relation (6.10), into the forward activation step. *It is clear that the forward activation itself will retain its adaptive character (with respect to the property P) only as long as these two meanings coincide*. In other words: if the actual chemical kinetic behavior of the pathway should cause the linkage between the initial substrate A_0 and the product A_{n-1} to depart from that expressed in (6.8), the forward activation step governed by (6.10) will no longer be adaptive with respect to P, but will rather have become *maladaptive*.

Stated another way: a particular linkage (namely (6.9)) between the predictor A_0 and the substrate A_{n-1} is “wired in” to the forward activation step of the pathway (6.4); this is expressed precisely by the relation (6.10). On the other hand, these quantities are *actually* linked by the detailed kinetic properties of the various reaction steps in the pathway. As long as the actual linkage between them is the same as that which is “wired in” to the forward activation step, the property P will be maintained, and the rate changes in E_n governed by (6.10) will be adaptive responses to fluctuations in A_0 with respect to the property P. However, if there should be a departure between the actual chemical kinetic linkage between A_0 and A_{n-1} from that expressed in (6.10), those same rate changes in E_n will represent maladaptive responses, to a degree measured by the departure of the actual linkage from that “wired in” to the activation.

How could such a deviation come about? Ultimately, any such deviation can be represented by changes in the apparent reaction rates k_i governing the individual steps in the pathway (6.10) (which, we recall, enter as constitutive parameters into the linkage (6.8)), or by the presence of sources and sinks for the intermediate reactants A_i in the pathway, beyond those which are explicitly represented in the rate equations (6.6). The latter would correspond to *side reactions*, competing with those occurring in the pathway; the former could arise from such things as changes in ambient temperature, or from changes in the rates of the intermediate enzymes E_i . Thus, the deviations we are contemplating arise because the pathway (6.6) is in fact immersed in a larger system, and is open to interactions with that system in ways which are not explicitly represented, either in the pathway (6.4), or in the kinetic equations (6.6) which describe that pathway.

But now the reader may observe that our discussion is taking a familiar form. In fact, it is nothing other than a reformulation of the results of Sect. 6.5 above, in the context of the special case we are now considering. In Sect. 6.5, we showed that there will always arise discrepancies between the behavior of a system predicted on the basis of a model, and the actual behavior of that system; these discrepancies arise precisely because the actual system is open to interactions to which the model is necessarily closed. Relative to some appropriate property P, these discrepancies result in errors or disordering; or in the present context, to a growing maladaptation of the pathway. And as usual, the character of the discrepancies, and indeed, the maladaptation itself, is unpredictable and invisible from within the pathway itself.

Thus, we can conclude that a forward activation pathway like (6.4) can only retain its adaptive character, or its competence to maintain the property P, for a

characteristic time. The time interval during which competence is retained does not depend on the pathway itself, but on the nature of the larger system in which it is embedded, and on the character of its interactions with that larger system. We may sum this up by concluding that the property P, characterizing the adaptive aspects of the pathway's behavior, is *temporally spanned*. The time interval during which it persists may be quite long, depending on the character of the ambient environment of the pathway, but it is inevitably *finite*.

The same may of course be said for any properties linked to P, including those pertaining to the larger system in which the pathway is embedded.

It must be emphasized that the temporal spanning of the property P which we have described does not arise from any localizable fault or failure attributable to any step within the pathway (6.4) itself. If we were to isolate each enzyme in the pathway, and each substrate, it would be found individually to be identical in its properties to the same enzyme or substrate at a time when the property P is maintained by the pathway. Likewise, the forward activation step, considered in isolation, is identical whether the property P is maintained or not. Thus, if we identify the loss of the property P with a mode of *system failure* in the pathway (6.4), that system failure cannot be attributed to any localized failure in any one of the individual steps in the pathway. Rather, it represents a kind of global failure, which has its roots in the very nature of the modeling relation. As we shall shortly see, this kind of temporal spanning, considered as a form of system failure, is an intrinsic property of anticipatory systems, which has no analog in non-anticipatory or reactive systems. As we shall see in subsequent chapters, it is responsible for the most far reaching characteristics of biological systems at all levels, and allows us to understand in a unified way many puzzling features of their behaviors.

6.3 General Characteristics of Temporal Spanning

We shall now proceed to recast the considerations of the preceding chapter into a more general form. This generalization will involve two rather distinct parts: (a) the replacement of the specific biosynthetic pathway (6.4) by a more general system, and concomitantly, the replacement of the forward activation of the pathway by a more general anticipatory signal, and (b) the replacement of the property P stabilized by the pathway by more general properties, pertaining not only to the system itself, but to larger systems in which the given one is embedded.

Let us begin by generalizing the relevant properties of the enzyme E_n in the pathway (6.4). In that pathway, E_n served as a transducer



which converted its substrate A_{n-1} to its product A_n at a definite rate. In its simplest terms, this transduction was described by the linkage relation

$$dA_{n-1}/dt = k_n A_n \quad (6.13)$$

which expresses how the rate of change of A_n is linked to the instantaneous value of the substrate concentration.

We shall generalize the enzymic transducer of chemical qualities diagrammed in (6.1) by the more general transducer.

$$\vec{u} \rightarrow \boxed{\varphi} \rightarrow \vec{v} \quad (6.14)$$

Here, we suppose that $\vec{u} = (u_1, \dots, u_r)$ is an *input vector* to the transducer (and as such, generalizing the substrate A_{n-1} of (6.12), and $\vec{v} = (v_1, \dots, v_s)$ is the corresponding *output vector* (hence generalizing the product A_n). As usual, the quantities u_i, v_j denote specific observables of the natural system on which the transducer acts, and we assume that their values are explicit functions of some appropriately defined dynamical time t .

The transducer φ itself, which generalizes the enzyme E_n , must be characterized by an appropriate *equation of state*, which links the input and output vectors. We will take this equation of state to be of the general form

$$\vec{v} = \varphi(\vec{u}) \quad (6.15)$$

which is an obvious generalization of (6.13). We leave open for the moment the question of the time dependence in (6.15), in the sense that we allow the possibility that (6.15) links the values of the input vector $\vec{u}$ at an instant with those of the output vector $\vec{v}$ at some later instant.

So far, we have not incorporated the forward activation step of our pathway. As we recall, the activation step was represented by making the rate constant k_n in (6.13) a specific function of a *predictor* A_0 . There were two aspects involved in doing this; first, there was the positing of a linkage between the present value of the predictor A_0 and a later value of the actual input A_{n-1} ; second, the *present* value of the predictor A_0 was able to *pre-adapt* the transducer E_n in a way which depended on the *predicted* value of the substrate A_{n-1} . This last was accomplished by expressing the rate constant k_n in (6.13) as an explicit function of the predictor; i.e. by writing

$$k_n = k_n(A_0). \quad (6.16)$$

Let us now incorporate these features into the more general situation we are constructing.

We accomplish the first of these steps by introducing a general predictor $\vec{a} = (a_1, \dots, a_m)$ to play the role of A_0 in our simple pathway. We assume as before that the *present* value of this predictor is linked with a future value of the input $\vec{v}$ to the transducer. The second step is accomplished by supposing that the *equation of state* (6.15) describing the transducer depends on the predictor $\vec{a}$. This will be denoted by rewriting (6.15) in the more general form

$$\vec{v} = \varphi_{\vec{a}}(\vec{u}). \quad (6.17)$$

For the sake of convenience, we have omitted from this equation of state the crucial time dependences; however, (6.17) is meant to be understood as follows: The value $\vec{a}(t)$ at some given instant predicts the input $\vec{v}(t + h)$ at some subsequent instant. At that instant, the transducer has been *pre-adapted* so that its activity on the *predicted input* $\vec{v}(t + h)$ is described by (6.17).

With these modifications, which incorporate the essential features of the forward activation of our biosynthetic pathway, we can enlarge our diagram (6.14) accordingly; we may write

$$\vec{u} \rightarrow \begin{array}{c} \text{---} \circ \text{---} \vec{a} \\ \boxed{\varphi_{\vec{a}}} \end{array} \rightarrow \vec{v} \quad (6.18)$$

where the symbol $\text{---} \circ$ is meant to indicate that the vector $\vec{a}$ represents a predictor, whose present value changes the *equation of state* describing the transducer.

As a final step, we shall now generalize the role of the intermediate steps of the biosynthetic pathway. We do this by embedding the diagram (6.18) into a *larger system* of unspecified structure. This larger system serves to generate both the input $\vec{u}$ and the predictor $\vec{a}$, and it also serves as a sink for the output $\vec{v}$. We need not, and indeed we cannot, be more specific about this larger system; the only assumption we can make about it is that its properties can be encoded into appropriate equations of state which generalize the relations (6.6) above. However, this unspecified larger system will play several key roles in our subsequent development, both in regard to the properties stabilized by the anticipatory step, and the manner in which these properties are maintained over time.

We have thus accomplished the first step we mentioned at the outset; we have seen how the essential features of our biosynthetic pathway can be formulated in completely general terms. We now turn our attention to the second step: namely, the formulation of the properties stabilized by the presence of the anticipatory step. We shall formulate these properties not only in terms of the diagram (6.18), but also in terms of the larger system in which the diagram is embedded. That is, we shall investigate how the anticipatory step is adaptive in terms of (6.18), and also in terms of the larger system containing it.

We shall suppose as usual that the property P can initially be taken as a linkage of the form

$$\Phi(\vec{u}, \vec{v}) = 0; \quad (6.19)$$

i.e. that it expresses the maintenance of a particular relation between the input vector $\vec{u}$ at some instant, and the output vector $\vec{v}$, generally at some other instant. The anticipatory step, governed by (6.17), is assumed to guarantee that (6.19) is satisfied, under the assumption that the prediction of the model embodied in the anticipatory step agrees with the actual input vector seen by the transducer.

Let us formulate this last condition more precisely. We have seen that the *present* value $\vec{a}(t)$ of the predictor $\vec{a}$ is linked to a future value $\vec{u}(t + h)$ of the input $\vec{u}$; this can be expressed by a linkage relation of the form

$$\vec{u}_p(t + h) = \psi(\vec{a}(t)). \quad (6.20)$$

We use the subscript p here to denote the fact that the value $\tilde{u}_p(t+h)$ is that predicted from the present value $a(t)$ of the predictor a . It is this predicted value to which the transducer is pre-adapted, according to (6.17). On the other hand, the actual input value to the transducer is $\tilde{u}(t+h)$; therefore, since the property P embodied by (6.19) pertains to the actual inputs, the maintenance of this property by the anticipatory step obviously requires that

$$\tilde{u}_p(t+h) = \tilde{u}(t+h). \quad (6.21)$$

That is, the maintenance of the property P invariant in time requires a concordance between the input value predicted from the model employed by the transducer, and the actual input presented to the transducer. In other words, the maintenance of the property P depends on the *fidelity* or stability of the model embodied in the anticipatory step.

Finally, we shall relate the property P embodied in (6.19), which pertains only to the system we have diagrammed in (6.18) above, to other properties of the larger system in which this diagram is embedded. We note in advance that if P' represents any property of this larger system, whose maintenance depends on the time-invariance of P , then P' also depends on the stability of the model which determines the anticipatory step.

To formulate such a property P' is very easy. In general, let y_1, y_2 be two observables of the larger system, which are linked to the predictor a and to the output v ; i.e. such that we can write

$$y_1 = y_1(\vec{a}, \vec{v}, \dots),$$

$$y_2 = y_2(\vec{a}, \vec{v}, \dots).$$

Suppose that $\psi(y_1, y_2) = 0$ is a linkage in the larger system which holds if and only if (6.19) is satisfied; i.e. if and only if the linkage

$$\varphi_a(\vec{u}) = \vec{v}$$

satisfies the property P . Then clearly ψ defines a property P' of the larger system, and by definition, this property P' is contingent upon the presence of the property P in the system (7). In this fashion, we can construct many such properties P' which are *linked* to P , in the sense that they are maintained time-invariant only if P is likewise.

We have thus reformulated all of the basic features of the forward activation pathway described in the preceding chapter, but in a quite general setting. It only remains to consider what happens to the property P , and to the properties P' of the larger system which are linked to it, as the model embodied in the anticipatory step becomes unstable. Once again, we will find that the consequences will simply be a paraphrase of situations we have already described, adapted to the more general context developed above. In particular, we shall find that all these properties are *temporally spanned* by the growing instability of the model.

In the present case, the instability of the model is reflected precisely by the growing discrepancy between the two sides of equation (6.21) above. That is, the

value $\bar{u}_p$ *predicted* on the basis of the model becomes increasingly different from the actual input u . Furthermore, the pre-adaptation step (6.17) is specifically geared to pertain to the predicted input u_p ; thus if $\bar{u} = \bar{u}_p$, the preadaptation will to this extent be maladaptive with respect to the property P , and will become increasingly so as the discrepancy between the predicted behavior and the actual behavior grows. Accordingly, the linkage (6.19) which expresses the property P departs more and more from the relation between $\bar{u}$ and $\bar{v}$ actually manifested by the system, until at some characteristic time it is lost entirely. Likewise, any property P' of the entire system which is linked to P will also increasingly depart from the actual system behavior, until these properties are also lost. Hence all such properties, and the linkages which represent them, are temporally spanned.

The discussion of temporal spanning which has been presented above is, of course, closely related to the notions of *error* which were developed in Sect. 5.6 above. We may recall that error always refers to the deviation between the actual behavior of a system and the behavior of some abstract model or encoding of that system. We are going to invoke this discussion now, in the context of the property P which characterizes our anticipatory step in (6.18).

We have seen that the property P is preserved invariant only when (6.21) holds; i.e. only when the actual input to the transducer coincides with the input predicted on the basis of the anticipatory model. If we suppose that an external observer of the system (6.18) identifies its “correct function” with the maintenance of the property P , he will in these circumstances conclude that the system is functioning correctly; i.e. without error. On the other hand, as the discrepancy between the predicted input $\bar{u}_p$ and the actual input $\bar{u}$ grows, the property P will eventually be lost, as we have seen. The external observer, who judges the “correct” functioning of the system by the persistence of the property, will under these circumstances conclude that the system is *malfunctioning*, or behaving *erroneously*. According to this criterion, we can see that the malfunction arises because of underlying errors in the predictive model employed in the anticipatory step. But it is clear that no *part* of the system (6.18) is itself malfunctioning in any sense; by any local criterion, each component of the system (6.18) is behaving “correctly” according to its equation of state. The ultimate source of the malfunction of (6.18) does not lie in any local failure in any part of the system; rather, it resides ultimately in a *linguistic* or *symbolic* feature, which may be formulated as the loss of *synonymy* between the predictor $a(t)$ and the transducer input which it is supposed to represent. Thus the malfunction of the system (6.18), as measured by the loss of the property P , will appear to the external observer to arise in spite of the fact that each constituent of the system is itself functioning correctly in any local sense. A malfunction of this type will be called a *global failure* of the system (6.18) it is to be distinguished from what we may call *local failures*, which would correspond to one or more of the constituents of the system itself departing from the behavior governed by its particular equation of state.

Since the property P is temporally spanned, we may identify the *lifetime* of the system (6.18) with the time at which the property P is lost. Thus, with respect to this property, the system (6.18) will have a definite lifetime. At the end of that time, the system will fail or malfunction in the global fashion we have described. It should be

emphasized that the lifetime of the system (6.18), in this sense, is not determined by the system (6.18) itself; rather, it is a property of the larger system in which (6.18) is embedded. Thus, as we have seen, this lifetime is unpredictable from within the system itself. However, we can say, in a general way, that the longer the system (6.18) has functioned correctly, and hence the longer it has interacted with the larger system which contains it, the closer it must be to its spanned time at which the property P is lost.

6.4 An Application: Senescence

In the present chapter, we wish to develop the thesis that the temporal spanning of properties in systems with anticipatory steps, which we have demonstrated above, may be casually implicated in the phenomena of *senescence* in complex systems. In order to do so, and to place this thesis in a general perspective, we will preface our discussion with a few general remarks concerning senescence, and briefly describe some previous approaches and hypotheses which bear upon it.¹

In biology, senescence is an almost ubiquitous property of organisms, but at present it remains one of the most poorly understood. The concept of senescence, being intimately concerned with mortality, has proved as difficult to define as life itself. Senescence is also closely related to the concepts of time and age, which we saw in Sect. 6.4 above to be exceedingly complex in themselves. Perhaps the most satisfactory working definition, which has been adopted by most of those concerned with senescence phenomena, is the following: *senescence is an increase in the probability of death per unit time with chronological age*. Put somewhat more picturesquely, senescence is an increase in the force of mortality with time. Such a definition is incomplete and unsatisfactory in many ways, as we shall see; but it will serve as a point of departure. Moreover, it explicitly calls attention to the basic conceptual notions involved: time, age, mortality.

In order to relate our discussion of temporal spanning of properties in anticipatory systems to this preliminary definition of senescence, it will be necessary to analyze the components of that definition further. The first step in this analysis will involve the “force of mortality”; i.e. the probability of death per unit time. Let us briefly consider what is meant by this notion.

Intuitively, we wish our definition of senescence to apply to *individual* organisms. On the other hand, the “force of mortality” is a population concept. To see this, let us notice that in order to determine empirically the probability of death of an individual organism per unit time, we must imagine a large population of identical organisms, for which we can count the number of deaths per unit time as a function of the ages of the organisms which have died. In this way, we can form ratios which approximate to the probability that a representative individual in the population will in fact die at a particular age. If this probability increases with age, then we may say that the *individuals* in the population exhibit senescence. The age at which this probability becomes unity can then be interpreted as the *intrinsic life span* of the individuals comprising the population.

Thus, at the very outset, we can see that the very definition of senescence, and of life span, is an *ensemble* concept, of the type which we have met before; e.g. in statistical mechanics. On the other hand, we desire to refer these properties, initially defined in terms of ensemble or population, to the individual members of such a population. We can only do this on the basis of some kind of reductionistic approach, in which we suppose that the statistical properties of the ensemble reflect some underlying properties of the individuals which comprise the ensemble. But this raises difficulties of its own, since we initially supposed that these individuals were *identical*; it would clearly make no sense to attempt to define the probabilities of death per age in a heterogeneous population. On the other hand, if there were no heterogeneity, the individuals in the population would all behave identically, and there would be no probabilistic sense to the “force of mortality”.

Thus, we have to consider a population which is at least initially identical with respect to some quality, but which we allow to be heterogeneous with respect to another. We also need to visualize the individuals in the population changing with respect to time in some fashion, which is ultimately to be reflected in the mortality of the individuals. Let us see if we can visualize this kind of situation. To do so, we shall employ the same sort of picture which we used earlier in our treatment of statistical mechanics, in a fashion originally suggested by G. Sacher and E. Trucco.²

Let us suppose that we can encode the instantaneous states of the individuals in our population into some suitable manifold X . Thus, at any instant of time t (according to some suitable external clock) the state of such an individual will be denoted by $x(t)$, and is a definite point of X .

Thus, we envisage a dynamics imposed on X , according to which any point in X determines an entire trajectory in X .

We shall suppose further that initially (i.e. at $t = 0$) our population is represented by a set of points $\Omega(0)$ in X . We shall suppose that this set of points is such that the distance between any pair of them in X is small; e.g. that they all lie within a sphere of small radius in X . Thus, these initial states are essentially identical in our encoding, but there is some small amount of initial heterogeneity.

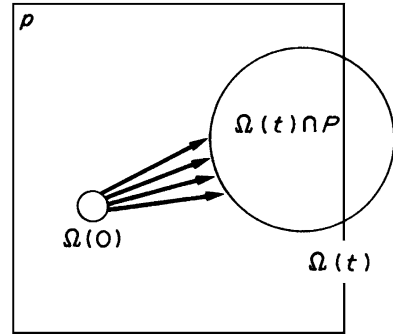
We shall also suppose that the set $\Omega(0)$ of initial states of our population is contained within some larger set $P \subset X$. We shall identify this set P with the totality of *viable states* of our individuals; a state outside this region is considered *dead*. In this fashion, we replace the vague concept of viability with some arbitrary property; which we further identify with the set of all points which satisfy this property.

Now as time increases from zero, each point in $\Omega(0)$ will move along its corresponding trajectory with X . Thus, the entire set $\Omega(0)$ will move or flow in X , and will at each instant $t < 0$ define a new set $\Omega(t)$.

The set used here is essentially that which we used in discussing Liouville’s Theorem in Sect. 3.3 above.

At each instant t , we can consider the set $\Omega(t) \cap P$. This can be interpreted as specifying the totality of individuals in our population which are still viable at the instant t . Let us define the quantity

Fig. 6.2



$$p(t) = \frac{\mu(\Omega(t) \cap P)}{\mu\Omega(t)}$$

Here μ denotes the measure or volume of subsets of X . For any instant t , we will have $0 \leq p(t) \leq 1$; i.e. $p(t)$ is essentially a probability. We may interpret $p(t)$ as the probability that a state initially in $\Omega(0)$ will still be viable at the instant t . The situation we are describing is shown diagrammatically in Fig. 6.2 below.

The intrinsic lifetime of a state in $\Omega(0)$ will thus be that time T for which its trajectory leaves P . It will be seen that these considerations depend on (a) the dynamics imposed on X , and (b) the nature of the property P .

It should be noted explicitly that although we have taken the source of variability to lie in the specification of initial state, we could just as easily incorporate the variability into the imposed dynamics. This would involve simply moving the appropriate constitutive parameters from the state description to the dynamics; it is entirely arbitrary where they are put. However, for pictorial purposes, it is more convenient to place them in the state space rather than the dynamics, and thus we shall continue to do so.

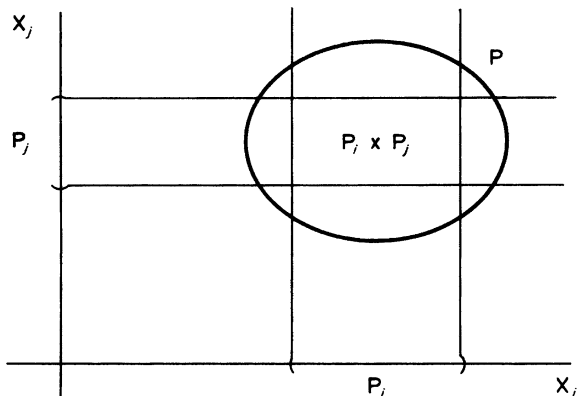
Let us also note that in this picture we have identified mortality or death with the loss of some property P by an individual in our population. In this sense, mortality is regarded as a kind of *system failure*, and the intrinsic lifetime of an individual is identified with the *maximum time to failure*, always determined with respect to the property P under consideration.

Now let us go a step further. Let us regard the individuals in our ensemble as complex systems, which can all be resolved into a common spectrum of subsystems. We shall take this to mean that our original state space X , into which the states of these individuals are encoded, are themselves cartesian products of the state spaces which characterize the subsystems. Thus, we can write

$$X = \prod_{i=1}^n X_i \quad (6.22)$$

where the X_i are the manifolds into which particular sub-systems are encoded.

Fig. 6.3



Just as we identified the viability of an individual in our population with a property P in X , so we can identify the “viability” of each subsystem X_i with a corresponding property P_i . This property P_i will as usual be identified with the set of points in X_i which satisfy the property. Thus, as long as a point in X_i lies in P_i , we shall say that the *subsystem* is viable; when the point leaves P_i , we shall say that the subsystem has become inviable, or has *failed*. Such a failure in a subsystem X_i will be called a *local failure*. Thus, the occurrence of a local failure in a subsystem X_i is a matter intrinsic to X_i alone; such a failure can be determined by looking at X_i , without reference to what is happening in the other subsystems.

We now inquire what is the relation between the *global* criterion of viability, namely the property P in X , and the *local* criteria of viability; i.e. the properties P_i in the individual X_i . Such a relation specifies, of course, the extent to which a *global* failure in an individual organism reflects a *local* failure in one or more of the subsystems into which the organism has been analyzed. Initially, there are three possibilities which can be entertained:

$$(a). \quad \prod_{i=1}^n P_i \subseteq P;$$

$$(b). \quad \prod_{i=1}^n P_i \supset P;$$

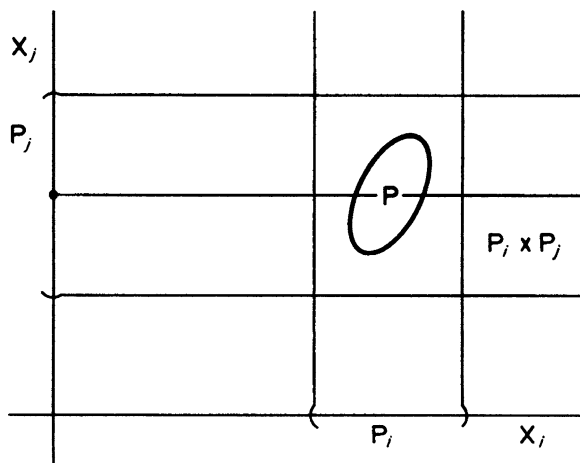
(c). Neither (a) nor (b) holds.

Let us consider the implications of these various possibilities.

The possibility (a) is indicated in diagrammatic form in Fig. 6.3 below.

Comparing this situation to Fig. 6.2 we see that it is impossible for any initial ensemble $\Omega(0)$ to leave P in X without simultaneously leaving at least one of the P_i in the corresponding X_i . In other words, in this situation *it is impossible for a global failure to occur without the associated appearance of at least one local failure in one of the constituent subsystems*. Thus, the two kinds of failure are in this case necessary concomitants of one another; if we wish, we can in fact *attribute* the

Fig. 6.4



global failure of the total system to the corresponding local failure which must occur in one or more of its subsystems in this case. That is, we can say that *global failures are in fact caused by local failures*.

The possibility (b) is sketched diagrammatically in Fig. 6.4 below:

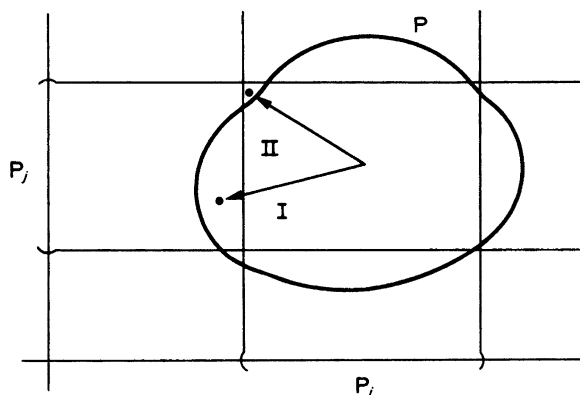
In this case, there is a complete separation between the notion of local failures in the subsystems and global failures in the total system. For, on comparing Figs. 6.4 to Fig. 6.2, we see that a global failure (i.e. inviability or death) of the organism is *never* associated with a local failure in any of the subsystems. Thus, in this case, if there is a local failure, we may perhaps say that such a local failure is *caused* by global failure, but never conversely.

The final possibility (c) is diagrammed in general form in Fig. 6.5 below. In this case, on comparison with Fig. 6.2, we see that there are two classes of ways in which an organism in our population can become inviable: (i) we can leave the region P as a result of local failures, as in the trajectory marked I in the figure; (ii) we can leave the region P without any local failures, as in the trajectory marked II. In this case, then, a global failure *may, but need not*, be associated with a local failure in a subsystem.

Now let us see what these general considerations mean in the context of senescence. Briefly, we shall show that all previous approaches to senescence have tacitly supposed that the hypothesis (a) is satisfied; i.e. that the senescence or loss of viability of an individual organism is causally related to local failures directly identifiable in particular subsystems of the organism. However much these approaches differ in detail, they are thus alike in supposing that global failures all necessarily arise as a consequence of local subsystem failures, which can be independently recognized.

The theories which have been heretofore proposed to account for senescence in organisms fall logically into two distinct classes, which we may call *error theories* and *program theories*. In intuitive terms, the error theories all suppose

Fig. 6.5



that normal physiological processes have superimposed upon them some source of randomness or fluctuation, which causes them to behave *erroneously*. As we have seen (cf. Chap. 5 above) the imposition of such fluctuations merely reflects the fact that physiological subsystems are more open than their physiologically significant interactions require; errors in this sense are measured against our own models or ideas as to what constitutes physiological interaction. To one degree or another, the error theories regard such fluctuations, occurring in specific subsystems, as antagonizing or opposing the maintenance of a local property P_i in the subsystem; senescence and death occur through the causal relation between such local properties P_i and the global property P . Thus, an error theory of senescence typically involves two aspects: (a) a specification of a source of fluctuation in a particular physiological subsystem, and (b) a specification of the effects of these fluctuations on the total viability of the organism.

As examples, let us consider a few of the well-known error theories. Perhaps the simplest of these is the *somatic mutation theory*.³ Here, the essential locus of fluctuation is asserted to be the primary genetic material itself; i.e. DNA in its role as template for protein synthesis. According to this theory, the base sequence in DNA is constantly subject to perturbation, through interaction with ambient radiation or mutagenic substances, or through a variety of other causes. Such perturbations constitute somatic mutation, and result in a corresponding modification of amino acid sequences for which these genes code. Since the primary sequence of amino acids in a protein is the determinant of the protein's conformation, and hence of its biological activity, such modifications in primary structure will result in corresponding alterations of rate and specificity at the protein level; it is these alterations within cells, and their effects on intercellular interactions, which we collectively call senescence.

Another popular error theory is the "error catastrophe" of Orgel which was briefly mentioned in Sect. 5.6 above. Here, the source of the fluctuations is placed in the translation step of protein synthesis. Briefly, Orgel observed that the translation step itself involves protein in an essential way (e.g. the synthetase enzymes, which

specifically attach individual amino acids to the correct species of transfer RNA). If the specificities of these enzymes is initially altered, through some random mechanism, then *all* proteins synthesized by the cell will be modified in primary sequence, including these synthetases themselves. Thus, the next “generation” of synthetase molecules will tend to be more erroneous than the first. By iterating the argument Orgel claimed that a positive feedback loop was effectively established, through which the initial synthetase errors are successively amplified until the cell becomes inviable; this is the “error catastrophe”.

Other error theories involve different sources of fluctuation; e.g. the random extinction of crucial stem cell lines,⁴ or the random cross-linking of collagenous supporting matrices. Still others implicate such systems as the immune system,⁵ which may be the source of fluctuations (e.g. by the random appearance of cells in the system which attack “self proteins”), or the effector (e.g. by recognizing aberrant protein, generated through somatic mutation or other errors, as foreign). The catalog of such theories is virtually limitless, both as regards source of fluctuation and its physiological effects. However, we can see immediately that all the error theories are alike in proposing that the ultimate cause of senescence lies in localized (random) failures in specific physiological sub-systems.

At the other logical extreme appear the “program theories” of senescence. In these theories, it is asserted in general that the primary causes of senescence are not of a random character, but rather represent the accurate expression of genetic programs similar to those expressed during development. Essential ingredients in these theories are various types of clocks, counters and timers, which cause lethal events to occur when a particular time interval, or a particular sequence of preceding events, has elapsed. Here again, the locus of the initial lesion, and the propagation of its physiological effects, can be of the widest latitude.

Ironically, despite the apparent logical polarity between the error theories and the program theories of senescence, they are in fact indistinguishable from one another on any empirical ground. Worse than this: not only are these theories inseparable from one another in logical terms, but it is empirically impossible to distinguish between theories belonging to the same class. It is safe to say that at the present time, every observation can be made compatible with any theory; conversely, every theory is equally well supported by the available empirical data.

This fact means one of two things: either senescence simply comprises a catalog of the unlimited number of ways in which physiological subsystems can locally fail, or else some radically different approach to senescent phenomena is indicated.

The former possibility has been advocated e.g. by Medawar⁶; according to such a view, senescence is simply the sum total of all physiological defects which appear after reproductive maturity, and hence cannot be selected against. There are strong reasons for believing that this is not so, but it would at least explain the featureless character of the empirical data concerning senescent organisms, and the multiplicity of plausible hypotheses regarding senescence which may be entertained on the basis of that data. However, it would also imply that there could be no *theory* of senescence; rather there would only be an encyclopedia of local failures.

The second possibility is far more interesting; namely, that a radically different approach to senescence, both conceptually and empirically, is called for. The essence of such an approach may be found in the recognition of the possibilities sketched in Figs. 6.4 and 6.5 namely, it is possible for complex systems to exhibit global modes of failure which are not causally related to local subsystem failures. So far, these possibilities exist only formally; to make them significant for senescence, we would need to show by example that such possibilities can actually arise in complex systems.

To do this, however, we need only recall the preceding sections in the present chapter. In those sections, we showed precisely that *any system possessing an anticipatory step exhibits properties which are temporally spanned*; and moreover, that *such a temporal spanning represents a kind of global failure which is independent of any local subsystem failures*. But these are precisely the characteristics we have seen to be required, if there is to be any comprehensive theory of senescence phenomena.

We have thus seen that any anticipatory system can in effect exhibit senescence, in the sense that it will automatically possess temporally spanned properties; and this temporal spanning is independent of any local subsystem failure. The next important question then becomes: are anticipatory systems widespread in nature, or at least are they sufficiently so to make it reasonable to seek a comprehensive theory of senescence on this basis? To this question, we have already given an affirmative answer in Sect. 1.1; indeed, we have argued that anticipatory behavior is in fact *ubiquitous* at all levels of biological organization. If this is so, and if temporal spanning is a universal concomitant of anticipatory systems, then it becomes at least plausible to construct a general theory of senescence in terms of such spanning.

Once this much is admitted, a number of interesting and suggestive conclusions immediately follow. One of these is the following: if indeed senescence arises from the temporal spanning of an anticipatory system, it instantly becomes clear why it has been so difficult to learn about senescence through empirical studies of physiological subsystems. Indeed, we may say that every technique of empirical analysis pertains precisely to such a subsystem, and the essence of our argument is that no local property of such a system, considered as an isolated system in its own right, is pertinent to the global failure of the larger system in which it is embedded. On these grounds, then, we would *expect* that an exclusive pre-occupation with the local empirical study of subsystems must fail to clarify the “mechanism” of senescence, since that mechanism is independent of the subsystems, and will be called into play even though no local subsystem failure occurs. An immediate corollary of this point of view is the following: an effective empirical study of senescence must proceed from an entirely different basis than has heretofore been the case; present techniques which are based entirely on analysis into physiological subsystems, are simply not adequate to deal with temporal spanning based on anticipatory loops. It is not possible to decide at the moment what the most effective approaches will in fact be; only experience with the behaviors of anticipatory systems in general, initially based on the careful study of model systems, is likely to clarify this question.

One further tantalizing corollary to these considerations may be mentioned. We have seen above that the temporal spanning of anticipatory systems arises from a growing discrepancy between the predictions of the internal model, and the actual inputs processed by the anticipatory step. We have also seen that this discrepancy can be eliminated by *recalibrating* the model; i.e. by applying an external dynamics which will bring the predicted and actual inputs once again into concordance as they are initially. In the present situation, such a recalibration would amount to a *rejuvenation* of the entire system. It is conceivable, for example, that the function of mitosis is to effect precisely such a recalibration at the cellular level. Here too, only the careful study of model systems will indicate how these kinds of possibilities may be further explored in an effective manner.

We conclude by observing once again that our arguments regarding the temporal spanning of properties are perfectly general; they apply to any complex system which contains anticipatory steps. Thus, they may be also applied to social and cultural systems, in which anticipations are likewise ubiquitous. We may recall in particular that it was the thesis of historians such as Spengler and Toynbee⁷ that human cultures and civilizations exhibit many of the properties of biological organisms, including phases which may at least be analogized with senescence. Such considerations lead us back towards the view of societies as superorganisms, which was briefly discussed in Sect. 1.1 above. These analogies, taken together with the ubiquitous properties of temporal spanning in anticipatory systems, suggest that a comprehensive study of all these phenomena is possible; from such a study, it is certain that much of great value will be learned.

References and Notes

1. Some good general references on senescence are the following:

Comfort, A., *The Biology of Senescence*. Rinehart (1956).

Strehler, B., *Time, Cells and Aging*. Academic Press (1962).

Burch, P. R. J., *An Inquiry Concerning Growth, Disease and Aging*. Oliver & Boyd (1962).

For a critical review of the various theoretical and experimental approaches to senescence, see e.g. Rosen, R., *Int. Rev. Cyt.* 54, 161–191 (1978). See also Note 1, Sect. 4.8 and Note 8, Sect. 5.6.

2. See Sacher, G. and Trucco, E., *Ann. N.Y. Acad. Sci.* 96, 985–999 (1962). See also Trucco, E., *Bull. Math. Biophys.* 25, 303–324; 343–366 (1963).
3. Although somatic mutation has long been considered as a mechanism for carcinogenesis, it does not seem to have been implicated in senescence before 1956 (cf. Danielli, J. F., *Experientia* Supp. #4, 55–81 (1956)). This is not surprising, since senescing cells and malignant cells seem to manifest opposing properties. A general theory of senescence based on somatic mutation was developed by

Szilard in 1959 (*PNAS* 45, 30–45); similar ideas were later developed by many others.

4. The two most popular kinds of cells involved in these random extinction hypotheses are those of the hemopoietic system and the neurons in the brain; see the general references cited in Note 1 above.
5. The great advocate of this idea has been M. Burnet. See for instance *The Clonal Selection Theory of Acquired Immunity*, Cambridge University Press (1959); *Immunological Surveillance*, Pergamon (1970). See also Walford, R. L., *The Immunologic Theory of Aging*, Munsckgarrrd (1969).
6. See Medawar, P. B., *Modern Quart.* 2, 30–38 (1945).
7. See Spengler, O., *Der Untergang des Abendlandes*. Beck (Munich) (1922); Toynbee, A., *A Study of History: Reconsiderations*. Oxford University Press (1961).

6.5 Adaptation, Natural Selection and Evolution

Throughout the foregoing sections, we have had ample occasion to mention the concept of *adaptation*. This is an idea utterly basic to the biological realm, and which is becoming increasingly important in an understanding of the properties and control of social systems and the human sciences. Indeed, the main point of our entire development has been that systems which anticipate are thereby rendered adaptive, and in fact are more adaptive, in some appropriate sense, than systems which simply react.

It is now time to consider the concept of adaptation in more detail. As we shall see, the circle of ideas on which this concept depends are very rich indeed¹; this very richness accounts in large measure for the confusion, controversy and acrimony with which the concept has been associated. During the course of this development, we shall not only be able to obtain a better understanding of what adaptation means, but we shall find that anticipatory behavior is in some sense a corollary of the general mechanisms by which adaptations are generated. During the course of our previous development, we did not ask *why* so much biological behavior should be of an anticipatory rather than a reactive character; the simple fact that it was so provided sufficient grounds for studying such systems in depth. It is thus of great interest to find that a study of adaptation automatically bears on the question of why anticipatory mechanisms are so prevalent in biology.

In order to best motivate and focus our discussion of adaptation, it is convenient to proceed *metaphorically*, as we did for example in our treatment of morphogenesis in Sect. 3.3 above. To do this, we shall initially elaborate on one of the first and simplest examples of adaptation, which we introduced at the outset in Sect. 1.1; namely, the example of a simple tropism, such as the negative phototropism or photophobia which is so common in simple organisms. We shall elaborate our metaphor by idealizing this kind of behavior.

To begin, let us imagine an organism which is free to move on a two-dimensional surface. Thus, at any instant of time, the organism can be located at a definite point of this surface, and thus its position may be characterized by a pair of numbers (x,y) relative to some convenient system of co-ordinates in the surface.

We shall also assume that the organism can perceive some environmental quality E (such as light intensity). This quality is assumed to have a definite value at every point (x,y) of the surface, and thus defines a *scalar field* thereon. We further suppose that this scalar field is such that a gradient ∇E may be defined at every point of the surface. Thus, the environmental quality E also defines for us a *vector field* on the surface in which our organism moves.

For our purposes, the *behavior* of our organism will be specified by determining the *direction* of its motion at any point on the surface. The components of the vector which specify the direction are just dx/dt and dy/dt , and we suppose that these components depend on *both* the gradient ∇E of the imposed vector field, and on the internal constitution of the organism itself. Thus, the behavior of the organism is determined by equations of state of the form

$$\begin{cases} dx/dt = \varphi(\alpha, \nabla E), \\ dy/dt = \psi(\alpha, \nabla E) \end{cases} \quad (6.23)$$

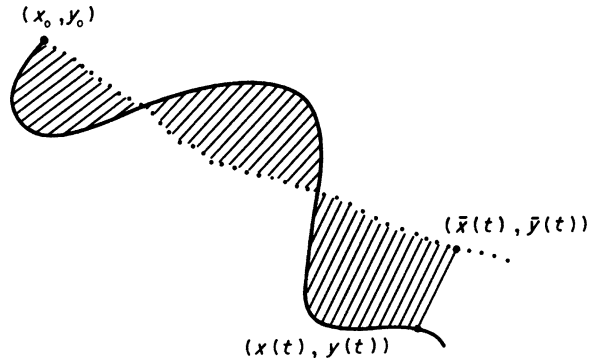
where we have denoted by α a set of constitutive parameters, characteristic of the organism, which we shall call the *genome* for these equations. This is in accord with our earlier discussion (cf. Sect. 3.5, Example 2C), in which we saw that the quantities entering into any equation of state can generally be characterized as genome, environment and phenotype; here the variable α denotes genome, the gradient ∇E denotes environment, and the resulting direction of motion (i.e. the components of the tangent vector at a point of our surface) is the phenotype. From these equations of state, we can infer, as usual, the path of our organism on the surface as a function of the genome, and the scalar field $E = E(x, y)$.

For our purposes, we shall assume that *different organisms are characterized by different values assigned to the genome α* . Thus, different organisms will generally follow different paths through our surface, even though they start initially at the same point, and are exposed to the same scalar field $E(x, y)$. The question is: how can we say whether the path followed by a particular organism is adaptive or not? When is one such path more adaptive than another?

In point of fact, there is as yet no way of characterizing adaptation within the confines of the description introduced so far. Indeed, all we have done is to characterize the manner in which an organism's state (i.e. its position on the surface) changes as a function of genome and environment. In order to specify whether such a change of state is adaptive or not, an entirely different element of structure must be introduced. This is what we now proceed to do.

Let us then suppose that we are given another scalar field $U = U(x, y)$ on our surface; one which is initially entirely independent of the environmental modality E to which our organism is responding. This scalar field likewise determines a vector

Fig. 6.6



field by forming its gradient ∇U . This vector field in general fibers out surface into a family of integral curves, which are trajectories of the associated dynamical system which can be written in the form

$$\begin{cases} dx/dt = \partial U / \partial x; \\ dy/dt = \partial U / \partial y. \end{cases} \quad (6.24)$$

What we are now going to do is to compare the curves arising in this way, as solutions of the system (6.24), with the paths followed by our organisms *in the same surface*, which are the solutions of the system (6.23).

In Fig. 6.6 below, we indicate pictorially two such curves. The heavy curve is assumed to represent the path of an organism, as determined by (6.23), the dashed curve is the corresponding path through the same initial point of our surface which is determined by (6.24). There is of course no reason why there should be any particular relation between these paths.

However, what we want to do now is to introduce a numerical measure of the degree of discordancy between the two paths we have shown. There are many ways in which this can be done. Because of the metaphorical character of our discussion, our subsequent conclusions will be essentially independent of our specific choice of a measure of discordancy between the paths; accordingly, we are free to choose the simplest. This simplest measure, which is also indicated in Fig. 6.6 above, is the area of the region bounded by the two curves, and (if necessary) by the geodesic of the surface connecting the corresponding points on the curve at a common instant t . For a given initial point (x_0, y_0) in the surface, the resulting measure $A(t)$ is clearly a monotonically increasing function of t . As noted, this function $A(t)$ is clearly a measure of how far the actual path of an organism deviates from a corresponding integral curve of (6.24); the greater the deviation, the bigger is $A(t)$.

The measure $A(t)$ just introduced, or rather its inverse $F(t) = 1/A(t)$, is what we shall call the *fitness* of a path. Thus, given any solution of (6.23), and any fixed time T , we can associate with the resulting path a *number*. Thus, *the fitness we have defined can be regarded as an observable of paths*. As usual, we can utilize such an

observable to impute properties to the paths on which the observable is defined. In particular, we shall say that *one path is more adaptive than another if its fitness is larger*. In particular, we can see that the most adaptive paths, in this sense, are the integral curves of (6.24); their fitness is infinite.

Let us pause here to stress that, in order to introduce a notion of adaptation, we require: (a) a way of associating numbers with paths, in such a way that two paths can be compared; (b) this number must be introduced in a manner initially independent of the specific mechanism (6.23) by which the compared paths are actually generated. The condition in effect introduces a *common currency* in terms of which any paths can be compared, regardless of how they are generated; the condition (b) means that the organism does not have any intrinsic mechanism for perceiving the fitness, and hence for determining whether the path it is following is adaptive or not. The point to stress is that, at this stage, there is no linkage whatever between the mechanism for generating specific paths, and the mechanism whereby their fitnesses, and hence their relative adaptive values, are assessed.

It must be emphasized that it is in fact meaningless to characterize a phenotype, or behavior, as adaptive apart from a postulated measure of fitness. Therefore, for example, the autonomous behavior of non-biological systems cannot be considered as adaptive in our sense. A familiar example of such behavior which is often regarded as intrinsically adaptive is expressed by the well-known Principle of Le Chatelier²; if a system in equilibrium is disturbed by the variation of one of the environmental quantities which determine that equilibrium, the system will move to a new equilibrium in such a way as to oppose the initial variation. In this example, there is no postulated measure of fitness, and hence no adaptation. The same is true of the homeostasis exhibited by open systems in the vicinity of a stable steady state, which we earlier (cf. Sect. 3.5) considered as a metaphor for the biological phenomenon of equifinality; and indeed, of any kind of homeostasis whatsoever. It must be stressed that, until a measure of fitness is introduced, no simple state can be meaningfully characterized as adaptive or maladaptive; and this includes all the cybernetic mechanisms which have been proposed as examples of adaptation.³ This is a subtle point, but it cannot be emphasized too strongly.

At this stage, however, we can already frame such questions as: what are the maximally adaptive paths in our surface? In mathematical terms, such questions are simply problems in *optimality*; they can be framed whenever a mechanism like (6.23) is proposed for generating a space of possible paths, and a criterion like (6.24) is available to associate numbers with these paths. In the light of our discussion so far, the solution of such an optimality problem can be regarded as *the determination of those genomes α in (6.23) whose paths deviate minimally from the integral curves of (6.24)*. However, we must note that, so far, the solution of such an optimality problem possesses no necessary relevance to our problem, because as we have stressed, there is so far no relation between the procedure by which paths are generated and the procedure by which their adaptive character is determined.

The final step in our metaphorical discussion is then to introduce precisely such a relation, which will be called *selection*. As we shall see, in its broadest terms, selection links the generation of paths to their fitnesses by, in effect, *allowing*

phenotypes (i.e., paths) to act back on the genomes which generated them. This is the fundamental property of selection, which we shall describe, and then interpret in biological terms.

Our first observation is the following: *we can utilize the fitness F defined above to define a scalar field on the space A of genomes.* Namely, to each α in A , we can associate the number $F(\alpha)$, the fitness of the path generated according to (6.23) with the genome α , evaluated as some definite fixed instant T of time. Next: *from this scalar field on A , we can construct the vector field ∇F on A .* Finally: *we can construct a dynamics on A itself,* of the general form

$$d\alpha/d\theta = K\nabla F(\alpha) \quad (6.25)$$

It is this dynamics (6.25) which embodies the essence of a selection mechanism. As we shall see in a moment, it is crucial that the time parameter θ entering into this dynamics be *different* from the time parameter t entering into the dynamics (6.23), in terms of which phenotypes (i.e. paths) are generated.

The essential feature of the selection dynamics (6.25) is that *it moves the genome in the direction of increasing fitness.* The stable steady states for this dynamics are thus those genomes for which fitness is *maximal*, in the ordinary mathematical sense. Thus, not only does fitness serve to link genomes to their adaptive consequences, but through the action of selection, the genome itself is controlled so as to generate phenotypes which are maximally adaptive. It is for this reason that the abstract optimality problem which can already be posed in terms of (6.23) and (6.24) can be claimed to have biological significance; selection forces organisms towards genomes for which fitness is maximal. This is the essential content of *the Principal of Optimal Design*⁴ as a tool for probing biological phenotypes, as proposed by N. Rashevsky many years ago; and for the appearance of optimality problems in any discussion of adaptation.

Now let us discuss briefly the biological basis for postulating a dynamics of the form (6.25). To the biologist fitness is invariably related to fecundity; and is measured by the size of the progeny left by any particular organism in a given set of environmental conditions.⁵ Let us suppose then that the fixed time T which we used in defining the scalar field $F(\alpha)$ is interpreted as the *generation time*. Next, we shall suppose with the biologist that the progeny left by an organism of a given genome are not genetically identical but rather populate some neighborhood of α in A . The picture is thus rather like treating A as an excitable medium, and α as a point source of excitation. Intuitively, some of the progeny α' of this genome α will be more fit than was α , according to how close they are to the gradient vector ∇F evaluated at α . Translating fitness into differential fecundity or reproductive rates means that the propagation of “excitation” in A away from the source α will be attenuated in all directions *except in the direction of ∇F* . This is precisely what is asserted by the dynamics described in (6.25), when all inessential transients and attenuations are neglected.

We now draw attention to the time scale for the dynamics (6.25), which we emphasized *must* be different from that occurring in (6.23). In some sense the “time

unit” for the dynamics (6.25) can be regarded as the generation time T . If fitness and selection are to be meaningful, this generation time must be very long with regard to the time unit in the dynamics (6.23), through which the behavior evaluated with the fitness function F is actually generated. On the other hand, this unit of generation time for (6.25) must itself be treated as an infinitesimal in describing a dynamics on the space A of genomes. These requirements are incompatible, in much the same sense that the time scales governing irreversible macroscopic processes and the underlying micro-dynamics are incompatible (cf. Sect. 4.3 above); they cannot be combined into a common temporal frame which encompasses both the adaptive dynamics (6.23) and the selective dynamics (6.25). The fundamental incompatibility of these two time scales has troubled many authors, and has even led a number of them to claim that any understanding of selection as a dynamical process is impossible.⁶ However, the basic difficulty here (as it was in the relation between irreversible macro-dynamics and the underlying micro-dynamics) arises from the complexities of temporal encodings in general, and from the fact that not all such encodings can be reduced to one another.

The dynamics (6.25), which occurs in a space of *genomes* is thus of quite a different character from the dynamics (6.23) which generates phenotypes or behaviors. In any equation of state, an observable appearing as genome may be regarded as pertaining to the essential *identity* of the specific system with which we are dealing; a change in the value of such a quality means precisely that a change of identity, or a *change in species*, has occurred. For instance, in the case of an equation of state like the van der Waals equation for non-ideal gases (cf. equation (3.55) above, and the related discussion) a change in value assigned to one of the constitutive parameters a , b , r amounts to changing the *species* of gas with which we are dealing. On the other hand, a change in the value assigned to an environmental or phenotypic quantity in such an equation of state is not to be regarded as a change of identity; it is simply a change of state in a system whose identity is remaining fixed.

In order to emphasize the difference between dynamical processes in which genome or identity is held fixed, and dynamical processes in which genome is changing, it is well to refer to them by different names. A dynamics like (6.23), in which genome is held fixed and only environment and phenotype are changing will accordingly be called *developmental dynamics*. All of the specific dynamical encodings with which we have heretofore been concerned have been of this character; as we have noted, they involve simple changes of state in a system whose identity is remaining constant. On the other hand, a dynamics like (6.25), which encodes precisely a change in system identity, will be called an *evolutionary dynamics*, or an *evolution*. This terminology is in exact accord with biological intuition, since evolution is exactly concerned with the appearance of new species. In these terms, it is not surprising to find that developmental dynamics, dealing entirely with change of state within a fixed species, and evolutionary dynamics, dealing with change of species, require entirely different temporal encodings.

Let us rephrase the above comments in a slightly different language. We have seen repeatedly above (cf. Sect. 3.3, Example 1) that the separation of variables oc-

curing in an equation of state into the categories of genome, environment and phenotype amounts to the construction of a fiber space, in which the genomes constitute the base space, and the remaining quantities constitute the fiber. In this picture, developmental dynamics refers to dynamical processes in a particular fiber; evolutionary dynamics refers to dynamical processes in the base space itself. In these terms, the distinction between the two kinds of dynamical processes emerges most clearly.

Let us pause here to recapitulate the essential features of the metaphor we have drawn. We have seen that, in order to characterize a particular phenotype or behavior (in our case, a path through a surface) as adaptive requires a numerical measure through which different behaviors (paths) can be compared with one another. We called this measure fitness; the higher the fitness, the more adaptive the behavior with which it is associated. The processes by which behavior is generated and through which fitness is assessed are independent of one another; they involve entirely different observables of both the environment and the organism. However, they are coupled through the imposition of a *selection mechanism*; this, as we have seen, allows phenotypes to act as genomes through the fitnesses with which they are associated. A selection mechanism thus generates an evolution in the space of genomes, which is of such a character that the adaptive value, or fitness, of the behaviors arising in the course of the evolution continually increases. Thus, in our terms, evolution is a mechanism for generating behaviors which are increasingly adaptive. Indeed, what we have shown is that, within the confines the validity of the metaphor we have been using, the concepts of adaptation, fitness, selection and evolution are themselves *linked*; none of them can be completely understood unless all of the others are taken into account. The specific expression of the linkage between these concepts is given by the relations (6.23), and (6.24) and (6.25) above.

Now let us proceed to a discussion of the proposition which we enunciated at the beginning of this chapter: namely, *that a behavior or phenotype which is adaptive necessarily is of an anticipatory character*. We shall now proceed to show how this comes about.

Let us recall two basic facts about the scheme which we have developed above:

1. Our organism is by hypothesis capable of perceiving only the environmental quality we called E, the scalar field defined on the surface on which the organism moves. On the other hand, the adaptive character of the organism's behavior is determined not by E, but rather by a different environmental modality, which we called U.
2. What the organism does at a given instant (i.e. which direction it turns in the surface at a given point) contributes to the evaluation of the fitness of the behavior *at a later instant*. Thus, the organism's present change of state has an important effect on what *will* happen to the organism subsequently.

Let us now suppose that the environmental quality E, which the organism sees, and which generates its behavior according to (6.23), is *linked* in some fashion to the environmental quality U, which the organism does not see, but which determines the extent to which its behavior is adaptive. For sake of illustration, let us suppose that E represents light intensity, and U represents something like a predator density,

or a nutrient density. Thus, we are supposing that there is an equation of state of the form relating the two vector fields defined

$$\Phi(\nabla E, \nabla \sigma) = 0 \quad (6.26)$$

on our surface by these qualities. If we knew this equation of state, we could predict the value of the field U at a point of our surface from a knowledge of the field E at the point; likewise, we could predict the gradient ∇U from a knowledge of the gradient ∇E at any point of the surface.

Now according to (6.24), the characterization of a behavior of our organism as adaptive is measured by the extent to which its path through the surface follows the gradient of U . The selection mechanism, embodied in (6.25), is defined in such a way that, by controlling the genome α , we generate organisms which will respond to E in such a way that they must closely follow the gradient of U . Thus, these adapted organisms treat E as an indicator or predictor for U ; by orienting themselves properly with respect to the vector field ∇E , they *automatically* follow the integral curves of the vector field ∇U . Their instantaneous behaviors at any point of our surface thus in effect embody the equation of state (6.26) through which these two vector fields are related. This is one basic aspect of how selection for increased adaptation generates a model.

A second, and perhaps more important aspect, arises through the tacit prediction that, *by orienting themselves appropriately with respect to ∇E , at any specific instant of time, they will thereby be maximizing their fitness, when evaluated at some later time.* That is, the retrospective or reactive mode through selection generates adaptation, becomes converted in the adapted organism to a *prediction* about how present behavior will affect future behavior.

All of this is transparently visible in the case of our negatively phototropic organism, which we may suppose to be fully adapted to an environment in which light intensity E is correlated to, say, predator density U . Thus, by following negative gradients of light intensity (i.e. moving towards regions of darkness) the organism automatically follows negative gradients of predator density, even though it cannot directly perceive the predator gradient. It also simultaneously maximizes its fitness; by moving to dark *now*, it guarantees maximal fecundity *later*. Indeed, it is now trivial to show that our organism now satisfies the five conditions given in Sect. 6.1 above, which characterize an anticipatory system. We leave the explicit verification as an exercise for the reader.

Moreover, we see from this kind of discussion the relatively minor role played by the specific *mechanism* by which the fully adapted behavior is actually generated; i.e. of the manner in which the organism actually orients itself in a scalar field of light intensities. In this regard let us recall the description of phototropism as given by Wiener (cf. p. 41 above), which as we saw dealt *entirely* with mechanism. In this case, the fully adapted behavior is simply *described* independent of the selection mechanism which generated it, and excised from the context of adaptation and selection from which it arose. Such a discussion is carried out entirely within the confines of a specific equation of state of the form (6.23); as we have seen,

from (6.23) alone we can have no concept of adaptation at all, let alone a concept of selection which produces an adaptive organism from imperfectly adaptive precursors. Consequently, the anticipatory features of the fully adaptive organism disappear, and with them, any chance to understand the nature and character of adaptation itself.

The picture of adaptation, selection and evolution which we have drawn, and which is embodied in the relations (6.23)–(6.26) above, is an exceedingly rich one, even when expressed within the confines of the particular metaphor we have been employing so far. It is well to stress again that none of these concepts can be properly understood unless all the others are taken into account. Failure to recognize this fact clearly has contributed more than anything else to the strife and controversy characterizing the literature pertaining to adaptation and evolution, closely akin to the “nature-nurture” controversies which clutter the literature on behavior.

Even from this limited discussion, we can draw quite a number of general conclusions. One simple illustration will suffice for us here.

Let us imagine a family of different kinds of organisms moving in our surface, each kind responding to a different kind of sensory modality E_i (for instance, to temperature, or to the concentration of some chemical species) according to an equation of state

$$\begin{aligned} dx/dt &= \varphi_i(\alpha_i, \nabla E_i) \\ dy/dt &= \psi_i(\alpha_i, \nabla E_i) \end{aligned} \tag{6.27}$$

analogous to (6.23) above. The forms of these equations (i.e. the functions φ_i , ψ_i) can be completely different from one kind of organism to another; i.e. the mechanisms by which the motions are generated can be entirely dissimilar. We suppose only that the same quantity U determines the fitnesses of the paths followed by each kind of organism (i.e. that the relation (6.24) is imposed on all organisms), and that there is for each E_i an equation of state of the form

$$\Phi_i(\nabla E_i, \nabla U) = 0$$

analogous to (6.26), where again the forms of these relations can be different for the different kinds of organisms. Then the imposition of the selection mechanism (6.25) will clearly generate, for each kind of organism, a genome α_i^* for which the equations (6.27) yield paths which follow the integral curves of (6.24). That is, selection will yield adapted organisms which behave similarly, even though each of them is responding to a different sensory modality, and even though the mechanism by which paths are generated are entirely dissimilar. This conclusion can be interpreted as what the biologist calls *convergence*, or *convergent evolution*.⁷ In other words: different kinds of organisms experiencing the same kind of selection mechanism will tend towards the same kind of behavior, independent of the specific mechanisms by which the behaviors themselves are generated.

Before proceeding further, let us pause to extract from the special metaphor developed above the essential aspects which serve to characterize any system in

which adaptation, and consequently associated notions of fitness and selection, can be formulated. This will involve nothing more than a restatement of the basic relations (6.23)–(6.26) in a general context.

The first pre-requisite, of course, is a family of behaviors or phenotypes, which are generated according to some equation of state analogous to (6.23). Let us suppose then that S is any natural system, which can be described in such a way that an equation of state of the form

$$\Phi(E_i, \alpha_i, v_i) = 0 \quad (6.28)$$

holds. Here the observables v_i correspond to phenotypic quantities, the E_i are environmental quantities, and the α_i constitute genome for the equation of state (6.28). It should be emphasized that this equation of state characterizes only one possible description of the natural system S ; i.e. pertains to a particular encoding of that system. There will, of course, in general be many other equations of state characterizing S , depending on which observables of S (and the environment) are chosen to generate our encoding.

The next prerequisite, as we have seen, is the introduction of a measure of *fitness*, with respect to which phenotypes or behaviors generated by equations of state like (6.28) can be compared with one another. As we have noted, such a notion of fitness amounts to the introduction of a common currency, adequate for the comparison not only of alternate phenotypes of a single system, but of phenotypes arising from diverse systems governed by entirely different equations of state. The stipulation of such a common currency for comparison of utterly diverse systems is not by any means unique to biology. For example, in physics, such a common currency is found in the concept of *energy*. All motions, and all interactions occurring between physical systems of any character, can be canonically formulated into propositions about the flow, exchange and conversion of energy. Indeed, it is this fact above all which allows us to characterize physics as a discipline, and to recognize the diverse phenomena of mechanics, electrodynamics, acoustics, optics, and a host of others as amenable to physical description in a unified way. It is perhaps the fact that the common universal currency of physics, which is energy, is no longer adequate as a measure of biological activities which distinguishes biological phenomena from the purely physical ones which are associated with them. In another realm, the science of economics, which is concerned with the diverse forms of wealth, begins with the introduction of a common currency (value) which does for economics what energy does for physics. The concept of fitness plays an analogous role in biology, as we have seen, in establishing the basis for discussing adaptation, selection and evolution.

Let us then suppose that there is a quantity F (the fitness) which is determined by each particular phenotype (v_i) arising from (6.28), and a family (U_i) of environmental qualities. This hypothesis may be expressed as another equation of state relating the system S and its environment, and will generally be of the form

$$\psi(F, U_i, v_i) = 0. \quad (6.29)$$

Thus, given any phenotype (v_i) generated according to (6.28), and any environment characterized by particular values of the qualities U_i , a definite value of fitness is associated.

It should be noted that the relation (6.29) defining the fitness is itself contingent upon a particular encoding of the natural system S . It states in effect that *if* the phenotype of our system is encoded through the observables v_i , and *if* only the environmental qualities U_i enter into the interaction between S and its environment through which fitness is determined, then that fitness would be expressed by (6.29).

In any case, once the common currency F is postulated, we may compare the adaptedness of any phenotype to any environment by substituting the (v_i) defined by (6.28) into the relation (6.29). In this way, the concept of fitness may be extended from phenotypes to the genotypes and environments which generated them.

We thus have formulated two of the basic pre-requisites, which allow us to generate phenotypes, and to compare them with respect to a common currency. This allows us to speak of phenotypes (or their genotypes) as adaptive to a particular degree under a given set of environmental conditions. The third pre-requisite is a notion of *selection*, which as we have seen, allows us to act on genotypes through their corresponding phenotypes. Such a selection mechanism induces a dynamics in a space of genomes; it is thus a mechanism for modifying the *identity* of the natural system with which we are dealing, and hence generates an *evolution* of the system. Thus, analogous to (6.25), we shall write

$$d(\alpha_i)/d\theta = k \nabla F(\alpha_i). \quad (6.30)$$

In general, the trajectories of this dynamics will take us to *stationary values* of the fitness in a fixed set of environmental conditions. As before, we emphasize that the time scale (defined by the differential $d\theta$) is *necessarily* different from the time scales appearing in the relations governing the generation of phenotypes through (6.28); it were not so, we would not be dealing with evolution.

Our final pre-requisite is the existence of a definite linkage between the qualities U_i which determines the fitness, and the qualities E_i which generate the behaviors or phenotypes. For generality, we shall write this linkage in the form

$$\Gamma(E_i, U_i, V_i) = 0 \quad (6.31)$$

where E_i and U_i are as in (6.28) and (6.29) respectively, and the V_i represent other environmental qualities on which the relation between the E_i (which are the qualities our system perceives) and the U_i (which are the qualities which define the fitness or adaptive character of system behaviors or phenotypes) may depend. Under the special hypothesis of $V_i = \text{constant}$, we recover the situation described explicitly in our previous metaphor. However, if the V_i are not constant, then the linkage relation between the U_i and the E_i will not be fixed; accordingly, the nature of the selection mechanism (6.30) will change, and with it the character of the evolutionary dynamics (6.30).

It is now very easy to see, by means of essentially the same argument as before, that the effect of the selection mechanism (6.30) is twofold: (a) it serves to generate within the evolving systems an image of the linkage between the perceived qualities E_i and those qualities U_i on which fitness depends; i.e. it generates an image of the linkage (6.31) within the system; and (b) because of the fact that the time scale for the selection dynamics is *slower* than that for the behavior dynamics, a *predictive model* for the fitness F (i.e. an image of the equation of state (6.29) is likewise generated within the system. Indeed, if in (6.29) we replace the U_i by the E_i according to relation (6.31), and we replace the v_i by the specific function of E_i and α_i obtained from (6.28), then the fitness F becomes a function of the α , and the E_i alone; clearly, the selection dynamics (6.30) will drive us towards that genome for which the resultant phenotype maximizes fitness, and possesses all of the other properties we have described.

It would take us far beyond the scope of the present volume to consider the various implications of the selection picture we have drawn.⁸ We have merely attempted to indicate how it is that selection and adaptation in fact generate specific predictive models, in such a way that the behavior of an organism at an instant of time bears a definite relation to an internal prediction about a *later* instant. In fact, a general theory of macroevolution can readily be built on the framework we have introduced, incorporating all of the traditional biological features of the Darwinian picture; the reader should explore for himself how such a general macroevolutionary theory would look. In particular, the reader should consider how the properties of the linkage (6.31) bear upon the *stability* of the models generated by selection, and how this in turn manifests itself on the effect of selection in changing environments. We shall consider some aspects of this circle of ideas in more detail in Sect. 6.7 below.

References and Notes

1. There are at least three distinct spheres in biology in which the term “adaptation” plays an essential role: evolution, physiology, and behavior. In evolution (and more generally, in population genetics) the whole thrust of natural selection is to generate what we may loosely call adaptation to definite environments or niches. Almost any book about evolution is a treatise about adaptation and how it is generated; for specific discussions and further references, see e.g.

Leigh, E. G., *Adaptation and Diversity in Natural History and the Mathematics of Evolution*. Freeman (1971).

Williams, G. G., *Adaptation and Natural Selection: A Critique*. Princeton University Press (1966).

In physiology, a host of adaptive mechanisms are studied, such as the “pupillary servomechanism”, which controls the amount of light admitted to the eye (cf. Sect. 1.1, and Note 6 to that chapter). At one time, there was intensive study of “adaptation” in bacteria, by which was meant their capacity to rapidly adjust their internal chemistry to utilize whatever carbon source was available;

this ultimately led to the discovery of inducible and repressible enzyme synthesis, the control of gene expression, and the operon hypothesis of Jacob and Monod (cf. Sect. 3.5, Note 15), among other things. On the behavioral side, which overlaps with physiology to an indefinite extent, depending on the author, we have the entire spectrum of learning and conditioning, which we shall discuss separately in Sect. 6.6 below. All of this sprawling biological literature is related in two ways: (a) one can argue that the capacity to manifest physiological and behavioral adaptations must itself evolve; and (b) the encodings appropriate for the study of *all* adaptive mechanisms are metaphorically related. Indeed, in all cases, the problem of adaptation involves a functional relation or linkage $\Phi(x_1, \dots, x_n) = 0$ between systematic and environmental observables; the problem is to transform the systematic variables so as to maintain the linkage when the environmental variables are modified. This is the essence of our previous assertion that adaptation (i.e. something being modified) and homeostasis (something remaining invariant) are two sides of the same coin; adaptation is concerned with what is changing, and homeostasis with what is invariant.

It might be remarked parenthetically that one aspect of adaptation studies which has become clinically popular is the study of “stress” (cf. Selye, H., *Stress*, McGraw-Hill 1950); here, the main interest lies in the untoward effects of physiological adaptation to “stressful” environmental modifications, and how they may be controlled. This has led to quite an extensive literature in which the term “adaptation” plays an important role; cf. Helson, H., *Adaptation-Level Theory*, Harper & Row (1964), or Coelho, G. V., *Coping and Adaptation*, Basic Books (1974).

Adaptation can also be studied in areas remote from biology, as in the context of human systems (cf. Day, R. H. and Groves, T., *Adaptive Economic Models*, Academic Press 1975), or in terms of mechanical artifacts and their control (cf. the extensive literature on adaptive control. In technical terms, an *adaptive* control system is one in which controller characteristics are modified according to the performance of the system being controlled. There is thus a close relation between adaptive control and the idea of a differential game; cf. Note 4 to Sect. 4.6 above). See for instance Yakowitz, S. J., *Mathematics of Adaptive Control Processes*, Elsevier (1969).

Finally, adaptation can be studied in quite abstract terms, yielding results which simultaneously illuminate all the various topics we have mentioned above. One example of this may be found in a long series of papers by Bremermann, in which a frank mutation-selection model is utilized to solve general optimization problems (cf. Bremermann, H., in *Biogenesis, Evolution, Homeostasis* (A. Locker, ed.), Springer-Verlag 1973, pp. 29–37). Another extensive formalism of this character was developed by John Holland (*Adaptation in Natural and Artificial Systems*, University of Michigan Press 1975).

Thus, it is clear that the spectrum of studies of adaptation is broad indeed.

2. In particular, this Principle refers to the equilibria of closed systems, and the response of such an equilibrium to an externally imposed perturbation.

Specifically, if such an equilibrium is disturbed through a change in *one* of its state variables, the system will move to a new equilibrium; in moving to that equilibrium, the state variable which was originally perturbed will move in the direction opposite to the one in which it was originally perturbed.

3. However, it may be argued that in a cybernetic homeostat, the controller imposes a measure of fitness on the states of the controlled system; this fitness is measured by the deviation of a state of the controlled system from the set-point in the controller.
4. For a fuller discussion of the Principle of Optimal Design and its correlates, see N. Rashevsky 1960. *Mathematical Biophysics*, Dover (1960).
5. The identification of fitness with fecundity is probably due to R. A. Fisher. Strictly speaking, it is most correctly used in population genetics as a measure of how the frequency of a particular gene is increasing or decreasing per generation in the gene pool under consideration. See e.g. Fisher, R. A., 1930. *Genetic Theory of Natural Selection*. Reprinted Dover 1958.
6. For an extreme statement along these lines, see e.g. Williams, M. B., *Lecture Notes in Biomathematics* #13, 226–240, Springer-Verlag (1977).
7. In evolutionary biology, convergence is usually restricted to morphological rather than behavioral characteristics; for a typical discussion, see e.g. Rensch, B., *Evolution Above the Specific Level*, Wiley (1959). However, it is clear that our discussion can be entirely recast into morphological terms. Similar ideas of convergence appear from time to time in the social and political sciences, with respect to bureaucracies, firms and even entire states.

6.6 Learning

In the present chapter, we wish to explore the following proposition: that the apparently disparate phenomena of evolution and of learning are in fact linked to each other, in the sense that a metaphor for the one is, at the same time, a metaphor for the other.¹ In fact, we can translate an evolutionary metaphor into a learning metaphor by means of a specific mapping process in which observables of the former are simply re-interpreted, or translated, into observables of the latter. From this it will immediately follow, from the arguments of the preceding chapter, that *learning processes generate predictive models*.

To see this, it is most transparent to proceed by developing a specific learning metaphor, and show how it comprises a realization of the evolutionary formalism developed in the preceding chapter. We will develop this metaphor within the context of the neural networks described in Example 4 of Sect. 3.5 above, which, it will be recalled, were themselves metaphors for the activity of the central nervous system in organisms.² What we will describe is a class of devices commonly called *perceptrons*, and which received a great deal of attention not so many years ago.

Let us then suppose that we are given an arbitrary neural network, as indicated schematically in Fig. 6.7 below:

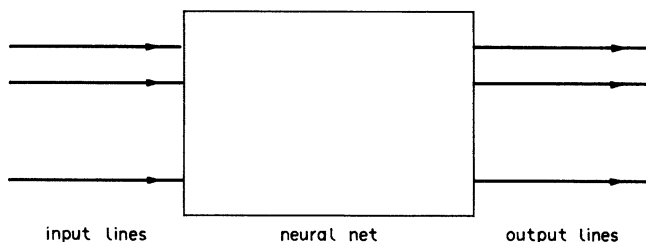


Fig. 6.7

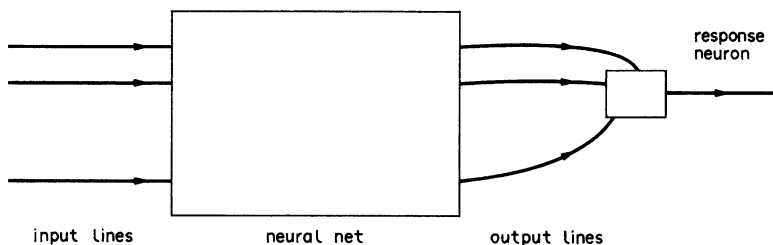


Fig. 6.8

We will consider the inputs coming into the network from the left as *sensory inputs*, whose states of excitation at any instant is determined completely by the environment of the network. These are the afferent lines to the network. Likewise, the output lines, leaving the network on the right, are the efferent lines, which may intuitively be regarded as driving specific responses to the sensory inputs.

Let us imagine that all of these efferent lines innervate a single organ, which can be in one of two states (“on” or “off”). If this is so, we may as well consider the innervated organ to itself be represented by a formal neuron which we shall call the *response neuron*. Thus, the network of Fig. 6.7 can be enlarged to give the network shown in Fig. 6.8 below:

Let us suppose at some initial instant t_0 the neural network N is in some definite state; e.g., the state in which no neuron in the network is on. At that instant, we impose a specific pattern of excitation on the afferent in input lines to the network. It is easy to see that, if there are k such input lines, there are 2^k distinct patterns of excitation which can be so imposed. According to the theory of neural networks as we developed it previously, at some subsequent instant, the response neuron will either fire or not. Whether it fires depends on both the nature of the input pattern, and on the constitutive parameters of the network (i.e. on its wiring diagram, the neural thresholds, and the magnitudes of the weights assigned to each of the internal axons of the network). If the response neuron fires at a subsequent instant under these conditions, we shall say that the network *recognizes* the initial input pattern; otherwise, we shall say that the network does not recognize the pattern.

Let us denote by ξ the set of possible input patterns to our network; as noted above, ξ contains 2^k elements, where k is the number of input or afferent lines

to the net. It is clear that our network necessarily decomposes ξ into two disjoint subsets which together exhaust ξ ; namely, the class of patterns which are recognized by the network, and the class of patterns which are not recognized. The specific set of input patterns which are recognized is thus a function of the constitutive parameters of the network we are using; if these are changed, so is the partition of ξ defined by the network. Any such network can thus be regarded as a meter, which defines a particularly simple kind of observable on ξ .

Now let us suppose that we arbitrarily choose a subset $\xi_r \subset \xi$, which we wish the network to recognize. That is, we wish the response neuron to fire at some subsequent instant if and only if the input pattern imposed on it belongs to ξ_r . In general, of course, the class of patterns recognized by the network will not coincide with ξ_r . However, we can *modify* the class of patterns recognized by the network by *changing the constitutive* parameters which define the network; i.e. by changing the pattern of interconnection and/or the thresholds and/or the weights which characterize the network. Let us focus our attention on the weights. Imagine that at an initial instant t_0 we impose some input pattern E from ξ on the input lines to the network. We shall say that the network responds *correctly* to this pattern if either: (a) $E \in \xi_r$, and the output neuron of the network fires at a subsequent instant, or (b) $E \notin \xi_r$, and the output neuron of the network does not fire at that instant. If the network has responded correctly, we make no change in it. If the network has not responded correctly, however, we shall modify the weights of its internal connections, according to some fixed algorithm, called a reinforcement algorithm, whose precise nature need not concern us here. We then repeat the process, by resetting the network to zero, picking another input pattern from ξ , and ascertaining whether the response of the network is correct or not. Then, it is a theorem that, under these conditions, there exists a reinforcement algorithm such that, if all the patterns in ξ are shown to the network as indicated a sufficient number of times, the reinforcement algorithm will force the network to “converge”; i.e. so that the correct response will thereafter be made to any pattern in ξ . That is, the network will “learn” to recognize precisely those patterns which belong to the pre-assigned set ξ_r . Moreover, the reinforcement algorithm in question does not depend on ξ_r ; i.e. the *same* algorithm will force convergence to any ξ_r . This is the principal theorem on which the study of perceptrons as paradigms for learning is based.

A great many experiments were performed at one time in constructing perceptrons of the above type which could be “trained” to recognize visual patterns. This was done by connecting the afferent inputs to a network of threshold elements to an array of photocells, which could be differentially illuminated; such an array of photocells was termed the “retina” of the perceptron. Of particular interest was the training of perceptrons to recognize relatively abstract classes of patterns characterized by the presence of specific qualitative *features*; e.g. to train the device to recognize the letter “E” regardless of its size or orientation on the retina.

Obviously, by increasing the number of response neurons, we can partition the set ξ of input stimuli into any number of classes. An analogous result to that stated above holds in this more general case; the same network can be trained to *discriminate* between the patterns belonging to each of these classes. Thus, we could

in principle train a network to recognize (i.e. to discriminate) between, say, the 26 characters comprising the English alphabet. However, we shall restrict attention to the case of a single response neuron, because of its simplicity, and because it already exhibits the essential features to which we wish to draw attention.

Let us now reformulate the situation we have described, which is clearly a metaphor for learning (in this case, learning to discriminate between afferent excitation patterns belonging to different classes) in the language developed in the preceding chapter.

First, let us note that for any neural network, there is a definite equation of state of the form

$$\Phi(E, \alpha, v) = 0 \quad (6.32)$$

where $E \in \xi$ is a particular pattern of excitation imposed on the input lines to the network at an initial instant t_0 , the quantity α denotes the totality of constitutive parameters characterizing the network (and in particular, the weights of the internal axons), and v is the state of the response neuron at some subsequent instant. Thus in this equation of state, E is environment, α is genome, and v is phenotype.

Next, we suppose that we have a distinguished observable $U : \xi \rightarrow \{0, 1\}$, which has the property that $U^{-1}(1) = \xi_r$, the set of inputs we wish the network to recognize. We shall define the *fitness* $F(v)$ of a phenotype v , arising from a given environment $E \in \xi$, to be 1 if $v = U(E)$, and zero otherwise.

Third, we observe that the imposition of a reinforcement algorithm precisely embodies a *selection mechanism*, which allows phenotypes to act on the genomes which generated them, according to the fitness of the phenotype. In this discrete setting we cannot of course talk about gradients; however, if we denote by α' the genome arising from the genome α from the implementation of a reinforcement algorithm, it is clear that there is an equation of state relating α and α' which is of the form

$$\Psi(\alpha', \alpha, F(v)) = 0. \quad (6.33)$$

It is clear that the time scale governing the transition $\alpha \rightarrow \alpha'$ has no relation to the time scale in which the phenotype v is actually generated by the network. In the present context, the “time scale” in which the genotype transitions take place is given simply by the *sequence* in which the rules of the reinforcement algorithm are applied. The only constraint on this time scale is clearly that the “time unit” for the application of the reinforcement algorithm must in some sense be *slower* than the time unit governing the dynamical processes by which the phenotype v is generated.

Finally, we note that there is a definite relation between the input E “seen” by the network and the quantity U which determines the fitness of the resulting phenotype. This relation can be expressed by another equation of state of the form

$$\Gamma(E, U(E)) = 0. \quad (6.34)$$

It is now clear that there is an exact analogy between our discussion of selection in the preceding chapter, and our metaphor for the generation of a particular discriminatory capability (i.e. learning) in perceptrons. Accordingly, the selection

mechanism which “trains” the perception thereby creates a predictive model of the selection mechanism itself. Insofar as defining a class ξ_r of inputs to be recognized can be thought of as specifying a “feature”, which is common to all the elements of ξ_r but is missing from all other inputs, the selection mechanism embodied in the reinforcement algorithm converges to a genome which *extracts* this feature, and uses it as a predictor. It is indeed correct to say that the “training” of a perceptron is equivalent to the establishment of a *tropism* in the system.

The simple metaphor for learning embodied in the perceptron is of course capable of infinite variation and extension. This has been one of the major driving forces in the study of “self-organizing” systems, both in their own right, and as metaphors for biological activities characterized by learning, evolution, intelligence, etc. Indeed, the perceptrons themselves constituted one of the early pillars of the field of “artificial intelligence”; an attempt to construct alternate realizations (in the form of machines) of behaviors manifested in organisms through metabolic, genetic and neural mechanisms. There is an enormous and sprawling literature devoted to these problems, but the field is still in a state of relative immaturity. Part of this stems from a failure to take account of the distinction between a metaphor and a model; part of it involves the absence of a sufficiently comprehensive viewpoint (including a failure to clearly articulate the separate ingredients which must enter into a coherent theory of adaptation).

Perhaps the most general formulation of perceptron-type metaphors was suggested by Norbert Wiener in one of the appendices to the second edition of his book *Cybernetics*. This suggestion uses the fact, which we alluded to at the outset, that any behavior which can be described can be simulated. For definiteness, suppose that a particular behavior can be represented as a function $f(x_1, \dots, x_n)$ of n variables. Under very general conditions, such a function can be represented in terms of a suitable basis set; say polynomials. Thus we can write

$$f(x_1, \dots, x_n) = \sum a_i \varphi_i(x_1, \dots, x_n).$$

The coefficients a_i appearing in such an expansion can be considered as analogous to constitutive parameters, or genome for the behavior in question; in a precise sense, these coefficients *code* for that behavior.

Let us now suppose that we possess devices which can generate the basis functions φ_i . If we think of the coefficients a_i as *weights* to be associated with the outputs of these devices, then we can imagine a selection mechanism which will “tune” these coefficients in such a way that our array of devices precisely simulates the original behavior. This is an obvious generalization of the ideas embodied in the perceptron; it was used by Wiener to provide a metaphor for, among other things, “self-reproduction” through the employment of the original system as a kind of template to which the selection mechanism must converge. The reader may find it a useful exercise to express these ideas in terms of the formalism we have developed in the preceding chapters.

For our purposes, the essential point to emphasize is that any such selection mechanism can be regarded as creating a system which contains a predictive model

of the selection which gave rise to it; i.e. to a system which changes state in the present in accord with some internal model of the subsequent effect of the change of state on fitness. It is of great interest that the selection mechanism itself, which as we have formulated it proceeds entirely within a reactive paradigm, generates a system which thereby falls outside that paradigm. Namely, if the adapted system is considered in isolation, its behavior will generally be anticipatory rather than reactive. These facts lend a most important perspective to any attempt to understand the behaviors of any system whose behaviors have been shaped by selection.

References and Notes

1. In fact, the sprawling literature on learning runs closely parallel to that on adaptation. Depending on one's viewpoint, it can be regarded either as a part of adaptation, or as a metaphor for it. For instance, the enormous body of work devoted to establishing the relation between neuro-physiology and learning relates to psychology in the same way that population genetics relates to descriptive evolutionary biology. In both cases, it seems hardly fortuitous that under very general conditions, inter-unit interactions automatically generate the right kind of gross behavior. Moreover, as we have seen, learning can be extrapolated from biology to mechanical artifacts (cf. our earlier discussion of "Artificial Intelligence"; e.g. Note 19 to Sect. 3.5 above); this is analogous to the study of adaptiveness in non-biological systems which was discussed in the preceding section. In any case, we can only hint here at the vast literature, which runs in an enormous stream, with many deep canyons, north from neurophysiology; the few samples we cite are hardly representative, but they perhaps connect most closely with the main thread of our discussion:

Anokhin, P. K., *Biology and Neurophysiology of the Conditioned Reflex and its Role in Adaptive Behavior*. Pergamon (1974).

Hebb, D. O., *The Organization of Behavior*. Wiley (1949).

Pribram, K., *The Languages of the Brain*. Prentice-Hall (1971).

Arbib, M. A., *The Metaphorical Brain*. Wiley (1972).

Rosenzweig, M. R. and Bennett, E. L. (eds.) *Neural Mechanisms of Learning and Memory*. MIT Press (1976).

2. The original concept of the perceptron was developed in

Rosenblatt, F., *Principles of Neurodynamics*. Spartan (1962). Perceptrons constituted an initial paradigm for the area of pattern recognition, which in many ways overlaps with Artificial Intelligence, as well as with learning theory, and with applied technology. A more recent and abstract treatment may be found in Minsky, M. and Papert, S., *Perceptrons*. MIT Press (1969).

6.7 Selection in Systems and Subsystems

In the present section, we shall consider some implications of the phenomena of adaptation and selection, when these are viewed from the standpoint of *subsystems* of a given system. Such subsystems may be thought of as individual cells, or organs, of a multicellular organism; or as individuals or institutions in a society. As we shall see, a number of surprising implications emerge when the machinery we have described, which pertains to whole systems, is projected down onto constituent subsystems. Conversely, the same circle of ideas bears on the inverse situation; namely, the possibility that selection mechanisms acting on a number of initially independent systems may establish definite linkages between them, leading to the emergence of a new system at a higher level.

It is convenient to begin our discussion with a particular example, which we have already explored in some detail. Thus, let us consider the perceptrons described in the preceding section. As we have seen, these are systems for which a measure of adaptation is at hand, and for which a selection mechanism (the reinforcement algorithm) for generating adaptive behavior can be defined.

Each neuron in the basic neuron net which constitutes the perceptron may be regarded as a subsystem of the perceptron. As such, each neuron has its own equation of state, whose variables can be categorized as environment, genome and phenotype for that particular neuron. It may happen that the environmental quantities for such a neuron are a subset of the input pattern imposed on the system as a whole, in which case the neuron “sees” a portion of the overall stimulus imposed on the entire network; or it may be that these quantities are entirely internal to the network, and hence linked to the input pattern only through relations involving other neurons; in this case such a neuron “sees” only the properties of other neurons afferent to it. Likewise, the genomic qualities of the neuron may be regarded as contributing to the overall genome of the entire perceptron, or as linked to that genome through relations involving other neurons. Finally, the phenotype of a constituent neuron is linked to the phenotype of the entire perceptron only through an equation of state involving all the other neurons.

As we have seen, the entire perceptron allows a notion of fitness to be defined on it, with respect to which changes in its structure may be regarded as adaptive. In particular, the imposition of a reinforcement algorithm causes precisely such changes in structure. With respect to such a definition of fitness, which obviously pertains only to the system as a whole, we may characterize any change occurring in an individual neuron of the system as adaptive or not, depending on whether the fitness of the entire system is increased or decreased by that change. Thus we may conclude that *a notion of fitness defined for the system as a whole, with respect to which behavior can be characterized as adaptive or not, induces a corresponding notion of adaptation on the behaviors of its subsystems.*

Now let us consider each of these subsystems as if it were an independent system subject to the selection mechanism induced on it, as we described, by the selection mechanism imposed upon the entire system. As the entire system

converges to a maximally adaptive structure, so will each subsystem converge to some structure which will be maximally adaptive with respect to the *induced* selection mechanism. Consequently, as we have seen, each of these subsystems will automatically generate a predictive model of the selection induced on it, and in terms of the environment imposed on it. However, from the standpoint of the individual subsystems, *these induced selection mechanisms are all different from one another, as are the imposed environments which they experience*. As a result, the predictive models which they generate are also all different from one another, and from the total model generated in the total system. Thus, if we consider these subsystems individually, their predictive models will all be different; *i.e. they will bifurcate from one another*. Indeed, not only will they bifurcate from each other, but also from the model generated in the system as a whole.

From these simple observations, a number of conclusions can be drawn which are of considerable interest and importance. For one thing: it is immediately obvious that we cannot in general hope to reconstruct the properties of the predictive models embodied in a fully adapted system by considering the corresponding models generated in subsystems. In a sense, this conclusion is another form of the failure of reductionism as a universal mode of system analysis, which we have already considered at great length above. Our previous discussions of reductionism were cast entirely within the context of the reactive paradigm; what we can conclude now is that the same kinds of difficulties arise (perhaps even more sharply) in the anticipatory paradigm as well.

This kind of situation is quite general; it pertains to any situation in which a system, containing a family of subsystems, is exposed to a selection mechanism. Since each of the subsystems sees only a fragment of the total situation, it can only form a model of that fragment through the selection imposed on the whole system. We can see this most clearly from our own experience, considering ourselves as subsystems of an evolving social structure. Each of us occupies a different position within the structure; thus each of us generates predictive models about the structure as a whole and utilizes them for generating his behavior. These models are all different, and depend at least in part upon the information we receive about the overall behavior of the structure; since our positions in it are generally different, so too will be our information, our models and our behaviors.

So far, we have considered the situation in which the *only* selection mechanism imposed on the subsystems is the one induced from that acting on the whole system. Let us suppose, however, that each of the subsystems has a separate selection mechanism imposed upon it, apart from that induced on it by the system to which it belongs. That is, for each subsystem, we suppose that we can separately define a local notion of *fitness*. Accordingly, the behaviors of each subsystem can be classified as adaptive or not *with respect to its own local measure of fitness*. Let us further suppose that this local measure of fitness or adaptation imposes a selection mechanism, whereby the phenotype of the subsystem can act upon its genome. We can then ask: what will happen to the system as a whole under these circumstances? Let us briefly discuss some ramifications of this kind of situation.

First, there is of course no reason to expect that the local selection mechanisms acting on the subsystems will be concordant with the mechanism induced on the subsystems by the global selection acting on the entire system. Stated another way, the two selection dynamics imposed on the subsystem genomes, considered separately, will *bifurcate* from each other. This means that a behavior in a subsystem which is adaptive from the global standpoint will be maladaptive from the local one, and conversely. In such circumstances, it is clear that the evolutionary process will depend on the relation between the *time scales* characterizing the local and global selection dynamics. Specifically, the selection with the faster dynamics will dominate the behavior of the subsystems, at least over the short term. If the local selection mechanisms are faster, the subsystems will each evolve *independently*, to generate behavior which is adaptive with respect to the local fitness measure, but generally maladaptive with respect to global fitness. If the global selection is faster, the subsystems will evolve so as to generate behavior which is adaptive from the standpoint of the whole system, but maladaptive with respect to the local fitness measure. Likewise the predictive models generated within the subsystems will correspondingly be determined by the faster selection dynamics.

There are any number of intermediate cases which can be envisioned, in which some subsystems are evolving according to a local selection mechanism, and others evolve according to the induced global mechanism, once again depending on which is the faster.

Attention is thus drawn inevitably to the *rates* at which these various selection mechanisms operate. These rates are themselves time-dependent in general, so we may expect that whether evolution is directed by local or global mechanisms in the subsystems is itself time-dependent. In particular, we may expect that there will be definite intervals in which the evolution of the system as a whole is dominated by the local mechanisms, and other intervals in which that evolution is dominated by the global mechanism. The situation emerging here is somewhat reminiscent of the kinds of phase transitions which characterized the Ising models (cf. Example 2B, Sect. 3.5 above), although strictly speaking these are not evolving systems in our sense. It will be recalled that, in the Ising models, under certain sets of environmental conditions, the interaction between neighboring elements (i.e. subsystems) was such that their states became correlated; under these circumstances the system as a whole could be thought of as “ordered”. Conversely, under other circumstances, the environment favored uncorrelated states between neighboring elements, and hence a “disordered” character for the system as a whole. It is precisely the transition between the global character of “order” and “disorder” which is manifested as a phase transition. Similarly, in the evolutionary situation we have been considering, we may expect transitions between a situation in which the global selection mechanism dominates (i.e. is faster) and those in which local selection mechanisms dominate; these will appear very similar to phase transitions between an ordered or co-operative phase, and a disordered or individualistic phase.

The kind of “phase transition” in evolving systems which we have just described has been graphically described by the biologist Garrett Hardin,¹ in a special case which is termed “the tragedy of the Common”. The situation he envisaged

concerned a common pasture, on which any member of a community could bring a flock to graze. When the number of individuals using the Common, and the sizes of their flocks, are small, the situation is stable, and can persist indefinitely. The situation is individually beneficial to each member of the community, and this is manifested by an increase in the size of each individual flock, and by the number of flocks which utilize the Common; this increase tacitly embodies the selection rule imposed on the community, and the model of the situation which each individual forms to guide his actions. However, at some point the utilization of the Common exceeds its carrying capacity. At this point, a new local selection mechanism comes into play, which acts faster than that imposed by the limited carrying capacity of the Common. Specifically, over the short term, each individual will find it advantageous to utilize the Common as heavily as he can, and increase the size of his flock as fast as he can, as Hardin has so vividly described. The result will be, of course, as increasingly maladaptive behavior generated in each individual, as judged by the (unfortunately slow) selection mechanism imposed by the limited carrying capacity of the Common on the community. Ultimately, of course, the community will crash completely; the Common will be destroyed, and all the flocks will be completely and uniformly wiped out.

Such a situation arises whenever the selection mechanism imposed on a system depends on an environmental resource which is depleted by increasing adaptation of the system as a whole. In another form, it appears in the “technological catastrophe” which has been described by numerous authors.² Consider, for instance, the evolution of a fishing industry. Initially, the fish population is large; this is the Common. As the number of fishermen, and the sizes of their catch, increases, the resource supporting them becomes scarcer. At this point, an apparent local advantage goes to those fishermen who employ technological means to find fish and maintain their catch; radar, mechanized trawling equipment, etc. As the resource becomes even more scarce, the associated technology is driven to become ever more refined by this local selection mechanism; in the limit, the technology becomes maximally sophisticated just as the resource becomes completely extinct. Here again, the local selection, acting rapidly, drives the population to behavior which is increasingly maladaptive from the viewpoint of the slower global selection; but it is the global mechanism which is ultimately decisive.

Speaking in purely biological terms, the generation of maladaptive behavior in subsystems, created by a fast-acting local selection mechanism of the kind we have described, results in the establishment of a *parasitic* relation between the subsystems and the total system. Such a relation is characterized precisely by the fact that what appears to be an adaptive response by the subsystem, according to its own local selection criterion, actually decreases the fitness of the total system to which the subsystem belongs. Under appropriate circumstances, as we have seen, the more adaptive the subsystem (i.e. the more effectively parasitic its behavior) the greater is the decline of fitness of the overall system, including the parasite. Because of the disparity of time scales in the local and global selection mechanisms, and precisely because the observables determining global fitness are generally not

directly perceptible by the subsystems, the catastrophic effects ultimately generated thereby are not manifested until it is too late for the subsystems to re-adapt.

Such situations, in which fast-acting local selection rules overshadow slower global selection mechanisms, are also unfortunately familiar in physiological contexts as well. For instance, the concepts of "health" and "pathology" largely pertain to such situations. A malignancy, for instance, is pathological precisely because it represents the manifestation of a local selection rule which becomes increasingly maladaptive from the standpoint of the total system to which it belongs; the malignancy is indeed a parasite, which ultimately causes the collapse of the entire system to which it belongs, and of itself as well. Conversely, the state of "health" is manifested when any local selection rules which may exist are too slow to affect behavior in a maladaptive fashion, or when they act in the same direction as global selection (i.e. so as to increase the fitness of the system as a whole).

The situation we have been describing can be expressed in another language. In effect, the generation of a parasitic relation between a system and its subsystems arises because local selection and global selection bifurcate from one another, so that a behavior adaptive from the local standpoint becomes maladaptive from the global one. We may express this by saying that such a bifurcation manifests a *conflict* between the local and global selection rules. It is a kind of conflict which arises entirely from the differential effects of selection mechanisms on systems and their subsystems. It can only be "resolved" in one of two ways: (a) by arranging matters so that the global selection mechanism always acts faster than local mechanisms, or (b) by eliminating local selection altogether. This last amounts to the imposition of specific constraints on the subsystems, which eliminates their capacity to respond to any local measures of fitness, apart from those induced on them by the system as a whole. The circumstances under which this can be done are, of course, not yet known; however, that it can be done is rendered plausible by the persistence of multicellular organisms, and indeed, the successive growth of their complexity, through biological evolution; and by the similar emergence and persistence of complex social structures, which have independently evolved many times in a number of different phyla. Such organisms and societies are characterized precisely by the fact that at least under most circumstances, selection does not act locally on the individual cells or members respectively, but globally on the system as a whole. On the other hand, the ecosystems to which such organisms and societies belong are characterized by the absence of global selection, and the predominance of local selection mechanisms on the individual constituents. Thus, for instance, it is hard to imagine a measure of fitness which would pertain to an ecosystem as a whole; and hence we cannot characterize a behavior of a constituent organism as "adaptive" from the standpoint of the ecosystem to which it belongs. On the contrary, the concept of adaptation prevalent in biology is defined completely in terms of the fitness of *individual* organisms.

It might be relevant to point out that elimination of local selection mechanisms according to (a) or (b) above, while eliminating the conflicts that lead to parasitism of subsystems on the total system, comes at a high cost. Namely, it leaves the total system vulnerable to the phenomena of *senescence* which were described in

the preceding chapters. For once an internal model is generated, the absence of local selection mechanisms means that there is no way to change it. As we saw abundantly, there must be in general a growing discrepancy between the predictions of such a model and the actual properties of the organism; consequently, the behavior generated by the model must necessarily become increasingly maladaptive. For instance, the model incorporated in the forward activation step described in Sect. 6.2 above, once genetically encoded into a particular individual, cannot be changed by selection mechanisms imposed on that individual. The enzyme involved in this step cannot become parasitic on the cell which contains it, but on the other hand, we have seen that its behavior becomes increasingly maladaptive, ultimately leading to a generalized system failure which can be identified with senescence. In biology, of course, the kind of failure is at least partially circumvented by *proliferation*, which keeps resetting the model to zero in the progeny. But they too, in their time, must necessarily senesce in this fashion.

This kind of conflict we have been describing, which involves a bifurcation between local and global selection rules, is only one of the kinds of conflict which can emerge in evolving systems. A somewhat different kind of conflict can arise *between* the constituent subsystems of such a system, and is based on the fact alluded to above, that each such subsystem generates a *different model* of the global selection mechanism imposed on the system as a whole. Under appropriate circumstances, these models can themselves bifurcate from one another, and we can thus speak of “conflicts” between these models. Such a conflict between models becomes important when the subsystems involved are both required to generate some phenotypic behavior which pertains to the system as a whole. To put the matter somewhat anthropomorphically, when the models employed by the subsystems to generate behavior bifurcate, each subsystem will “decide” that a particular phenotypic behavior is adaptive, and the two decisions will be different. Indeed, this is the source of conflict which is most visible in human societies, either between individuals, or between institutions. Under such circumstances, the same objective situation will be perceived differently by the subsystems, and different courses of action will be advocated and pursued. In a complex society, for which by definition many models are possible, there are accordingly many such sources of conflict.

Such situations of conflict can, of course, also arise within a single organism possessing a number of sensory modalities. Thus, a stimulus imposed on such an organism, processed through several such modalities, may simultaneously activate contradictory impulses to effector or motor organs. For instance, an organism which is both chemotropic and phototropic may be placed in an environment in which the phototropic response requires movement in one direction, and the chemotropic response requires movement in the opposite direction. In such organisms, there generally exists specific mechanisms which serve to resolve these conflicts, and commit the organism to a single response in the presence of ambiguous environments; these are collectively called “integrating mechanisms”.³ The point to stress here is that such an integrating mechanism is required precisely because different subsystems of a complex system necessarily generate different models of that system, which will typically bifurcate from one another; when such models both

effect a common response, conflict will arise which must be resolved before any response can be made. Most conflict in human societies appears to be of this form, and the integrating mechanisms available to resolve them are unfortunately not always adequate. For these purposes, an understanding of the nature of bifurcation between alternate models, which we have considered in great detail in the preceding sections, must play a crucial role.

So far, we have considered the relations which may exist between an evolving system and its subsystems. Let us now take up the converse question; namely, if we start with a number of initially independent (i.e. unlinked) systems, under what circumstances can they interact so as to form a composite system which itself can evolve?

Such a question is of considerable importance, not only for dealing with biological questions bearing on the evolution of multicellularity or social organization, but also for social technologies within our own society. For instance, it is a common belief that the institutions of our society are in some sense not sufficiently adaptive; that the various crises with which our society is increasingly faced arise from a state of non-adaptation or maladaptation. Implicit in this view is the idea that we must treat our institutions as if they were *individuals* in the biological sense; we must intrinsically define a measure of fitness for them, and a selective mechanism which would allow them to evolve in the direction of increasing fitness; as we have seen, that is what making them adaptive means. On the basis of the preceding discussion, however, two essential aspects of this situation must be considered: (a) as we have pointed out above, fast local adaptive mechanisms (e.g. those which might be imposed on societal institutions, as opposed to society as a whole) will generally bifurcate from each other, as well as from any slower selective mechanism operating on the whole society. Under such circumstances, unless very special (and very imperfectly understood) conditions are satisfied, an integrated social structure may not be possible at all. (b) The perceived "malfunctioning" of social institutions, far from arising because they are insufficiently adaptive, may well occur because they are *already too adaptive*. That is, they are already behaving as biological individuals, at the expense of the social structure to which they belong; i.e. they are already parasitic on that structure, in the sense we have described above. In such a case, a causal increase in their adaptive capabilities will only render them more parasitic, and again make any overarching social structure impossible.

In any case, we see that questions of adaptation and maladaptation must be approached with great care. Moreover, they cannot be meaningfully discussed apart from general notions of fitness, which give them meaning and from selection mechanisms, which serve to generate adaptive behavior. These in turn depend in the last analysis entirely upon modeling relations, and especially on the manner in which predictive models can bifurcate from one another. Thus, the questions in which an understanding of social organization must be based, as well as the basic questions of biology, all ultimately devolve on those with which we have been concerned at such great length above. It thus appears that the concept of a model, and the relationships which exist between models, lies at the root of everything which we need to know.

References and Notes

1. Hardin, G., *Science* 131, 1292–1297 (1960).
2. See for example
 Ellul, J., *The Technological Society*. Knopf (1973).
3. The problem of establishing coherent behavior in a population of semi-autonomous units is one of the great problems facing any reductionistic analysis of complex systems, or any attempt to generate them from structural subunits. In principle, the easiest way to generate coherent behavior is to provide the population with some external cue (e.g. the conductor of an orchestra); this is what generates coherent behavior in systems like the Ising models. Somewhat more interesting are situations in which the units generate the coherence internally, as in populations of weakly coupled oscillators; these will generally entrain to a common frequency (see e.g. Winfree, S., *The Geometry of Biological Time*, Springer-Verlag 1979). At a somewhat higher level, coherence-generating mechanisms are often called integrating mechanisms, and are much studied from the standpoint of the control of complex neuromotor activities (see for example Kilmer, W., McCulloch, W. and Blum, J., *Int. J. Man-Machine Studies*, 1, 279–309 1979). At still higher levels, coherence is often called *consensus*; its absence is called *conflict*. One of the great mysteries is how to generate consensus; one approach, applicable in very limited situations, has been called “Delphi”; see for instance Linstone, H. A. and Turoff, J., *The Delphi Method: Techniques and Applications*, Addison-Wesley (1975). In general, however, the ontogenesis of coherent behavior is shrouded in obscurity, though it is obviously a matter of some importance.

6.8 Perspectives for the Future

It is now time to pull together the various threads we have woven; to see where they have taken us so far, and where they direct us for the future. To that end, we will review the basic ideas which have entered into our development, and specifically pose a few of the many open questions which remain.

The point of departure for our entire development was the recognition that most of the behavior we observe in the biological realm, if indeed not all of the behavior which we consider as characteristically biological, is of an anticipatory rather than a reactive character. In fact, if it were necessary to try to characterize in a few words the difference between living organisms and inorganic systems, such a characterization would not involve the presence of DNA, or any other purely structural attributes; but rather that organisms constitute the class of systems which can behave in an anticipatory fashion. That is to say, organisms comprise

those systems which can make predictive models (of themselves, and of their environments) and use these models to direct their present actions.

We saw very early that the behaviors of systems which employ predictive models are vastly different from those which do not. At the most fundamental level, anticipatory systems appear to violate those principles of causality which have dominated science for thousands of years. It is for this reason that the study of anticipatory systems *per se* has been excluded routinely from science, and that therefore we have had to content ourselves with *simulations* of their behavior, constructed in purely reactive terms. Restriction to such simulations has limited us to the study of the *mechanisms* whereby biological actions are effected, thereby distorting our approach to biological processes, and in fact precluding from the outset any basic understanding of how these processes actually work.

Once we recognized that anticipatory behavior is the general rule in biological systems, and that it depends essentially on the presence of predictive models, our attention was inexorably drawn to the nature of the modeling relation itself. What, indeed, is a model? The better part of our development was concerned with attempting to clarify this question. In its most general terms, we found that a modeling relation between systems is established through an encoding of qualities pertaining to one of them into corresponding qualities of the other, in such a way that the linkages between these qualities are preserved. In science, we generally attempt to encode qualities of natural systems into purely formal (i.e. mathematical) ones, in such a way that the rules of inference of the mathematical system correspond to causal relations, and particularly dynamical relations in the natural system. We found that the same natural system generally admits many models, depending on which of its qualities are thus encoded; indeed, the *complexity* of a natural system is perceived through the number of distinct modeling relations into which it can enter. Conversely, we found that many different natural systems can admit the same model; this provided the basis for the fundamental concept of *analogy* between systems, and the use of analogy as a powerful scientific tool. We sought to illustrate these ideas with many examples, drawn from the widest possible variety of scientific disciplines; not only to show their universality, but also to demonstrate that the concept of a model is not something exotic or unusual, but rather of the broadest currency imaginable.

The raw material for the construction of modeling relations is, and must be, the result of observation. In the broadest sense, observation provides the means by which the qualities of natural systems are defined and represented. As we developed it, observation involves the dynamical *interaction* of natural systems, and the employment of the change of state induced in one of the interacting systems as a label for the value of some corresponding quality of the other system. Thus, we stressed that the making of observations, which is generally considered the hallmark of empirical or environmental science, already involves the concept of a model in an essential way, and thus theoretical science (i.e. the study of models) is simply a kind of extension of the process of observation itself.

We also pointed out in this connection that an act of observation is a quintessential act of abstraction; the observation of a single quality of a natural system is indeed

the greatest kind of abstraction which can be made of that system. From this point of view, the development of theoretical science is an attempt to combine observations in such a way that our view of systems becomes *less* abstract than it could be if we were restricted to observation alone. Thus, we stressed what should be a commonplace; that there is no antagonism between “theory” and “experiment”; it is unfortunately not a commonplace, because it has been obscured by the antagonism between “theorists” and “experimentalists”.

Even though models are, in this sense, less abstract than observations, they are nevertheless abstractions. Thus it becomes of great importance to understand how different models of the same system are interrelated. The crucial concept here was that of bifurcation. Indeed, a modeling relation can generally be formulated in mathematical terms as a conjugacy; the failure of a modeling relation, which is a logical independence between two modes of description, thus becomes exactly a bifurcation in the mathematical sense. We saw how this concept of bifurcation was related, on the one hand, to phenomena of emergence, and on the other hand, to the concept of error. As we formulated it, error is not a stochastic phenomenon, but rather indicates a discrepancy between the behavior of a natural system and the corresponding behavior of a particular model of that system.

Since the point of our development was the consideration of *predictive* models, it was necessary to describe the concept of time in some detail. We found that time itself is complex, in the sense that it admits many different kinds of encodings, and these encodings can themselves bifurcate from one another. Indeed, it is only within a class of systems which are themselves already similar in some non-temporal sense that a concept of time can be uniformly introduced which is valid for all the systems in the class. It was in fact the non-comparability of time scales generated by different kinds of dynamical processes which was at the heart of our discussion of selection and adaptation, which is the process by which predictive models are actually generated in biological systems.

Armed with a deeper understanding of the modeling relation itself, how then shall we approach the study of systems whose behavior is controlled by models? As we saw, there are several different kinds of questions we can ask about such systems, each with its own emphasis and point of departure. Let us illustrate a few of them.

First, we may ask how best to study the “physiology” of an anticipatory system. That is, we suppose a given system which contains some predictive model, and uses that model to determine its present behavior. We do not ask how that model was generated, either ontogenetically or phylogenetically, but rather seek to understand the behavior of the system as a whole. Certain aspects of the behavior of such a system can be understood without knowing the specific nature of the model employed by the system, but follow from the general character of modeling relations. For instance, we know that the model, as an abstraction, must ultimately bifurcate from what it models; thus any such system is in a sense *spanned*, and must undergo a characteristic form of senescence, as we have seen. We indicated how the character of this senescence can tell us something about the nature of the model, and how it is linked to the other qualities of the system.

However, for detailed understanding of such a system, we need to know specifically what the model is which the system is employing to generate its behavior. Thus the basic question arises: *how can we determine this model, from observations performed on the system itself?* This is the basic question underlying the “physiology” of anticipatory systems; it is one which has no counterpart in the theory of purely reactive systems, and raises a host of entirely new problems of both a theoretical and a practical character. Analogous problems which can be formulated in a formal context should be of great help in telling us how to approach this basic question; for instance, how do we need to observe a computer, or a Turing machine, in order to determine its program? It should be noted that the reactive paradigm itself, based on developments initiated in particle mechanics, presume that there are in general effective procedures for extracting system laws (i.e. linkages between observables, or equations of state) from appropriate observations of the observables themselves; we are now asking explicitly how (and indeed, whether) this can be effectively done in the context of anticipatory systems. Such questions have received relatively scant attention in the past; they now are seen to be of the essence.

Another kind of basic question we can ask about anticipatory systems concerns their “ontogeny”; the manner in which they are generated. We have suggested in the preceding sections of the present chapter that the ontogeny of the models, and their role in control of anticipatory behavior, is the natural consequence of selection mechanisms; that these in turn involve the closely linked ideas of fitness, adaptation and a mechanism through which genome can act on phenotype. Here, of course, the terms “genome” and “phenotype” are to be understood in very general terms, they are ways of classifying the qualities which appear in an equation of state. Thus, the operation of a selection mechanism is a metaphor both for biological evolution and for the structurally quite different phenomena usually associated with the notion of learning. It is quite clear that a study of the ontogenesis of anticipatory systems, governed by the requirement that adaptation (as measured by fitness) increases, can throw some light on the physiological problems articulated above. Of particular interest in this regard, as we sketched in the preceding chapter, is the circle of ideas relating the several subsystems of an adaptive system, and the manner in which a study of subsystems might bear upon the determination of an anticipatory model in the system as a whole. However, here too, the necessary ideas have only just begun to be formulated, and are themselves still in an essentially embryonic state.

A third circle of questions concerns the basic question of how we can hope to apply an understanding of anticipatory systems to develop a technology of anticipatory control. Such a technology would be of vast importance, in many vital areas. For instance, in a medical context, it is clear that many of the so-called metabolic diseases, which are as yet so imperfectly understood, can be thought of as derangements of anticipatory mechanisms. Similarly, as we have suggested, senescence can be regarded as a generalized maladaptation arising from a growing discrepancy between what the system’s internal models are predicting, and what the system itself is actually doing. As we saw, the hallmark of this type of senescence is a kind of generalized maladaptation, without any localizable failure in specific subsystems. Likewise, hormonal control seems to be of an essentially anticipatory

character; a hormone is thus to be regarded as a predictor, representing some anticipated future state of the organism. To approach endocrine mechanisms in this way is to cast an entirely new light on endocrine disorders, and points up once again the need to extract from an anticipatory system some information about the character of the models employed by the system. Exactly the same may be said about other vital physiological subsystems, such as the central nervous system and the immune system, which in some sense seem to work entirely off models. Hence a theory of anticipatory systems seems bound to find crucial applications in medicine.

Finally, of course, we must return to the circle of applications which were the initial point of departure for our entire development; namely, the management of our own societies. Here, too, we feel that a deep understanding of anticipatory systems in general, and the character of the modeling relations which direct them, will be central. Furthermore, the ubiquitous character of anticipatory mechanisms in biology, and their emergence through selection mechanisms, provides for us a vast encyclopedia for how to solve complex problems of the type with which we are presently confronted (and also, equally usefully, of how not to solve them). This encyclopedia represents a natural biological resource to be harvested; a resource perhaps ultimately more important to our survival than the more tangible resources of food and energy. To learn to exploit this resource to the full involves an understanding of the metaphoric relations between biology and social systems, which we are only beginning to be able to grasp. Indeed, it was for this purpose that we went so deeply into the character of metaphorical relationships between systems in the above pages. The formal tools arising from these considerations represent a beginning, but still only a beginning, in these directions.

One other problem regarding anticipatory systems, unique to the human realm, may also be mentioned. The basic situation with which we have dealt so far involves the interaction of an anticipatory system with an environment that is non-anticipatory; i.e. is describable entirely within a reactive paradigm. We may, however, ask what happens when an anticipatory system must interact with an environment which is itself anticipatory. This is the situation embodied most graphically in the illustration below:



The behavior of such systems is characteristic of human interactions. The closest approach to a theory of such interactions is found, of course, in the Theory of Games. But this theory is in many ways phenomenological and unsatisfactory; it is like a probabilistic theory of error, and awaits a more basic theory arising from fundamental principles. The development of such a theory of interacting anticipatory systems represents yet another direction for future research.

Finally, we come full circle to the ideas of Robert Hutchins, with which we started. His basic question, it will be recalled, was: "What ought we to do now"? The crucial word here is "ought"; a word which has traditionally been regarded as foreign to science. Indeed, if we stay entirely within a reactive paradigm, this word never arises. Perhaps this is the fundamental reason why Hutchins was so suspicious of scientists, and hence of science; in his view, there had to be something fundamentally missing in a field in which the word "ought" was excluded as a matter of principle. However, in the study of anticipatory systems, we find that "ought" is of the essence; the character of a predictive model assumes almost an ethical character, even in a purely abstract context. We might even say that the models embodied in an anticipatory system are what comprise its individuality; what distinguish it uniquely from other systems. As we have seen, a change in these models is a change of identity; this is perhaps why, for human beings, the preservation of models becomes identical with the preservation of self. The identification of one's self with one's models explains, perhaps, why human beings are so often willing to die; i.e. to suffer biological extinction, rather than change their models, and why suicide is so often, and so paradoxically, an ultimate act of self-preservation.

The study of anticipatory systems thus involves in an essential way the subjective notions of good and ill, as they manifest themselves in the models which shape our behavior. For in a profound sense, the study of models is the study of man; and if we can agree about our models, we can agree about everything else.

Chapter 7

Appendix

7.1 Prefatory Remarks

In this final part of the book, we shall briefly sketch some of the more recent developments which have grown out of the material presented in the preceding sections.

In their way, these newer developments are very revolutionary. Their revolutionary character lies in the questions they raise about the class of mathematical systems which can be the images of natural systems. Throughout the present book, we have stressed that the essence of science lies in the modeling relation; the establishment of a correspondence between percepts and the relations which link them, and the ingredients which make up a formal or mathematical system. This correspondence must match what we call the causal properties of the natural system with the inferential properties of the formal one, as we have amply seen. Also, as we have seen, we can and do use the characteristics of a formal model to *impute* corresponding properties back to the natural system being modeled.

One fundamental imputation, which has been preserved essentially intact since the time of Newton, is the idea that nature is to be represented in such a way that there is a partition into *states* and *dynamical laws*, or into *propositions* and *production rules*. However much our system descriptions may differ from each other technically, they all, to this day, share or partake of this dual structure. Even in quantum mechanics, which seems to be the most radical departure from the Newtonian picture, the primary difference between the two is the argument over what constitutes a state or state description; thus, even here the duality between states and dynamical laws persists intact.

Thus, the class of mathematical images of natural systems, or the class of mathematical structures which could be images of natural systems, is tacitly assumed to be some kind of category of general dynamical system. What I assert in the Appendix to follow, and what is the main revolutionary content of that Appendix, is that this class of mathematical images is too small; it is not enough to do physics in, let alone biology.

I try to make it plausible that this category of general dynamical systems, in which all science has hitherto been done, is only able to represent what I call *simple systems* or *mechanisms*. Natural systems which have mathematical images lying outside of this category and which accordingly do not admit a once-and-for-all partition into states plus dynamical laws, are thus not simple systems; they are *complex*. However, as we argue, complex systems can be approximated, locally and temporarily, by appropriately chosen simple ones; indeed, Chaps. 5 and 6, above were essentially concerned with the nature of such approximations.

The reader will also find a number of unsuspected relationships between the notion of complexity and the old Aristotelian categories of causation. In particular, we will argue that the category of Final Cause, excluded from science for so long because of its incompatibility with the science of simple systems or mechanisms, is not excluded from the science of complex systems. A corollary is: an anticipatory system must be complex; a complex system may be anticipatory. I believe that the reader will find a number of other such surprises. But then, this is the essence of complexity.

I hope that the reader will indulge my closing this Preface with a few personal remarks about the material to follow. It was never my hope or ambition or expectation to embark on the kind of epistemological investigations whose results are reported here. What I wanted to do was to get a better insight into the material basis of organic phenomena. I entered into this endeavor with a complete faith in the generality of physical laws, and in the explanatory powers of mathematics. This kind of faith is not abandoned lightly, and in fact I have never abandoned it. What I have had to abandon, indeed have been forced to abandon, is the conceptual framework in which physics and biology have heretofore been done, and limitations this framework imposes on the relation it admits between mathematics and the material world. One does not lightly or gladly abandon the traditions of centuries and millennia; it would be much nicer and more convenient to be able to work within those traditions than to have to move outside them. I have tried to indicate why and how they must be departed from. I hope that the reader will agree that these departures are indeed forced upon us; that there is nothing whimsical, speculative, or fanciful about them; and that to at least some extent they can reflect Newton's own words: *Hypothesis non fingo*.

7.2 Introduction

We shall introduce the rather wide-ranging considerations which follow with a discussion of the concept of *information* and its role in scientific discourse. Ever since Shannon began to talk about "Information Theory" (by which he meant a probabilistic analysis of the deleterious effects of propagating signals through "channels"; cf. Shannon, 1949) this concept has been relentlessly analyzed and re-analyzed. The time and effort expended on these analyses must surely rank as one of the most unprofitable investments in modern scientific history; not only has there

been no profit, but the currency itself has been debased to worthlessness. Yet, in biology, for example, the terminology of information intrudes itself insistently at every level; code, signal, computation, recognition. It may be that these informational terms are simply not scientific at all; that they are an anthropomorphic stopgap; a *façon de parler* which merely reflects the immaturity of biology as a science, to be replaced at the earliest opportunity by the more rigorous terminology of force, energy, and potential which are the province of more mature sciences (i.e. physics) in which “information” is never mentioned. Or, it may be that the informational terminology which seems to force itself upon us bespeaks something fundamental; something which is missing from physics as we now understand it. We shall take this latter viewpoint, and see where it leads us.

In human terms, information is easy to define; it is anything which is or can be the answer to a question. Therefore we shall preface our more formal considerations with a brief discussion of the status of interrogatives, in logic and in science.

The amazing fact is that interrogation is not ever a part of formal logic, including mathematics. The symbol “?” is not a logical symbol, as for instance are “v”, “^”, “∃”, or “∀”; nor is it a mathematical symbol. It belongs entirely to informal discourse, and as far as I know, the purely logical or formal character of interrogation has never been investigated. Thus, if “information” is indeed connected in an intimate fashion with interrogation, it is not surprising that it has not been formally characterized in any real sense. There is simply no existing basis on which to do so.

I do not intend to go deeply here into the problems of extending formal logic (always including mathematics in this domain) so as to include interrogatories. What I want to suggest here is a relation between our informal notions of interrogation and the familiar logical operation “ $\Rightarrow$ ”; the conditional, or the implication operation. Colloquially, this operation can be rendered in the form “If A, then B”. My argument will involve two steps. First, I will argue that *every* interrogative can be put into a kind of conditional form:

If A, then B?

(where B can be an indefinite pronoun like “who”, “what”, etc., as well as a definite proposition). Second, and most important, I will argue that every interrogative can be expressed in a more special conditional form, which can be described as follows. Suppose I know that some proposition of the form

If A, then B

is true. Suppose I now change or vary A; i.e. replace A by a new expression which I will call δA . The result will be an interrogative, which I can express as

If δA , then δB ?

Roughly, I am treating the true proposition “If A, then B” as a reference, and I am asking what happens to this proposition if I replace the reference expression A by the new expression δA . I could of course do the same thing with B in the reference proposition; replace it by a new proposition δB and ask what happens to A. I assert that every interrogative can be expressed this way, in what I shall call a *variational form*.

The importance of these notions for us will lie in their relation to the external world; most particularly in their relation to the concept of *measurement*, and to the notions of causality to which they become connected when a formal or logical system is employed to represent what is happening in the external world; i.e. to describe some physical or biological system or situation.

Before doing this, I want to motivate the two assertions made above regarding the expression of arbitrary interrogatives in a kind of conditional form. I will do this by considering a few typical examples, and leaving the rest to the reader for the moment.

Suppose I consider the question

“Did it rain yesterday?”

First, I will write it in the form

“If (yesterday), then (rain)?”

which is the first kind of conditional form described above. To find the variational form, I presume I know that some proposition like

“If (today), then (sunny)”

is true. The general variational form of this proposition is

“If δ (today), then δ (sunny)?”.

In particular, then, if I put

δ (today) = (yesterday),

δ (sunny) = (rain)

I have indeed expressed my original question in the variational form. A little experimentation with interrogatives of various kinds taken from informal discourse (of great interest are questions of classification, including existence and universality) should serve to make manifest the generality of the relation between interrogation and the implicative forms described above; of course this cannot be *proved* in any logical sense, since as noted above, interrogation sits outside logic.

It is clear that the notions of observation and experiment are closely related to the concept of interrogation. That is why the results of observation and experiment (i.e. data) are so generally regarded as being information. In a formal sense, simple observation can be regarded as a special case of experimentation; intuitively, an observer simply determines what *is*, while an experimenter systematically perturbs what is, and then observes the effects of his perturbation. In the conditional form, then, an observer is asking a question which can generally be expressed as:

“If (initial conditions), then (meter reading)?”

In the variational form, this question may be formulated as follows: assuming the proposition

“If (initial conditions = 0), then (meter readings = 0)”

is true (this establishes the reference, and corresponds to calibrating the meters), our question becomes

“If δ (initial conditions = 0), then δ (meter readings = 0)?”

where simply

$$\delta(\text{initial conditions} = 0) = (\text{initial conditions})$$

and

$$\delta(\text{meter readings} = 0) = (\text{meter readings}).$$

The experimentalist essentially takes the results of observation as his reference, and thus, basically asks the question which in variational form is just

“If δ (initial conditions), then δ (meter readings)?”

The theoretical scientist, on the other hand, deals with a different class of question; namely, with the questions which arise from assuming a δB (which may be B itself) and asking for the corresponding δA . This is a question which an experimentalist cannot approach directly, not even in principle. It is mainly the difference between the two kinds of questions which marks the difference between experiment and theory, as well as the difference between the explanatory and predictive roles of theory itself; clearly, if we give δA and ask for the consequent δB , we are predicting, whereas if we assume a δB and ask for the antecedent δA , we are explaining.

It should be noted that exactly the same duality arises in mathematics and logic themselves; i.e. in purely formal systems. Thus a mathematician can ask (*informally*): If (I make certain assumptions), then (what follows)? Or, he can start with a conjecture, and ask: If (Fermat’s Last Theorem is true), then (what initial conditions must I assume to explicitly construct a proof)? The former is analogous to prediction, the latter to explanation.

When formal systems (i.e. logic and mathematics) are used to construct images of what is going on in the world, then interrogations and implications become associated with ideas of causality. Indeed, the whole concept of natural law depends precisely on the idea that causal processes in natural systems can be made to correspond with implication in some appropriate descriptive inferential system (e.g. Sect. 7.3 above, where this theme is developed at great length).

But the concept of causality is itself a complicated one; this fact has been largely overlooked in modern scientific discourse, to its cost. That causality is complicated was already pointed out by Aristotle. To Aristotle, all science was animated by a specific interrogative: *why*? He said explicitly that the business of science was to concern itself with “the *why* of things”. In our language, these are just the questions of *theoretical* science: if (B), then (what A)? and hence we can say B *because* A . Or, in the variational form, δB *because* δA .

But Aristotle argued that there were four distinct categories of causation; four ways of answering the question *why*. These categories he called *material cause*, *formal cause*, *efficient cause*, and *final cause*. These categories of causation are not

interchangeable. If this is so (and I will argue below that indeed it is) then there are correspondingly *different kinds of information*, associated with different causal categories. These different kinds of information have been confused, mainly because we are in the habit of using the same mathematical language to describe all of them; it is from these inherent confusions that much of the ambiguity and murkiness of the concept of information ultimately arises. Indeed, we can say more than this: the very fact that the same mathematical language does not (in fact, cannot) distinguish between essentially distinct categories of causation means that the mathematical language we have been using is in itself somehow fundamentally deficient, and that it must be extended by means of supplementary structures to eliminate those deficiencies.

7.3 The Paradigm of Mechanics

The appearance of Newton's *Principia* towards the end of the 17th century was surely an epochal event. Though nominally the theory of physical systems of mass points, it was much more than this. In practical terms, by showing how the mysteries of the heavens could be understood on the basis of a few simple universal laws, it set the standards for explanation and prediction which have been accepted ever since. It unleashed a feeling of optimism almost unimaginable today; it was the culmination of the entire Renaissance. More than that; in addition to providing a universal explanation for specific physical events, it also provided a language and a way of thinking about systems which has persisted essentially unchanged to the present time; what has changed has only been the technical manifestation of the language and its interpretation. In this language, the word "information" never appears in any formal technical sense; we have only words like "energy", "force", "potential", "work", and the like.

It is important to recognize the twin roles played by Newtonian mechanics in science; as a reductionistic ultimate, and as a paradigm for representation of systems not yet reduced to systems of interacting particles. The essential feature of this paradigm is the employment of a mathematical language with a built-in duality, which we may express as the distinction between *internal states* and *dynamical laws*. In Newtonian mechanics, the internal states are represented by points in some appropriate manifold of phases, and the dynamical laws represent the internal or impressed forces. The resulting mathematical image is thus nowadays what is called a *dynamical system*. However, the dynamical systems arising in mechanics are mathematically rather special ones, because of the way phases are defined (they possess a symplectic structure). Through the work of people like Poincaré, Birkhoff, Lotka, and many others over the years, however, this dynamical system paradigm, or its numerous variants, has come to be regarded as the universal vehicle for representation of systems which could not be technically described mechanically; systems of interacting chemicals, organisms, ecosystems, and many others. Even the most radical changes occurring within physics itself, like relativity and quantum

theory, manifest this framework; in quantum theory, for instance, there was the most radical modification of what constitutes a *state*, and how it is connected to what we can observe and measure; but otherwise, the basic partition between states and dynamical laws is relentlessly maintained. Roughly, this partition embodies a distinction between what is inside or intrinsic (the states) and what is outside (the dynamical laws, which are formal generalizations of the mechanical concept of impressed force).

This, then, is our inherited *mechanical paradigm*, which in its many technical variants or interpretations has been regarded as a universal language for describing systems and what they do. The variants take many forms; automata theory, control theory, and the like, but they all conform to the same basic framework first exhibited in the *Principia*.

Among other things, this framework is regarded as epitomizing the concept of causality. We will look at this closely, because it will become important to us momentarily, when we turn to the concept of information in this framework.

Mathematically, a dynamical system can be regarded simply as a vector field on a manifold of states; to each state, there is an assigned velocity vector (in mechanics it is in fact an acceleration vector). A given state (representing what the system is intrinsically like at an instant) together with its associated tangent vector (which represents what the effect of the external world on the system is like at an instant) uniquely determines how the system will change state, or move in time. This translation of environmental effects into a unique tangent vector is already a causal statement, in some sense; it translates into a more perspicuous form through a process of *integration*, which amounts to “solving” the equations of motion. More precisely, if a dynamical system is expressed in the familiar form

$$dx_i/dt = f_i(x_1, \dots, x_n) \quad i = 1, \dots, n \quad (7.1)$$

in which time does not generally appear as an explicit variable (but only implicitly through its differential or derivation dt), the process of integration manifests the explicit dependence of the state variables $x_i = x_i(t)$ on time:

$$x_i(t) = \int_{t_0}^t f_i(x_1(\tau), \dots, x_n(\tau))d\tau + x_i(t_0). \quad (7.2)$$

This is a more traditional kind of causal statement, in which the “state at time t ” is treated as an *effect*, and the right-hand sides of (7.2) above are the *causes* on which this effect depends.

Before going further, let us take a look at the integrands in (7.2), which are the velocities or rates of change of the state variables. The mathematical character of the entire system is determined entirely by the *form* of these functions. Hence, we can ask: what is it that expresses this form (i.e. which determines whether our functions are polynomials, or exponentials or of some other form? And given the general form (polynomial, say), what is it that picks out a specific function and distinguishes it from all others of that form?

The answer, in a nutshell, is: *parameters*. As I have written the system (7.1) above, no such parameters are explicitly visible, but they are at least tacit in the very writing of the symbol “ f_i ”.

Mathematically, what these parameters do is to serve as coordinates for function spaces; just as any other kinds of coordinates do, they label or identify the individual members of such spaces. They thus play an entirely different role from the state variables which constitute the arguments or domains of the functions which they identify.

Here we see the first blurring. For the parameters which specify the form of the functions f_i can *mathematically* be thrown in as arguments of the functions f_i themselves; thus we could (and in fact always do) write

$$f_i = f_i(x_1, \dots, x_n, a_1, \dots, a_r) \quad (7.3)$$

where the a_i are *parameters*. We could even extend the dynamical equations (7.1) by writing

$$da_i/dt = 0$$

(if the a_i are indeed independent of time); thus mathematically we can entirely eradicate any distinction between the parameters and the state variables.

There is still one further distinction to be made. We pointed out above that these parameters a_i represent the effects of the “outside world” on the intrinsic system states. These effects involve *both* the system *and* the outside world. Thus some of the parameters must be interpreted as intrinsic too (the so-called *constitutive* parameters), while the others describe the “state of the outside world”. These latter obey their own laws, not incorporated in (7.1), so they are, from that standpoint, simply regarded as *functions of time* and must be posited independently. They constitute what are variously called *inputs* or *controls* or *forcings*. Indeed, if we regard the states ($x_i(t)$), or any mathematical function of them, as corresponding *outputs* (that is, output as a function of input rather than just of time) we pass directly to the world of control theory.

So let us see where we are. If we divide the world into state variables plus dynamical laws, this amounts to dividing the world into state variables plus parameters, where the role of the parameters is to determine the *form* of the functions which in fact define the dynamical laws. The state variables are the arguments of these functions, while the parameters are coordinates in function spaces. Further, we must additionally partition the parameters themselves into two classes; those which are *intrinsic* (the constitutive parameters) and those which are extrinsic, i.e. which reflect the nature of the environment. The intrinsic parameters are intuitively closely connected with what we might call *system identity*; i.e. with the specific nature or character of the system itself. The values they assume might, for example, tell us whether we are dealing with oxygen, carbon dioxide, or any other chemical *species*. The values assumed by these parameters, therefore, cannot change without our perceiving that a *change of species* has occurred. The environmental parameters, as well as the state variables, however, can change without affecting the species of the system with which we are dealing.

These distinctions cannot be accommodated with the simple language of vector fields on manifolds; that language is too abstract. We can only recapture these distinctions by (a) superimposing an informal layer of *interpretation* on the formal language, as we have essentially done above, or (b) by changing the language itself, to make it less abstract. Let us see how this can be done.

Just to have names for the various concepts involved, I will call the constitutive parameters, which specify the *forms* of the dynamical laws, and hence the species of system with which we are dealing, the system *genome*; I will call the remaining parameters, which reflect the nature of the external world, the system *environment*. The state variables themselves I will call *phenotypes*. This rather provocative terminology is chosen in deliberate reflection of corresponding biological situations; in particular, I have argued (cf. Sect. 3.3 above) that, viewed in this light, the genotype-phenotype dualism which is regarded as so characteristically biological has actually a far more universal currency.

The mathematical structure appropriate to reflect the distinctions we have made is that of genome-parameterized mappings from a space of environments to a space of phenotypes; i.e. mappings of the form

$$f_g : E \rightarrow P$$

specified in such a way that given any initial phenotype, environment plus genome determines a corresponding trajectory. Thus we have no longer a simple manifold of states, but rather a fiber-space structure in which the basic distinctions between genome, environment, and phenotype are embodied from the beginning. Some of the consequences of this picture are examined in Rosen (1978, 1983); we cannot pause to explore them here.

Now we are in a position to discuss the actual relation between the Newtonian paradigm and the categories of causation which were described earlier. In brief, if we regard “phenotype of the system at time t ” as *effect*, then

- (a) *Initial phenotype is material cause;*
- (b) *Genome g is formal cause;*
- (c) $f_g(a)$, *as an operator on initial phenotype, is efficient cause.*

Thus, the distinctions we have made between genome, environment, and phenotype turn out to be directly related to the old Aristotelian categories of causation. As we shall soon see, that is why these distinctions are so important.

We may note in passing that one of the Aristotelian categories is missing from the above; there is no *final cause* visible in the above picture. Ultimately, this is the reason why final cause has been banished from science; the Newtonian paradigm simply has no room for them. Indeed, it is evident that any attempt to superimpose a category of final causation upon the Newtonian picture would effectively destroy the other categories in this picture.

In a deep sense, the Newtonian paradigm has led us to the notion that we may effectively *segregate the categories of causation* in our system descriptions. Indeed, the very concept of system state segregates the notion of material cause from the

other categories of causation, and tells us that it is all right to deal with all aspects of material causation independent of the other categories. Likewise with the concept of genome and environment. I would in fact claim that *this very segregation into independent categories of causation is the heart of the Newtonian paradigm*. When we put it this way, however, the universality of the paradigm perhaps no longer appears so self-evident. Indeed, that it is not universal at all will be one of our main conclusions as we now turn back to the concept of information with which we began.

7.4 Information

We said above that information is, or can be, the answer to a question, and that a question can generally be put into what we called the variational form: If δA , then δB ? This is going to serve as the connecting bridge between “information” and the Newtonian paradigm we have described. In fact, it has played an essential role in the historical development of Newtonian mechanics and its variants, under the rubric of *virtual displacements*.

In mechanics, a virtual displacement is a small, imaginary change imposed on the *configuration* of a mechanical system, while the expressed forces are kept fixed. The animating question is: “If such a virtual displacement is made under given circumstances, then what happens?” The answer, in mechanics, is the well-known *Principle of Virtual Work*: If a mechanical system is in equilibrium, then the virtual work done by the impressed forces as a result of the virtual displacement must vanish. This is a static (equilibrium) principle, but it can readily be extended from statics to dynamics where it is known as *D’Alembert’s Principle*. In the dynamic case, it leads then directly to the differential equations of motion of a mechanical system when the impressed forces are known. Details can be found in any text on classical mechanics.

In what follows, we are going to explore the effect of such virtual displacements on the apparently more general class of dynamical systems of the form

$$dx_i/dt = f_i(x_1, \dots, x_n), \quad i = 1, \dots, n \quad (7.4)$$

(In fact, however, there is a close relationship between the general dynamical systems (7.4) and those of Newtonian mechanics; indeed, the former systems can be regarded as arising out of the latter by the imposition of a sufficient number of non-holonomic constraints.¹)

As we have already noted, the language of dynamical systems, like that of Newtonian mechanics, does not include the word “information”. Rather, the study of such systems revolves around the various concepts of *stability*. However, in one of his analyses of oscillations in chemical systems, J. Higgins (1967) drew attention to the quantities

$$u_{ij}(x_1, \dots, x_n) = \partial/\partial x_j(dx_i/dt).$$

These quantities, which he called “cross-couplings” if $i \neq j$ and “self-couplings” if $i = j$, arise in an essential way in the conditions which govern the existence of oscillatory solutions of (7.4). In fact, it turns out that it is not so much the magnitudes, as the signs, of these quantities which are important. In order to have a handy way of talking about the signs of these quantities, he proposed that we call the j th state variable x_j an *activator* of the i th, at a state $(x_1^0, \dots, x_n^0)$, whenever the quantity

$$u_{ij}(x_1^0, \dots, x_n^0) = \frac{\partial}{\partial x_j} \left(\frac{dx_i}{dt} \right) (x_1^0, \dots, x_n^0) > 0$$

and an *inhibitor* whenever

$$u_{ij}(x_1^0, \dots, x_n^0) < 0.$$

Now activation and inhibition are *informational* terms. Thus, Higgins’ terminology provides an initial hint about how dynamical language might be related to informational language, through the Rosetta Stone of stability.

Now let us see what Higgins’ terminology amounts to. If x_j activates x_i at a state, it means that a (virtual) increase in x_j increases the *rate of change* of x_i , or alternatively, that a (virtual) decrease of x_j decreases the rate of change of x_i . It is eminently reasonable that this is what an activator should do intuitively. Conversely, if x_j inhibits x_i at a state, it means that an increase in x_j decreases the rate of change of x_i , etc.

Thus the n^2 functions $u_{ij}(x_1, \dots, x_n)$, $i, j = 1, \dots, n$ constitute a kind of informational description of the dynamical system (7.4), which I have elsewhere (Rosen, 1979) called an *activation-inhibition pattern*. As we have noted, such a pattern concisely represents the answers to the variational questions: “If we make a virtual change in x_j , what happens to the rate of production of x_i ?”

There is no reason to stop with the quantities u_{ij} . We can, for instance, go one step further, and consider the quantities

$$u_{ijk}(x_1, \dots, x_n) = \partial / \partial x_k (\partial / \partial x_j (dx_i / dt)).$$

Intuitively, these quantities measure the effect of a (virtual) change in x_k on the *extent* to which x_j activates or inhibits x_i . If such a quantity is positive at a state, it is reasonable to call x_k an *agonist* of x_j with respect to x_i ; if it is negative, an *antagonist*. That is, if u_{ijk} is positive, a (virtual) increase in x_k will increase or facilitate the activation of x_i by x_j , etc. The quantities u_{ijk} thus define another layer of informational interaction, which we may call an *agonist-antagonist pattern*.

We can iterate this process, in fact to infinity, producing at each state r a family of n^r functions $u_{ij \dots r}(x_1, \dots, x_n)$. Each layer in this increasing sequence describes how a (virtual) change of a variable at that level modulates the properties of the preceding level.

In the above considerations, we have considered only the effects of virtual changes in state variables x_j on the velocities dx_i/dt at various informational levels. We could similarly consider the effects of virtual displacements at these various levels on the second derivatives d^2x_i/dt^2 (i.e. on the *accelerations* of the x_i), on the third derivatives d^3x_i/dt^3 , and so on. Thus, we have a doubly infinite web of informational interactions, defined by functions

$$u_{ijk\dots r}^m(x_1, \dots, x_n) = \frac{\partial}{\partial x_r} \left(\dots \frac{\partial}{\partial x_j} \left(\frac{dx_i^m}{dt^m} \right) \dots \right)$$

If we start from the dynamical equations (7.4), then nothing new is learned from these circumlocutions, beyond perhaps a deeper insight into the relations between dynamical and informational ideas. Indeed, given any layer of informational structure, we can pass to succeeding layers by mere differentiation, and to antecedent layers by mere integration. Thus in particular, knowledge of any layer in this infinite array of layers determines all of them, and in particular the dynamical equations themselves. For instance, if we know the activation-inhibition pattern $u_{ij}(x_1, \dots, x_n)$, we can reconstruct the dynamical equations (7.4) through the relations

$$df_i = \sum_{j=1}^n u_{ij} dx_j \quad (7.5)$$

(note in particular that the differential form on the right-hand side is like a generalized *work*), and then putting the function $f_i(x_1, \dots, x_n)$ so determined equal to the rate of change dx_i/dt of the i th state variable.

However, *our ability to do all this depends in an absolutely essential way on the exactness of the differential forms which arise at every level of our web of informational interaction*, and which relate each level to its neighbors. For instance, if the forms in (7.5) are not exact, there are no functions $f_i(x_1, \dots, x_n)$ whose differentials are given by (7.5), and hence *no rate equations of the form (7.4)*. In fact, in such a situation, the simple relations between the levels in our web (namely, that each level is the derivative of the preceding level and the integral of the succeeding one) breaks down completely; and levels become independent of each other, and must be posited separately. Thus, for instance, two systems could have the same activation-inhibition patterns, but vastly different agonist-antagonist patterns, and hence manifest entirely different behaviors.

Just to fix ideas, let us see what is implied by the requirement that the differential forms

$$\sum_{j=1}^n u_{ij} dx_j$$

defined by the activation-inhibition pattern be exact. The familiar necessary conditions for exactness here take the form

$$\frac{\partial}{\partial x_k} u_{ij} = \frac{\partial}{\partial x_j} u_{ik}$$

for all $i, j, k = 1, \dots, n$. Intuitively, these conditions mean precisely that *the relations of agonism and activation are entirely symmetrical* (commutative); that x_k as an agonist of the activator x_j is exactly the same as x_j as an agonist of the activator x_k . And likewise for all other levels.

Clearly, such situations are extremely degenerate in informational terms. They are so because the requirement of exactness is highly nongeneric for differential forms. Thus, these very simple considerations suggest a most radical conclusion: that *the Newtonian paradigm, with its emphasis on dynamical laws, restricts us from the outset to an extremely special class of systems, and that the most elementary informational considerations force us out of that class*. We shall explore some of the implications of this situation in the subsequent section.

Meanwhile, let us consider some of the ramifications of these informational ideas, which hold even within the confines of the Newtonian paradigm. These will concern the distinctions we made in the preceding section between environment, phenotype, and genome, the relations of these distinctions to different categories of causation, and the correspondingly different categories of information which these causal categories determine.

First, let us recall what we have already asserted above, namely, that according to the Newtonian paradigm, every relation between physical magnitudes (i.e. every equation of state) can be represented as a genome-parameterized family of mappings

$$f_g : E \rightarrow P$$

from environments to phenotypes. It is worth noting specifically that, in particular, every dynamical law or equations of motion are of this form, as can be seen by writing

$$d\vec{x}/dt = f_g(\vec{x}, \vec{a}). \quad (7.6)$$

Here in traditional language, $\vec{x}$ is a vector of states, $\vec{a}$ is a vector of “external controls” (which together with states constitutes *environment*) and the phenotype here is precisely the tangent vector $d\vec{x}/dt$ attached to the state $\vec{x}$.² In this case, then, tangent vector or phenotype, constitutes *effect*; genome g is identified with formal cause, state x with material cause, and the operator $f_g(\dots, \vec{a})$ with efficient cause.

By analogy with the activation-inhibition networks and their associated informational structures which were described above, we are going to consider formal quantities of the form

$$\frac{\partial}{\partial(\text{cause})} \left(\frac{d}{dt} (\text{effect}) \right) \quad (7.7)$$

As always, such a formal quantity represents an answer to a question: If (cause is varied), then (what happens to effect)? This is exactly the same question we asked in

connection with the definition of activation-inhibition networks and their correlates, but now set into the wider context to which our analysis of the Newtonian paradigm has led us. That is, we may now virtually displace *any* magnitude which affects our relation (7.6), whether it be a genomic magnitude, an environmental magnitude, or a state variable. In a precise sense, the effect of such a virtual displacement is measured precisely by the quantity (7.7).

In particular, then, it follows that there are indeed different *kinds* of information. What kind of information we are dealing with depends on whether we apply our virtual displacement to a genomic magnitude (associated with formal cause), an environmental magnitude (efficient cause), or state variable (material cause). Formally, then, we can initially distinguish at least the following three cases:

1. Genomic information:

$$\frac{\partial}{\partial(\text{genome})} \left(\frac{d}{dt} (\text{effect}) \right); \quad (7.8)$$

2. Phenotypic information:

$$\frac{\partial}{\partial(\text{state})} \left(\frac{d}{dt} (\text{effect}) \right);$$

3. Environmental information:

$$\frac{\partial}{\partial(\text{control})} \left(\frac{d}{dt} (\text{effect}) \right).$$

We shall confine ourselves to these three for present purposes, which generalize only the activation-inhibition patterns described above.

We now come to an important fact; namely, *the three categories defined above are not equivalent*. Before justifying this assertion, we must spend a moment discussing what is meant by “equivalent”. In general, the mathematical assessment of the effects of perturbations (i.e. of real or virtual displacements) is the province of *stability*. The effect on subsequent dynamical behavior of modifying or perturbing a system state is the province of Lyapunov stability of dynamical systems; the effect of perturbing a control is part of control theory; the effect of perturbing a genome is the province of structural stability. To fix ideas, let us consider genomic perturbations, or *mutations*. A virtual displacement applied to a genome g replaces the initial mapping f_g determined by g with a new mapping $f_{g'}$. Mathematically, we say that the two mappings f_g , $f_{g'}$ are equivalent, or similar, or conjugate, if there exist appropriate transformations

$$\alpha : E \rightarrow E,$$

$$\beta : P \rightarrow P$$

such that the diagram

$$\begin{array}{ccc} E & \xrightarrow{f_g} & P \\ \alpha \downarrow & & \downarrow \beta \\ E & \xrightarrow{f_{g'}} & P \end{array}$$

commutes; i.e. if

$$\beta(f_g(e)) = f_{g'}(\alpha(e))$$

For every e in E . Intuitively, this means that a mutation $g \rightarrow g'$ can be offset, or annihilated, by imposing suitable *coordinate transformations* on the environments and phenotypes. Stated yet another way, a virtual displacement of genome can always be counteracted by corresponding displacements of environment and phenotype so that the resultant variation on effect vanishes.

We have elsewhere (see e.g. Sect. 3.3 above) shown at great length that this commutativity may not always obtain; i.e. that there may exist genomes which are bifurcation points. In any neighborhood of a bifurcating genome g , there exist genomes g' for which f_g and $f_{g'}$ fail to be conjugate.

With this background, we can return to the question of whether the three kinds of information (genomic, phenotypic, and environmental) which we have defined above are equivalent. Intuitively, equivalence would mean that the effect of a virtual displacement δg of genome, say with everything else held fixed, could equally well be produced by a virtual displacement δa of environment, or by a virtual displacement δp of phenotype. Or stated another way, the effect of a virtual displacement δg of genome can be annihilated by virtual displacements $-\delta a$, $-\delta p$ of environment and phenotype respectively. This is simply a restatement of the definition of conjugacy or similarity of mappings.

If all forms of information are equivalent, it would follow that there could be no bifurcating genomes. We note in passing that the assumption of equivalence of the three kinds of information we have defined above thus creates terrible ambiguities when it comes to *explanation* of particular effects. We will not consider that aspect here, except to note that it is perhaps very fortunate for us that, as we have seen, they are not equivalent.

Let us look at one immediate consequence of the non-equivalence of genomic, environmental, and phenotypic information, and of the considerations which culminate in that conclusion. Long ago (cf. von Neumann, 1951; Burks, 1966) von Neumann proposed an influential model for a “self-reproducing automaton”, and subsequently, for automata which “grow” and “develop”. This model was based on a famous theorem of Turing (1936) establishing the existence of a universal computer (universal Turing machine). From the existence of such a universal computer, von Neumann argued that there must also exist a universal constructor. Basically, he argued that computation (i.e. following a program) and construction (following a blueprint) are both algorithmic processes, and that anything holding for one

class of algorithmic processes necessarily holds for any other class. This universal constructor formed the central ingredient of the “self-reproducing automaton”.

Now a computer acts, in the language we have developed above, through the manipulation of efficient cause. A constructor, if the term is to have any shred of its intuitive meaning, must essentially manipulate material cause. The inequivalence of the two categories of causality, in particular manifested by the non-equivalence of environmental and phenotypic information, means that we cannot blithely extrapolate from results pertaining to efficient causation into the realm of material causation. Indeed, in addition to invalidating von Neumann’s specific argument, we learn that great care must be exercised in general when arguing from purely logical models (i.e. from models pertaining to efficient cause) to any kind of physical realization, such as developmental or evolutionary biology (which, as noted, pertain to material cause).

Thus, we see how significant are the impacts of informational ideas, even within the confines of the Newtonian paradigm. In this paradigm, as we have shown, the categories of causation are essentially segregated into separate packages. We will now turn to the question of what happens when we leave the comforting confines of that paradigm.

7.5 An Introduction to Complex Systems

I am going to call any natural system for which the Newtonian paradigm is completely and eternally valid a *simple system*, or *mechanism*. Accordingly, a *complex system* is one which, for one reason or another, falls outside this paradigm. We have already seen a hint of such systems in the preceding section; e.g. systems whose activation-inhibition patterns u_{ij} do not give rise to exact differentials $\sum u_{ij} dx_j$. However, some further words of motivation must precede an immediate conclusion that such systems are truly complex (i.e. fall fundamentally outside the Newtonian paradigm). We must also justify our very usage of the term “complex” in this context.

What I have been calling the Newtonian paradigm ultimately devolves upon *the class of distinct mathematical descriptions* which a system can have, and the relations which exist between these descriptions. As we have noted extensively above, the touchstone of system description arising in this paradigm is the fundamental dualism between states and dynamical laws. Thus, in this paradigm, the mathematical objects which can describe natural systems comprise a category, whose objects may be called general dynamical systems. In a formal sense, it looks as if any mathematical object falls into this category, because the Newtonian partition between states and dynamical laws exactly parallels the partition between propositions and production rules (rules of inference) which presently characterize all logical systems and logical theories. However, as we shall soon see, although this category of general dynamical systems is indeed large, it is not everything (and indeed, as we shall argue, it is far from large enough).

As noted, the Newtonian paradigm asserts much more than simply that every image of a natural system must belong to a given category. It asserts certain relationships between such images. In particular, (and this is the reductionistic content of the paradigm) it asserts that among these images there is a biggest one, which effectively maps on all the others. Intuitively, this is the master description or ultimate description, in which every shred of physical reality has an exact mathematical counterpart; in category-theoretic terms, it is much like a free object (a generalization of the concept of free semigroup, free group, etc.)³

There is still more. The ingredients of this ultimate description, by their very nature, are themselves devoid of internal structure; the only things about them which can change are their relative positions and velocities. Given the forces acting between them, as Laplace noted long ago, everything that happens in the external world is in principle predictable and understandable. From this perspective, everything is determined; there are no mysteries, no surprises, no errors, no questions, and no information. This is as much true for quantum theory as for classical; only the nature of state description has changed. And it applies to everything, from atoms to organisms to galaxies.

Let us look now at how this universal picture manifests itself in biology. First from the standpoint of the physicist, biology is concerned with a rather small class of extremely special (indeed, inordinately special) systems. The theoretical physicist, in his quest for general and universal laws, has thus never had much to do with organisms. As far as he is concerned, what makes organisms special is not that they transcend his paradigms, but rather that their specification within the paradigm requires a plethora of special constraints and conditions, which must be superimposed on the universal canons of system description and reduction. The determination of these special conditions is an empirical task; essentially someone else's business. But it is not doubted that the relation between physics and biology is the relation between general and particular.

The modern biologist, in general, avidly embraces this perspective.⁴ Historically, in fact, biology has only recently caught up with the Newtonian revolution which swept the rest of Natural Philosophy in the 17th century. The three-century lag arose because biology has no analog of the solar system; no way to make immediate and meaningful contact with the Newtonian paradigm. Not until physics and chemistry had elaborated the technical means to probe microscopic properties of matter (including organic matter) was the idea of a "molecular biology" even thinkable. And this did not happen until the 1930's.

At present, the fact is that there is still no single inferential chain which leads from anything important in physics to anything important in biology. This is in fact; a datum; a piece of information. How are we to understand it? There are various possibilities. Kant, long ago, argued that organisms could only be properly understood in terms of Final Causes or intentionality; hence from the outset he suggested that organisms fall completely outside the canons of Newtonian science which work for everything else. Indeed, the essential telic nature of organisms precluded even the possibility that a "Newton of the grassblade" would come along, and do for biology what Newton had done for physics. Another possibility is the

one we have already mentioned; we have simply not yet characterized all those special conditions which are necessary to bring biology fully within the scope of universal physical principles. Still a third possibility has grown up within biology itself, as a consequence of evolutionary ideas; it is that much of biology is the result of *accidents* which are *in principle* unpredictable and hence governed by no laws at all.⁵ In this view, biology is as much a branch of history as of science. At present, this last view sits in a kind of double-think relation with reductionism; the two are quite inconsistent, but do allow modern biologists to enjoy the benefits of vitalism and mechanism together.

Still a fourth view was expressed by Einstein, who said in a letter to Leo Szilard: "One can best appreciate, from a study of living things, how primitive physics still is".

As we have noted, the present prevailing view in biology is that the Newtonian canons are indeed universal, and we are lacking only knowledge of the special conditions and constraints which distinguish organisms from other natural systems within those canons. One way of saying this with a single word is to assert that organisms are *complex*. This word is not well defined, but it does connote several things. One of them is that complexity is a system property, no different from any other property. Another is that the *degree* to which a system is complex can be specified by a number, or set of numbers. These numbers may be interpreted variously as the dimensionality of a state space, or the length of an algorithm, or as a cost in time or energy incurred in solving system equations.

On a more empirical level, however, complexity is recognized differently. If a system surprises us, or does something we have not predicted, or responds in a way we have not anticipated; if it makes errors; if it exhibits emergence of unexpected novelties of behavior, we also say that the system is complex. In short, complex systems are those which behave "counter-intuitively".

Sometimes, of course, surprising behavior is simply the result of incomplete characterization; we can then hunt for what is missing, and incorporate it into our system description. In this way, the planet Neptune was located from unexplained deviations of Uranus from its expected trajectory. But sometimes, this is not the case; in the apparently analogous case of the anomalies of the trajectory of the planet Mercury, for instance, no amount of fiddling within the classical picture would do, and only a massive readjustment of the paradigm itself (General Relativity) could avail.

From these few words of introduction, we see that the identification of "complexity" with situations where the Newtonian paradigm fails is in accord with the intuitive connotation of the term, and is the alternative to regarding as complex any situation which merely is technically difficult within the paradigm.

Now let us see where "information" fits into these considerations. We recall once more that "information" is the actual or potential response to an interrogative, and that every interrogative can be put into the variational form: "if δA , then δB ?". The Newtonian paradigm asserts, among other things, that the answers to such interrogatives follow from dynamical laws superimposed on manifolds of states. In their turn, these dynamical laws are special cases of what we have elsewhere called

equations of state, which link or relate the values of system observables. Indeed, the concept of an observable was the point of departure for our entire treatment of system description and representation (cf. Rosen, 1978); it was the connecting link between the world of natural phenomena and the entirely different world of formal systems which we use to describe and explain.

However, the kinds of considerations we have developed above suggests that this world is not enough. We require also a world of variations, increments, and differentials of observables. It is true that every linkage between observables implies a corresponding linkage between differentials, but as we have seen, the converse is not true. We are thus led to the notion that a differential relation is a generalized linkage, and that a differential form is a kind of generalized observable. A differential form which is not the differential of an observable thus is an entity which assumes no definite numerical value (as an observable does), but which can be incremented.

If we do think of differential forms as generalized observables, then we must correspondingly generalize the notion of equation of state. A generalized equation of state thus becomes a linkage or relation between ordinary observables and differentials or generalized observables. Such generalized equations of state are the vehicles which answer questions of our variational form: If δA , then δB ?

But as we have repeatedly noted, such generalized equations of state do not generally follow from systems of dynamical equations, as they do in the Newtonian paradigm. Thus we must find some alternative way of characterizing a system of this kind. Here is where the informational language which we have introduced above comes to the fore. Let us recall, for instance, how we defined the activation-inhibition network. We found a family of functions u_{ij} (i.e., of observables) which could be thought of in the dynamical context as modulating the effect of an increment dx_j on that of another increment, df_i . That is, the values of each observable u_{ij} measure precisely the extent of activation or inhibition which x_j exerts on the rate at which x_i is changing.

In this language, a system falling outside the Newtonian paradigm (i.e. a complex system) can have an activation-inhibition pattern, just as a dynamical (i.e. simple) system does. Such patterns are still families of functions (observables) u_{ij} , and the pattern itself is manifested by the differential forms

$$\omega_i = u_{ij}dx_j$$

But in this case, there is no global velocity observable f_i which can be interpreted as the rate of change of x_i ; there is only a velocity *increment*. It should be noted explicitly that the u_{ij} which define the activation-inhibition pattern need not be functions of the x_i alone, or even functions of them at all. Thus, the differential forms which arise in this context are different from those with which mathematicians generally deal, and which can always be regarded as cross-sections of the cotangent bundle of a definite manifold of states.

The next level of information is the agonist-antagonist pattern u_{ijk} . In the category of dynamical systems, this is completely determined by the activation-inhibition pattern, and can be obtained from the latter by differentiation:

$$u_{ijk} = \frac{\partial}{\partial x_k} u_{ij}.$$

In our world of generalized observables and linkages, the u_{ijk} are independent of the u_{ij} , and must be posited separately; in other words, complex (non-Newtonian) systems can have identical activation-inhibition patterns but quite different agonist-antagonist patterns.

Exactly the same considerations can now be applied to every subsequent layer of the informational hierarchy; each of them is now independent of the others, and must hence be posited separately. Hence a complex system requires an *infinite* mathematical object for its description.

We cannot go into the mathematical details of the considerations we have sketched so briefly above. Suffice it to say that a complex system, defined by a hierarchy of informational levels of the type we have described, is quite a different kind of object than is a dynamical system. For one thing, it is quite clear that there is no such thing as a set of *states*, assignable to such a system once and for all. From this alone, we might expect that the nature of causality in such systems is vastly different than it is in the Newtonian paradigm; we shall come to this in a moment.

The totality of mathematical structures of the type we have defined above forms a category. In this category the class of general dynamical systems constitutes a very small subcategory. We are suggesting that the former provides a suitable framework for the mathematical imaging of complex systems, while the latter, by definition, can only image simple systems or mechanisms. If these considerations are valid (and I believe they are), then the entire epistemology of our approach to natural systems is radically altered, and it is the basic notions of information which provide the natural ingredients for this.

There is, however, a profound relationship between the category of general dynamical (i.e. Newtonian) systems, and the larger category in which it is embedded. This can only be indicated here, but it is important indeed. Namely, there is a precise sense in which an informational hierarchy can be *approximated*, locally and temporarily, by a general dynamical system. With this notion of approximation there is an associated notion of *limit*, and hence of topology. Using these ideas, it can in fact be shown that what we can call the category of complex systems is the completion, or limiting set, of the category of simple (i.e. dynamical) systems.

The fact that complex systems can be approximated (albeit locally and temporarily) by simple ones is a crucial one. It explains precisely why the Newtonian paradigm has been so successful, and why, to this day, it represents the only effective procedure for dealing with system behavior. But in general, we can also see that it can supply *only* approximations in general, and in the universe of complex systems, it amounts to replacing a *complex* system with a *simple subsystem*. Some of the profound consequences of doing this are considered in detail in Chap. 5 above.

This relationship between complex systems and simple ones is, by its very nature, without a reductionistic counterpart. Indeed, what we presently understand as “physics” is seen in this light as *the science of simple systems*. The relation between physics and biology is thus not at all the relation of general to particular; in fact, quite the contrary. It is not biology, but physics, which is too special. We

can see from this perspective that biology and physics (i.e. contemporary physics) grow as two divergent branches from a *theory of complex systems* which as yet can be glimpsed only very imperfectly.

The category of simple systems is, however, still the only thing we know how to work with. But to study complex systems by means of approximating simple systems puts us in the position of early cartographers, who were attempting to map a sphere while armed only with pieces of planes. Locally, and temporarily, they could do very well, but globally, the effects of the topology of the sphere become progressively important. So it is with complexity; over short times and only a few informational levels, we can always make do with a simple (i.e. dynamical) picture. Otherwise, we cannot; we must continually replace our approximating dynamics by others as the old ones fail. Hence another characteristic feature of complex systems; they appear to possess a multitude of partial dynamical descriptions, which cannot be combined into one single complete description. Indeed, in earlier work we took this as the defining feature of complexity (cf. Notes to Sects. 5.5–5.7).

We shall add one brief word about the status of causality in complex systems, and about the practical problem of determining the functions which specify their informational levels. As we have already noted, complex systems do not possess anything like a state set which is fixed once and for all. And in fact, in complex systems, the categories of causality become intertwined in a way which is not possible within the Newtonian paradigm. Intuitively, this follows from the independence of the infinite array of informational layers which constitutes the mathematical image of a complex system. The variation of any particular magnitude with such a system will typically manifest itself independently in many of these layers, and thus reflect itself partly as material cause, partly as efficient cause, and even partly as formal cause in the resultant variation of other magnitudes. We feel that it is, at least in large part, this involvement of magnitudes simultaneously in each of the causal categories which make biological systems so refractory to the Newtonian paradigm.

Also, this intertwining of the categories of causation in complex systems makes the interpretation of experimental results of the form “If δA , then δB ” extremely difficult to interpret directly. If we are correct in what we have said so far, such an observational result is far too coarse as it stands to have any clear-cut meaning. In order to be meaningful, an experimental proposition of this form must isolate the effect of a variation δA on a single informational level, keeping the others clamped. As might be appreciated from what has been said so far, this will in general not be an easy thing to do. In other words, the experimental study of complex systems cannot be pursued with the same tools and ideas as are appropriate for simple systems.

Our final conceptual remark is also in order. As we pointed out above, the Newtonian paradigm has no room for the category of final causation. This category is closely tied up with the notion of anticipation, and in its turn, with the ability of systems to possess internal predictive models of themselves and their environments, which can be utilized for the control of present actions. We have argued at great length above that anticipatory control is indeed a distinguishing feature of the organic world, and developed some of the unique features of such anticipatory systems. In the present discussion, we have in effect shown that, in order for a system

to be anticipatory, it must be complex. Thus, our entire treatment of anticipatory systems becomes a corollary of complexity. In other words, complex systems can admit the category of final causation in a perfectly rigorous, scientifically acceptable way. Perhaps this alone is sufficient recompense for abandoning the comforting confines of the Newtonian paradigm, which has served us so well over the centuries. It will continue to serve us well, provided that we recognize its restrictions and limitations as well as its strengths.

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Appendix: Notes

Note 1, Sect. 7.4

Newton's original *particle mechanics*, or *vectorial mechanics*, is hard to apply to many practical problems, and was very early, through the work of people like Euler and Lagrange, transmuted into another form, generally called *analytical mechanics*. This latter form is what is usually used to deal with extended matter; e.g. with rigid bodies. From the standpoint of particle mechanics, the rigidity of a macroscopic body is a consequence of interparticle forces, which must be explicitly taken into account in describing the system. Thus, if there are N particles in the system (however large N may be) there is a phase space of $6N$ dimensions, and a set of

dynamical equations which expresses for each particle the resultant of *all* forces seen by that particle. In analytical mechanics, on the other hand, any rigid body can be completely described by giving only five configurational coordinates (e.g. the coordinates of the center of mass, and two angles of rotation about the center of mass), however many particles it contains. From the particulate point of view, the internal forces which generate rigidity are replaced by *constraints*; supplementary conditions on the configuration space which must be identically satisfied. Thus, the passage from particle mechanics to analytical mechanics involves a partition of the forces in an extended system into two classes: (a) the internal or *reactive* forces, which hold the system together, and (b) the *impressed* forces, which push the system around. The former are represented in analytical mechanics by algebraic constraints, the latter by differential equations in the configuration variables (five for a rigid body).

A system in analytical mechanics may have additional constraints imposed upon it by specific circumstances; e.g. a ball may roll on a table top. It was recognized long ago that these additional constraints (which, like all constraints, are regarded as expressing the operation of reactive forces) can be of two types, which were called by Hertz *holonomic* and *non-holonomic*. Both kinds of constraints can be expressed locally, in infinitesimal form, as

$$\sum_{i=1}^n u_i(x_1, \dots, x_n) dx_i = 0$$

where $x_1, \dots, x_n$ are the configuration coordinates of the system. For a holonomic constraint, the above differential form is exact; i.e. is the differential of some global function $\varphi(x_1, \dots, x_n)$ defined on the whole configuration space. Thus the holonomic constraint translates into a global relation

$$\varphi(x_1, \dots, x_n) = \text{constant}.$$

This in turn means that the configurational variables are no longer independent, and that one of them can be expressed as a function of the others. The constraint thus reduces the dimension of the configuration space by *one*, and therefore reduces the dimension of the phase space by *two*.

A non-holonomic constraint, on the other hand, does not allow us to eliminate a configurational variable in this fashion. However, since it represents a relation between the configuration variables and their differentials, it does allow us to eliminate a coordinate of *velocity*, while leaving the dimension of the configuration space unaltered. That is, a non-holonomic constraint serves to eliminate one degree of freedom of the system. It thus also eliminates one dimension from the space of impressed forces which can be imposed on the system without violating the constraint.

Similarly, if we impose r independent non-holonomic constraints on our system, we (a) keep the dimension of the configuration space what it was originally; (b) eliminate r coordinates of velocity, and thus reduce the dimensionality of the phase space by r ; (c) likewise, we reduce by r the dimensionality of the set of impressed forces which can be imposed on the system.

Let us express these facts mathematically. A non-holonomic constraint can be expressed locally in the general form

$$\left(x_1, \dots, x_n, \frac{dx_1}{dt}, \dots, \frac{dx_n}{dt} \right) = 0$$

which can (locally) be solved for one of the velocity coordinates (dx_1/dt say). Thus, it can be written in the form

$$\begin{aligned} \frac{dx_1}{dt} &= \left(\psi \ x_1, x_2, \dots, x_n, \frac{dx_2}{dt}, \dots, \frac{dx_n}{dt} \right) \\ &= \psi(x_1, \vec{\alpha}) \end{aligned}$$

where we have written $\vec{\alpha} = (x_2, \dots, x_n, dx_2/dt, \dots, dx_n/dt)$. (At this point the reader is invited to compare this relation with (7.6) above.)

Likewise, if there are r non-holonomic constraints, these can be expressed locally by the r equations

$$dx_i/dt = \psi_i(x_1, \dots, x_r, \vec{\alpha}) \quad i = 1, \dots, r$$

where now $\vec{\alpha}$ is the vector $(x_{r+1}, \dots, x_n, dx_{r+1}/dt, \dots, dx_n/dt)$. These equations of constraint, which intuitively arise from the *reactive* forces holding the system together, now become more and more clearly the kind of equations we always use to describe general dynamical or control systems.

Now what happens if $r = n$? In this case, the constraints leave us only *one degree of freedom*; they determine a vector field on the configuration space. There is in effect only one *impressed* force that can be imposed on such a system, and its only effect is to get the system moving; once moving, the motion is determined entirely by the *reactive* forces, and not by the *impressed* force. Mathematically, the situation is that of an autonomous dynamical system, whose manifold of states is the *configuration* space of the original mechanical system.

This relation between dynamics and mechanics is quite different from the usual one, in which the manifold of states is thought of as generalizing the mechanical notion of *phase*, and the equations of motion generalize *impressed* force. In the above interpretation, however, it is quite otherwise; the manifold of states correspond now to mechanical *configurations*, and the equations of motion come from the *reactive* forces.

Note 2, Sect. 7.4

The reader should be most careful not to confuse two kinds of propositions, which are equivalent mathematically but completely different epistemologically and causally. On the one hand, we have a statement like

$$d\vec{x}/dt = f_g(\vec{x}, \vec{\alpha}).$$

This is a local proposition, linking a tangent vector or velocity $d\vec{x}/dt$ to a state $\vec{x}$, a genome g , and a control $\vec{a}$. Each of these quantities is built up out of observables assuming definite numerical values at any instant of time, and it is *their values at a common instant* which are related by this proposition.

On the other hand, the integrated form of these dynamical relations looks like

$$\vec{x}(t) = \int_{t_0}^t f_g(\vec{x}, \vec{a}(\tau)) d\tau.$$

This is a relation which involves time *explicitly*, and links the values of observables *at one instant* with values (assumed by these and other observables) *at other instants*.

Each of these epistemically different propositions has its own causal structure. In the first of them, we can treat the tangent vector $d\vec{x}/dt$ as effect, and define its causal antecedents as we have done. In the integrated form, on the other hand, we take $\vec{x}(t)$ as effect, and find a correspondingly different causal structure. In general, the mathematical or logical equivalence of two expressions of linkage or relationship in physical systems does not at all connote that their causal structures are identical. This is merely a manifestation of what we said earlier, that the mathematical language we use to represent physical reality has abstracted away the very basis on which such causal discriminations can be made.

Note 3, Sect. 7.5

It should be recognized that this reductionistic part of the Newtonian paradigm can fail for purely mathematical reasons. If it should happen that there is no way to effectively map the master description, then this is enough to defeat a reductionistic approach to those system behaviors with which the partial description deals. This is quite a different matter from the one we are considering here, in which no Newtonian master description *exists*, and the program fails for *epistemological* reasons, rather than mathematical ones.

Note 4, Sect. 7.5

This statement is not simply my subjective assessment. In 1970 there appeared a volume entitled “Biology and the Future of Man”, edited by Philip Handler, then president of the National Academy of Sciences of the U.S.A. The book went to great lengths to assure the reader that it spoke for biology as a science; that in it biologists spoke with essentially one voice. At the outset, it emphasized that the

volume was not prepared as a (mere) academic exercise, but for serious pragmatic purposes:

Some years ago, the Committee on Science and Public Policy of the National Academy of Sciences embarked on a series of 'surveys' of the scientific disciplines. Each survey was to commence with an appraisal of the 'state of the art'. . . . In addition, the survey was to assess the nature and strength of our national apparatus for continuing attack on those major problems, e.g., the numbers and types of laboratories, the number of scientists in the field, the number of students, the funds available and their sources, and the major equipment being utilized. Finally, each survey was to undertake a projection of future needs for the national support of the discipline in question to assure that our national effort in this regard is optimally productive. . . .

To address these serious matters, the Academy proceeded as follows:

. . . . Panels of distinguished scientists were assigned subjects. . . . Each panel was given a general charge. . . . as follows:

The prime task of each Panel is to provide a pithy summary of the status of the specific sub-field of science which has been assigned. This should be a clear statement of the prime scientific problems and the major questions currently confronting investigators in the field. Included should be an indication of the manner in which these problems are being attacked and how these approaches may change within the foreseeable future. What trends can be visualized for tomorrow? What lines of investigation are likely to subside? Which may be expected to advance and assume greater importance? Are the questions themselves. . . . likely to change significantly? Having stated the major questions and problems, how close are we to the answers? The sum of these discussions, panel by panel, should constitute the equivalent of a complete overview of the highlights of current understanding of the Life Sciences.

There were twenty-one such Panels established, spanning the complete gamut of biological sciences and the biotechnologies. The recruitment for these Panels consisted of well over 100 eminent and influential biologists, mostly members of the Academy. How the panelists themselves were chosen is not indicated, but there is no doubt that they constituted an authoritative group.

In due course, the Panels presented their reports. How they were dealt with is described in colorful terms:

. . . . In a gruelling one week session of the Survey Committee. . . . each report was *mercilessly* exposed to the criticism of all the other members. . . . Each report was then rewritten and subjected to the *searching, sometimes scathing*, criticisms of the members of the parent Committee on Science and Public Policy. The reports were again revised in the light of this exercise. Finally, the Chairman of the Survey Committee. . . . devoted the summer of 1968 to the final editing and revising of the final work.

Thus we have good grounds for regarding the contents of this volume as constituting a true authoritative consensus, at least as of 1970. There are no minority reports; no demurrals; biology does indeed seem guaranteed here to speak with one voice.

What does that voice say? Here are a few characteristic excerpts:

The theme of this presentation is that life can be understood in terms of the laws that govern and the phenomena that characterize the inanimate, physical universe and, indeed, that at its essence life can be understood *only* (emphasis added) in the language of chemistry.

A little further along, we find this:

Until the laws of physics and chemistry had been elucidated, *it was not possible even to formulate* (emphasis added) the important, penetrating questions concerning the nature of life... The endeavors of thousands of life scientists... have gone far to document the thesis... (that) living phenomena are indeed intelligible in the physical terms. And although much remains to be learned and understood, and the details of many processes remain elusive, those engaged in such studies hold *no doubt* (emphasis added) that answers will be forthcoming in the reasonably near future. Indeed, *only two major questions* (emphasis added) remain enshrouded in a cloak of *not quite* (emphasis added) fathomable mystery: (1) the origin of life... and (2) the mind-body problem... yet (the extent to which biology is understood) even now constitutes a satisfying and exciting tale.

Still further along, we find things like this:

While *glorying* (emphasis added) in how far we have come, these chapters also reveal how large is the task that lies ahead... If (molecular biology) is exploited with vigor and understanding... a shining, hopeful future lies ahead.

And this:

Molecular biology provides the closest insight man has yet obtained of the nature of life – and therefore, of himself.

And this:

It will be evident that the huge intellectual triumph of the past decade will, in all likelihood, be surpassed tomorrow – and to the everlasting benefit of mankind.

It is clear from such rhapsodies that the consensus reported in this volume is not only or even mainly a scientific one; it is an emotional and aesthetic one. And indeed, anyone familiar with the writings of Newton's contemporaries and successors will recognize them.

The volume to which we have alluded was published in 1970. But it is most significant that nothing fundamental has changed since then.

Note 5, Sect. 7.5

In the inimitable words of Jacques Monod (“Chance and Necessity”, pp. 42–43):

We can assert today that a universal theory, however completely successful in other domains, could never encompass the biosphere, its structure and its evolution as phenomena deducible from first principles... The thesis I shall present... is that the biosphere does not contain a predictable class of objects or events but constitutes a particular occurrence, compatible with first principles but not deducible from these principles, and therefore *essentially unpredictable* (emphasis added).

Chapter 8

Relational Science: Towards a Unified Theory of Nature

8.1 Introduction

In over 300 years of science in the West we have perfected a method of “seeing” nature as a dynamical system of *efficient* and *material* causes (using the terms of natural causality established by Aristotle), constrained by the unchanging parameters (constants and metrics) of a single, universal context that supplies *formal cause* (which we accordingly attempt to represent in a complete mathematical structure or “formalism”). This view is generally referred to as the Newtonian, classical, or mechanistic view of nature, and it has dominated science up to the post-modern era. Aristotle’s *final cause* was left out of this view entirely as belonging to the realm of the mystical, and (before we accepted duality in physics) formal cause was thought to be fully describable as one closed system of logic applying to the entire universe.

In Rosen’s collective works, he showed clearly that this mechanistic view of science is associated with a category of general mathematics that allows, and restricts us to, creating simple approximations to nature’s complex behaviors. With this volume, *Anticipatory Systems*, Rosen explored the implications of formal and final cause, which seem to be critically involved in establishing, and thus defining, complex systems and especially in characterizing living systems. And yet formal and final causes have not been well understood within the scientific community. In fact, it can be said that we have not had a useful theory of these causes in the sciences to date. To a very large extent, Rosen’s methodical developments in this and other works were aimed at developing a theory that would account for the effects of formal and final causes in our scientific images of nature, and thereby provide a more rigorous foundation for complexity science. Exploring these implications, as attempted in summary form here, may provide a more integrating view of Rosen’s relational theory and hopefully serve as a guide to not only its critical value in science, but to future directions for its development.

There is no suggestion in Rosen’s work that the approach of Western science has been generally wrong; only that it has lacked consideration of these important aspects, and that it has consequently misrepresented life. He described a “simple”

system as one that can have a complete model; a criterion that characterizes non-complex, computable, dynamical or mechanistic systems. Simple systems involve efficient and material causes under the presumption that they exist within a single formalism; that is, implicitly, within a completely describable system of formal cause in nature. Rosen described such assumptions as “too impoverished” mathematically to allow modeling complexity or life, and yet “... *still the only thing we know how to work with*”. The aim of much of Rosen’s work was to provide tools to allow us to work with more.

We can infer from the sum of Rosen’s work that the inadequacy of the dynamical view is rooted in its simplification of formal and final causes; its attempt to overlook or reduce them to efficient and material laws that might be made complete in themselves. In contrast, Rosen’s relational theory suggests that it is possible to expand science by re-including the contextual causes that mechanism left out. If we do so, we see nature itself establishing and realizing its own formal paradigms. In other words, systems can establish the context for other systems and therefore formal cause can vary, ensuring that no single descriptive formalism can be complete.

In this volume, Rosen brings us from the limitations of the dynamical system description to the edge of complexity, which he saw as existing mathematically in the infinite limit beyond what any combination of mechanistic system descriptions is capable of representing. The mechanistic approach has been a powerful method for gaining knowledge about certain aspects of our natural world and for supporting the development of technology precisely because it selects phenomena that have a simple explanation. It presumes that events have a single past configuration and a single future one, such that the future configuration of a system can be predicted completely on the basis of past events and the fixed laws, or as it was modified, fixed probabilistic laws, of a universal context. Any finite combination of such system descriptions may give approximations or simulations of events, but the cost of assuming either deterministic or probabilistic simplicity is to be unable to represent the full range of entailment possibilities of a whole, complex system.

We can see that cost in terms of the presumed separation between an “observer” of nature – a scientist, perhaps – and what an observer can observe. In effect, that epistemological separation is what establishes the mechanistic view, because it uses a natural system – ourselves – to describe nature as a set of external events, which then appear to belong to a universe that has been separated from the context in which we perceive it. The more we formalize that view, the more nature appears to be separated from all information contexts, because, ultimately, ours is natural. In other words, what we observe is what can be seen, not what sees. If we do not then reason, from ample evidence, that nature “sees” by modeling itself, we will not include that aspect of nature in our scientific models. By “see” or “perceive” is meant an information relation that is fundamentally different from the kinds of interactions we can observe of material objects; one that is, in contrast, *contextual*. Indeed the mechanistic interactions describe an important aspect of natural process, but there is more to nature (and implicitly science) than can be seen from a single perspective; there is also the context and the relation between the two. It has not been immediately obvious, of course, how to describe the larger view; what

analytical method can be used to describe relations without either simplifying them to mechanisms, or making them seem mysterious. Rosen's work addressed that middle ground that has eluded us for centuries.

Translation between externalized events and internalized representations (notably ours) is the essence of science and scientific philosophy; that is, how we will relate "*the only thing we know how to work with*" (perceptions of an outer world) to inferences that necessarily exist in an internalized information context. Rosen claimed that the translation that is appropriate for mechanisms is not appropriate for complex systems. When analyzing a complex system the assumption that there are no natural information contexts other than our own can no longer be maintained, because that assumption is too highly selective of what we can thus describe or model. Systems that are viewed in the absence of their own internal information relations are seen only through their pre-determined events; for what exists before an event and between events is the essence of information and formal cause.

What seems confusing to us is that some systems retain self-determining relations and some do not. In the latter case, they are what gives nature its persistent forms, and what mechanistic science was focused on. The descriptive simplicity of that view has been extremely useful with regard to predictable aspects of nature and with regard to machine technology. Machines are designed to separate generative models from their realizations; to, in effect, place their operation in different systems than their production, so that the operation of the machine will reflect the original design without undue influences from its application environment. In complex systems, the context of production is co-mingled with the context of operation, such that a change in one will result in a change in the other. In a living system that principle results in adaptation and evolution. However, when we selectively observe mechanical systems in nature, we then see nature only as a pre-designed artifact. The process of *realization* of models, which is the natural process of definition, is therefore of primary interest in Rosen's theories, but so is the more implicit process of *contextualization*.

Rosen therefore found it necessary to "retreat" to a broader foundation; to relax the restrictions that define mechanistic or dynamical systems and thereby to develop a more general mathematics for complex systems in which "*the entire epistemology of our approach to natural systems is radically altered.*" To understand Rosen's approach and its departure from the definitions that restrict us to dynamical systems theory, we must first understand, from his own characterization, that the "ingredients" of his approach are "*the basic notions of information*". And although Rosen himself did not complete this theory development, he showed how to put the ingredients together.

8.2 R-Theory

There were two avenues of development in Rosen's work that were evident in *Anticipatory Systems*. One avenue was associated with the mapping of causality using category theory, the more general foundation that he referred to in mathematics;

and the other avenue was associated with modeling relations, which are explicitly formed of information relations between a model and a natural system that it models. It is the combination of these two tracks that gives us a complete view of relational theory. Rosen focused primarily on laying the mathematical and epistemological foundation, leaving the full synthesis for others to pursue. Indeed there were hints in his writings that he did not believe we were yet ready as a society to accept that synthesis without considerable re-thinking. As we explore the full implications of what might now be called *R-theory* its revolutionary character begins to emerge.

We might understand R-theory best by considering the idea of *insides* and *outsides* of systems; that every system or event viewed externally (the basis of the mechanical/dynamical view), must also exist on the inside of another system that constitutes a context. The relationship between externally realized (interactive) properties and internalized or contextualized conditions then establishes the relational view. Its greater generality with respect to the mechanistic view is obvious when we realize that mechanism assumes one and only one causally conditioning interior, or context, whereas relational theory assumes that every system has both aspects with respect to other systems and therefore contextual definitions (which will be identified as natural models) may be as numerous and varied as natural systems. The question of simplicity is then one of the extent to which contextual determinations agree. The view from a single context reveals nature as a collection of parts with exclusively defined properties, locations, and dynamic changes within that context. The view of the inside of any contextual system reveals it as a set of attributes and intersecting conditions that act like models for the potential existence of parts (i.e., events and phenomena). In the total view there cannot be externality without internality, and nature can thus be described by the exchange of information between *realized* and *contextualized* domains. Science itself is but one such context and even science specifies and changes the outer world as it is put into practical application.

By attempting to formalize science on the basis of observational realities alone, mechanistic science was attempting to standardize on one formal cause context with a set of fixed definitions, parameters, and metrics for efficient laws. The crisis of post-modern science was precisely the result of discovering formal cause variation – that contexts matter. Goedel’s incompleteness proof for number theory was also a demonstration that the effect of context, formal cause, cannot be universally reduced to one common system. Instead there are known variations in the parameters of a dynamical system under conditions that isolate events from general interactions. For example, the smooth relativistic change in the Lorentz scaling of space-time (the “shape” of space) is a continuous variation in the formal metrics of space-time that corresponds with the degree of isolation between events separated by speed of light communication. The discontinuous differences in quantum reality occur explicitly when phenomena are isolated from general interactions such that observer interactions are more defining. Wave-particle duality also suggests different contextual constraints on the distribution and interaction of events. However, living systems establish formal cause differences with much greater facility as a consequence

of how they are organized, extending and maintaining their functions via self-produced, internalized models that open the system to adaptation and evolution.

In relational theory the fundamental causes of complexity are the same for all systems, regardless of scale. That similarity between physical and biological worlds, however, is not what was hoped for by classical scientists; it is not the explanation of biology from the assembly of explicit dynamical behaviors, from the “bottom-up”; but instead it is the explanation of physical systems from reduction of their natural complexity. That complexity also forms the basis for living systems, which are actually better examples of it because instead of reducing complexity, they employ and enhance it. It is just for this reason that Rosen often wrote that biology has something to teach physics.

The epistemological separation of an observing context from nature-as-observed in external events creates a paradox in which we represent nature as something that itself can have no natural representations. That, in the clearest terms, is what makes the mechanistic view non-complex. By removing information and its causalities from our concept of nature we indeed obtain a view of nature as an independent object of description. But, as a consequence, our own experiences with information cannot then be considered natural and what living systems do best, modeling, cannot be studied as a natural phenomenon. As a consequence science has been struggling, as it matures, to recapture the fullness of nature in some form that will not destroy the predictive and technological advantages of knowing simple mechanisms. It is likely for that reason that Rosen discussed the mechanistic/dynamical system view in depth as a prelude to introducing the differences that characterize complex systems. It was perhaps in hopes of forming a bridge between the two worlds. Even so, he concluded that there is no *syntactic* way to cross that bridge; that is, from the dynamical side. It must be crossed from the general, contextual (*semantic*) side, from a new science of complex and anticipatory systems to the science of simple systems. Complexity, while perhaps approached in the ultimate limit of infinite mechanisms, is not reachable by any finite addition of them. In contrast, however, the finite world of mechanisms is reachable by a reduction of the complex.

These epistemological and methodological issues become most critical in considering the problem of anticipation, which Rosen dealt with directly in this book. The entire discussion of dynamical systems, and how to exceed them to understand complexity, underlies the consideration of anticipation. Given the very large sociological gap associated with just the introduction of relational complexity to science, the discussion of anticipation seems almost out of reach. But if we accept Rosen’s foundational arguments about complexity, we can then begin the discussion of biology, which is characterized by anticipation. In other words, we must first understand that complex systems are not special in any sense; they are the general case of any system for which we consider context dependency. It is then possible to understand the truly special nature of dynamical systems or mechanisms, and it is also possible to understand what is unique about life forms, primarily the special nature of organisms as metabolism-repair (M-R) systems. Unraveling the layers of misunderstanding about what is special and what is general in nature is thus crucial

to understanding Rosen and R-theory. Complexity is general; mechanism and life are special and quite opposite ways of organizing otherwise complex relations.

If mechanistic science was originally meant to be a picture of reality, it was taken that way only naively, with the intention that it might represent an incremental process of gaining knowledge that might not have a limit. Rosen's philosophical arguments are clear on this point; that indeed an infinite series of mechanistic models is implied by any complex system, but unfortunately it is not an infinite increase in knowledge even if all those mechanisms could be specified. It is instead like Zeno's paradox, where the goal is approached by an infinite series of half-measures that assure it cannot be reached at all. The half measure, in this case, is half of Aristotle's four-part causality. In fact, the problem is worse than Zeno's paradox, because as we make incremental additions of efficient and material entailments to approximate complex phenomena, the approximations can reach an absolute limit to their accuracy beyond which further additions will not decrease the error. This is because a complex system does not imply just one end-point. Once uncertainty is encountered in the addition of mechanistic models, it cannot be removed by any similar additions: the other half of causality that takes place via context must be added. Context refers to the formal causes (which may be seen as attractors), by which the parameters of a dynamic system and their events are established, and necessarily the final causes by which context is established. This last step, the closure with final cause, was not directly described in Rosen's work (nor by Aristotle), but remains as a logical and profound conclusion of relational theory that Rosen left to for his students and colleagues to infer. We find in Rosen's writings, however, very clear descriptions of the problem of causal closure, and a pathway toward its resolution for those who wish to follow it.

The equipment we need to do so is a basic understanding of causality, more or less as it was divided into four parts by Aristotle, but closed within nature in a way that Aristotle did not perceive. We need also to relate the causes to each other, which Aristotle did only hierarchically and with some ambiguity that has prevented their integration ever since. The full and recursive integration of these causes is how we can define and describe whole systems, which may then be seen as the objective relational units of nature that we can analyze. It was Rosen's conclusion that the answer to understanding complexity and life was to dissolve the divide between objective and subjective domains by objectifying the relations between them. Those relations are natural instances of modeling relations, consisting of a recursive hierarchy of the causes. That unique organization of causes defines wholeness and gives identity to systems, which then interact by sharing or substituting causes. The picture of reality that emerges is one that is less rigorously familiar to ecologists; it is that of a fully interconnected system-dependent reality in which wholes are more than the sum of their parts, precisely because of relations. The view is approached in physics as "model-dependent realism," although current descriptions in physics do not adequately explore the realism.

Perhaps the greatest stumbling block has been that in a true modeling relation, final cause must be part of wholeness, and yet it has been shunned as external to nature. In R-theory it is part of the naturally complex world completing or "closing"

the other causes by providing natural exemplars. A modern re-interpretation of causality may thus close the causality loop by defining final cause as *imminent* in nature, rather than *immanent* from a mystical domain, as was the accepted cosmology in Aristotle's time. Interestingly, mysticism is not eliminated in this re-interpretation, but it is brought within our understanding in terms of creative potentials that are the origin of otherwise describable events. The hope of finding an "exact" science led us to place finality outside of and prior to nature (and thus to reinforce the uniquely Western view of an unnatural external creator), whereas returning it to nature accepts that nature is fundamentally complex, and thereby gains us the ability to consider the origin and identity of systems.

As must be expected, anticipation and the formation of anticipatory systems involve final cause. But relational logic leads to the conclusion that final cause is a prerequisite for anticipation not a definition of it: final cause is part of general complexity and wholeness in all of nature, living or not. Final cause steers (or attracts) the development of a system toward pre-defined exemplars; but anticipation is a special use of final cause in which the exemplars are selected on the basis of system sustainability. That drives evolution, which is an exploration and selection of various model realizations that best sustain the system's functions. While final causes themselves may provide end-points of system change, anticipation associates those end points with meaning in terms of persistence of a system's functions. In other words, end-directedness itself might explain why any system moves toward a future condition, whereas anticipation is a meaningful selection from alternative futures with respect to conserving whole aspects of the system; a selection of self-reinforcing functions. Anticipation is thus defined by *adaptive* final cause.

It seems that Rosen was quite aware that the deep and inescapable implications of relational thinking would be hard for many to accept. Final cause, after all, had been purged from science and labeled nearly as its antithesis. Nevertheless, he seemed also quite aware that the incentives for following his path would increase in future years, because if the limitations that were built into mechanistic science are rigorously applied extensively enough they must eventually cause problems of incompleteness. At first the discovery of regularities in nature was the hallmark of modern science, but post-modern science has been more characterized by the discovery of nature's irregularities, particularly irregularities in the classical descriptions established through the modern era. The implications of uncertainty were thought to be damaging to the very foundation of science. But the classical alternative was actually worse because it clearly implied that either we too are machines, or that the one thing we *must* take for granted, our own experiential existence, has no explanation at all. It seems bizarrely paradoxical that the way to gain knowledge about the natural world should be to describe it as separate from all knowledge: we have to be suspicious of what kind of knowledge such a program can produce. Quite plainly, it produces descriptive knowledge about outcomes, which is only prescriptive if the system under consideration is already scripted; that is, if it appears fully constrained by its contexts. It provides no knowledge about how the scripting takes place; the causal origins of a system.

8.3 Cause vs. Probability

Science has been at a cross-roads for over a century now. The post-modern/post-normal era of science has been one characterized by a combination of immiscible theories and hopes for an acceptable synthesis; that is, one that could be demonstrated to be consistent with prior facts, parsimonious in its formulation, general in its terms of reference, universal in its scope, applicable as an analytical method, and capable of improving our knowledge in testable ways. We began searching for a scientific “theory of everything” but especially we were searching for a theory that could explain how indeterminism in nature could arise at all from a deterministic background. Evidence for self-determination (which Humberto Maturana and Francisco Varela called *autopoiesis*) in living, conscious and thinking organisms was shrouded in epistemological and other confusion, but the evidence for uncertainty in physics became unavoidable. Evidence at many scales revealed apparently self-generating behavior that we could not comprehend, and that hinted at greater levels of organization. Certainly living systems involve complex organization, but it was a shock to discover that the most fundamental physical systems do too. Most scientists could not accept that a theory of mind, while perhaps providing analogies and metaphors for physics, could replace its equations. Theories of information were limited to the thermodynamic concept of entropy, which itself is limited to the efficient and material world of energetically closed (causally open) systems. Mechanistic thinking, by itself, does not represent levels of organization that are not law-like. And yet we naively thought that all organization in nature would turn out to be governed by fixed efficient laws. We did not understand the nature of the problem, which was that a system cannot describe itself without setting up an infinite regress of exteriors, or “larger systems” as Rosen called them. Machines are made; they have their origin in another system. Therefore the machine metaphor requires an external maker, a fixed causality that is outside the natural world of that description.

Perhaps for that very reason, scientists and even philosophers have been unwilling to explore very far outside the mechanical box; fearing that science would be in chaos if its laws admitted to any variations at their origin. The discovery of wave-particle duality was thus precisely the crisis everyone had been working to prevent, and mainly as an emergency measure they decide to patch the problem with probability theory, thus preserving the use of number theory but allowing for duality in the difference between probability and certainty. However, it was an unsatisfying solution for many because it essentially assigned ontological status to a subjective or at least conditional measure. Neither Einstein nor Schrodinger accepted it as a natural philosophy, arguing that nature is not governed by chance; that there must be a more fundamental set of principles.

Probability is based on counting how often something occurs under given conditions and then expecting it to be that way in the aggregate under the same conditions. It abandons the idea of predicting individual events and instead defines limits to knowledge (i.e., ignorance) of such specifics. It was not meant to be about

natural causality but about empirically determined regularities in collective data, the specific causes of which may be unknown (or unknowable). A contextual domain of possibilities can indeed be described by a set of empirical probabilities referencing it to past exemplars (say, actual distributions or logical possibilities, as in the flipping of a coin). However, even a probability model depends on how conditions are evaluated; in other words it ultimately assumes some necessary relation that links conditions with events. There is always an implicit or explicit underlying model of how supposedly random events *should* distribute, and an implicit assumption that there are fixed laws of probability and randomness. In effect, then, using a probability model shifts the original assumption that we can have an exact model for predicting events, to the assumption that we can have an exact model for predicting the random distribution of events (the probability density function), which is thus an expression of formal causation. In the case of quantum mechanics, the underlying quantum wave function implicitly acts as a model of nature, no matter how much one might insist that it is just a way of describing results. As formal cause, one step removed from efficient cause, it brings us one step closer to dreaded final cause.

Quantum probability waves are implicitly a propagation of natural model-based information about nature, essentially representing formal cause as a necessary consideration in our picture of the physical world. The success of this theory confirms the need to include the idea of natural contextual entailments in science; in other words, to consider contextual system dependency, which is formal cause. To the degree that the underlying probability models (quantum wave functions) were thought to have a natural referent of some kind, the use of probability theory in physics was a step toward relational thinking. Both theories are based on information relations associating events with conditional models and both specify potential existences, thus softening the connection between prior conditions and expected results by predicting only the tendency, frequency, or suitability of occurrence of events. But most theorists attempt to explain these probabilities in terms of fixed background potentials without considering the cause of those potentials themselves.

Nevertheless, taking this first step and introducing uncertainty between the model and observations covers a multitude of sins. Not only does it change the idea of Natural law to a tendency (a potential occurrence) instead of a necessity, it also covers the problems of lack of knowledge or just plain measurement error. With exact science in question, probability theory has invaded nearly every branch of science; but because it conflates natural indeterminism and measurement error it is often easiest not to make the distinction, which would require going to deep theoretical foundations. The result has been that the concept of causality itself (which was defined to mean something deterministic) has fallen out of favor. Rosen, like Einstein, was seeking to find a deeper causal basis for nature, but as it became increasingly unpopular to speak of causality, Rosen was accused of “answering questions nobody wants to ask.” The final cause origin of models that necessarily underlie probabilities should be explored with the same realist implications that led us to accept formal cause; in other words to ask what the cause of formal cause might be. Nevertheless, this reluctant acceptance of formal cause in physics did not mean a further exploration of final cause.

Consequently, there is a major difference between relational theory and the general use of probability theory in science where we may ask neither the specific causes of individual events, nor, in many cases, the cause of the probabilities. It has become sufficient to quantify a potential and treat that as a fact, even though science does not have to halt there. Since it was acceptable, with minor exceptions, to arrogate final and formal causes to human cognition alone, probability theory was designed for just that descriptive philosophy. Its implicit adoption as a quasi-reality in quantum mechanics was thus unexpected and shocking; but it spoke of a greater reality, not a dark secret to hide. Without a theory of how the probabilities originate, there is no way to distinguish uncertainty in nature from uncertainty in human perception, and today a great deal is being made of that fact to suggest that there may be no difference. That hides the discovery and even returns us to some very extreme forms of anthropocentrism. But we have only taken the first step toward complexity, consideration of formal cause; and that step necessarily conflates the two uncertainties.

The next step is to describe final cause to explain the natural origin of probabilities; that is, to consider them as generated rather than pre-existing potentials. Relational theory, when fully developed, gives a causal theory once again, explaining how mechanisms are produced from a larger relational system, and how that system can even increase its organization to produce living systems. As such it necessarily accords model-based behavior to all of nature, and it thus allows us to explore organizational differences that account for the different kinds of natural systems we observe. Once this view is worked out sufficiently, we will regain all that was lost, because the more deterministic modeling exercises remain, where they are accurate, as system-specific reductions of the complex; while living systems represent an enhanced organization that introduces anticipation and adaptation. But to arrive there we must assume not that there are fixed underlying potentials for anything, but that the underlying distribution model for potential occurrence of an event is determined by exemplars (final cause) within, not prior to or outside of nature.

Probability theory was adopted widely for its ability to analyze knowledge, and its realist interpretations were safely confined to special areas of sub-atomic physics and thermodynamics. To a great extent the rest of science was insulated from its potentially meaningful implication of contextual causality. Ecology, for example, has struggled without a central theory since its beginning and, aside from very limited dynamical models; it consists mainly of statistical interpretations of data to determine emergent patterns and dynamic sub-processes of an otherwise mysteriously complex system. And yet, of all the sciences, ecology is probably the most quintessentially relational. It is potentially broader in its causal foundation than quantum physics. Quantum mechanics is a more restricted version of relational theory because it has yet to fully extend the concept of the observer (and thus contextual causes) to all systems, and thus to incorporate the idea of natural exemplars into causality.

While it may seem theory-neutral to view the natural world through a probability filter it is not, because in the absence of attributing the cause of underlying potentials

to natural conditions, one merely succeeds in substituting the observer's assumptions. When restricted philosophically to a theory of knowledge, a probability model is about the process of observing. However, it may well be that the theory works in the quantum world because events behave exactly like observers, and natural systems *are* observers. In relational theory systems have dual aspects as material objects and contextual models (of self and other systems). We may indeed be limited in our ability to distinguish incomplete knowledge from natural indeterminism, but with a realist interpretation of contextual causes, analysis can be aimed at experiments to make that distinction. The instrumentalist view of model-dependent phenomena (which is currently popular) erects a barrier to knowledge by assuming that knowledge itself cannot be part of nature; that an arrogated and undefined entity called "we" constitutes observers and all else does not.

Analytically, translating prior exemplary occurrence (distributions of events sampled statistically and interpreted probabilistically) into a contextual potential, involves the modeling relation as a meta-model for nature's own way of establishing attractive potentials from prior exemplars. Theoretically, this step is the essence of ecological niche modeling, if that exercise is taken primarily to be about modeling adaptation (and not merely about statistical correlations). The fields of landscape and geographical ecology, for example, are currently experiencing a rapid rise of interest in niche modeling, but most of the work still focuses on making strictly statistical predictions of occurrence. In a bizarre reversal of epistemology, that commits two sins: First, it bases the modeler's choice of factors on mere correlations instead of experimental science to infer adaptation; and second, it ignores the realized domain where ecological dynamics are responsible for realizing the niche. The reason for this discontinuity in ecology and other living system sciences is, again, the taboo against taking formal and final causes seriously. In other words, the result of the false claims of physicists that nature does not establish its own formal and final causes is that other sciences that have deferred to physics are crippled in their ability to consider the full causality of the living system.

To summarize: analysis of complexity requires consideration of both realized and contextual domains; i.e., all four of Aristotle's causes reinterpreted as a recursive holarchy within nature. Science must therefore involve two kinds of models: one describing system-dependent (contextual) potentials and the other describing dynamics of system realization. The two kinds of models cannot be merged or reduced except when modeling a mechanism; and otherwise they must themselves be coupled as a modeling relation.

8.4 Context: The Final and Formal Causes

Formal cause is the implicit system dependency of nature's laws, which makes it fundamentally complex, impredicative, and non-computable. It is the formative potential of a contextual model that may have multiple realizations. The externally attributed properties of a system are events. They occur on the causal "inside" of

another system, which is their context. There cannot be an event without a context for the event, and yet the context is not itself a local object; it is the organization of events that conditions and attributes the existence of additional events. For example the metrics of space-time constitute formal cause for the organization of the event world, attributing events with local coordinates and the “shape” of that coordinate system. The grand assumption of classical science was that all system contexts have the same effect on the organization of internal events; that there exists one and only one formal cause. We can no longer maintain that assumption, but if we admit formal cause variation into the epistemology of science, the question of what causes differences in formal cause can then be answered as final cause: the effect of exemplars from prior realization, thus implying self-reference and self-similarity.

Neither Aristotle nor his followers had a natural answer for where final cause originates. As a result final cause has not been dealt with in science, and formal cause variations had to be treated as exceptions at the “edge” of reality, with mysterious origins. Unfortunately that placed living systems at the same edge, hardly where they belong in a world that needs answers about how to manage them. The consequence of these decisions about the organization of science has been that it could advance society considerably in abilities to make efficient changes and to employ efficient processes for specific ends, but not at all in the ability to comprehend broader systemic effects and relations, or the origins of various kinds of systems.

Duality is our great clue in this mystery of the causes. At its root it involves the relation between foregrounds and backgrounds or outsides and insides of a system. It is the relation between what is singled out (abstracted) from nature and the context in which it is perceived or interacted with. For example, we observe discrete events of the mechanistic world in a space-time context that is defined on the principle that two things cannot occupy the same space at the same time and one event cannot be in two different locations. Only if the rules of space-time itself are altered can there be an exception. However, such exceptions *have* occurred under controlled laboratory conditions that in effect alter the formal cause system of space-time. In other words, space-time itself must now be considered relative and produced; not a fixed prior reference system. With that modification, the strangeness of uncertainty becomes understandable as the effect of different space-time selections.

Aristotle’s view of final cause did not allow for its inclusion in natural science because he and many others saw the ultimate cause of everything as the act of an external creator. That concept is retained in mechanistic science in terms of the big-bang origin of the universe; but there it could be kept historical: one act of creation after which everything is presumed to operate mechanically (except that it doesn’t). In Aristotle’s view nature is produced from an undivided whole that is an immanent cause of the world (final cause), after which its operation is governed by the descending hierarchy of formal, efficient, and material causes. Philosophers and scientists who followed Aristotle realized the problem this hierarchy posed for science: external intervention with implicit purpose toward unknowable ends. Mechanistic science was thus a grand compromise, resolving that external cause as an historical event. But consequently it cannot deal with the origins of any

system, let alone itself; only the conservative reconfiguration of one system that unexplainably originated from nothingness 14 billion years ago.

Mechanistic science, thus steeped in as much theology as any other view, nevertheless left us with some ability to understand and even control the temporal world. Perhaps not too surprisingly, then, we retained Aristotle's hierarchy culturally but distinguished half for science and half for the humanities, arts and "soft" sciences. To the early Western philosophers of science dealing with the politics of the time there could be no other solution than to divide the causal levels in this way, with obvious advantages for establishing a new discipline and, in some cases, saving people's lives.

8.5 Causal Closure

We can read amid the detailed mathematical treatment in *Anticipatory Systems*, that Rosen's proposed solution to the scientific and epistemological questions of complexity, final cause, and life was indeed *causal closure*, which ultimately leads us to the implications described above. In other words, if Aristotle's four causes are a hierarchy, as he said, they must constitute a closed hierarchy within nature in order to avoid the problem of external origin. If that is the case, it is then possible to describe systems that are causally closed with each other. Rosen presented this idea as *efficient closure* between natural systems that therefore produce each other. He used familiar mechanistic terms but nevertheless challenged the limitations that mechanism places on our concept of the natural world, limitations that would preclude such causal loops. Efficient causes are processes by which a current system configuration is produced. From the view of purely efficient/material (mechanistic) cause, a single future system is produced from a single prior one, on back to original creation and external context for the universe. But, if different efficient causes can produce each other, system origins can then be entailed from within nature. Rosen suggested that is exactly what characterizes complexity and thus allows for life.

In mechanism, laws govern the dynamics of material objects, but material objects do not govern the creation of efficient laws. Thus mechanism allows for only one form of end-directedness: It is a one-way descent from immanent (supreme) causation toward material death (entropic "heat death") from which nothing recovers. But if this is only half of causality, what happens in the other half that was left out of the mechanistic picture? If Aristotle's hierarchy is a circular hierarchy, then the contextual side that includes final and formal cause is an ascendant causality, generating new functions from prior structures. As Rosen wrote (citing Erwin Schroedinger), it is the case of an "*inertial object*" (a material result) acting as a "*gravitational object*" (a cause of dynamics). Clearly the explanations for life cannot exist only on the descendent side of these causes, but by treating causality as a closed loop it is possible to bring the ascendent, contextual causes into natural science. Rosen is thus clear that there is nothing mystical about relational theory,

although from its implications we can certainly speculate differently than before about the mystical, which relational theory places within nature rather than without.

Causal closure means that there is an inverse causality (referred to here as context) that runs opposite to the apparent end of mechanism. This result is indicated by the fact that a system of efficient closures is a paradox in the mechanistic view; it should not exist and yet it describes the behavior of complex systems, which obviously do exist. That paradox is only resolved by incorporating both sides of the causal hierarchy, both contextual and phenomenal realities, into a larger view. To describe *life itself* mathematically Rosen thus adopted the more general formalism of category theory, which broadens the mathematics of natural description by generalizing objects as sets and their morphisms. The entailment diagram that shows a solid headed arrow (an efficient cause) implicating a hollow-headed arrow (a material cause) characterizes this analysis. That mapping is a basic picture of mechanistic cause, except that in category theory the result of a material map itself can become a morphism; that is, the result of one process can be the cause of another, thus removing the requirement that laws cannot be system dependent. When that is allowed, a natural inverse contextual category is implied that accounts for the generation of new systems and new functions from existing nature; and there is a corresponding increase in the number of unrealized possibilities in nature, which is a decrease in entropy. While from the limited perspective of the mechanistic view any reversal of the second law of thermodynamics must be a local reversal, this larger view suggests that such reversal is a natural consequence of contextual entailment at any level; that is, any system that is closed in all four causes and thus whole.

In this view, complexity is natural and the theoretical problem facing relational science is then to explain mechanism and the apparent existence of a self-consistent classical world. As it turns out, that is easy: Classical mechanism is a fully reduced modeling relation; it is the limit to which a complex system can be constrained. With these new terms of reference we can say that natural functions (which can be associated with nature's laws) change states of a system (which describe its physical structure), and it is also true, as allowed in category theory, that natural structures change functions. With that freedom added back into our thinking, it is then possible to develop descriptions of a natural system that include the apparent intervention into mechanistic formalism mentioned earlier, which quite reasonably comes from causal loops between relatively isolated systems. But with regard to the expansion of science, if closed hierarchical loops cannot exist in a mechanistic domain (of mathematics or as attributed to nature) then we must define a broader domain of science and nature where they can exist.

8.6 Modeling Relations

The work that Rosen presented in *Anticipatory Systems* came at a point of development of these ideas where the critical role of modeling relations was being revealed. Even at this stage, however, Rosen gave strong hints that modeling relations might

be a fundamental reality, as a true picture of science but also, by implication, of how natural systems relate to each other through natural models. He also wrote that they are not exclusively about biology, that *“the concept of a model is not something exotic or unusual, but rather of the broadest currency imaginable in all disciplines.”* A modeling relation is an information relation between a model, which we might see more generally as the contextual aspect of a system, and the realized aspect of a different system (or the same system as an identity relation) that is abstracted and modeled within that context. The modeling relation allows neither complete agreement nor complete disagreement between the two (there is always similarity and error, or a “discrepancy” in the relation), and therefore it establishes a principle whereby nature can communicate with images of itself. It is also an explicit representation of the mind-body problem, thus implying information *in* nature, as, for example, Gregory Bateson also claimed. Modeling relations, as Rosen described them, turn out to be the critical idea in forming an analytical and theoretical synthesis for relational science. However, these broader and more controversial implications were not highlighted in Rosen’s mathematical underpinning in which he left the conclusions to be drawn.

Rosen applied the concept of a relation between nature and models of nature as a central element of his arguments about complexity and anticipation. There were two levels of that application. First there was a description of science itself as a program to understand nature by representing it in surrogate (or analogous) systems with similar entailments. In such an exercise scientists attempt to get a model of the system’s entailments to commute with nature, or at least to get a simulation to commute with certain behaviors. The important question for science, then, is to what degree can one system represent another system? In distinguishing approaches in science that involve more complex relational thinking from those that reduce nature to mechanisms, Rosen made the bold assertion that scientific models should themselves be entailed, that is organized, in the same way that we believe nature is organized. Implicitly that means considering all possible causes. When science does not do that, for example when it focuses only on efficient and material causes (mechanisms), it is actually not applying a model at all in Rosen’s terms, but a simulation. Most of current science is thus based on simulation. However, the importance of this observation goes beyond mere labeling. By this criterion, a true model must then be a natural system; only then can it be said to be fully entailed like nature. Without delving into the philosophical arguments surrounding this issue, which are extensive, we can understand the depth of Rosen’s theory best in this concept of a model. If a modeling relation involves all the known causes then nature must be describable in terms of modeling relations. A system of analysis must therefore exist in which nature is seen to comprise nothing but modeling relations.

Unfortunately, in all of Rosen’s work, a complete synthesis of the ideas of modeling relations and the ideas of relational causality in terms of category theory mappings was not presented. In effect he presented two views of complexity, one of the mechanistic paradox in which closed loops of efficient causation cannot technically exist; and the other of the implicit incompleteness of all descriptions when considered from the perspective of modeling relations. Rosen thus reasoned

quite legitimately from Godel's incompleteness to the logical incompleteness of any system of description in which "realized" components of nature are the exclusive elements of analysis. His view was perhaps most clearly stated in his later book, "Life Itself" in which he wrote that instead of objectifying the efficient and material aspects of a system, the aim of a science of complex systems should be to "*objectify impredicative loops*"; i.e. modeling relations themselves.

Modeling relations are both information decodings from contextual models into realized behavior, and information encodings from realized behavior into contextual models. As such they define the concept of wholeness and whole systems in the ideal sense, as closed hierarchies (holarchies) of all four causes. Again, we have to read somewhat between the lines in Rosen's writings (or actually to put the lines of reasoning together) to arrive at the conclusion that the implied existence is a complex reality constituted of modeling relations. That result corrects the blunder in Western science (albeit perhaps a willing one, as mentioned earlier), of imagining that final causation comes from outside of nature. It changes our idea of nature from a causally open and materially closed reality, to the other way around.

Most significantly a causally closed system allows us to define system identity, wholeness, and relatedness; placing causation in the framework of a modeling relation and merging the two theory tracks. The realized domain is one of exclusive system properties as traditionally analyzed. Its exclusivity is what allows measurement and concepts of natural interaction. But in the contextual domain, system potentials are non-exclusive, overlapping as in Venn diagrams, with three possible results. Their non-intersection defines unrelated identities of different systems; their union identifies a larger system in which both systems are implied; and their intersection defines a new system specified by the mutual constraints of the original system models. The contextual intersection thus implicates an emergent new system in the realized world. Context is therefore the domain where emergence of new systems takes place: It comprises the natural conditions that act as models with respect to realized systems. A more specific model thus originates a new system that may introduce a host of new functional relations in nature. Clearly, then, excluding that domain from science excludes all consideration of origins and any causes that can be called creative. However, how tightly (mechanistically) constrained, or ascendantly organized a system is, depends on the relative isolation of its internal relations from general interactions with the environment that would bring its models into a more strictly reactive correspondence with the general causality.

8.7 M-R Systems and Anticipation

If modeling relations are not unique to living systems but ubiquitous in nature, and closure is also natural, then something additional is required to characterize biology. We find that additional criterion in the M-R system of internalized entailments, which characterizes cells and organisms. In other words, this special arrangement of internalized models that establish each other (are closed to efficient cause) seems

to distinguish life forms from all other kinds of systems. As M-R systems were examined more deeply in later and related work, Rosen showed that they represent an efficient closure of three functions necessary for life; *metabolism*, *repair* and *replication*, which emerge from four natural components. This unique organization of internal components and their functions can be shown to have its own modeling relations with the outer environment representing behavior and selection, thus forming an identity and ensuring that an M-R system is adaptive and evolutionary. In this volume, Rosen identified M-R systems with *anticipatory systems* as a unique category of systems that generate their own internal predictive models (the most commonly referenced diagram of the M-R system was presented later, in *Life Itself*).

Describing nature in terms of modeling relations means that no system is truly separate from everything else, and that no two systems are exactly alike; it is a world view of qualified separation and qualified connection that, therefore, is complex and relational. But it was also clear that mere possession of an internal model is an insufficient criterion for defining life. The mere existence of models is identified with fundamental complexity in nature, whereas it is the production and use of an “internal predictive model” that is identified with living systems, and the main criterion that addresses Rosen’s primary question, “What is life?”

Rosen defined an anticipatory system as one that changes its behavior according to an internally produced model of the future *for a desired result*. The last part of this criterion is important but often overlooked because in science it is not considered appropriate to attribute intention to natural systems. However, merely responding to a model of the future means nothing if it is not an adaptive response. We can translate the term “desired” into natural terms by associating it with formal cause and thus distinguishing anticipation from pre-adaptation. We can then say that living systems are distinguished from non-living systems by their ability, through adaptation and evolution, to employ and develop complex system models in a unique way (via the M-R system) that prolongs system functions, often by quite sophisticated means. Thus the living system anticipates an internally predicted future in the sense that adaptation to predicted conditions prolongs existence; whereas merely responding to a future prediction, perhaps randomly, can at most result in pre-adaptation.

Anticipation thus involves *selective* response to possible futures represented in the present. The very neatly predictable mechanistic world, where “the future” is imagined as the one and only possible outcome of the past, cannot be preserved if there are systems that anticipate, for anticipation involves symbolizing multiple unrealized possibilities and selecting from those choices. *Impredicativity*, as part of a temporal sequence, means that the future states of a system are not uniquely predicated on its past, as would be required of a mechanism; and therefore it is not uniquely predicated on general laws that dictate temporal change. Systems that are capable of symbolizing and responding to multiple possible futures, that is, systems driven by internal models, are thus capable of selecting or being selected by alternative futures through their present and subsequent behavior. Furthermore, these effects are cumulative across all systems, thus conflating prediction of the future with collectively creating it. Once present behavior is predicated on a

symbolized outcome, we are in a domain of causal feedback in which the present is, in some part, governed by conditions that will, at least in part, become the future.

It may be worth noting that even stronger ideas of “retrocausation” may also be treated in a relational framework. For example, modeling relations can represent multiple historical pathways that might equivalently arrive at the present. In other words, the formal representation of the past may be determined by present models, and in a complex world multiple non-contradictory models may provide alternate histories. Furthermore, if we consider potential futures and plausible pasts modeled in the present to thus “exist” in the sense of a model, we might go one step farther, as some physicists do, and suggest that such multiple realities are in a sense actual. Instead of imagining that present models are the result of history, we can imagine they are merely selecting which alternatives will be experienced by the “observer.” The application of R-theory to explain temporal sequences (normal dynamics), for example, requires that events be realized from a context and that they are in some sense discontinuous. That discontinuity allows multiple sequences (multiple histories) to be collectively true for their associated set of events where no specific events are in contradiction.

We thus enter a reality that is best described as a relationship between that which exists in the present as a measurable set of conditions, and that which cannot be said to exist as such, but is nevertheless causal. It is a modeling relation between existence in a locally defined and attributed domain, the world of realized, measurable systems, and existence in a non-locally distributed context where models are formed and can combine to originate new systems. In this sense both domains exist. In the domain of local existence, which is defined by space-time coordinates, we measure *what has already happened*; whereas in the contextual domain of non-local potentials, models define *what might happen* as nature’s possibilities. As each of these domains is formed from the other, the two are mutually attractive, resulting in complexity and even directional change. On the other hand, to the extent that the system being considered is not closed, that it is open and interactive, interactions will establish a common context that we can recognize as the realized world and that can be successfully modeled as a mechanism. It is thus the case in this view that both the living and the material world are emergent from the complex.

8.8 Organization, Entropy, and Time

In R-theory, whether a system is complex, simple, or living depends on how its modeling relations are organized. When a system contains macroscopically complex components like living cells, neurons, organisms, species, etc., its complexity and uncertainty relations are the same in principle as those of a sub-atomic system that has microscopically complex components. Complexity is not a function of scale, it is associated with causal isolation of components of a system, which occurs naturally, but which living systems also excel at establishing.

In the cases of living (presumably, M-R) systems, there can also be multiple organization possibilities that have the same general realization. For example, there can be multiple kinds of M-R system closure that differ in their contextual organization and, as a consequence, have different behavioral characteristics. These organizational possibilities are the result of how the inverse entailments are organized on the contextual side of the system's causality; that is, its final and formal causes. The generation of different organizational possibilities from contextual differences alone allows novel behaviors and thus new systems to emerge. True relational complexity also alters the standard model of entropy. Rosen pointed out, for example, that the 2nd law of thermodynamics applies to ideally closed systems (of which there are none in nature). When systems are partially open (energetically and causally) such that they are interactive identities (thus complex) they will then exhibit entropy increase and entropy decrease in their simple and complex aspects, respectively.

Entropy was invented as a classical measure of total system order, which can be related to "organization" in the following way. If entropy is increased or decreased it is understood to indicate a corresponding decrease or increase of order, respectively. That change in entropy (and degree of order) is also associated with changes in the flow of energy through the system. However, as a result of complexity, there may be alternative patterns of organization that decrease entropy as distinguished from the increase in entropy resulting from metabolic processes. When models establish their own internal time frames, as Rosen described, or alter their environments, it is a similar change in formal cause and organization of the system.

Our choice of world views has much to do with how we view change through time. We see, through Rosen's work, that the mechanical view in science is one in which we assume, incorrectly, that we can learn all possible behaviors from those that have already occurred, and that what will occur in the future is a predictable reconfiguration of the past. It is a view devoid of anything truly creative, anything truly new or alive, and it corresponds with the basic physical (and Western theological) assumption that creation occurred at one unapproachable point in projected history (and thus by an external agent), after which the universe has run automatically, using up its energy and increasing its entropy toward eventual heat death. However, that implicit end is only one of several possible mythologies.

The goal of traditional Western science has been to figure out the mechanism(s) of presumed automation. On the other hand, the relational view suggests that existence as we measure it is a realization of models that establish events and also the parameters of space and time that will be interpolated between those events. It suggests that the event world is discrete and contextually relative, not continuous and uniquely defined; and that a temporal sequence of events is not really a change but a re-creation of the event at different times and locations. In other words the event world is quantized by its events and organized by its formal models. If this is the more correct view, it opens many possibilities for investigation of currently anomalous phenomena that seem to violate not only the laws of thermodynamically closed systems, but those of space and time. But the true value of relational theory is not necessarily to present a new cosmology or to challenge proven modes of understanding. It is to introduce a broader theoretical framework where these and other questions can be asked.

8.9 Conclusion

A comprehensive re-structuring of the foundations of science, primarily its epistemological assumptions, is needed to expand science so that it can address complex phenomena that are pressing upon us at an astonishing rate, with equally astonishing failures in our understanding of them – phenomena that call into question the nature of existence and the future of humanity. Nothing can be more important in science today than to develop these new theoretical lines. Indeed, we must explore different futures than the machine metaphor alone can imply.

The basic complex relation is between two aspects of an otherwise unified whole. These aspects are: (1) the realized (actual) aspect that can be locally observed, and (2) the contextualized (potential) aspect that can only be inferred as a non-local potential. These complementary aspects never exist separately but they act differently and have different causal properties that we can know. Contextual and realized complements of nature are never “fractioned” in a relational analysis as they are in a mechanistic analysis, as neither side is discarded. The knowable aspects of this relation – the terms of reference for scientific models – are “structure” (measured state and change) and “function” (inferred potential, or “law”), more or less as understood previously, but adding their contextual aspects. Whereas the fundamental modeling relation represents our ontological view of nature (the “man behind the curtain”); structure and function are the epistemological units that can constitute an empirical study and analysis.

Relational theory remains completely inside the domain of scientific epistemology by the fact of relating fully natural domains distinguished only by their contextual relation as insides and outsides of related systems. The boundary between living and non-living systems, in this theory, involves a categorical difference in how systems are entailed (the M-R system vs. complex modeling relations or mechanisms not organized as self-producing units). Systems entailed as one interactive system produce a classical world; whereas systems entailed with causally isolated components, separated from their environment by internal entailments, are complex. Because events in the realized domain are distributed and separated by space and time coordinates, and the conditions for their viability and identity (their models) are separately realized, all systems are out of equilibrium with their potential existence, which drives them dynamically and attractively. We need only the interaction of modeling relations to describe the realized domain, both its indeterminate nature and its collective classical nature. But to describe the biological domain we arrive at a special self-entailment that internalizes and isolates the cause of modeling relations themselves, thus establishing ecological and evolutionary M-R systems (described in much greater detail in Rosen’s later book, *Life Itself*). Living systems, owing to their closed causal entailments, are capable of constructing sophisticated internal models that are necessarily involved in anticipation, adaptation and evolution.

Rosen’s method of inferring the greater reality by combining paradoxical results from two theory tracks was precisely the method describe by Einstein in his layman’s explanation of how he came up with relativity theory, by combining two paradoxical conclusions of Newtonian physics. He wrote: “... *by systematically*

holding fast to both these laws, a logically rigid theory could be arrived at.” Rosen’s theory of relational complexity, applying the same method of synthesis by holding fast to complex *entailments* between phenomena and *information relations* between entailments, also results in a logically rigid theory that should change science and our perception of the natural world. Rosen may well be considered the Einstein of biology, having provided what may be an even more comprehensive world-view than Einstein did in physics.

While this discussion may seem to reach far beyond the present volume, *Anticipatory Systems*, the book does not stand alone. One must see it as one very methodical step in a carefully constructed theory of life and nature that Rosen assembled over the course of his lifetime. This commentary drew from the depths of that work as a whole and attempted to reach beyond it toward new directions for development. Hopefully, some useful directions for further exploration have been identified, even if only crudely. In any case, Rosen’s work seems to remain relatively flawless and the necessity of advancing the theory on the basis of the original work should be strongly emphasized. The logical consistency and profound implications of that work have survived extremely well the test of professional critique, and it is now time to seriously explore the full extent of its applicability.

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About the Author

Dr. Kineman is a research scientist combining physics and ecology to study the relation between living systems and the environment. He received his B.S. degree in Earth Physics in 1972 from the University of California Los Angeles and his Ph.D. in Ecosystem Science, Policy, and Ethics in 2007 from the University of Colorado, Boulder. He completed a 27 year career in public service as an officer and scientist with the National Oceanic and Atmospheric Administration, and Senior Research Warden with the Kenya Wildlife Service and US Peace Corps. In semi-retirement Dr. Kineman continues his research as a Senior Research Scientist with the Cooperative Institute for Research in Environmental Sciences (Ecosystem Sciences Division), Boulder; Honorary Adjunct Fellow of the Ashoka Trust for Research in Ecology and Environment, Bangalore; and visiting faculty at the Indian Institute of Sciences, Bangalore, and the Sri Sathya Sai University, Prashanthi Nilayam. His current research is in relational theory, ecological modeling and informatics, physics of life, and ancient philosophies of Greece and the Far East. His research led him to the work of Dr. Robert Rosen in 1997, shortly before Rosen’s death in 1998, after which he has been working closely with Dr. Rosen’s daughter, Judith Rosen, and through the Relational Science Special Integration Group of the International Society for the Systems Sciences, to further develop relational theory. Dr. Kineman lives in Boulder, Colorado, and spends part-time in India.

Autobiographical Reminiscences

By Robert Rosen (Copyright, Judith Rosen)

I have never enjoyed writing as an activity in itself, though over the course of time I have done a considerable amount of it. Already as a graduate student, I begrudged the time and effort it required; begrudged it because I already knew what I was merely now repeating and expositing, and because I felt the effort expended on mere repetition could more profitably be invested in trying to find out something new. I still feel that way.

I was early persuaded to act otherwise by my mentor Nicolas Rashevsky, then my Major Professor at the University of Chicago. He did not tell me that I was being “impractical” in such an attitude; that my scientific career and status would depend on a burgeoning publication list. He must have known that such arguments would cut little ice with me. He did not merely demand it, as he was in a position to do. Rather, he invoked the Categorical Imperative; he pointed out that if others had acted as I was proposing to act, then I could have no access to their accumulated knowledge and wisdom, and therefore could not learn from them. I had no answer to this, so I conceded, even while admiring his artistry in choosing that one particular argument to which I would have to acquiesce.

So I thereby acknowledged a duty to report. That is how I view my scientific writing; as reporting. It is not proselytizing; it is not advocacy; it is not even instruction. And it is in that light that I have prepared the present article, even though it is about me, and not so much about what I know. I hope the reader will appreciate this spirit from the outset.

Though the reporting of my scientific work, including the material to follow, is simply the discharge of a Kantian duty, I feel quite otherwise about the work itself. I have never regarded my attachment to science as constituting in any conventional sense a “career” or vocation or job. To me, it is an Imperative in itself, more akin to what theologians refer to as a “calling”; something which would be corrupted and defiled by being subordinated to any such personal considerations as constitute professional aggrandizement. Indeed, it has always seemed to me a kind of miracle that people were willing to pay me to do what I wanted to do, and would have done, anyway. On the rare occasions when, at the urging of

others, I have violated this Imperative for parochial “career” considerations, I have invariably come to grief. Whatever scientific powers I possess cannot be employed to such personal ends; like witchcraft, they can only be directed outward, and cannot be invoked on one’s own behalf. Therein lies their strength, and also, in another sense, their curse.

I must spend some time in explaining this Imperative, since it constitutes, as it were, the invisible steel skeleton which has guided and which supports the otherwise perhaps inexplicable diversity of my individual scientific activities. To me, on the other hand, these activities comprise a self-evident unity, each one forced on me by the preceding ones, and by that underlying skeleton, of which I am never unaware.

Einstein has reported how his scientific instincts were galvanized in early childhood by a compass needle. What the compass needle did for Einstein was accomplished for me by humble living things; beetles and crickets and caterpillars. Among my earliest memories are walks through wild and overgrown vacant lots which dotted the asphalt Brooklyn landscape into which I was born. Under ever rock was a new and thrilling universe of living things. From these experiences was born an eternal passion, a lust, to understand why these things, in their separate ways, were alive, while the rock was not. The rocks were themselves mildly interesting, but in a bland, impersonal way; it was the life which was the compelling challenge to me. If I could find out what the life was, I would know what the rocks were, but, as it even then seemed to me, not the other way around.

When I was five or six, I was taken to see the Disney film “Fantasia”. I remember being mesmerized by the panoply of life through the eons, which the Disney cartoonists set to Stravinsky’s “Rite of Spring”. This was worth spending a lifetime with. Though I did not even know the word at the time, had already determined to become a Biologist.

By that age, I had long since learned not to ask complicated questions of the adults around me, either family or teachers, because they did not know. Although I had no idea then where they came from, books seemed more authoritative, so I began reading anything I could find dealing with life and the living. Unconsciously, I was casting about for information, not only about this life which fascinated me, but on how one best went about understanding it; information on how to be the kind of Biologist I increasingly aspired to be.

As I read, and assimilated, and integrated, my perspective continually shifted. At first, I thought I would be what was then called a “naturalist”; continuing to study and observe organisms in themselves. But lots of other people had been doing that for a long time, and they did not know (or even care) what the life was; perhaps the answer was not there, however much fun such studies might be. Then I thought I would be a paleontologist, going back and back to the historical beginnings. That phase lasted somewhat longer, until I realized the answers might not lie in historical records either. By then I was reading about genetics and biochemistry, about metabolism and physiology and embryology and the intriguing possibility disclosed itself that, in the inner workings of what was alive would reside the best way to get at what made it alive.

Thus, I entered into a prolonged empirical phase, essentially a reductionist phase, dominated by biochemistry. Although it may surprise some people, I acquired a fairly extensive laboratory capability during those years. By this time I was in high school, a "Biology major" at Stuyvesant, taking elective courses in analytical and organic chemistry, and using the laboratory facilities for my own purposes when they were unoccupied. I became rather notorious for these activities, but became good enough to be utilized informally as a laboratory assistant at faculty demonstrations of techniques.

It was the attempt to understand what I was doing in these empirical activities (basically, to understand what a molecule was) that led me to instruct myself in physical chemistry, and then to the physics which underlay it, and then, fatefully, into the mathematics in which the physics was expressed. I somehow came quickly to the conclusion that, wherever the life was, the avenue for finding it was somewhere in there. That abruptly ended my empirical phase, and I decided that henceforth, I must become proficient enough in that mathematical language to understand, to the root, what realities were being, or indeed could be, expressed through it.

Up to that point, I had had only the most perfunctory interest in the sciences of the inanimate; these were the rocks again, and not the life. Suddenly, it now seemed a matter of urgent necessity to master these things. To facilitate acquiring such a mastery, it seemed the most natural thing in the world to change my major. So I blithely shifted out of biology and into mathematics. It felt perfectly right to do so, and I regarded it as the merest tactical device in the service of the unchanging strategy I was groping for.

I could not explain to anyone that I was not "abandoning" biology for mathematics. I well remember vainly trying to explain it to the Guidance Counselor, who regarded us as high-strung, unstable adolescents, and to whom any change in behavior patterns was an ominous portent of disaster. Somehow, I managed to convince her that there was nothing sinister in what I was doing, but henceforth I had the feeling of being watched closely. I did not like it.

Thus began a long period of total immersion, in both pure mathematics and in mathematical physics, which lasted almost unbroken until the end of my student days. I was accepted by these constituencies as one of them, but at the cost of not disclosing my ulterior motivations for being there. I felt much like the Englishman who visited Mecca during the Hadj; disguised as an Arab, and knowing he would be torn to pieces if his true identity were disclosed. Indeed, except for the required year of college biology, I took no more formal courses in the subject until almost done with graduate school, and then only to satisfy formal degree requirements. But to me, this posed no hardship; biology was "my" subject, which I could pick up again at any time, whenever my extended tactical detour was through. In any case, there was nothing in any of those college biology courses which I did not already know, many times over.

I quickly came to recognize that my instincts had been correct; that the mathematical universe had much of value to offer me, which could not be acquired in any other way. I saw that mathematical thought, though nominally garbed in

sylogistic dress, was really about patterns; you had to learn to see the patterns through the garb. That was what they called “mathematical maturity”. I learned that it was from such patterns that the insights and theorems really sprang, and I learned to focus on the former rather than the latter. More of this in a moment.

After a few years of such acclimation I came to focus my interest on the theory of operators and of operator algebras. This was beautiful in itself; but it was also the language of quantum mechanics, then the last and most exciting word in theoretical physics. The science of the rocks, and hence, it was impressively argued, of everything. I resolved to do my graduate work at the University of Chicago, because its Department of Mathematics was then the strongest in the country in this field; get my PhD there; and then would turn back to my Imperative, apparently in the form of getting the life to emerge from the rocks.

As it happened, I did not go to Chicago immediately after graduating from Brooklyn College, for familial reasons. While growing up, I had come to love New York and its infinite diversities, and was fond of boasting that there was nothing which could not be found in that city, if one only knew where and how to look. My parents, who were somehow terrified of my “abandoning” New York for Chicago, threw these words in my teeth; why go to Chicago when everything was already in New York? We came to a compromise: I would spend my first graduate year in New York; if I found it unsatisfactory, I could leave for Chicago unopposed.

I investigated several possibilities. One was the Courant Institute. They were horrified by even the suggestion of biology, and offered me instead a PhD program in fluid dynamics, which in their view, exhausted the universe. I settled rather on Columbia University, which at least on paper had a bit of a program in operator theory. In some ways, it was fortunate for me that I did so, as I will explain below. But in general, the year was one of intense academic frustration; I was learning little there, and I came to hate the sterile ambience of the place.

There was on the Columbia faculty one person who was described as a “biophysicist.” I went to see him, hoping to get first-hand advice, about what biophysicists did. In fact, this person was a muscle physiologist, who had a little laboratory in the attic on the 14th floor of the physics building, Pupin Hall. It was a true attic; dark, damp, and disorderly. I found the little cubbyhole which housed this person, knowing already I was on a fruitless errand. But I went through with it, trying to explain my intentions, however imperfectly. To this day, I remember his contemptuous retort: “We don’t do any of that theoretical stuff around here; we keep our feet on the ground.” It was all I could do to keep from laughing in his face, at the sudden vision of ourselves, fourteen stories up in a corner of an attic, keeping our feet on the ground. It is a picture which often comes to mind when dealing with experimentalists, even now. In any case, I stuck out the remainder of the academic year, picked up a perfunctory MA, and left for a new life in Chicago.

After the sterility of Columbia, and indeed of the four years of college which had preceded it, the University of Chicago was like an explosion of light. The sheer intellectual ferment of the place was like nothing else in my experience, filled with the excitement of new things to learn. But things were to turn even better for me, from a completely unforeseen quarter. I had long known of the existence

of a Committee on Mathematical Biology at the University of Chicago. I knew Rashevsky's book, "Mathematical Biophysics". And I knew that this was very far from the sorts of things I had in mind. In my view, all these activities were focused entirely on epiphenomena of life, and not on the life itself. Blood flow in arteries? Propagation of action potentials? This is not the stuff of life; this was back to the rocks again. Indeed, such concerns seemed diametrically opposite to my own; knowing about them only strengthened my resolve to persist in my own strategy and begin afresh.

Nevertheless, soon after I arrived in Chicago, and almost by accident, I obtained an appointment to see Rashevsky himself. I expected it would be like my encounter with the Columbia "bio-physicist". But it was not. Rashevsky offered me something I had never received or solicited or expected from any external quarter: encouragement. He informed me that his own views had changed radically over the past few years, and had led him to a new approach which he had christened Relational Biology. He gave me his keystone paper, then only two years old and entitled, provocatively, "Topology and Life", to read. This turned out to be the only thing I had ever come across that was in my ballpark; consonant with my own Imperative. After a few more discussions, Rashevsky offered me a small fellowship, an office of my own, and absolute *carte blanche* in preparation of a dissertation, in return for taking my PhD in his Committee.

After a day or two of reconsideration of my alternatives, I accepted this offer. My feeling was that I had already accomplished my purpose in studying Mathematics; I regarded myself as fully independent and fluent in that language. To persist in Mathematics, in the face of Rashevsky's offer, would gain me little and would in fact slow me down. So I transferred out of the Department of Mathematics, and into the Committee on Mathematical Biology. Once again, everyone thought I must have lost my mind, and once again, I could not explain. But by my lights, it was the only correct thing to do. I received my PhD in Mathematical Biology two years later.

Upon entering the Committee, in the fall of 1957, I at last felt fully free to unleash myself in the pursuit of my Imperative, too inexperienced to be daunted by what I was proposing to undertake. I felt, what I still feel, that I had at least an even chance of success; that my inherent intellectual equipment and the scientific capabilities I had accumulated gave me perspectives which no one else had, and that if I failed, it would only be my own fault. In short, I was already the Biologist I had aspired to become, and it was now time to put those arts to the test.

The next two years consisted of an absolute orgy, a frenzy, of activity. I simultaneously embarked along at least a half-dozen fronts. Much of this work was only published years later, if at all. Early in 1957, I had discovered the (M,R)-systems, and developed some of their extraordinary properties; this work, published in 1958 and 1959, became my dissertation. I began to explore the quantum-mechanical dictum that material events consisted of observables being evaluated on states, as the tangible bridge between the rocks and the life. I became aware of the strange epistemological status of Church's Thesis, and began to explore its actual implications. I did some (abortive) work on algebraic aspects of biological coding schemes, and decided on morphogenesis, a uniquely biological phenomenon, as

my testing ground for general theoretical ideas in biology. Most of my subsequent scientific work has been based, in large part, on the foundations I established in those two years.

I felt then, and continue to feel, that none of this work was in any way “speculative”. Indeed, I believe that theory is the antithesis of “speculation”, despite the confusion between the two in the minds of those who do speculate. Nor have I ever believed that theory and “practice” were in any way adversarial. What I do believe is that “practice,” in the form of observation and experiment, cannot constitute or replace theory, and that most of the basic questions of science, especially in Biology, fall quite outside the ken of “practice”, in the usual sense. My own life would have been made much simpler if empirics alone would suffice for my Imperative.

It might be well to spend a moment on the general scientific ambience of those years, since they were exciting in a way which can barely be dreamed of today. On the “theoretical” side there was Schrodinger’s little book, “What is Life?”, in which, however Schrodinger did little but repeat the words and outlook of his student Max Delbruck. From my viewpoint, Schrodinger did not begin to answer his question; he rather equated “life” with a kind of stability, and asserted that “life” must be molecular because molecules are stable too. In the late 1940s appeared Norbert Weiner’s book “Cybernetics”, invoking a new technological language the Cartesian equation between animal and machine. There was the confluence then crystallizing between foundational work in mathematics itself (exemplified primarily in the Turing machine and the execution of algorithms) and digital computation, and the brain, embodied in the neural networks proposed decades earlier by Rashevsky himself; all this roughly constituted the province of “Automata Theory”. There was the Theory of Information of Shannon. There was Game Theory. And in Biology itself, there was the increasing inroad of digital thought, of hardware and software, which were the concomitants of “molecular biology”. And of course, part and parcel of all this, was the newly emerging strain of General Systems Theory, associated especially with names like Bertalanffy and Ashby. A yeasty mix indeed.

To me, though, and in the light of my own Imperative, all these things were potential colors for my palette, but not the palette itself. I regarded them as monochromes, individually perhaps lovely in themselves, but not to be applied when a different hue was required. I could not share the prevailing sentiment that these developments, either individually or collectively, would paint themselves into the picture I was striving after. Rather, I felt it was the picture which would illuminate them.

Indeed, my own scientific work of those years was pushing me against these currents. Consider, for example, the discovery which most shocked me in those days, when I still had unlimited faith in the physicists’ Quantum Mechanics as the ultimate bridge between the rocks and the life. I had long been puzzled by the fact that the state spaces they posited for every material system were mathematically indistinguishable; abstractly identical; isomorphic (they are all separable Hilbert spaces, and there is objectively only one such). The perceptible differences between material systems must thus lie only in a “choice of co-ordinates”, and in how

the observables, the Hermitian operators on states, were labelled; hence in what, mathematically, constituted the subjective. This in turn implied that we could get from one system to any other by relabelling these observables; by calling one of them, say, a Hamiltonian instead of another. Hence that any system would appear to be any other system, if only we looked at them with the “right eyes”. The only escape from this disturbing conclusion seemed to be to limit the universality of Quantum Mechanics itself. . . or what is the same thing, enlarge what can constitute an observable or an observation, or a state.

I was unprepared to do this for a long time. But I was forced to it by the following considerations, which I discovered in 1959. As I have already noted, whatever else Quantum Mechanics say, it asserts that “information” about any material phenomenon consists of observables evaluated on states. Hence, *a fortiori*, “genetic information” must be of that character too, and this must provide the material, physical basis of the formal “coding schemes” which then so preoccupied everyone. So I tried to find what the observables had to be in order to manifest this “information”. The shock was in discovering that the families of observables I characterized in that way could not contain anything which behaved like a Hamiltonian. And, of course, without a Hamiltonian, you cannot even get started in doing traditional Quantum Mechanics. In a sense, what I then showed was that Quantum Theory and Quantum Mechanics do not coincide, and that the former was much bigger than the latter.

At the root of these considerations is the indissoluble dependence of Quantum Mechanics upon energy conservation; that is what a Hamiltonian expresses. What happens in rocks seems to fall within such structures; what happens in life, as I showed then, and more sharply later, need not. There was an immediate parallel with the “open systems” of Bertalanffy and their devastating challenge to the Second Law of Thermodynamics; it was not that the Law was wrong; it simply did not apply. I would say that, today, there is still no satisfactory “physics” of open systems, primarily because people persist in thinking of closed systems as fundamental, and of open ones as simply closed ones canonically perturbed.

At any rate, such considerations provided the soil for a constant preoccupation with when, and under what circumstances, two systems could be considered in any sense identical; such studies ran a gamut from the physics of the Gibbs paradox, and the objectivity of entropy, to considerations of similitude and conjugacy.

Such considerations, and many others like them, from many different perspectives, led me away from the facile Reductionisms which almost all of my colleagues were rushing to embrace, and which they identified with science itself. From my perspectives, physics could not swallow Biology; rather, any attempt to do so would have to radically transform physics.

Fortunately, I had a positive alternative to such negative, pessimistic conclusions, in the spirit of Rashevsky’s Relational Biology, and manifested in my own (M,R)-systems. As I have characterized this spirit, it involves “throwing away the physics and keeping the organization,” instead of the reverse. What remains then is an abstract pattern of functional organization, which has properties of its own, independent of any particular way it might be materially *realized*. Indeed, it is

what remains invariant in the class of all such material realizations, and hence characterizes that class. It is my ultimate object of study; it, and not those material objects which happened to be available to realize it.

To me, such patterns, and the elements and relations which comprise them, are as real and objective and perceptible as the products of any Reductionistic fragmentation; indeed, in some ways more so. In my view, a science too narrowly construed to encompass them from the outset is too narrow to do Biology in, just as narrow identification of mathematics with computability excludes thereby almost all of mathematics. More of this later.

The study of these (M,R)-systems brought my mathematical training and instincts to uses I could not have foreseen. For one thing, the diagrams which expressed them were in themselves an immediate invocation of the Theory of Categories. I had started to imbibe this theory during my otherwise wasted year at Columbia University. The graduate algebra course I took during that year was taught by Samuel Eilenberg, and was really a course in Category Theory; sets, operations, and structure-preserving transformations. Eilenberg, of course, was one of the creators of Category Theory. The other creator, Saunders MacLane, was at Chicago, where I imbibed much more. I became intrigued by the historical roots of the Theory, which had grown out of the attempt to make algebraic “models” of geometric objects in order to discriminate between them. It expressed in a purely mathematical realm the patterns of relations, between objects and models, and between one model and another, which I was trying to find in the realm of the living. The numbers (e.g. Betti numbers) which came out of Algebraic Topology were like the observables of material nature, but there was much more underneath them. It has been one of my primary ongoing concerns to make all this clear.

My adaptation of Category Theory to the (M,R)-systems, and indeed my utilization of Category Theory itself as a kind of framework for the notion of modeling in general, is typical of how I have used my mathematical tools over the years. Not so much in the making of particular kinds of models of particular biological phenomena (although I have done a substantial amount of that), or the invocation of specific theorems from specific mathematical domains (although I have done that too) but rather an invocation of the entailment patterns from which the theorems arise, or sometimes do not arise. So I seldom have occasion to invoke a particular theorem from Algebraic Topology (say); what is more germane to me is the relation established between a space and its models, and between one model and another; and why such relations hold.

Indeed, I have come to regard models in general as a natural but profound extension of the concept of observability, as the physicist understands it. A model indeed represents to me an inherent adjective, or property, or quality, or attribute, of the system being modeled; what the old philosophers called an essence, no less than any measured value of some magnitude does. But rather than trying to reduce every model to such measured values, or alternatively, trying to syntactically build every model out of such numerical observables, I have had to proceed in quite a different way. Indeed, it has turned out that most qualities of interest to me, were simply not expressible in such limited terms. One must follow one’s “observables”

to assume values other than mere numbers; to assume values in inferential patterns (in models, in short), and at the same time allow the referents of such observables to be other than conventional reductionistic fragments. Once again, none of this seems to me in any way “speculative”; it is as firmly grounded in observation as any reductionistic scheme. But it involves a notion of “observation” far more broadly conceived than has been usual, and tailored to the demands of Biology; traditional concepts of observability, and the kinds of models which can be based on them, appear in this light as very, very “special” indeed.

Thus, I have come to partition Biology into that which depends on an underlying relational pattern (e.g. an (M,R)-system independent of how it is realized; and that which depends upon the material details of a particular realization (and of course, that which depends on both). And of course, the word “realization” admits a great deal of latitude. For instance, I have come to believe that social structures, as things in themselves, realize many of the relational patterns which individual organisms to. To that extent, we can learn deep things about each by treating the one as a surrogate for the other, however different they may appear in exclusively material terms. It was in exactly that spirit that I undertook, for example, a long-term study of “anticipatory systems”, which is still going on.

My concerns with “anticipation”, in which what is happening now seems determined by something about the future, are worth describing in more detail. Anticipatory behavior is in fact damned as “acausal”, because causality is construed precisely as allowing only the past to affect the present. I initially softened this by interposing a “predictive model” as a transducer between now and later. But nevertheless, the presumed telic or finalistic aspects of anticipation seemed to violate the one-way causal flow on which “objective science” itself is presumed to rest. And I noticed that my own (M,R)-systems have an inherent anticipatory aspect, built into their very organization.

Once again, my mathematical experience served to illuminate this situation, mathematics, the quintessence of what is objective. In mathematics, the analog of anticipation is *impredicativity*; a situation in which what is defined depends essentially on having it available from the outsel. The associated “self-references”, in which something is getting outside a single, one-way, coherent time-frame, can lead (and have led) to devastating paradoxes. Russel called them “vicious circles”, and it was believed that the salvation of mathematics itself depended on eliminating them; somehow straightening out the impredicative loops, and proceeding in a purely syntactic way only from “past” through “present” to “future”. Indeed, it was part of the allure of the algorithm, embodied in the machine, that it could only manifest this kind of flow, from input to output, and impredicativities, by their very nature, could not arise in them.

The trouble with this is that by thus “saving” mathematics from impredicativity by indentifying it with what machines can do (i.e. with pure syntax, or symbol manipulation, or word processing) the cost is relinquishing most of mathematics itself. In a certain suggestive language, there are more things in the “mathematical universe” than can be projected down predicatively into a single coherent time-

frame. This is a very Platonic thing to say, but it is still true. And I believe Biology shows that it is likewise true in the causal realms of material reality as well.

My (M,R)-systems inherently manifest such an impredicative loop; one which cannot be straightened out without losing everything. They are thus not approachable via “machines” in the usual sense; they are not purely syntactic objects. They are what I call *complex*; they must have non-computable models. I would argue that, precisely by excluding temporally closed causal loops, and indeed by indentifying this exclusion with science itself, we have lost not only life, in my sense, but most of its material basis, its physics, as well. To invoke a parallel mentioned earlier: just as the “closed system” is too impoverished, too special, to be a basis for (say) the physics of morphogenesis, exactly so is the simple system, one which can be described entirely as software to a machine, too impoverished to accommodate the living. In fact, these two situations are closely related, but it would take too long to explain that relation here.

Now, let me turn to some other matters which merit reporting. As Rashevsky pointed out to me all those years ago, I am not in the game alone. If I have made myself the scientist, the biologist, I originally aspired to be, I cannot take the entire credit, though I must entirely assume any blame. I have received much assistance and support from the communities to which I have necessarily belonged, including that of some very great minds. This in turn leads me to talk a bit about the scientific community itself, such as it is, and about the institutions which are supposed to house and support them.

I have generally regarded the University as my natural habitat, and my interests and capabilities of sufficient breadth so that I could fit in smoothly, and to mutual advantage, almost anywhere. All this seems to be becoming less and less true as time goes on. Nevertheless, I have had the benefit of participating in at least three extraordinary communities, organized around extraordinary personalities. The first of them, of course, was the Committee on Mathematical Biology at the University of Chicago, created and maintained by Rashevsky. All told, I spent about a decade in this community, first as a student, then as a Research Associate, then as Assistant Professor. For me, the Committee stopped existing when Rashevsky was driven out of it. For a long time now, it has not existed in any form at all.

I was fortunate to find for a while another congenial habitat, in the complex of activities which nucleated around the Center for Theoretical Biology at the State University of New York at Buffalo. The personality here was that of James F. Danielli. That lasted another decade, until Danielli was driven out, and the Center abolished. At Buffalo, I also had the opportunity to create and administer a graduate program in what was called “Biomathematics,” although I have always disliked that word (much as I have also disliked being nicknamed “Bob”, incidentally, which nearly everyone who has ever known me uses, and which my parents and older relatives lengthened to an even more ignominious version; “Bobby”... In my opinion, “Robert” has much more poetry to it. However, I suppose it is, alas, rather too late to make such stipulations.). I look back with some pride in this program, since it was the best of its kind in North America; best because it was the most

cohesive, the most comprehensive, and at the same time the most individualized. So I will take a moment to describe it.

The program was open to anyone, in any of the dozen or so participating Departments, who wanted to work in the area; anything from population dynamics to biochemical control. There was a core curriculum, which everyone was expected to take, and which consisted of those concepts I felt basic to any specialization. That curriculum consisted of five courses, of which I taught four myself, organized around the concept of *stability*. The basic course was about dynamical systems; mostly what was then called qualitative theory of systems of first-order differential equations. The second course, built specifically on the first, dealt with (mostly linear) input-output analysis, control theory, and optimal controls. The third was about discrete-time systems, in those days primarily automata theory, regarded as a paraphrase of continuous-time dynamics to discrete situations. The fourth was concerned with spatially extended systems, described by partial differential equations. The fifth, which was in fact never taught because I find the subject uncongenial, was supposed to deal with stochastics. My expositions were built around many examples, as many as possible taken from biological situations, and the emphasis was on making the underlying unifying patterns as conspicuous as I could.

When a student enrolled in the program, I would organize an individual curriculum most consonant with his or her interests. If the student had no interests, I would put him on a reading program of broad scope, until one emerged. Then, and only then, an appropriate curriculum would, so to speak, organize itself around that interest.

I began this program around 1967. At that time, there were almost no coherent text materials I could rely on. So I conceived the idea of turning my course notes into text-books, a digression which I viewed as innocent public service. The notes for the first course were published by Wiley in 1970, under the title "Dynamical System Theory in Biology". The ideas and viewpoints expressed therein have become utterly commonplace today but it was then met with such virulent hostility, especially on the biological side, that I cancelled my plans for the remaining volumes, and vowed never to waste my time on exposition again. What expository work I have done since then has been confined to editing (e.g. a three-volume series, "Foundations of Mathematical Biology" for Academic Press, and the biannual series "Progress in Theoretical Biology", of which seven volumes ultimately appeared.) One of the main thrusts of the latter series was to acquaint English-speaking scientists with the work being done in Eastern Europe and the Far East.

Indeed, the Center itself was always the core of an extensive publication program of its own. The offices of the "Journal of Theoretical Biology" were located there since Danielli had founded it in 1962, and served as its Editor in Chief until his death over a decade later. I remained connected through this time with the "Bulletin on Mathematical Biology", which Rashevsky had founded, and then later, when I got to know Richard Bellman, with his "Mathematical Biosciences". I was also heavily involved with Danielli's authoritative "International Review of Cytology" in those Buffalo years. For various reasons, I have dissociated myself from many of

these editorial activities, as these publications, and the policies they now implement, have become less and less congenial to me.

But for a while, Buffalo was paradise. It was at the Center, for instance, that I came to know people like von Bertalanffy, and many others who came through for longer or shorter periods of residence. The Center was destroyed, however, in 1975, in a brutal upwelling of resentments, jealousies, and low parochial politics. But I am sure we all have our academic horror stories to tell. Nevertheless, I continue to regard what happened then as a tragedy for both the field and for innovative university research in general, and it certainly bespoke a catastrophe for SUNYAB, from which it has never recovered.

At the moment, and indeed for at least a decade past, there has been no coherent, broadly-based graduate program in this area anywhere in North America. The field seems to prosper, not because of new cohorts of students trained in the area, but by the accretion and immigration of people trained in other areas. Many of these people are technically very adept, but it seems to me that they are producing little in the way of new ideas; what appears in the journals now is primarily the reworking of old ideas, often dating back thirty years or more. Reading them is a dreary exercise, and that is one of the main reasons I have disaffiliated myself, both from the journals which publish them, and from the organizations these journals represent. How can I endorse, for instance, the present editorial policy of the "Journal of Theoretical Biology", when its current co-editor publicly derides the whole endeavor as "trivial", and at best an exercise in combinatorics?

I left Buffalo in 1975, with the closing of the Center, and took up an appointment as "Killam Professor" (so named because it was funded in memory of a wealthy benefactor; Isaac Walton Killam) at Dalhousie University in Halifax, Nova Scotia. This was like a five year Sabbatical, which released me for that duration from the strictures of academic politics, and left me free during that time to continue pursuit of my Imperative in good company. I shall always be grateful to Dalhousie for the haven it provided, even though circumstances are much different today from what they were then.

In general, I believe that under presently prevailing circumstances, the best thing I can do, for myself, and for my field, is to pursue my Imperative in my own way, and continue to report. I feel I am [as this was being written, in the early 1990s] much closer to my ultimate goals than I have ever been, and that I can only get stronger as I advance. As I said at the outset, I am not by nature a proselytizer; but my reports are out there, for others to make of it what they will.

If, as I believe, my scientific work comprises a single unity, then that unity reflects the mandates of the underlying unified problem with which I have been concerned. I have tried to listen only to what that problem tells me, and to follow its exigencies. That is the key to how I perceive science itself, and why I have never allowed anyone to tell me how science in general, and Biology in particular, "ought" to be done. Only the problem itself can do that.

If nothing else, I hope to have shown that mathematics and life are not opposites. Most Biologists, I dare say, believe that where mathematics is, there life cannot be, and vice versa. Most (pure) mathematicians, for quite different reasons, feel the

same way. But I rather believe that the corpus of mathematics is the only other thing which shares the organic qualities of life, and provides the only hope for articulating these qualities in a coherent way. But that way is quite different from what has hitherto seemed to suffice for the rocks in this world.

Quite early in my professional life, a colleague said to me in exasperation, "The trouble with you, Rosen, is that you keep trying to answer questions nobody wants to ask." This is doubtless true. But I have no option in this; and in any event, the questions themselves are real, and will not go away by virtue of not being addressed. This attitude, I know, has estranged me from many of my colleagues in the scientific enterprise, and has put me far from today's "main stream". But sooner or later, if I am at all correct, that "stream" will flow my way. In the meantime, I must continue to do what the problem demands of me; as I see it now, it consists of finding a (relational) model, an essence, *all* of whose material realizations must be counted as alive. I think I have indeed found at least one such model; the trick now is to find the objective grounds by which such an assertion can be demonstrated.

As I suspect you have ascertained by now, my relations to General System Theory follow no direct, straight-line trail. There were many sources which fed into it, prepared in many cases by my own independent work, before I had ever heard of General System theory *per se*.

For instance, I had early been much taken with the "Mechano-Optical Analogy" of William Rowan Hamilton, which seemed to me so different in character from anything else in theoretical physics. Hamilton (whom I consider one of the most original minds of the 19th century; perhaps only Poincaré and a few others are even comparable to him) did not try to "reduce" optics to mechanics, nor *vice versa*, as Maxwell fruitlessly tried to do later, but rather related them through mathematically homologous action principles. This was an incredibly fertile thing to do; among other things, it led Schrodinger to his Wave Mechanics (which Hamilton himself had all but derived), and, in a completely different direction, to all modern approaches to optimal control. I have found many occasions to invoke it myself, in many contexts. Thus, when (much later) I heard General System Theory characterized as comprising anything bearing directly on independent disciplines, I thought of it as an attempt to do on a broad scale what Hamilton had done in Physics; as a way of relating apparently diverse kinds of systems in a way different from simply trying to reduce them both to a set of common parts. I was also independently familiar with Bertalanffy's development of the "open system" metaphor, which I always viewed as similar in spirit to Hamilton's. That is, diverse systems behaved as they did simply because they were open; not because of irrelevant structural details. In fact, it was largely this metaphor which led me to think of stability as a basic organizing concept, and ultimately to my text on Dynamical Systems mentioned above.

When I came to Chicago, I learned, of course, of Anatol Rapoport's work on random neural networks, done during the decade when Rapoport was a member of the Committee on Mathematical Biology. This too turned out to have application to many diverse subject areas; originally developed to show how specific architectural features could be robustly generated through simple statistical biases (thus freeing

these features from the burdens of specifying precise wirings), it gave insight into e.g. the nature of epidemics, the spread of rumors, and many other things.

So it was that my own independent work and study were leading me precisely in the direction marked out by General System Theory. I became aware of the Society for General System Research, however, only around 1962, when I was asked to have some of my early papers reprinted in one of the SGSR Yearbooks (I think it was the third).

I looked at some of the other papers in these Yearbooks, and must confess I found them disappointing. They tended to start from a premise that "General System Theory" was about something they called a "General System" and spent a great deal of effort trying to characterize what that was. It was not an activity I found particularly germane. Indeed, I regarded the field as General (System Theory), not as (General System) Theory.

In addition, I was by then beginning to travel to meetings and conferences, at many of which people who called themselves "system theorists" were in attendance. By and large, I did not find much common ground with these people, or with what they were doing. That distanced me to some extent from the field itself, though I continued to keep an eye on it.

The situation changed considerably when I met Ross Ashby in 1967, at a six-week Workshop on Theoretical Biology in Fort Collins, Colorado. On those picturesque surroundings, we had many provocative discussions, and discovered many commonalities of interests and inclinations.

The situation changed still further when, in 1968, Ludwig von Bertalanffy moved to the State University of New York at Buffalo, and was providentially quartered in the Center for Theoretical Biology (though his appointment was in one of the Social Science departments). I vividly remember meeting him for the first time. When I introduced myself, he impulsively embraced me, like a long-lost brother. A rich and, I think, mutually rewarding symbiotic relation developed between Bertalanffy and myself, and with the CTB at large; it provided a natural home for him, as it had for me. However, he arrived when things were going very sour at SUNY-Buffalo, as noted above, and I have no doubt that the anxieties and uncertainties generated by repudiation of firm agreements by his department played a major role in precipitating the heart attacks which killed him.

I got to understand Bertalanffy's view of System Theory, not only as a scientific, but also as a social instrument. Until then, I had always found the term "General Systems *Movement*" uncomfortable; but that is exactly how von Bertalanffy perceived it. Whereas I viewed the reductionisms and materialisms rampant in biology merely as scientifically inadequate, von Bertalanffy saw them as evil and dehumanizing; in the deepest sense immoral. Animated by his profound love of both science and of humanity, he was inspired to project his view of Systems, governed by ways of relating things rather than stressing differences, into an alternate world view; a paradigm, as he called it, which would offer both science and mankind something better. He viewed his vision, then, not merely as something to be reported, but as a Gospel to be preached.

Von Bertalanffy radiated a simple goodness, a largeness of mind, and a dignity notably absent in those who attacked him so violently, such as molecular biologist Jacques Monod. Monod was typical of the excessively positivist, algorithmic, brute-force people who naturally cluster around the idea that reductionism (or as Monod preferred, “analysis”) is all there is to science. Their political counterparts were then called “action intellectuals” who, in practicing their self-styled pragmatism and *realpolitik*, only succeeded in committing blunder upon blunder.

Let me tell one story about Monod, who liked to say of himself “Je cherche ‘a comprendre’”. I met him only once, in 1964, when I attended the International Biophysics Congress in Paris. Monod gave one of the big plenary lectures, and it was about operons. The operon was a functional genetic unit, proposed by Monod and his co-worker Francois Jacob in 1959. These authors had proposed that networks of such operons could account for differential gene expression (i.e. for differentiation) in higher organisms, and had illustrated their thesis with a few simple networks which manifested these behaviors. I was interested in these operons from the outset, for two reasons: one, they were *functional* units, not structural ones; you could not isolate an operon per se and put it in a test tube. Thus, it seemed to me that the molecular biologists were leading themselves into a realm they claimed not to exist; a realm which transcended “analysis”. Second, because the operon itself is basically a switch, just like a neuron; an operon network is thus very like a neural network. But instead of axons or any other material channels for signals, operon networks relied on invisible channels governed by specificities. Moreover, I had shown that the simple “operon networks” proposed by Jacob and Monod to explain differentiation were *identical* with the two-factor nets Rashevsky had published decades earlier, to illustrate how “brainlike” behaviors such as discrimination, learning and memory could arise in networks of neuronlike elements. At any rate, in his talk, Monod stressed precisely these networks, and lamented openly that there was as yet no “theory” of them. This encouraged me to approach him after his talk, to suggest the above to him. He listened with obvious irritation for a minute or two, then cut me off with the statement “I am not an embryologist!”, turned on his heel and walked away. I was amazed by this; he really *did not* want to know. This is why, in my eyes, Monod and his ilk are little, and why Bertalanffy was great.

Experiences like that outlined above with Monod, repeated a hundredfold, convinced me that it is useless to preach to those who *will not* hear, whatever one’s Gospel, and equally useless to preach to those who already believe. Besides, my nature is not that of a preacher or advocate. I am, and remain, a practitioner of von Bertalanffy’s paradigm, and preach it only implicitly, through that practice. And I practice it because my problem tells me that I must.

There was one exception; I *did* proselytize once. It was around 1974, when George Klir and his dean Walter Lowen, came up to Buffalo from Binghamton to talk to us at the CTB. At that time, CTB itself was undergoing its terminal demolition, leaving many first-level people without faculty lines. George, by contrast, then had faculty lines without people. So the obvious arrangements were made, despite malevolent attempts by the highest levels of the Buffalo administration to prevent it. Since then, George Klir and I have had many fruitful interactions. In

1981, I believe it was, he persuaded me to assume the presidency of the SGSR for one year, assuring me it was only a ceremonial gesture. However, it turned out there was one small string attached – namely to organize the Annual Meeting. My proselytizing on that occasion was aimed at the system theorists; my message, that they be aware of cognate developments in the sciences. I invited only congenial scientists to speak, and I think it went well. But that has been the extent of my overt attempts at advocacy.

My recent book, “Life, Itself”, published by Columbia University Press and released in August 1991, could have been subtitled, “Why I am not a Mechanist.” I knew that Francis Crick had published a book with my title about a decade earlier, taking exactly a mechanist stance, but I saw no need to cede the title, nor indeed anything else, to Crick. In fact, I had decided to someday write a book with this title when I was still in my teens, after reading a strange little story by Poe called “The Oval Portrait.”

The theme of my book, “Life, Itself”, is that Mechanism and Vitalism pose a false dichotomy. Roughly, I argue that the external, public, material world is full of closed causal loops, just as the internal, mathematical world is full of closed inferential ones (impredicativities). The “world” of the mechanism, or machine (or, as I call it, the simple systems), and which I believe is an artificial human limitation on reality, does not allow such loops. Accordingly, as a class, these simple systems are extremely poor, or limited, in entailment and hence extremely nongeneric. I pose this in a number of different languages, each bearing on a different part of system theory. In particular, I pose it in a causal language, and show that a closed loop of entailment permits a perfectly rigorous notion of *final cause*.

I call a system which is not simple “*complex*”. Complex systems cannot be exhausted by any finite number of simple (mechanical) models; they cannot be described as software to a “machine”. Life itself is tied up irretrievably with this notion of complexity, which differs from conventional uses of this word, but I could think of no other.

Complex systems constitute, to me, a perfectly objective and rigorous universe, in which there are “enough” entailments for life, anticipation, and many other things to exist. In the simple, mechanistic one, by contrast, they cannot exist; their basis has been eliminated at the outset. Clearly I cannot distill three hundred pages to a few paragraphs; indeed the three hundred pages are highly distilled already. In a sense, the book is as much about System Theory as it is about Biology; the two are so closely entwined that I cannot, and would not, separate them. It is no accident that the initiative for System Theory itself came mostly from Biology; of its founders, only Kenneth Boulding came from another realm, and he told me he was widely accused of “selling out” to biologists. I know that Ludwig von Bertalanffy would be pleased by the effort; I hope he would be pleased with the result of that effort.

Epilogue

By Judith Rosen

After my father finished writing this paper, he gave it to me to read, saying, “Here, Jude. Give me your honest opinion, kid. I cannot be objective. I do not really feel comfortable working on this kind of thing. It seems almost a conceit writing anything autobiographical.” I told him I did not think so and, after I had read it, said that it may someday help someone a great deal because everyone has to start somewhere; finding dead-ends or having to turn around and change direction are not failures. But a lot of people just coming up and feeling their way like he did over those early years might mistakenly feel that they are not capable because of similar obstacles. The fact that my father details some of his own dead ends and detours in this paper makes it clear, for posterity, that it is something that happens – even to someone as focused and determined as he was all his life. Therefore, my strong opinion is that his professional life’s experiences detailed in this paper give anyone reading this the reassurance that life is just like that for all of us.

This paper also shows that his strategy, which I think is highly original and extremely effective, was to not only scrutinize each dictum that was offered as a given in any of these disciplines, but look all the way back at what the original creator of the dictum was trying to accomplish and then follow the logic (or “illogic”) of the origins of it. He did this with many of the accepted traditions in science. What he discovered by doing so is that a large number of the seemingly ironclad tenets, or rules, of science were merely habits based on flawed premises. That was, in my mind, one of his greatest talents and one of the more unusual aspects to his perspective on the universe.

What his paper does not talk about or illustrate is that his life was so much more well-rounded than consisting of just his work. Even though he would say that his “Imperative” was the core of who he was, the truth is that his curiosity and his unusual ability to see the big picture AND the details all at the same time were aspects of him that were applicable to every other aspect of life. His astonishing “sticky-fly-paper-memory” (as he called it) was so much fun to explore; he could retrieve facts from anywhere inside his mind, in an instant. If you asked what was the gestation of an elephant, he knew. If you asked when was JS Bach born, he knew. If you asked any obscure homework question, he knew, and what is even more amazing is that he not only knew the detail you were looking for but all the background and the context, including dates, places, quotes, connections, and consequences... and he could just pull it out of his memory at will. I have to rummage, at the best of times, saying, “I KNOW the word I need is in here SOMEWHERE!”... But he could recite pages of a Shakespeare play he had to read for high school and had not looked at since. He was a killer at Trivial Pursuit (we had to change the rules for him because otherwise the game would be too short). He was a blast to travel with, because he knew at least a half-dozen languages and was so seasoned a traveller that he was able to handle any and all bizarre travel-related situations, without getting the least bit flustered. He loved to “play”. Whether playing involved literal things

like piano or organ (Bach fugues were his favorite) or figurative things like hiking up a mountain, going to a fine restaurant, watching an old Pink Panther movie on tv, or a million other things, he was enthusiastic and great company. As far as his ability to be a friend goes, well, speaking from personal experience, I think that was perhaps his greatest talent of all.

The Devil's Advocate

A short story about Robert Rosen and his work, by Judith Rosen

NOTE: All scientific ideas described in the following fictional setting are non-fictional. This may be an unusual way to translate my father's work, but what can I say? I am, among other things, a fiction writer. This is more fun for me; I hope it will be more fun for the reader, too. I am confident I can accurately predict what my father would think of it: He would laugh! He might also be astonished. And, I suspect, he would be *very* pleased.

My intentions in writing this piece were twofold: (1) To present the work in relatively non-technical terms, which I frequently heard my father do when discussing the ideas with people outside his field (as, in fact, he did with me in the beginning). I feel it is important, almost a Kantian duty in fact, to pass on to other people the same benefits I had in comprehending the work. And; (2) To give readers some sense of having "met" the man – the fully articulated human being. There was a much larger personality which animated the scientific mind visible in Robert Rosen's published work. Now that he's gone, it is no longer possible for anyone to get to know my father as a human being, directly. The fraction of his mind represented in the scientific work, as important as the work is, does not do him justice as someone with enormous range of interests and capacities, as well as a playful sense of humor and deep, restless creativity. I was once asked how I would characterize my father, if I could only use one word. That turned out to be an easy task for me: He was "*FUN*"! I hope I have managed here (using a few more words than that) to introduce you to the living man – as I knew him – in an entertaining vehicle like this rather mischievous short story. Fiction has the power, I think, to convey many different types of information simultaneously and involve much more of the mind in integrating all of that information. I leave it up to the reader to decide whether I have succeeded or not.

The Devil's Advocate

Judith Rosen

“Uh – Excuse me, but... is this... *Hell?*”

“Mmm Hmm,” the big, ugly guy with the horns said, leaning against an odd looking kiosk with his muscular arms folded across his chest; the very picture of other-worldly insouciance. “I’m afraid so. And I’m the one in charge of it. Most people refer to me as ‘The Devil’.”

“Oh! How INTERESTING! I have to admit I really didn’t think this place *existed!*”

The Devil raised an eyebrow. “That’s an unusual reaction,” he said. “Most souls start crying around this point. Tell me; What gave you the first clue that you have arrived in Hell?”

Robert shrugged, concentrating on the question. “Well, aside from your appearance, which is sort of a *dead* giveaway...” He was pleased that the Devil seemed to appreciate the humor of a pun. “I think it was the general volcanic ferment of the landscape.”

“Volcanic?”

“Yes. The pervasive smell of various sulfurous compounds, the clouds of steam, the dark, yellowish cast of the place, not to mention the occasional bursts of extreme heat – It kind of reminds me of a few parts of Yellowstone National Park, actually.”

The Devil was looking at him rather strangely and leaned over to pick up a clipboard, consulting it. “What’s your name?”

“Robert Rosen.”

“Ahhhh!” he said, putting the clipboard down again. “That explains it! So, *you’re* Rosen... Nice to finally meet you.” He extended his hand with a smile – then seemed a bit startled when Robert did not hesitate to reach out and grasp it in a firm handshake.

Robert tried not to be distracted by curiosity over the shape of the Devil’s hands or his fingernails... which looked more like claws. Or by the fact that the Devil’s hand was warm! “Uh, you’ve heard of me?!” he said, reluctantly letting go.

The Devil nodded, a twinkle in his yellow eyes. “Oh yes. Lately we’ve had quite a run of angry souls from Academia around here, complaining at length – about YOU. “*Damn that ROSEN,*” they say, along with something about you always answering

the questions they didn't want to ask! You won't be able to avoid them. There are entirely too many!"

Robert could not help it; he actually laughed out loud, surprising the Devil.

"You don't seem very concerned to be stuck here in Hell with a lot of angry souls who really don't like you."

Robert shrugged. "Yeah, well, somehow it's... really not so different from my *life*, that way," he said. "Certainly sounds just like my experience in Academia, regardless of which university! Besides," he shrugged again. "I'm from Brooklyn. But, uh," He cleared his throat. "Do you mind if I ask what I did in my life that has consigned me to Hell?"

The Devil grinned widely, displaying teeth which Robert had to admit were QUITE impressive in their jagged sharpness. There was something vaguely familiar about that jumble of pointy teeth. Wait a minute – Ah – that was it: *Ferengi*! They looked just like *Quark*'s teeth.

"Oh, I don't mind at all," the Devil said, startling Robert out of his reverie. "In fact, I'm happy to tell you that you're a first for me."

Robert was rather pleased. "Really? How so?"

"I've never had an inmate who committed this particular sin."

"What was it?"

"Are you SURE you don't know?"

"Ummmm... Well, as far as I know, I've lived a good life. Except..."

The Devil was delighted. "You've figured it out now?" he said, waiting expectantly.

Robert looked sideways at the Devil. "Is... God Jewish?"

The Devil's grin disappeared, replaced by incredulity. "*What?*"

"No? Then, Catholic perhaps? No? Moslem?"

"ARE YOU NUTS?!" the Devil thundered. "This has nothing to do with RELIGION!"

"Oh. Well, I just thought, since I was raised in an orthodox Jewish family and stopped living according to the religion pretty much as soon as I left home for college, maybe that was the reason..."

"NO!"

"OK then can you give me a hint? Does it have anything to do with the Ten Commandments?"

"Why? Did you break some?"

"Oh, absolutely! If they're really the same ten I was taught."

"Which ones did you break?"

"Uh, well... I haven't kept the Sabbath since I left home, really. And... I've definitely coveted – more than once, in fact. And..."

"Coveted!? What do you think this is – Confession?! Those aren't SINS, damn it!"

Robert shrugged in apology. "Well, you did ask about *Commandments*. I always thought they were linked, though! Could you, maybe, define what you mean by "sin" for me and I'll..."

The Devil scowled down at him. "All right, here's a hint... You were nosing around in areas that you had no business delving into!"

Robert was perplexed. "What areas? Is this about sex? No? Marital fidelity? Or...?"

"You really don't have a clue, do you," the Devil said, studying him, listening to his thoughts. "Here, let me enlighten you: You were committing the same sin that Eve committed in the biblical story, except there was no Eve. You're actually the first to do it."

Robert stroked the stubble on his chin, thinking. "And it's not about sex or Commandments?" at the shake of the Devil's head, he probed further. "OK. Well, it's probably also safe to assume this has nothing to do with befriending snakes or eating apples or bad fashion choices like fig leaves..." At the Devil's snort of derision, he continued. "So, my guess is, it must involve knowledge in some way..."

"Right! Now you're catching on!"

"Really?! So, what knowledge did I pursue that pissed God off?"

The Devil rolled his eyes in exasperation. "You wrote a book entitled 'Life, Itself: A Comprehensive Inquiry Into the Nature, Origin, and *FABRICATION* of Life'..."

Robert was astounded. "God... *read my book?*"

"NO!" the Devil thundered again, pounding the top of the kiosk with his fist. "God has no need to READ your book to know what was in it! What's the matter with you?!"

Robert chuckled at the strangeness of his predicament. "Well, I guess I'm still digesting the fact that there's apparently a real HELL – and someone called "*The Devil*" is in charge of it! If THIS is true, then it might also be possible that..."

"A-ha!" the Devil said, wagging a finger at him. "There you go again – *If this, then that!* Once a theoretical scientist, always a theoretical scientist, eh Robert? Y'know, that's what ultimately got you sent to Hell! Don't you think maybe it's time for a change in your approach to things?"

"Would it get me an executive pardon?" Robert countered.

"No."

"Then, why should I bother?"

The Devil blinked, then grinned again. "Good point! You know, I'm beginning to like you."

"Thank you." He looked doubtfully at the Devil's grin. "You know, those really are very impressive teeth."

"I'm glad you like them."

"Why do you have such long, sharp, pointed teeth? If you don't mind my asking, that is?"

"Why should I mind?" the Devil asked, grinning wider. "Is this where I get to say; '*The better to EAT you with!*'?"

"Well... *ARE* you carnivorous?"

The Devil was dumbfounded. "Robert, what the Hell kind of question is THAT?!" he finally said, tilting his head to the side. "*Am I carnivorous?*" he repeated, absently fingering one of his horns in bemusement. "I can't believe you just asked me that!"

Robert shrugged. "Well, no offense intended. But it just seems odd to me if you *are*, because. . . I mean – where's the MEAT?"

"The *WHAT*?"

"The thing is, we're just *souls* here, right? Or something equivalent? I think my body was cremated! So, even if you do consume souls, why would you need meat-tearing teeth like that to. . ."

"Robert!" the Devil interrupted, looking at him with wide eyes. "You are the most unexpected mind. . . You ask the *strangest*. . ." He started chuckling, shaking his head. "You're not like anyone else here! In fact, I have never met anyone even remotely *like* you!"

Robert's eyebrows flicked up once and he sighed. "I've heard *THAT* before."

The Devil paused, then laughed softly – with real amusement this time. "So have I," he confided. "You should hear it when GOD says it to you! He really has no sense of humor. At ALL." He stared at Robert for a moment, considering. "Tell you what, Robert. . . Perhaps you could help me with something."

"What?"

"Well, I've been ordered by our humorless Commander-In-Chief to take a full inventory of Hell. And. . . I really *don't feel like it*! Certainly not all by myself. It's going to be a lot of work! I could use some assistance – especially with the math. I've always hated math! There are, of course, plenty of inmates here who could handle it for me but, frankly, until now there was no one down here I could stand to have working with me. Cowardly buggers, the lot of 'em! You're different. You're not afraid but you're not a pain in the ass – at least, not yet. You're not disrespectful but you don't grovel, either. I can't *stand* souls who beg! Word to the wise," the Devil leaned down until he and Robert were eye to eye; reptilian into human. "No matter what I do to you, don't *BEG*." He straightened up again and smiled. "So, what do you say? If you help me with the inventory, you'll get the grand tour of Hell. . . be able to see it all. . . Are you game?" He leaned back, watching Robert intently.

"Sure, why not?" Robert said.

"And?" The Devil prompted.

Robert gazed at him, confused. "And. . . what?"

"Just like that?!"

"Um. . . Just like what?"

"You're not even going to try and cut a deal, first? . . . Ask me for special favors before agreeing to help?"

"Why would I try to do that *before* I've impressed you with my math skills? Besides, I don't really expect it would do me much good to ask for favors. . . this being Hell, after all."

"Right. So, you just agreed to help, out of the goodness of your heart – because I asked you to?" the Devil said in a deeply suspicious tone.

Robert chuckled, finally recognizing the concern. "Not really, no. If it makes you feel any better, I'm generally not this helpful most of the time! In fact, I tend to prefer solitude to being helpful. But. . . I've done a lot of traveling, see? And I've learned from all of that what to do and what not to do, when traveling. This is my

first time in Hell. It's a new place. I don't know my way around. In my experience, it's always best to have a native show you the ropes in a new environment. The way I see it, you're probably the best guide I'm ever going to be able to find in Hell. And, besides... it's not like I have anything better to do right now – as far as I know, anyway."

The Devil nodded. "Good points – all of them. I'm impressed."

"Yeah? Good! So? When shall we start?"

After a quick preliminary tour of Hell, the Devil took Robert to his palace, where they could begin planning the process of *counting the damned sinners*, as the Ruler of Hell put it. Apparently, the inventory order had specified that several different measurements were required: First, an overall tally, to be broken down into categories of sin, which would each be further broken down into various demographics (such as premeditation prior to commission of sin, number of times of commission, etc.). It didn't really surprise Robert to find out that God was apparently a Reductionist – at least, in His approach to this Inventory.

After a brief but heated exchange, which ensued because Robert asked the Devil why God needs an inventory of hell if He's omniscient... whereupon the Devil became quite upset over the lack of a suitably logical answer. Robert began to see that perhaps this was not the time to question the reasons for the task before them, but to organize the task, itself.

"Before we get started, then," Robert said, "is there anything that YOU want to add to this inventory?"

"Like what?"

"Well, I dunno," Robert shrugged. "But it just seems to me that if you're going to have to go to all the trouble of this big, comprehensive inventory, just because somebody else wants their numbers, you might as well build your own agenda into the project and get what you want out of it at the same time. That way, you're doing it as much for yourself as for whomever ordered the inventory in the first place. Frankly, that's how I got all the way through school."

The Devil stared at him. "You think like I do, Robert."

"I do?"

"Yes. That kind of thinking is what got me into trouble."

Robert digested that piece of information. "So, uh, does that mean you... just want to follow orders – as written?"

The Devil grinned. "HELL, NO!" He clapped Robert on the back, conspiratorially. "Just give me a few minutes to give this some thought. Hmmm. yes.... I can think of several things...."

Robert waited. Time ticked by, slowly.

Robert was fascinated by the evidence that time was every bit as much a part of the organization of Hell as it had been in life. He was also delighted to realize that he apparently didn't need to eat! After twenty years of enslavement to an insulin syringe, this was pure emancipation.

"What are you smiling about, Robert?" The Devil said, coming around from behind him.

"Me? Oh, I was just thinking how nice it is not to have to worry about an insulin reaction anymore."

"Hmmm. . ." The Devil nodded slowly. "I appreciate your honesty. Most souls try to lie to me – they're afraid I'm going to conjure up whatever it was they had in their mind they were pleased to be free of."

"Do you?"

"Yes! I DO, as a matter of fact! But don't worry, I only do that when I'm lied to." He fixed Robert with a sideways glare. "And I always *know*."

"I am perfectly willing to believe that," Robert said. "Seems like a reasonable skill for you to have, given the circumstances."

"You're a quick study." The Devil smiled at him. "You know what, Robert? I think you were much better off being sent here. You are too independent-minded for heaven. Too used to thinking for yourself. They really don't appreciate that kind of thing, there."

Robert shrugged. "I'll take your word for it. I have to admit; I can't really see any other useful way to go about things. I learned early in my life that if I don't think for myself, I can't always count on others to do adequate thinking on my behalf. In fact," he said, "I found as I got older that what most people consider to be adequate thinking is really not adequate at all for my needs! My needs always seem to be different from just about everybody else's. So, it was a survival mechanism. The pattern is so ingrained I think it's not really something I DO, anymore, if it ever was. Instead, it's more like who I AM."

"You definitely wouldn't fit in, in heaven, then! The rules would drive you nuts!"

Robert chuckled at the prospect. "Just might be more like *Hell* – for me – than this is, you think?"

The Devil grinned. "I think God may have miscalculated, sending you to me, though. . ."

"Why?"

"Because. . . I'm having *fun*! And you're giving me all sorts of wonderful ideas! Hey, y'wanna play a little prank and get back at Him?!"

"Not particularly, no."

"Why not??? I'll help you with it! He shafted us both, y'know!"

"It would seem so," Robert admitted. "But I just don't enjoy spending my time on things like that. I can think of so much more I'd rather do."

"Like what, for instance?"

Robert gestured to the panorama of Hell spread out before them from their vantage point on the stone terrace of the Devil's palace, which was perched high above Hell on a rocky formation of cliffs. . . Reminiscent of the Amalfi coast, he thought. . . but without the sunshine. Or the fresh air. "Well, for example," Robert said, "it would seem that there are as many questions to be answered about *the WHY of things* after death as there were in life. I wonder. . . has anyone ever studied the epistemology of Hell?"

"No! No one has ever thought to *study anything* about this place – they're usually too busy freaking out!" The Devil threw his head back and laughed. "Ah, that's beautiful! The epistemology of Hell. . . Is that what you're going to do?"

"I just might."

"Gonna write a book?"

"Well, now... I wouldn't say THAT..."

"You could call it *Damnation, Itself!*"

"Yeah," Robert said, inspired by the funny idea, "*Damnation Itself; A Comprehensive Inquiry Into the Nature, Origin, and Fabrication of HELL!*"

"Tell me, Robert," the Devil said, once they stopped laughing. "Do you think you figured out all the secrets about LIFE, with your various books?"

"Hardly!" Robert shook his head and chuckled, contemplating that idea. "No, I barely scratched the surface." His brown eyes twinkled up at the Devil. "And apparently that was enough to get me sent to Hell! But at least I got the answers to my OWN question."

"Which was?"

"*Why are living organisms ALIVE?*"

"And, what was your answer?"

"Well, that's not something I could sum up in a sentence or two. It would take a conversation and probably not a short one..."

"Do you think I wouldn't be able to understand it?" There was an implicit threat hanging in the question.

"Of course you could understand it!" Robert said, oblivious as always to the mind-states of others. "It's easy to understand once you have the context! But that's what takes time... Are you sure you want to hear it? It would take us away from our task."

"Oh." The Devil considered this a moment, then made a dismissive gesture with his fingers. "The inventory can wait. I've procrastinated on it for this long, I can put it off for a while longer!"

Robert laughed. At the Devil's inquiring glance, he explained. "I'm beginning to see what you mean – about us thinking alike. We have a quote, where I come from: '*What is worth putting off until tomorrow is worth putting off until next week.*' I don't know who said it first, though."

"Ah well, whoever it was, if he's dead, he's probably here, somewhere," the Devil told him. "Sounds like my kind of guy!"

The situation soon became even more interesting, as far as Robert was concerned. The Devil had decided that they should be comfortable while they were talking about life and living, so he conjured up what appeared to be a large wooden table, some deep, overstuffed leather chairs, a bottle of some unidentified amber liquid in an ornate crystal decanter, and two matching snifters.

Robert accepted his glass after the Devil poured him a stiff libation and obligingly held his glass aloft when the Devil lifted his own.

"To HELL with EVERYTHING!" the Devil toasted, and took a swig of his drink.

"To interesting conversations," Robert toasted and raised the glass tentatively to his lips for a sip. Much to his surprise, it was a delicious – albeit fiery – alcoholic liquor; As fine as any Cognac, Armagnac, or Scotch he'd ever had (and he had made

it a project in life to taste and compare many different brands of all three). “Hey! This is really *good*,” he said, taking his seat and making himself comfortable.

The Devil scoffed at his amazement. “What did you expect? I don’t drink crap!”

“Wait... how can I be drinking this at all?” Robert wondered out loud. “I’m dead.”

“There you go – thinking again,” the Devil said with a rueful smile, sinking into his own comfy chair. “Just accept it as a Welcome-To-Hell gift and get on with describing your answers. So, tell me; why ARE living organisms ALIVE?”

Robert took another sip of his drink and cleared his throat. “Well, it has to do with a circle of ideas which I call *Complexity*,” he said. “But the way I define that word is somewhat different from the way most other people have been using it. For me there is an innate *relational* aspect to complexity which ultimately ends up dealing with matters of organization of systems. Most people using the word “complexity” have generally used it to refer to anything that I would describe as merely complicated. That means intricate enough in some way to not be easily comprehended, perhaps just by sheer size or magnitude of detail. But complicatedness is not the same thing as complexity. My definition is based on my own observations and, I’d characterize this as a relational, interactive universe. Interactions are what drive causality and the WAY interactions happen, meaning the context under which they occur, has enormous influence on the effects. My conclusion, very early on in my career, was that context MATTERS.”

“Are you saying that context creates matter?”

Robert gestured with his snifter. “Well, there’s some truth to that, actually, although indirectly... But what I meant was that there’s an assumption in science that *Matter – the material particles and energy that make up the material aspects of the universe* – is what’s important to study and aspects of context are just incidental or are based on something intrinsic to the material parts and particles, themselves. I’m saying that context, as a whole, is sort of a *ding an sich* – a causal “thing in itself” – as are individual aspects of context such as relations. Relational changes can, in fact, change everything about an expressed outcome, behavior, or effect resulting from an interaction, even with all the same material players interacting. So, this is why I’ve concluded that it is the organization of a system which we need to study. System organization can confer definite behaviors and properties upon any system so organized.

“So... that’s IT? The way something is configured is more important than what it’s configured with?”

Robert shook his head. “Not if, by configuration, you mean the physical or tangible aspects; the material structure. It’s important not to confuse organization with material structure. Science does this all the time but it’s a mistake because the two are not the same. It’s possible to kill an organism without damaging the physical structure, right? So, our organism is, in the material sense, intact... but, where’s the life?”

“So... where IS it? In the organization alone?”

“Well, you see, organization is not something which is fractionable from any other aspect of the system, including the material structure – which happens to be

part of the organization. In fact, non-fractionability is one of the hallmarks of a complex system. So, we have to talk a little bit about what organization is and what it can mean, in terms of overall system behaviors and properties.” Robert paused to take another sip of his drink, enjoying the pleasant burn after the swallow. “Science currently has a very hard time with this,” he continued, “But, the inescapable truth is that there ARE important behaviors and properties of complex systems which cannot be analyzed by any reductive means or approaches. Life and mind represent two of them.”

“How can you be sure of that?” asked the Devil.

“Follow the logic in the entailment: To fractionate any system is to change or even destroy its organization. Smashing an atom, say. Or killing and dissecting an organism to study it. If we destroy the organization, we lose all the causal information contained in, or indeed created by, that organization. . . . Complex organization – and especially complex LIVING organization – does, in fact, create and carry within itself the capacity for massive amounts of information we need in order to understand why it is the way it is. I contend that most natural systems in the universe are “complex” in this relational sense. So, IF there is important information about the system which is entailed by organization rather than by the nature of the material components alone, then that is information that cannot be learned from any amount of study of those components. You see?”

“All right, I get that much,” said the Devil. “Assuming you’re right. . . why do you suppose that you were able to see this aspect of the universe when everybody else was looking at it differently? And, incidentally, Robert – that doesn’t explain why organization can make something be alive.”

Robert smiled at him. “I’m getting to that,” he said. “But, to answer your other question, first. . . I think I was able to see it because I was trying to use physics to answer questions that crop up outside of the “field” of *Physics*. I don’t see those two as the same thing, you see. . . physics with a small “p” is a set of tools with which to do science and a physicist is one who is an expert in wielding those tools – either mentally, as a theoretical physicist, or in applications, as a practical physicist. . . or both. In contrast; *Physics* with a capital “P” is supposedly a branch of science, but it often behaves more like a community of believers, much like a religion, in that one must agree to the precepts of the orthodoxy in order to be a member.”

The Devil shook a fist. “Damn them! I HATE orthodoxies!” he grumbled.

“I’m not particularly fond of them myself,” Robert admitted. “And they’re not fond of me, either. Which is why you have had a run of angry dead members of Academia complaining about *Rosen*!”

“Oh!” The Devil was pleased. “You were raining on their Academic parade, were you?”

“Not on purpose, no.”

“Are you saying you just. . . *accidentally* rained all over their parade?”

Robert grinned. “I didn’t even realize there *was* a parade. Or that the truth could be defined as rain in relation to it. *I am a scientist*, dammit! All I did was voice certain findings of mine that I felt should be of interest to anyone in the sciences because these findings have very serious implications for *science in general*.” Robert

took a deep breath, slowly letting it back out again as he shook his head. “I guess I had a few rosy illusions about the nature of Academia, when I started. I thought the truth was more important than maintaining some status quo or scent-marking various boundaries between disciplines... or winning prestigious awards. I’ve seen people win Nobel Prizes over *nothing*! Nothing important, anyway... It has become more of a popularity contest and a political campaign than a way to recognize breakthroughs in science. It rewards whoever can play the game the best. Who the hell cares about THAT stuff?! That’s not why I became a scientist! I wanted to know the answers to my OWN questions. In the process of finding them, I also found a few other *rather important things* – that’s all. But, for members of the orthodoxy, it was intolerable because the answers I was getting proved certain beloved traditions in science are either incomplete, counterproductive, or else beside the point.”

“Ah,” the Devil said, tilting his head back with his eyes half closed, remembering the quote. “Hence: ‘*Damn that Rosen – always answering questions that nobody wants to ASK!*’ It makes sense, now. Well done, Robert! Bravo!”

“Hey, no offense – but no thanks! I don’t want any credit and I also refuse to accept any blame for other people’s discomfort. The truth IS what it IS. If I am right about it, it behooves my brethren in Academia and Science to figure it out sooner rather than later. They can’t say I didn’t warn them: I did my duty and published my findings. My conscience is clear! Uh, that is... the fact that I’m sitting here in Hell having a drink with the Devil, notwithstanding!” He chuckled at the beauty of that irony, saluted the Devil with his snifter, and took another sip of his drink. “Oh, this is damn good!”

The Devil was grinning. “That’s why they say alcohol is ‘*of the Devil*’, you know...”

Robert finished the thought, grinning himself. “Because it’s DAMN good?”

“Yep!” The Devil tipped his head back to empty his glass, roaring in delight as the fire of it burned all the way down to his gullet. He grabbed up the decanter and deftly refilled his snifter, leaning over to do the same for Robert. “All right then, so tell me why the orthodoxy of Physics would get their shorts all in a twist over this organizational business, Robert.”

“It’s simple really... no pun intended. Although, that’s a *good* one, even if I... *DO say so myself!*” Robert said, using a Woody Allenesque voice.

“He’s on his way here, you know,” the Devil told him.

“Oh really? Good! That should keep things lively around here; another New Yorker in Hell. We have quite a bit in common, culturally, y’know.” Robert settled deeper into his chair, gathering up his thoughts to finally get back to the business at hand. “So, anyway... the situation was that Physics, as an orthodoxy in science, has always seen itself as the most general science and, along with thermodynamics, as the bedrock foundational set of ideas from which nearly all modes of analysis have been created. Collectively, these are the basic tools of science, the use of which then generates information considered to be applicable to (and able to tell us something important about) all systems in the universe, including biological ones. However, this turns out not to be the case AT ALL. That’s what I discovered by trying to use

physics to answer questions in biology. I can even pinpoint *why* this situation has developed, if you're interested?"

The Devil shrugged, "Sure, I'll bite... as it were..."

Robert raised an appreciative eyebrow along with his snifter at the joke. "Historically, the field of Physics has always dealt far better analyzing phenomena coming from systems that have an organization which is non-complex – what I call *simple*."

"Like what? What are some examples?"

"Anything that has an organization which is entirely computable, meaning that it can be converted to pure syntax – like zeroes and ones – without loss of necessary information. So, for example, orbital mechanics of a solar system; missile trajectories, conversions in cycles like the water cycle; molecular organization; the organization of any machine... and *there's the rub*: Science committed itself, via René Descartes, to using the concept of a human construction; a machine, as its working mental model for all systems. This is Descartes' *Machine Metaphor*: which presumes that all systems in the universe are *just like machines* and therefore will conform to fundamental information we learn from dealing with machines. But most systems in the universe are NOT simple! Natural, self-organizing systems tend towards complex organization – of varying degrees. And, therefore, many of the approaches and techniques that were developed based on the concept of the machine metaphor are actually inappropriate for trying to learn certain fundamental things about complex organization."

"Uh... wait a minute. We're not any closer to talking about why living organisms are alive, Robert" the Devil said.

Robert grinned. "I'm still getting to that! I warned you this was a set of ideas not easily summed up in a few sentences. They tend to ramify off in all directions and apply to many, many different things."

The Devil nodded, conceding the point. "Carry on, then."

"Ah, thank you," Robert said, leaning back in his chair once again with his snifter. "Now, where was I?"

"The machine thing," the Devil prompted.

"Ah! Right... So, this built-in presumption – that all systems in the universe are like machines – has really hamstrung the sciences, but very few scientists realize just how bad it is. Or how dangerous. The reason they haven't is that, as long as one stays within the realm of systems which have simple organization or which are nominally complex, one can do an enormous amount of good work, using science based on this and, therefore, never realize that the foundations are inherently flawed. In that realm, the limitations don't get in the way. They remain *invisible*. This kind of science can also be used to do a great deal of analysis of *components* of complex systems which have been fractionated. So the flaws and limitations have been well camouflaged, all these years with only a few troubling paradoxes arising, here and there, to give us any clue that something is awry. In that realm we can ignore the little inconsistencies that crop up and not run into the brick wall. But... that does not change the fact that it IS a brick wall. Only when we try to apply the reductionistic principles and techniques derived from the machine metaphor to studying or explaining the

behavior of *intact complex systems* – like weather systems, organisms, ecosystems, social systems, and so on. . . well. . . that’s when the flaws become not only apparent, but insurmountable.”

“Where does the explanation for life come in, Robert?”

Robert smiled at him, leaning forward. “All right: Life, as we know and recognize it, is a consequence of complex organization of a certain type – a category I call “Anticipatory” – in a material system.”

“But that doesn’t tell me anything!” the Devil protested.

“Well, it will. It doesn’t make sense YET because you need to know a bit more of the context before the meaning becomes clear to you. See, that’s what I’ve been trying to give you – the *dictionary* by which to interpret the meaning and the primer for understanding the references in what I’m saying. It’s important because otherwise, everybody just goes by whatever meaning and definitions they’ve already got in their heads and fill in the blank spots with more of the same or just ignore them and, almost invariably, they come up with either a wrong understanding of what I’m saying or else it is simply incomprehensible to them. You need the context to see how it fits together and why certain accepted ideas won’t ever fit and must be discarded.”

The Devil shrugged, conceding the point. “OK so I jumped the gun. Let’s go back to this: Why would organization make such a difference?”

“Ah,” said Robert, settling more deeply into his chair. “That’s what I spent the second half of my career learning about!”

“What was the first half, then?”

“The first half was spent figuring out why it doesn’t work to try to apply principles from contemporary Physics to most biological situations. It *SHOULD* work. In principle, if biological systems exist in the same universe with orbital mechanics and molecular configurations, etc., then what we learn from a study of one should be applicable to studying the others.”

“But it’s not?”

“It *WOULD BE*, except for the fact that the field of Physics, as it currently exists, has misinterpreted certain basic aspects about systems – all systems, including simple systems and including the universe in its entirety as a system. It’s this basic and fundamental misinterpretation which has allowed the machine metaphor to stand, unchallenged, all these centuries. I see these as flaws in the foundations of science, not necessarily flaws in physics, per se. The good news is that it’s fixable and, once it’s fixed, Physics really can be the general science it claims to be. But. . . only if these limitations, these flaws, are fixed. They cannot be allowed to remain there at the foundations.”

“What, in your opinion, has the field of Physics misinterpreted about the universe, Robert?”

“The impact of relations on causality.”

“Well, now,” the Devil said, “I’m no physicist, but we have plenty of physicists here in Hell, and I happen to know that Physics, the Orthodoxy, is aware of the notion that the result of any measurement is relative to the measurement taker. . .”

“Yes, that’s Einstein’s contribution. But Physics still believes that causality in the universe has to do with states and dynamical equations. As if it’s only a matter of initial conditions, particles, coordinates and measurements, about states and phase transitions, chronicles in a linear progression in one direction only, from one state to the next state, and so on. Even Quantum Physics has inherited innumerable assumptions and precepts, as well as many basic approaches, essentially intact from Newtonian Mechanics; the concept of states and recursion, for example, now encoded as *wave functions*. These things are only true inside their model and, once again, are close enough to the observable reality of *simple systems* to *seem true* via experiments. What they fail to realize is the sheer magnitude of the implications stemming from the fact that there is information *in the relation between* the observer and the observed which cannot be dispensed with. That has been *proven* – to everyone’s satisfaction, including mine. *That fact changes everything*, yet even Einstein didn’t realize the full importance of what it means! What he discovered was that *this is a RELATIONAL universe*; a universe where a set of relations create every interaction and every measurement creates another set of relations. Once we agree that relations can have causal importance, then nothing can remain the same because we must also agree that anything which impacts relations is going to impact all. Right?”

“I . . . guess I’m with you so far,” the Devil said.

“So, it *must* be true that the organization of systems, which creates, constrains, and maintains all relations within any system itself, is something we need to study if we want to fully understand any system.”

“Ah, but what you’re saying would be true even in simple systems,” the Devil said, in a tone that conveyed his certainty that he had detected a flaw in Robert’s logic. . .

“Yes. It *IS* true in simple systems.”

“Ah HAH! But you just said that science has been able to ignore this with simple systems and not see the flaws!”

“Yes I did. And you think that is inconsistent, do you?”

“I would have to say that I do, Robert.”

“Well, it isn’t and here’s why. . . they’ve been lucky.”

“Lucky?!”

“Yep. If Newton had been a biologist, he would either have come up with something completely different, or he would have gone stark, raving mad out of sheer frustration! To reiterate: complex organization is any organization in which the relations contribute critical causal impact to the system as a whole. Simple organization means that relations supply fairly minimal causal impact and therefore simple organization can be reduced to, or equated with, physical structure without generating insurmountable problems. So, *they were lucky*. They were dealing mostly with simple systems or aspects of complex systems which are, separated from the whole, simple in organization on their own. But the level of success achieved has given Physics the impression that the reason they have achieved this success is because the machine metaphor and the models, methods, and approaches based on it will work for ALL systems in the universe. Think about it: If you spent your

career working with types of system organization which contribute almost no causal impact on the overall system behavior. . . and you were unacquainted with systems where this is not so, would you have any reason to believe that there are, in fact, organizational types which CAN contribute *the bulk of system behaviors*?"

"Ahhhhh, Hmmmm. . . So, this is how come science has been able to get as far as they have, not having realized the true importance of this organizational aspect?"

"Exactly. Because in a machine, the organizational impact is nominal and you don't lose much information reducing organization to structure. The machine metaphor is what really spawned the reductionist paradigm in science. The way science goes about analyzing everything is by breaking stuff down into smaller and smaller subsystems, which destroys the organizational entailments before they even begin their analysis. When they do this to a complex system, they cannot see what's missing. They can only see that what they have in terms of information, about human physiology, say, is not adding up to the whole. That's why it is common to say that *Complex systems are more than the sum of their parts*. However, what the presumption of the machine tells us to do in that instance is learn more and more about the parts, and the parts of the parts, reducing ever smaller, amassing ever more data. . . because in a machine, that really CAN tell us almost everything we need to know about the whole. The entire way we currently do science encourages this perspective, which is highly distorting. . . particularly from within. It can be very difficult to see anything outside of it if you are used to operating from deep inside this kind of mindset – *and used to being successful using it*."

"So, how come biologists haven't seen it, if they're outside of it?"

"But they're not outside of it. The subject is but the science is not. Physics underlies biology. Biologists tend to base their thinking on the rules and methods generated by Physics. I did! When I started out, I was perfectly sure that I could achieve what I intended to achieve, using physics. No one was more surprised than I was when it finally became clear to me that it would never work."

The Devil leaned forward and pinned Robert with his eyes. "So how come you were able to recognize the flaws when so many other Biologists didn't? I bet you think you're smarter than everybody else!"

Robert laughed. "Whether I'm smarter or not isn't for me to decide. Don't get me wrong; I have a healthy self-confidence in my own ability to learn and think. Furthermore, it's kind of become obvious through my lifetime that nobody else sees things the way I do and that has turned out to be a useful thing. However, I think it is impossible to judge a quality like "smarter than", outside of any particular, distinct comparison. It's the sort of judgment which relies on the context; namely, the relation between my smart-ness and someone else's. Even then, it would be very difficult to say, categorically, that one person is smarter than another. How do you measure it? Smarter in what sense? Someone I might consider an idiot in science may, at the same time, be a genius in something else – some area of human activity or thought where I, myself, might be an idiot. So, this is one of the many cases where the mode of measurement we choose is going to create information that may or may not accurately reflect the true situation. And so will our criteria for employing it: For example how would you differentiate between intelligence and talent? Are they

the same? Or are they not... and if they are not the same, then is one of them more important than the other? When? Why? It's all very subjective."

The Devil leaned back in his chair, sipping his drink. "Hmmm... I think I just reinforced part of your argument for you – that thing about the importance of context..."

Robert grinned. "Yes, I think so too... What can I say? Except... how about a little more o' that elixir of yours?" He held up his nearly empty snifter, which induced the Devil to survey the contents of his own glass and find it lacking as well.

After both snifters were refilled, the Devil settled back in his chair and put his enormous claw-like feet up on the table, encouraging Robert to do the same. Once Robert had obliged and they both contributed a few jokes about the comparison between the common stereotype of the Devil possessing *hooves* and his actual feet, the Devil said, "Well, Robert... as interesting as all this is, you still have not elucidated how it is that organizational qualities can be the cause of life in an organism."

"We are quickly getting there," he assured the Devil. "However, there is a loose end left over from the issues that occupied the first part of my career: You asked how I recognized the flaws when others did not."

The Devil nodded. "So?"

"I've wondered that, myself. I do think it partly has to do with a certain combination of abilities and interests, like my fluency in physics and mathematics, for example, coupled with my main passion, which is biology. Very few biologists that I know are also fluent or even interested in the more technical sciences... and very few physicists that I know are really interested in issues peculiar to biological systems. I'm also a theorist interested in foundational issues, yet I've had extensive experience in empirical, or applied, science, which is a combination equally unusual among my colleagues. So that kind of range would seem to be a logical part of the explanation. But, I also think it has something to do with certain habits of thought; a way of looking, a way of *seeing*. When I observe phenomena in the universe, I don't see "things": I see patterns. Analogies. Similarities. I have always thought in comparative terms and patterns, ever since I was a kid. My curiosity about phenomena has always been to wonder; *Why does THIS happen?* I wanted to know what *entails* some phenomenon, so my study of the phenomenon itself was always geared to understanding why that particular phenomenon is the way it is or happens the way it does – following the causal pathways to find out *what ENTAILS it?* This sort of question turns out to be a very comprehensive question to answer; much more detailed and exacting than, say, the question; *What is this phenomenon?* Or even; *What causes this phenomenon?* Entailment is a far bigger concept than causality, just as organization is a far bigger concept than physical configuration or structure."

"I see... So you looked at living organisms and wondered; *Why are they alive?* Right?"

Robert nodded, sipping.

"So, tell me then... how DID you come to the conclusion that it was an organizational property which *entails* life?"

“Partly because there are so many different types of living organisms: Enormous variation of form, physical structure, behavior, abilities, reproductive modes, etc. The ingredients are all different, the parts are different, the habitats are different, and yet... somehow we can easily recognize them all as a UNITY. They are all *living organisms*. Why can we recognize that?”

“You asked *why* about THAT?!”

“Yes. Don’t you think it’s a reasonable thing to wonder?”

“Well... I dunno...” the Devil said, trying to imagine.

“It seemed a significant fact to me. It’s also a significant fact that, whatever it is which *causes* life, it must not be directly connected to the form, or structure, or the components, or the ingredients except for the more indirect value of the roles those aspects play within the whole system. But that’s a similarity in functional terms, though, you see? So, for example fur and feathers are very different, but they provide some of the same functionality to the organism. Reproducing by laying eggs is radically different from what either fruit trees or human beings do but all of these are modes of reproduction. Thus, the differences between various organisms can be seen as similarities when viewed in terms of the functional roles those things embody and contribute to something ELSE – something which all living organisms must necessarily share. The one commonality all living organisms share is the nature of their entailment patterns. It’s an Anticipatory pattern! All living organisms represent Anticipatory Systems – THAT’S what we are recognizing when we can see a unity among living systems. That anticipatory signature, which is so different from the reactive patterns of behavior generated by non-living systems. And it is system organization which is responsible for the pattern.”

“But how did you come to the realization that it was their organization that is responsible for that? And how did you uncouple the notion of organization from that of physical structure?”

“*By recognizing the power in a relation.* A relation between two things can change every observable about the two things – this is a fact. The evidence is clear and it is indisputable! Even in a simple system like a molecular bond... No theoretical chemist can explain why it is that sodium and chlorine, two poisonous elements, can combine to create sodium chloride – table salt – which is absolutely essential for human life. They can tell you *what* happens, they can *make it happen*, they can *undo it and remake it* again. All perfectly simple... But they cannot tell you WHY. Indeed, the more I looked at the effect a single change in a single relation could have on the flow of causality in a system, the more I realized that causality, itself, is the result of relational interactions. Indeed; nothing happens until something interacts with something else. Space and time interact to create aspects of our universe. Together, they create the potential for causality – for change.”

“Hmmm,” the Devil said softly, “I guess that’s really true... everything is static until they interact with one another in some way.”

“Indeed, and yet, even the state of non-interaction between two objects or entities can sometimes involve a set of relations which have critical information in them, as well. Like a bombardier beetle keeping the two chemicals it uses for its weapon separated until there’s a requirement to defend itself. These are all, potentially,

aspects of *context*; both in a direct way and in a larger, less direct sense. For example, non-interaction can be a context for interaction and interaction can also be a context for non-interaction. . . .”

“What?”

Robert chuckled at the puzzled look on the Devil's face. “What this means is that there is no such value as a “state” – never mind a *generic* state. We can only create a temporary, artificial version of the theoretical notion of “state” . . . and even then, we can only do so with simple systems like machines without destroying information in the process. Reductionism was built using the machine metaphor; a model based on *the idea* of what a machine IS as a system. But what about a system such as an atom, which I contend is a complex system – Can we fractionate an atom without losing information? Have we destroyed the system if we fractionate it? Are atoms static? What natural system is ever truly static? Nothing in the universe actually exists *outside of* time or space, and therefore is always in a condition of constant interaction with the combined system of space/time, even if nothing else. I argue that we can never truly detach or fractionate any natural system from all contextual interaction and to try is not even desirable.”

“Why not?”

“Because there is information in those relations between system and environment as well – information we need to understand the system, itself. Nowhere is this more true than in the biological realm! Living organisms encode aspects of their environment into their own organization to such a degree that the continuation of those relations becomes essential for continued existence. This is why space travel poses such difficulties; we have to bring enough of our context with us to survive away from our planet. The way we are organized specifies a context. Indeed, I contend that our organization includes information *which can act as a set of models, predicting* our context.”

“So, let me see if I got this straight, Robert. . . If there is no such thing as a timeless state, then what's the scientific alternative for a complex system?”

“Constant interactive relational causality. Have you ever heard the expression; “*Change is the only constant*”? There's a grain of truth in that, although it is also true that if a system maintains homeostasis, it's a way of NOT changing, at least, not too far outside of necessary organizational constraints, but you can bet that any system maintaining homeostasis is doing so via. . . a whole lotta. . . *constant interactive relational causality, deh!*” The last five words came out in Archie Bunker's thick Bronx mis-pronunciation style, causing the Devil's eyebrows to rise. Robert, for whom such quick expressions of the mischief in his soul were normal, didn't notice. . . he just sailed on past, switching back to his own voice again. “It's common for people to say that “life is a process, not a thing”. There's a grain of truth in that, too. . . But it's far more than JUST a process. That's like saying that a living system is *just a system with a process*. How does that explain life? What is it about living systems which entails the aspect of system behavior that they're interpreting as a process? People who call life “*a process*” can't answer that question.”

“Why do you believe that is?”

"I think it's because they fail to look beyond the process itself for what entails the process. They may recognize that any single process is actually a series of relational interactions whereby the effects of each step's relations become part of the next step's relational interactions, and so forth. But so what? According to that, crystal formation can be described as a process. Oxidation can be described as a process. Rust! Metabolism is often described as a process, with sub-processes and sub-sub-processes, etc. But no amount of examination into the process of metabolism is going to entail life for us. It goes the other way. Life entails metabolism. In order to understand the process of metabolism, one has to include the larger organization. In other words; the contextual relations for the subsystem we call *metabolism*. That's where we'll find the missing entailments. We can't use the reductionist method of looking at ever smaller pieces, like we could do in a machine or other simple system. Science doesn't seem to have a clue why not, and this is a clear example of the toxic effects of the machine metaphor on the foundations, again. Those effects are very subtle and subversive, but lethal. We presume such things with living systems and we poison ourselves and our environment. We still approach them as if they are just like machines; killing and dismantling living organisms in order to study them. My view is that as soon as we kill an organism, we've already lost too much information to understand why it was alive."

"So, do you think it's a waste of time to study the subsystems and the parts and pieces of the parts?" the Devil asked him.

Robert shook his head. "No, not a waste of time... but you have to know what you want to find out and where to look for it. It IS a waste of time to look for life in the pieces! You can learn about specific physiologies and specific sub-processes, etc, by using a combination of reductionist methods and relational methods, but the two must work together and there must be a sound theoretical foundation underlying the entire enterprise, or else the experimental activity is flying blind. If you don't realize where to look for what you are trying to find, you may just blow right past it, or toss it out. Like killing a living organism to try and study life! The first aspect of the organism they are sacrificing, *the organization*, is the very place they need to be looking. But, Science has built a tradition of analysis which always proceeds from larger to smaller, from wholes to parts. That's what scientists are used to."

"Because of the machine metaphor..."

Robert nodded. "Exactly. You can approach machines or cosmology or molecular pieces of living organisms that way and get away with it – in fact you can be very, spectacularly, successful. But you will never be able to learn why living systems are alive, that way, or anything else about the universe that is a consequence of complexity. Simple organization means that the entailments *can* be found in the parts and the direct relations between the parts, which basically equals *structure*. Complex organization, on the other hand, can create the same behaviors in two distinctly different systems with entirely different parts and different structures from each other and yet these two will still have the same entailment patterns –."

"Because the entailments are organizational," the Devil finished his sentence for him, causing Robert to raise his glass in salutation of his accuracy. "And... living

organisms, in all their variety, are the proof of that,” the Devil added, saluting back with his own sniffer.

“Correct! So if we want to know what entails life, we need to look at what is common to all living organisms. The only things they all have in common with one another are organizational.”

“OK! So I’m getting it! Uh... Wait a minute. Did you just say organization was a “thing”? But... didn’t you say earlier that organization isn’t a thing, it’s a quality.”

“Ahhhh,” Robert said, a smile spreading across his face. “This is where complexity gets even more interesting! I actually referred to organizational *aspects* as ‘things,’ which is a bit different than saying organization, itself, is a thing... But the truth is that *organization* can be viewed in many different ways with regards to many different aspects of a system – all at the same time. One example would be to view system organization as a *collection* of specific relations between parts, subsystems, and other relations – *and relational effects*. So, in that sense, a collection of any kind is a “thing”... The important idea to keep in mind is the multiplicity of dimension. At the same time that organization is “a collection” it is also definable in myriad other ways. It would be a mistake to ever view organization as a single entity, like a material object. That’s what I’m cautioning against when I say *it’s not a thing*. It all depends on contextual constraints. Context is absolutely critical and it is created by relations and relational effects. Thus, any system has internal contexts, external contexts, and the contextual fact that these two are in constant interaction with one another. In a living system, the concept goes dimensions beyond that... One aspect of what I refer to as Natural law (the bedrock set of entailments which drive all others, and all causality, in the universe) is that context can change everything. In fact, by doing so, it can ultimately change *itself* – the way internal context of a living system can do with modifying external context. The first beaver builds a dam in a stream, driven to do so by some aspect of internal context, and thereby changes the evolutionary path of its species which evolve in concert with such external contextual changes and interactions. So, you see? Given enough contextual conditions, *up* becomes *down*, *black* becomes *white*, *yin* becomes *yang*, *good* becomes *evil*, *death* becomes *life*.”

“You’re waxing very philosophical, Robert.”

“Here’s a riddle for you: When is a relation not a *relation*?”

“Beats the hell out of ME,” the Devil admitted.

“When it’s a *thing*.”

“I think you have had too much of my... what did you call it? My *elixir*!” the Devil said, showing his teeth again. However, this time he didn’t look all that dangerous, somehow... in fact, Robert thought it was probably as close to a goofy grin as the Devil would be able to achieve with his demonic features.

“Are you suggesting I’m inebriated?”

The Devil nodded. “Yeah... that, too!”

“Nah... I don’t get drunk,” Robert told him. “I don’t know why. It’s a shame, really. I’ve tried, on occasion. My physiology has always been peculiar in relation to most of my fellow humans. Then again, I must say, I’ve never actually tried to get

drunk *without* my physiology before!” He started to laugh at the lunacy of that idea. “Hey, maybe I can finally become a drunken bum here in Hell.”

The Devil chuckled too. “I will do all I can to encourage it, I promise. Here... lemme give us a refill... Right! So, tell me... When is a relation a THING, Robert?”

Robert raised his glass in silent thanks for the replenished level of drink. “A relation can be seen as a *thing*, when the organization treats it as such. In other words, when the total effect of some relation – or process, even – figures into the larger organization as if *that effect* were a single aspect, then it can be said that a relation or process can act, within the overall organization, as a THING.”

“Oh. I see! So, it’s just *acting like it’s a thing*... Hmmmm... That figures into the “*more than the sum of the parts*” bit, right?”

“Right! The organization can’t tell what’s a material object and what’s a relation or a relational effect... In essence, this has to do with organizational definitions differing from our own, as observers. But, what also must be borne in mind is that, just as it is with the multiplicity inherent in the concept of organization... at the same time as some relation is acting as a *thing* in one part of the organization, it is likely to be acting in a different capacity in another part or in multiple aspects of the same overall organization. In other words, any single aspect we could point to in a complex system may be ‘wearing many different hats’, as the saying goes; playing multiple roles within multiple different functional entailment relations, in multiple scales of time perhaps... all happening simultaneously, sequentially, and various combinations of all of the above... within the organization of the system.”

“Wait, Robert... how is anybody supposed to be able to sort out what’s what in that case?”

“Not by taking it apart, that’s for sure,” Robert said. “It is hopefully clear now why I say that trying to understand such a system solely from a study of the material structure, or parts and pieces, is not going to give science the kinds of information we need to understand some of the most important aspects of these systems. So, now you’re getting some idea of just how much more than the sum of the material parts a complex system can really be. In fact, there is no way to sum a system like this. Complex organization has aspects of infinity to it which is also what makes it impossible to reduce such a system to anything finite and expect to still have all the same information. Hence the non-computability of its organization. In order to compute something, you must be able to reduce it to syntax. Complexity does not fit into that category. Hell, we can’t even do that to mathematics!”

“Well, what’s your suggestion for science, then? Are you able to envision any alternate means of studying these things? Or is it more fun to just be a party pooper!”

“A what?!” Robert said, laughing.

The Devil shrugged, gesturing with his drink. “Haven’t you heard? Every party needs a pooper.”

“Is that a law of the universe?”

“Sure seems that way t’me!”

Robert was laughing. “Like Murphy’s Law? And the Peter Principle?”

"Yeah," the Devil said, finding it hard to take a swig out of his glass when he was grinning so broadly. "The Peter Principle... that describes heaven pretty good, I can tell you that right now!"

Robert adopted Arnold Schwarzenegger's thick accent. "*Hello. I'm the pahty poopah!*"

The Devil was giggling by this time. "HELL-oh, I'm the pahty POOPAH," he intoned, doing a creditable impression of Robert's impression of the Austrian accent.

"Is that how God would define YOU, do you think?"

For half a minute, Robert worried he might have gone too far in asking the question. The shocked silence was absolute. But then the Devil exploded in deep, rumbling guffaws that vibrated the air and the furniture, traveling through the stone floor of the terrace. It was strangely infectious, particularly after it mellowed to a dull roar of convulsive laughter.

When the Devil could finally breathe normally and it was safe to take a swig of his drink without danger of having it come out his nose, the Devil emptied his glass and picked up the decanter to refill both glasses. "Damn it all, Robert. I haven't laughed like that in... BLOODY HELL... how long has it been?! ... Seems like forever!"

"I'm glad I could be of service," Robert said. "I must say I haven't had much to laugh about in the last couple years, myself."

The Devil looked over at him. "You laugh easily, it seems to me. Even when you're meeting the Devil at the Gates of Hell!"

Robert grinned. "Yeah, I did, didn't I... Probably has something to do with the fact that I'm from Brooklyn! Hey, did you ever see the movie 'Angel Heart'?"

"No... I'm waiting for it to come out on cable. OF COURSE I DIDN'T SEE THE MOVIE!"

Robert started laughing hard again, realizing what a crazy situation this was. "Sorry! I forgot! The Devil had some good lines in that movie... I'd think you'd appreciate it."

"Who played me?"

"Uh... An actor named Robert De Niro."

"Oh. OK." The Devil nodded, pleased.

"How do you know who he is?"

"Never mind."

Robert laughed again. "OK, well anyway, I brought it up because the Devil asks the main character two different questions and the answer to both of them was; 'No. I'm from Brooklyn.'"

"What were the two questions?"

"The first one was; *Are you religious?* And the second one was; *Do you speak French?*"

The Devil looked a bit perplexed. "*No, I'm from Brooklyn?*"

"Maybe one has to be from Brooklyn to get the joke, but it made ME laugh, both times!"

"Yeah, but... Robert, we've already agreed that you're a really strange guy!" The Devil sent a toothsome smile at Robert who smiled back and sketched a general "*whatever you say*" with his free hand. "Ah! So... where were we here? I had asked you a question... what was it?"

"Uh... it was the beginning of the pahty poopah thing," Robert reminded him, a little concerned that mention of it might set them both off again, but fortunately it was only gently amusing after its earlier violent reaction.

"Right! That was it! What are the alternatives for science... if they can't take complex systems apart to study the, uh... what... the *organization*! That was it."

"Well, I think we need to expand the paradigm to include the relational side of things and then we can use a combination of reductionist and relational approaches to study complex systems. There are contexts where the only way to usefully model aspects of a complex system is in relational terms. In other words, mappings of entailment relations, mappings of functional relations, and so forth. So much remains to be done in all areas; from expanding the paradigm to developing new tools and modes of approach... These are all areas of science that desperately require study and development. I was not able to do much more in my lifetime than document the situation and point the way. I tried, wherever possible, to sketch whatever ideas I had for potential development, all through my published work. However, once again, the machine metaphor in the foundations of Physics disallows both the notion of function and the notion of a relation as being capable of carrying the same level of importance of entailment, within the overall organization, as a material part of a system does. In short; the machine metaphor makes it explicit that a relation must *never* be considered a *thing* in scientific terms."

"But why? And why deny the concept of function? Humans create machines to perform functions all the time. It's not exactly an alien concept!"

"True, but there is a *very* good reason, the irony of which I think you will really appreciate, if I'm not mistaken..." Robert's eyes were again full of mischief.

"What is it?"

"When the entailment patterns which generate human-created machines are used as the model for all natural systems in the universe, including living organisms, what happens?"

"I'm... not sure what you're getting at..."

Robert readjusted his position in his chair, leaning forward. "This has to do with the concept of "*Finality*" in the study of entailment patterns. Are you familiar with it?"

The Devil showed his teeth. "Finality as in '*the Final Solution*'?"

Robert laughed in spite of himself. "*No*. Finality as in Final Causation. Aristotle developed these modes of analysis, initially, and I have found them to be extremely useful. They help us look at "*the why of things*". There are four Categories of Causation in Aristotle's toolkit, which constitute four different modes to be used for analyzing the entailment patterns of any system. We can ask "Why does this particular system exist?" using each of the four categories as our mode of approach. When we're done, we will have arrived at four sets of answers; all four of them consisting of entirely different types of information from each other – and not

reducible to one another – yet, all of which are *equally accurate*... That is an extremely significant set of facts.”

“How d’you know they’re facts?”

“Because I have proven all of it, in my work. The proofs mathematical and otherwise, are all in the books and papers I published.”

“Oh. OK. Well, continue,” the Devil said.

“All right. Now, the categories are: Formal Causation, Efficient Causation, Material Causation, and Final Causation. What’s interesting here is that the first three of the categories are incomplete to the point of inadequacy, in explaining any system, when used as a single source of information. The fourth one, Final Causation, is far more complete because it refers to functional requirements which entail the existence of the system. It’s relational in a comprehensive way and because of that, the category of Final Causation CAN be used entirely by itself in analysis... and will give us essential information about the entire system. In fact, the kind of information we get by analysis via the category of Final Causation is information that trumps the other three categories.”

“Wait, Robert...” the Devil rubbed his eyes in fatigue. “You said I would find this amusing, didn’t you? What’s funny about this?!”

“All right, you’ve only got a little more to go and you will have the whole contextual background. We’re almost there... If a machine serves some function or purpose, then that function or purpose can be said to be the “Final Causation” answer – it explains the entailment for why the machine exists, according to that category of causation. So, let’s use your palace as our system and analyze it this way. If we ask the question: *Why does this palace exist?* The Final Causation answer would be: *Because the Devil needs a home.* That’s the functional entailment of this system called “a palace”. We’re not dealing with what the palace is made out of, which is a Material Causation answer; or how the palace was constructed, which is an Efficient Causation answer; or what the architectural designs of the palace specified, which is a Formal Causation answer... None of those answers are as complete in terms of information as the Final Causation answer. Do you see?”

“Um, I’m not sure...”

“It’s perfectly logical, perfectly rigorously scientific, to analyze a system in terms of the functional entailments. However, the big problem that the use of this category creates for science is that *the functional entailments of all human-made machines always point a finger at the humans who made the machines.* There was some functional need, be it transportation or a place to live or a way to cook food, or whatever... and someone created a new system in order to serve that need. So, the Final Causation answer to; *Why does this system exist?* – when the system is a machine – will always be *Because the one who made the system needed it for some functional purpose.* Well, can you see what the legacy of the machine metaphor will be when we try to analyze living systems using the category of Final Causation?”

“Uh... not quite...”

“We will end up with what looks like the *hand and mind of God.*”

The Devil was astounded. “You’re kidding!!”

“Nope.”

“Science thinks God makes machines?!”

“Uh... Well, it's not so much that they think God makes *machines* – not directly at least! But that's what they see coming out of this analysis if we use Aristotle's Fourth Category of Causation to analyze naturally occurring, self-organizing systems. See, if we truly believe that all systems in the universe are *like machines* and the Final Causation answer to *Why does any machine exist?* points to the functional requirements of *the machine-MAKER*... Then, what is the only logical answer to the Final Causation analysis applied to living systems? *Why do living organisms exist?*...”

The Devil started laughing. “How the Hell did you get around that one, Robert? Did you kill God with Science?!”

“Of course not – although that *does* sort of describe the intentions of the knee-jerk reaction on the part of science during Descartes' time, which continues on today! You see, by disallowing the Fourth Category of Causation and, along with it, ALL notions of *functional entailments* in any scientific analysis... they thought they could protect science from encroachment by religion. The feeling was that religion would kill science, otherwise. So, they tried to kill religion first – sort of a pre-emptive strike.”

The Devil nodded. “I can tell you that I've had lots and lots of new inmates come here to me out of that mutual hatred and fear, over the past couple millennia!”

“I'll bet!” Robert agreed. “They have often been mortal enemies, particularly with the rise of Christianity and the wide reaching power of the Catholic Church. Scientists were occasionally burned at the stake just for doing science. So, once the church's attention was diverted by the reformation and all the other developments around the world, and scientists could finally be engaged in science openly without fear... imagine their horror upon discovering what looked like ‘*proof of God's existence*’ right at the cornerstone of the scientific foundations.”

“I AM,” the Devil said, chuckling. “I am!”

“Their solution was; they decided that “Final Causation” created anomalous information and had to go. They declared anything having to do with it was *unscientific*... never realizing that it is not only *completely unnecessary*, but very bad for science, itself, to have done that. They were intent on getting God out of the scientific equation, but there is no need to forbid an entire mode of analysis in order to accomplish it! All that is required is to recognize that the machine metaphor is inappropriate as a model for natural systems. *That's all*.”

“That's ALL they need to do?”

“Yes!”

“What is the net effect of doing that?”

“Well, let's use it and see... When we analyze any deliberately constructed system like your palace, we still get the same Final Causation answer, right? *The palace exists because the Devil needs a place to live*. The Final Causation answer comes from outside the system, which is absolutely correct. And, as I said; it's worth noting that we clearly don't need the other three types of analysis to understand functional entailment relations. In fact, the details of the other three categories could all be changed: It could be made out of some other material; or built by a different

company; or designed by a different architect. . . and you would still have a home. Your functional need for a home could be served by a log cabin or an igloo or what-have-you. It might be an empty cave that you found and turned into a home, instead of something designed and built specifically to be a home. Thus, we can see that the functional entailments don't specify any of the details about the other categories of causation. The functional entailments could be achieved any number of ways."

"So, are you saying that this is why living organisms are so diverse? They are alike in their functional entailment patterns but not in their details?"

"Exactly. You're getting it!"

"No, I'm not. Because God is still in there. . ."

"That's an illusion. Forget the idea that all systems must be constructed with some functional purpose from an outside source. That's how machines are brought into existence. But if nobody "made" the system we're analyzing. . . If we use the category of Final Causation to answer the question; *Why does any particular organism exist?* What's our functional entailment answer?"

". . . There isn't one."

"Look again."

"I don't understand what you mean", the Devil grumbled in irritation.

"What you're saying when you say that *there isn't one* is really that there are no functional entailments driving the process *from outside the system*. Right?"

"Um. . ."

"Look at this another way: Why does *the universe* exist? What's the Final Causation answer?"

"There isn't one! It just DOES!" the Devil thundered.

"You're soooo close," Robert said, unperturbed. "The universe exists and yet *nothing from outside made it*. . . That means that all the functional entailments *are generated from within*. In other words: *IT ENTAILS ITSELF*."

The Devil was thunderstruck. "OH."

"So, you see? There is no need to kill God. Those who want to believe that God created the universe should have no objection to the notion that God is rather more clever than humans, surely? They can say that God is able to create incredibly complex, self-entailed – and self-entailing – systems, whereas human beings have only created simple systems. . . which have to be entailed from outside. On the other hand, those who prefer to leave the unknowables, like God, out of the analysis will appreciate that the Category of Final Causation – and functional entailments along with it – do not have anything whatsoever to do with God. They can finally remove that stupid prohibition, and with it the terrible limitation it created, from the enterprise of science."

"Uh, that is, unless it were to turn out that the universe really IS like a big machine!"

"I'm satisfied that I've proven, with my work, that it is NOT."

"Really?"

"Mmmm Hmmm."

"Well, Robert, there's just one question left to answer for me, then."

"What's that?"

“What do you tell people who ask you how the universe was created in the first place? Or how did the first living organism get here? Hmmm?”

“I tell them the truth: that I don’t have answers to those questions.”

“Ah-HAH!” the Devil said. “So are you saying that science is of no help with finding those kinds of answers, then?!”

“Not necessarily, no. I’m just saying that I’ve never attempted to develop answers to those questions. It wasn’t my area of interest. Remember... my question was; *Why are living organisms ALIVE?*... not; *How did living organisms come to be?* These are vastly different questions: The former deals with epistemological issues and the latter deals with ontological issues.”

“Are you saying that the two aren’t related?”

“No. Although I will say I can show plenty of proof that, where *complex* systems are concerned, epistemological answers don’t specify what the ontological answers may be.”

“What about the Categories of Causation? If you use all four, the answers together specify the ontology of machines... All categories together point to the machine makers.”

“Yes, they do! Very insightful of you!”

“Thanks,” the Devil said, feeling rather pleased with himself.

“However,” Robert told him, “that is only true with systems possessing simple organization.”

“How do you know?”

“Try it. Use the four categories to analyze a complex system; doesn’t have to be a living organism. There are many different complex organizational types or categories of types. The category of living systems leaves a lot of other complex systems out.”

“Name one,” the Devil challenged.

“The atom,” Robert said.

“Now, hold on there... you said that Physics deals with simple systems!”

“That’s not quite what I said. What I said was: *Historically, the field of Physics has always dealt far better analyzing phenomena which derive from systems which have organizations that are non-complex – what I call simple.* Physics has had – and still has – a great deal of trouble trying to account for the anomalous aspects of atomic behavior that contradict the assumptions about how such systems work, so those things are ignored; relegated to the status of *some minor detail that will be sorted out once we can really assess all the particles in an atom.* In short, they presume that atoms conform to the same models and laws of Physics derived from the machine metaphor. But that doesn’t mean that atoms are the way they think they are. So, what I said was that Physics does better with simple systems or aspects of complex systems which are simple.”

The Devil waved a hand. “Same basic thing,” he said.

Robert shook his head. “No, it’s not.”

“Are you contradicting me?” the Devil asked, his voice ominously quiet.

Robert was unperturbed. “When it comes to MY words and their meaning... you better believe it! Even God couldn’t put words in my mouth and get away with it. Not if I have free will!”

The Devil was mollified. “Hmmmph,” he snorted. “I’d love to see that encounter! And while you’re at it, you can ask Him to explain *why the bloody Hell He needs me to do a comprehensive inventory when He’s supposed to be all-knowing!*”

“All right,” Robert solemnly agreed, meaning every word, “It’s a deal. If He ever shows himself to me, *I will ask Him*. I have quite a few questions of my own I’m saving up for Him, as a matter of fact. Quite a few.”

“I’ll bet!” The Devil muttered and chuckled to himself, reaching one long arm out to grasp his bottle of fiery liquid, which never seemed to have any less in it, regardless of how much the two of them drank. He poured some more of the liquid into his snifter, sloshing a little over both sides before finding the general center of the glass, where the opening was. He then gestured towards Robert’s glass with the bottle and a questioning grunt.

Robert leaned over, holding his glass for the Devil to refill, following the swaying movement of the bottle so that none was spilled. “Thanks,” he said, leaning back once again, cradling his snifter in his hands, studying the swirling liquid in between taking occasional sips. “Ah, this is marvelous!”

The Devil looked over at him and attempted to say “Don’t get too comfortable” . . . except that he ran aground trying to enunciate “comfortable” and, after several attempts, abandoned the word, muttering, “. . . aw, y’knowudd-I-mean. . .”

Robert nodded solemnly. “I believe I do.” The Devil recognized Robert’s utter lack of disrespect and was soothed back into good humor.

They sat, contemplating their sniffers and sipping the contents in companionable silence for quite a while. The sounds of Hell were muffled at this altitude into a gentle wooshing noise that ebbed and flowed on the breeze across the open terrace. It was comfortably cool now, in fact. Robert studied the vast landscape stretching out below until it disappeared on the dark horizon. He admired the way the various shades of yellow, orange, red, gray, purple, and black all somehow blended and melded into a strangely beautiful vista. He had learned on his brief tour earlier that, like everything else in his experience, the landscape of Hell could look very different up close and Robert did not mind admitting he was really not in much of a hurry to study it in that kind of detail.

As he finished the last swallow in his glass, he looked over at the Devil, planning to ask for one more refill. He was surprised to see that the Devil had fallen sound asleep. As Robert watched, the Devil began snoring happily, his snifter caught in his knobby fingers and hanging askew, dribbling the contents in a thin stream across his belly and onto the stone floor of the terrace. For the first time, Robert became aware of the fact that the Devil had a belly button.

“Hmmm!” he said to himself. “Fascinating!” . . . *Why would the Devil have an umbilicus?* Robert tried to imagine what the Devil’s mother might have looked like. . . or what typical Devil parents might consider naughty behavior in their young. . . and realized he had to make himself stop thinking about it because it threatened to make him choke on his own mirth and start coughing. He did not particularly want to wake the Devil up, anytime soon. It was just too peaceful right now. Wonderfully peaceful, in fact.

Robert reached across the table to pick up the bottle, hoping it was OK to pour himself a refill. He raised his snifter towards the slumbering Devil in silent, cosmic thanks, and took another sip, savoring the sensation of warmth as it seemed to travel from mouth to stomach. He shook his head, bemused over the strangeness of his current situation, and concluded he would need a great deal more information before he could even begin to contemplate the nature of *life-after-death*. Not very much about it was making any sense at all. He looked over once more at the Devil, who had slid a little further down in his chair, his long, hairy legs disappearing under the table. The enormous head and neck looked to be bent at a strange angle because the horns were hitting the tall back of the chair. The total effect was that the snoring began to resonate with a slightly deeper, more nasal tone.

Smiling at the comical tableau, Robert quoted Scotty, from one of the original Star Trek episodes, including a perfect mimic of the exaggerated Scottish brogue; “Ach! Aye drrrank ‘im rrrright... underrr... the... tebble!”

The moment would be nearly perfect, he thought, if only he had his pipe... but he had given that up a few years ago, when the diabetic neuropathy had progressed too far to enjoy keeping the damn thing lit. No sooner had he finished the thought than a complete set of smoking materials suddenly appeared on the table next to him. It even included his favorite brand of pipe tobacco, which had been discontinued by the company around the same time he had given up smoking it. Robert happily filled himself a pipeful, expertly lit it, inhaled the aromatic perfume of the smoke as it swirled on the breeze, puffed once more and then blew a perfectly circular smoke ring on the out breath. Hah! *He could still do it!* The smoke ring stretched and morphed its shape, traveling across the patio, over the stone railing and out of sight. Robert sighed in quiet contentment and settled deeply into his leather chair; pipe in one hand, snifter in the other, feet up on the ottoman that had also miraculously appeared, just as he thought he needed one.

“I could get used to this,” he said, enjoying his own astonishment.

The End

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